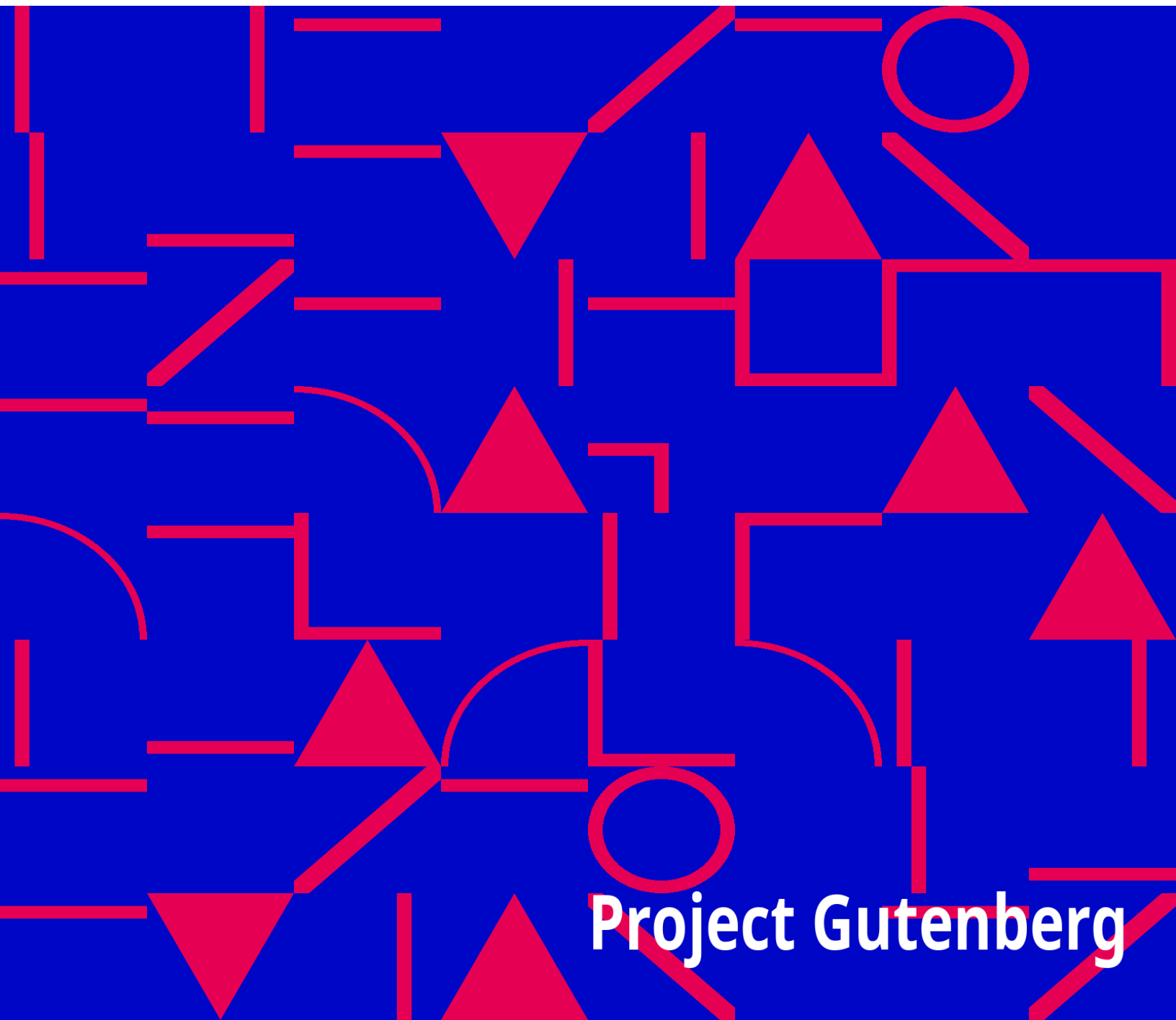


The New Gresham Encyclopedia. Amiel to Atrauli

Various

The background of the lower half of the page is a solid blue field. Overlaid on this field is a dense, repeating pattern of red geometric shapes. These shapes include various sizes of triangles, squares, circles, and lines, some of which are solid red and others which are red outlines. The pattern is abstract and modern, resembling a stylized architectural or mathematical design.

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The New Gresham Encyclopedia. Amiel to Atrauli

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*** START OF THE PROJECT GUTENBERG EBOOK THE NEW
GRESHAM ENCYCLOPEDIA. AMIEL TO ATRAULI ***

Transcriber's
note:

A few typographical errors have been corrected. They appear in the text like this, and the explanation will appear when the mouse pointer is moved over the

marked passage.

**THE
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VOLUME . I . PART . 2



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VOLUME I PART 2

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è, as in her.

ī, as in *pine*, or as *ei* in Ger. *mein*.

i, as in *pin*, also used for the short sound corresponding to ē, as in French and Italian words.

eu, a long sound as in Fr. *jeûne* = Ger. long ö, as in *Söhne*, *Goethe* (Goethe).

eu, corresponding sound short or medium, as in Fr. *peu* = Ger. ö short.

ō, as in *note*, *moan*.

o, as in *not*, *soft*—that is, short or medium.

ö, as in *move*, *two*.

ū as in *tube*.

u, as in *tub*: similar to è and also to *a*.

u", as in *bull*.

ü, as in Sc. *abune* = Fr. *û* as in *dû*, Ger. *ü* long as in *grün*, *Bühne*.

u', the corresponding short or medium sound, as in Fr. *but*, Ger. *Müller*.

oi, as in *oil*.

ou, as in *pound*; or as *au* in Ger. *Haus*.

CONSONANTS

Of the *consonants*, **b, d, f, h, j, k, l, m, n, ng, p, sh, t, v, z**, always have their common English sounds, when used to transliterate foreign words. The letter **c** is not used by itself in re-writing for pronunciation, **s** or **k** being used instead. The only consonantal symbols, therefore, that require explanation are the following:—

ch is always as in *rich*.

d, nearly as *th* in *this* = Sp. *d* in *Madrid*, &c.

g is always hard, as in *go*.

h represents the guttural in Scotch *loch*, Ger. *nach*, also other similar gutturals.

n', Fr. nasal *n* as in *bon*.

r represents both English *r*, and *r* in foreign words, which is generally much more strongly trilled.

s, always as in *so*.

th, as *th* in *thin*.

th, as *th* in *this*.

w always consonantal, as in *we*.

x = *ks*, which are used instead.

y always consonantal, as in *yea* (Fr. *ligne* would be re-written *lēny*).

zh, as *s* in *pleasure* = Fr. *j*.

Amiel', Henri Frédéric, French-Swiss philosophical writer and poet, born at Geneva, 1821, died there 1881. Educated at Geneva, he resided a considerable time abroad, especially in Germany, and was much influenced by German thought and science. On his return he first held the chair of æsthetics, and then that of philosophy. He published several volumes of poetry as well as other works, but he is best known by his *Journal Intime*, published after his death, and translated into English (1885), with a critical study by Mrs. Humphry Ward. It shows great critical and philosophical power, but is pessimistic.

Amiens (a' -mē-an'), a town of France, capital of the department of Somme, on the railway from Boulogne to Paris. It has a citadel, wide and regular streets, and several large open areas; a cathedral, one of the largest and finest Gothic buildings in Europe, founded in 1220 by Bishop Evrard, after designs made by the architect Robert de Luzarches. Having water communication with the sea by the Somme, which is navigable for small

vessels, it has a large trade and numerous important manufactures, especially cotton and woollen goods. It was taken by the Germans in 1870, and again in 1914, by General von Kluck. Pop. (1911) 93,207.—The *Peace of Amiens*, concluded between Great Britain, France, Spain, and the Batavian Republic, 27th March, 1802, put an end for a time to the great war which had lasted since 1793.

Amir. See *Emir*.

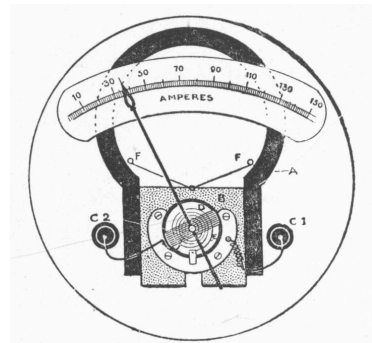
Amirante Islands (a' -mē-ra' n'tā), a group of eleven small islands in the Indian Ocean, lying south-west of the Seychelles, and forming a dependency of Mauritius.

Amlwch (am'löh), a seaport in North Wales, Island of Anglesey. Pop. (1921), 2694 (urb. dist.).

Ammana'ti, Bartolomeo, a sculptor and architect, born at Florence in 1511, died 1592; executed the *Leda* at Florence, a gigantic *Neptune* for St. Mark's Place at Venice, a colossal *Hercules* at Padua, and after the inundation in 1557, which destroyed all the bridges of the Arno, built the celebrated Trinity Bridge at Florence, finished in 1570. He was an imitator of Michael Angelo without his inspiration and genius.

Ammergau (a' m'er-gou), a district in Upper Bavaria, having its centre in the villages of Ober and Unter Ammergau. The former village is famous on account of the Passion Play which is performed there, at intervals usually of ten years.

Am'meter (short for ampere-meter), an instrument used for the measurement of electric currents. For commercial use the scale is marked so as to read amperes directly, but for experimental purposes it is usual to have a scale with divisions numbered in tens, in which case the reading multiplied by a suitable constant gives the value of the current in amperes. By employing suitable shunts this admits of the one instrument being used for a number of ranges.



Ammeter.—Front removed to show details.

A. Large magnet. B. Soft-iron keeper magnetized by

The types of ammeter and the principles upon which they work are as follows: (a) *Soft-iron type*, the action of a magnetic field on a piece of soft iron; (b) *moving-coil type* and *dynamometer type*, the action of a magnetic field on a current-carrying coil; (c) *hot-wire type*, the expansion of a conductor due to the heating produced by the current; (d) *induction type*, the action of a magnetic field on the eddy currents produced in a metal disc.

magnet and acting as resistance. D. Cylinder turning within B, and actuated by current entering at C1, and flowing through spiral wire (not shown) at base of D, and through coil on cylinder to terminal C2. E. Hair-spring regulating pointer. F. Pointer stops.

The "soft-iron" ammeter can be used for both direct and alternating currents, is inexpensive, and is sufficiently accurate for commercial use.

For direct-current measurements where a high degree of accuracy is of first importance, a "moving-coil" ammeter is invariably used.

In alternating-current circuits its place is taken by the dynamometer type, which reads both direct and alternating currents.

In cases where absence of inductance in the instrument is important, e.g. in the measurements in wireless-telegraph and telephone circuits, the "hot-wire" ammeter is used. It measures both direct and alternating currents, and, when properly used, has a high degree of accuracy.

The "induction" type cannot be used for direct currents, and has the limitation that with alternating currents it will read correctly only at the frequency for which it is calibrated.

Almost invariably an ammeter gives its full-scale reading when a small current, say of the order of one-tenth of an ampere, is passing through the instrument itself. In order to read larger currents a device is employed whereby a definite fraction of the current to be measured passes through the instrument.—BIBLIOGRAPHY: J. A. Fleming, *A Handbook for the Electrical Laboratory and Testing Room* (2 vols.); G. D. Aspinall Parr, *Electrical Measuring Instruments*.

Ammia' nus Marcelli' nus, a Roman historian, born at Antioch in Syria about 320, died about 390. He wrote in 31 books (of which the first 13 are lost) a history of the Cæsars, from Nerva to Valens, which was highly

thought of by Gibbon for its fidelity. His MS. was printed for the first time at Rome in 1474.

Am'mon (often called **Ammon-Ra**, i.e. Ammon-Sun), an ancient Egyptian deity, one of the chief gods of the country, identified by the Greeks with their supreme god Zeus, while the Romans regarded him as the representative of Jupiter; represented as a ram, as a human being with a ram's head, ornamented with the solar disc, or simply with the horns of a ram. There was a celebrated temple of Ammon in the Oasis of Siwah in the Libyan desert.



Ammon.

Ammon, Oasis of. See *Siwah*.

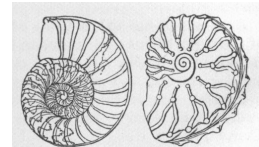
Ammo'nia, an alkaline substance, which differs from the other alkalis by being gaseous, and is hence sometimes called the *volatile alkali*. It is a colourless pungent gas, composed of nitrogen and hydrogen; formula, NH_3 . It was first prepared by Priestley, who termed it *alkaline air*. He obtained it from sal-ammoniac by the action of lime, by which method it is yet generally prepared. It is used for many purposes, both in medicine and scientific chemistry; not, however, in the gaseous state, but frequently in solution in water, under the names of *liquid ammonia*, *aqueous ammonia*, or *spirits of hartshorn*. It is generally prepared from the ammoniacal liquor obtained as a by-product on distilling coal. Combined with acids, ammonia forms salts which are of immense value to agriculture. The well-known odour of farmyard manure is very largely due to the formation of ammonia during the rotting of the dung. Many animal substances, such as bones, clippings and shavings of horn, hoof, &c., and certain vegetable matters yield ammonia when heated. Sal-ammoniac is ammonium chloride.

Ammoni'ăcum, a gum-resinous exudation from an umbelliferous plant, the *Dorēma ammoniăcum*. It has a fetid smell, is inflammable, soluble in water and spirit of wine; used as an antispasmodic, stimulant, and expectorant in chronic catarrh, bronchitic affections, and asthma; also used for plasters.

Ammo'niaphone, an instrument, consisting of a metallic tube containing some substance saturated with ammonia, peroxide of hydrogen, and a few flavouring compounds, fitted with a mouthpiece to breathe through, which is said to render the voice strong, clear, rich, and ringing by the inhalation

of the ammoniacal vapour. It was invented by Dr. Carter Moffat, and was suggested by the presence of ammonia in some quantity in the atmosphere of Italy—the country of fine singers.

Am´monites, a group of fossil cephalopods, now divided into a large number of genera, ranging from the Coal Measures (Texas) to the uppermost cretaceous strata. The ammonites differ from the nautili in having the tube connecting the chambers placed on the outer margin of the coiled shell, while the calcareous neck where it passes through the partitions is directed forwards. The partitions are much folded, producing markings like the fronds of ferns where they meet the inner wall of the shell. The name arises from confusion with a coiled gastropod, which was held to resemble the horns of the Egyptian deity Jupiter Ammon.



Ammonites obtusus.
Ammonites varians

Am´monites, a Semitic race frequently mentioned in Scripture, descended from Ben-Ammi, the son of Lot (*Gen. xix, 38*), often spoken of in conjunction with the Moabites. A predatory and Bedouin race, they inhabited the desert country east of Gad, their chief city being Rabbath-Ammon (Philadelphia). Wars between the Israelites and the Ammonites were frequent; they were overcome by Jephthah, Saul, David, Uzziah, Jotham, &c. They appear to have existed as a distinct people in the time of Justin Martyr, but have subsequently become merged in the aggregate of nameless Arab tribes.

Ammo´nium, the name given to the hypothetical radicle (formula, NH_4) of ammonium salts. It functionates as a metal, has not been isolated, but it is believed to exist in an amalgam with mercury.

Ammo´nius Sac´cas, a Greek philosopher who lived about A.D. 175-240. Originally a porter in Alexandria, he derived his epithet from the carrying of *sacks* of corn. The son of Christian parents, he abandoned their faith for the polytheistic philosophy of Greece. His teaching was historically a transition stage between Platonism and Neo-Platonism. Among his disciples were Plotinus, Longinus, Origen, &c. The books often attributed to him are by a Christian philosopher of the same name.

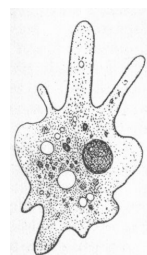
Ammuni'tion, another form of the word munition, with a more restricted meaning. It is now taken to include cartridges of all sorts for guns, howitzers, rifles, and all fire-arms. Ammunition comprises both cartridges in which explosive and missiles are combined to form one compact article, and also other forms of projectiles of which the explosive agent forms one portion and the actual missile the other. Bombs, grenades, shells, powder, and bullets are all included in the generic term ammunition. As a qualifying word used adjectivally it is found in ammunition-wagon, ammunition-carrier, ammunition-mules, ammunition-column, &c. In the British service the Royal Army Ordnance Corps is entrusted with the provision of supplies of ammunition generally, while the actual distribution in the field is the duty of the ammunition-column, a Royal Artillery organization.

Am'nesty (Gr. *amnestia*, forgetfulness), the releasing of a number of persons who have been guilty of political offences from the consequence of these offences. The earliest recorded amnesty in history is that of Thrasybulus at Athens, and the last act of amnesty passed in Britain was that of 1747, after the second Jacobite rebellion.

Am'nion, the innermost membrane surrounding the fetus of mammals, birds, and reptiles.—In botany, a gelatinous fluid in which the embryo of a seed is suspended, and by which it is supposed to be nourished.

Amo'aful, village near Kumassi, West Africa, at which the Ashanti were defeated by British troops under Wolseley, 31st Jan., 1874.

Amœ'ba, a microscopic genus of rhizopodous Protozoa, of which *A. difflūens*, common in freshwater ponds and ditches, is the type. It exists as a mass of protoplasm, and pushes its body out into finger-like processes or pseudopodia, and by means of these moves about or grasps particles of food. There is no distinct mouth, and food is engulfed within any portion of the soft sarcode body. Reproduction takes place by fission, or by a single pseudopodium detaching itself from the parent body and developing into a separate amœba.



Amœba
proteus.

Amœbe'an Poetry, poetry in which persons are represented as speaking alternately, as in some of Virgil's *Eclogues*.

Amol´, a town of Northern Persia, 76 miles N.E. of Teheran. Extensive ruins tell of former greatness, the most prominent being the mausoleum of Seyed Quam-u-deen, who died in 1378. Pop. in winter estimated at about 40,000.

Amo´mum, a genus of plants of the nat. ord. Zinziberaceæ (ginger, &c.), natives of warm climates, and remarkable for the pungency and aromatic properties of their seeds. Some of the species yield Cardamoms, others Grains of Paradise.

Amontilla´do, a dry kind of sherry wine of a light colour, highly esteemed.

Amoor. See *Amur*.

A´mor, the god of love among the Romans, equivalent to the Gr. *Erōs*.

Amor´go (ancient **Amorgos**), an island in the Grecian Archipelago, one of the Eastern Cyclades, 22 miles long, 5 miles broad; area, 106 sq. miles; it has a town of the same name, with a castle and a large harbour. Pop. 3561.

Am´orites, a powerful Canaanitish tribe at the time of the occupation of the country by the Israelites; occupied the whole of Gilead and Bashan, and formed two powerful kingdoms—a northern, under Og, who is called King of Bashan; and a southern, under Sihon, called King of the Amorites; first attacked and overthrown by Joshua; subsequently subdued, and made tributary or driven to mingle with the Philistines and other remnants of the Canaanitish nations.

Amorphous Rocks or **Minerals**, those having no regular structure, or without crystallization, even in the minutest particles.

Amorphozo´a, a term applied to some of the lower groups of animals, as the sponges and their allies, which have no regular symmetrical structure.

Amortiza´tion, in law, the alienation of real property to corporations (that is, in *mortmain*), prohibited by several English statutes.

A´mos, one of the minor prophets; flourished under the Kings Uzziah of Judah and Jeroboam II of Israel (810 to 784 B.C. by the common chronology). Though engaged in the occupations of a peasant he must have had a considerable amount of culture, and his book of prophecies has high

literary merits. It contains denunciations of Israel and the surrounding nations, with promises of the Messiah.

Amoy', an important Chinese trading port, on a small island off the south-east coast opposite Formosa; has a safe and commodious harbour, and its merchants are among the wealthiest and most enterprising in China; one of the five ports opened to foreign commerce by the treaty of Nanking in 1842. The privilege was confirmed and extended by the treaty of Tien-tsin in 1858, and the port is now open to all countries. Pop. 114,000.

Ampel'idæ. See *Chatterers*.

Ampère (a' n' -pār), André-Marie, a celebrated French mathematician and philosopher, founder of the science of electro-dynamics, born at Lyons in 1775, died at Marseilles in 1836; professor of mathematical analysis at the Polytechnic School, Paris, and of physics at the College of France. What is known as *Ampère's Theory* is that magnetism consists in the existence of electric currents circulating round the particles of magnetic bodies, being in different directions round different particles when the bodies are unmagnetized, but all in the same direction when they are magnetized.

Ampère, Jean-Jacques-Joseph-Antoine, historian and professor of French literature in the College of France; the only son of André-Marie Ampère; born at Lyons 1800, died 1864; chief works: *Histoire Littéraire de la France avant le 12^e siècle* (1839); *Introduction à l'Histoire de la Littérature française au moyen âge* (1841); *Littérature, Voyages et Poésies* (1833); *La Grèce, Rome et Dante, Études Littéraires d'après Nature*; *l'Histoire romaine à Rome* (4 vols. 8vo, 1856-64); *Promenades en Amérique* (1855); *César, Scènes historiques* (1859), full of hostile allusions to the French Empire.

Ampere (am'pār), in electricity, the unit employed in measuring the strength or intensity of an electric current, being equivalent to the current produced by the electro-motive force of one volt in a wire having the resistance of one ohm. The name (cf. *Farad, Coulomb, Watt, &c.*) is derived from that of the well-known physicist, Ampère. An *ampere-meter* or *ammeter* is an instrument by which the strength of an electric current is given in amperes.

Amphib'ia, a class of vertebrate animals, which in their early life breathe by gills or branchiæ, and afterwards partly or entirely by lungs. The Frog, breathing in its tadpole state by gills and afterwards throwing off these organs and breathing entirely by lungs in its adult state, is an example of the latter phase of amphibian existence. The Proteus of the underground caves of Central Europe exemplifies forms in which the gills of early life are retained throughout life, and in which lungs are developed in addition to the gills. A second character of this group consists in the presence of two occipital 'condyles', or processes by means of which the skull articulates with the spine or vertebral column; Reptiles possessing one condyle only. The class is divided into four orders: the Ophiomorpha (or serpentiform), represented by the Blindworms, in which limbs are wanting and the body is snake-like; the Urodela or 'Tailed' Amphibians, including the Newts, Proteus, Siren, &c.; the Anoura, or Tailless Amphibia, represented by the Frogs and Toads; and the Labyrinthodontia, which includes the extinct forms known as Labyrinthodons. The term Amphibia was originally employed by Linnæus in his *Systema Naturæ*, and adopted by Cuvier in his *Tableau Élémentaire*. See *Batrachia*.

Amphibol'ogy, in logic, an equivocal phrase or sentence, not from the double sense of any of the words, but from its admitting a double construction, as 'The duke yet lives that Henry shall depose'.

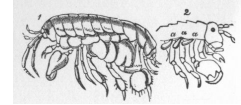
Amphic'tyonic League (or Council), in ancient Greece, a confederation of tribes for the protection of religious worship, but which also discussed questions of international law, and matters affecting their political union. The most important was that of the twelve northern tribes which met alternately at Delphi and Thermopylæ. The tribes sent two deputies each, who assembled with great solemnity; composed the public dissensions, and the quarrels of individual cities, by force or persuasion; punished civil and criminal offences, and particularly transgressions of the law of nations, and violations of the temple of Delphi. Its calling on the States to punish the Phocians for plundering Delphi caused the Sacred Wars, 595-586, 448-447, 357-346 B.C.

Amphi'on, in Greek mythology, son of Zeus and Antiöpē, and husband of Niöbē. He had miraculous skill in music, being taught by Mercury, or, according to others, by Apollo. In poetic legend he is said to have availed

himself of his skill when building the walls of Thebes—the stones moving and arranging themselves in proper position at the sound of his lyre. See *Zethus*.

Amphioxus. See *Lancelet*.

Amphip'oda, an order of sessile-eyed malacostracan crustaceans, with feet directed partly forward and partly backward. Many species are found in springs and rivulets, others in salt water. The sand-hopper and shore-jumper are examples.

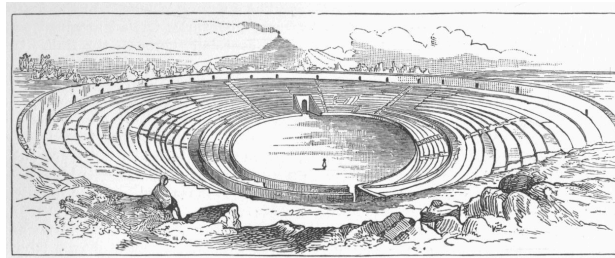


Amphipoda
1. Shore-jumper
(*Orchestia littoralis*), 2.
Portion showing
the respiratory
organs *a a a*.

Amphip'rostyle, in architecture, said of a structure having the form of an ancient Greek or Roman oblong rectangular temple, with a prostyle or portico on each of its ends or fronts, but with no columns on its sides or flanks.

Amphisbæ'na (Gr., from *amphis*, both ways, and *bainein*, to go), a genus of serpentiform, limbless, lacertilian reptiles; body cylindrical, destitute of scales, and divided into numerous annular segments; the tail obtuse, and scarcely to be distinguished from the head, whence the belief that it moved equally well with either end foremost. There are several species, found in tropical America. They feed on ants and earthworms, and were formerly, but erroneously, deemed poisonous. In Greek mythology the amphisbæna was a serpent believed to possess two heads.

Amphis'cii (Gr. *amphi*, on both sides, and *skia*, shadow), a term sometimes applied to the inhabitants of the intertropical regions, whose shadows at noon in one part of the year are cast to the north and in the other to the south, according as the sun is in the southern or northern signs.



Amphitheatre at Pompeii

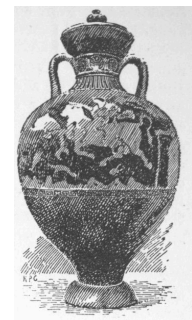
Amphitheatre, an ancient Roman building of an oval form without a roof, having a central area (the *arena*) encompassed with rows of seats, rising higher as they receded from the centre, on which people used to sit to view the combats of gladiators and of wild beasts, and other sports. The first amphitheatre at Rome was that constructed by C. Scribonius Curio, 59 B.C. The Colosseum at Rome is the largest of all the ancient amphitheatres, being capable of containing 100,000 persons, 87,000 of whom occupied numbered and reserved seats. That at Verona is one of the best examples remaining. Its dimensions are 502 feet by 401, and it is 98 feet high. The name means 'both-ways theatre', or 'theatre all round', the theatre forming only a semicircular building.

Amphitriṭē, in Greek mythology, daughter of Oceanus and Tethys, or of Nereus and Doris, and wife of Poseidon (or Neptune), represented as drawn in a chariot of shells by Tritons, with a trident in her hand. In the Homeric poems she is the personification of the Sea, and her marriage to Poseidon is alluded to in a number of scenes depicted on ancient monuments. Such are a bas-relief in the glyptothek at Munich and a mosaic in the museum at Naples.

Amphitryon, in Greek legend, King of Thebes, son of Alcæus, and husband of Alcmena. Plautus, and after him Molière, have made an amour of Jupiter with Alcmena the subject of amusing comedies.

Amphiuma, a genus of amphibians which frequent the lakes and stagnant waters of North America. The adults retain the clefts at which the gills of the tadpole projected.

Amphōra, a vessel used by the Greeks and Romans for holding liquids; commonly tall and narrow, with two handles and a pointed end which fitted into a stand or was stuck in the ground to enable it to stand upright; used also as a cinerary urn, and as a liquid measure—Greek = 9 gallons; Roman = 6 gallons.



Amphora

Amplexicaul, in botany, said of a leaf that embraces and nearly surrounds the stem.

Am'plitude, in astronomy, the distance of any celestial body (when referred by a secondary circle to the horizon) from the east or west points.

From a Roman
specimen in
the British
Museum

Ampt'hill, a market-town of England, Bedfordshire, about 7 miles south-west of Bedford. Pop. (1921), 2269.

Ampul'la, the Latin name for a vessel bellying out like a jug, which contained unguents for the bath; also a vessel for drinking at table. The ampulla has also been employed for ceremonial purposes, such as holding the oil or chrism used in various Church rites and for anointing monarchs at their coronation. The ampulla of the English sovereigns now in use is an eagle, weighing about 10 oz., of the purest chased gold, which passed through various hands to the Black Prince. The ampulla of the French kings, kept at Rheims in the tomb of St. Remy, was destroyed in 1793.

Amputa'tion, in surgery, that operation by which a member is separated from the body.

Amra'oti, a town of British India in Berár; it is celebrated for its cotton, and is a place of good trade. Pop. 35,000. The district has an area of 4733 sq. miles. Pop. 876,000.

Am'ritsir, or **Amritsar** ('the pool of immortality'), a flourishing commercial town of India, capital of a district of the same name, in the Punjab, the centre of the Sikh religion since the end of the sixteenth century. It has considerable manufactures of shawls and silks; and receives its name from the sacred pond constructed by Ram Das, the apostle of the Sikhs, in which the Sikhs and other Hindus immerse themselves that they may be purified from all sin. Pop. 152,756.—The district of Amritsir has an area of 1601 sq. miles. Pop. 900,000.

Am'ru, originally an opponent, and subsequently a zealous supporter of Mahomet, and one of the ablest of the Mahomedan warriors. He brought Egypt under the power of the Caliph Omar in 638, and governed it wisely till his death in 663. The burning of the famous Alexandrian Library has been generally attributed to him, though only on the authority of a writer who lived six centuries later.

Am'sterdam (that is, 'the dam of the Amstel'), one of the chief commercial cities of Europe, capital of Holland (but not the residence of the sovereign), situated at the confluence of the Amstel with the Y or IJ, an arm of the Zuider-Zee. On account of the lowness of the site of the city the greater part of it is built on piles. It is divided by numerous canals into about 90 islands, which are connected by nearly 300 bridges. Many of the streets have a canal in the middle with broad brick-paved quays on either side, planted with rows of trees; the houses are generally of brick, many of them six or seven stories high, with pointed gables turned to the streets. Among the public buildings are the old stadthouse, the work of Jacob van Kempen, commenced in 1648 and finished in 1655, which is now a royal palace, the interior being decorated by the Dutch painters and sculptors of the seventeenth century with their masterpieces; the justiciary hall, an imitation of a Greek temple; the town hall (fourteenth century); the exchange, a handsome building, constructed in 1836 on the site of the old bourse built in 1608; the Palace of National Industry; the national museum; and the central railway station. The old church is a structure of the fourteenth century with stained-glass windows painted by Digman in the fifteenth century. The chief educational institutions of the kingdom are here, including the city university, a free university, gymnasiums and other secondary schools, the national picture gallery or museum, containing many masterpieces of Dutch artists, &c. Among its numerous industries may be mentioned as a speciality the cutting and polishing of diamonds. It has also factories and workshops dealing with wool, cotton, silk, tobacco, leather, machinery, and metal goods, glass, liqueurs, cocoa, &c. The harbour, formed by the Y, lies along the whole of the north side of the city, and is surrounded by various docks and basins. The trade is very great, being much facilitated by the great ship-canal (15 miles long, opened 1876, admitting the largest vessels) connecting the Y directly with the North Sea at Y-Muiden, where the entrance is between two long piers projecting into the sea. Another canal of much less importance, the North Holland Canal (46 miles long, 20 feet deep), connects Amsterdam with the Helder. Between the harbour and the Zuider Zee the Y is now crossed by a great dam in which are locks to admit vessels and regulate the amount of water in the North Sea Canal. The oversea trade of Amsterdam has immensely increased since the opening of the great canal, and the foreign trade of the kingdom practically centres in Amsterdam and Rotterdam. There is also a large trade with the interior by

railway, river, and canal. In the beginning of the thirteenth century Amsterdam was but a fishing village. In the fourteenth and fifteenth centuries it had attained some importance, especially through the Baltic trade. The ruin of Antwerp through the troubles with Spain was greatly to its advantage, and during the seventeenth and eighteenth centuries Amsterdam was one of the wealthiest and most flourishing cities in the world. Its forced alliance with France ruined its trade, but since 1813 its commerce has revived. Amsterdam is the birthplace of Spinoza and of the painters van de Velde and van der Neer. Pop. (1919), 647,120.

Amsterdam, a town of New York State, United States, on the Mohawk River and the Erie Canal, 33 miles N.W. of Albany; a busy manufacturing town. Pop. (1920), 33,524.

Amsterdam, New, a town in British Guiana, on the east side of the River Berbice, near the sea, with some trade as a seaport. Pop. 8903.

Amsterdam Island, a small and almost inaccessible island in the Indian Ocean, about halfway in a direct line between the Cape of Good Hope and Tasmania. It is sparsely provided with vegetation, and inhabited only by sea-birds, but it was taken possession of by France in 1893, along with the neighbouring St. Paul. It was discovered by the Dutch in 1633.

Amstetten, a town in Lower Austria, on the Ybbs, and on the railway from Vienna to Linz. In 1805 a victory was gained here by the French under Murat over the Russians under Bagration. Pop. 3760.

Amuck', or **Amuk**, to run, a phrase applied to natives of the Eastern Archipelago, who are occasionally seen to rush out in a frantic state of temporary mental derangement, making indiscriminate and murderous assaults on all that come in their way.

Amu-Darya. See *Oxus*.

Amu-Darya, district. See *Turkestan*.

Am'ulet, a piece of stone, metal, &c., marked with certain figures or characters, which people in some countries wear about them, superstitiously deeming them a protection against diseases, enchantments, witchcraft, &c. According to Pliny the elder, the *bullā*, or amulet, was first hung by

Tarquinius Priscus on the neck of his son. Articles that archaeologists have decided to be amulets have been found dating from prehistoric times, and they were commonly worn in ancient times by the Jews, Greeks, and Romans, as they still are by Persians, Arabs, and many other peoples. See *Charms*.

Amundsen, Captain Roald, Norwegian polar explorer, born at Borge, Norway, 16th July, 1872. He was first-lieutenant on the *Belgica* during the Belgian south polar expedition, 1897-9. He then planned an expedition to the area of the north magnetic pole and a north-west passage by water. On 17th June, 1903, he embarked from Christiania on the small sailing vessel the *Gjøa*, with a company of six men, and reached King William Land, where the vessel remained for two years. Here he made his headquarters, and by numerous excursions was able to prove that the north magnetic pole has no stationary position, but is in continual movement. On 11th July, 1906, his vessel reached the Behring Strait, and on 30th August entered the Pacific. After his return Amundsen began his preparations for an Antarctic expedition, and on 9th August, 1910, he sailed from Norway on Nansen's ship, the *Fram*, and reached the South Pole on 7th March, 1912. He published an account of his North-West Passage expedition, entitled *Sydpolen. Den norske Sydpolsfaerd med Fram 1910-12*. An English translation was published in 1913. Amundsen started on a North Polar Expedition in 1918.

Amur', or **Amoor'**, one of the largest rivers of Eastern Asia, formed by the junction of the Rivers Shilka and Argun; flows first in a south-eastern and then in a north-eastern direction till it falls into an arm of the Sea of Okhotsk, opposite the Island of Sakhalin, after a course of 1500 miles. It forms, for a large portion of its course, part of the boundary-line between the Russian and the Chinese dominions, and is navigable throughout for four months in the year.—*Amoor Territory*. In 1858 Russia acquired from China the territory on the left bank of the Upper and Middle Amoor, together with that on both banks of the Lower Amoor. The western portion of the territory was organized as a separate province, with the name of the Amoor (area, 154,795 sq. miles. Pop. 261,500). The eastern portion was joined to the Maritime Province of Eastern Siberia.

Am'urath, or **Murad**, the name of several Ottoman sultans. See *Ottoman Empire*.

Amyclæ (a-mī'klē), a town of ancient Greece, the chief seat of the Achæans in Laconia, a short distance from Sparta, by which it was conquered about 800 B.C.

Amyg'daloid (Gr. *amygdalē*, an almond), meaning 'almond-shaped', a term used in anatomy and geology.

Amyg'dalus, the genus to which the almond belongs.

Am'yl, in chemistry, a hypothetic radicle believed to exist in many compounds, especially the fusel-oil series, and having the formula C_5H_{11} .—*Amyl Nitrite*, or *Nitrite of Amyl*, an amber-coloured fluid, smelling and tasting like essence of pears, which has been employed as an anæsthetic and also in relieving cardiac distress, as in angina pectoris.

Am'ylene (C_5H_{10}), an ethereal liquid with an aromatic odour, prepared from fusel-oil. It possesses anæsthetic properties, and has been tried as a substitute for chloroform, but is very dangerous.

Amyl'ic Alcohol, one of the products of the fermentation of grain, &c., commonly known by the name of fusel-oil (q.v.).

Amyot (ä-mi-ō), Jacques, French writer and scholar, whose translations from the Greek have themselves become classics, was born in 1513, and died Bishop of Auxerre in 1593, having been for twelve years a professor of classics at Bourges, and having enjoyed the patronage of Margaret of Navarre and Henry II. His chief translations are those of Plutarch's *Lives* and his *Morals*, the *Aethiopica* of Heliodorus, and the *Daphnis and Chloe* of Longus. Sir Thomas North's English translation of Plutarch (1575), of which Shakespeare made much use, was derived from that of Amyot.

Amyrida'ceæ, a nat. ord. of plants, consisting of tropical trees or shrubs, the leaves, bark, and fruit of which abound in fragrant resinous and balsamic juices. Myrrh, frankincense, and the gum-elemi of commerce are among their products. Among the chief genera of the order are *Amyris*, *Balsamodendron*, *Boswellia*, and *Canarium*.

A´na, the neuter plural termination of Latin adjectives in *-ānus*, often forming an affix with the names of eminent men to denote a collection of their memorable sayings—thus *Scaligeriana*, *Johnsoniana*, the sayings of Scaliger, of Johnson; or to denote a collection of anecdotes, or gossip matter, as in *boxiana*. Hence, as an independent noun, books recording such sayings; the sayings themselves.

Anabap´tists (from the Gr. *anabaptizein*, to rebaptize), a name given to a Christian sect by their adversaries, because, as they objected to infant baptism, they rebaptized those who joined their body. Their doctrine is based upon the words of Christ in *St. Mark*, xvi, 16. The founder of the sect appears to have been Nicolas Storch, a disciple of Luther's, who seems to have aimed also at the reorganization of society based on civil and political equality. Gathering round him a number of fiery spirits, among whom was Thomas Münzer, he incited the peasantry of Suabia and Franconia to insurrection—the doctrine of a community of goods being now added to their creed. This insurrection was quelled in 1525, when Münzer was put to the torture and beheaded. After the death of Münzer the sectaries dispersed in all directions, spreading their doctrines wherever they went. In 1534 the town of Münster in Westphalia became their centre of action. Under the leadership of Bockhold and Matthias their numbers increased daily, and being joined by the restless spirits of the adjoining towns, they soon made themselves masters of the town and expelled their adversaries. Matthias became their prophet, but he fell in a sally against the Bishop of Münster, Count Waldeck, who had laid siege to the city. Bockhold then became leader, assuming the name of John of Leyden, King of the New Jerusalem, and Münster became a theatre of all the excesses of fanaticism, lust, and cruelty. The town was eventually taken (June, 1535), and Bockhold and a great many of his partisans suffered death. This was the last time that the movement assumed anything like political importance. In the meantime some of the apostles, who were sent out by Bockhold to extend the limits of his kingdom, had been successful in various places, and many independent teachers, who preached the same doctrines, continued active in the work of founding a new empire of pure Christians. It is true that they rejected the practice of polygamy, community of goods, and intolerance towards those of different opinions which had prevailed in Münster; but they enjoined upon their adherents the other doctrines of the early Anabaptists, and certain

heretical opinions in regard to the humanity of Christ, occasioned by the controversies of that day about the sacrament. The most celebrated of those Anabaptist prophets were Melchior Hoffmann, the founder of the Hoffmannists or Millenarians; Galenus Abrahamssohn, from whom the sect of the Galenists were called; and Simon Menno, founder of various sects known as Mennonites. Menno's principles are contained in his *Principles of the True Christian Faith* (1556), a work which is held as authoritative on points of doctrine and worship among the Baptist communities at the present day. The application of the term Anabaptist to the general body of Baptists throughout the world is unwarranted, because these sects have nothing in common with the bodies which sprung up in various countries of Europe during the Reformation, except the practice of adult baptism. The Baptists themselves repudiate the name Anabaptist, as they claim to baptize according to the original institution of the rite, and never repeat baptism in the case of those who in their opinion have been so baptized. It is under the designation of Mennonites that they exist to-day, principally in Holland, Germany, and the United States.

An´abas. See *Climbing-perch*.

Anab´asis (Gr. *anabasis*, a march up country), the title of Xenophon's celebrated account of the expedition of Cyrus the Younger against his brother Artaxerxes, King of Persia. The title is also given to Arrian's work which records the campaigns of Alexander the Great.

An´ableps, a genus of fishes of the perch family, found in the rivers of Guiana, consisting of but one species, remarkable for a peculiar structure of the eyes, in which there is a division of the iris and cornea, by transverse ligaments forming two pupils, and making the whole eye appear double. The young are brought forth alive.

Anabolism (Gr. *ana*, up, and *bolé*, a throw), a biological term suggested by Michael Foster, and used by Gaskell in 1886, and meaning the building-up of organic life, or the process by which a substance is transformed into another which is more complex. Anabolism is the constructive phase of metabolism (q.v.).

Anacanthi´ni (Gr. neg. prefix *an*, and *akantha*, a spine), an order of osseous fishes, including the cod, plaice, &c., with spineless fins, cycloid or

ctenoid scales, the ventral fins either absent or below the pectorals, and ductless swim-bladder.

Anacardia'ceæ, a nat. ord. of plants, consisting of tropical trees and shrubs which secrete an acrid resinous juice, which is often used as a varnish. Mastic, Japan lacquer, and Martaban varnish are some of their products. The cashoo or cashew (genus *Anacardium*), the pistacia, sumach, mango, &c., are members of the order.

Anach'aris, a genus of plants, nat. ord. Hydrocharidaceæ, the species of which grow in ponds and streams of fresh water; water-thyme or water-weed. It appeared in Britain in the nineteenth century. *A. Alsinastrum* has been introduced from North America into European rivers, canals, and ponds, and by its rapid growth in dense tangled masses tends to choke them so as materially to impede navigation. The plants in our canals perfect no seed, their spread being due to vegetative vigour only.

Anach'ronism, an error of chronology by which things are represented as coexisting which did not coexist; applied also to anything foreign to or out of keeping with a specified time. Thus it is an anachronism when Shakespeare, in *Troilus and Cressida*, makes Hector quote Aristotle. There are anachronisms in the *Cid* and the *Nibelungenlied*, and also in Dante's *Inferno*, when the poet introduces pagan mythology into the Christian hell.

Anacolu'thon, a want of grammatical and logical sequence in the structure of a sentence.

Anacon'da, the popular name of two of the largest species of the serpent tribe, viz. a Ceylonese species of the genus *Python* (*P. tigris*), said to have been met with 33 feet long; and *Eunectes murinus*, a native of tropical America, allied to the boa-constrictor, and the largest of the serpent tribe, attaining the length of 40 feet. They frequent swamps and rivers, are without poison fangs, and kill their victims by constriction.

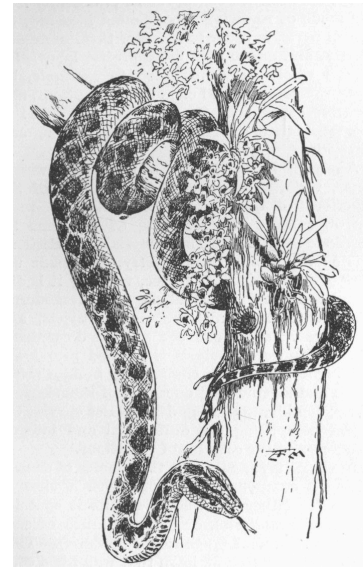
Anaconda, a town of the United States, Montana, with the largest copper-smelting works in the world. Pop. (1920), 11,668.

Anac'reon, an amatory lyric Greek poet of the sixth century B.C., native of Teos, in Ionia. Only a few fragments of his works have come down to us; the collection of odes that usually passes under the name of Anacreon is

mostly the production of a later time, the poetry of the real Anacreon being much less frivolous.

Anadyom'ēnē (Gr., 'she who comes forth'), a name given to Aphroditē (Venus) when she was represented as rising from the sea, as in the celebrated painting by Apelles, painted for the temple of Æsculapius at Cos, and afterwards in the temple of Julius Caesar at Rome.

Anadyr (a ' -nä'dēr), the most easterly of the larger rivers of Siberia and of all Asia; rises in the Stanovoi Mountains, and falls into the Gulf of Anadyr; length, 600 miles.



Anaconda (*Python tigris*)

Anæ'mia (Gr., 'want of blood'), a medical term applied to an unhealthy condition of the body, in which there is a diminution of the red corpuscles which the blood should contain. The principal symptoms are paleness and general want of colour in the skin, languor, emaciation, want of appetite, fainting, palpitation, &c.

Anæsthe'sia, or **Anæsthe'sis**, a state of insensibility to pain, produced by inhaling chloroform, or by the application of other anæsthetic agents.

Anæsthet'ics are medical agents chiefly used in surgical operations for the abolition of pain. They are divided into (1) *general anæsthetics*, those in which complete unconsciousness is produced; (2) *local anæsthetics*, those which act upon the nerves of a limited area alone.

The earliest record of attempts to produce anæsthesia is to be found in the thirteenth century. Since then many agents have been tried. The first scientific effort was in 1800, when Sir Humphry Davy experimented with nitrous oxide, but without practical result. In 1844 Wells, an American dentist, used nitrous oxide, also without result. In 1846 Morton, another American dentist, used ether, and from that time it was increasingly used in America. In the same year the first operation under ether was performed in University College Hospital, London. In 1847 Sir James Simpson (Edinburgh) introduced chloroform. Through his influence it was soon largely used throughout England and Scotland, and continued to be the

chief anæsthetic till about the end of the nineteenth century, when ether again became popular in England. To-day, in England, as always in the United States, ether is the most widely-used anæsthetic. Much controversy exists regarding the respective merits of ether and chloroform. The general opinion is, that ether is on the whole safer, but more liable, in the British climate, to be followed by bronchitis; while there are various conditions when chloroform is still preferable. They are frequently combined in use. *Nitrous-oxide gas* (laughing gas) is much used in dentistry. Lately, nitrous oxide has been used with ether; while ether and oxygen together were much used with the British Expeditionary Force in France during the European War (1914-8). The administration of all anæsthetics is helped when the patient is given a hypodermic injection of morphia shortly before. *Twilight sleep*, increasingly used in childbirth, is the production of a partial anæsthesia by the administration of scopolamin morphine. *Local anæsthetics* are much used in minor surgery, and with proper technique act effectively. Cocaine was the first of these, and is still widely used. Of later developments, eucaine and novocaine are best known. Spinal anæsthesia is the injection of stovaine or similar substance into the spinal cord, producing anæsthesia of a large part of the body, varying according to the site of the injection.

Anagal'lis, a genus of the nat. ord. Primulaceæ, to which belongs the Pimpernel, the 'poor man's weather-glass'. See *Pimpernel*.

Anagni (a' -nän'yē), a town of Italy, province of Rome; the seat of a bishopric erected in 487. Pop. 10,400.

An'agram, the transposition of the letters of a word or words so as to form a new word or phrase, a connection in meaning being frequently preserved; thus, *evil, vile*; *Horatio Nelson, Honor est a Nilo* (honour is from the Nile). The seventeenth century was the golden age of the anagram, but it was employed by the Hebrews and the Greeks.

Anahuac (a' -na' -wa' k'; Mex., 'near the water'), an old Mexican name applied to the plateau of the city of Mexico, from the lakes situated there, generally elevated from 6000 to 9000 feet above the sea.

An'akim, the posterity of Anak, the son of Arba, noted in sacred history for their fierceness and loftiness of stature. Their stronghold was Kirjath-arba

or Hebron, which was taken and destroyed by Caleb and the tribe of Judah.

Anakolu'thon. See *Anacoluthon*.

Analep'tic, a restorative or invigorating medicine or diet.

An'alogue, in comparative anatomy an organ in one species or group having the same function as an organ of different structure in another species or group, as the wing of a bird and that of an insect, both serving for flight. Organs in different animals having a similar anatomical structure, development, and relative position, independent of function or form, such as the arm of a man and the wing of a bird, are termed *homologues*.

Anal'ogy is the mode of reasoning from resemblance to resemblance. When we find on attentive examination resemblances in objects apparently diverse, and in which at first no such resemblances were discovered, a presumption arises that other resemblances may be found by further examination in these or other objects likewise apparently diverse. It is on the belief in a unity in nature that all inferences from analogy rest. The general inference from analogy is always perfectly valid. Wherever there is resemblance, similarity or identity of cause somewhere may be justly inferred; but to infer the particular cause without particular proof is always to reason falsely. Analogy is of great use and constant application in science, in philosophy, and in the common business of life.

Anal'ysis, the resolution of an object, whether of the senses or the intellect, into its component elements. The word was introduced by Boyle in the seventeenth century. In philosophy it is the mode of resolving a compound idea into its simple parts, in order to consider them more distinctly, and arrive at a more precise knowledge of the whole. It is opposed to *synthesis*, by which we combine and class our perceptions, and contrive expressions for our thoughts, so as to represent their several divisions, classes, and relations.

Analysis, in mathematics, is, in the widest sense, the expression and development of the functions of quantities by calculation; in a narrower sense the resolving of problems by algebraic equations. The analysis of the ancients was exhibited only in geometry, and made use only of geometrical assistance, whereby it is distinguished from the analysis of the moderns,

which extends to all measurable objects, and expresses in equations the mutual dependence of magnitudes. Analysis is divided into lower and higher, the lower comprising, besides arithmetic and algebra, the doctrines of functions, of series, combinations, logarithms, and curves, the higher comprising the differential and integral calculus, and the calculus of variations.

In chemistry, analysis is the process of decomposing a compound substance with a view to determine either (*a*) what elements it contains (*qualitative analysis*), or (*b*) how much of each element is present (*quantitative analysis*). Thus by the first process we learn that water is a compound of hydrogen and oxygen, and by the second that it consists of one part of hydrogen by weight to eight parts of oxygen. As a means of testing soils and feeding-stuffs, analysis has given important results; it has enabled Liebig to solve the problem of plant-nutrition.

Anam. See *Annam*.

Anamor'phosis, a term denoting a drawing executed in such a manner as to present a distorted image of the object represented, but which, when viewed from a certain point, or reflected by a curved mirror or through a polyhedron, shows the object in its true proportions.

An'ānas. See *Pine-apple*.

Anapa', a seaport of Russia in province Kuban, on the Black Sea, 50 miles south-east of Kertsh, constructed by the Turks in 1781, and formerly fortified. Pop. about 7000.

An'apæst, in prosody, a foot consisting of two short and one long syllable, or two unaccented and one accented syllable, e.g.


 The As-syr-ian came down like the wolf on the fold.
 —(Byron's *The Destruction of Sennacherib*.)

An'aplasty, a surgical operation to repair superficial lesions, or make up for lost parts, by the employment of adjacent healthy structure or tissue. Artificial noses, &c., are thus made.

Anarajapoo'ra, or **Anuradhapura**, a ruined city, the ancient capital of Ceylon, built about 540 B.C., and said to have covered an area of 300 sq. miles, doubtless a great exaggeration. There are still several dagobas in tolerable preservation, but the great object of interest is the sacred Bo-tree planted over 2000 years, and probably the oldest historical tree in the world, but shattered by a storm in 1887.

An'archists, a revolutionary sect or body setting forth as the social ideal the extreme form of individual freedom, holding that all government is injurious and immoral, and that the destruction of every social form now existing must be the first step to the creation of a new social system. According to Herbert Spencer, anarchism is the doctrine of *laissez faire*. Anarchists usually look upon Diderot as one of their pioneers, and quote his lines: "La nature n'a fait ni serviteurs ni maîtres. Je ne veux ni donner ni recevoir de lois." Historically, however, it is Proudhon who may be considered as the father of anarchism. The recognition of the anarchists as an independent sect may be dated from the secession of Bakunin and his followers from the Social Democrats at the congress of the Hague in 1872, since which they have maintained an active propaganda. Their principal journals have been *La Révolte* (Paris), the *Freiheit* (New York), *Liberty* (Boston), and the *Anarchist* (London). Among modern philosophers of anarchism are Elisée Reclus and Prince Kropotkin.

Anarthrop'oda, one of the two great divisions (the Arthropoda being the other) of the Annulosa, or ringed animals, in which there are no articulated appendages. It includes the leeches, earth-worms, tube-worms, &c.

A'nas, a genus of web-footed birds, containing the true ducks.

Anasarca. See *Dropsy*.

Anasta'sius I, Emperor of the East, succeeded Zeno, A.D. 491, at the age of sixty. He was a member of the imperial life-guard, and owed his elevation to Ariadne, widow of Zeno, whom he married forty days after the death of her husband. He distinguished himself by suppressing the combats between men and wild beasts in the arena, abolishing the sale of offices, building the fortifications of Constantinople, &c. His support of the heretical Eutychians led to a dangerous rebellion. He died A.D. 518.

Anastat´ica, a genus of cruciferous plants, including the Rose of Jericho (*A. hierochuntica*). See *Rose of Jericho*.

Anastatic Printing, a process by which the perfect facsimile of a page of type or an engraving, old or new, can be reproduced and printed in the manner of a lithograph. The print or page to be transferred is dipped in diluted nitric acid, and, while moist with dilute acid, it is laid face downwards on a polished zinc plate and passed through a roller-press. The zinc is immediately corroded by the acid contained in the paper, excepting on those parts occupied by the ink of the type or engraving. The ink, while rejecting the acid, is loosened by it, and deposits a thin film on the zinc, thus protecting it from the action of the acid. The result is that those parts are left slightly raised in relief and greasy. The plate is then treated as in ordinary lithographic printing (q.v.).—BIBLIOGRAPHY: F. H. Collins, *Authors' and Printers' Dictionary*; C. T. Jacobi, *Printing*; J. Southward, *Modern Printing*.

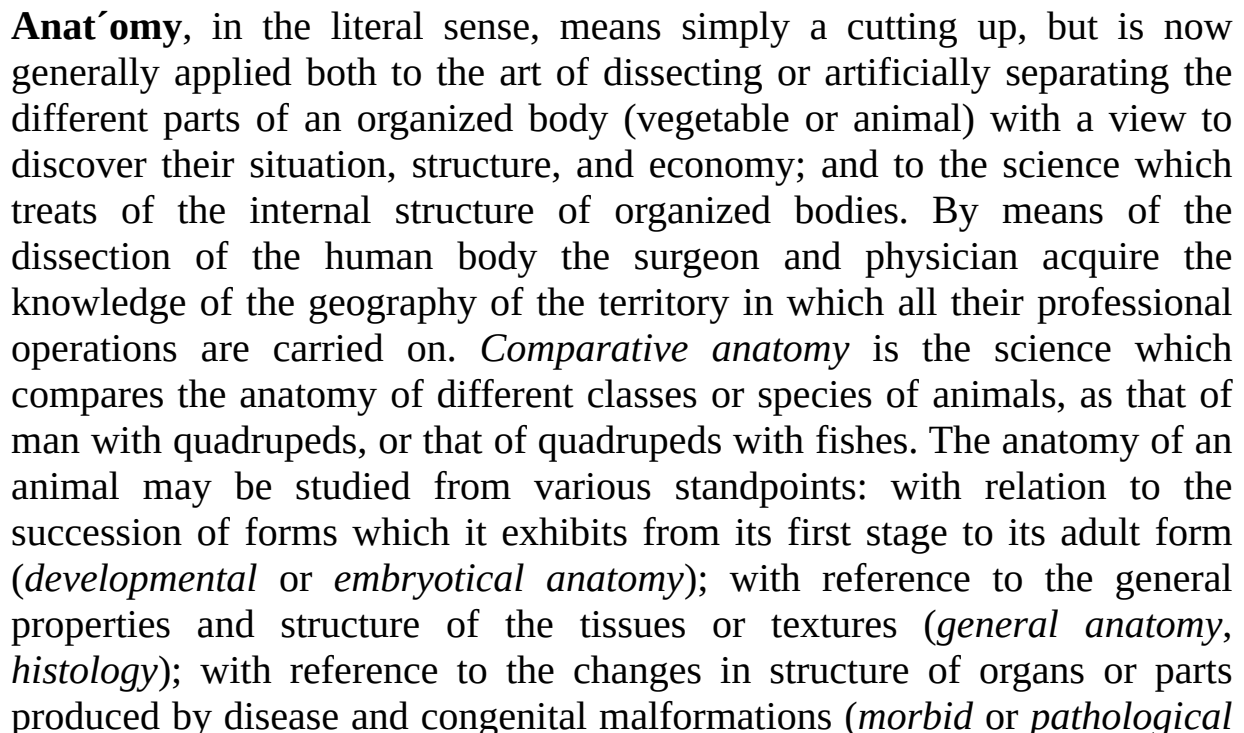
Anastomo´sis, in animals and plants, the inosculation of vessels, or the opening of one vessel into another, as an artery into another artery, or a vein into a vein. By means of anastomosis, if the course of a fluid is arrested in one vessel it can proceed along others. It is by anastomosis that circulation is re-established in amputated limbs, and in aneurism when the vessel is tied.

Anath´ema, originally a gift hung up in a temple (Gr., *anatithēmi*, to lay up), and dedicated to some god, a votive offering; but it gradually came to be used for *expulsion*, *curse*. The Roman Catholic Church pronounces the sentence of anathema against heretics, schismatics, and all who wilfully pursue a course of conduct condemned by the Church. The subject of the anathema is declared an outcast from the Church, all the faithful are forbidden to associate with him, and the utter destruction of his body and soul is foretold.

Anat´idæ, a family of swimming birds, including the Ducks, Swans, Geese, &c.

Anato´lia (from Gr. *anatolē*, the sunrise, the Orient), the modern name of Asia Minor (q.v.).

ANATOMY



anatomy); or with reference to the function, use, or purpose performed by the organs or parts (*teleological* or *physiological anatomy*). According to the parts of the body described, the different divisions of human anatomy receive different names; as, *osteology*, the description of the bones; *myology*, of the muscles; *arthrology*, of the ligaments and sinews; *splanchnology*, of the viscera or internal organs, in which are reckoned the lungs, stomach, and intestines, the liver, spleen, kidneys, bladder, pancreas, &c. *Angiology* describes the vessels through which the liquids in the body are conducted, including the blood-vessels, which are divided into arteries and veins, and the lymphatic vessels, some of which absorb matters from the bowels, while others are distributed through the whole body, collecting juices from the tissues and carrying them back into the blood. *Neurology* describes the system of the nerves and of the brain; *dermatology* treats of the skin.—Among anatomical labours are particularly to be mentioned the making and preserving of anatomical preparations. Preparations of this sort can be preserved (1) by macerating the body so as to obtain the bones of the skeleton; or (2) by treating the body or some part of it with alcohol, formalin, or other preservative, which renders its tissues imperishable.

Among the ancient writers or authorities on human anatomy may be mentioned Hippocrates the younger (460-377 B.C.), Aristotle (384-322 B.C.), Herophilus and Erasistratus of Alexandria (about 300 B.C.), Celsus (53 B.C.-A.D. 37), and Galen of Pergamus (A.D. 130-200), the most celebrated of all the ancient authorities on the science. From his time till the revival of learning in Europe in the fourteenth century anatomy was checked in its progress. In 1315 Mondino, professor at Bologna, first publicly performed dissection, and published a *System of Anatomy* which was a textbook in the schools of Italy for about 200 years. In the sixteenth century Fallopio of Padua, Eustachi of Venice, Vesalius of Brussels, Varoli of Bologna, and many others, enriched anatomy with new discoveries. In the seventeenth century Harvey discovered the circulation of the blood, Asellius discovered the manner in which the nutritious part of the food is conveyed into the circulation, while the lymphatic system was detected and described by the Dane T. Bartoline. Among the renowned anatomists of later times we can only mention Malpighi, Boerhaave, William and John Hunter, the younger Meckel, Bichat, Rosenmüller, Quain, Sir A. Cooper, Sir C. Bell, Carus, Joh. Müller, Gegenbaur, Owen, and Huxley.

Until 1832 the law of Great Britain made very insufficient provision for enabling anatomists to obtain the necessary supply of subjects for dissection. An Act of some years previously had, it is true, empowered a criminal court, when it saw fit, to give up to properly-qualified persons the body of a murderer after execution for dissection. This, however, was far from supplying the deficiency, and many persons, tempted by the high prices offered for bodies by anatomists, resorted to the nefarious practice of digging up newly-buried corpses, and frequently, as in the case of the notorious Burke and Hare of Edinburgh, to murder. To remedy these evils a statute was passed in 1832, which was intended to make provision for the wants of surgeons, students, or other duly-qualified persons, by permitting, under certain regulations, the dissection of the bodies of persons who die friendless in alms-houses, hospitals, &c. The Act also appointed inspectors of anatomy, regulated the anatomical schools, and required persons practising the operations to obtain a licence. Relatives may effectually object to the anatomical examination of a body even though the deceased had expressed a desire for it.—BIBLIOGRAPHY: D. J. Cunningham, *Textbook of Anatomy*; J. Quain, *Elements of Anatomy*; A. M. Buchanan, *Manual of Anatomy*; A. Thomson, *Anatomy for Art Students*.

Anaxagoras, an ancient Greek philosopher of the Ionic school, born at Clazomenæ, in Ionia, probably about 500 B.C. When only about twenty years of age he settled at Athens, and soon gained a high reputation, and gathered round him a circle of renowned pupils, including Pericles, Euripides, Socrates, &c. At the age of fifty he was publicly charged with impiety and condemned to death, but the sentence was commuted to perpetual banishment. He thereupon went to Lampsacus, where he died about 428. Anaxagoras belonged to the atomic school of Ionic philosophers. He held that there was an infinite number of different kinds of elementary atoms, and that these, in themselves motionless and originally existing in a state of chaos, were put in motion by an eternal, immaterial, spiritual, elementary being, *Nous* (Intelligence), from which motion the world was produced. His conception of *Nous* as the first cause of movement marks a great advance in the history of philosophical thought, for he thus placed spirit above matter. The stars were, according to him, of earthy materials; the sun a glowing mass, about as large as the Peloponnesus; the earth was

flat; the moon a dark, inhabitable body, receiving its light from the sun; the comets wandering stars.

Anaximan´der, an ancient Greek (Ionic) philosopher, was born at Miletus in 611 B.C., and died 547. The fundamental principle of his philosophy is that the source of all things is an undefined substance infinite in quantity. The firmament is composed of heat and cold, the stars of air and fire. The sun occupies the highest place in the heavens, has a circumference twenty-eight times larger than the earth, and resembles a cylinder, from which streams of fire issue. The moon is likewise a cylinder, nineteen times larger than the earth. The earth has the shape of a cylinder, and is placed in the midst of the universe, where it remains suspended. His philosophy is thus a step in advance of the theories of Thales, the conception of the Infinite, however vague, being superior to the idea of water constituting the first principle of all things. Anaximander occupied himself a great deal with mathematics and geography. To him is credited the invention of geographical maps and the first application of the *gnomon* or style fixed on a horizontal plane to determine the solstices and equinoxes.

Anaximenes (an-aks-im´e-nēz) **of Miletus**, an ancient Greek (Ionic) philosopher, according to whom air was the first principle of all things. Finite things were formed from the infinite air by compression and rarefaction produced by eternally existent motion; and heat and cold resulted from varying degrees of density of the primal element. He flourished about 550 B.C.

Anbury (an´be-ri) (called also **Club-root** and **Fingers and Toes**), a disease in turnips, in which knobs or excrescences are formed on the root, which is then useless for feeding purposes. Some authorities distinguish anbury proper from 'fingers and toes' in turnips, setting it down as a distinct disease due to a fungus, while in the other case the roots simply assume a bad habit of growth through some unknown influence.

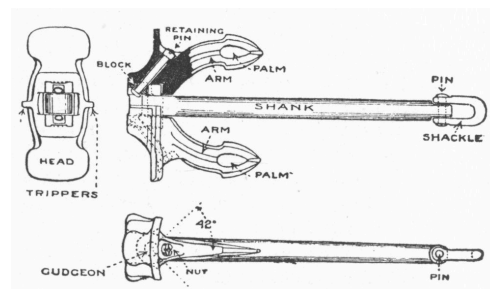
Ancachs (a´n-ka´ch´), a department of Peru, between the Andes and the Pacific; area, 16,562 sq. miles. Capital Hararaz. Pop. 500,000.

Ancestor-worship, an ancient and widespread practice, displayed in its most characteristic form in modern China and ancient Rome, which apparently was based upon the belief that dead parents or ancestors,

represented by images or 'ancestral tablets', could be revived by appropriate ceremonies, such as burning incense or offering libations, and give the benefit of their wisdom to their descendants who performed the vitalizing ritual and asked for their advice upon, or their sanction for, actions affecting the welfare of the family. The earliest deity was a dead king (Osiris), whose advice was sought by his son and successor. Hence in primitive religions, in which an endless variety of modifications of these more ancient beliefs has arisen, ancestor-worship may take the form of pious devotion to an actual ancestor or to a supernatural deity. As many of the most ancient gods were identified with animals, the dead ancestor, or his soul, is believed by many peoples to become incarnate in the appropriate animal, which is accorded the special veneration of a god or supernatural adviser, and set apart as sacred. Ancestor-worship still survives in a great variety of forms among various peoples.—BIBLIOGRAPHY: E. B. Tylor, *Primitive Culture*; F. B. Jevons, *Introduction to the History of Religion*; D. G. Brinton, *Religions of Primitive Peoples*.

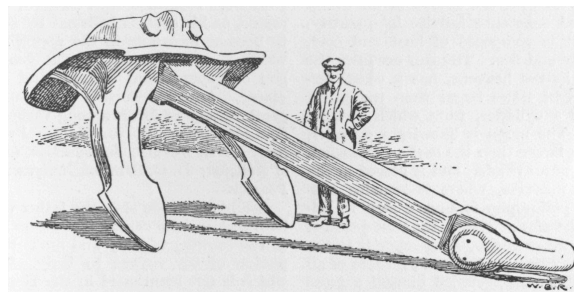
Anchises (an-kī'sēz), the father of the Trojan hero Æneas, who carried him off on his shoulders at the burning of Troy and made him the companion of his voyage to Italy. This voyage, which is not mentioned in the Homeric legend, is described by Virgil in his *Æneid*. He died at Drepanum, in Sicily.

An'chor, an implement for holding a ship or other vessel at rest in the water. In ancient times large stones or crooked pieces of wood heavily weighted with metal were used for this purpose. The anchor now used is of iron, formed with a strong *shank*, at one extremity of which is the *crown*, from which branch out two *arms*, terminating in broad *palms* or *flukes*, the sharp extremity of which is the *peak* or *bill*; at the other end of the shank is the *stock* (fixed at right angles to the plane of the arms), behind which is the *ring*, to which a cable can be attached. The principal use of the stock is to cause the arms to fall so as one of the flukes shall enter the ground. Many anchors are made nowadays without a stock. The anchors of the largest size carried by men-of-war are the *best* and *small bowers*, the *sheet*, and the *spare*, to which are added the



Modern Stockless Anchor (Hall's Patent)

stream and the *kedje*, which are used for anchoring in a stream or other sheltered place and for warping the vessel from one place to another. Many improvements and novelties in the shape and construction of anchors have been introduced within recent times. The principal names connected with those alterations are those of Lieutenant Rodgers, who introduced the *hollow-shanked anchor* with the view of increasing the strength without adding to the weight; Porter, who made the arms and flukes movable by pivoting them to the stock instead of fixing them immovably, causing the anchor to take a readier and firmer hold, and avoiding the chance of the cable becoming foul; Trotman, who further improved on Porter's invention; and M. Martin, whose anchor is of very peculiar form, and is constructed so as to be self-canting, the arms revolving through an angle of 30° either way, and the sharp points of the flukes being always ready to enter the ground.



Type of Anchor used on Lusitania, Mauretania,
&c.

An'chorites, or **An'chorets** (Gr. *anachorētai*, persons who have withdrawn themselves from the world), in the early Church a class of religious persons who generally passed their lives in cells, from which they never removed. Their habitations were, in many instances, entirely separated from the abodes of other men, sometimes in the depth of wildernesses, in pits or caverns; at other times several of these individuals fixed their habitations in the vicinity of each other, but they always lived personally separate. The continual prevalence of fierce wars, civil commotions, and persecutions at the beginning of the Christian era must have made retirement and religious meditation agreeable to men of quiet and contemplative minds. This spirit, however, soon led to fanatical excesses; many anchorites went without proper clothing, wore heavy chains, and we find at the close of the fourth century Simeon Stylites passing thirty years on the top of a column without ever descending from it, and finally dying there. In Egypt and Syria, where

Christianity became blended with the Grecian philosophy and strongly tinged with the peculiar notions of the East, the anchorites were most numerous; in Europe there were comparatively few, and on the development and establishment of the monastic system they completely disappeared. See *Asceticism*.

Anchovy (an-chō'vi), a small fish of the Herring family, all the species, with exception of the common anchovy (*Engraulis encrasicolus*) and *E. meletta* (both Mediterranean species), inhabitants of the tropical seas of India and America. The common anchovy, so esteemed for its rich and peculiar flavour, is not much larger than the middle finger. It is caught in vast numbers in the Mediterranean, and frequently on the coasts of France, Holland, and the south of England, and pickled for exportation. A favourite sauce is made by pounding the pickled fish in water, simmering for a short time, adding a little cayenne pepper, and straining the whole through a hair-sieve.

Ancho'vy-pear (*Grias cauliflōra*), a tree of the nat. ord. Myrtaceæ, a native of Jamaica, growing to the height of 50 feet, with large leaves and large white flowers, and bearing a fruit somewhat bigger than a hen's egg, which is pickled and eaten like the mango, and strongly resembles it in taste.

Anchu'sa. See *Alkanet*.

Anchylo'sis. See *Ankylosis*.

Ancient Lights, in English law, windows or other openings which have been in existence for at least twenty years, and during that time have enjoyed the access of light without interruption, so that a right is established against the obstruction of the light by a neighbouring proprietor.

Ancillon (a'n'-sē-yōn'), Jean Pierre Frédéric, an author and statesman of French extraction, born at Berlin in 1767 (where his father was pastor of the French reformed church); died there in 1837. He became professor of history in the military academy at Berlin, and in 1806 he was charged with the education of the crown-prince. He successively occupied several important offices of state, being at last appointed Minister of Foreign Affairs. He wrote on philosophy, history, and politics, partly in French, partly in German.

Anckarström. See *Ankarström*.

Anco'na, a seaport of Italy, capital of the province of the same name, on the Adriatic, 130 miles N.E. of Rome, with harbour works begun by Trajan, who built the ancient mole or quay. A triumphal arch of white marble, erected in honour of Trajan, stands on the mole. Ancona is a station of the Italian fleet, and the commerce is increasing. The town is indifferently built, but has some remarkable edifices, among others, the cathedral. There is a colossal statue of Count Cavour. Ancona is said to have been founded about four centuries B.C., by Syracusan refugees. It fell into the hands of the Romans in the first half of the third century B.C., and became a Roman colony. Pop. 68,430. The province has an area of 748 sq. miles. Pop. 333,381.

Ancona Fowl. See *Poultry*.

Ancre (än'-kr), Concino Concini, Marshal and Marquis d', was a native of Florence, and on the marriage of Marie de' Médici to Henri IV, in 1600, came in her suite to France, where he obtained rapid promotion, more especially after the assassination of the king (1610). He became successively Governor of Normandy, Marshal of France, and last of all, Prime Minister. Being thoroughly detested by all classes, at last a conspiracy was formed against him, and he was shot dead on the bridge of the Louvre in 1617.

Ancre, Battle of. This battle was the final one in the British offensive in France in 1916. It began on 13th Nov. after a two day's preliminary bombardment of the German salient, on both sides of the River Ancre, from Beaumont-Hamel to St. Pierre Divion. One area of extraordinary strength was the Y ravine which stretches from Beaumont-Hamel plateau towards the river. The assaults on both banks of the river were vigorous and determined. A fierce struggle was waged in the Y ravine, which Scottish troops ultimately cleared with the bayonet. Beaumont-Hamel having fallen, the British line was extended well beyond it. Further gains were made on the following day. The prisoners captured numbered 7200. This brilliant action paved the way for further successes in the spring.

An'cus Mar'cius, according to the traditionary history of Rome the fourth king of that city, who succeeded Tullus Hostilius, 638, and died 614 B.C. He was the son of Numa's daughter, and sought to imitate his grandfather by

reviving the neglected observances of religion. He is said to have built the wooden bridge across the Tiber known as the Sublician, constructed the harbour of Ostia, and built the first Roman prison.

Ancy´ra. See *Angora*.

Andalu´sia (Sp. *Andalucia*), a large and fertile district in the south of Spain, bounded N. by Estremadura and New Castile, E. by Murcia, S. by the Mediterranean Sea, and W. by Portugal and the Atlantic; area, about 33,777 sq. miles, comprising the modern provinces of Seville, Huelva, Cadiz, Jaen, Cordova, Granada, Almeria, and Malaga. It is traversed throughout its whole extent by ranges of mountains, the loftiest being the Sierra Nevada, many summits of which are covered with perpetual snow (Mulahacen is 11,678 feet). Minerals abound, and several mines have been opened by English companies, especially in the province of Huelva, where the Tharsis and Rio Tinto copper-mines are situated. The principal river is the Guadalquivir. The vine, myrtle, olive, palm, banana, carob, &c., grow abundantly in the valley of the Guadalquivir. Wheat, maize, barley, and many varieties of fruit grow almost spontaneously; besides which, honey, silk, and cochineal form important articles of culture. The horses and mules are the best in the Peninsula; the bulls are sought for bull-fighting over all Spain; sheep are reared in vast numbers. Agriculture is in a backward state, and the manufactures are by no means extensive. The Andalusians are descended in part from the Moors, of whom they still preserve decided characteristics. Andalusia is still famous for its bull-fighters. Pop. 3,828,916.

Andalusian Fowl. See *Poultry*.

An´damans, a chain of islands on the east side of the Bay of Bengal, the principal being the North, Middle, South, and Little Andamans. Middle Andaman is about 60 miles long, and 15 or 16 miles broad; North and South Andaman are each about 50 miles long. The Andamanese, about 1315 in number (1911), are mostly in a state of nature, living almost naked in the rudest habitations. They are small (generally much less than 5 feet), well-formed, and active, skilful archers and canoeists, and excellent swimmers and divers. These islands have been used since 1858 as a penal settlement by the Indian Government, the settlement being at Port Blair, on South

Andaman. Here rice, coffee, pineapples, nutmegs, &c., are grown, while the jungle has been cleared off the neighbouring hills. The natives in the vicinity of the settlement have become to some extent civilized. The climate is humid, but the settlement is healthy. Pop. 18,000.

Andante (a ' n-da ' n'tā; It., 'at a walking pace'), in music, denotes a movement somewhat slow, graceful, distinct, and soothing. The word is also applied substantively to that part of a sonata or symphony having a movement of this character. In Handel's music one often meets the expression *andante allegro*, which is equivalent to *andante con moto*.

Andelys, Les (lāz än'd-lēz), two towns in France called respectively Grand and Petit Andely, distant half a mile from each other, in the department of Eure, on the right bank of the Seine, 19 miles S.E. of Rouen. Grand Andely dates from the sixth century, its church, built in the thirteenth century, is one of the finest in the department. Petit Andely owes its origin to Richard Cœur de Lion, who, in 1195, built here the Château Gaillard, in its time one of the strongest fortresses in France, but now wholly a ruin. Pop. 5530.

Andenne', a town of Belgium, province of Namur, on the right bank of the Meuse and 10 miles east of Namur; manufactures delftware, porcelain, tobacco-pipes, paper, &c. Pop. 7803.

Andernach (a'n'der-na'ch), a town of Rhenish Prussia, on the left bank of the Rhine, 10 miles N.W. of Coblenz, partly surrounded with walls. Pop. 9800.

An'dersen, Hans Christian, a Danish novelist, poet, and writer of fairy tales, was born of poor parents at Odense, 2nd April, 1805. He learned to read and write in a charity school, from which he was taken when only nine years old, and was put to work in a manufactory in order that his earnings might assist his widowed mother. In his leisure time he eagerly read national ballads, poetry, and plays, and wrote several tragedies full of sound and fury. In 1819 he went to Copenhagen, but failed in getting any of his plays accepted, and in securing an appointment at the theatre, having to content himself for some time with unsteady employment as a joiner. His abilities at last brought him under the notice of Councillor Collin, a man of considerable influence, who procured for him free entrance into a Government school at Slagelse. From this school he was transferred to the

university, and soon became favourably known by his poetic works. Through the influence of Oehlenschläger and Ingermann he received a royal grant to enable him to travel, and in 1833 he visited Italy, his impressions of which he published in *The Improvvisatore* (1835)—a work which rendered his fame European. The scene of his following novel, *O. T.*, was laid in Denmark, and in *Only a Fiddler* he described his own early struggles. In 1835 appeared the first volume of his *Fairy Tales*, of which successive volumes continued to be published year by year at Christmas, and which have been the most popular and widespread of his works. Among his other works are *Picture-books without Pictures*—conversations of the author with the moon, who came to visit the poet in his garret; *A Poet's Bazaar*—the result of a voyage in 1840 to the East; and a number of dramas. In 1845 he received an annuity from the Government. He visited England in 1848, and acquired such a command of the language that his next work, *The Two Baronesses*, was written in English. In 1855 he published an autobiography, under the title *My Life's Romance*, an English translation of which, published in 1871, contained additional chapters by the author, bringing the narrative to 1867. Among his later works we may mention, *To Be or Not To Be* (1857); *Tales from Jutland* (1859); *The Ice Maiden* (1863). He died 4th Aug, 1875, having had the pleasure of seeing many of his works translated into most of the European languages.

Anderson, a town of the United States, Indiana, on the west branch of White River, 32 miles north-east of Indianapolis, with various manufacturing works. Pop. 23,856.

Anderson, Elizabeth Garrett, M.D., born in 1836, maiden name Garrett, married Mr. J. S. Anderson of the Orient Line of steamers. She studied medicine, but met with many obstacles, the study of medicine by women being then discouraged on all hands; at last she was licensed to practise by the Apothecaries' Society in 1865, and afterwards passed examinations at the University of Paris and obtained the degree of Doctor of Medicine. From 1866 to 1890 she was senior physician to the New Hospital for Women; from 1876 to 1898 lecturer on medicine in the London School of Medicine for Women. She did much to aid in opening the medical profession to women. In 1908 she was elected Mayor of Aldeburgh, being the first woman to hold the position of mayor in England. She died on 17th Dec., 1917. Her daughter Louisa Garrett Anderson, born in 1873, went to

France in 1914 as Joint Organiser of and Chief Surgeon to the Women's Hospital Corps, Voluntary Unit.

Anderson, James, a Scottish writer on political and rural economy, born at Hermiston in 1739, died in 1808. In 1790 he started the *Bee*, which ran to eighteen volumes, and contains many useful papers on agricultural, economical, and other topics. Some of his other publications, *Recreations in Agriculture, Natural History, &c.*, contain anticipations of theories afterwards propounded by Malthus and Ricardo.

Anderson, John, F.R.S., professor of natural philosophy in the University of Glasgow, born 1726, died 1796. By his will he directed that the whole of his effects should be devoted to the establishment of an educational institution in Glasgow, to be denominated *Anderson's University*, for the use of the unacademical classes. According to the design of the founder, there were to be four colleges—for arts, medicine, law, and theology—besides an initiatory school. As the funds, however, were totally inadequate to the plan, it was at first commenced with only a single course of lectures on natural philosophy and chemistry. The institution gradually enlarged its sphere of instruction, coming nearer and nearer to the original design of its founder, the medical school in particular possessing a high reputation. In 1886 it was incorporated with other institutions to form the Glasgow and West of Scotland Technical College (now Glasgow Royal Technical College), Anderson's College medical school, however, retaining a distinct position.

Anderson, Joseph, Scottish antiquary, born in 1832, became a school teacher, was for some years newspaper editor, and in 1870 was appointed keeper of the National Museum of Antiquities, Edinburgh. His chief works embody the lectures delivered by him as Rhind lecturer in archæology to the Society of Antiquaries of Scotland: *Scotland in Early Christian Times*, *Scotland in Pagan Times*, and the *Early Christian Monuments of Scotland*. He also edited *The Orkneyinga Saga*, *The Oliphants in Scotland*, and Drummond's *Ancient Scottish Weapons*. He died in 1916.

Anderson, Robert, M.D., Scottish biographical writer, born 1750, died 1830. He furnished biographical and critical notices for *A Complete Edition of the Poets of Great Britain* (1792-5), and was for a time editor of the *Edinburgh Magazine*.

Andersson, Carl Jan, an African traveller, born in Sweden in 1827, died in the land of the Ovampos, in Western Africa, in July, 1867. He published *Lake Ngami, or Discoveries in South Africa* (London, 2 vols., 1856), and *The Okavango River* (London, 1861). The observations of his last voyage were published in 1875 in *Notes of Travel in South Africa*.

Andes (an'dēz), or, as they are called in Spanish South America, **Cordilleras** (ridges) **de los Andes**, or simply **Cordilleras**, a range of mountains stretching along the whole of the west coast of South America, from Cape Horn to the Isthmus of Panama and the Caribbean Sea. In absolute length (4500 miles) no single chain of mountains approaches the Andes, and only a certain number of the higher peaks of the Himalayan chain rise higher above the sea-level; which peak is the highest of all is not yet settled. Several main sections of this huge chain are distinguishable. The Southern Andes present a lofty main chain, with a minor chain running parallel to it on the east, reaching from Tierra del Fuego and the Straits of Magellan, northward to about lat. 28° s., and rising in Aconcagua to a height of 23,080 feet. North of this is the double chain of the Central Andes, enclosing the wide and lofty plateaus of Bolivia and Peru, which lie at an elevation of more than 12,000 feet above the sea. The mountain system is here at its broadest, being about 500 miles across. Here are also several very lofty peaks, as Illampu or Sorata (21,484 feet), Sahama (21,054 feet), Illimani (21,024 feet). Farther north the outer and inner ranges draw closer together, and in Ecuador there is but a single system of elevated masses, generally described as forming two parallel chains. In this section are crowded together a number of lofty peaks, most of them volcanoes, either extinct or active. Of the latter class are Pichincha (15,918 feet), with a crater 2500 feet deep; Tunguragua (16,685 feet); Sangay (17,460 feet); and Cotopaxi (19,550 feet). The loftiest summit here appears to be Chimborazo (20,581 feet); others are Antisana (19,260 feet) and Cayambe (19,200 feet). Northward of this section the Andes break into three distinct ranges, the east-most running north-eastward into Venezuela, the westmost running north-westward to the Isthmus of Panama. In the central range is the volcano of Tolima (17,660 feet). The western slope of the Andes is generally exceedingly steep, the eastern much less so, the mountains sinking gradually to the plains. The whole range gives evidence of volcanic action, but it consists almost entirely of sedimentary rocks. Thus mountains

may be found rising to the height of over 20,000 feet, and fossiliferous to their summits (as Illimani and Sorata or Illampu). There are about thirty volcanoes in a state of activity. The loftiest of these burning mountains seems to be Gualateñiri, in Peru (21,960 feet). The heights of the others vary from 13,000 to 20,000 feet. All the districts of the Andes system have suffered severely from earthquakes, towns having been either destroyed or greatly injured by these visitations. Peaks crowned with perpetual snow are seen all along the range, and glaciers are also met with, more especially from Aconcagua southwards. The passes are generally at a great height, the most important being from 10,000 to 15,000 feet. Railways have been constructed to cross the chain at a similar elevation. The Andes are extremely rich in the precious metals, gold, silver, copper, platinum, mercury, and tin all being wrought; lead and iron are also found. The llama and kindred species—the guanaco, vicuña, and alpaca—are characteristic of the Andes. Among birds, the condor is the most remarkable. The vegetation necessarily varies much according to elevation, latitude, rainfall, &c., but generally is rich and varied. Except in the south and north little rain falls on the western side of the range, and in the centre there is a considerable desert area. On the east side the rainfall is heavy in the equatorial regions, but in the south is very scanty or altogether deficient. From the Andes rise two of the largest water systems of the world—the Amazon and its affluents, and the La Plata and its affluents. Besides which, in the north, from its slopes flow the Magdalena to the Caribbean Sea, and some tributaries to the Orinoco. The mountain chain pressing so close upon the Pacific Ocean, no streams of importance flow from its western slopes. The number of lakes is not great; the largest and most important is that of Titicaca on the Bolivian plateau. In the Andes are towns at a greater elevation than anywhere else in the world, the highest being the silver-mining town of Cerro de Pasco (14,270 feet), the next being Potosi.

An'desin, a kind of felspar containing both soda and lime, and named from being first obtained in the Andes.

An'desite, a name given to a crystalline volcanic rock or group of rocks of very wide occurrence, consisting mostly of felspar mixed with other ingredients, especially hornblende and augite, often also hypersthene and mica, the four chief varieties being named accordingly. Andesite is often porphyritic in character, with large crystals of felspar scattered through it.

These rocks are commonly eruptive products of volcanoes of the tertiary or more recent periods, and the name was given by C. L. von Buch on account of their prevalence in the lavas of volcanoes of the Andes. The Ochils and other hills of middle Scotland largely consist of andesite.

Andijan´, a town of Russian Turkestan, Ferghana, south of the Syr-Darya, a terminus of the Transcaspian Railway, 73 miles north-east of Khokand. Pop. 82,235.

Andi´ra, a genus of leguminous American trees, with fleshy plum-like fruits. The wood is suitable for building purposes. The bark of *A. inermis*, or cabbage tree, is narcotic, and is used as an anthelmintic under the name of *worm-bark* or *cabbage bark*. The powdered bark of *A. ararōba* is used as a remedy in certain skin diseases, as herpes.

Andiron (and´ī-ěrn), a horizontal iron bar raised on short legs, with an upright standard at one end, used to support pieces of wood when burning in an open hearth, one andiron being placed on either side of the hearth.

Andkhoo, or **Andkhoui** (a´nd-hö´, a´nd-hö´i), a town of Afghanistan, about 200 miles south of Bokhara, on the commercial route to Herat. Pop. estimated at 15,000.

Andocides (an-dos´i-dēz), an Athenian orator, born about 440 B.C., died about 393 B.C. He took an active part in public affairs, and was four times exiled; the first time along with Alcibiades, for profaning the Eleusinian mysteries. Several of his orations are extant, one called *On the Mysteries* being the best.

Andor´ra, or **Andorre´**, a small nominally independent State in the Pyrenees, south of the French department of Ariège, with an area of about 191 sq. miles. It has been a separate State for six hundred years, is governed by its own civil and criminal codes, and has its own courts of justice, the laws being administered by two judges, one of whom is chosen by France, the other by the Bishop of Urgel, in Spain. The little State pays an annual due of 960 francs to France, and 460 pesetas to the Bishop of Urgel. The chief industry is the rearing of sheep and cattle. The commerce is largely in importing contraband goods into Spain. The inhabitants, who speak the Catalan dialect of Spanish, are simple in their manners, their wealth

consisting mainly of cattle and sheep. The village of Old Andorra is the capital. Pop. 5231.

An'dover, a town in England, in Hants, 12 miles north by west of Winchester, with a fine church, and a trade in corn, malt, &c. Interesting Roman remains have been found in the vicinity. Pop. (1921), 8569.

An'dover, a town in Massachusetts, 25 miles N.N.W. of Boston, chiefly remarkable for its literary institutions—Phillip's Academy, founded in 1778; the Andover Theological Seminary, founded in 1807; and Abbot Academy, a girls' school, founded in 1829. Pop. 7300.

Andrassy (a ' n-drä'shē), Count Julius, Hungarian statesman, born 1823, died in 1890. He took part in the revolution of 1848, was condemned to death, but escaped and went into exile. He was appointed Premier when self-government was restored to Hungary in 1867; became imperial Minister for Foreign Affairs in 1871, but retired from public life in 1879.

Andrassy, Julius, Hungarian statesman, son of the preceding. He was born in 1860, and entered the Reichstag in 1884. He became Minister of the Interior in 1906, and retained that office until 1909. In 1912 he represented Austria at the conference on the Balkan question. In 1918 he was appointed Minister for Foreign Affairs, but soon resigned.

André (an'drā), Major John, adjutant-general in the British army during the American revolutionary war. Employed to negotiate the defection of the American general Arnold, and the delivery of the works at West Point, he was apprehended in disguise, 23rd Sept., 1780, within the American lines; declared a spy from the enemy, and hanged 2nd Oct., 1780. His remains were brought to England in 1821 and interred in Westminster Abbey, where a monument has been erected to his memory.

Andrea del Sarto. See *Sarto*.

Andreæ (a ' n'dre-ā), Johann Valentin, German author, born 1586, died 1654. He was the author of numerous tracts, several of them of an amusing and satirical character. He was long believed to be the founder of the celebrated Rosicrucian order, an opinion that received a certain support from some of his works, but in all probability the real intention of the writer was to ridicule the folly of contemporary alchemists.

An´dreasberg, St., a mining town of the Harz Mountains, in Prussia, 57 miles S.S.E. of Hanover. Pop. about 4000.

Andreev, Leonid Nicolaievitsh, Russian author, born in 1871, died in 1919. He studied law at the Universities of Moscow and Petrograd, but finding his practice unremunerative he became a police-court reporter for a daily paper. At the age of twenty-three he attempted suicide, driven to it by his miserable circumstances and struggle for existence. His first story, *About a Poor Student*, based upon his own experiences, attracted but little attention, and his literary career really began when Gorky discovered his talent. He was one of the most prolific Russian writers, the short story being his speciality. He was a mystic and a fatalist, like so many of his compatriots. His works include: *The Red Laugh* (1905); *The Seven who were Hanged* (1909); *Judas Iscariot and the Others* (1910); *A Dilemma* (1910); *Silence and Other Stories*, &c. His works have been translated into many European languages.

Andrew, St., brother of St. Peter, and the first disciple whom Christ chose. He is said to have preached in Scythia, in Thrace and Asia Minor, and in Achaia (Greece), and according to tradition he was crucified by order of the Roman governor Ægeas at Patræ, now Patras, in Achaia, on a cross of the form **X** (decussate cross), now known as a St. Andrew's cross. The Russians revere him as the apostle who brought the gospel to them; the Scots, as the patron saint of their country. The day dedicated to him is 30th Nov. The Russian order of St. Andrew was instituted by Peter the Great in 1698. For the Scottish Knights of St. Andrew or the Thistle, see *Thistle*.

An´drewes, Lancelot, an eminent and learned bishop of the English Church, born in London in 1555, died at Winchester 1626; was high in favour both with Queen Elizabeth and James I. In 1605 he became Bishop of Chichester; in 1609 was translated to Ely, and appointed one of the king's privy-councillors; and in 1618 he was translated to Winchester. He was one of those engaged in preparing the authorized version of the Scriptures. He left sermons, lectures, and other writings.

An´drews, St., an ancient city and parliamentary burgh in Fifeshire, Scotland, 31 miles north-east from Edinburgh; was erected into a royal burgh by David I in 1140, and after having been an episcopal, became an

archiepiscopal see in 1472, and was for long the ecclesiastical capital of Scotland. The cathedral, now in ruins, was begun about 1160, and took 157 years to finish. The old castle, founded about 1200, and rebuilt in the fourteenth century, is also an almost shapeless ruin. In it James III was born and Cardinal Beaton assassinated, and in front of it George Wishart was burned. There are several other interesting ruins. The trade and manufactures are of no importance, but the town is in favour as a watering-place. Golf is much played here. Pop. 7597.—The *University of St. Andrews*, the oldest of the Scottish universities, founded in 1411, consists of the united colleges of St. Salvator and St. Leonard and the college of St. Mary, both at St. Andrews, and embraces also University College, Dundee. In 1579 the colleges of St. Salvator and St. Leonard were restricted to the teaching of arts and medicine, and that of St. Mary to theology. In 1747 the two former colleges were united by Act of Parliament. University College, Dundee, was founded in 1880. The united college of St. Salvator and St. Leonard has a principal (who is also principal of the university) and twelve professors, and the college of St. Mary has a principal and four professors. Degrees, open to women as well as men, are conferred in arts, divinity, science, medicine, and law; and the university also confers the diploma and title of L.L.A. (Lady Literate in Arts). The number of students is 420. In connection with the university is a library, founded in 1612 and containing about 150,000 printed volumes and 150 MSS. The university unites with the other three Scottish universities in returning three members to Parliament. Madras College or Academy, founded by Dr. Bell of Madras, the principal secondary school of the place, provides accommodation for upwards of 1500 scholars.

An'drews, Thomas, chemist, was born at Belfast in 1813; studied chemistry at Glasgow under Thomas Thomson, and for a short time in Paris; then medicine at Belfast, Dublin, and Edinburgh, taking the degree of M.D. at the last place. After practising and teaching chemistry for ten years in Belfast, he became vice-president of the Northern College there, which in 1849 was converted into Queen's College, and Andrews now became professor of chemistry in the college, a post which he held till 1879. He died in 1885, having received various academic distinctions in the course of his life. His name is associated with valuable researches on the heat of chemical combustion, and on the nature of ozone, but especially with the

discovery of the existence of a critical temperature for every gas, above which it cannot be liquefied by any pressure, however great. He wrote many scientific papers, which have been published in a collective form by P. G. Tait and A. Crum Brown.

An'dria, a town of South Italy, province of Bari, with a fine cathedral, founded in 1046; the Church of Sant' Agostino, with a beautiful Pointed Gothic portal; a college; manufactures of majolica, and a good trade. Pop. 53,274.

Androclus, or **Androcles**, a Roman slave who once pulled a thorn out of a lion's paw and dressed the wound. Androclus was afterwards condemned to be thrown to the lions in the Circus Maximus, and encountered the same lion that he had helped; the beast, instead of attacking him, fawned on him and caressed him. The story is told by Aulus Gellius, *Noctes Atticæ*, v, 14.

Andrœ'cium, in botany, the male system of a flower; the aggregate of the stamens.

Andromache (an-drom'a-kē), in Greek legend, wife of Hector, and one of the most attractive women of Homer's *Iliad*. The passage describing her parting with Hector, when he was setting out to battle, is well known and much admired (*Iliad*, vi, 369-502). Euripides and Racine have made her the chief character of tragedies.

Androm'ēda, in Greek mythology, daughter of the Ethiopian king Cepheus and of Cassiopeia. Cassiopeia having boasted that her daughter surpassed the Nereids, if not Hēra (Juno) herself, in beauty, the offended goddesses prevailed on their father, Poseidōn (Neptune), to afflict the country with a horrid sea-monster, which threatened universal destruction. To appease the offended god, Andromeda was chained to a rock, but was rescued by Perseus; and after death was changed into a constellation. The legend forms the subject of tragedies by Euripides and Sophocles, and Ovid introduced it into his *Metamorphoses*.

Androm'eda. See *Ericaceæ*.

Androni'cus, the name of four emperors of Constantinople.—**Andronicus I**, Comnenus, born 1110, murdered 1185.—**Andronicus II**, Palæologus, born 1258, died 1332. His reign is celebrated for the invasion of the Turks.

—**Andronicus III**, Palæologus the Younger, born 1296, died 1341.—**Andronicus IV**, Palæologus, reigned in the absence of John IV. In 1373 he gave way to his brother Manuel, and died a monk.

Androni'cus, Livius, the most ancient of the Latin dramatic poets; flourished about 240 B.C.; by origin a Greek, and long a slave. A few fragments of his works have come down to us.

Androni'cus of Rhodes, a peripatetic philosopher who lived at Rome in the time of Cicero. He arranged Aristotle's works in much the same form as they retain in present editions.

Androni'cus Cyrrhestes (sir-es'tēz), a Greek architect about 100 B.C., who constructed at Athens the Tower of the Winds, an octagonal building, still standing. On the top was a Triton, which indicated the direction of the wind. Each of the sides had a sort of dial, and the building formerly contained a clepsydra or water-clock.

Andropo'gon, a large genus of grasses, mostly natives of warm countries. *A. Schænanthus* is the sweet-scented lemon-grass of conservatories. Others also are fragrant.

An'dros (now **Andro**), one of the islands of the Grecian Archipelago, the most northerly of the Cyclades; about 25 miles long and 6 or 7 broad; area, 100 sq. miles. A considerable trade is done in silk, wine, olives, figs, oranges, and lemons. Andro or Castro, the capital, has a good port. Pop. 18,809.

Andros Islands, a group of isles belonging to the Bahamas, lying southwest of New Providence, not far from the east entrance to the Gulf of Florida. The passages through them are dangerous. Pop. 7545.

Andrussovo, a Russian village in the government of Smolensk. A treaty was signed here between Poland and Russia (1667).

Andujar (a' n-dö-här'), a town in Spain, in Andalusia, 50 miles E.N.E. of Cordova, on the Guadalquivir, which is here crossed by a fine bridge; manufactures a peculiar kind of porous earthen water-bottles and jugs (*alcarazas*). Pop. 16,500.

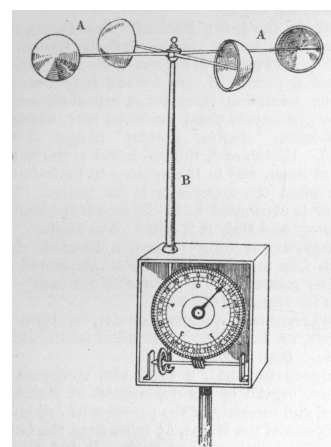
An´ecdote, originally some particular about a subject not noticed in previous works on that subject; now any particular or detached incident or fact of an interesting nature; a single passage of private life.

Anega´da, a British West Indian island, the most northern of the Virgin group, 10 miles long by $4\frac{1}{2}$ broad; contains numerous salt ponds, from which quantities of salt are obtained. Pop. 200.

Anelectric, a body not easily electrified.

Anelectrode, the positive pole of a galvanic battery.

Anemom´eter (Gr. *anēmos*, wind, *metron*, measure), an instrument for measuring the force and velocity of the wind. This force is usually measured by the pressure of the wind upon a square plate attached to one end of a spiral spring (with its axis horizontal), which yields more or less according to the force of the wind, and transmits its motion to a pencil which leaves a trace upon paper moved by clockwork. Various instruments have been devised for this purpose, but the one most commonly adopted by meteorological stations is after the type invented by Dr. Robinson of Armagh. It consists of four hemispherical cups A attached to the ends of equal horizontal arms, forming a horizontal cross which turns freely about a vertical axis B. By means of an endless screw carried by the axis a train of wheelwork is set in motion; and the indication is given by a hand which moves round a dial; or in some instruments by several hands moving round different dials like those of a gas-meter. It is found that the centre of each cup moves with a velocity which is almost exactly one-third of that of the wind. There are various other forms of the instrument, one of which is portable, and is especially intended for measuring the velocity of currents of air passing through mines, and the ventilating spaces of hospitals and other public buildings. The direction of the wind as indicated by a vane can also be made to leave a continuous record by various contrivances; one of the most common being



Beckley's Improved
Robinson Cup
Anemometer

a pinion carried by the shaft of a vane, and driving a rack which carries a pencil.

Anem'ōnē (Gr. *anēmos*, wind), wind-flower, a genus of plants belonging to the Buttercup family (Ranunculaceæ), containing about ninety species, found in temperate regions, three of them occurring in Britain: the white-flowered (*A. nemorōsa*), the only one truly native; the blue-flowered (*A. apennīna*); and the yellow-flowered (*A. ranunculoides*), a common European species naturalized in some parts of Britain. Several species are cultivated as florists' flowers.

Anemoph'ilous, said of flowers that are fertilized by the wind conveying the pollen.

Anem'oscope, any contrivance indicating the direction of the wind; generally applied to a vane which turns a spindle descending through the roof to a chamber where, by means of a compass-card and index, the direction of the wind is shown.

Aneroid Barometer. See *Barometer*.

Ane'thum, a genus of plants; dill.

Aneu'rin, a poet and prince of the Cambrian Britons who flourished in the seventh century, author of an epic poem, the *Gododin*, relating the defeat of the Britons of Strathclyde by the Saxons at the battle of Catteraeth. See *Celtic Literature*.

An'eurism, or **Aneurysm** (Gr. *aneurysma*, a widening), the dilatation or expansion of some part of an artery. Aneurisms arise partly from the too violent motion of the blood, and partly from degenerative changes occurring in the coats of the artery, diminishing their elasticity. They are therefore more frequent in the great branches; in particular, in the vicinity of the heart, in the arch of the aorta, and in the extremities, where the arteries are exposed to frequent injuries by stretching, violent bodily exertions, thrusts, falls, and contusions. An internal aneurism may burst and cause death.

Angara', a Siberian river which flows into Lake Baikal at its N. extremity, and leaves it near the s.w. end, joining the Yenisei as the Lower Angara or

Upper Tunguska.

Angel (Gr. *angelos*, a messenger), one of those spiritual intelligences who are regarded as dwelling in Heaven and employed as the ministers or agents of God. To these the name of good angels is sometimes given, to distinguish them from bad angels, who were originally created to occupy the same blissful abode, but lost it by rebellion. The Old Testament represents them as messengers of the Divine will, and Christ spoke of them more than once (*St. Matt.* xviii, 10; *St. Luke*, xv, 10). Generally, however, Scripture speaks of angels with great reserve, Michael and Gabriel alone being mentioned by name in the canonical books, while Raphael is mentioned in the Apocrypha. The angels are represented in Scripture as in the most elevated state of intelligence, purity, and bliss, ever doing the will of God so perfectly that we can seek for nothing higher or better than to aim at being like them. There are indications of a diversity of rank and power among them, and something like angelic orders—Michael, Gabriel, Raphael, Uriel, &c., seraphim and cherubim. They are represented as frequently taking part in communications made from heaven to earth, as directly and actively ministering to the good of believers, and shielding or delivering them from evils incident to their earthly lot. That every person has a good and a bad angel attendant on him was an early belief, and is held to some extent yet. Roman Catholics, since St. Ambrosius, who died in 397, show a certain veneration or worship to angels, and beg their prayers and their kind offices. The New Testament, however, formally forbidding such veneration (*Col.* ii, 18, &c.), Protestants consider this unlawful.

Angel, a gold coin introduced into England in the reign of Edward IV, and coined down to the Commonwealth, so named from having the representation of the archangel Michael piercing a dragon upon it. It had different values in different reigns, varying from 6s. 8d. to 10s.



Angel of Queen
Elizabeth

Angel-fish, a fish, *Squatina angelus*, nearly allied to the sharks, very ugly and voracious, preying on other fish. It is from 6 to 8 feet long, and takes its name from its pectoral fins, which are very large, extending horizontally like wings when spread. This fish connects the rays with the sharks, but it differs from both in having its mouth placed at the

extremity of the head. It is common on the south coasts of Britain, and is also called *Monk-fish* and *Fiddle-fish*.

Angel'ica, a genus of umbelliferous plants, one of which, *A. sylvestris*, a tall plant bearing large umbels of white flowers tinged with pink, is common in wet places in Britain, and was formerly believed to possess *angelic* properties as an antidote to poison, a specific against witchcraft, &c. The name is also given to an allied plant, the *Archangelica officinālis*, found on the banks of rivers and ditches in the north of Europe, once generally cultivated as an esculent, and still valued for its medicinal properties. It has a large fleshy aromatic root, and a strong-furrowed branched stem as high as a man. It is cultivated for its agreeable aromatic odour and carminative properties. Its blanched stems, candied with sugar, form a very agreeable sweetmeat, possessing tonic and stomachic qualities.

Angelico (a`n-jel'i-kō), **Fra**, the common appellation of *Fra Giovanni da Fiesole*, one of the most celebrated of the early Italian painters. Born 1387, he entered the Dominican order in 1407, and was employed by Cosmo de' Medici in painting the monastery of St. Mark and the church of St. Annunziata with frescoes. These pictures gained him so much celebrity that Pope Nicholas V invited him to Rome to ornament his private chapel in the Vatican, and offered him the archbishopric of Florence, which Angelico declined. He died at, Rome 1455. He has been called the 'painter of seraphic dreams'. His works were considered unrivalled in finish and in sweetness and harmony of colour, and were made the models for religious painters of his own and succeeding generations.

Angeln (a`ng'eln), a district in Schleswig of about 300 sq. miles, bounded N. by the Bay of Flensburg, S. by the Schlei, E. by the Baltic, the only continental territory which has retained the name of the Angles.

Angelo (a`n'je-lō), Michael. See *Buonarotti*.

An'gelus, in the Roman Catholic Church, a short form of prayer in honour of the incarnation, consisting mainly of versicles and responses, the angelic salutation three times repeated, and a collect, so named from the word with which it commences, '*Angelus Domini*' (Angel of the Lord). Hence, also, the bell tolled in the morning, at noon, and in the evening to indicate the

time when the angelus is to be recited. The prayer is attributed to St. Bonaventura, and in Germany and Italy it is called 'Ave Maria'.

Ångermann (ong´er-ma´n), a Swedish river which falls into the Gulf of Bothnia, noted for its fine scenery. It is navigable for nearly 70 miles for vessels of 600 tons.

Angermünde (a´ng´er-mün-de), a town in Prussia, on Lake Münde, 42 miles north-east of Berlin. Pop. 8200.

Angers (a´n´-zhā), a town and river-port of France, capital of the department of Maine-et-Loire, and formerly of the province of Anjou, on the banks of the Maine, 5½ miles from the Loire, 150 miles south-west of Paris. It has an old castle, built by Louis IX, once a place of great strength, now used as a prison, barrack, and powder-magazine; a fine cathedral of the twelfth and thirteenth centuries, with very fine old painted windows; is the seat of a bishop, and has a school of arts and manufactures; a public library, an art-gallery, a large modern hospital, the remains of a hospital founded by Henry II of England in 1155; courts of law, theatre, &c. It manufactures sail-cloth, hosiery, leather, and chemicals; foundries, &c. In the neighbourhood are immense slate-quarries. Pop. 83,786.

Angevins (an´je-vins), natives of Anjou, often applied to the race of English sovereigns called Plantagenets (q.v.). Anjou became connected with England by the marriage of Matilda, daughter of Henry I, with Geoffrey V, Count of Anjou. The Angevin kings of England were Henry II, Richard I, John, Henry III, Edward I, Edward II, Edward III, and Richard II.

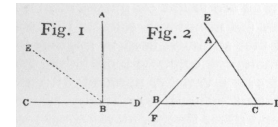
Angilbert, St., the most celebrated poet of his age, secretary and friend of Charlemagne, whose daughter, Bertha, he married. In the latter part of his life he retired to a monastery, of which he became abbot. Died 814.

Angina Pectoris (an´ji-na pek´to-ris), or **Heart-spasm**, a disease characterized by an extremely acute constriction, felt generally in the lower part of the sternum, and extending along the whole side of the chest and into the corresponding arm, a sense of suffocation, faintness, and apprehension of approaching death: seldom experienced by any but those with organic heart-disease. The disease rarely occurs before middle age, and is more frequent in men than in women. Those liable to attack must lead a

quiet, temperate life, avoiding all scenes which would unduly rouse their emotions. The first attack is occasionally fatal, but usually death occurs as the result of repeated seizures. The paroxysm may be relieved by opiates, or the inhalation, under due precaution, of anæsthetic vapours.

Angiosperm (an'ji-o-spèrm), a term for any plant which has its seeds enclosed in a seed-vessel. Exogens are divided into those whose seeds are enclosed in a seed-vessel, and those with seeds produced and ripened without the production of a seed-vessel. The former are *angiosperms*, and constitute the principal part of the species; the latter are *gymnosperms*, and chiefly consist of the Coniferæ and Cycadaceæ.

Angle, the point where two lines meet, or the meeting of two lines in a point. A *plane rectilineal angle* is formed by two straight lines which meet one another, but are not in the same straight line; it may be considered the degree of opening or divergence of the two straight lines which thus meet one another. A *right angle* is an angle formed by a straight line falling on another perpendicularly, or an angle which is measured by an arc of 90 degrees. When a straight line, as A B (fig. 1), standing on another straight line C D, makes the two angles A B C and A B D equal to one another, each of these angles is called a *right angle*. An *acute angle* is that which is less than a right angle, as E B C. An *obtuse angle* is that which is greater than a right angle, as E B D. Acute and obtuse angles are both called *oblique*, in opposition to right angles. *Exterior* or *external angles*, the angles of any rectilineal figure without it, made by producing the sides; thus, if the sides A B, B C, C A of the triangle A B C (fig. 2) be produced to the points F D E, the angles C B F, A C D, B A E are called *exterior* or *external angles*. A *solid angle* is that which is made by more than two plane angles meeting in one point and not lying in the same plane, as the angle of a cube. A *spherical angle* is an angle on the surface of a sphere, contained between the arcs of two great circles which intersect each other.



Angler (*Lophius piscatorius*), also from its habits and appearance called **Fishing-frog** and **Sea-devil**, a remarkable fish often found on the British coasts. It is from 3 to 5 feet long; the head is very wide, depressed, with protuberances, and bearing long separate movable tendrils; the mouth is capacious, and armed with formidable teeth. Its voracity is extreme, and it

is said to lie concealed in the mud, and attract the smaller fishes within its reach by gently waving the filamentous appendages on its head.

Angles, a Low German tribe who in the earliest historical period had their seats in the district about Angeln, in the duchy of Schleswig, and who in the fifth century and subsequently crossed over to Britain along with bands of Saxons and Jutes (and probably Frisians also), and colonized a great part of what from them has received the name of England, as well as a portion of the Lowlands of Scotland. The Angles formed the largest body among the Germanic settlers in Britain, and founded the three kingdoms of East Anglia, Mercia, and Northumbria.

Anglesey (ang'gl-sē), or **Anglesea** ('the Angles' Island'), an island and county of North Wales, in the Irish Sea, separated from the mainland by the Menai Strait; 20 miles long and 17 miles broad; area, 176,630 acres. The surface is comparatively flat, and the climate is milder than that of the adjoining coast. The chief agricultural products are oats and barley, wheat, rye, potatoes, and turnips. Numbers of cattle and sheep are raised. Anglesey yields a little copper, lead, silver, ochre, &c. The Menai Strait is crossed by a magnificent suspension-bridge, 580 feet between the piers and 100 feet above high-water mark, and also by the great Britannia Tubular Railway Bridge. The chief market-towns are Beaumaris, Holyhead, Llangefni, and Amlwch. The county returns one member to Parliament. Pop. (1921), 51,695.

Anglesey, Henry William Paget, Marquess of, English soldier and statesman, was the eldest son of Henry, first Earl of Uxbridge, and was born in 1768. He was educated at Oxford, and in 1790 entered Parliament as member for the Carnarvon boroughs. In 1793 he entered the army, and in 1794 he took part in the campaign in Flanders under the Duke of York. In 1808 he was sent into Spain with two brigades of cavalry to join Sir John Moore, and in the retreat to Coruña commanded the rear-guard. In 1812 he became, by his father's death, Earl of Uxbridge. On Napoleon's escape from Elba he was appointed commander of the British cavalry, and at the battle of Waterloo, by the charge of the heavy brigade, overthrew the Imperial Guard. For his services he was created Marquess of Anglesey. In 1828 he became Lord-Lieutenant of Ireland and made himself extremely popular, but was recalled in consequence of favouring Catholic Emancipation. He

was again Lord-Lieutenant in 1830, but lost his popularity by his opposition to O'Connell and his instrumentality in the passing of the Irish Coercion Acts; and he quitted office in 1833. From 1846-52 he was Master-General of the Ordnance. He died in 1854.

Anglicanism, the term is capable of a wider use, but is usually employed as descriptive of the type of doctrine formulated by the Church of England in the period of the Protestant Reformation. The two most notable formularies of that period are the Confession of Faith, known as the Thirty-nine Articles, which assumed its present shape in 1571, and the Liturgy, known as the Book of Common Prayer, issued in 1559 in what was substantially its present shape. By the Clerical Subscription Act of 1865 assent is required to both Prayer Book and Articles. The Articles are not and never were binding upon laymen.—BIBLIOGRAPHY: Mgr. Moyes, *Aspects of Anglicanism*; F. Y. Kinsman, *Principles of Anglicanism*.

Angling, the art of catching fish with a hook or *angle* (A. Sax. *angel*) baited with worms, small fish, flies, &c. We find occasional allusions to this pursuit among the Greek and Latin classical writers; it is mentioned several times in the Old Testament, and it was practised by the ancient Egyptians. The first reference to angling in England is a passage in a tract, entitled *Piers Fulham*, supposed to have been written about the year 1420. The oldest work on the subject in English is the *Treatyse of Fysshinge with an Angle*, printed by Wynkyn de Worde in 1496, along with a treatise on hunting and hawking, the whole being ascribed to Dame Juliana Berners or Barnes, prioress of a nunnery near St. Albans. Walton's inimitable discourse on angling was first printed in 1653. The chief appliances required by an angler are a rod, line, hooks, and baits. Rods are made of various materials, and of various sizes. The cane rods are lightest, and where fishing-tackle is sold they most commonly have the preference; but in country places the rod is often of the angler's own manufacture. Rods are commonly made in separate joints, so as to be easily taken to pieces and put up again. They are made to taper from the butt end to the top, and are usually possessed of a considerable amount of elasticity. In length they may vary from 10 feet to more than double that length, with a corresponding difference in strength—a rod for salmon being necessarily much stronger than one suited for ordinary burn trout. The *reel*, an apparatus for winding up the line, is attached to the rod near the lower end, where the hand grasps it while

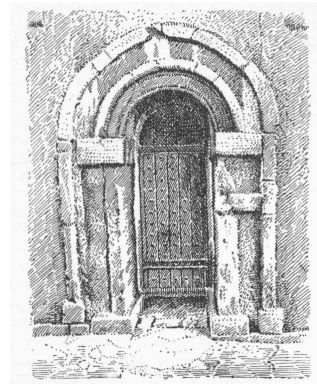
fishing. The best are usually made of brass, are of simple construction, and so made as to wind or unwind freely and rapidly. That part of the line which passes along the rod and is wound on the reel is called the *reel line*, and may vary from 20 to 100 yards in length, according to the size of the water and the habits of the fish angled for; it is usually made of twisted horse-hair and silk, or of oiled silk alone. The casting line, which is attached to this, is made of the same materials, but lighter and finer. To the end of this is tied a piece of fine gut, on which the hook, or hooks, are fixed. The casting or gut lines should decrease in thickness from the reel line to the hooks. The hook, of finely-tempered steel, should readily bend without breaking, and yet retain a sharp point. It should be long in the shank and deep in the bend; the point straight and true to the level of the shank; and the barb long. Their sizes and sorts must of course entirely depend on the kind of fish that is angled for. Floats formed of cork, goose and swan quills, &c., are often used to buoy up the hook so that it may float clear of the bottom. For heavy fish or strong streams a cork float is used; in slow water and for lighter fish quill floats. *Baits* may consist of a great variety of materials, natural or artificial. The principal natural baits are worms: common garden worms, brandlings, and red worms, maggots, or gentles (the larvæ of blow-flies such as are found on putrid meat), insects, small fish (as minnows), salmon roe, &c. The artificial flies so much used in angling for trout and salmon are composed of hairs, furs, and wools of every variety, mingled with pieces of feathers, and secured together by plaited wire, or gold and silver thread, marking-silk, wax, &c. The wings may be made of the feathers of domestic fowls, or any others of a showy colour. Some angling authorities recommend that the artificial flies should be made to resemble as closely as possible the insects on which the fish is wont to feed, but experience has shown that the most capricious and unnatural combinations of feather, fur, &c., have been often successful where the most realistic imitations have failed. Artificial minnows, or other small fish, are also used by way of bait, and are so contrived as to spin rapidly when drawn through the water in order to attract the notice of the fish angled for. Angling, especially with the fly, demands a great deal of skill and practice, the casting of the line properly being the initial difficulty. Nowhere is the art pursued with greater success and enthusiasm than in Britain and the United States.—
BIBLIOGRAPHY: *Fishing* (vol. i), *Salmon and Trout* (vol. ii), *Pike and Coarse Fish* (Badminton Library); H. G. Hutchinson, *Fishing* (2 vols., Country Life

series); Viscount Grey, *Fly Fishing*; Gathorne-Hardy, *The Salmon*; Marquess of Granby, *The Trout*; H. T. Sheringham, *Elements of Angling*; W. M. Gallichan, *The Complete Fisherman*.

Anglo-Catholic, a term sometimes used to designate those churches which hold the principles of the English Reformation, the Anglican or Established Church of England and the allied churches. The term is also applied to that party in the English Church which favours doctrines and religious forms closely approaching those of the Roman Catholic Church, objects to be called Protestant, and corresponds closely with the *Ritualistic* section of the Church.

Anglo-Egyptian Sudan. See *Sudan*.

Anglo-Saxons, the name commonly given to the nation or people formed by the amalgamation of the Angles, Saxons, and Jutes, who settled in Britain in the fifth and sixth centuries after Christ, the Anglo-Saxons being simply the English people of the earlier period of English history. The tribes who were thus the ancestors of the bulk of the English-speaking nationalities came from North Germany, where they inhabited the parts about the mouths of the Elbe and Weser, and the first body of them who gained a footing in Britain are said to have landed in 449, and to have been led by Hengist and Horsa. From the preponderance of the Angles the whole country came to be called *Engla-land*, that is, the land of the Angles or English.



Saxon Architecture.
Doorway, Earl's Barton,
Northampton

Many scholars object to the term 'Anglo-Saxon' as being inaccurate and open to misinterpretation. Correctly used, Anglo-Saxon means *English-Saxon*, as distinguished from the Old-Saxon of the Continent; incorrectly used, as it has been too frequently, it is taken as = Angle + Saxon, a union of Angle and Saxon. Camden (1551-1623) is responsible for the widespread use of the term; ignorance is responsible for the misuse. Many scholars prefer to apply the term 'Old English' to the language and people of England before A.D. 1100, partly because this term is more accurate and partly because its use helps to emphasize the essential continuity of the language.

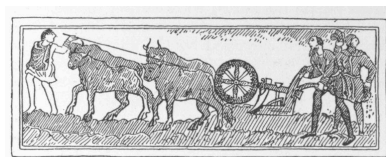
The whole Anglo-Saxon community was frequently spoken of as consisting of the *eorls* and the *ceorls*, or the nobles and common freemen. The former were the men of property and position, the latter were the small landholders, handicraftsmen, &c., who generally placed themselves under the protection of some nobleman, who was hence termed their *hláford* or lord. Besides these there was the class of the serfs or slaves (*theówas*), who might be either born slaves or freemen who had forfeited their liberty by their crimes, or whom poverty or the fortune of war had brought into this position. They served as agricultural labourers on their masters' estates, and were mere chattels, as absolutely the property of their master as his cattle.

The king (*cyning*, *cyng*) was at the head of the State; he was the highest of the nobles and the chief magistrate. He was not looked upon as ruling by any Divine Right, but by the will of the people, as represented by the *witan* (wise men) or great council of the nation. The new king was not always the direct and nearest heir of the late king, but one of the royal family whose abilities and character recommended him for the office. He had the right of maintaining a standing army of household troops, the duty of calling together the *witan*, and of laying before them public measures, with certain distinctions of dress, dwelling, &c., all his privileges being possessed and exercised by the advice and consent of the *witena-gemót* or parliament (literally, 'meeting of the wise'). Next in rank and dignity to the king were the *ealdormen*, who were the chief *witan* or counsellors, and without whose assent laws could not be made, altered, or abrogated. They were at the head of the administration of justice in the shires, possessing both judicial and executive authority, and had as their officers the *scír-geréfan* or sheriffs. The ealdormen led the *fyrð* or armed force of the county, and the ealdorman, as such, held possession of certain lands attached to the office, and was entitled to a share of fines and other moneys levied for the king's use and passing through his hands. The whole executive government may be considered as a great aristocratical association, of which the ealdormen were the members, and the king little more than the president. The ealdorman and the king were both surrounded by a number of followers called *thegnas* or thanes, who were bound by close ties to their superior. The king's thanes were the higher in rank; they possessed a certain quantity of land, smaller in amount than that of an ealdorman, and they filled offices connected with the personal service of the king or with the administration of justice. The *scir-geréfa* (shire-reeve or sheriff) was also an important functionary. He presided at the county court along with the ealdorman and bishop, or alone in their absence; and he had to carry out the decisions of the court, levy fines, collect taxes, &c. The shires were divided into hundreds and tithings, the latter consisting of ten heads of families, who were jointly responsible to the State for the good conduct of any member of their body. For the trial and settlement of minor causes there was a hundred court held once a month. The place of the modern Parliament was held by the *witena-gemót*. Its members, who were not elected, comprised the *æthelings* or princes of the blood royal, the bishops and abbots, the ealdormen, the thanes, the sheriffs, &c.

One of the peculiar features of Anglo-Saxon society was the *wergyld*, which was established for the settling of feuds. A sum, paid either in kind or in money, was placed upon the life of every freeman, according to his rank in the State, his birth, or his office. A corresponding sum was settled for every wound that could be inflicted upon his person; for nearly every injury that could be done to his civil rights, his honour, or his domestic peace, &c. From the operation of this principle no one from king to peasant was exempt.

Agriculture, including especially the raising of cattle, sheep, and swine, was the chief occupation of the Anglo-Saxons. Gardens and orchards are frequently mentioned, and vineyards were common in the southern counties. The forests were extensive, and valuable both from the mast they produced for the swine, and from the beasts of the chase

which they harboured. Hunting was a favourite recreation among the higher ranks, both lay and clerical. Fishing was largely carried on, herrings and salmon being the principal fish caught; and the Anglo-Saxon whaling vessels used to go as far as Iceland. The manufactures were naturally of small moment. Iron was made to some extent, and some cloth, and saltworks were numerous. In embroidery and working of gold the English were famous over Europe. There was a considerable trade at London, which was frequented by Normans, French, Flemings, and the merchants of the Hanse towns. Our Anglo-Saxon forefathers were notorious for their excess in eating and drinking, and in this respect formed a strong contrast to their Norman conquerors. Ale, mead, and cider were the common beverages, wine being limited to the higher classes. Pork and eels were favourite articles of food. The houses were rude structures, but were often richly furnished and hung with fine tapestry. The dress of the people was loose and flowing, composed chiefly of linen, and often adorned with embroidery. The men wore their hair long and flowing over their shoulders. Christianity was introduced among the Anglo-Saxons in the end of the sixth century by St. Augustine, who was sent by Pope Gregory the Great, and became the first Archbishop of Canterbury. Kent, then under King Ethelred, was the first place where it took root, and thence it soon spread over the rest of the



Ploughing

From an Anglo-Saxon Calendar
in the British Museum.

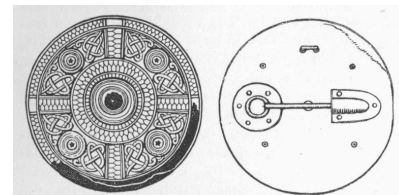
country. The Anglo-Saxon Church long remained independent of Rome, notwithstanding the continual efforts of the Popes to bring it under their power. It was not till the tenth century that this result was brought about by Dunstan. Many Anglo-Saxon ecclesiastics were distinguished for learning and ability, but the Venerable Bede holds the first place.

Anglo-Saxon Language.—The Anglo-Saxon language, which is simply the earliest form of English, claims kinship with Dutch, Icelandic, Danish, Swedish, and German, especially with the Low German dialects (spoken in North Germany). It was called by those who spoke it *Englisc* (English). The existing remains of Anglo-Saxon literature show different dialects, of which the northern and the southern were the principal. The former was the first to be cultivated as a literary language, but afterwards it was supplanted in this respect by the southern or that of Wessex. It is in the latter that the principal Anglo-Saxon works are written. The Anglo-Saxon alphabet was substantially the same as that which we still use, except that some of the letters were different in form, while it had two characters either of which represented the sounds of *th* in *thy* and in *thing*. Nouns and adjectives are declined much as in German or in Latin. The pronouns of the first and second person had a dual number, 'we two' or 'us two' and 'you two', besides the plural for more than two. The infinitive of the verb is in *-an*, the participle in *-ende*, and there is a gerund somewhat similar in its usage to the Latin gerund. The verb had four moods—indicative, subjunctive, imperative, and infinitive, but only two tenses, the present (often used as a future) and the past. Other tenses and the passive voice were formed by auxiliary verbs. Anglo-Saxon words terminated in a vowel much more frequently than the modern English, and altogether the language is so different that it has to be learned quite like a foreign tongue. Yet, notwithstanding the large number of words of Latin or French origin that our language now contains, and the changes it has undergone, its framework, so to speak, is still Anglo-Saxon. Many chapters of the New Testament do not contain more than 4 per cent of non-Teutonic words, and as a whole it averages perhaps 6 or 7.

The existing remains of Anglo-Saxon literature include compositions in prose and poetry, some of which must be referred to a very early period, one or two perhaps to a time before the Angles and Saxons emigrated to England. The most important Anglo-Saxon poem is the ancient epic of

Beowulf, extending to more than 6000 lines. Beowulf is a Scandinavian prince, who slays a monster named Grendel, after encountering supernatural perils, and is at last slain in a contest with a frightful dragon. Its scene appears to be laid entirely in Scandinavia. Its date is uncertain; parts of it may have been brought over at the emigration from Germany, though in its present form it is much later than this. The poetical remains include a number of religious poems, or poems on sacred themes; ecclesiastical narratives, as lives of saints and versified chronicles; psalms and hymns; secular lyrics; allegories, gnomic poems, riddles, &c. The religious class of poems was the largest, and of these Cædmon's (flourished about 660) are the most remarkable. His poems consist of paraphrases of considerable portions of the Bible history, and treat of the creation, the temptation, the fall, the exodus of the Israelites, the story of Daniel, the incarnation, and the harrowing of hell, or release of the ransomed souls by Christ. Other most interesting poems are those ascribed to Cynewulf, the *Christ*, *Elene*, and *Juliana*, the subjects respectively being Christ, the finding of the cross by the Empress Helena, and the life of Juliana. Rhyme was not used in Anglo-Saxon poetry, alliteration being employed instead, as in the older northern poetry generally. The style of the poetry is highly elliptical, and it is full of harsh inversions and obscure metaphors.

The Anglo-Saxon prose remains consist of translations of portions of the Bible, homilies, philosophical writings, history, biography, laws, leases, charters, popular treatises on science and medicine, grammars, &c. Many of these were translations from the Latin. The Anglo-Saxon versions of the Gospels, next to the Moeso-Gothic, are the earliest scriptural translations in any modern language. The Psalms are said to have been translated by Bishop Aldhelm (died 709), and also under Alfred's direction; and the *Gospel of St. John* by Bede; but it is not known who were the authors of the extant versions. A translation of the first seven books of the Bible is believed to have been the work of Ælfric, who was Abbot of Ensham and lived about the beginning of the eleventh century. We have also eighty homilies from his pen, several theological treatises, a Latin grammar, &c.



Anglo-Saxon Brooch
Ornament on front (left) is formed by means of plates of thin gold and wire, with bosses of ivory and red glass.

King Alfred was a diligent author, besides being a translator of Latin works. We have under his name translations of Boethius' *De Consolatione Philosophiæ*, the *Universal History* of Orosius, Bede's *Ecclesiastical History*, the *Pastoral Care* of Gregory the Great, &c. The most valuable to us of the Anglo-Saxon prose writings is the *Saxon Chronicle*, as it is called, a collection of annals recording important events in the history of the country, and compiled in different religious houses. Of this *Chronicle* there are seven MSS. in existence, and the latest text comes down to 1154. A considerable body of laws remains, as well as a large number of charters. The whole of the literature has never yet been printed. For Anglo-Saxon history, see *England*.—BIBLIOGRAPHY: (History) H. M. Chadwick, *The Origin of the English Nation* (Cambridge); (Language) Sweet, *Anglo-Saxon Primer and Reader*; (Literature) B. ten Brink, *Geschichte der Englischen Litteratur*; Stopford A. Brooke, *English Literature, from the beginning to the Norman Conquest*; Henry Morley, *English Writers* (vols. i and ii).

Anglo-Saxon Law. Series of laws written in the vernacular, and unique among Teutonic peoples, were issued from the seventh century onwards by Æthelberht, Hlothhere, Eadric, and Withraed, Kings of Kent, by Ine, King of Wessex, by Alfred, Edward the Elder, Athelstan, Edmund, Edgar, Æthelred, and Canute, in addition to a number of important by-laws and regulations of various kinds, which bear no king's name. We hear, also, of laws issued by other kings which have been lost, and there must have been a considerable body of traditional law which was never committed to writing. What laws are extant, show us a society mainly agricultural, divided by birth into a noble, a free peasant, and a servile class. There was also a dependent class in Kent, intermediate between the freeman and the slave. The strongest social ties were those of the kindred, and the relationship between lord and man.

The laws were issued by the king and his councillors. Cases were decided by the production of oaths which varied in value according to the rank of the swearer, or by the several forms of the ordeal. No distinction was made between civil and criminal law, and even homicide could be atoned for by payment of a sum varying according to the social status of the dead man. The object of the laws was to restrict private vengeance, to prevent and punish theft (primarily of cattle), to stop dishonest trading, to secure the persons and residences of the people, to enforce the mutual obligations of

relatives, and masters and men, to provide adequate defence for the kingdom by means of garrisoned boroughs and a well-armed mounted infantry, to protect the helpless, and to safeguard the rights of the Church and its servants.

The early laws present considerable difficulty owing to their antiquity. The laws of Æthelberht are the earliest records in the English language, though, like many of the other laws, they are only preserved in a MS. of the twelfth century. The standard edition is Liebermann's *Gesetze der Angelsachsen* (Halle, A. S. Max Niemeyer).—BIBLIOGRAPHY: Pollock and Maitland, *History of English Law*; H. M. Chadwick, *Studies in Anglo-Saxon Institutions*.

Ango'la, a Portuguese territory in Western Africa, south of the Congo, extending from the sea to Rhodesia, and from about lat. 6° s. to lat. 17° s. (area, 484,800 sq. miles; pop. 4,119,000). It is a country of varied features, and its resources, as yet hardly known, are probably very great. The province is rich in malachite, copper, iron, petroleum, and salt; rubber supplies are becoming exhausted. The Coanza (Kwanza) is the largest river. The capital is the seaport of Loanda; other ports are Benguella (or Benguela) and Mossamedes. Three railways now run inland from these places. It exports coffee, hides, gum, rubber, wax, &c.

Angola Pea (*Cajānus indicus*). See *Pigeon Pea*.

Ango'niland, a district of South Africa, lying to the west of the southern half of Lake Nyassa, and partly in British Central Africa, partly in Rhodesia. It is a plateau with an average height of 4000 feet, the name being derived from the Angoni, a race of mixed Zulu blood, who used to make murderous raids on their neighbours, and have given much trouble to the missionaries and others.

Ango'ra (ancient, **Ancy'ra**), a town in Asia Minor, 215 miles E.S.E. of Constantinople, with considerable remains of Byzantine architecture, and relics of earlier times, both Greek and Roman, such as the remnants of the Monumentum Ancyranum, raised in honour of the Emperor Augustus, and giving us much valuable information about his public life and work. All the animals of this region are long haired, especially the goats (see *Goat*), sheep, and cats. This hair forms an important export as well as the fabric

called camlet here manufactured from it; other exports being goats' skins, dye-stuffs, gums, honey and wax, &c. A railway connects it with Skutari. Pop. 32,000. In 1920 Kemal Pasha set up a National Government at Angora, and refused to recognize the Treaty of Sèvres. A treaty concluded with France was ratified by the Angora Government on 23rd Oct., 1921.

Angostu´ra, or **Ciudad Bolivar**, a city of Venezuela, capital of the province of Bolivar, on the Orinoco, about 240 miles from the sea, with governor's residence, a college, a handsome cathedral, and a considerable trade, steamers and sailing-vessels ascending to the town. Exports: gold, cotton, indigo, tobacco, coffee, cattle, &c.; imports: manufactured goods, wines, flour, &c. Pop. 17,535.

Angostura Bark, the aromatic bitter medicinal bark obtained chiefly from *Galipēa officinālis*, a tree of 10 to 20 feet high, growing in the northern regions of South America; nat. ord. Rutaceæ. The bark is valuable as a tonic and febrifuge, and is also used for a kind of bitters. From this bark being adulterated, indeed sometimes entirely replaced, by the poisonous bark of *Strychnos Nux-Vomica*, its use as a medicine has been almost given up.

Angoulême (a`n`-gö-lām), an ancient town of Western France, capital of department Charente, on the Charente, 60 miles N.N.E. of Bordeaux, on the summit of a rocky hill. It has a fine old cathedral, built in the twelfth century and restored in 1875, a beautiful modern town hall, built in 1858, a lyceum, public library, natural history museum, &c. There are manufactures of paper, woollens, and linens; distilleries, sugar-works, tanneries, &c. Calvin lived here for three years (1527-30). Pop. 38,211.

Angra do Heroismo, the chief seaport of Terceira, one of the Azores, with the only convenient harbour in the whole group. It has a cathedral, a military college and arsenal, &c., and is the residence of the Governor-General of the Azores, and of the foreign consuls. Pop. 10,057.

Angra Pequena (a`n`gra` pe-kā`na`; Port. 'little bay'), a bay on the coast of former German S.W. Africa, where the Bremen commercial firm Lüderitz in 1883 acquired a strip of territory and established a trading station. In 1884, notwithstanding some weak protests of the British, Germany took under her protection the whole coast territory from the Orange River to 26° s. lat., and soon after extended the protectorate to the Portuguese frontier, but not

including the British settlement of Walvis Bay. Angra Pequena, called by the Germans Lüderitzbucht, was captured by the South African forces in Sept., 1915. See *South-West Africa*.

Angri (a`n`grē), a town of Southern Italy, 12 miles N.W. of Salerno, in the centre of a region which produces grapes, cotton, and tobacco in great quantities. In the vicinity of Angri, Teias, King of the Ostrogoths, was defeated by Narses. Pop. 11,574.

Anguilla (an-gwil`la). See *Eel*.

Anguilla (ang-gil`a), or **Snake Island**, one of the British West India Islands, 60 miles N.E. of St. Kitts; about 20 miles long, with a breadth varying from 3 to 1¼ miles; area, 35 sq. miles. A little sugar, cotton, tobacco, and maize is grown. There is a saline lake in the centre, which yields salt. Pop. 4075, of whom 100 are white.

Anguis (ang`gwis). See *Blind-worm*.

Angus (ang`gus), a name of Forfarshire.

An`halt, formerly a duchy of North Germany, lying partly in the plains of the Middle Elbe, and partly in the valleys and uplands of the Lower Harz, and almost entirely surrounded by Prussia; area, 888 sq. miles. All sorts of grain, wheat especially, are grown in abundance; also flax, rape, potatoes, tobacco, hops, and fruit. Excellent cattle are bred. The inhabitants are principally occupied in agriculture, though there are some iron-works and manufactures of woollens, linens, beet-sugar, tobacco, &c. The dukes of Anhalt traced their origin to Bernard (1170-1212), son of Albert the Bear. In time the family split up into numerous branches, and the territory was afterwards held by three dukes (Anhalt-Köthen, Anhalt-Bernburg, and Anhalt-Dessau). In 1863 the Duke of Anhalt-Dessau became sole heir to the three duchies. The united principality, incorporated in the German Empire, had one vote in the Bundesrath and two in the Reichstag. The executive power, previous to the changes resulting from the European War, was vested in the duke, and the legislative in a Diet of thirty-six members. The reigning duke in 1918 was Eduard, who succeeded his brother on 21st April, 1918. With the outbreak of the revolution in Germany in 1918 Anhalt became a republic, but its status in the German Republic still remains to be

determined. Pop. (1919), 331,258, almost all Protestants. The chief towns are Dessau, Bernburg, Köthen, and Zerbst.

An'holt, an island belonging to Denmark, in the Cattegat, midway between Jutland and Sweden, 7 miles long, $4\frac{1}{2}$ broad, largely covered with drift-sand, and surrounded by dangerous banks and reefs. Pop. 300.

Anhy'dride, a chemical term synonymous with acidic oxide (see *Chemistry*) and applied to those oxides which unite with water to form acids. They were formerly called *anhydrous acids*.

Anhy'drite, anhydrous sulphate of calcium, a mineral presenting several varieties of structure and colour. The *vulpinite* of Italy possesses a granular structure, resembling a coarse-grained marble, and is used in sculpture. Its colour is greyish-white, intermingled with blue.

Ani (ä'nē), a ruined city in Armenia, formerly the residence of the Armenian dynasty of the Bagratidæ, having in the eleventh century a population of 100,000 and 1000 churches. In the thirteenth century it was taken by the Tartars, and was destroyed by an earthquake in 1319.

Aniche (a' -nēsh), a town or village in the French department Nord, arrondissement Douai, with coal-mines, glass-works, chemical-works, &c. Pop. 6927.

Aniene (a' -nē-ā'nā). See *Anio*.

An'iline, $C_6H_5NH_2$, is an extremely important substance as it forms the starting-point in the preparation of a large number of substances. It was first prepared by Unverdorben, in 1826, by distillation of indigo. Aniline is present in small quantity in coal-tar, and is prepared commercially from benzene by transforming it by means of nitric acid into nitro-benzene and reducing this with scrap-iron and hydrochloric acid. The substance can also be prepared by reducing nitro-benzene electrolytically. It is a liquid of peculiar odour, boiling at $182^\circ C.$, colourless when quite pure, but rapidly darkening in colour on standing, so that commercial aniline is usually dark-brown. It is a basic substance, and forms crystalline salts with acids. The salts, like aniline itself, become coloured on exposure to air. Aniline contains the characteristic chemical group NH_2 , the amino group, and

substances containing this group react with nitrous acid at 0° C., forming diazonium compounds; these combine readily with phenols, naphthols, and other amino compounds to form azo compounds, highly-coloured compounds many of which are dyes. Many dyes are prepared from aniline, e.g. rosaniline, magenta, methylene blue, aniline blue, &c., also some explosives, e.g. tetranitraniline, which is a powerful explosive prepared by nitrating aniline and the substance tetranitromethylaniline, "tetryl", used in detonators. Several medicinal substances are also prepared from aniline, for instance, antifebrin and atoxyl.

An'ilism, aniline poisoning, a name given to the aggregate of symptoms which often show themselves in those employed in aniline works, resulting from the inhalation of aniline vapours. It may be either acute or chronic. In a slight attack of the former kind, the lips, cheeks, and ears become of a bluish colour, and the person's walk may be unsteady; in severe cases there is loss of consciousness. Chronic anilism is accompanied by derangement of the digestive organs and of the nervous system, headaches, eruptions on the skin, muscular weakness, &c.

Animal, an organized and sentient living being. Life in the earlier periods of natural history was attributed almost exclusively to animals. With the progress of science, however, it was extended to plants. In the case of the higher animals and plants there is no difficulty in assigning the individual to one of the two great kingdoms of organic nature, but in their lowest manifestations the vegetable and animal kingdoms are brought into such immediate contact that it becomes almost impossible to assign them precise limits, and to say with certainty where the one begins and the other ends. From *form* no absolute distinction can be fixed between animals and plants. Many animals, such as the sea-shrubs, sea-mats, &c., so resemble plants in external appearance that they were, and even yet popularly are, looked upon as such. With regard to *internal structure* no line of demarcation can be laid down, all plants and animals being, in this respect, fundamentally similar; that is, alike composed of molecular, cellular, and fibrous tissues. Neither are the chemical characters of animal and vegetable substances more distinct. Animals contain in their tissues and fluids a larger proportion of nitrogen than plants, whilst plants are richer in carbonaceous compounds than the former. In some animals, moreover, substances almost exclusively confined to plants are found. Thus the outer wall of the Sea-squirts contains

cellulose, a substance largely found in plant-tissues; whilst *chlorophyll*, the colouring-matter of plants, occurs in Hydra and many other lower animals. *Power of motion*, again, though broadly distinctive of animals, cannot be said to be absolutely characteristic of them. Thus many animals, as oysters, sponges, corals, &c., in their mature condition are rooted or fixed, while the embryos of many plants, together with numerous fully-developed forms, are endowed with locomotive power by means of vibratile, hair-like processes called cilia. The distinctive points between animals and plants which are most to be relied on are those derived from the *nature and mode of assimilation of the food*. Plants feed on *inorganic matters*, consisting of water, ammonia, carbonic acid, and mineral matters. They can only take in food which is presented to them in a *liquid* or *gaseous* state. The exceptions to these rules are found chiefly in the case of plants which live *parasitically* on other plants or animals, in which cases the plant may be said to feed on organic matters, represented by the juices of their hosts. Animals, on the contrary, require *organized* matters for food. They feed either upon plants or upon other animals. But even carnivorous animals can be shown to be dependent upon plants for subsistence; since the animals upon which Carnivora prey are in their turn supported by plants. Animals, further, can subsist on *solid* food in addition to liquids and gases; but many animals (such as the Tapeworms) live by the mere imbibition of fluids which are absorbed by their tissues, such forms possessing no distinct digestive system. Animals require a due supply of *oxygen gas* for their sustenance, this gas being used in respiration. Plants, on the contrary, require *carbonic acid*. The animal exhales or gives out carbonic acid as the part result of its tissue-waste, whilst the plant, taking in this gas, is enabled to decompose it into its constituent carbon and oxygen. The plant retains the former for the uses of its economy, and liberates the oxygen, which is thus restored to the atmosphere for the use of the animal. Animals receive their food into the interior of their bodies, and assimilation takes place in their internal surfaces. Plants, on the other hand, receive their food into their external surfaces, and assimilation is effected in the external parts, as is exemplified in the leaf-surfaces under the influence of sunlight. Cf. T. J. Parker and W. A. Haswell, *Text-Book of Zoology; Cambridge Natural History*.

Animal Chemistry. See *Chemistry*.

Animalcule (an-i-mal'kūl), a general name given to many forms of animal life from their minute size. The larger examples are just visible to the naked eye, but most of them are strictly microscopic. Some are pigmented, but the majority are colourless. The term is not applicable to a particular zoological type, but it is customary to confine it to the 'Protozoa', 'Rotifera', or 'Wheel Animalcules'.

Animal Heat. All animals possess a certain amount of heat or temperature which is necessary for the performance of vital action. The only classes of animals in which a constantly-elevated temperature is kept up are birds and mammals. The bodily heat of the former varies from 39.4° to 43.9° C., and of the latter from 35.5° to 40.5° C. The mean or average heat of the human body is about 99° F., and it never falls much below this in health. Below birds, animals are named 'cold-blooded', this term meaning in its strictly-physiological sense that their temperature is usually that of the medium in which they live, and that it varies with that of the surrounding medium. The temperature of 'warm-blooded' animals is remarkably constant, although there are individual variations. In man this variation is slight, amounting only to fractions of a degree. The cause of the evolution of heat in the animal body is referred to the union (by a process resembling ordinary combustion) of the carbon and hydrogen of the system with the oxygen taken in from the air in the process of respiration.

Animal Magnetism. See *Hypnotism*, *Mesmer*.

Animals, Cruelty to, an offence against which societies have been formed and laws passed in England and other countries. According to English law, if any person shall cruelly beat, ill-treat, overdrive, abuse, or torture any domestic animal, he shall forfeit a sum not exceeding £5 for every such offence. Bull-baiting, cock-fighting, and the like are also prohibited. Provision is also made for the punishment of persons unlawfully and maliciously killing, maiming, or wounding cattle, dogs, birds, beasts, and other animals.

Animal Worship, a practice found to prevail, or to have prevailed, in the most widely-distant parts of the world, both the Old and the New, but nowhere to such an amazing extent as in ancient Egypt, notwithstanding its high civilization. Nearly all the more important animals found in the

country were regarded as sacred in some part of Egypt, and the degree of reverence paid to them was such that throughout Egypt the killing of a hawk or an ibis, whether voluntary or not, was punished with death. The worship, however, was not, except in a few instances, paid to them as actual deities. The animals were merely regarded as sacred to the deities, and the worship paid to them was symbolical.

An'ima Mun'di. See *Pantheism*.

Animé (an'i-me), a resin obtained from the trunk of an American tree (*Hymenæa Courbaril*). It is of a transparent amber colour, has a light, agreeable smell, and is soluble in alcohol. It strongly resembles copal, and, like it, is used in making varnishes. See *Copal*.

An'imism, the system of medicine propounded by Stahl, and based on the idea that the soul (*anima*) is the seat of life. In modern usage the term is applied to express the general doctrine of souls and other spiritual beings, and especially to the tendency, common among savage races, to attribute souls or spirits to inanimate things, and to explain phenomena not due to obvious natural causes by attributing them to spiritual agency. Amongst the beliefs of animism is that of a human apparitional soul, bearing the form and appearance of the body, and living after death a sort of semi-human life. —BIBLIOGRAPHY: Sir J. G. Frazer, *The Golden Bough*; Andrew Lang, *Myth, Ritual, and Religion*.

Anio (now **Aniēne** or **Teverōne**), a river in Italy, a tributary of the Tiber, which it enters from the east a short distance above Rome, renowned for the natural beauties of the valley through which it flows, and for the remains of ancient buildings there situated, as the villas of Mæcenæ and the Emperor Hadrian.

Anise (an'is; *Pimpinella Anīsum*), an annual plant of the nat. ord. Umbelliferæ, a native of Eastern Asia, Egypt, and the Mediterranean coasts, and cultivated in Spain, France, Italy, Malta, &c., whence the fruit, popularly called *aniseed*, is imported. This fruit is ovate, with ten narrow ribs, between which are oil-vessels. It has an aromatic smell, and is largely employed to flavour liqueurs (aniseed or anisette), sweetmeats, &c. *Star-anise* is the fruit of an evergreen Asiatic tree (*Illicium anisātum*), nat. ord. Magnoliaceæ, and is brought chiefly from China. Its flavour is similar to

that of anise, and it is used for the same purposes. An essential oil is obtained from both kinds of anise, and is used for scenting soaps and in the preparation of cordials.

Anjou (a´n´-zhö), an ancient province of France, now forming the department of Maine-et-Loire, and parts of the departments of Indre-et-Loire, Mayenne, and Sarthe; area, about 3000 sq. miles. In 1060 the province passed into the hands of the House of Gatinais, of which sprang Count Godfrey V, who, in 1127, married Matilda, daughter of Henry I of England, and so became the ancestor of the Plantagenet kings. Anjou remained in the possession of the English kings up to 1204, when John lost it to the French king Philip Augustus. In 1226 Louis VIII bestowed this province on his brother Charles; but in 1328 it was reunited to the French Crown. John I raised it to the rank of a duchy, and gave it to his son Louis. Henceforth it remained separate from the French Crown till 1480, when it fell to Louis XI.

Ankarström (a´n´ka´r-streum), Jan Jakob, the murderer of Gustavus III of Sweden, was born about 1762, and was at first a page in the Swedish Court, afterwards an officer in the royal bodyguard. He was a strenuous opponent of the sovereign's measures to restrict the privileges of the nobility, and joined Counts Horn and Ribbing in a plot to assassinate Gustavus. The assassination took place on 15th March, 1792. Ankarström was tried, tortured, and executed in April, dying boasting of his deed.

Anker, an obsolete measure used in Britain for spirits, beer, &c., containing 8½ imperial gallons. A measure of similar capacity was used in Germany and elsewhere in Europe.

An´klam, a town in Prussia, province of Pomerania, 47 miles north-west of Stettin, on the River Peene, which is here navigable. Shipbuilding, woollen and cotton manufactures, soap-boiling, tanning, &c., are carried on. Pop. 15,280.

Anko´bar, or **Anko´ber**, a town in Abyssinia, former capital of Shoa, on a steep conical hill 8200 feet high. Pop. 2000.

Ankylo´sis, or **Anchylo´sis**, stiffness of the joints caused by a more or less complete coalescence of the bones through ossification, often the result of

inflammation or injury. False ankylosis is stiffness of a joint when the disease is not in the joint itself, but in the tendinous and muscular parts by which it is surrounded.

Ankylostomi'asis, a 'worm disease' to which miners are subject in some localities, is caused by vast numbers of small parasitic worms (*Ankylostoma* or *Anchylostoma duodenale*) in the duodenum or upper portions of the intestinal canal. Deriving their sustenance from the system, these worms produce anæmia or bloodlessness (that is, deficiency of the red corpuscles of the blood), the sufferers being pallid, feeble, short-breathed, liable to faint, and unequal to any laborious work, and death may result if a cure is not effected. Fortunately the disease is not difficult to cure if the remedies are applied—remedies such as will expel the worms from the intestine. The disease is said to be common in tropical and sub-tropical countries all over the world. In Europe it was perhaps first observed in 1879 in the case of workmen engaged in excavating the St. Gothard tunnel. Since 1896 it has been well known in some of the German mines; and in 1903 it was detected among the miners engaged in the Dolcoath mine in Cornwall. The eggs of the worms are carried from the body with the fæces; under favourable circumstances they develop into larvæ, which may gain entrance again into the human body by the mouth (perhaps in drinking-water), to attain full development in the intestine. Careful sanitary arrangements are a preventive of the disease, which is also known as 'miner's worm', 'miner's anæmia', &c.

Ann, or **Annat**, in Scottish law, the half-year's stipend of a living, after the death of the clergyman, payable to his family or next of kin. The right to the ann is not vested in the clergyman himself, but in his representatives; and, accordingly, it can neither be disposed of by him nor attached for his debts.

Anna, an Anglo-Indian money of account, the sixteenth part of a rupee, and of the value of one penny; it is divided into four pice.

An'naberg, a town in Saxony, 47 miles south-west of Dresden. Mining (for silver, cobalt, iron, &c.) is carried on, and there are manufactures of lace, ribbons, fringes, buttons, &c. Pop. 17,025.

Anna Comne'na, daughter of Alexius I, Comnenus, Byzantine emperor. She was born 1083, and died 1148. After her father's death she endeavoured to secure the succession for her husband, Nicephorus Briennius, but was

baffled by his want of energy and ambition. She wrote (in Greek) a life of her father Alexius (*The Alexiad*, a work in fifteen books). She is a character in Sir Walter Scott's *Count Robert of Paris*.

Anna Ivanov'na, Empress of Russia, born in 1693, the daughter of Ivan, the elder half-brother of Peter the Great. She was married in 1710 to the Duke of Courland, in the following year was left a widow, and in 1730 ascended the throne of the tsars on the condition proposed by the senate, that she would limit the absolute power of the tsars, and do nothing without the advice of the council composed of the leading members of the Russian aristocracy. But no sooner had she ascended the throne than she declared her promise null, and proclaimed herself autocrat of all the Russias. She chose as her favourite Ernest John von Biren or Biron, who was soon all-powerful in Russia, and ruled with great severity. Several of the leading nobles were executed, and many thousand men exiled to Siberia. In 1737 Anna forced the Courlanders to choose Biren as their duke, and nominated him at her death regent of the empire during the minority of Prince Ivan (of Brunswick). Anna died in 1740. See *Biren*.

An'nals, a history of events in chronological order, each event being recorded under the year in which it occurred. The name is derived from the first annual records of the Romans, which were called *annāles pontificum* or *annāles maxīmi*, drawn up by the *pontifex maximus* (chief pontiff). The practice of keeping such annals was afterwards adopted also by various private individuals, as by Fabius Pictor, Calpurnius Piso, and others. The name hence came to be applied in later times to historical works in which the matter was treated with special reference to chronological arrangement, as to the *Annals* of Tacitus.

Annam', a country of Asia occupying the east side of the South-eastern or Indo-Chinese Peninsula, along the China Sea. It comprises Tonquin in the north, Annam (in a narrower sense), and Cochin-China farther south; with the inland territory of the Laos tribes: together, area, 170,000 sq. miles; pop. 15,000,000, 9,000,000 being in Tonquin. In the narrow sense Annam now denotes the country between Tonquin and French Cochin-China, under the nominal rule of a native king (the present ruler, Khai-Dinh, succeeded to the throne in 1916). Annam has an area of 52,100 sq. miles. Pop. (1919), 5,952,000, including 2117 Europeans. The coast is considerably indented,

especially at the mouths of the rivers, where it affords many commodious harbours. Tonquin is mountainous on the north, but in the east is nearly level, terminating towards the sea in an alluvial plain yielding good crops of rice, cotton, fruits, ginger, and spices, and a great variety of varnish trees, palms, &c. The principal river is the Song-ka, which has numerous tributaries, many of them being joined together by canals, both for irrigation and commerce. Tonquin is rich in gold, silver, copper, and iron. Annam (in the narrow sense) is, generally speaking, unproductive, but contains many fertile spots, in which grain, leguminous plants, sugar-cane, cinnamon, &c., are produced in great abundance. Agriculture is the chief occupation, but many of the inhabitants are engaged in the spinning and weaving of cotton and silk into coarse fabrics, the preparation of varnish, iron-smelting, and the construction of ships or junks. The inhabitants are said to be the ugliest of the Mongoloid races of the peninsula, being under the middle size and less robust than the surrounding peoples. Their language is monosyllabic, and is connected with the Chinese. The religion of the majority is Buddhism, but the educated classes hold the doctrines of Confucius. The principal towns are Hanoi, the capital of Tonquin, and Huë, the capital of the kingdom and formerly of the whole empire. Annam was conquered by the Chinese in 214 B.C., but in A.D. 1428 it completely won its independence. The French began to interfere actively in its affairs in 1847 on the plea of protecting the native Christians. By the treaties of 1862 and 1867 they obtained the southern and most productive part of Cochin-China, subsequently known as French Cochin-China; and in 1874 they obtained large powers over Tonquin, notwithstanding the protests of the Chinese. Finally, in 1883, Tonquin was ceded to France, and next year Annam was declared a French protectorate. After a short period of hostilities with China the latter recognized the French claims, and Tonquin is now a French colony, while the kingdom of Annam is, since 1886, entirely under French direction. Cf. F. R. Eberhardt, *Guide de l'Annam*.

Annamaboe (-bō'), a seaport in Western Africa, on the Gold Coast, 10 miles east of Cape Coast Castle, with some trade in gold-dust, ivory, palm-oil, &c. Pop. about 5000.

An'nan, a royal and police burgh in Scotland, on the Annan, a little above its entrance into the Solway Firth, one of the Dumfries district of burghs. Pop. 3928.—The River *Annan* is a stream 40 miles long running through

the central division of Dumfriesshire, to which it gives the name of *Annandale*.

Annap´olis, the capital of Maryland, United States, on the Severn, near its mouth in Chesapeake Bay. It contains a college (St. John's), a state-house, and the United States Naval Academy. Pop. (1920), 11,214.

Annap´olis, a small town in Nova Scotia, on an inlet of the Bay of Fundy, with an important traffic by railway and steamboat. It is one of the oldest European settlements in America, dating from 1604.

Ann Arbor, a town of Michigan, United States, on the Huron River, about 40 miles west of Detroit; the seat of the State university. It has flour-mills, and it manufactures woollens, iron, and agricultural implements. Pop. 19,516.

Annates (an´nāts), a year's income claimed for many centuries by the Pope on the death of any bishop, abbot, or parish priest, to be paid by his successor. In England they were at first paid to the Archbishop of Canterbury, but were afterwards appropriated by the Popes. In 1532 the Parliament gave them to the Crown; but in 1703 Queen Anne restored them to the Church by applying them to the augmentation of poor livings. See *Queen Anne's Bounty*.

Annat´to, or **Annato**, an orange-red colouring matter, obtained from the pulp surrounding the seeds of *Bixa Orellāna*, a shrub native to tropical America, and cultivated in Guiana, St. Domingo, and the East Indies. It is sometimes used as a dye for silk and cotton goods, though it does not produce a very durable colour, but it is much used in medicine for tinging plasters and ointments, and to a considerable extent by farmers for giving a rich colour to milk, butter, and cheese. The colour given by annatto approaches very nearly the natural colouring matter of milk fat. It is guaranteed to preserve the same colour throughout the year, and is considered to be a legitimate colouring matter.



Annatto (*Bixa Orellāna*)

Anne, Queen of Great Britain and Ireland was born at Twickenham, near London, 6th Feb., 1664. She was the second daughter of James II, then Duke of York, and Anne, his wife, daughter of the Earl of Clarendon. With her father's permission she was educated according to the principles of the English Church. In 1683 she was married to Prince George, brother of King Christian V of Denmark. On the arrival of the Prince of Orange in 1688, Anne wished to remain with her father; but she was prevailed upon by Lord Churchill (afterwards Duke of Marlborough) and his wife to join the triumphant party. After the death of William III in 1702 she ascended the English throne. Her character was essentially weak, and she was governed first by Marlborough and his wife, and afterwards by Mrs. Masham. Most of the principal events of her reign are connected with the war of the Spanish Succession. The only important acquisition that England made by it was Gibraltar, which was captured in 1704. Another very important event of this reign was the union of England and Scotland under the name of Great Britain, which was accomplished in 1707. Anne seems to have long cherished the wish of securing the succession to her brother James, but this was frustrated by the internal dissensions of the cabinet. Grieved at the disappointment of her secret wishes, she fell into a state of weakness and lethargy, and died, 20th July, 1714. The reign of Anne was distinguished not only by the brilliant successes of the British arms, but also on account of the number of admirable and excellent writers who flourished at this time, among whom were Pope, Swift, and Addison. Anne bore her husband many children, all of whom died in infancy, except one son, the Duke of Gloucester, who died at the age of twelve.

Anne (of Austria), daughter of Philip III of Spain, was born at Madrid in 1602, and in 1615 was married to Louis XIII of France. Richelieu, fearing the influence of her foreign connections, did everything he could to humble her. In 1643 her husband died, and she was left regent, but placed under the control of a council. But the Parliament overthrew this arrangement, and entrusted her with full sovereign rights during the minority of her son Louis XIV. Having brought upon herself the hatred of the nobles by her boundless confidence in Cardinal Mazarin, she was forced to flee from Paris during the wars of the Fronde. She ultimately quelled all opposition, and was able in 1661 to transmit to her son unimpaired the royal authority. She spent the remainder of her life in retirement, and died 20th Jan., 1666.

Annealing (an-ēl'ing), a process to which many articles of metal and glass are subjected after making, in order to render them more tenacious and which consists in heating them and allowing them to cool slowly. When the metals are worked by the hammer, or rolled into plates, or drawn into wire, they acquire a certain amount of brittleness, which destroys their usefulness, and has to be remedied by annealing. The tempering of steel is one kind of annealing. Annealing is particularly employed in glass-houses, and consists in putting the glass vessels, as soon as they are formed and while they are yet hot, into a furnace or oven, in which they are suffered to cool gradually. The toughness is greatly increased by cooling the articles in oil.

Annecy (a'n-sē), an ancient town in France, department of Haute-Savoie, situated on the Lake of Annecy, 21 miles s. of Geneva; contains a cathedral and a ruinous old castle once the residence of the counts of Genevois; manufactures of cotton, leather, paper, and hardware. Pop. 15,622.—The lake is about 9 miles long and 2 broad.

Annel'ida, or **Annulata**, an extensive division or class of Annulosa or articulate animals, so called because their bodies are formed of a great number of small rings, the outward signs of internal segmentation. The earth-worm, the lobworm, the nereis, and the leech belong to this division. They have red, rarely yellow or green, blood circulating in a double system of contractile vessels, a double ganglionated nervous cord, and respire by external branchiæ, internal vesicles, or by the skin. Their organs of motion consist of bristles or *setæ*, which are usually attached to the lateral surfaces of each segment, the bristles being borne on 'foot processes' or *parapodia*. The number of body segments varies. As many as 400 may be found in some sea-worms. A complete digestive system is developed, consisting of a mouth—armed with horny jaws and a protrusible proboscis—gizzard, stomach, and intestine. See *Earth-worm*, *Leech*, &c.



Lobworm (one of the Annelida)

Annexation, a term applied to the acquisition by a State of territory previously belonging to another Power, or independent. It is applicable not only to the extension of a State's sovereignty over adjoining territory, but also to an acquisition of a remote territory. The inhabitants of the annexed

territory are absolved from their allegiance to their former sovereign. Such annexations in modern history were those of Alsace-Lorraine by Germany in 1871, of California by the United States, of Bosnia and Herzegovina by Austria in 1908, and of the Boer Republics by Great Britain.

Annfield Plain, a straggling colliery town (urban district) of England, Durham, about 7 miles south-west of Gateshead. Pop. (1921), 16,524.

Annobon', or **Annobom**, a beautiful Spanish island of Western Africa, south of the Bight of Biafra, about 4 miles long by 2 miles broad, and rising abruptly to the height of 3000 feet, richly covered with vegetation. Pop. 2000.

Annonay (a'n-o-nā), a town in southern France, department of Ardèche, 37 miles s.s.w. of Lyons, in a picturesque situation. It is the most important town of Ardèche, manufacturing paper and glove leather to a large extent, also cloth, felt, silk stuffs, gloves, hosiery, chemical manures, glue, gelatine, brushes, chocolate, and candles. There is an obelisk in memory of Joseph Montgolfier of balloon fame. Pop. 16,660.

An'nual, in botany, a plant that springs from seed, grows up, produces seed, and then dies, all within a single year or season. Among annual grasses may be noted all our cereals, barley, wheat, rye, and oats.

An'nual, in literature, the name given to a class of publications which at one time enjoyed an immense yearly circulation, and were distinguished by great magnificence both of binding and illustration, which rendered them much sought after as Christmas and New Year presents. Their contents were chiefly prose tales and ballads, lyrics, and other poetry. The earliest was the *Forget-me-not*, started in 1822, and followed next year by the *Friendship's Offering*. The *Literary Souvenir* was commenced in 1824, and the *Keepsake* in 1827. Among the names of the editors occur those of Alaric A. Watts, Mrs. S. C. Hall, Harrison Ainsworth, Lady Blessington, Mary Howitt, &c. The popularity of the annuals reached its zenith about 1829, when no less than seventeen made their appearance; in 1856 the *Keepsake*, the last of the series, ceased to exist.

Annual Register, an English publication commenced in 1758 by Dodsley, the publisher, and since continued in yearly volumes down to the present

day. There was also an *Edinburgh Annual Register*, the historical part of which was for several years contributed by Sir Walter Scott and afterwards by Robert Southey. It commenced in 1808 and came to a close in 1827.

Annu'ity, a periodical payment, made annually, or at more frequent intervals, and continuing either a certain number of years, or for an uncertain period, to be determined by a particular event, as the death of the recipient or annuitant, or that of the party liable to pay the annuity; or the annuity may be perpetual. An annuity is usually raised by the present payment of a certain sum as a consideration whereby the party making the payment, or some other person named by him, becomes entitled to an annuity, and the rules and principles by which this present value is to be computed have been the subjects of careful investigation. The present value of a perpetual annuity is evidently a sum of money that will yield an interest equal to the annuity, and payable at the same periods; and an annuity of this description, payable quarterly, will evidently be of greater value than one of the same amount payable annually, since the annuitant has the additional advantage of the interest on three of the quarterly payments until the expiration of the year. In other words, it requires a greater present capital to be put at interest to yield a given sum per annum, payable quarterly, than to yield the same annual sum payable at the end of each year. The present value of an annuity for a limited period is a sum which, if put at interest, will at the end of that period give an amount equal to the sum of all the payments of the annuity and interest; and, accordingly, if it be proposed to invest a certain sum of money in the purchase of an annuity for a given number of years, the comparative value of the two may be precisely estimated, the rate of interest being given. But annuities for uncertain periods, and particularly life annuities, are more frequent, and the value of the annuity is computed according to the probable duration of the life by which it is limited. Such annuities are often created by contract, whereby the Government or a private annuity office agrees, for a certain sum advanced by the purchaser, to pay a certain sum in yearly, quarterly, or other periodical payments to the person advancing the money, or to some other named by him, during the life of the annuitant. Or the annuity may be granted to the annuitant during the life of some other person, or during two or more joint lives, or during the life of the longest liver or survivor among a number of persons named. If a person having a certain capital, and

intending to spend this capital and the income of it during his own life, could know precisely how long he should live, he might lend his capital at a certain rate during his life, and by taking every year, besides the interest, a certain amount of the capital, he might secure the same annual amount for his support during his life in such manner that he should have the same sum to spend every year, and consume precisely his whole capital during his life. But since he does not know how long he is to live, he agrees with the Government or an annuity office to take the risk of the duration of his life, and they agree to pay him a certain annuity for life in exchange for the capital which he proposes to invest in this way. The probable duration of his life therefore becomes a subject of computation; and for the purpose of making this calculation tables of longevity are made by noting the proportion of deaths at certain ages in the same country or district. Founding on a comparison of many such tables, the British Government has empowered the Postmaster-General to grant annuities at the following rates, which are probably more closely adjusted to their actual value than those of insurance companies and other dealers in annuities: To secure an immediate annuity of £100, the cost is, for males of 20 years, £2279, 3s. 4d.; for females of same age, £2482, 10s.; for males of 30 years, £2045, 8s. 4d., for females, £2258, 6s. 8d.; for males of 40 years, £1789, 6s. 8d.; for females, £1990; for males of 60, £1148, 6s. 8d.; females, £1275, 8s. 4d.; and so on. *Deferred* annuities, that is, such as have their first payments postponed for a greater or less number of years, are also granted. We give the rates for an annuity of £100 deferred 20 years: Males aged 20, £848, 6s. 8d.; females, £1014, 13s. 4d.; males aged 35, £557, 1s. 8d.; females, £697, 1s. 8d.; and so on. If a person on whose life the deferred annuity is to depend should die before payment commences, the purchase-money may be returned to his or her representatives, provided that an agreement to that effect had been made in the first instance, but in this case the purchase-money is necessarily higher. See *Insurance*.—BIBLIOGRAPHY: Baily, *Life Annuities and Assurances*; J. Henry, *Government Life Annuity Commutation Tables*.

Annuloi'da, one of Professor Huxley's eight primary groups, a division (sub-kingdom) of animals, including the Rotifera, Scolecida (tape-worms, &c.), all which are more or less ring-like in appearance, and the Echinodermata, whose embryos show traces of annulation.

Annulo'sa, a division (sub-kingdom) of animals regarded by some as synonymous with the Arthropoda or Articulata; according to other systematists, including both the Articulata and Annulata or worms.

Annunciation, the declaration of the angel Gabriel to the Virgin Mary informing her that she was to become the mother of our Lord.—*Annunciation* or *Lady Day* is a feast of the Church in honour of the annunciation, celebrated on the 25th of March.—The Italian order of *Knights of the Annunciation* was instituted by Amadeus VI, Duke of Savoy, in 1360. The king is always grand-master. The knights must be of high rank, and must already be members of the order of St. Mauritius and St. Lazarus. The decoration of the order consists of a golden shield suspended from a chain or collar of roses and knots, the letters F. E. R. T. being inscribed on the roses, and standing for *Fortitudo ejus Rhodum tenuit* (its bravery held Rhodes).—There are two orders of *nuns of the Annunciation*, one originally French, founded in 1501 by Joanna of Valois; the other Italian, founded in 1604 by Maria Vittoria Fornari of Genoa.

Annunzio (a'n-nu'nt'syō), Gabriele d', Italian poet, novelist, and dramatist, born at Pescara in 1863, his patronymic being Rapagnetta. He was educated at Prato and in Rome, and early took to literature and journalism. In 1898 he was elected a member of the Italian Chamber of Deputies, in which he joined the Socialist party. He came before the public when a schoolboy with a volume of verse called *Primo Vere*, to which others—naturally much more mature—were subsequently added. Several of his novels have been published in English, as: *The Child of Pleasure*, *The Victim*, *The Triumph of Death*, *The Virgin of the Rocks*, *The Fire of Life*. Some of these have been very successful, though disfigured to some extent by coarse realism and voluptuousness. He began to write plays later in life. Among them *Gioconda*, *The Dead City*, and *Francesca da Rimini* may be read in English versions, and *Gioconda* and *Francesca* have been performed on the English stage. His more recent works include: *Le Martyr de Saint Sebastien* (1911), *Le Chèvrefeuille* (1914), *La Beffa di Buccari* (1918), *Notturmo* (1918). D'Annunzio is the most prominent Italian writer of the present day, and in wealth of language and distinction of style stands far ahead of all others. He served in the European War from 1915-18, and was wounded. In Sept., 1919, he led a raid and occupied the port of Fiume. See *European War* and *Fiume*.

An´oa, an animal (*Anoa depressicornis*) closely allied to the buffalo, about the size of an average sheep, very wild and fierce, inhabiting the rocky and mountainous localities of the Island of Celebes. The horns are straight, thick at the root, and set nearly in a line with the forehead.

Ano´bium, a genus of coleopterous insects, the larvae of which often do much damage by their boring into old wood. By means of their heads they produce a loud, ticking sound in the wood, the so-called *death-watch* ticking. *A. striātum*, a common species, when frightened, is much given to feigning death.

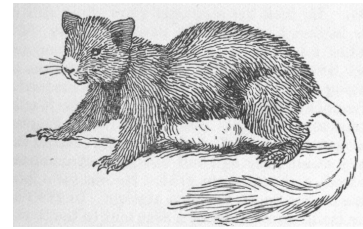
An´ode, (Gr. *ana*, up, *hodos*, way), the positive pole of the voltaic current, being that part of the surface of a decomposing body which the electric current enters: opposed to *cathode* (Gr. *kata*, down, *hodos*, way), the way by which it departs.

An´odon, or **Anodon´ta**, a genus of lamellibranchiate bivalves, including the fresh-water mussels, without or with very slight hinge-teeth. See *Mussel*.

An´odyne, a medicine, such as an opiate or narcotic, which allays pain.

Anointing, rubbing the body or some part of it with oil, often perfumed. From time immemorial the nations of the East have been in the habit of anointing themselves for the sake of health and beauty. The Greeks and Romans anointed themselves after the bath. Wrestlers anointed themselves in order to render it more difficult for their antagonists to get hold of them. In Egypt it was common to anoint the head of guests when they entered the house where they were to be entertained. In the Mosaic law a sacred character was attached to the anointing of the garments of the priests, and things belonging to the ceremonial of worship. The Jewish priests and kings were anointed when inducted into office, and were called the *anointed of the Lord*, to show that their persons were sacred and their office from God. In the Old Testament also the prophecies respecting the Redeemer style him *Messias*, that is, the *Anointed*, which is also the meaning of his Greek name Christ. The custom of anointing still exists in the Roman Catholic Church in the ordination of priests and the confirmation of believers and the sacrament of extreme unction. The ceremony is also frequently a part of the coronation of kings.

Anom'alure (*Anomalūrus*), a genus of rodent animals inhabiting the west coast of Africa, resembling the flying-squirrels, but having the under surface of the tail furnished for some distance from the root with a series of large horny scales, which, when pressed against the trunk of a tree, may subserve the same purpose as those instruments with which a man climbs up a telegraph pole to set the wires.

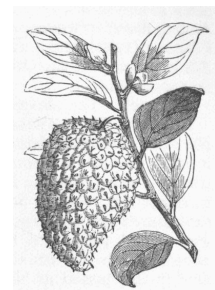


Anomalure (*Anomalurus Peli*)

Anom'aly, a deviation from the common rule. In astronomy, the angle which a line drawn from a planet to the sun has passed through since the planet was last at its perihelion or nearest distance to the sun. The *anomalistic year* is the interval between two successive times at which the earth is in perihelion, or 365 days 6 hours 13 minutes 48 seconds. In consequence of the advance of the earth's perihelion among the stars in the same direction as the earth's motion, and of the precession of the equinoxes, which carries the equinoxes back in the opposite direction to the earth's motion, the anomalistic year is about 4 minutes 40 seconds longer than the sidereal year, and about 25 minutes longer than the tropical or common year. The time of a complete revolution of the perihelion is computed at 108,000 years.

Anomu'ra, a section of the crustaceans of the ord. Decapoda, with irregular tails not formed to assist in swimming, including the hermit-crabs and others.

Ano'na, a genus of plants, the type of the nat. ord. Anonaceæ. *A. squamōsa* (sweet-sop) grows in the West Indian Islands, and yields an edible fruit having a thick, sweet, luscious pulp. *A. muricāta* (sour-sop) is cultivated in the West and East Indies; it produces a large pear-shaped fruit, of a greenish colour, containing an agreeable slightly-acid pulp. The genus produces other edible fruits, as the common custard-apple or bullock's heart, from *A. reticulāta*, and the cherimoyer of Peru, from *A. Cherimolia*.



Anona or Sour-sop (*Anōna muricāta*)

Anona´ceæ, a nat. ord. of trees and shrubs, having simple, alternate leaves, destitute of stipules, by which character they are distinguished from the Magnoliaceæ, to which they are otherwise closely allied. They are mostly tropical plants of the Old and the New World, and are generally aromatic. See *Anona*.

Anoplothe´rium, an extinct genus of the Ungulata or Hoofed Quadrupeds, forming the type of a distinct family, which were in many respects intermediate between the swine and the true ruminants. These animals were pig-like in form, but possessed long tails, and had a cleft hoof, with two rudimentary toes. Some of them were as small as a guinea-pig, others as large as an ass. Six incisors, two canines, eight premolars, and six molars existed in each jaw, the series being continuous, no interval existing in the jaw. *A. commūne*, from the Eocene rocks, is a familiar species.

Anoplu´ra, an order of apterous insects, of which the type is the genus *Pedicūlus* or louse,

Anopshehr. See *Anupshahr*.

Anorexia. See *Appetite*.

Anos´mia, a disease consisting in a diminution or destruction of the power of smelling, sometimes constitutional, but most frequently caused by strong and repeated stimulants, as snuff, applied to the olfactory nerves.

Anoura. See *Anura*.

Anquetil-Duperron (a´nk-tēl-du´-pā-ron´), Abraham Hyacinthe, a French orientalist, born 1731, died 1805. He studied theology for some time, but soon devoted himself to the study of Hebrew, Arabic, and Persian. His zeal for the Oriental languages induced him to set out for India, where he prevailed on some of the Parsee priests to instruct him in the Zend and Pehlevi and to give him some of the Zoroastrian books. In 1762 he returned to France with a valuable collection of MSS. In 1771 he published his *Zend-Avesta*, a translation of the *Vendidad*, and other sacred books, which aroused much interest. Among his other works are *L'Inde en rapport avec l'Europe* (1790), and a selection from the *Vedas*. His knowledge of the Oriental languages was by no means exact.

Ansbach. See *Anspach*.

An'selm, St., a celebrated Christian philosopher and theologian, born at Aosta, in Piedmont, in 1033, died at Canterbury 1109. At the age of twenty-seven (1060) he became a monk at Bec, in Normandy, whither he had been attracted by the celebrity of Lanfranc. Three years later he was elected prior, and in 1078 he was chosen abbot, which he remained for fifteen years. During this period of his life he wrote his first philosophical and religious works: the dialogues on *Truth* and *Free-will*, and the treatises *Monologion* and *Proslogion*; and at the same time his influence made itself so felt among the monks under his charge that Bec became the chief seat of learning in Europe. In 1093 Anselm was offered by William Rufus the archbishopric of Canterbury, and accepted it, though with great reluctance, and with the condition that all the lands belonging to the see should be restored. William II soon quarrelled with the archbishop, who would show no subservience to him, and would persist in acknowledging Pope Urban II in opposition to the antipope Clement. William ultimately had to give way. He acknowledged Urban as Pope, and conferred the pallium upon Anselm. The king became his bitter enemy, however, and so great were Anselm's difficulties that in 1097 he set out for Rome to consult with the Pope. Urban received him with great distinction, but did not venture really to take the side of the prelate against the king, though William had refused to receive Anselm again as archbishop, and had seized on the revenues of the see of Canterbury, which he retained till his death in 1100. Anselm accordingly remained abroad, where he wrote most of his celebrated treatise on the atonement, entitled *Cur Deus Homo* (*Why God was made Man*). When William was succeeded by Henry I Anselm was recalled; but Henry insisted that he should submit to be reinvested in his see by himself, although the Popes claimed the right of investing for themselves alone. Much negotiation followed, and Henry did not surrender his claims till 1107, when Anselm's long struggle on behalf of the rights of the Church came to an end. Anselm was a great scholar, a deep and original thinker, and a man of the utmost saintliness and piety. Anselm's great achievement in philosophy was his ontological argument for the existence of God; and his importance in the ecclesiastical history of England cannot be exaggerated. The chief of his writings are the *Monologion*, the *Proslogion*, and the *Cur Deus Homo*. The first is an attempt to prove inductively the existence of God by pure reason

without the aid of Scripture or authority; the second is an attempt to prove the same by the deductive method; the *Cur Deus Homo* is intended to prove the necessity of the incarnation. Among his numerous other writings are more than 400 letters. His life was written by his domestic chaplain and companion, Eadmer, a monk of Canterbury, and is edited by M. Rule for the 'Rolls Series'. See *Scholasticism*. Cf. Père Ragey, *Histoire de Saint Anselme*; J. M. Rigg, *Anselm of Canterbury*.

Ans'gar, or **Anshar**, called the *Apostle of the North*, born in 801 in Picardy, died in 864 or 865. He took the monastic vows while still in his boyhood, and in the midst of many difficulties laboured as a missionary in Denmark and Sweden. He died with the reputation of having made, if not the first, the most successful attempts to propagate Christianity in the North.

An'son, George, Lord, celebrated English navigator, born 1697, died 1762. He entered the navy at an early age and became a commander in 1722, and captain in 1724. He was for a long time on the South Carolina station. In 1740 he was made commander of a fleet sent to the South Sea, directed against the trade and colonies of Spain. The expedition consisted of five men-of-war and three smaller vessels, which carried 1400 men. After much suffering and many stirring adventures he reached the coast of Peru, made several prizes, and captured and burned the city of Païta. His squadron was now reduced to one ship, the *Centurion*, but with it he took the Spanish treasure galleon from Acapulco, and arrived in England in 1744 with treasure to the amount of £500,000, having circumnavigated the globe. His adventures and discoveries are described in the well-known *Anson's Voyage*, compiled from materials furnished by Anson. A few days after his return he was made rear-admiral of the blue, and not long after rear-admiral of the white. His victory over the French admiral Jonquière, near Cape Finisterre in 1747, raised him to the peerage, with the title of Lord Anson, Baron of Soberton. Four years afterwards he was made First Lord of the Admiralty. In 1758 he commanded the fleet before Brest, protected the landing of the British at St. Malo, Cherbourg, &c., and received the repulsed troops into his vessels. Finally, in 1761, he was appointed to convey the queen of George III to England.

Anso'nia, a town of the United States, Conn., on the Nangatuck, with manufactures of brass and copper, and especially clocks. Pop. 17,643.

Anspach (a'n'spa'h), or **Ansbach**, a town in Bavaria, at the junction of the Holzbach with the Lower Rezat, 24 miles south-west of Nürnberg. Anspach gave its name to an ancient principality or margravate, which had a territory of about 1300 sq. miles, with 300,000 inhabitants. in the end of the eighteenth century. The last margrave sold his possessions in 1791 to Prussia. It was occupied by the French in 1806, and transferred by Napoleon to Bavaria. The town has manufactures of trimmings, buttons, straw-wares, &c. Pop. 19,995.

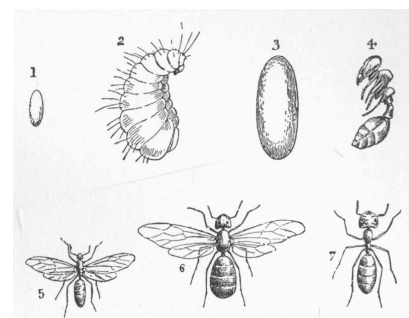
An'sted, David Thomas, an English geologist, born 1814, died 1880. He was professor of geology at King's College, London, and assistant-secretary to the Geological Society, whose quarterly journal he edited for many years.

An'ster, John, LL.D., professor of civil law in the University of Dublin, born in County Cork, 1793, died 1867. He published a volume of poems, but is chiefly known by his fine translation of Goethe's *Faust*, Part I, 1835; Part II, 1864.

An'stey, Christopher, an English poet, born 1724, died 1805. He was author of *The New Bath Guide*, a humorous and satirical production describing fashionable life at Bath in the form of a series of letters in different varieties of metre, which had a great reputation in its day, but is now almost forgotten.

Anstruther (an'struth-ër; popularly an'stër), Easter and Wester, two small royal and police burghs of Scotland, in Fifeshire, forming, with the contiguous royal burgh of Cellardyke or Nether Kilrenny, one fishing and seaport town. Total pop. (1921), 4641.

Ant, the common name of hymenopterous (or membranous-winged) insects of various genera, of the family Formicidæ, of which there are numerous species, at least 2000 being known. They are found in most temperate and tropical regions. They are small but powerful insects, and have long been noted for their remarkable intelligence and interesting habits. They are social insects, living in communities regulated by definite laws, each member of the



The Wood-ant (*Formica rufa*)

1. Egg. 2. Larva. 3. Cocoon of fine white silk. 4. Young ant,

society bearing a well-defined and separate part in the work of the colony. Each community consists of males; of females much larger than the males; and of barren females, otherwise called neuters, workers, or nurses. The neuters are wingless, and the males and females only acquire wings for their 'nuptial flight', after which the males perish, and the few females which escape the pursuit of their numerous enemies divest themselves of their wings, and either return to established nests, or become the foundresses of new colonies. The neuters perform all the labours of the ant-hill or abode of the community; they excavate the galleries, procure food, and feed the larvæ or young ants, which have not got organs of motion. In fine weather they carefully convey them to the surface for the benefit of the sun's heat, and as attentively carry them to a place of safety either when bad weather is threatened or the ant-hill is disturbed. In like manner they watch over the safety of the nymphs or pupæ about to acquire their perfect growth. Some communities possess a special type of neuters, known as 'soldiers', from the duties that specially fall upon them, and from their powerful biting jaws. There is a very considerable variety in the materials, size, and form of ant-hills, or nests, according to the peculiar nature or instinct of the species. Most of the British ants form nests in woods, fields, or gardens, their abodes being generally in the form of small mounds rising above the surface of the ground and containing numerous galleries and apartments. Some excavate nests in old tree-trunks. One little yellow ant (*Myrmica domestica*) is common in houses in Britain in some localities. Some ants live on animal food, very quickly picking quite clean the skeleton of any dead animal they may light on. Others live on saccharine matter, being very fond of the sweet substance, called honey-dew, which exudes from the bodies of aphides, or plant-lice. These they sometimes keep in their nests, and sometimes tend on the plants where they feed; sometimes they even superintend their breeding. By stroking the aphides with their antennæ they cause them to emit the sweet fluid, which the ants then greedily sip up. Various other insects are looked after by ants in a similar manner, or are found in their nests. It has been observed that some species, like the European Red Ant (*Formica sanguinea*), resort to violence to obtain working ants of other species for their own use, plundering the nests of suitable kinds of their larvæ and pupæ, which they carry off to their own nests to be carefully reared and kept as slaves. In

taken out of cocoon. 5. Male ant.
6. Female ant. 7. Worker ant.
(All magnified.)

temperate countries male and female ants survive, at most, till autumn, or to the commencement of cool weather, though a very large proportion of them cease to exist long previous to that time. The neuters pass the winter in a state of torpor, and of course require no food. The only time when they require food is during the season of activity, when they have a vast number of young to feed. Some ants of Southern Europe feed on grain, and store it up in their nests for use when required. Some species have stings as weapons, others only their powerful mandibles, or an acrid and pungent fluid (formic acid) which they can emit. The name ant is also given to the neuropterous insects otherwise called Termites (q.v.). BIBLIOGRAPHY: Lord Avebury (Sir John Lubbock), *Ants, Bees, and Wasps*; H. W. Bates, *A Naturalist on the Amazons*; *Cambridge Natural History*.

Antac'id, an alkali, or any remedy for acidity in the stomach. Dyspepsia and diarrhœa are the diseases in which antacids are chiefly employed. The principal antacids in use are magnesia, lime, and their carbonates, and the carbonates of potash and soda.

Antæ'us, the giant son of Poseidon (Neptune) and Gē (the Earth), who was invincible so long as he was in contact with the earth. Heracles (Hercules) grasped him in his arms and stifled him suspended in the air, thus preventing him from touching the earth.

Antakieh, or **Antakia**. See *Antioch*.

Antal'kali, a substance which neutralizes an alkali, and is used medicinally to counteract an alkaline tendency in the system. All true acids have this power.

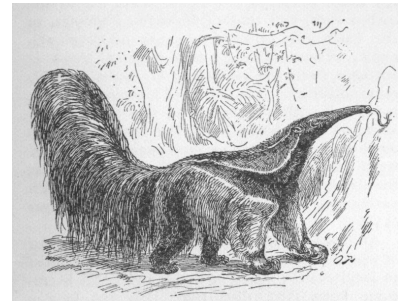
Antananarivo (an-tan-an-a-rē'vō), the capital of Madagascar, situated in the central province of Imérina, on rocky eminences rising from a plain. Until 1869 all buildings within the city were of wood or rush, but since the introduction of brick and stone, the whole city has been rebuilt. It contains two former royal palaces, immense timber structures; a Protestant and a Roman Catholic cathedral, mission churches, schools, &c. Antananarivo is the residence of the French governor of Madagascar, and there is a strong French garrison. It has manufactures of metal work, cutlery, silk, &c. Pop. (exclusive of the troops) 63,115.

An'tar, an Arabian warrior and poet of the sixth century, author of one of the seven Moallakas (poems) hung up in the Kaaba at Mecca; hero of a romance analogous in Arabic literature to the Arthurian legend of the English. The romance of *Antar* is composed in rhythmic prose interspersed with fragments of verse, many of which are attributed to Antar himself, and has been generally ascribed to Asmai (born A.D. 740, died about A.D. 830), preceptor to Harun-al-Rashid. It has been published in 32 vols. at Cairo (1889).

Antarctic (ant-ärk'tik), a term signifying the opposite of *Arctic*, and therefore relating to the southern pole or to the regions near it. The *Antarctic Circle*, which of course corresponds to the *Arctic Circle*, is a circle parallel to the equator and distant from the south pole $23^{\circ} 28'$, marking the area within which the sun does not set when on the tropic of Capricorn. The Antarctic Circle has been arbitrarily fixed on as the limits of the Antarctic Ocean, it being the average limit of the pack-ice; but the name is often extended to embrace a much wider area. The lands within or near the Antarctic Circle are but imperfectly known, and a very large area around the south pole is altogether unknown. The work of exploration has been hitherto baffled at various points by what seems an unsurmountable ice-barrier, which in some places is connected with masses of land and may as a whole belong to a great Antarctic continent. Among land-masses that have long been known to exist in the Antarctic Ocean, though our knowledge of them is very imperfect, are those to which have been attached the names Graham Land, Victoria Land, Wilkes Land, Enderby Land, South Shetland Islands, &c. The Antarctic regions are even colder and more inhospitable than the Arctic, and partly on account of their remoteness from the maritime nations there have been far fewer efforts at their exploration, the south pole being far less a goal of discovery than the north. See *South Polar Expeditions*.

Ant-eater, a name given to mammals of various genera that prey chiefly on ants, but usually confined to the genus *Myrmecophaga*, ord. Edentata. In this genus the head is remarkably elongated, the jaws destitute of teeth, and the mouth furnished with a long, extensile tongue covered with glutinous saliva, by the aid of which the animals secure their insect prey. The eyes are particularly small, the ears short and round, and the legs, especially the fore-legs, very strong, and furnished with long, compressed, acute nails,

admirably adapted for breaking into the ant-hills. The most remarkable species is the *Myrmecophāga jubāta*, or ant-bear, a native of the warmer parts of South America. It is from 4 to 5 feet in length from the tip of the muzzle to the root of the black bushy tail, which is about 2 feet long. The body is covered with long hair, particularly along the neck and back. It is a harmless and solitary animal, and spends most of its time in sleep. Some are adapted for climbing trees in quest of the insects on which they feed, having prehensile tails. All are natives of South America. The name ant-eater is also given to the pangolins and to the aardvark. The echidna of Australia is sometimes called *porcupine ant-eater*.



Ant-eater (*Myrmecophāga jubāta*)

Antece´dent, in grammar, the noun to which a relative or other pronoun refers; as, Solomon was the *prince who* built the temple, where the word *prince* is the antecedent of *who*.—In logic, that member of a hypothetical or conditional proposition which contains the condition, and which is introduced by *if* or some equivalent word or words; as, if the sun is fixed, the earth must move. Here the first and conditional proposition is the *antecedent*, the second the *consequent*.

Antedilu´vian, before the flood or deluge of Noah's time; relating to what happened before the deluge. In geology the term has been applied to organisms, traces of which are found in a fossil state in formations preceding the Diluvial (Glacial epoch), particularly to extinct animals such as the palæotherium, the mastodon, &c.

An´telope, the name given to the members of a large family of Ruminant Ungulata or Hoofed Mammalia, closely resembling the Deer in general appearance, but essentially different in nature from the latter animals. They are included with the Sheep and Oxen in the family of the Cavicornia or 'Hollow-horned' Ruminants. Their horns, unlike those of the Deer, are not deciduous, but are permanent; are never branched, but are often twisted spirally, and may be borne by both sexes. They are found in greatest number and variety in Africa. Well-known species are the chamois

(European), the gazelle, the addax, the eland, the koodoo, the gnu, the springbok, the sasin or Indian antelope, and the prongbuck of America.

Anten'nae, the name given to the movable jointed organs of touch and hearing attached to the heads of insects, myriapods, &c., and commonly called horns or feelers. They present a very great variety of forms.

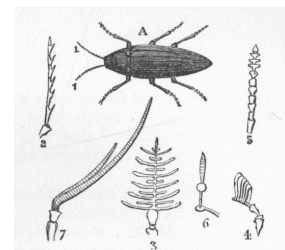
Antequera (a' n-te-kā'ra'), a city of Andalusia, in Spain, in the province of Malaga, a place of some importance under the Romans, with a ruined Moorish castle. It manufactures woollens, leather, soap, &c. Pop. 32,360.

Ant'eros, in Greek mythology, the god of mutual love. According to some, however, Anteros is the enemy of love, or the god of antipathy; he was also said to punish those who did not return the love of others.

Anthe'lion, pl. **Anthelia**, a luminous ring, or rings, seen by an observer, especially in alpine and polar regions, around the shadow of his head projected on a cloud or fog-bank, or on grass covered with dew, 50 or 60 yards distant, and opposite the sun when rising or setting. It is due to the diffraction of light.

Anthelmin'thics, or **Anthelmin'tics**, a class of remedies used to destroy worms when lodged in the alimentary canal; classed as vermicides or vermifuges, according as the object is to kill the worms, or to expel them by purgation.

An'them, originally a hymn sung in alternate parts; in modern usage, a sacred tune or piece of music set to words taken from the Psalms or other parts of the Scriptures, first introduced into church service in Elizabeth's reign; a developed motet. The anthem may be for one, two, or any number of voices, but seldom exceeds five parts, and may or may not have an organ accompaniment written for it.



Antennæ
1,1. Filiform
Antennæ of Cucujo
Firefly of Brazil
(*Pyrophorus
luminosus*). 2.
Denticulate Antenna;
3. Bipinnate; 4.
Lamellicorn; 5.
Clavate; 6. Geniculate;
7. Antenna and
Antennule of
Crustacean.



Anthemion

Anthe´mion, an ornament or ornamental series used in Greek and Roman decoration, which is derived from floral forms, more especially the honeysuckle. It was much used for the ornamentation of friezes and interiors, for the decoration of fictile vases, the borders of dresses, &c.

An´themis, a genus of composite plants, comprising the camomile or chamomile.

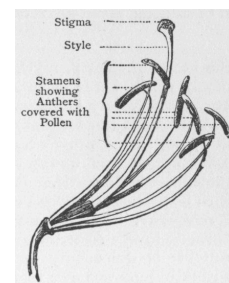
Anthe´mius, a Greek mathematician and architect of Lydia; designed the church of St. Sophia at Constantinople, and is credited with the invention of the dome; died A.D. 534.

An´ther, the male organ of the flower; that part of the stamen which is filled with pollen.

Anthe´ste´ria, an annual Greek festival held in honour of all the gods, but especially in honour of Dionysus. It celebrated the beginning of spring, and the season when the wine of the previous vintage was considered fit for use.

Anthocy´anin, the blue colour of flowers, a pigment obtained from those petals of flowers which are blue, by digesting them in spirits of wine.

Anthol´ogy (Gr. *anthos*, a flower, and *legein*, to gather), the name given to several collections of short poems which have come down from antiquity. The first to compile a Greek anthology was Meleager, a Syrian, about 60 B.C. He entitled his collection, which contained selections from forty-six poets besides many pieces of his own, the *Garland*; a continuation of this work by Philip of Thessalonica in the age of Tiberius was the first entitled *Anthology*. Later collections are that of Constantine Cephalas, in the tenth century, who made much use of the earlier ones, and that of Maximus



The Reproductive
Organs of the
Lily

Planudes, in the fourteenth century, a monk of Constantinople, whose anthology is a tasteless series of extracts from the *Anthology* of Cephalas, with some additions. The treasures contained in both, increased with fragments of the older poets, idylls of the bucolic poets, the hymns of Callimachus, epigrams from monuments and other works, have been published in modern times as the *Greek Anthology*. There is no ancient Latin anthology, the oldest being that of Scaliger (1573).

An'thon, Charles, LL.D., an American editor of classical school-books, and of works intended to facilitate the study of Greek and Latin literature; born 1797, died 1867. He was long a professor in Columbia College, New York.

An'thony, St. the founder of monastic institutions, born near Heraclea, in Upper Egypt, A.D. 251. Giving up all his property he retired to the desert, where he was followed by a number of disciples, who thus formed the first community of monks. He died at the age of 105.—*St. Anthony's Fire*, a name given to erysipelas.

An'thracene ($C_{14}H_{10}$) occurs in coal-tar in small quantity, about 0.25-0.45 per cent. During the distillation of tar a high-boiling fraction, boiling above $270^{\circ} C.$, is obtained; this is crude anthracene oil, a greenish oily substance which, on further distillation, yields a crystalline mass, 50 per cent anthracene. This is carefully purified by distillation and chemical treatment to separate the anthracene from the other substances occurring with it, and the product obtained is finally purified by crystallization. When pure it forms colourless crystalline scales melting at $216^{\circ} C.$, and having a violet fluorescence. It forms a series of derivatives, the most important being anthraquinone and alizarine and the numerous derivatives of these. Anthracene was originally a useless product in coal-tar distillation, but it became valuable as soon as it was discovered that alizarine—from which many dyes are manufactured directly or indirectly—could be prepared from it.

An'thracite, glance or blind coal, a non-bituminous coal of a shining lustre, approaching to metallic, and which burns without smoke, with a weak or no flame, and with intense heat. It consists of, on an average, 90 per cent carbon, 2 oxygen, 3 hydrogen, and 5 ashes. It has some of the properties of coal or charcoal, and, like that substance, represents an extreme alteration

of vegetable substances by loss of gases, either during conditions of decay or after entombment among stratified rocks. It is found in England, Scotland, and Ireland, and in large quantities in the United States (Pennsylvania), and near Swansea (South Wales). See *Coal*.

An'thrax, a fatal disease to which animals are subject, always associated with the presence of an extremely minute micro-organism (*Bacillus anthrācis*) in the blood. It attacks cattle more frequently than other animals; sheep, horses, pigs, dogs, and fowls are liable to anthrax, but not cats. The mode of infection in animals is chiefly by ingestion. It may also be contracted through a wound or scratch in the skin, but this mode of infection is commoner in human beings than in animals. It frequently assumes an epizootic form, and extends over large districts, affecting all classes of animals which are exposed to the exciting causes. It is also called splenic fever, and is communicable to man, appearing as carbuncle, malignant pustule, or wool-sorter's disease.

Anthropol'atry, the worship of man, a word always employed in reproach; applied by the Apollinarians, who denied Christ's perfect humanity, towards the orthodox Christians.

Anthro'polite, a petrification of the human body or skeleton, or of parts of the body, by the encrusting action of calcareous waters, and hence hardly to be considered fossil or sub-fossil.

Anthropol'ogy, the science of man, including the study of man's place in nature, that is, of the measurement of his agreement with and divergence from other animals and the history of the emergence of human characteristics; of the distinctive features and geographical distribution of the races of mankind, their customs and beliefs; of the remains of extinct types of the human family and the evidence relating to their modes of life. It puts under contribution all sciences which have man for their object, as anatomy, palæontology, psychology, archæology, history, and comparative religion. All the races of mankind that are now living, much as they differ in external appearance, such as colour of skin, character of hair, form of skull, face, and body, and stature, belong to one species, *Homo Sapiens*; but an earlier species of more brutal type, *H. neanderthalensis*, now completely extinct, is known from fossil remains found in Germany, Belgium, France,

Gibraltar, and Croatia. Three more ancient and primitive types, probably representing distinct genera of the human family, have been discovered respectively at Piltdown, in Sussex (Eoanthropus), at Mauer, near Heidelberg (Palæanthropus), and in Java, the Ape-man (Pithecanthropus). The Piltdown man may represent the very remote, but direct, ancestor of modern man; but the Heidelberg man and the Ape-man were probably divergent 'sports' whose descendants never became men of the modern type.

In structure the gorilla reveals a close kinship with the human family, and was probably derived from a common ancestry which probably differentiated into man's forerunner and the gorilla's in Miocene times. Of existing races the aboriginal Australian is much the most primitive, and represents the survival of the earliest type of *Homo Sapiens* soon after this species became differentiated from men of the Neanderthal species. The negro, whose home is tropical Africa, is primitive in some respects, but in others is highly specialized. He is distinguished by his black skin, flat nose, prominent jaws and thick everted lips, and so-called 'woolly' or 'peppercorn' hair. In stature he shows a wider range of variation than any other race, including, as he does, the tallest and the shortest varieties of mankind. The Bushman is a peculiarly distinct racial type now restricted to the deserts of South Africa. Though his skin is yellowish rather than black, he is akin to the negro. The Mongolian race probably assumed its distinctive features, yellowish skin, coarse black hair, and characteristic facial and bodily traits, in Eastern Asia; and the aboriginal population of America was sprung mainly from the less-specialized branch of this race. The so-called white races include three main stocks, a people of short stature, olive complexion, and long heads, the Mediterranean race; a taller people with fair hair and long heads, the Nordic race; and a short, thick-set, black-haired, broad-headed Alpine race, which made its way from Asia into Europe many centuries after the other two chief components of Europe's population. For long ages in every part of the world intermixture has been taking place in varying degrees between the different races of mankind, so that to-day probably no pure race exists. See *Ethnography*, *Ethnology*, *Man*, &c.—BIBLIOGRAPHY: E. B. Tylor, *Anthropology*; D. G. Brinton, *Races and Peoples*; W. Z. Ripley, *The Races of Europe*; E. Carpenter, *Anthropology*; G. Elliott Smith, *The Migrations of Early Culture*; H. G. F. Spurrell, *Modern Man and his Forerunners*; *Dictionnaire des Sciences*

Anthropologiques; The Journal of the Anthropological Institute of Great Britain.

Anthropom'etry, the systematic examination of the height, weight, and other physical characteristics of the human body. It was shown in the British Association Report of 1888 that variations in stature, weight, and complexion, existing in different districts of the British islands, are chiefly due to difference of racial origin.

The Scotch male adults stand first in height (68.71 inches), the Irish second (67.90 inches), the English third (67.66 inches), and the Welsh last (66.66 inches). In weight the Scotch take the first place (165.3 lb.), the Welsh the second (158.3 lb.), the English the third (155.0 lb.), and the Irish the last (154.1 lb.). The average height of adult females is 4.71 inches less than the male average, and their average weight 32.2 lb. under that of the males. The average height of the adult males of the principal races or nationalities of the world may be given as under; but it is acknowledged that more numerous measurements might alter some of the figures considerably: Polynesians 69.33 inches, Patagonians 69 inches, Negroes of the Congo 69 inches, Scotch 68.71 inches, Iroquois Indians 68.28 inches, Irish 67.90 inches, United States (whites) 67.67 inches, English 67.66 inches, Norwegians 67.66 inches, Zulus 67.19 inches, Welsh 66.66 inches, Danes 66.65 inches, Dutch 66.62 inches, American Negroes 66.62 inches, Hungarians 66.58 inches, Germans 66.54 inches, Swiss 66.43 inches, Belgians 66.38 inches, French 66.23 inches, Berbers 66.10 inches, Arabs 66.08 inches, Russians 66.04 inches, Italians 66 inches, Spaniards 65.66 inches, Esquimaux 65.10 inches, Papuans 64.78 inches, Hindus 64.76 inches, Chinese 64.17 inches, Poles 63.87 inches, Finns 63.60 inches, Japanese 63.11 inches, Peruvians 63 inches, Malays 62.34 inches, Lapps 59.2 inches, Bosjesmans 52.78 inches. General average, 65.25 inches.

Anthropomor'phism, the representation or conception of the Deity under a human form, or with human attributes and affections. *Anthropomorphism* is based upon the natural inaptitude of the human mind for conceiving spiritual things except through sensuous images, and in its consequent tendency to accept such expressions as those of Scripture when it speaks of the eye, the ear, and the hand of God, of his seeing and hearing, of his remembering and forgetting, of his making man in his own image, &c., in a

too literal sense. In a general sense anthropomorphism is the assumption of man that his own characteristics are present in beings or things widely different from himself, more particularly in forces of nature and gods. The term is, therefore, also applied to that doctrine which attributes to animals mental faculties of the same nature as those of man, though much lower in degree: strictly called *biological anthropomorphism*, to distinguish it from anthropomorphism proper, or *theological anthropomorphism*. Cf. E. Caird, *Evolution of Religion*; J. R. Illingworth, *Personality, Human and Divine*.

Anthropoph'agi, man-eaters, cannibals. Cannibalism was practised in very ancient times; and though some peoples, as the New Zealanders and Fijians, have given it up in recent times, it is still practised over a wide area in Central Africa, where human flesh is a common article of food. Superstitious ideas are often associated with cannibalism among those who practise it. The Caribs were cannibals at the time of the Spanish conquest, and the word 'cannibal' is derived from their name. See *Cannibalism*.

Anthus. See *Pipit*.

Anti-aircraft guns are guns so mounted that they may be pointed upward to fire directly against objects in the air. During the European War these guns, fitted with special appliances and ammunition, were used for defence against air-raids of the enemy, against Zeppelins and Gothas. The anti-aircraft guns are of various types, ranging from light machine-guns up to batteries of 3-inch and 6-inch guns. Some of them have brought down enemy machines flying at a height of 10,000 or 12,000 feet.

Antibes (a 'n ' -tēb) (ancient **Antipolis**), a fortified town and seaport of France, department Alpes-Maritimes, on the Mediterranean, 11 miles S.S.W. of Nice; founded about 340 B.C. Pop. 12,198.

Anti-burgher Synod, a section of the Scottish Secession Church, which held its first meeting in Edinburgh in the house of Adam Gib on 10th April, 1747. It was formed in consequence of a breach resulting from a controversy respecting the religious clause of the oath taken by burgesses in Edinburgh, Glasgow, and Perth. Those in favour of the oath were designated Burghers, whilst their opponents, who condemned the oath of the burgesses, became known as Anti-burghers. The union of the burgher and anti-burgher sections was brought about in 1820 through the exertions

of John Jamieson, minister at the Anti-burgher Church in Nicholson Street, Edinburgh. See *United Free Church of Scotland*.

An'tichlor, the name given to any chemical substance, such as hyposulphite of sodium, employed to remove the small quantity of chlorine which obstinately adheres to the fibres of the cloth when goods are bleached by means of chlorine.

An'tichrist, a word occurring in the first and second *Epistles of St. John*, and nowhere else in Scripture, in passages having an evident reference to a personage real or symbolical mentioned or alluded to in various other passages both of the Old and New Testaments. The *idea itself*, however, of Antichrist can be traced back to the second century B.C., and appears first of all in the *Book of Daniel*. In every age the Church has held through all its sects some definite expectation of a formidable adversary of truth and righteousness prefigured under this name. Thus Roman Catholics have found Antichrist in heresy, and Protestants in Romanism. In one point the sects have generally been agreed, namely, in regarding the various intimations on this subject in the Old and New Testaments as a homogeneous declaration or warning, inspired by the spirit of prophecy, of danger to the true religion from some disaffection and revolt organized in the latter days by Satan. Most modern critics take a different view of the matter. They do not regard the various Scriptural writers who have dealt with this subject as having had any common inspiration or design. They believe that each writer from his own point of view, guided by mere human sagacity, gives expression in his predictions to his own individual apprehensions, or narrates as prediction what he already knows. Originally Antichrist is nothing else than the incarnate devil, and the idea of the battle of God with a human opponent, endowed with devilish wickedness, arose under the influence of historical conditions. It is the near political horizon which suggests the danger, or contemporary history the substance of the prophecy; thus the Antichrist of Daniel is Antiochus Epiphanes, that of St. John Nero, that of St. Paul some adversary of Christianity about to appear in the time of the Emperor Claudius.—BIBLIOGRAPHY: S. Baring-Gould, *Curious Myths of the Middle Ages*; W. Bousset, *Antichrist*.

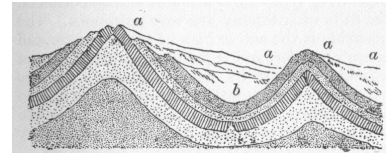
Anticli'max, a sudden declension of a writer or speaker from lofty to mean thoughts or language, as in the well-known lines, quoted in Pope's *Martinus*

Scriblerus on the Art of Sinking in Poetry as from an anonymous author:

And thou, Dalhousie, the great god of war,
Lieutenant-colonel to the Earl of Mar.

Pope, Addison, and Fielding were masters in this art of sudden descent.

Anticli´nal line or axis, in geology, the ridge of a wave-like curve made by a series of superimposed strata, the strata dipping from it on either side as from the ridge of a house: a *synclinal line* runs along the trough of such a wave.



a, a. Anticlinal line. b.
Synclinal line

Anti-Corn-Law League, an association formed in England in 1836 to procure the repeal of the laws regulating or forbidding the importation of corn. The object of the league was attained in 1846.

Anticos´ti, an island of Canada, in the mouth of the St. Lawrence, 125 miles long by 30 miles broad. The interior is mountainous and wooded, but there is much good land, and it is well adapted for agriculture.

Anticy´clone, a phenomenon presenting some features opposite to those of a cyclone. It consists of a region of high barometric pressure, the pressure being greatest in the centre, with light winds flowing outwards from the centre, and not inwards as in the cyclone, accompanied with great cold in winter and with great heat in summer.

Anticyra (an-tis'i-ra), the name of two towns of Greece, the one in Thessaly, the other in Phocis, famous for hellebore, which in ancient times was regarded as a specific against insanity and melancholy. Hence various jocular allusions in ancient writers (*Naviga Anticyram*, sail to Anticyra).

An´tidote, a medicine to counteract the effects of poison.

Antietam (an-tē'tam), a small stream in the United States which falls into the Potomac about 50 miles N.W. of Washington; scene of an indecisive battle between the Federal and Confederate armies, 17th Sept., 1862.

Anti-Federalists, the political party in the United States which after the formation of the Federal constitution in 1787 opposed its ratification. Whilst the Federalists were striving to turn the federation into a united nation, and stood for a strong Government and centralizing tendencies, their opponents, the Anti-Federalists, either more democratic, or pretending that a strong Government meant a 'disguised' monarchic power, endeavoured to preserve a loose disintegrated federation. The Anti-Federalist party was gradually transformed into the Democratic-Republican party, led by Jefferson.

Antifriction Metal, a name given to various alloys of tin, zinc, copper, antimony, lead, &c., which oppose little resistance to motion, with great resistance to the effects of friction, so far as concerns the wearing away of the surfaces of contact. Babbitt's metal (50 parts tin, 5 antimony, 1 copper) is one of them.

Antigone (an-tig'ō-nē), in Greek mythology, the daughter of Œdipus and Jocasta, celebrated for her devotion to her brother Polynices, for burying whom against the decree of King Creon she suffered death. She is the heroine of Sophocles' *Œdipus at Colonus* and his *Antigone*; also of Racine's tragedy *Les Frères Ennemis*.

Antig' onish, a town in the E. of Nova Scotia, in county of the same name; the seat of a Roman Catholic bishop, with a cathedral, a college, and a good harbour. Pop. 1787.

Antig' onus, one of the generals of Alexander the Great, born about 382 B.C. In the division of the empire, after the death of Alexander, Antigonus obtained Greater Phrygia, Lycia, and Pamphylia as his dominion. But he soon managed to extend his power, being assisted by his warlike son, Demetrius Poliorcētēs. Ptolemy, Seleucus, and Lysimachus, who had also been generals of Alexander, alarmed by his ambition, united themselves against him; and a long series of contests ensued in Syria, Phœnicia, Asia Minor, and Greece, ending in 301 B.C. with the battle of Ipsus in Phrygia, in which Antigonus was defeated and slain, his dominions being divided among the conquerors.

Antigonus Gon' atas, son of Demetrius Poliorcētēs, and grandson of the above, succeeded his father in the kingdom of Macedon and all his other

European dominions, but did not obtain actual possession of them for some years. He died, after a reign of forty-four years, 239 B.C.

Antigua (an-tē'gwa), one of the British West Indies, the most important of the Leeward group; 28 miles long, 20 broad; area, 108 sq. miles. Its shores are high and rocky, and much indented by creeks and inlets furnishing several good harbours. The surface is diversified by hill and dale, but nowhere rises to a greater height than 1500 feet. A considerable portion of it is fertile, and the climate is healthy, but there is a scarcity of water, there being no streams and few springs, droughts are not infrequent, and hurricanes are apt to cause serious loss and damage. Chief products are sugar, cotton, and pineapples. The island has fairly good shipping connections with the United Kingdom, the United States, and Canada. Antigua is governed as a crown colony, the Islands of Barbuda and Redonda being attached to it. The capital, St. John, the residence of the governor of the Leeward Islands, stands on the shore of a well-sheltered harbour in the north-west part of the island. Falmouth (English Harbour) in the south has also an excellent harbour with a dockyard. The island was discovered by Columbus in 1493; the first settlement was made by the English in 1632. Since then, except for a short period of occupation by the French, it has been a British possession. Pop. 32,269 (1911).

Anti-Jac'obin, a famous magazine (1797-1818), the original object of which was to satirize the Jacobin principles of the Fox section of Whigs; principal contributors: Gifford, Canning, Frere, and Ellis.

Anti-Lebanon, the eastern of the two parallel ranges known as the Mountains of Lebanon in Palestine. See *Lebanon*.

Antilegom'ena (things spoken against or objected to), a term applied by early Christian writers to the *Epistle to the Hebrews*, *2 Peter*, *James*, *Jude*, *2 and 3 John*, and the *Apocalypse*, which, though read in the churches, were not for some time received into the canon of Scripture.

Antilles (an-til'ēz), another name for the West Indian Islands (excluding Bahamas). See *West Indies*.

Antilochus (an-til'o-kus), in Greek legend, a son of Nestor, distinguished among the younger heroes who took part in the Trojan War by beauty,

bravery, and swiftness of foot. He was slain by Memnon, but Achilles avenged his death.

Antimacass'ar, a covering for chairs, sofas, couches, &c., made of open cotton or worsted work, to preserve them from being soiled, as by the oil applied to the hair.

Antimachus (an-tim'a-kus), a Greek poet who lived about 400 B.C., and wrote an epic called the *Thebais* on the mythical history of Thebes, and a long elegy called *Lydē*, inspired by a mistress or wife of that name. Both works were full of mythological details. Only fragments of his writings remain, and from these it can be gathered that his style was rather laboured and artificial. Yet the Alexandrian grammarians ranked him next to Homer.

An'timony (chemical symbol, Sb, from Lat. *stibium*; sp. gr. 6.7, atomic wt. 120.2), a brittle metal of a bluish-white or silver-white colour and a crystalline or laminated structure. It melts at 630.6° C., and burns with a bluish-white flame. The mineral called stibnite or antimony-glance, is a trisulphide (Sb_2S_3), and is the chief ore from which the metal is obtained. It is found in many places, including France, Spain, Hungary, Italy, Canada, Australia, and Borneo. The metal, or, as it was formerly called, the *regulus of antimony*, does not rust or tarnish when exposed to the air. When alloyed with other metals it hardens them, and is therefore used in the manufacture of alloys, such as Britannia-metal, type-metal, and pewter. In bells it renders the sound more clear; it renders tin more white and sonorous as well as harder, and gives to printing types more firmness and smoothness. The salts of antimony are very poisonous. The protoxide is the active base of tartar emetic and James's powder, and is justly regarded as a most valuable remedy in many diseases.—*Yellow antimony* is a preparation of antimony of a deep yellow colour, used in enamel and porcelain painting. It is of various tints, and the brilliancy of the brighter hues is not affected by foul air.

Antino'mianism ('opposition to the law'), the name given by Luther to the inference drawn by John Agricola (1492-1566), from the doctrine of justification by faith, that the moral law is not binding on Christians as a rule of life. The term antinomian has since been applied to all doctrines and practices which seem to contemn or discountenance strict moral obligations. The Lutherans and Calvinists have both been charged with

antinomianism, the former on account of their doctrine of justification by faith, the latter both on this ground and that of the doctrine of predestination. The charge is, of course, vigorously repelled by both.

Antinomy, the opposition of one law or rule to another law or rule; in the Kantian philosophy, that natural contradiction which results from the law of reason, when, passing the limits of experience, we seek to conceive the complex of external phenomena, or nature, as a world or cosmos.

Antinous (an-tin'ō-us), a young Bithynian whom the extravagant love of Hadrian has immortalized. He drowned himself in the Nile in A.D. 122. Hadrian set no bounds to his grief for his loss. He gave his name to a newly-discovered star, erected temples in his honour, called a city after him, and caused him to be adored as a god throughout the empire. Statues, busts, &c., of him are numerous.

Antioch (an'ti-ok), a town in Syria, famous in ancient times as the capital of the Greek Kings of Syria, on the left bank of the Orontes, about 21 miles from the sea, in a beautiful and fertile plain. It was founded by Seleucus Nicator in 300 B.C., and named after his father Antiochus. In Roman times it was the seat of the Syrian governors, and the centre of a widely-extended commerce. It was called the 'Queen of the East' and 'The Beautiful'. Antioch is frequently mentioned in the New Testament, and it was here that the disciples of our Saviour were first called Christians (*Acts*, xi, 26). In the first half of the seventh century it was taken by the Saracens, and in 1098 by the Crusaders. They established the principality of Antioch, of which the first ruler was Bohemond, and which lasted till 1268, when it was taken by the Mameluke Sultan of Egypt. In 1516 it passed into the hands of the Turks. The modern Antioch, or *Antakieh*, has recently grown from a small place to a flourishing town. Pop. estimated at 30,000.—There was another Antioch, in Pisidia, at which St. Paul preached on his first missionary journey.

Antiochus (an-ti'ō-kus), a name of several Græco-Syrian kings of the dynasty of the Seleucidæ.—**Antiochus I**, called *Sōtēr* ('saviour'), was the son of Seleucus, general of Alexander the Great, and founder of the dynasty. He was born about 324 B.C., and succeeded his father in 280 B.C. During the greater part of his reign he was engaged in a protracted struggle with the

Gauls who had crossed from Europe, and by whom he was killed in battle, 261 B.C.—**Antiochus II**, surnamed *Theos* (god), succeeded his father, lost several provinces by revolt, and was murdered in 246 B.C. by Laodicē, his wife, whom he had put away to marry Berenīcē, daughter of Ptolemy.—**Antiochus III**, surnamed the *Great*, grandson of the preceding, was born 242 B.C., succeeded in 223 B.C. The early part of his reign embraced a series of wars against revolted provinces and neighbouring kingdoms, his expeditions extending to India, over Asia Minor, and afterwards into Europe, where he took possession of the Thracian Chersonese. Here he encountered the Romans, who had conquered Philip V of Macedon, and were prepared to resist his further progress. Antiochus gained an important adviser in Hannibal, who had fled for refuge to his Court; but he lost the opportunity of an invasion of Italy while the Romans were engaged in war with the Gauls, of which the Carthaginian urged him to avail himself. The Romans defeated him by sea and land, and he was finally overthrown by Scipio at Mount Sipylus, in Asia Minor, 190 B.C., and very severe terms were imposed upon him. He was killed while plundering a temple in Elymais to procure money to pay the Romans.—**Antiochus IV**, called *Epiphānes*, youngest son of the above, is chiefly remarkable for his attempt to extirpate the Jewish religion, and to establish in its place the polytheism of the Greeks. This led to the insurrection of the Maccabees, by which the Jews ultimately recovered their independence. He died 164 B.C.



Medal of
Antiochus
Epiphanes

Antioquia (a`n-tē-ō-kē`a`), a town of South America, in Colombia, on the River Cauca; founded in 1542. Pop. 8730. It gives name to a department of the republic; area, 22,752 sq. miles. Pop. 739,434. Capital, Medellin.

Antip`aros (ancient, **Oliāros**), one of the Cyclades Islands in the Grecian Archipelago, containing a famous stalactitic grotto or cave. It lies south-west of Paros, from which it is separated by a narrow strait, and has an area of 10 sq. miles, and about 700 inhabitants.

Antip`ater, a general and friend of Philip of Macedon, father of Alexander the Great. On the death of Alexander, in 323 B.C., the regency of Macedonia was assigned to Antipater, who succeeded in establishing the Macedonian rule in Greece on a firm footing. He died 317 B.C., at an advanced age.

Antip´athy, a special dislike exhibited by individuals to particular objects or persons, usually resulting from physical or nervous organization. An antipathy is often an unaccountable repugnance to what people in general regard with no particular dislike, as certain sounds, smells, articles of food, &c., and it may be manifested by fainting or extreme discomfort.

Antiphlogis´tic, a term applied to medicines or methods of treatment that are intended to counteract inflammation, such as bloodletting, purgatives, diaphoretics, &c.

An´tiphon, a Greek orator, born near Athens; founder of political oratory in Greece. His orations are the oldest extant, and he is said to have been the first who wrote speeches for hire. He was put to death for taking part in the revolution of 411 B.C., which established the oligarchic government of the Four Hundred. Antiphon seems to have specialized in homicide cases; his most celebrated speech is *On the Murder of Herodas*. Cf. Sir R. C. Jebb, *Attic Orators*; J. F. Dobson, *The Greek Orators*.

Antiphon, or **Antiph´ony** ('alternate song'), in the Christian Church a verse first sung by a single voice, and then repeated by the whole choir; or any piece to be sung by alternate voices.

Antipodes (an-tip´o-dēz), the name given relatively to people or places on opposite sides of the earth, so situated that a line drawn from one to the other passes through the centre of the earth and forms a true diameter. The longitudes of two such places differ by 180°. The difference in their time is about twelve hours, and their seasons are reversed.

Antipodes Islands, a group of small uninhabited islands in the South Pacific Ocean, about 460 miles S.E. by E. of New Zealand; so called from being nearly antipodal to Greenwich. Antipodes Island rises to 1300 feet, and is largely covered with coarse grass; huts have been fitted up to shelter castaways.

An´tipope, the name applied to those who at different periods have produced a schism in the Roman Catholic Church by opposing the authority of the Pope, under the pretence that they were themselves Popes. The Roman Church cannot admit that there ever existed two Popes; but the fact is, that in several cases the competitors for the papal chair were equally

Popes; that is to say, the claims of all were equally good. Each was frequently supported by whole nations, and the schism was nothing but the struggle of political interests. Twenty-nine antipopes are enumerated in Church history; the last of them is Felix V, 1439-49.

Antipyret'ics, medicines given for the purpose of reducing fever by lowering the patient's temperature, whether by causing perspiration or otherwise. Quinine, antipyrin, phenacetin, are common antipyretics. An aperient or purgative often serves the same purpose.

Antipy'rin, a drug obtained from coal-tar products, valuable in reducing fever and in relieving pain, being much used in nervous headache and neuralgia.

An'tiquaries, those devoted to the study of ancient times through their relics, as old places of burial, remains of ancient habitations, early monuments, implements or weapons, statues, coins, medals, paintings, inscriptions, books, and manuscripts, with the view of arriving at a knowledge of the relations, modes of living, habits, and general condition of the people who created or employed them. Societies or associations of antiquaries have been formed in all countries of European civilization. In Britain the Society of Antiquaries of London was founded in 1572, revived in 1717. and incorporated in 1751. The Society of Antiquaries of Scotland was founded in 1780, incorporated in 1783, and has the management of a large national antiquarian museum in Edinburgh. One of the best-known antiquarian societies in Europe is the *Société Royale des Antiquaires du Nord* at Copenhagen.

Antiques (an-tēks'), a term specifically applied to the remains of ancient art, as statues, paintings, vases, cameos, and the like, and more especially to the works of Grecian and Roman antiquity.

Antirrhinum (an-ti-rī' num) (from *anti*, instead of, and *rhis*, snout), a genus of annual or perennial plants of the nat. ord. Scrophulariaceæ, commonly known as *snapdragon*, on account of the peculiarity of the blossoms, which, by pressing between the finger and thumb, may be made to open and shut like a mouth. They all produce showy flowers, and are much cultivated in gardens. Many varieties of some of them, such as the great or common snapdragon (*Antirrhinum majus*), have been produced by gardeners. The

lesser snapdragon grows in sandy soil, and is found in cornfields in the south of England and Ireland.

Antisana (a´n-tē-sä´na´), a volcano in the Andes of Ecuador, 35 miles S.E. by E. of Quito. Whymper, who ascended it in 1880, makes its height 19,260 feet.

Antis´cians (Gr. *anti*, over against, *skia*, a shadow), those who live under the same meridian, at the same distance N. and S. of the equator, and whose shadows at noon consequently are thrown in contrary directions.

Antiscorbu´tics, remedies against scurvy. Lemon-juice, ripe fruit, milk, salts of potash, green vegetables, potatoes, fresh meat, and raw or lightly-boiled eggs, are some of the principal foodstuffs containing antiscorbutic vitamins.

Anti-Sem´itism, hostility to the Jews (Semites), actively exhibited in severities and attacks of various kinds. The movement assumed vast proportions about 1880 and manifested itself in various countries, especially Russia, Austria-Hungary, Germany, Roumania, and France. It may be attributed to different motives in different countries, but on the whole owed its origin not only to the fact of the Jews being a 'peculiar people' by race and religion, but also to the comparatively high position won by them in modern times in the financial and political worlds. The religious element is quite prominent in the popular attacks on the Jew, although modern anti-Semitism is essentially social and economic. In Western Russia there was a great outburst against the Jews in 1881, in which men, women, and children were slaughtered. The Government of the Tsar, by its anti-Jewish policy, may be said to have sanctioned this murderous outbreak, which was followed by harsh laws and actual persecutions, though afterwards there was a mitigation of the severity shown towards the Jews. Yet in 1903 the world was startled by a terrible massacre of Jews at Kishinev, in Bessarabia, connived at by the authorities on the spot; and towards the end of 1905, in connection with the Russian revolutionary movement, there were dreadful massacres of Jews in Odessa, Kishinev, and other towns, the authorities being similarly involved. In Roumania, until 1919, the position of the Jews resembled what it was elsewhere in mediæval times, and was less favourable than it was even

under the Turks. In Germany the movement has been worked chiefly by politicians for their own ends, though the racial and religious question has also had some influence; and among the ignorant the belief that the Jews murder Christian children for ritual purposes has been revived, as also in Austria and in Hungary. In these countries the movement has been partly political, partly social and economic, partly religious. In France anti-Semitism has been employed chiefly as a weapon by monarchists and clericals as against republicanism, and by the Socialists as against capitalism, racial antipathy having also its influence on the movements. It reached its height in 1895 at the time of the Dreyfus affair. In Britain, too, anti-Semitism has of late made itself felt.—BIBLIOGRAPHY: A. Leroy-Beaulieu, *Israel among the Nations*; Bernard Lazare, *L'anti-sémitisme, son histoire et ses causes*.

Antisep'tic (Gr. *anti*, against, and *sēpein*, to rot), an agent which destroys the germs of putrefaction or suppuration is called an antiseptic. Many substances act thus, e.g. chlorine, iodine, hypochlorous acid, sulphurous acid, camphor, creosote, iodoform, nascent oxygen ('Sanitas'), corrosive sublimate, formaldehyde ('Formalin'), potassium permanganate ('Condy's Fluid'), carbolic acid (Lysol, Izal, Cyllin); lately aniline dyes have become prominent: of these flavine has proved the most useful addition to surgery of recent years. It was much used in the European War (1914-8).—*Antiseptics* are also used for purifying surgical instruments, &c., and commercially as disinfectants. When introduced by Lister into surgical practice they led to revolutionary advances in surgery. The tendency of late years has been to abandon antiseptic for aseptic (sterile) mode of technique, but during the war (1914-8) there was a general return to antiseptic methods in surgery.

Antispasmod'ic, a medicine for the cure of spasms and convulsions; such belong largely to the class of ethers, as sulphuric ether, chloric ether, nitric ether, &c.

Antisthenes (an-tis'the-nēz), a Greek philosopher and the founder of the school of Cynics, born at Athens about 444 B.C. He was first a disciple of Gorgias and then of Socrates, at whose death he was present. His philosophy was a one-sided development of the Socratic teaching. He held virtue to consist in complete self-denial and in disregard of riches, honour,

or pleasure of every kind. He himself lived as a beggar. He died in Athens at an advanced age.

Antis'trophe. See *Strophe*.

Anti-submarine. See *Submarine*.

Anti-Taurus, a mountain range of Anatolia, Asia Minor, extending from the Sicilian Taurus towards the north-east, and connecting the Taurus mountain system with Mount Ararat, Mount Elbruz, and the Caucasus. See *Taurus*.

Antith'esis (opposition), a figure of speech consisting in a contrast or opposition of words or sentiments; as, 'When our vices *leave us*, we flatter ourselves we *leave them*'; 'The prodigal *robs his heir*, the miser *robs himself*'.

Antitoxin, the name given to a class of bodies of unknown nature having the capacity of neutralizing the poisonous substances (toxins) by which certain bacteria produce disease. If such a toxin be introduced every few days in increasing doses, into, e.g., the horse, and if, after some months of this treatment, the animal be bled, its serum contains the antitoxin to the toxins used. The use of the antitoxin to the toxin of diphtheria is most efficacious in curing that disease, and the treatment has caused a great fall in the death-rate. It ought to be applied as soon as possible after signs appear in the throat. An antitoxic treatment is also applicable in cases of tetanus (lock-jaw), a disease liable to follow any wound contaminated with dirt, especially with manured soil. Less success has been achieved when the disease is fully established, but if the antitoxin be injected immediately after the wound has been incurred, then the subsequent development of the disease is prevented. This preventive treatment has been attended with marked effect in the case of wounds received in war, which it is almost impossible to keep free from contamination. Antitoxins were extensively used during the European War. (1914-8). In bacterial diseases other than those mentioned, sera have been produced by injecting into large animals dead and living bacteria, e.g. the organisms of epidemic cerebro-spinal meningitis (spotted fever), pneumonia, blood-poisoning, &c., and these sera probably depend for their action on the presence of bodies similar to antitoxins. See *Bacteria*, *Diphtheria*.

Anti-trade Winds, a name given to any of the upper tropical winds which move northward or southward in the same manner as the trade-winds which blow beneath them in the opposite direction. These great aerial currents descend to the surface after they have passed the limits of the trade-winds, and form the south-west or west-south-west winds of the north temperate, and the north-west or west-north-west winds of the south temperate zones.

Antitrinita'rians, all who do not receive the doctrine of the divine Trinity, or the existence of three persons in the Godhead; especially applied to those who oppose such a doctrine on philosophical grounds, as contrasted with Unitarians, who reject the doctrine as not warranted by Scripture.

An'titype, that which is correlative to a type; by theological writers the term is employed to denote the reality of which a *type* is the prophetic symbol.

An'tium, in ancient Italy, one of the most ancient and powerful cities of Latium, the chief city of the Volsci, and often at war with the Romans, by whom it was finally taken in 338 B.C. It was 38 miles distant from Rome, a flourishing seaport, and became a favourite residence of the wealthy Romans. It was destroyed by the Saracens, but vestiges of it remain at Porto d'Anzo, near which many valuable works of art have been found.

Antivari (a'n-tē'va'-rē), a seaport town on the eastern shore of the Adriatic, ceded to Montenegro by the Treaty of Berlin (1878). Antivari was opened as a free port on 23rd Oct., 1909. It was occupied by the Austrians in 1916, and by the Italians in Nov., 1918. Pop. 2500.

Antlers, the horns of the deer tribe, or the snags or branches of the horns.

Ant-lion, the larva of a Neuropterous insect (*Myrmelëon formicārius*), which in its perfect state greatly resembles a dragon-fly; curious on account of its ingenious method of catching the insects—chiefly ants—on which it feeds. It digs a funnel-shaped hole in the driest and finest sand it can find, and when the pit is deep enough, and the sides are quite smooth and sloping, it buries itself at the bottom with only its formidable mandibles projecting, and waits till some luckless insect stumbles over the edge, when it is immediately seized, its juices sucked, and the dead body jerked out. It inhabits Southern Europe.

Antofagas'ta, a Chilean seaport on the Bay of Morena, and a territory of the same name taken from Bolivia in the war of 1879-82, and definitely ceded to Chile in 1885. The territory has an area of 46,408 sq. miles; pop. (1919), 235,506. The port is connected by railway with the silver and other mines lying inland, and exports silver, copper, cubic nitre, &c., partly from Bolivia. Pop. (1919), 69,175.

Antoinette (a'n'-twa'-net), Marie. See *Marie Antoinette*.

Antokolski, Mark, Russian sculptor of Jewish extraction, born at Vilna in 1843. He studied at the Petrograd Academy of Fine Arts, and his earliest success was a wooden statue *The Jewish Tailor* (1864). In 1868 he received a grant for travelling, and whilst in Italy he finished his famous statue, *Ivan the Terrible*. He was made an academician, and in 1878 was awarded the first prize for sculpture at the Paris International Exhibition. In 1888 he settled permanently in Paris, where he died in 1902. His works include: *Peter the Great* (1872), *Christ before the People* (1874), *The Death of Socrates* (1876), *Spinoza* (1882), *Yermak* and *The Sleeping Beauty* (1900).

Antommarchi (-mär'kē), Carlo Francesco, Italian physician, born in Corsica in 1780, died in Cuba 1838. He was professor of anatomy at Florence when he offered himself as physician to Napoleon at St. Helena. Napoleon at first received him with reserve, but soon admitted him to his confidence, and testified his satisfaction with him by leaving him a legacy of 100,000 francs. On his return to Europe he published *Les Derniers Moments de Napoléon* (2 vols., 8vo, 1823).

Antonell'i, Giacomo, cardinal, born 1806, died 1876. He was educated at the Grand Seminary of Rome, where he attracted the attention of Pope Gregory XVI, who appointed him to several important offices. On the accession of Pius IX in 1846 Antonelli was raised to the dignity of cardinal-deacon; two years later he became president and Minister of Foreign Affairs, and in 1850 was appointed Secretary of State. During the sitting of the Œcumenical Council (1869-70) he was a prominent champion of the papal interest. He strongly opposed the assumption of the united Italian crown by Victor Emanuel.

Antonell'o (of Messina), an Italian painter who died at Venice, probably in 1493, and is said to have introduced oil-painting into Italy, having been

instructed in it by Jan Van Eyck. Three works by him are in the National Gallery, London.

Antoni´nus, Itinerary of. See *Itinerary*.

Antoni´nus, Marcus Aurelius. See *Aurelius*.

Antoni´nus, Wall of, a barrier erected by the Romans across the isthmus between the Forth and the Clyde, in the reign of Antoninus Pius. Its western extremity was at or near Dunglass Castle, its eastern at Carriden, and the whole length of it exceeded 27 miles. It was constructed A.D. 140 by Lollius Urbicus, the imperial legate, and consisted of a ditch 40 feet wide and 20 feet deep, and a rampart of stone and earth on the south side 24 feet thick and 20 feet in height. It was strengthened at either end and along its course by a series of forts and watch-towers. It may still be traced at various points, and is commonly known as *Graham's Dyke*.

Antoni' nus Pius, Titus Aurelius Fulvus, Roman emperor, was born at Lavinium, near Rome, A.D. 86, died A.D. 161. In 120 A.D. he became consul, and he was one of the four persons of consular rank among whom Hadrian divided the supreme administration of Italy. He then went as proconsul to Asia, and after his return to Rome became more and more the object of Hadrian's confidence. In A.D. 138 he was selected by that emperor as his successor, and the same year he ascended the throne. He speedily put down the persecutions of the Christians, and carried on but a few wars. In Britain he extended the Roman dominion, and, by raising a new wall (see preceding article), put a stop to the invasions of the Picts and Scots. The senate gave him the surname *Pius*, that is, dutiful or showing filial affection, because to keep alive the memory of Hadrian he had built a temple in his honour. He was succeeded by Marcus Aurelius, his adopted son.



Coin of
Antoninus Pius

Anto'nius, Marcus (Mark Antony), Roman triumvir, born 83 B.C., was connected with the family of Cæsar by his mother. Debauchery and prodigality marked his youth. To escape his creditors he went to Greece in 58, and from thence followed the consul Gabinius on a campaign in Syria as commander of the cavalry. He served in Gaul under Cæsar in 52 and 51. In 50 he returned to Rome to support the interests of Cæsar against the aristocratical party headed by Pompey, and was appointed tribune. When war broke out between Cæsar and Pompey, Antony led reinforcements to Cæsar in Greece, and in the battle of Pharsalia he commanded the left wing. He afterwards returned to Rome with the appointment of Master of the Horse and Governor of Italy (47). In 44 B.C. he became Cæsar's colleague in the consulship. Soon after Cæsar was assassinated, Antony, by the reading of Cæsar's will, and by the oration which he delivered over his body, excited the people to anger and revenge, and the murderers were obliged to flee. After several quarrels and reconciliations with Octavianus, Cæsar's heir (see *Augustus*), Antony departed to Cisalpine Gaul, which province had been conferred upon him against the will of the Senate. But Cicero thundered against him in his famous *Philippics*; the Senate declared him a public enemy, and entrusted the conduct of the war against him to Octavianus and the consuls Hirtius and Pansa. After a campaign of varied fortunes Antony fled with his troops over the Alps. Here he was joined by

Lepidus, who commanded in Gaul, and through whose mediation Antony and Octavianus were again reconciled. It was agreed that the Roman world should be divided among the three conspirators, who were called *triumviri*. Antony was to take Gaul; Lepidus, Spain; and Octavianus, Africa and Sicily. They decided upon the proscription of their mutual enemies, each giving up his friends to the others, the most celebrated of the victims being Cicero the orator. Antony and Octavianus departed in 42 for Macedonia, where the united forces of their enemies, Brutus and Cassius, formed a powerful army, which was, however, speedily defeated at Philippi. Antony next visited Athens, and thence proceeded to Asia. In Cilicia he ordered Cleopatra, Queen of Egypt, to apologize for her insolent behaviour to the *triumviri*. She appeared in person, and her charms fettered him for ever. He followed her to Alexandria, where he bestowed not even a thought upon the affairs of the world, till he was aroused by a report that hostilities had commenced in Italy between his own relatives and Octavianus. A short war, followed, which was decided in favour of Octavianus before the arrival of Antony in Italy. A reconciliation was effected, which was sealed by the marriage of Antony with Octavia, the sister of Octavianus. A new division of the Roman dominions was now made (in 40), by which Antony obtained the East, Octavianus the West. After his return to Asia Antony gave himself up entirely to Cleopatra, assuming the style of an Eastern despot, and so alienating many of his adherents and embittering public opinion against him at Rome. At length war was declared at Rome against the Queen of Egypt, and Antony was deprived of his consulship and government. Each party assembled its forces, and Antony lost, in the naval battle at Actium, 31 B.C., the dominion of the world. He followed Cleopatra to Alexandria, and on the arrival of Octavianus his fleet and cavalry deserted, and his infantry was defeated. Deceived by a false report of her death which Cleopatra had disseminated, he fell upon his own sword (30 B.C.).—BIBLIOGRAPHY: Mommsen, *Roman History*; Plutarch, *Lives* (translated by Langhorne); De Quincey, *Essay on the Cæsars*.

Antonoma'sia, in rhetoric, the use of the name of some office, dignity, profession, science, or trade instead of the true name of the person, as when *his majesty* is used for a king, *his lordship* for a nobleman; or when, instead of Aristotle, we say, *the philosopher*; or, conversely, the use of a proper noun instead of a common noun; as, a *Solomon* for a wise man.

Antony, Mark. See *Antonius (Marcus)*.

Antony, St. See *Anthony*.

An'trim, a county of Ireland, province of Ulster, in the north-east of the island; area, 702,654 acres, of which about a third are arable. The eastern and northern districts are comparatively mountainous, with tracts of heath and bog, but no part rises to a great height. The principal rivers are the Lagan and the Bann, which separate Antrim from Down and Londonderry respectively. The general soil of the plains and valleys is strong loam. Flax, oats, and potatoes are the principal agricultural produce. Cattle, sheep, swine, and goats are extensively reared. There are salt-mines and beds of iron-ore, which is worked and exported. A range of basaltic strata stretches along the northern coast, of which the celebrated Giant's Causeway is the most remarkable portion, the vast aggregates of natural rock pillars there being very striking. The interior also contains some scenes of picturesque beauty, particularly the fertile valley of the Lagan, between Belfast and Lisburn. Much of the scenery of the county, however, is dreary and monotonous. Lough Neagh, the largest lake of the United Kingdom, is principally in Antrim. Its waters are carried to the sea by the Bann, which is of no use for navigation, being obstructed by weirs and rocks. The spinning of linen and cotton yarn, and the weaving of linen and cotton, are the staple manufactures, but the cotton manufacture is small compared with that of linen. The principal towns are Belfast, Lisburn, Ballymena, Larne, and Carrickfergus. In 1898 Belfast, the former capital, was constituted into a county borough. About fifty per cent of the inhabitants are Presbyterians, being the descendants of Scottish immigrants of the seventeenth century. The county sends four members to Parliament; Belfast returns nine. Pop. (excluding Belfast) 193,864 (1911).—The town of Antrim, at the north end of Lough Neagh, is a small place with a pop. of 1826.

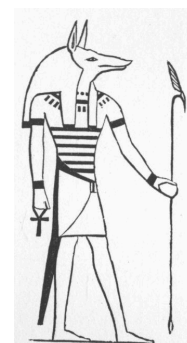
Ant-thrush, a name given to certain passerine or perching birds having resemblances to the thrushes and supposed to feed largely on ants. They all have longish legs, short wings, and a short tail. The true ant-thrushes of the Old World belong to the genus *Pitta*. They chiefly inhabit southern and south-eastern Asia and the Eastern Archipelago, but are also found in Africa and Australia, and are birds of brilliant plumage, exhibiting black, white, scarlet, blue, and green in vivid contrast, there being generally no blending

of colours by means of intermediate hues. These birds are not now regarded as allied to the thrushes, nor are they allied to the ant-birds, or ant-thrushes of the New World, which live among close foliage and bushes. Some of these are called ant-shrikes and ant-wrens. They belong to several genera.

Ant´werp (Du. and Ger. *Antwerpen*, Fr. *Anvers*), the chief port of Belgium, and one of the first on the Continent, the capital of a province of the same name, on the Scheldt, about 50 miles from the open sea. It lies in a fertile plain at an abrupt turn of the river, which is here from 160 to 280 yards wide, and has a depth varying from 25 to 50 feet. It is strongly fortified, being completely surrounded on the land side by a semicircular inner line of fortifications, the defences being completed by an outer line of forts and outworks. Fine quays have been constructed along the river banks. The general appearance of the city is exceedingly picturesque, an effect produced by the numerous churches, convents, and magnificent public buildings, the stately antique houses that line its older thoroughfares, and the profusion of beautiful trees with which it is adorned. The older streets are tortuous and irregular, but those in the newer quarters are wide and regular. Some of the squares are very handsome. The cathedral, with a spire 400 feet high, one of the largest and most beautiful specimens of Gothic architecture in Belgium, contains Rubens' celebrated masterpieces, *The Descent from the Cross*, *The Elevation of the Cross*, and *The Assumption*. The other churches of note are St. James's, St. Andrew's, and St. Paul's, all enriched with paintings by Rubens, Vandyck, and other masters. Among the other buildings of note are the exchange, the town hall, the palace, the theatre, academy of the fine arts, picture and sculpture galleries, &c. The harbour accommodation is extensive and excellent, large new docks and quays having been recently built, and other works being under construction or contemplated. The shipping trade is now very large, Antwerp being a great centre of the world's commerce, and the goods being largely in transit. The entries of vessels in a year aggregate over 13,000,000 tons. Much of the trade is with Britain. There are numerous but not very important industries. Antwerp is mentioned as early as the eighth century, and in the eleventh and twelfth it had attained a high degree of prosperity. In the sixteenth century it is said to have had a pop. of 200,000, and it had then an extensive foreign trade. The wars between the Netherlands and Spain greatly injured its commerce, which was almost ruined by the closing of the

navigation of the Scheldt in accordance with the peace of Westphalia (1648). It was only in the nineteenth century that its prosperity revived. In the European War (1914-8), the Germans, under General von Beseler, entered Antwerp on 7th Oct., 1914, and remained there until Nov., 1918. Pop. (1919), 322,857.—The province consists of a fertile plain 1093 sq. miles in area, and has a pop. of over 1,000,000.

Anu'bis (*Anepo* on the monuments), one of the deities of the ancient Egyptians, the son of Osiris by Isis. The Egyptian sculptures represent him with the head, or under the form, of a jackal, with long pointed ears. His office was to conduct the souls of the dead from this world to the next, and in the lower world he weighed the actions of the deceased previous to their admission to the presence of Osiris.



Anubis

Anúshahr (*a-nöp'shär*), a town of Hindustan, United Provinces, on the Ganges, 75 miles S.E. of Delhi, a resort of Hindu pilgrims who bathe in the Ganges. Pop. 15,000.

Anu'ra, or **Anou'ra** (Gr. *an*, negative, *oura*, a tail), an ord. of Batrachians which lose the tail when they reach maturity, such as the frogs and toads.

Anuradhapura. See *Anarajapura*.

A'nus, the opening at the lower or posterior extremity of the alimentary canal through which the excrement or waste products of digestion are expelled.

An'vil, an instrument on which pieces of metal are laid for the purpose of being hammered. The common smith's anvil is generally made of seven pieces, namely, the core or body; the four corners for the purpose of enlarging its base; the projecting end, which contains a square hole for the reception of a set or chisel to cut off pieces of iron; and the beak or conical end, used for turning pieces of iron into a circular form, &c. These pieces are each separately welded to the core and hammered so as to form a regular surface with the whole. When the anvil has received its due form, it is faced with steel, and is then tempered in cold water. The smith's anvil is generally placed loose upon a wooden block. The anvil for heavy

operations, such as the forging of ordnance and shafting, consists of a huge iron block deeply embedded, and resting on piles of masonry.

Anville, Jean Baptiste Bourguignon d' (jăn´ ba´ p-tēst bōr-gē-nyōn´ dăn´-vēl), a celebrated French geographer, born 1697, died 1782; published a great number of maps and writings illustrative of ancient and modern geography.

Anynaks, a negro tribe inhabiting the banks of the Upper Sobat (a tributary of the White Nile), between the Egyptian Sudan and Abyssinia. They rebelled against British authority in 1912.

Anzacs, a composite word used as the name of the British colonial troops in the Gallipoli undertaking. The men being from Australia and New Zealand, their organization was officially known as the Australian-New Zealand Army Corps. The full title, however, was much too cumbersome, and a clerk in one of the head-quarters offices at Zeitoun, where the troops were in training, hit upon the word *Anzacs*, formed from the initial letters of the long title. The Anzacs landed near Gaba Tepe, Gallipoli, on the morning of 25th April, 1915, and had their first encounter with the Germans on the Western Front on 6th May, 1916. In 1916 the word Anzac was officially adopted by the War Office.

Anzin (a´ n´-zan´), a town of France, department of Nord, about 1 mile north-west of Valenciennes, in the centre of an extensive coal-field, with blast-furnaces, forges, rolling-mills, foundries, &c. Pop. 14,325.

Aonia, in ancient geography a name for part of Bœotia in Greece, containing Mount Helicon and the fountain Aganippe, both haunts of the muses.

A´orist, the name given to one of the tenses of the verb in some languages (as the Greek), which expresses indefinite past time.

Aor´ta, in anatomy, the great artery or trunk of the arterial system, proceeding from the left ventricle of the heart, and giving origin to all the arteries except the pulmonary. It first rises towards the top of the breast-bone, when it is called the *ascending aorta*; then makes a great curve, called the transverse or *great arch of the aorta*, whence it branches off to the head and upper extremities; thence proceeding towards the lower extremities,

under the name of the *descending aorta*, it branches off to the trunk; and finally divides into the two iliacs, which supply the pelvis and lower extremities.

Aosta (a´-os´ta´; ancient **Augusta Prætoria**), a town of north Italy, 50 miles N.N.W. of Turin, on the Dora-Baltea, with an ancient triumphal arch, remains of an amphitheatre, &c. Pop. 7000.

Aoudad (a-ö´dad), the *Ammotrăgus tragelăphus*, a quadruped allied to the sheep, most closely to the mouflon, from which, however, it may be easily distinguished by the heavy mane, commencing at the throat and falling as far as the knees. It is a native of North Africa, inhabiting the loftiest and most inaccessible rocks.

Apaches (a´-pä´chez), a warlike race of North-American Indians, numbering between 5000 and 6000, and inhabiting Arizona, New Mexico, and Oklahoma. The final surrender of the tribe took place in 1886, but a few in Mexico still maintain their independence and hostility to the whites. The name *Apache* was assumed by Parisian hooligans, notorious for their criminal outrages.

Ap´anage, an allowance which the younger princes of a reigning house in some European countries receive from the revenues of the country, generally together with a grant of public domains, that they may be enabled to live in a manner becoming their rank.

Ap´atite, a translucent but seldom transparent mineral, which crystallizes in a regular six-sided prism, usually terminated by a truncated six-sided pyramid. It passes through various shades of colour, from white to yellow, green, blue, and occasionally red, scratches fluor-spar but is scratched by felspar, and has a specific gravity of about 3.5. It is a compound of calcium phosphate with calcium fluoride or chloride. It occurs principally in igneous rocks, particularly diorites. The very coarse-grained granites of Ontario contain apatite crystals of corresponding size, which have been picked out as a source of artificial phosphate manures. Apatite supplies to soils almost all the phosphorus available for plants in a state of nature.

Ape, a common name of a number of quadrumanous animals inhabiting the Old World (Asia and the Asiatic Islands, and Africa), and including a

variety of species. The word *ape* was formerly applied indiscriminately to all quadrumanous mammals; but it is now limited to the anthropoid or man-like monkeys. The family includes the chimpanzee, gorilla, orang-outang, &c., and has been divided into three genera, *Troglodytes*, *Simia*, and *Hylobates*. See *Chimpanzee*, *Gibbon*, *Gorilla*, *Monkey*, *Orang*, &c.

Apeldoorn (ä'pel-dörn), a town of Holland, province of Guelderland, 17 miles north of Arnhem, with manufactures of paper. The royal palace Loo is here. Pop. 44,474.

Apelles (a-pel'ēz), the most famous of the painters of ancient Greece and of antiquity, was born in the fourth century B.C., probably at Colophon. Ephorus of Ephesus was his first teacher, but attracted by the renown of the Sicyonian school he went and studied at Sicyon. In the time of Philip he went to Macedonia, and there a close friendship between him and Alexander the Great was established. The most admired of his pictures was that of Venus rising from the sea and wringing the water from her dripping locks. His portrait of Alexander with a thunderbolt in his hand was no less celebrated. He died about the end of the century. Among the anecdotes told of Apelles is the one which gave rise to the Latin proverb, 'Ne sutor ultra crepidam'—'Let not the shoemaker go beyond the shoe'. Having heard a cobbler point out an error in the drawing of a shoe in one of his pictures he corrected it, whereupon the cobbler took upon him to criticize the leg, and received from the artist the famous reply.

Apennines (Lat. *Mons Apenninus*), a prolongation of the Alps, forming the 'backbone of Italy'. Beginning at Savona, on the Gulf of Genoa, the Apennines traverse the whole of the peninsula and also cross over into Sicily, the Strait of Messina being regarded merely as a gap in the chain. The average height of the mountains composing the range is about 4000 feet, and nowhere do they reach the limits of perpetual snow, though some summits exceed 9000 feet in height. Monte Corno, called also Gran Sasso d'Italia (Great Rock of Italy), which rises among the mountains of the Abruzzi, is the loftiest of the chain, rising to the height of 9541 feet, Monte Majella (9151) being next. Monte Gargano, which juts out into the Adriatic from the *ankle* of Italy, is a mountainous mass upwards of 5000 feet high, completely separated from the main chain. On the Adriatic side the mountains descend more abruptly to the sea than on the western or

Mediterranean side, and the streams are comparatively short and rapid. On the western side are the valleys of the Arno, Tiber, Garigliano, and Volturno, the largest rivers that rise in the Apennines, and the only ones of importance in the peninsular portion of Italy. They consist almost entirely of limestone rocks, and are exceedingly rich in the finest marbles. On the south slopes volcanic masses are not uncommon. Mount Vesuvius, the only active volcano on the continent of Europe, is an instance. The lower slopes are well clothed with vegetation, the summits are sterile and bare.

Apenrade (ä'pen-rä-de), a seaport in Schleswig-Holstein, on a fiord of the Little Belt, beautifully situated, and carrying on a considerable fishing industry. Pop. 7800.

Ape'rient, a medicine which, in moderate doses, gently but completely opens the bowels: examples, castor-oil, Epsom salts, senna, &c.

Apet'alous, a botanical term applied to flowers or flowering-plants which are destitute of petals or corolla.

Aphanip'tera, an order of wingless insects, composed of the different species of fleas. See *Flea*.

Apha'sia (Gr. *a*, not, and *phasis*, speaking), in pathology, a symptom of certain morbid conditions of the nervous system, in which the patient loses the power of expressing ideas by means of words, or loses the appropriate use of words, the vocal organs the while remaining intact and the intelligence sound. There is sometimes an entire loss of words as connected with ideas, and sometimes only the loss of a few. In one form of the disease, called *aphemia*, the patient can think and write, but cannot speak; in another, called *agraphia*, he can think and speak, but cannot express his ideas in writing. In a great majority of cases, where post-mortem examinations have been made, morbid changes have been found in the left frontal convolution of the brain.

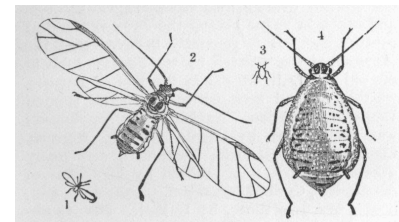
Aphe'lion (Gr. *apo*, from, and *hēlios*, the sun), that point of the orbit of the earth or any other planet which is remotest from the sun.

Aphe'mia. See *Aphasia*.

Aphides (af'i-dēz). See *Aphis*.

Aphis, a genus of insects (called plant-lice) of the ord. Hemiptera, the type of the family Aphidēs. The species are very numerous and destructive. The *A. rosæ* lives on the rose; the *A. fabæ* on the bean; the *A. humūli* is injurious to the hop, the *A. granaria* to cereals, and *A. lanigěra* or woolly aphis equally so to apple trees. The aphides are furnished with an inflected beak, and feelers longer than the thorax. In the same species some individuals have four erect wings and others are entirely

without wings. The feet are of the ambulatory kind, and the abdomen usually ends in two horn-like tubes, from which is ejected the substance called honey-dew, a favourite food of ants. (See *Ant.*) The aphides illustrate parthenogenesis; hermaphrodite forms produced from eggs produce viviparous wingless forms, which again produce others like themselves, and thus multiply during summer, one individual giving rise to millions. Winged sexual forms appear late in autumn, the females of which, being impregnated by the males, produce eggs.



Aphides
Cabbage-leaf Plant-louse
(*Aphis brassicæ*)—1, 2. Male,
natural size and magnified. 3, 4,
Female, natural size and
magnified.

Apho'nia (Gr. *a*, not, and *phōnē*, voice), in pathology, the greater or less impairment, or the complete loss of the power of emitting vocal sound. The slighter and less permanent forms often arise from extreme nervousness, fright, and hysteria. Slight forms of structural aphonia are of a catarrhal nature, resulting from more or less congestion and tumefaction of the mucous and submucous tissues of the larynx and adjoining parts. Severer cases are frequently occasioned by serous infiltration into the submucous tissue, with or without inflammation of the mucous membrane of the larynx and of its vicinity. The voice may also be affected in different degrees by inflammatory affections of the fauces and tonsils; by tumours in these situations; by morbid growths pressing on or implicating the larynx or trachea; by aneurisms; and most frequently by chronic laryngitis and its consequences, especially thickening, ulceration, &c.

Aph'orism, a brief, sententious saying, in which a comprehensive meaning is involved, as 'Familiarity breeds contempt'; 'Necessity has no law'.

Aphrodite (af-ro-di'tē), the goddess of love among the Greeks; counterpart of the Roman Venus. A festival called Aphrodisia was celebrated in her honour in various parts of Greece, but especially in Cyprus. See *Venus*.

Aphthæ (af'thē), a disease occurring especially in infants, but occasionally seen in old persons, and consisting of small white ulcers upon the tongue, gums, inside of the lips, and palate, resembling particles of curdled milk: commonly called *thrush* or *milk-thrush*.

A'pia, the chief place and trading centre of the Samoa Islands, on the north side of the Island of Upolu. It has a wireless station.

A'piary (Lat. *apis*, a bee), a place for keeping bees. The apiary should be well sheltered from strong winds, moisture, and the extremes of heat and cold. The hives should face the south or south-east, and should be placed on shelves 2 feet above the ground, and about the same distance from each other. There is no place for handling bees like the open air in suitable weather, and for this reason bee-houses, or bee-sheds, formerly in use, are not much in vogue now. As to the form of the hives and the materials of which they should be constructed there are great differences of opinion. The old dome-shaped straw *skep* is still in general use among the cottagers of Great Britain. Its cheapness and simplicity of construction are in its favour, while it is excellent for warmth and ventilation; but it has the disadvantage that its interior is closed to inspection, and the honey can only be got out by stupefying the bees with the smoke of the common puff-ball or chloroform, or by fumigating with sulphur, which entails the destruction of the swarm. Wooden hives of square box-like form are now gaining general favour among bee-keepers. They usually consist of a large breeding chamber below and two sliding removable boxes called 'supers' above for the abstraction of honey without disturbing the contents of the main chamber. It is of great importance that the apiary should be situated in the neighbourhood of good feeding grounds, such as gardens, clover-fields, or heath-covered hills. When their stores of honey are removed, the bees must be fed during the winter and part of spring with syrup or with a solution consisting of 2 lb. loaf-sugar to a pint of water. In the early spring slow and continuous feeding (a few ounces of syrup each day) will stimulate the queen to deposit her eggs, by which means the colony is rapidly strengthened and throws off early swarms. New swarms may make their

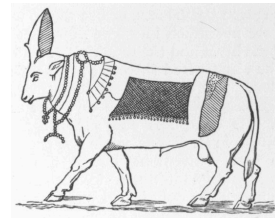
appearance as early as May and as late as August, but swarming usually takes place in the intervening months. See *Bee-keeping, Hives*.

Apic'ius, Marcus Gabius, a Roman epicure in the time of Augustus and Tiberius, who, having exhausted his vast fortune on the gratification of his palate, and having *only* about £80,000 left, poisoned himself that he might escape the misery of plain diet. The book of cookery published under the title of *Apicius* was written by one Cælius, and belongs to a much later date.

A'pion, a Greek grammarian, born in Egypt, lived in the reigns of Tiberius, Caligula, and Claudius, A.D. 15-54, and went to Rome to teach grammar and rhetoric. Among his works, one or two fragments only of which remain, was one directed against the Jews, which was replied to by Josephus.

A'pios, a genus of leguminous climbing plants, producing edible tubers on underground shoots. An American species (*A. tuberōsa*) has been used as a substitute for the potato, but its tubers, though numerous, are small.

A'pis, a bull to which divine honours were paid by the ancient Egyptians, who regarded him as a symbol of Osiris. At Memphis he had a splendid residence, containing extensive walks and courts for his entertainment, and he was waited upon by a large train of priests, who looked upon his every movement as oracular. He was not suffered to live beyond twenty-five years, being secretly killed by the priests and thrown into a sacred well. Another bull, characterized by certain marks, as a black colour, a triangle of white on the forehead, a white crescent-shaped spot on the right side, &c., was selected in his place. His birthday was annually celebrated, and his death was a season of public mourning. See *Animal Worship*.



Apis

A'pis, a genus of insects. See *Bee*.

A'pium, a genus of umbelliferous plants, including celery.

Aplacen'tal. See *Placenta, Marsupialia*, and *Echidna*.

Aplanat'ic. See *Optics, Photography*.

Aplysia. See *Sea-hare*.

Apoc´alypse (Gr. *apokalypsis*, a revelation), the name given to the last book of the New Testament, in the English version called *The Revelation of St. John the Divine*. Although a Christian work, the *Apocalypse* belongs to a class of literature dealing with eschatological subjects and much in vogue among the Jews of the first century B.C. It is generally believed that the *Apocalypse* was written by the apostle John in his old age (A.D. 95-97) in the Isle of Patmos, whither he had been banished by the Roman Emperor Domitian. Anciently its genuineness was maintained by Justin Martyr, Irenæus, Clement of Alexandria, Tertullian, and many others; while it was doubted by Dionysius of Alexandria, Cyril of Jerusalem, Chrysostom, and, nearer our own times, by Luther. The *Apocalypse* has been explained differently by almost every writer who has ventured to interpret it, and has furnished all sorts of sects and fanatics with quotations to support their creeds or pretensions. The modern interpreters may be divided into three schools—namely, the *historical school*, who hold that the prophecy embraces the whole history of the Church and its foes from the time of its writing to the end of the world; the *Præterists*, who hold that the whole or nearly the whole of the prophecy has been already fulfilled, and that it refers chiefly to the triumph of Christianity over Paganism and Judaism; and the *Futurists*, who throw the whole prophecy, except the first three chapters, forward upon a time not yet reached by the Church—a period of no very long duration, which is immediately to precede Christ's second coming. See *Bible*.—BIBLIOGRAPHY: R. H. Charles, *Studies in the Apocalypse*; F. C. Burkitt, *Jewish and Christian Apocalypses*.

Apocalyptic Number, the mystic number 666 found in Rev. xiii, 18. As early as the second century ecclesiastical writers found that the name Antichrist was indicated by the Greek characters expressive of this number. By Irenæus the word *Lateinos* was found in the letters of the number, and the Roman Empire was therefore considered to be Antichrist. Protestants generally believe it has reference to the Papacy, and, on the other hand, Catholics connect it with Protestantism. It is, however, almost certain that the number refers to Nero, for by transliterating the Greek *Kaisar Neron* into Hebrew, and adding together the sums denoted by the Hebrew letters, we obtain the number 666.

Apocar´pous, in botany, a term applied to such fruits as are the produce of a single flower, and are formed of one carpel, or a number of carpels free

and separate from each other.

Apoc'rypha (Gr., 'things concealed or spurious'), a term applied in the earliest churches to various sacred or professedly inspired writings, sometimes given to those whose authors were unknown, sometimes to those with a hidden meaning, and sometimes to those considered objectionable. The term is specially applied to the fourteen undermentioned books, which were written during the two centuries preceding the birth of Christ. They were written, not in Hebrew, but in Greek, and the Jews never allowed them a place in their sacred canon. They were incorporated into the Septuagint, and thence passed to the Vulgate. The Greek Church excluded them from the canon in 360 at the Council of Laodicea. The Latin Church treated them with more favour, but it was not until 1546 that they were formally admitted into the canon of the Church of Rome by a decree of the Council of Trent. The Anglican Church says they may be read for example of life and instruction of manners, but that the Church does not apply them to establish any doctrine. All other Protestant churches in Britain and America ignore them. The following fourteen books form the Apocrypha of the English Bible: The first and second *Books of Esdras*, *Tobit*, *Judith*, the rest of the *Book of Esther*, the *Wisdom of Solomon*, the *Wisdom of Jesus the son of Sirach*, or *Ecclesiasticus*, *Baruch the Prophet*, the *Song of the Three Children*, *Susanna and the Elders*, *Bel and the Dragon*, the *Prayer of Manasses*, and the first and second *Books of Maccabees*. Besides the Apocryphal books of the Old Testament there are many spurious books composed in the earlier ages of Christianity, and published under the names of Christ and his apostles, or of such immediate followers as from their character or means of intimate knowledge might give an apparent plausibility to such forgeries. These writings comprise: 1st, the *Apocryphal Gospels*, which treat of the history of Joseph and the Virgin before the birth of Christ, of the infancy of Jesus, and of the acts of Pilate; 2nd, the *Apocryphal Acts of the Apostles*; and 3rd, the *Apocryphal Apocalypses*, none of which have obtained canonical recognition by any of the churches. —BIBLIOGRAPHY: Wace, *The Apocrypha*; Porter, in *Hastings' Bible Dict.*, i, pp. 111-23; W. D. F. Oesterley, *Book of the Apocrypha*; R. H. Charles, *Religious Development between the Old and the New Testaments*.

Apocyna'ceæ, a nat. ord. of dicotyledonous plants, having for its type the genus *Apocynum* or dog-bane. The species have opposite or sometimes

whorled leaves without stipules; the corolla monopetalous, hypogynous, and with the stamens inserted upon it; fruit two-celled. The plants yield a milky juice, which is generally poisonous; several yield caoutchouc, and a few edible fruits. The bark of several species is a powerful febrifuge. To the order belongs the periwinkle (*Vinca*). See *Cow-tree*, *Periwinkle*, *Oleander*, *Tanghin*.

Ap'oda. See *Proteolepadidæ*.

Ap'odal Fishes, the name applied to such malacopterous fishes as want ventral fins. They constitute a small natural family, of which the common eel is an example.

Apo'dōsis, in grammar, the latter member of a conditional sentence (or one beginning with *if*, *though*, &c.) dependent on the condition or *protāsis*; as, if it rain (*protasis*) I shall not go (*apodosis*).

Ap'ogee (-jē; Gr. *apo*, from, and *gē*, the earth), that point in the orbit of the moon or a planet where it is at its greatest distance from the earth; also the greatest distance of the sun from the earth when the latter is in *aphelion*.

Apol'da, a town of Germany, in Saxe-Weimar, at which woollen goods are extensively manufactured. Pop. 22,610.

Apollina'rians, a sect of Christians who maintained the doctrine that Christ had a human body and a human sensitive soul, but no human rational mind, the Divine Logos (the Word) taking the place of the mind, and that God was consequently united in him with the human body and the sensitive soul. Apollinaris, the author of this opinion, was, from A.D. 362 till at least A.D. 382, Bishop of Laodicea, in Syria, and a zealous opposer of the Arians. As a man and a scholar he was highly esteemed, and was among the most popular authors of his time. He formed a congregation of his adherents at Antioch, and made Vitalis their bishop. The *Apollinarians*, or *Vitalians*, as their followers were called, soon spread their settlements in Syria and the neighbouring countries, established several societies, with their own bishops, and one even in Constantinople; but many adherents drifted away to Monophysitism, and the sect soon became extinct.

Apollina'ris Water, a natural aerated water, belonging to the class of acidulated soda waters, and derived from the Apollinarisbrunnen, a spring

in the valley of the Ahr, near the Rhine, in Rhenish Prussia, forming a highly-esteemed beverage.

Apol'lo, son of Zeus (Jupiter) and Leto (Latona), who, being persecuted by the jealousy of Hera (Juno), after tedious wanderings and nine days' labour, was delivered of him and his twin sister, Artēmis (Diana), on the Island of Delos. Skilled in the use of the bow, he slew the serpent Python on the fifth day after his birth; afterwards, with his sister Artēmis, he killed the children of Niobē. He aided Zeus in the war with the Titans and the giants. He destroyed the Cyclopes, because they forged the thunderbolts with which Zeus killed his son and favourite Asklepios (Æsculapius). According to some traditions he invented the lyre, though this is generally ascribed to Hermes (Mercury). The brightest creation of polytheism, Apollo is also the most complex, and many aspects of the people's life were reflected in his cult. He was originally the sun-god; and though in Homer he appears distinct from Helios (the sun), yet his real nature is hinted at even here by the epithet Phœbus, that is, the radiant or beaming. In later times the view was almost universal that Apollo and Helios were identical. From being the god of light and purity in a physical sense, he gradually became the god of moral and spiritual light and purity, the source of all intellectual, social, and political progress. He thus came to be regarded as the god of song and prophecy, the god that wards off and heals bodily suffering and disease, the institutor and guardian of civil and political order, and the founder of cities. His worship was introduced at Rome at an early period, probably in the time of the Tarquins. Among the ancient statues of Apollo that have come down to us, the most remarkable is the one called *Apollo Belvedere*, from the Belvedere Gallery in the Vatican at Rome. This statue was discovered at Frascati in 1455, and purchased by Pope Julian II, the founder of the Vatican museum. It is a copy of a Greek statue of the third century B.C., and dates probably from the reign of Nero.



Apollo Belvedere
(Vatican, Rome)

Apollodo'rus, a Greek writer who flourished 140 B.C. Among the numerous works he wrote on various subjects, the only one extant is his *Bibliothecē*,

which contains a concise account of the mythology of Greece down to the heroic age.

Apollo'nius of Perga, Greek mathematician, called the 'great geometer', flourished about 240 B.C., and was the author of many works, only one of which, a treatise on *Conic Sections*, partly in Greek and partly in an Arabic translation, is now extant.

Apollo'nius of Rhodes, a Greek rhetorician and poet, flourished about 230 B.C. Of his various works we have only the *Argonautica*, an epic poem of considerable merit, though perhaps written with too much care and labour. It deals with the story of the Argonautic expedition.

Apollo'nius of Ty'ana, in Cappadocia, a Pythagorean philosopher who was born in the beginning of the Christian era, early adopted the Pythagorean doctrines, abstaining from animal food and maintaining a rigid silence for five years. He travelled extensively in Asia, professed to be endowed with miraculous powers, such as prophecy and the raising of the dead, and was on this account set up by some as a rival to Christ. His ascetic life, wise discourses, and wonderful deeds obtained for him almost universal reverence, and temples, altars, and statues were erected to him. He died at Ephesus about the end of the first century. A narrative of his strange career, containing many fables, with, perhaps, a kernel of truth, was written by Philostratus about a century later.

Apollo'nius of Tyre, the hero of a tale which had an immense popularity in the Middle Ages and which indirectly furnished the plot of Shakespeare's *Pericles, Prince of Tyre*. The story, originally in Greek, first appeared in the third century after Christ.

Apoll'os, a Jew of Alexandria, who learned the doctrines of Christianity at Ephesus from Aquila and Priscilla, became a preacher of the gospel in Achaia and Corinth, and an assistant of Paul in his missionary work. Some have regarded him as the author of the *Epistle to the Hebrews*.

Apoll'yon ('the Destroyer'), a name used in Rev. ix, 11 for the angel of the bottomless pit.

Apologetics (-jet'iks), this term, as used in Christian theology, does not carry with it the idea of excuse or regretful acknowledgment, but signifies a

defensive or vindictory statement, which accords with its meaning in the original Greek. In the conventional division of systematic theology apologetics comes first in order, and is followed by the disciplines of dogmatics and ethics, which expound Christian belief and Christian duty respectively. There is a tendency, however, in the more recent treatment of systematic theology, to include the defence or vindication of the various Christian doctrines within the dogmatic scheme, leaving to apologetics—in so far as it may be regarded as a separate discipline from dogmatics—the discussion of such general themes as religion and revelation, authority and inspiration, and the essence and truth of Christianity. Such discussions belong essentially to what is now often called philosophy of religion. The preference of the term philosophy of religion to that of apologetics is indicative at once of the wider theological outlook of our time and of the conciliatory, adaptable, and more sympathetic spirit in which the Christian apologist approaches the new thought and culture.—BIBLIOGRAPHY: A. B. Bruce, *Apologetics*; R. Mackintosh, *First Primer of Apologetics*; J. R. Illingworth, *Reason and Revelation*; A. E. Garvie, *A Handbook of Christian Apologetics*.

Apologue (ap'ō-log), a story or relation of fictitious events intended to convey some useful truths. It differs from a parable in that the latter is drawn from events that take place among mankind, whereas the apologue may be founded on supposed actions of brutes or inanimate things. Æsop's fables are good examples of apologues.

Apol'ogy, a term at one time applied to a defence of one who is accused, or of certain doctrines called in question. Of this nature is the *Apology of Socrates* written by Plato; also a work with the same title sometimes attributed to Xenophon. The name passed over to Christian authors, who gave the name of apologies to the writings which were designed to defend Christianity against the attacks and accusations of its enemies, particularly the pagan philosophers, and to justify its professors before the emperors. Of this sort were those by Justin Martyr, Athenagoras, Tertullian, Tatian, and others.

Aponeuro'sis, in anatomy, a name of certain greyish-white shining membranes, composed of interlacing fibres, sometimes continuous with the muscular fibre, and differing from tendons merely in having a flat form.

They serve several purposes, sometimes attaching the muscles to the bones, sometimes surrounding the muscle and preventing its displacement, &c. See *Anatomy*.

Apophthegm (ap'ō-them), a short pithy sentence or maxim. Julius Cæsar wrote a collection of them, and we have a collection by Francis Bacon.

Apoph'yllite, a species of mineral of a foliated structure and pearly lustre, called also fish-eye stone. It belongs to the Zeolite family, and is a hydrated silicate of lime and potash, containing also fluorine.

Ap'oplexy, sudden abolition of consciousness, followed after recovery of consciousness by persistent disturbance of sensation or voluntary motion, from suspension of the functions of the cerebrum, resulting from blocking or rupture of the blood-vessels of the brain. In a complete apoplexy the person falls suddenly, is unable to move his limbs or to speak, gives no proof of seeing, hearing, or feeling, and the breathing is stertorous or snoring, like that of a person in deep sleep. The premonitory symptoms of this dangerous disease are drowsiness, giddiness, dulness of hearing, frequent yawning, disordered vision, noise in the ears, vertigo, &c. It is most frequent between the ages of fifty and seventy. A large head, short neck, full chest, sanguine and plethoric constitution, and corpulency are generally considered signs of predisposition to it; but the state of the heart's action, with a plethoric condition of the vascular system, has a more marked influence. Out of sixty-three cases carefully investigated only ten were fat and plethoric, twenty-three being thin, and the rest of ordinary habit. The common predisposing causes are disease or senile changes in the blood-vessels and affections of the valves of the heart; but other factors may possibly play some part either as exciting or predisposing causes, such as long and intense thought, continued anxiety, habitual indulgence of the temper and passions, sedentary and luxurious living, sexual indulgence, intoxication, &c. More or less complete recovery from a first and second attack is common, but a third is almost invariably fatal.—Cf. Grasset, *Traité du système nerveux*.

Aposiope'sis, in rhetoric, a sudden break or stop in speaking or writing, usually for mere effect or a pretence of unwillingness to say anything on a

subject; as, 'his character is such—but it is better I should not speak of *that*', or Virgil's "Quos ego—sed motos praestat componere fluctus" (*Aen.* I, 135).

Apos'tasy (Gr. *apostasis*, a standing away from), a renunciation of opinions or practices and the adoption of contrary ones, usually applied to renunciation of religious opinions. It is always an expression of reproach. What one party calls *apostasy* is termed by the other *conversion*. Catholics, also, call those persons *apostates* who forsake a religious order or renounce their religious vows without a lawful dispensation.

A posterio'ri. See *A priori*.

Apos'tles (literally, persons sent out, from the Gr. *apostellein*, to send out), the twelve men whom Jesus selected to attend him during his ministry, and to promulgate his religion. Their names were as follows: Simon Peter, and Andrew his brother; James, and John his brother, sons of Zebedee; Philip; Bartholomew; Thomas; Matthew; James, the son of Alphaeus; Lebbaeus his brother, called *Judas* or *Jude*; Simon, the Canaanite; and Judas Iscariot. To these were subsequently added Matthias (chosen by lot in place of Judas Iscariot) and Paul. The Bible gives the name of apostle to Barnabas also, who accompanied Paul on his missions (*Acts*, xiv, 14). In a wider sense those preachers who first taught Christianity in heathen countries are sometimes termed apostles; for example, St. Denis, the apostle of the Gauls; St. Boniface, the apostle of Germany; St. Augustine, the apostle of England; Francis Xavier, the apostle of the Indies; Adalbert of Prague, apostle of Prussia Proper. During the life of the Saviour the apostles more than once showed a misunderstanding of the object of His mission, and during His sufferings evinced little courage and firmness of friendship for their great and benevolent Teacher. After His death they received the Holy Ghost on the day of Pentecost, that they might be enabled to fulfil the important duties for which they had been chosen. According to one interpretation of *Matthew*, xvi, 18, Christ seems to appoint St. Peter the first of the apostles; and the Pope claims supreme authority from the power which Christ thus gave to St. Peter, of whom all the Popes, according to the Catholic dogma, are successors in an uninterrupted line.

Apostles' Creed, a well-known formula or declaration of Christian belief, formerly believed to be the work of the apostles themselves, but it can only

be traced to the fourth century. See *Creed*.

Apostol'ic, or **Apostol'ical**, pertaining or relating to the apostles.—*Apostolic Church*, the Church in the time of the apostles, constituted according to their design. The name is also given to the four churches of Rome, Alexandria, Antioch, and Jerusalem, and is claimed by the Roman Catholic Church, and occasionally by the Episcopalians.—*Apostolic Constitutions and Canons*, a collection of regulations attributed to the apostles, but generally supposed to be spurious. They appeared in the fourth century, are divided into eight books, and consist of rules and precepts relating to the duty of Christians, and particularly to the ceremonies and discipline of the Church.—*Apostolic fathers*, the Christian writers who during any part of their lives were contemporary with the apostles. There are five—Clement, Barnabas, Hermas, Ignatius, Polycarp.—*Apostolic king*, a title granted by the Pope to the kings of Hungary, first conferred on St. Stephen, the founder of the royal line of Hungary, on account of what he accomplished in the spread of Christianity.—*Apostolic see*, the see of the Popes or Bishops of Rome: so called because the Popes profess themselves the successors of St. Peter, its founder.—*Apostolic succession*, the uninterrupted succession of bishops, and, through them, of priests and deacons (these three orders of ministers being called the *apostolical orders*), in the Church by regular ordination from the first apostles down to the present day. All Episcopal churches hold theoretically, and the Roman Catholic Church and many members of the English Church strictly, that such succession is essential to the officiating priest, in order that grace may be communicated through his administrations.

Apostol'ics, **Apostolici**, or **Apostolic Brethren**, the name given to certain sects who professed to imitate the manners and practice of the apostles. The last and most important of these sects was founded about 1260 by Gerhard Segarelli of Parma. They went barefooted, begging, preaching, and singing throughout Italy, Switzerland, and France; announced the coming of the kingdom of heaven and of purer times; denounced the papacy, and its corrupt and worldly church; and inculcated the complete renunciation of all worldly ties, of property, settled abode, marriage, &c. This society was formally abolished, 1286, by Honorius IV. In 1300 Segarelli was burned as a heretic, but another chief apostle appeared—Dolcino, a learned man of Milan. In self-defence they stationed themselves in fortified places whence

they might resist attacks. After having devastated a large tract of country belonging to Milan they were subdued, A.D. 1307, by the troops of Bishop Raynerius, in their fortress Zebello, in Vercelli, and almost all destroyed. Dolcino was burned. The survivors afterwards appeared in Lombardy and in the south of France as late as 1368.

Apo'strōphē (Gr., 'a turning away from'), a rhetorical figure by which the orator changes the course of his speech, and makes a short impassioned address to one absent as if he were present, or to things without life and sense as if they had life and sense. The same term is also applied to a comma when used to contract a word, or to mark the possessive case, as in 'John's book'.

Apothecaries' weight, the weight used in dispensing drugs, in which the pound (lb.) is divided into 12 ounces (℥), the ounce into 8 drachms (℥), the drachm into 3 scruples (℥), and the scruple into 20 grains (grs.), the grain being equivalent to that in avoirdupois weight.

Apoth'ecary, in a general sense, one who keeps a shop or laboratory for preparing, compounding, and vending medicines, and for the making up of medical prescriptions. In England the term was long applied (as to some little extent still) to a regularly licensed class of medical practitioners, being such persons as were members of, or licensed by, the *Apothecaries' Company* in London. The apothecaries of London were at one time ranked with the grocers, with whom they were incorporated by James I in 1606. In 1617, however, the apothecaries received a new charter as a distinct company. They were not yet regarded as having the right to prescribe, but only to dispense, medicines; but in 1703 the House of Lords conferred that right on them, and they afterwards became a well-established branch of the medical profession. In 1815 an Act was passed providing that no person should practise as an apothecary in any part of England or Wales unless after serving an apprenticeship of five years with a member of the society, and receiving a certificate from the society's examiners. As in country places every practitioner must be to some extent an apothecary, this Act gave the society an undue influence over the medical profession. Dissatisfaction therefore long prevailed, but nothing of importance was done till the Medical Act of 1858, which brought the desired reform. The Apothecaries' Society, governed by a master, two wardens, and twenty-two

assistants, has prescribed a course of medical instruction and practice which candidates for the licence of the society must pass through. Since 1874 apprenticeship has not been necessary.

Apothe´cium, in botany, the receptacle of lichens, consisting of the spore-cases or asci, and of the paraphyses or barren threads.

Apotheo´sis (deification), a solemnity among the ancients by which a mortal was raised to the rank of the gods. The custom of placing mortals, who had rendered their countrymen important services, among the gods was very ancient among the Greeks. The Romans, for several centuries, deified none but Romulus, and first imitated the Greeks in the fashion of frequent apotheosis after the time of Cæsar. From this period apotheosis was regulated by the decrees of the senate, and accompanied with great solemnities. Almost all the Roman emperors were deified.

Appalachian Mountains (ap-pa-lā´chi-an), also called **Alleghanies**, a vast mountain range in N. America extending for 1300 miles from Cape Gaspé on the Gulf of St. Lawrence, s.w. to Alabama. The system has been divided into three great sections: the *northern* (including the Adirondacks, the Green Mountains, the White Mountains, &c.), from Cape Gaspé to New York; the *central* (including a large portion of the Blue Ridge, the Alleghanies proper, and numerous lesser ranges), from New York to the valley of the New River; and the *southern* (including the continuation of the Blue Ridge, the Black Mountains, the Smoky Mountains, &c.), from the New River southwards. The chain consists of several ranges generally parallel to each other, the altitude of the individual mountains increasing on approaching the south. The highest peaks rise over 6600 feet (not one at all approaching the snow-level), but the mean height is about 2500 feet. Lake Champlain is the only lake of great importance in the system, but numerous rivers of considerable size take their rise here. Magnetite, hematite, and other iron ores occur in great abundance, and the coal-measures are among the most extensive in the world. Gold, silver, lead, and copper are also found in small quantities, while marble, limestone, fire-clay, gypsum, and salt abound. The forests covering many of the ranges yield large quantities of valuable timber, such as sugar-maple, white birch, beech, ash, oak, cherry tree, white poplar, white and yellow pine, &c., while they form the haunts of large numbers of bears, panthers, wild cats, and wolves.

Appalachicola (-chi-cō'la), a river of the United States, formed by the Chattahoochee and Flint Rivers, which unite near the northern border of Florida; length, about 100 miles; it flows into the Gulf of Mexico, and is navigable.

Appam, the name of a British merchant ship of the Elder-Dempster line captured by the German raiding cruiser *Moewe* (Sea-gull) on 16th Jan., 1916. A German prize crew succeeded in bringing the *Appam* westward, and was able to pass the British cordon off Chesapeake Bay and to reach Norfolk, Virginia. The vessel was carrying, among others, an ex-governor of Sierra Leone and some military officers from the west coast of Africa, but the passengers were at once released and allowed to return to England.

Appanage. See *Apanage*.

Appa'rent, among mathematicians and astronomers, applied to things as they appear to the eye, in distinction to what they really are. Thus they speak of apparent motion, magnitude, distance, height, &c. The *apparent magnitude* of a heavenly body is the angle subtended at the spectator's eye by the diameter of that body, and this, of course, depends on the distance as well as the real magnitude of the body; *apparent motion* is the motion a body seems to have in consequence of our own motion, as the motion of the sun from east to west, &c.

Appari'tion, according to a belief held by some, a disembodied spirit manifesting itself to mortal sight; according to the common theory an illusion involuntarily generated, by means of which figures or forms, not present to the actual sense, are nevertheless depicted with a vividness and intensity sufficient to create a temporary belief in their reality. Such illusions are now generally held to result from an over-excited brain, a strong imagination, or some bodily malady. In perfect health the mind not only possesses a control over its powers, but the impressions of the external objects alone occupy its attention, and the play of imagination is consequently checked, except in sleep, when its operations are relatively more feeble and faint. But in an unhealthy state of the mind, when its attention is partly withdrawn from the contemplation of external objects, the impressions of its own creation, or rather reproduction, will either overpower or combine themselves with the impressions of external objects,

and thus generate illusions which in the one case appear alone, while in the other they are seen projected among those external objects to which the eyeball is directed. This theory explains satisfactorily a large majority of the stories of apparitions; still there are some which it seems insufficient to account for.—See *Crystal Gazing, Hypnotism, Spiritualism*.—BIBLIOGRAPHY: F. Podmore, *Modern Spiritualism*; F. W. H. Myers, *Human Personality, and its survival of bodily Death*.

Appeal, in legal phraseology, the removal of a cause from an inferior tribunal to a superior, in order that the latter may revise, and if it seem needful reverse or amend, the decision of the former. The supreme court of appeal for Great Britain is the House of Lords. Certain defects in connection with the settlement of appeals by this body were remedied by the Appellate Jurisdiction Act, 1876, while a new court of appeal was established as a division of the Supreme Court of Judicature. In Ireland there is also a Court of Appeal similar to that in England; while in Scotland the highest court is the Court of Session. From the decisions of the Indian and all colonial courts, and the courts of the Isle of Man and the Channel Islands, appeal may be made to the Privy Council. Appeals from the decisions of justices of a borough or county may be made to the quarter-sessions of either respectively, in cases of summary jurisdiction, or upon a point of law to divisional courts of the High Court of Justice, which was established at the same time as the Court of Appeal; from quarter-sessions, county and other inferior courts, to the High Court. In Scotland the Court of Session reviews the decisions of the county courts, there being an appeal from its decisions to the House of Lords.—In France the court of final appeal in all cases is the *cour de cassation*.—In the United States the system of appeals differs in different States.

Appearance in law is the first formal act incumbent on a defendant who intends to resist the claim in the writ or action served upon him. It consists usually in lodging in court a written notice stating simply that the defendant intends to dispute the claim. Failure to enter appearance within the prescribed time entails decree passing against the defendant in absence, but procedure exists in all courts for enabling such decrees to be recalled within a limited period. Appearance should be entered under protest if it is desired to dispute the jurisdiction of the court or the regularity of the writ.

Appendicitis, a disease which has become well known in recent times through the more accurate methods of diagnosis and the increased safety of surgical operation. It is caused by inflammation of the vermiform appendix, a narrow, hollow, worm-like body from 2 to 4 inches long, opening at one end into the large intestine and forming a cul-de-sac at the other. In appendicitis the inflammation begins in the appendix and frequently spreads to the neighbouring parts, causing inflammation of the cæcum, a condition known as perityphlitis. The most frequent cause of appendicitis is a hard piece of insufficiently-digested food becoming lodged in the appendix. Occasionally orange pips, grape stones, &c., are the cause, though not so often as is popularly supposed.

The symptoms are: abdominal pain (especially low down at the right side), fever, nausea, vomiting, constipation; these varying according to the intensity of the attack. Three types are recognized:

1. A mild type, when the symptoms subside in a few days and the patient soon *appears* to be in normal health.
2. A severe type, in which, if left alone, the appendix bursts into the abdominal cavity and death from general peritonitis results.
3. Another type, in which the inflammation in the appendix leads to the formation of a localized abscess, sometimes of great size.

The treatment for the severe and for the abscess-forming types is essentially immediate operation; while for the mild type operation may either be performed at once or after the attack has passed off. Anyone who has had one attack of appendicitis is liable to have it repeated in a much severer form, hence the advisability of having the appendix removed after the first attack, however slight. During an attack, prior to surgical interference, complete rest in bed is essential. Abdominal pain should be treated with frequent hot fomentations, and the diet should be reduced to small quantities of fluid.

Appenzell (a`p`pen-tsel), a Swiss canton, wholly enclosed by the canton of St. Gall; area, 162 sq. miles. It is divided into two independent portions or half-cantons, Outer-Rhoden, which is Protestant, and Inner-Rhoden, which is Catholic. It is an elevated district, traversed by branches of the Alps;

Mount Säntis in the centre being 8250 feet high. It is watered by the Sitter and by several smaller affluents of the Rhine. Glaciers occupy the higher valleys. Flax, hemp, grain, fruit, &c., are produced, but the wealth of Inner-Rhoden lies in its herds and flocks—that of Outer-Rhoden in its manufactures of embroidered muslins, gauzes, cambrics, and other cotton stuffs; also of silk goods and paper. The town of Appenzell (Ger. *Abtenzelle*, abbot's cell) is the capital of Inner-Rhoden, on the Sitter, with about 4300 inhabitants. Trogen is the capital of Outer-Rhoden, Herisau the largest town (pop. 11,000). Pop. Outer-Rhoden, 60,000; Inner-Rhoden, 15,000.

Apperception. See *Metaphysics*.

Ap'petite, in its widest sense, means the natural desire for gratification, either of the body or the mind; but is generally applied to the recurrent and intermittent desire for food. A healthy appetite is favoured by work, exercise, plain living, and cheerfulness; absence of this feeling, or defective appetite (*anorexia*), indicates diseased action of the stomach, or of the nervous system or circulation, or it may result from vicious habits. Depraved appetite (*pica*), or a desire for unnatural food, as chalk, ashes, dirt, soap, &c., depends often in the case of children on vicious tastes or habits; in grown-up persons it may be symptomatic of dyspepsia, pregnancy or chlorosis. Insatiable or canine appetite or voracity (*bulimia*) when it occurs in childhood is generally symptomatic of worms; in adults common causes are pregnancy, vicious habits, and indigestion caused by stomach complaints or gluttony, when the gnawing pains of disease are mistaken for hunger.

Ap'pian, a Roman historian of the second century after Christ, a native of Alexandria, was governor and manager of the imperial revenues under Hadrian, Trajan, and Antoninus Pius, in Rome. He compiled in Greek a Roman history, from the earliest times to those of Augustus, in twenty-four books, of which only eleven have come down to us. Appian's style is not attractive, but he gives us much valuable information.

Appia'ni, Andrea, a painter, born at Milan in 1754, died in 1817. As a fresco-painter he excelled every contemporary painter in Italy. He displayed his skill particularly in the cupola of Santa Maria di S. Celso at Milan, and in the paintings representing the legend of Cupid and Psyche prepared for

the walls and ceiling of the villa of the Archduke Ferdinand at Monza (1795). Napoleon appointed him royal court painter, and portraits of almost the whole of the imperial family were painted by him.

Appian Way, called *Regina Viarum*, the Queen of Roads: the oldest and most renowned Roman road, was constructed during the censorship of Appius Claudius Cæcus (313-310 B.C.). It was built with large square stones on a raised platform, and was made direct from the gates of Rome to Capua, in Campania. It was afterwards extended through Samnium and Apulia to Brundisium, the modern Brindisi. It was partially restored by Pius VI, and between 1850 and 1853 it was excavated by order of Pius IX as far as the eleventh milestone from Rome.

Appius Claudius, surnamed *Cæcus*, or the blind, a Roman patrician, elected censor 312 B.C., which office he held four years. While in this position he made every effort to weaken the power of the Plebs, and constructed the road and aqueduct named after him. He was subsequently twice consul, and once dictator. In his old age he became blind, but in 280 B.C. he made a famous speech in which he induced the senate to reject the terms of peace fixed by Pyrrhus. He is the earliest Roman writer of prose and verse whose name we know.

Appius Claudius Crassus, one of the Roman *decemvirs*, appointed 451 B.C. to draw up a new code of laws. He and his colleagues plotted to retain their power permanently, and at the expiry of their year of office refused to give up their authority. The people were incensed against them, and the following circumstances led to their overthrow. Appius Claudius had conceived an evil passion for Virginia, the daughter of Lucius Virginius, then absent with the army in the war with the Æqui and Sabines. At the instigation of Appius, Marcus Claudius, one of his clients, claimed Virginia as the daughter of one of his own female slaves, and the *decemvir*, acting as judge, decided that in the meantime she should remain in the custody of the claimant. Virginius, hastily summoned from the army, appeared with his daughter next day in the forum, and appealed to the people; but Appius Claudius again adjudged her to Marcus Claudius. Unable to rescue his daughter, the unhappy father stabbed her to the heart. The *decemvirs* were deposed by the indignant people 449 B.C., and Appius Claudius died in prison or was strangled.

Apple (*Pyrus Malus*), the fruit of a well-known tree of the nat. ord. Rosaceæ, or the tree itself. The apple belongs to the temperate regions of the globe, over which it is almost universally spread and cultivated. The tree attains a moderate height, with spreading branches; the leaf is ovate; and the flowers are produced from the wood of the former year; but more generally from very short shoots or spurs from wood of two years' growth. The original of all the varieties of the cultivated apple is the wild crab, which has a small and extremely sour fruit, and is a native of most of the countries of Europe. Apples have been used as food and cultivated for upwards of 4000 years, and were probably introduced into Britain by the Romans. The greater number of the varieties now grown have, however, been cultivated only within the last century or so. To the facility of multiplying varieties by grafting is to be ascribed the amazing extension of the sorts of apples, the number of varieties known being over 2000. Many of the more marked varieties are known by general names, as pippins, codlins, rennets, &c. The oldest apple in cultivation is a variety called 'the lady', which originated in Britain early in the seventeenth century. Apples for the table are characterized by a firm juicy pulp, a sweetish acid flavour, regular form, and beautiful colouring; those for cooking by the property of forming by the aid of heat into a pulpy mass of equal consistency, as also by their large size and keeping properties; apples for cider must have a considerable degree of astringency, with richness of juice. The propagation of apple trees is accomplished by seeds, cuttings, suckers, layers, budding, or grafting, the last being almost the universal practice. The tree thrives best in an open situation where it will receive the maximum amount of sunshine and protection from cold winds. The protection is particularly necessary in districts where cold winds and frosts prevail during the flowering season. The wood of the apple tree or the common crab is hard, close-grained, and often richly coloured, and is suitable for turning and cabinet work. The fermented juice (*verjuice*) of the crab is employed in cookery and medicine. Apples are largely imported into Great Britain from the Continent and the United States and Canada. The designation apple, with various modifying words, is applied to a number of fruits having nothing in common with the apple proper, as alligator-apple, love-apple, &c.—Cf. A. E. Wilkinson, *The Apple*.

Ap'pleby, county town of Westmorland, England, on the Eden, 28 miles S.S.E. of Carlisle. Disfranchised in 1832, it gave its name to a parliamentary division of the county until 1918. It has an old castle, the keep of which, called Cæsar's Tower, is still fairly well preserved. Pop. (1921), 1786.

Apple of discord, according to the story in Greek mythology, the golden apple thrown into an assembly of the gods by the goddess of discord (Eris) bearing the inscription 'for the fairest'. Aphrodītē (Venus), Hera (Juno), and Athēnē (Minerva) became competitors for it, and its adjudication to the first by Paris so inflamed the jealousy and hatred of Hera to all of the Trojan race (to which Paris belonged) that she did not cease her machinations till Troy was destroyed.

Apple of Sodom, a fruit described by old writers as externally of fair appearance, but turning to ashes when plucked; probably the fruit of *Solānum sodomēum*.

Ap'pleton, a city of Wisconsin, United States, 100 miles N.W. of Milwaukee by rail. It has many flour, paper, saw, and woollen mills, and other manufactories, and is the seat of a collegiate institute and of the Lawrence University. Pop. (1920), 19,561.

Appliqué, in needlework or metal-work, a design or feature having the appearance of being independently made and attached to the surface of the object it adorns. When the ornament is sunk into the body of the object it is called *inlay*.

Appoggiatura (a ' p-poj-a ' -tō'ra '), in music, a small additional note of embellishment preceding the note to which it is attached, and taking away from the principal note a portion of its time.

Appoint' ment, a term in English law signifying the exercise of some power, reserved in a conveyance or settlement, of burdening, selling, or otherwise disposing of the lands or property conveyed. Such a reserved power is termed a *power of appointment*.

Appomatt'ox Court-house, a village in Virginia, United States, 20 miles E. of Lynchburg. Here, on 9th April, 1865, General Lee surrendered to General Grant, and thus virtually concluded the American Civil War.

Apponyi, Albert, Count, Hungarian statesman, born in 1846. Leader of the Conservative National party, he joined the Liberal party in 1899, and in 1901 was elected President of the Chamber of Deputies. He was Minister of Education in 1906 and 1917, and again in 1918. In 1920 he was at the head of the delegation which came to Paris to settle the peace-terms with Hungary.

Apposi'tion, in grammar, the relation in which one or more nouns or substantive phrases or clauses stand to a noun or pronoun, which they explain or characterize without being predicated of it, and with which they agree in case; as Cicero, the *orator*, lived in the first century before Christ; the opinion, *that a severe winter is generally followed by a good summer*, is a vulgar error.

Apprai'ser, a person employed to value property, and duly licensed to do so by licence taken out every year. The valuation must be duly set down in writing, and there is a certain fixed scale of charges for the appraiser's services.

Apprehen'sion, the seizing of a person as a criminal whether taken in the act or on suspicion, and with or without a warrant, a warrant being necessary when the person apprehending is not present at the commission of the offence. See *Arrest*.

Appren'tice, one bound by indenture to serve some particular individual or company of individuals for a specified time, in order to be instructed in some art, science, or trade. In England a person under the age of twenty-one cannot bind himself apprentice, and accordingly the usual way is for a relation or friend to become a contracting party to the indenture, and engage for the faithful performance of the agreement. An infant cannot be bound apprentice by his friends without his own expressed consent. In Scotland a boy under fourteen or a girl under twelve years of age cannot become a party to an indenture without the concurrence of a parent or guardian; above that age they may enter into an indenture of themselves, and thereby become personally bound. An indenture is determinable by the consent of the parties to it, and also by the death, bankruptcy, or retirement from business of the master. *Parish apprentices* are bound out by the guardians of the poor to suitable persons, and in this case the consent of the apprentice

is not necessary. The system of apprenticing by indenture is now much less common than formerly.—Cf. R. A. Bray, *Boy Labour and Apprenticeship*.

Approach'es, in field-engineering, an old-fashioned name for what are now called 'communication trenches'.

Appropria'tion. See *Impropriation*.

Appro'ver (ap-prö'ver), in English law, any accomplice in a crime who is allowed by the judges of jail-delivery to become king's evidence, that is, to be examined in evidence against his accomplices, it being understood that the approver will himself be pardoned upon making a full and open confession.

Approxima'tion, a term used in mathematics to signify a continual approach to a quantity required, when no process is known for arriving at it exactly. Although, by such an approximation, the exact value of a quantity cannot be discovered, yet, in practice, it may be found sufficiently correct; thus the diagonal of a square, whose sides are represented by unity, is $\sqrt{2}$, the exact value of which quantity cannot be obtained; but its approximate value may be substituted in the nicest calculations.

Appuleius. See *Apuleius*.

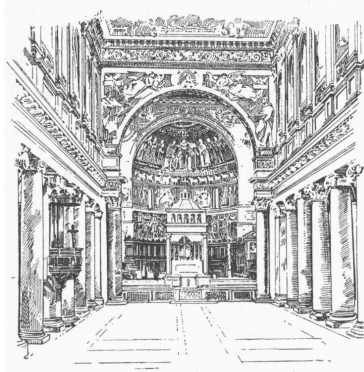
Ap'ricot (*Prunus Armeniäca*), a fruit of the plum genus which was introduced into Europe from Asia more than three centuries before Christ, and into England from Italy in 1524. It is a native of Armenia and other parts of Asia and also of Africa. The apricot is a low tree, of rather crooked growth, with somewhat heart-shaped leaves and sessile flowers. The fruit is sweet, more or less juicy, of a yellowish colour, about the size of a peach, and resembling it in delicacy of flavour. Some of the best varieties are 'Frogmore Early', 'Moorpark', 'Royal', &c. The wood is coarsely grained and soft. Apricot trees are chiefly raised against walls, and are propagated by budding and grafting.

Apries (ā'pri-ēz), Pharaoh-Hophra of Scripture, the eighth king of the twenty-sixth Egyptian dynasty. He succeeded his father Psammetichus in 590 or 589 B.C. The Jews under Zedekiah revolted against their Babylonian oppressors and allied themselves with Apries, who was, however, unable to raise the siege of Jerusalem, which was taken by Nebuchadnezzar. A still

more unfortunate expedition against Cyrene brought about revolt in his army, in endeavouring to suppress which Apries was defeated and slain about 570 B.C.

A´pril (Lat. *Aprīlis*, from *aperire*, to open, because the buds open at this time), the fourth month of the year. The strange custom of making fools on 1st April by sending people upon errands which end in disappointment, and raise a laugh at the expense of the person sent, prevails throughout Europe. It has been connected with the miracle plays of the Middle Ages, in which the Saviour was represented as having been sent, at this period of the year, from Annas to Caiaphas and from Pilate to Herod. This explanation, however, is perhaps itself a piece of April fooling. In France the party fooled is called *un poisson d'avril*, 'an April fish'; in Scotland, a 'gowk', or cuckoo.

A prio´ri ('from what goes before'), a phrase applied to a mode of reasoning by which we proceed from general principles or notions to particular cases, as opposed to *a posteriori* ('from what comes after') reasoning, by which we proceed from knowledge previously acquired. Mathematical proofs are of the *a priori* kind; the conclusions of experimental science are *a posteriori*. It is also a term applied to knowledge independent of all experience.



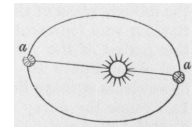
Apse—Church of Sta Maria
in Trastevere, Rome

Apse, a portion of any building forming a termination or projection semicircular or polygonal in plan, and having a roof forming externally a semi-dome or semi-cone, or having ridges corresponding to the angles of the polygon; especially such a semicircular or polygonal recess projecting

from the east end of the choir or chancel of a church, in which the altar is placed. The apse was developed from the somewhat similar part of the Roman basilicæ, in which the magistrate (*prætor*) sat.

Ap'sheron, a peninsula on the western shore of the Caspian Sea formed by the eastern extremity of the Caucasus Mountains. It extends for about 40 miles, and terminates in Cape Apsheron. It yields immense quantities of petroleum. See *Baku*.

Apsis, pl. **Ap'sides** or **Apsi'des**, in astronomy, one of the two points of the orbit of a heavenly body situated at the extremities of the major axis of the ellipse formed by the orbit, one of the points being that at which the body is at its greatest and the other that at which it is at its least distance from its primary. In regard to the earth and the other planets, these two points are called the aphelion and perihelion; and in regard to the moon they are called the apogee and perigee. The line of the apsides has a slow forward angular motion in the plane of the planet's orbit, being retrograde only in the case of Venus. This in the earth's orbit produces the anomalistic year. See *Anomaly*.

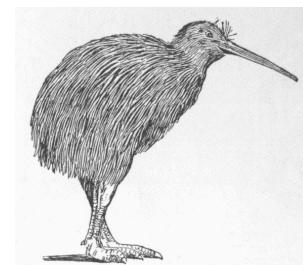


aa, Apsides

Apt (ät; ancient **Apta Julia**), a town of Southern France, department Vaucluse, 32 miles east by south of Avignon, with an ancient Gothic cathedral. Pop. 6336.

Ap'tera (Gr. *apteros*, wingless), wingless insects, such as lice and certain others, popularly called *Spring-tails*, and composed of two groups, Collembola and Thysanura.

Ap'teryx, a nearly extinct genus of cursorial birds, distinguished from the ostriches by having three toes with a rudimentary hallux, which forms a spur. They are natives of the South Island of New Zealand; are totally wingless and tailless, with feathers resembling hairs; about the size of a small goose; with long curved beak something like that of a curlew. They are entirely nocturnal, feeding on insects, worms, and seeds.—*A. austrālis*, called *Kiwi-kiwi* from its cry, is the best-known species.



Apteryx (*Apteryx Mantelli*)

Apuleius, or **Appuleius** (ap-ū-lē'us), author of the celebrated satirical romance in Latin called the *Golden Ass*, born at Madaura, in Numidia, about A.D. 125; the time of his death is unknown. He studied at Carthage, then at Athens, where he became warmly attached to the Platonic philosophy, and finally at Rome. Returning to Carthage he married a rich widow, whose relatives accused him of gaining her consent by magic, and the speech by which he successfully defended himself is still extant. Besides his *Golden Ass* (which is also known as the *Metamorphoses*, and which was translated into English by W. Adlington in 1566), with its fine episode of Cupid and Psyche, he was also the author of many works on philosophy and rhetoric, some of which are still extant.

Apu'lia, a department or division in the south-east of Italy, on the Adriatic, composed of the provinces of Foggia, Bari, and Lecce; area, 7376 sq. miles. Pop. 2,237,791.

Apure (a'-pö'rā), a navigable river of Venezuela, formed by the junction of several streams which rise in the Andes of Colombia; it falls into the Orinoco.—*Apure*, one of the States of Venezuela, has a pop. of 30,008.

Apurimac (a'-pö-rē-ma'k'), a river of South America, which rises in the Andes of Peru, and, being augmented by the Vilcamayu and other streams, forms the Ucayale, one of the principal head-waters of the Amazon.—The department of Apurimac in Peru has an area of 8187 sq. miles, and a pop. of 177,887.

Aq'ua (Lat. for water), a word much used in pharmacy and old chemistry.—*Aqua fortis* (= strong water), a weak and impure nitric acid. It has the power of eating into steel and copper, and hence is used by engravers, etchers, &c.—*Aqua marina*, a fine variety of beryl. See *Aquamarine*.—*Aqua regia*, or *aqua regalis* (= royal water), a mixture of nitric and hydrochloric acids, with the power of dissolving gold and other precious metals.—*Aqua Tofana*, a poisonous fluid made about the middle of the seventeenth century by an Italian woman Tofana or Toffania, who is said to have procured the death of no fewer than 600 individuals by means of it. It consisted chiefly, it is supposed, of a solution of crystallized arsenic.—*Aqua vitæ* (= water of life), or simply *aqua*, a name familiarly applied to the

whisky of Scotland, corresponding in meaning with the *usquebaugh* of Ireland, the *eau de vie* (brandy) of the French.

Aq´uamarine, a name given to some of the finest varieties of beryl of a sea-green or blue colour. Varieties of topaz are also so called.

Aqua´rium, a vessel or series of vessels constructed wholly or partly of glass and containing salt or fresh water in which are kept living specimens of marine or fresh-water animals along with aquatic plants. In principle the aquarium is based on the interdependence of animal and vegetable life; animals consuming oxygen and exhaling carbonic acid, plants reversing the process by absorbing carbonic acid and giving out oxygen. The aquarium must consequently be stocked both with plants and animals, and for the welfare of both something like a proper proportion should exist between them. The simplest form of aquarium is that of a glass vase; but aquaria on a larger scale consist of a tank or a number of tanks with plate-glass sides and stone floors, and contain sand and gravel, rocks, sea-weeds, &c. By improved arrangements light is admitted from above, passing through the water in the tanks and illuminating their contents, while the spectator is in comparative darkness. The most important aquarium is at the zoological station at Naples. There is also one, on a smaller scale, at Plymouth, maintained by the Marine Biological Association of the United Kingdom. Aquaria on a large scale have been constructed in connection with public parks or gardens, and the name is also given to places of public entertainment in which large aquaria are exhibited.—Cf. G. C. Bateman, *Fresh-water Aquaria*; M. J. Newbigin, *The Aquarium*.

Aquarius (Lat., the Water-bearer), a sign of the zodiac which the sun enters about the 21st of Jan.: it now enters the formerly coincident constellation Aquarius about a month later.

Aquatint, a method of etching on copper by which a beautiful effect is produced, resembling a fine drawing in sepia or Indian ink. The special character of the effect is the result of sprinkling finely-powdered resin or mastic over the plate, and causing this to adhere by heat, the design being previously etched, or being now traced out. The nitric acid (aqua fortis) acts only in the interstices between the particles of resin or mastic, thus giving a slightly granular appearance.

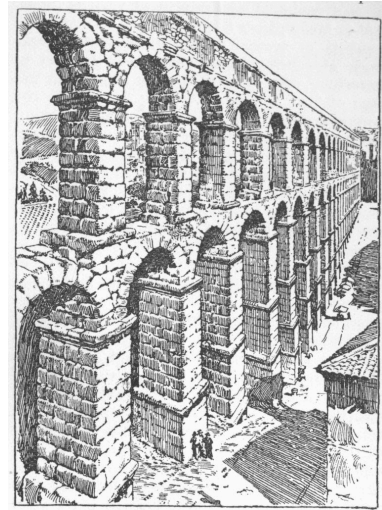
Aqua Tofa'na. See *Aqua*.

Aqua vitæ. See *Aqua*.

Aq'ueduct (Lat. *aqua*, water, *duco*, to lead), an artificial channel or conduit for the conveyance of water from one place to another: more particularly applied to structures for conveying water from distant sources for the supply of large cities. Aqueducts were extensively used by the Romans, and many of them still remain in different places on the Continent of Europe, some being still in use. The Pont du Gard in the south of France, 14 miles from Nîmes, is still nearly perfect, and is a grand monument of the Roman occupation of this country. The ancient aqueducts were constructed of stone or brick, sometimes tunnelled through hills, and carried over valleys and rivers on arches. The Pont du Gard spans the River Gard, and was built to convey to Nîmes the water of springs rising in the neighbourhood of the modern Uzès. It is built of great blocks of stone; its height is 160 feet; length of the highest arcade, 882 feet. The aqueduct at Segovia, originally built by the Romans, has in some parts two tiers of arcades 100 feet high, is 2921 feet in length, and is one of the most admired works of antiquity. One of the most remarkable aqueducts of modern times is that constructed by Louis XIV for conveying the waters of the Eure to Versailles. The extensive application of metal pipes has rendered the construction of aqueducts of the old type less necessary; but what may be called aqueduct bridges are still frequently constructed in connection with canals and also with water-works for the supply of towns. Where canals exist canal aqueducts are common, since the water in any section of a canal must be kept on a perfect level.

Many large towns now derive a supply of water from sources at a great distance, and in bringing the water to the place where it is required much tunnelling is often necessary as well as digging and excavating in the open. A tunnel furnishing a water channel may be driven through miles of rock strata of various kinds, and in many places it may have to be lined with concrete or cement wholly or partially, brick-work also being much employed. Instead of tunnelling, the channel may be formed on the plan of 'cut and cover', being first cut in the ground and then covered over, leaving the surface much in the same state as before. And, of course, iron piping is often used in connection with such tunnels, the water being conveyed so far in an aqueduct of one kind, and so far in one of another kind, according as

is deemed most suitable. In the Thirlmere aqueduct, which brings water to Manchester, there are 45 miles of cast-iron pipes, 37 miles of cut-and-cover work, and 14 miles of tunnels proper. Pipes are naturally laid where valleys occur, and the water simply enters the pipes at one end and flows out at the other by the influence of gravity, there being a suitable chamber constructed at either end of the pipe line where there is a junction with a section of tunnel. Aqueduct bridges were first introduced into England in the eighteenth century, the first being the aqueduct at Barton Bridge conveying the Bridgewater Canal across the Irwell. In such



Aqueduct at Segovia

bridges the water-channel may be made of cast iron. There are great aqueduct bridges on some of the Indian canals, such as the Nadrai bridge on the Lower Ganges Canal. In America water is often carried long distances in *flumes* or open wooden channels, supported, where necessary, on trestles. Great wooden pipes are also common there, built of large staves and hooped round with iron or steel. These often rest on the surface of the ground without any covering.—BIBLIOGRAPHY: C. Herschel, *Frontinus*; Wegmann, *Water-supply of City of New York*; J. F. Bateman, *The Manchester Waterworks*; J. M. Gale, *The Glasgow Waterworks*; A. Prescott Folwell, *Water Supply Engineering*.

Aq'ueous humour, the limpid watery fluid which fills the space between the cornea and the crystalline lens in the eye.

Aqueous rocks, composed of matter deposited by water from suspension or solution. Called also *sedimentary rocks*. See *Geology*.

Aquifolia'ceæ, a nat. ord. of plants; the holly tribe. The species consist of trees and shrubs, and the order includes the common holly (*Ilex Aquifolium*) and the *I. paraguayensis*, or Paraguayan tea tree.

Aquila (a`k`wē-la`), a town in Italy, capital of the province of Aquila, 55 miles north-east of Rome, the seat of a bishop, an attractive and interesting town with spacious streets and handsome palaces. In 1703 and 1706 it

suffered severely from earthquakes. Pop. 22,050.—The province has an area of 2493 sq. miles. Pop. 422,634.

Aq'ula, a companion of St. Paul (*Acts*, xviii, 2, 3). Expelled from Rome, he and his wife, Priscilla, settled in Corinth, where Paul stayed with them. They were converted to Christianity by the Apostle.

Aq'ula, a native of Pontus, flourished about A.D. 130. He became a Jewish proselyte, and made a close and accurate translation of the Hebrew Scriptures into Greek, extant only in fragments.

Aq'ula, name of a constellation in the northern hemisphere. See *Constellations*.

Aquila'ria. See *Aloes-wood*.

Aquile'gia. See *Columbine*.

Aquileia (ak-wi-lē'ya), an ancient city near the head of the Adriatic Sea, in Upper Italy, built by the Romans in 182 or 181 B.C. Commanding the N.E. entrance into Italy, it became important as a commercial centre and a military post, and was frequently the base of imperial campaigns. In 425 it was destroyed by Attila. The modern Aquileia or Aglar is a small place of some 1700 inhabitants, consisting chiefly of fishermen.

Aquinas (a-kwī'nas; i.e. of Aquino), St. Thomas, a celebrated scholastic divine, born in 1225 or 1227, most probably at the castle of Rocco Secca, near Aquino. His father was Count of Aquino, in the kingdom of Naples. He was educated at the Benedictine monastery of Monte Casino, and at the University of Naples, where he studied for six years. About the age of seventeen he entered a convent of Dominicans, much against the wishes of his family. He attended the lectures of Albertus Magnus at Cologne, in whose company he visited Paris in 1245 or 1246. Here he became involved in the dispute between the university and the Begging Friars as to the liberty of teaching, advocating the rights claimed by the latter with great energy. In 1257 he received the degree of doctor from the Sorbonne, and began to lecture on theology, rapidly acquiring the highest reputation. In 1263 he is found at the Chapter of the Dominicans in London. In 1268 he was in Italy, lecturing in Rome, Bologna, and elsewhere. In 1271 he was again in Paris lecturing to the students; in 1272 he was professor at Naples. In 1263 he

had been offered the archbishopric of Naples by Clement IV, but refused the offer. He died, in 1274, on his way to Lyons to attend a general council for the purpose of uniting the Greek and Latin Churches. He was called, after the fashion of the times, the *angelic doctor*, and was canonized by John XXII. The most important of his numerous works, which are all written in Latin, are the *Summa Theologica*, which, although only professing to treat of theology, is in reality a complete and systematic summary of the knowledge of the time, and the *Summa Philosophica*. The work of St. Thomas consisted in an effort to harmonize the new scientific teachings of the age—derived from Arabian and Byzantine sources—with the doctrine of the Church, and to refute heresy. His disciples were known as *Thomists*. See *Thomism*.—Cf. P. Conway, *St. Thomas Aquinas*; and article in *Hastings' Encyclopedia of Religion and Ethics*.

Aquita´nia, later **Aquitaine**, a Roman province in Gaul, which comprehended the countries on the coast from the Garonne to the Pyrenees, and from the sea to Toulouse. It was brought into connection with England by the marriage of Henry II with Eleanor, daughter of the last Duke of Aquitaine. The title to the province was for long disputed by England and France, but it was finally secured by the latter (1453).

Arabah´, a deep rocky valley or depression in north-western Arabia, between the Dead Sea and Gulf of Akabah, a sort of continuation of the Jordan valley.

Arabesque (ar´a-besk), a species of ornamentation for enriching flat surfaces, often consisting of fanciful figures, human or animal, combined with floral forms. There may be said to be three periods and distinctive varieties of arabesque—(a) the Roman or Græco-Roman, introduced into Rome from the East when pure art was declining; (b) the Arabesque of the Moors as seen in the Alhambra, introduced by them into Europe in the Middle Ages; (c) Modern Arabesque, which took its rise in Italy in the Renaissance period of art. The arabesques of the Moors, who are prohibited by their religion from representing animal forms, consist essentially of complicated ornamental designs based on the suggestion of plant-growth, combined with extremely complex geometrical forms.

Arabgir (a´-ra´b-gēr´), or **Arabkir´**, a town in Asia, 147 miles w.s.w. of Erzerum, noted for its manufacture of silk and cotton goods. Pop. between 20,000 and 30,000.

Ara´bia, a vast peninsula in the S.W. of Asia, bounded on the N. by the great Syro-Babylonian plain, N.E. by the Persian Gulf and the Sea of Oman, S. or S.E. by the Indian Ocean, and S.W. by the Red Sea and Gulf of Suez. Its length from N.W. to S.E. is about 1800 miles, its mean breadth about 600 miles, its area approximately 1,200,000 sq. miles, its population probably less than 5,000,000. Roughly described, it exhibits a central table-land surrounded by a series of deserts, with numerous scattered oases, while around this is a line of mountains parallel to and approaching the coasts, and with a narrow rim of low grounds (*tehāma*) between them and the sea. In its general features Arabia resembles the Sahara, of which it may be considered a continuation. Like the Sahara, it has its wastes of loose sand, its stretches of bare rocks and stones, its mountains devoid of vegetation, its oases with their wells and streams, their palm-groves and cultivated fields—*islands of green amidst the surrounding desolation*. Rivers proper there are none. By the ancients the whole peninsula was broadly divided into three great sections—Arabia Petræa (containing the city Petra), Deserta (desert), and Felix (happy). The first and last of these answer roughly to the modern divisions of the region of Sinai in the N.W. and Yemen in the S.W., while the name *Deserta* was vaguely given to the rest of the country. (See *Explorations, Modern*.) The principal divisions at the present are Madian in the north-west; south of this, Hejaz, Assir, and Yemen, all on the Red Sea, the last named occupying the south-western part of the peninsula, and comprising a *tehāma* or maritime lowland on the shores of the Red Sea, with an elevated inland district of considerable breadth; Hadramaut on the south coast; Oman occupying the south-east angle; El-Hasa and Koveït on the Persian Gulf; El-Hamad (Desert of Syria), Nefûd, and Jebel Shammar in the north; Nejd, the Central Highlands, which occupies a great part of the interior of the country, while south of it is the great unexplored Dahkna or sandy desert. Between 1902-5 a joint commission of British and Turkish officers laid down a boundary line defining the limits between Turkish territory and that of the independent Arab tribes in political relations with Great Britain. Nearly the whole of Southern Arabia came within the sphere of British influence. Madian belongs to Egypt; the Hejaz, Yemen, Bahr-el-

Hasa, Koveit, &c., were more or less under the suzerainty of Turkey until 1914. The rest of the country is ruled by independent chiefs—sheikhs, emirs, and imâms—while the title of sultan has been assumed by the chief of the Wahabis in Nejd, the sovereign of Oman (who has a subsidy from the Indian Government), and some petty princes in the south of the peninsula. On 9th June, 1916, the Grand Shereef of Mecca declared himself independent of the Turkish Government, and an Arab revolt spread rapidly. The Grand Shereef Hussein then announced to the Moslem world that the Shereefate of Mecca was henceforth independent, and on 4th Nov., 1916, he had himself formally proclaimed King, or Sultan, of Arabia. The status of the whole of Arabia was determined by the Peace Conference. (See *Hejaz, Mesopotamia, Syria, Sykes-Picot Treaty*.) The chief towns are Mecca, the birthplace of Mahomet; Medina, the place to which he fled from Mecca (A.D. 622), and where he is buried; Hodeida, a seaport exporting Mocha coffee; Aden, on the s.w. coast, belonging to Britain; Sana, the capital of Yemen; and Muscat, the capital of Oman. The chief towns of the interior are Hail, the residence of the Emir of Jebel Shammar; Oneizah, under the same ruler; and Rijadh, capital of Nejd and Hasa. The most flourishing portions of Arabia are in Oman, Hadramaut, and Nejd. In the two former are localities with numerous towns and villages and settled industrious populations like that of India or Europe.

The climate of Arabia in general is marked by extreme heat and dryness. Aridity and barrenness characterize both high and low grounds, and the date-palm is often the only representative of vegetable existence. There are districts which in the course of the year are hardly refreshed by a single shower of rain. Forests there are few or none. Grassy pastures have their place supplied by steppe-like tracts, which are covered for a short season with aromatic herbs, serving as food for cattle. The date-palm furnishes the staple article of food; the cereals are wheat, barley, maize, and millet; various sorts of fruit flourish; coffee and many aromatic plants and substances, such as gum-arabic, benzoin, mastic, balsam, aloes, myrrh, frankincense, &c., are produced. There are also cultivated in different parts of the peninsula, according to the soil and climate, beans, rice, lentils, tobacco, melons, saffron, colocynth, poppies, olives, &c. Sheep, goats, oxen, the horse, the camel, ass, and mule supply man's domestic and personal wants. Among wild animals are gazelles, ostriches, the lion,

panther, hyena, jackal, &c. Among mineral products are saltpetre, mineral pitch, petroleum, salt, sulphur, and several precious stones, as the carnelian, agate, and onyx. The people of Arabia, according to their own traditions, are derived from two stocks, the pure Arabs and the naturalized Arabs or Mustarab. They are leading either a settled agricultural life or a nomadic existence. In Southern Arabia the Jews form a large element in the towns' population. Commerce is largely in the hands of foreigners, among whom the Jews and Banians (Indian merchants) are the most numerous.

The history of Arabia previous to Mahomet is obscure. The earliest inhabitants are believed to have been of the Semitic race. Jews in great numbers migrated into Arabia after the destruction of Jerusalem, and, making numerous proselytes, indirectly favoured the introduction of the doctrines of Mahomet. With his advent the Arabians revolted and united for the purpose of extending the new creed; and under the caliphs—the successors of Mahomet—they attained great power, and founded large and powerful kingdoms in three continents. (See *Caliphs*.) On the fall of the caliphate of Bagdad in 1258 the decline set in, and on the expulsion of the Moors from Spain the foreign rule of the Arabs came to an end. In the sixteenth century Turkey subdued Hejaz and Yemen, and received the nominal submission of the tribes inhabiting the rest of Arabia. The allegiance of Hejaz was renounced early in the European War; but Yemen achieved its independence in the seventeenth century, and maintained it till 1871, when the territory again fell into the hands of the Turks. In 1839 Aden was occupied by the British. Oman early became virtually independent of the caliphs, and grew into a well-organized kingdom. In 1507 its capital, Maskat or Muscat, was occupied by the Portuguese, who were not driven out till 1659. The Wahabis appeared towards the end of the eighteenth century, and took an important part in the political affairs of Arabia, but their progress was interrupted by Mohammed Ali, pasha of Egypt, and they suffered a complete defeat by Ibrahim Pasha. He extended his power over most of the country, but the events of 1840 in Syria compelled him to renounce all claims to Arabia. The Hejaz thus again became subject to Turkish sway, and until 1914 Turkey continually extended its rule not only over Yemen, but also over the district of El-Hasa on the Persian Gulf.—BIBLIOGRAPHY: Sir R. F. Burton, *Pilgrimage to Medina and Mecca*; E. Reclus, *Les Arabes*; C. M. Doughty, *Arabia Deserta*,

and *Wanderings in Arabia*; G. W. Bury, *Arabia Infelix*; S. M. Zwemer, *Arabia, the Cradle of Islam*.

Arabian Language and Literature.—The Arabic language belongs to the Semitic dialects, among which it is distinguished for its richness, softness, and high degree of development. By the spread of Islam it became the sole written language and the prevailing speech in all South-Western Asia and Eastern and Northern Africa, and for a time in Southern Spain, in Malta, and in Sicily; and it is still used as a learned and sacred language wherever Islam is spread. Almost a third part of the Persian vocabulary consists of Arabic words, and there is the same proportion of Arabic in Turkish. The Arabic language is written in an alphabet of its own, which has also been adopted in writing Persian, Hindustani, Turkish, &c. As in all Semitic languages (except the Ethiopic), it is read from right to left. The vowels are usually omitted in Arabic manuscripts, only the consonants being written.

Poetry among the Arabs had a very early development, and before the time of Mahomet poetical contests were held and prizes awarded for the best pieces. The collection called the *Moallakât* contains seven pre-Mahommedan poems by seven authors. Many other poems belonging to the time before Mahomet, some of equal age with those of the *Moallakât*, are also preserved in collections. Mahomet gave a new direction to Arab literature. The rules of faith and life which he laid down were collected by Abu-Bekr, first caliph after his death, and published by Othman, the third caliph, and constitute the *Koran*—the Mahommedan Bible. The progress of the Arabs in literature, the arts and sciences, may be said to have begun with the government of the caliphs of the family of the Abbassides, A.D. 749, at Bagdad, several of whom, as Harun al Rashid and Al Mamun, were munificent patrons of learning: and their example was followed by the Omniades in Spain. In Spain were established numerous academies and schools, which were visited by students from other European countries; and important works were written on geography, history, philosophy, medicine, physics, mathematics, arithmetic, geometry, and astronomy. Most of the geography in the Middle Ages is the work of the Arabs, and their historians since the eighth century have been very numerous. The philosophy of the Arabs was of Greek origin, and derived principally from that of Aristotle. Numerous translations of the scientific works of Aristotle and other Greek philosophers were made, principally by Christian scholars who resided as

physicians at the Courts of the caliphs. These were diligently studied in Bagdad, Damascus, and Cordova, and, being translated into Latin, became known in the west of Europe. Of their philosophical authors the most celebrated are Alfarabi (tenth century), Ibn Sina or Avicenna (died A.D. 1037), Alghazzali (died 1111), Ibn Roshd or Averroes (twelfth century), called by pre-eminence The Commentator, &c. In medicine they excelled all other nations in the Middle Ages, and they are commonly regarded as the earliest experimenters in chemistry. Their mathematics and astronomy were based on the works of Greek writers, but the former they enriched, simplified, and extended. It was by them that algebra was introduced to the Western peoples, and the Arabic numerals were similarly introduced. Astronomy they especially cultivated, for which famous schools and observatories were erected at Bagdad and Cordova. The *Almagest* of Ptolemy in an Arabic translation was early a textbook among them. Alongside of science poetry continued to be cultivated, but after the ninth or tenth centuries it grew more and more artificial. Among poets were Abu Nowas, Asmai, Abu Temmam, Motenabbi, Abul-Ala, Busiri, Tograi, and Hariri. Tales and romances in prose and verse were written. The tales of fairies, genii, enchanters, and sorcerers in particular passed from the Arabians to the Western nations, as in *The Thousand and One Nights*. Some of the books most widely read in the Middle Ages, such as *The Seven Wise Masters*, the *Fables of Pilpay* (or Bidpai), and the *Romance of Antar* found their way into Europe through the instrumentality of the Arabs. At the present day Arabic literature is almost confined to the production of commentaries and scholia, discussions on points of dogma and jurisprudence, and grammatical works on the classical language. There are a few newspapers published in Arabic.—BIBLIOGRAPHY: C. Huart, *History of Arabic Literature*; R. A. Nicholson, *Literary History of the Arabs*.

Arabian Architecture. See *Moorish Architecture*, *Saracenic Architecture*.

Arabian Gulf. See *Red Sea*.

Arabian Nights, or **The Thousand and One Nights**, (Ar. Alf Layla wa-Layla), a celebrated collection of Eastern tales, based upon an old work, called *Hazar Afsana*, long current in the East, and supposed to have been derived by the Arabians from India, through the medium of Persia. They were first introduced into Europe in the beginning of the eighteenth century

by means of the French translation of Antoine Galland. Of some of them no original MS. is known to exist; they were taken down by Galland from the oral communication of a Syrian friend. The story which connects the tales of *The Thousand and One Nights* is as follows: The Sultan Shahriyar, exasperated by the faithlessness of his bride, made a law that every one of his future wives should be put to death the morning after marriage. At length one of them, Sheherazade, the generous daughter of the grand-vizier, succeeded in abolishing the cruel custom. By the charm of her stories the fair narrator induced the sultan to defer her execution every day till the dawn of another, by breaking off in the middle of an interesting tale which she had begun to relate. In the form we possess them these tales belong to a comparatively late period, though the exact date of their composition is not known. Lane, who published a translation of a number of the tales, with valuable notes, is of opinion that they took their present form some time between 1475 and 1525. Sir Richard Burton's complete English translation was issued in 16 vols. (1885-8).

Arabian Sea, the part of the Indian Ocean between Arabia and India.

Arabic Figures, the characters 1, 2, 3, 4, 5, 6, 7, 8, 9, 0; of Indian origin, introduced into Europe by the Moors. They did not come into general use till after the invention of printing.

Ara'bi Pasha, Egyptian soldier and revolutionary leader, born 1839. In Sept., 1881, he headed a military revolt, and was for a time virtually dictator of Egypt. Britain interfered, and after a short campaign, beginning with the bombardment of Alexandria and ending with the defeat of Arabi at Tel-el-Kebir, he surrendered, and was banished to Ceylon, being pardoned in 1900. He died in obscurity in Cairo in 1911.

Arable Land, land which is fit for ploughing, and capable of being cultivated, as distinguished from grass-land, wood-land, common pasture, mountains, forests, morasses, and waste. In Government returns the term is applied to land that is actually under regular cultivation. The land capable of being cultivated amounts in England and Wales to about 25 per cent, and in Ireland to about 13 per cent. In the course of the last thirty or forty years there has, however, been a considerable diminution in the area of land

actually cultivated, as a result of large foreign imports of grain and other agricultural products.

Arabs. The Arabs, as a race, are of middle stature, of a powerful though slender build, and have a skin of a more or less brownish colour; in towns and the uplands often almost white. Their features are well cut, the nose straight, the forehead high. They are naturally active, intelligent, and courteous; and their character is marked by temperance, bravery, and hospitality. The first religion of the Arabs, the worship of the stars, was supplanted by the doctrines of Mahommedanism, which succeeded rapidly in establishing itself throughout Arabia. Besides the two principal sects of Islam, the Sunnites and the Shiites, there also exists, in considerable numbers, a third Mahommedan sect, the Wahabis, which arose in the latter half of the eighteenth century, and for a time possessed great political importance in the peninsula. The mode of life of the Arabs is either nomadic or settled. The nomadic tribes are termed Bedouins (or Bedawins), and among them are considered to be the Arabs of the purest blood.

Aracacha, or **Arracacha** (ar-a-kä'cha), a genus of umbelliferous plants of Southern and Central America. The root of *A. esculenta* is divided into several lobes, each of which is about the size of a large carrot. These are boiled like potatoes and largely eaten in South America.

Aracan (ar-a-kan'), the most northern division of Lower Burmah, on the Bay of Bengal; chief town and seaport Akyab. It was ceded to the English in 1826, as a result of the first Burmese war.

Araçari (a' -ra' -sä'rē), native name of a genus of brilliant birds (*Pteroglossus*) closely allied to the toucans, but generally smaller; natives of the warm parts of South America.

Aracati (a' -ra' -ka' -tē'), a Brazilian river-port, State of Ceará, on the River Jaguaribe, about 10 miles from its mouth. Exports hides and cotton. Pop. about 10,000.

Ara'ceæ, a nat. ord. of monocotyledonous plants, mostly tropical, having the genus *Arum* as the type. Most of the species have tuberous roots abounding in starch, which forms a wholesome food after the acrid juice has been washed out. See *Arum*, *Caladium*, *Dumb-cane*.

Arachis (ar'a-kis), a genus of leguminous plants much cultivated in warm climates, and esteemed a valuable article of food. The most remarkable feature of the genus is that when the flower falls the stalk supporting the small undeveloped fruit lengthens, and bending towards the ground pushes the fruit into the ground, when it begins to enlarge and ripen. The pod of *A. hypogæa* (popularly called ground, earth, or pea nut) is of a pale-yellow colour, and contains two seeds the size of a hazel-nut, in flavour sweet as almonds, and yielding when pressed an excellent oil.

Arachnida (a-rak'ni-da; Gr. *arachnē*, a spider), a class of Arthropoda or higher Annulose animals including the Spiders, Scorpions, Mites, Ticks, &c. They have the body divided into a number of segments or *somites*, some of which have always articulated appendages (limbs, &c.). There is often a pair of nervous ganglia in each somite, although in some forms (as spiders) the nervous system becomes modified and concentrated. They are oviparous and somewhat resemble insects, but they have a united head and thorax, and do not undergo a metamorphosis similar to insects. They respire by tracheæ, by pulmonary sacs, or by the skin.

Ar'ack, or **Ar'rack**, a spirituous liquor manufactured in the East Indies from a great variety of substances. It is often distilled from fermented rice, or it may be distilled from the juice of the coco-nut and other palms. Pure arack is clear and transparent, of a yellowish or straw colour, and with a peculiar but agreeable taste and smell; it contains at least 52 to 54 per cent of alcohol.

Arad (o'rod), a town of the former kingdom of Hungary, on the Maros, 30 miles north of Temeswar, divided by the river into O (Old) Arad and Uj (New) Arad, connected by a bridge; it has a fortress, and is an important railway centre, with a large trade and manufactures. The town is now within the confines of Roumania, Uj Arad being called Arodul Neo. Population of Old and New Arad together, 63,166.

Ar'adus (now **Ruad**), an inlet about a mile in circumference lying 2 miles off the Syrian coast, 35 miles N. of Tripolis; the site of the Phœnician stronghold Arvad, a city second only to Tyre and Sidon; now occupied by about 3000 people, mainly fishermen.

Arafat', or **Jebel er Rahmeh** ('Mountain of Mercy'), a hill in Arabia, about 200 feet high, with stone steps reaching to the summit, 15 miles south-east of Mecca; one of the principal objects of pilgrimage among Mahommedans, who say that it was the place where Adam first received his wife Eve after they had been expelled from Paradise and separated from each other 120 years. A sermon delivered on the mount constitutes one great ceremony of the *Hajj*; or pilgrimage to Mecca, and entitles the hearer to the name and privileges of a *Hajji* or pilgrim.

Ar'ago, Dominique François, a French physicist, born in 1786, died at Paris in 1853. After studying in the Polytechnic School at Paris, he was appointed a secretary of the Bureau des Longitudes. In 1806 he was associated with Biot in completing in Spain the measurements of Delambre and Méchain to obtain an arc of the meridian. Before he got back to France he had been shipwrecked and narrowly escaped being enslaved at Algiers. In 1809 he was elected to the Academy of Sciences and appointed a professor at the Polytechnic School. He distinguished himself by his researches in the polarization of light, galvanism, magnetism, astronomy, &c. His discovery of the magnetic properties of substances devoid of iron, made known to the Academy of Sciences in 1824, procured him the Copley medal of the Royal Society of London in 1825. A further consideration of the same subject led to the equally remarkable discovery of the production of magnetism by electricity. He took part in the revolution of 1848, and held the office of Minister of War and Marine in the provisional Government. At the *coup d'état* of Dec., 1852, he refused to take the oath to the Government of Louis Napoleon, but the oath was not pressed. His works, which were posthumously collected and published, consist, besides his *Astronomie Populaire*, chiefly of contributions to learned societies, and biographical notices (*éloges*) of deceased members of the Academy of Sciences.

Arago, Emmanuel, son of Dominique François, French advocate and politician, was born at Paris in 1812; called to the bar 1837; took part in the revolution of 1848; renounced politics after the *coup d'état* of Dec., 1852, but continued to practise at the bar. After the fall of the Empire he again took a prominent part in public affairs, and held several important offices. He is author of a volume of poems and many theatrical pieces. He died in 1896.

Arago, Étienne, brother of Dominique Arago, born 1802, died 1892. He founded the journals *La Réforme* and *Le Figaro*; was director of the Théâtre du Vaudeville, 1829; took part in the revolution of 1848; was condemned to transportation, 1849; fled from France, but returned in 1859; was mayor of Paris during the Franco-Prussian war, and appointed archivist to the École des Beaux Arts, 1878. He was author of upwards of 100 dramas, *La Vie de Molière*, *Les Bleus et les Blancs*, and other works.

Aragon´, Kingdom of, a former province or kingdom of Spain, now divided into three provinces of Teruel, Huesca, and Saragossa; bounded on the N. by the Pyrenees, N.W. by Navarre, W. by Castile, S. by Valencia, and E. by Catalonia; length about 190 miles, average breadth 90 miles; area, 18,298 sq. miles. It was governed by its own monarchs until the union with Castile on the marriage of Ferdinand and Isabella (1469).

Arago´na, a town in Sicily, 8 miles N.N.E. of Girgenti. Pop. 16,000. In the neighbourhood is the mud volcano of Macculuba.

Aragonite, a mineral formed of calcium carbonate crystallized in the rhombic system; specific gravity 2.94 (compare *Calcite*). Aragonite passes into calcite in the course of geological time, but is important as the mineral precipitated to form the oolitic limestones of warm seas, and from being the material of most molluscan shells. It was first found in Aragon.

Araguaya (a´ -ra´ -gwĩ'a´), a Brazilian river, principal affluent of the Tocantins; rises about the 18th degree of S. lat.; in its course northwards forms the boundary between the two States of Matto Grosso and Goyaz, and falls into the Tocantins near lat. 6° S.; length, about 1300 miles, of which over 1000 are navigable.

A´ral, a salt-water lake in Asia, in Russian territory, about 150 miles W. of the Caspian Sea, between 43° 42´ and 46° 44´ N. lat., and 58° 18´ and 61° 46´ E. long.; length 270 miles, breadth 165; area, 26,650 sq. miles (or not much smaller than Scotland). It stands 240 feet above the level of the Caspian, and 160 feet above the Mediterranean. It receives the Amu Darya or Oxus and the Syr Darya or Jaxartes, and contains a multitude of sturgeon and other fish. It is encircled by desert sandy tracts, and its shores are without harbours. It has no outlet. The Aral contains a large number of small islands; steamers have been placed on it by the Russians.

Ara'lia, a genus of plants with small flowers arranged in umbels and succulent berries, the type of the nat. ord. Araliaceæ, which is nearly related to the Umbelliferæ, but the species are of a more shrubby habit. They are natives chiefly of tropical or sub-tropical countries, and in Britain are represented by the ivy; ginseng belongs to the order. From the pith of *A. papyrifera* is obtained the Chinese rice-paper.

A'ram, Eugene, a self-taught scholar whose unhappy fate has been made the subject of a ballad by Hood and a romance by Lord Lytton, was born in Yorkshire, 1704, executed for murder, 1759. In 1734 he set up a school at Knaresborough. About 1745 a shoemaker of that place, named Daniel Clarke, was suddenly missing under suspicious circumstances; and no light was thrown on the matter till full thirteen years afterwards, when an expression dropped by one Richard Houseman, respecting the discovery of a skeleton supposed to be Clarke's, caused him to be taken into custody. From his confession an order was issued for the apprehension of Aram, who had long quitted Yorkshire, and was at the time acting as usher at the grammar-school at Lynn. He was brought to trial on 3rd Aug., 1759, at York, where, notwithstanding an able and eloquent defence which he made before the court, he was convicted of the murder of Clarke, and sentenced to death. He was among the first to recognize the affinity of the Celtic to the other European languages, and under favourable circumstances might have done some valuable work in philological science.—Cf. W. Bristow, *The Genuine Account of the Life and Trial of Eugene Aram*.

Aramæ'an, or **Aramaic**. See *Semitic Languages*, *Syriac*.

Ar'an, an island lying off the W. coast of Donegal, Ireland, has an area of 4335 acres, a lighthouse, and a pop. of 1308, chiefly engaged in fishing.—Also called *North Island of Aran*, or *Arranmore*.

Arane'idæ, the spider family.

Aran Islands, or **South Islands of Aran**, three islands at the mouth of Galway Bay, off the W. coast of Ireland. The largest, Aranmore or Inishmore, comprises 7635 acres, and has a pop. of 2592; the next, Inishmaan, 2252 acres, pop. 473; and the least, Inishere, 1400 acres, pop. 456. They are remarkable for a number of architectural remains of a very early date. The inhabitants are chiefly engaged in agriculture and fishing.

Aranjuez (a´-ra´n-hu¨-eth´), a small town and palace in Spain, 30 miles from Madrid, with splendid gardens laid out by Philip II. The Court used to reside here from Easter till the close of June, when the number of people increased from 4000 to 20,000. It has a wireless station. Pop. 12,000.

Arany (o-ron´y), Janos, Hungarian poet, born 1817, died 1882. He was for some time a strolling player, but became professor of Latin at the Normal School of Szalonta, professor of Hungarian literature at Nagy Körös, and secretary of the Hungarian Academy. Author of *The Lost Constitution*, *Katalin*, and a series of three connected narrative poems on the fortunes of Toldi.

Arap´ahoes, a tribe of American Indians located near the head-waters of the Arkansas and Platte Rivers. They number in all about 2000.

Arapaima (a-ra-pĩ´ma), a genus of South American fresh-water fishes, ord. Physostomi, family Osteoglossidæ, one species of which (*A. gigas*) grows to the length of 15 or 16 feet, and forms a valuable article of food in Brazil and Guiana. It is covered with large bony scales, and has a bare and bony head.

Ar´arat, a celebrated mountain in Armenia, an isolated volcanic mass showing two separate cones known as the Great and Little Ararat, resting on a common base and separated by a deep intervening depression. The elevations are: Great Ararat, 16,916 feet; Little Ararat, 12,840 feet; the connecting ridge, 8780 feet. Vegetation extends to 14,200 feet, which marks the snow-line. According to the Bible Mount Ararat was the resting-place of the Ark when the waters of the Flood abated.

Araro´ba, or **Arraroba**, the powdered bark of *Andīra ararōba*. See *Andira*.

A´ras (the ancient **Araxes**), a river of Asia Minor, rising s. of Erzerum at the foot of the Bingol-dagh; it flows for some miles through South Caucasia, turning eastwards to the Erivan plain n. of Ararat. It then sweeps in a semi-circle mostly between Caucasia and Persia round to its confluence with the Kur, 60 miles from its mouth in the Caspian; length, 500 miles.

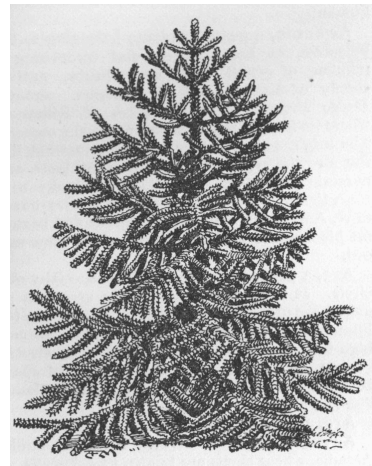
Ara´tus, a Greek poet, born at Soli in Cilicia; lived about 270 B.C.; was a favourite of Ptolemy Philadelphus. His poem *Phænomena* is a version of a

prose work on astronomy by Eudoxus; one verse of it is quoted by St. Paul in his address to the Athenians (*Acts*, xvii, 28).

Ara'tus of Sicyon, a statesman of ancient Greece, born 272 B.C. In 251 B.C. he overthrew the tyrant of Sicyon and joined that city to the Achæan League, which he greatly extended. He accepted the aid of Antigonos Doson, King of Macedon, against the Spartans, and became in time little more than the adviser of the Macedonian king, who had now made the League dependent on himself. He is said to have been poisoned by Philip V of Macedon, 213 B.C.

Arauca'nians, a South American native race in the southern part of Chile, occupying a territory stretching from about 37° to 40° of s. lat. They are warlike and more civilized than many of the native races of S. America, and maintained almost unceasing war with the Spaniards from 1537 to 1773, when their independence was recognized by Spain, though their territory was much curtailed. Their early contests with the Spaniards were celebrated in Ercilla's Spanish poem *Araucana*. With the Republic of Chile they were long at feud, and in 1861 had at their head a French adventurer named Antoine de Tounens, who claimed the title of king. In 1882 they submitted to Chile. The Chilian province of Arauco receives its name from them.

Arauca'ria, a genus of trees of the coniferous or pine order, indigenous to Australasia and South America. The species are large evergreen trees with pretty large, stiff, flattened, and generally imbricated leaves, verticillate spreading branches, and bearing large cones, each scale having a single large seed. The species *A. imbricāta* (the Chile pine or monkey-puzzle), with hard, sharp, pointed leaves, was introduced into Britain in 1796. It is a native of the mountains of Southern Chile, where it forms vast forests and yields a hard durable wood. Its seeds are eaten when roasted. The Moreton Bay pine of New South Wales (*A. Cunninghamii*) supplies a valuable timber used in house and boat building, in making furniture, and in other carpenter work. A species, *A. excelsa*, or Norfolk Island pine, abounds



Chile pine (*Araucaria imbricāta*)

in several of the South Sea Islands, where it attains a height of 220 feet with a circumference of 30 feet, and is described as one of the most beautiful of trees. Its foliage is light and graceful, and quite unlike that of *A. imbricata*, having nothing of its stiff formality. Its timber is of some value, being white, tough, and close-grained.

Arau'co, a province of Chile, named from the Araucanian Indians; area, 2189 sq. miles; pop. 73,260; capital, Lebu.

Araval'li Hills, a range of Indian mountains running N.E. and S.W. across the Rajputána country, which they separate into two natural divisions—desert plains on the N.W. and fertile lands on the S.E.; highest point, Mount Abu (5653 feet).

Araxes. See *Aras*.

Ar'băces, one of the generals of Sardanapălus, King of Assyria. He revolted and defeated his master, and became the founder of the Median Empire in 846 B.C.

Ar'balist. See *Cross-bow*.

Arbe'la (now **Erbil**), a place in the vilayet of Bagdad, giving name to the decisive battle fought by Alexander the Great against Darius, at Gaugamela, about 50 miles distant from it, 1st Oct., 331 B.C.

Arbitrage (a`r`bi-tra`zh), or **Arbitration of Exchanges**, an operation or calculation by which the currency of one country is converted into that of another through the medium of intervening currencies, for the purpose of ascertaining whether direct or indirect drafts and remittances are preferable. —*Arbitrageur* (a`r`bi-tra`-zheur) is one who makes calculations of currency exchanges. See *Stock Exchange*.

Arbitra'tion, the hearing and determination of a cause between parties in controversy, by a person or persons chosen by the parties. This may be done by one person, but it is common to choose more than one. Frequently two are nominated, one by each party, with a third, the *umpire* (or, in Scotland, sometimes the *oversman*), who is called on to decide in case of the primary arbitrators differing. In such a case the umpire may be agreed upon either by the parties themselves, or by the arbitrators when they have received

authority from the parties to the dispute to settle this point. The determination of arbitrators is called an *award*. By the law of England the authority of an arbitrator cannot be revoked by any of the parties without the leave of the court or of a judge.—BIBLIOGRAPHY: Russell, *Arbitration*; Redman, *Arbitration*; Scots Law, see Bell, *On Arbitration*; American Law, see Morse, *Law of Arbitration*; R. G. Morris, *International Arbitration*.

Ar'blast. See *Cross-bow*.

Arbo'ga, an old Swedish city, province of Westmannland; once an important commercial town, now only of historical interest from having been at one time a residence of the family of Vasa, the scene of Church assemblies and national diets, and for the antiquities in its neighbourhood. Pop. 5050.

Arbois (a' r-bwä), a town of France, department of Jura; famous for its wines. Pop. 5000.

Arbor Day, a day officially set apart in the United States for the annual planting of trees by the people, and especially by school-children. The custom was instituted in 1872.

Arbore'tum (Lat. *arbor*, a tree), a place in which a collection of different trees and shrubs is cultivated for scientific or educational purposes. The largest arboretum in Britain, perhaps the finest in the whole world, is that of the Royal Gardens, at Kew, inaugurated in 1762, to which 180 acres are now devoted. Next in celebrity are the arboreta at Edinburgh (Inverleith) and at Dublin (Glasnevin), the Botanical Gardens at Oxford, and the Botanic Gardens at Glasgow. Other arboreta are that of the Jardin des Plantes, Paris, and the Arnold Arboretum, at Jamaica Plain, Boston. The term arboretum has also been applied in a restricted sense, as in the *Arboretum Fruticetum Britannicum*, the monumental work by J. C. Loudon.

Ar'boriculture includes the culture of trees and shrubs, as well as all that pertains to the preparation of the soil, the sowing of the seeds, and the treatment of the plants in their young state, the preparation of the land previous to their final transplantation, their just adaptation to soil and situation, their relative growth and progress to maturity, their management during growth, and the proper season and period for felling them.

Arbor Vitæ (literally, 'tree of life'), the name of several coniferous trees of the genus *Thuja*, allied to the cypress, with flattened branchlets, and small imbricated or scale-like leaves. The name is derived from valuable medicinal properties having formerly been ascribed to the aromatic resin they mostly yield. Those generally cultivated in Britain are: the common Arbor Vitæ (*Thuja occidentālis*), a native of North America, where it grows to a height of 40 or 50 feet, introduced into Britain about 1566; the giant Arbor Vitæ or Red Cedar (*Thuja gigantea*), introduced in 1854; and the Chinese Arbor Vitæ (*Thuja orientālis*).

Arbroath (ar-brōth'), or **Aberbrothock**, a royal municipal and police burgh and seaport in the county of Forfar, Scotland, at the mouth of the small River Brothock. Its ancient abbey, founded by William the Lion in 1178, and dedicated to Saints Mary and Thomas à Becket, is now a picturesque ruin. There are numerous flax and hemp spinning-mills and factories, and much canvas and linen is made; also tanning, shoemaking, and fishing, and a small shipping trade, but the harbour is bad. Pop. 19,499. It unites with Montrose, Forfar, Brechin, and Inverbervie (the Montrose burghs) in sending a member to Parliament.

Arbuth'not, John, an eminent physician and distinguished wit, born at Arbuthnot, Kincardineshire, Scotland, 1667, died 1735. He received the degree of Doctor of Medicine at the University of St. Andrews, and went to London, where he soon distinguished himself by his writings and by his skill in his profession. In 1704 he was chosen Fellow of the Royal Society, and soon after he was appointed physician extraordinary, and then physician in ordinary to Queen Anne. About this time he became intimate with Swift, Pope, Gay, and other wits of the day. His writings, other than professional or scientific, include his contributions (in conjunction with Swift and Pope) to the *Memoirs of Martinus Scriblerus*, *History of John Bull*, *Art of Political Lying*, &c. He was conspicuous not only for learning and wit, but also for worth and humanity.

Ar'butus, a genus of plants belonging to the Ericaceæ, or heath order, and comprising a number of small trees and shrubs, natives chiefly of Europe and N. America. *Arbŭtus Unědo* abounds near the lakes of Killarney, where its fine foliage adds charms to the scenery. The bright red or yellow berries, somewhat like the strawberry, have an unpleasant taste and narcotic

properties. The Corsicans make wine from them. The trailing arbutus or may-flower of N. America, a plant with fragrant and beautiful blossoms, is *Epigæa repens*, of the same nat. ord.

Arc, a portion of a curved line, especially of a circle. It is by means of circular arcs that all angles are measured.—*Electric* or *Voltaic arc*, the luminous arc of intense brightness and excessively high temperature which is formed by an electric current in crossing over the interval of space between the carbon points of an electric lamp. See *Arc-light*.

Arc, Jeanne d'. See *Joan of Arc*.

Ar'ca, a genus of bivalve molluscs, family Arcadæ, whose shells are known as *ark-shells*.

Arcachon (a'r-ka'-shōn'), a town of s.w. France, department Gironde, on the almost landlocked basin of Arcachon, a much-frequented bathing-place, with great oyster-breeding establishments. It is connected by railway with Bordeaux. Pop. 10,266.

Arcade, a series of arches supported on piers or pillars, used generally as a screen and support of a roof, or of the wall of a building, and having beneath the covered part an ambulatory as round a cloister, or a footpath with shops or dwellings, as frequently seen in old Italian towns. Sometimes a porch or other prominent part of an important building is treated with arcades. At the present day Bologna, Padua, and Berne have fine examples of mediæval arcaded streets, and among more modern work various streets in Turin, and the Rue de Rivoli, Paris, are lined with arcades, with shops underneath. In mediæval architecture the term arcade is also applied to a series of arches supported on pillars forming an ornamental dressing or enrichment of a wall, a mode of treatment of very frequent occurrence in the towers, apses, and other parts of churches. In modern use the name arcade is often applied to a passage or narrow street containing shops arched over and covered with glass, as for example the Burlington Arcade, London, the Royal Arcade at Newcastle, and the Gallería Vittorio Emmanuele in Milan.

Arca'dia, the central and most mountainous portion of the Peloponnesus (Morea), the inhabitants of which in ancient times were celebrated for

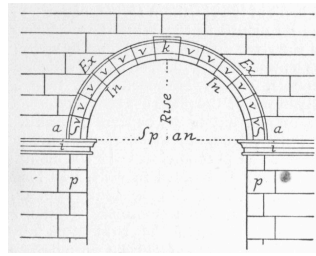
simplicity of character and manners. Their occupation was almost entirely pastoral, and thus the country came to be regarded as typical of rural simplicity and happiness. At the present day Arcadia forms a nomarchy of the kingdom of Greece. Area, 2028 sq. miles. Pop. 162,324.

Arca'dius, born in 377, died 408; son of the Emperor Theodosius, on whose death in 395 the empire was divided, he obtaining the East, and his brother Honorius the West. He proved a feeble and pusillanimous prince.

Arcanum, The Great (meaning secret), a term applied in the Middle Ages to the highest problems of alchemy and the discovery of the supposed great secrets of nature, such as the philosopher's stone and the elixir of life. See *Alchemy*.

Arcature, in architecture, a small arcade built into a wall or applied against it, decorative rather than structural. Arcatures occur in Anglo-Norman churches of the twelfth and thirteenth centuries.

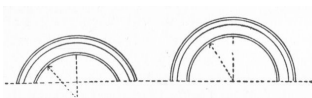
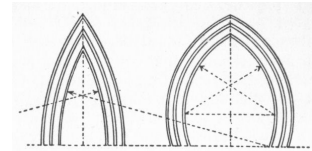
Arcesilaus (a`r-ses-i-lā`us), a Greek philosopher, the founder of the second or middle academy, was born about 315 B.C., died 239 B.C. He left no writings, and of his opinions so little is known that it has been doubted whether he was a strict Platonist or a sceptic.



Parts of an Arch
a. Abutments. *i.*
 Impost. *p.* Piers. *v.*
 Voussoirs or arch-
 stones. *k.* Keystone. *s.*
 Springers. *In.* Intrados.
Ex. Extrados.

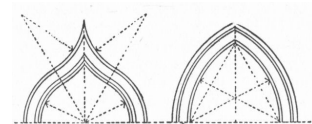
Arch, a structure composed of separate pieces, such as stones or bricks, having the shape of truncated wedges, arranged on a curved line, so as to retain their position by mutual pressure. The separate stones which compose the arch are called *voussoirs*

or *arch-stones*; the extreme or lowest voussoirs are termed *springers*, and the uppermost or central one is called the *keystone*. The under or concave side of the voussoirs is called the *intrados*, and the upper or convex side the *extrados* of the arch. The supports which afford resting and resisting points to the arch are called *piers* and *abutments*. The upper part of the pier or



abutment, where the arch rests—technically where it *springs from*—is the *impost*. The *span* of an arch is in circular arches the length of its chord, and generally the width between the points of its

opposite imposts whence it springs. The *rise* of an arch is the height of the highest point of its intrados above the line of the imposts; this point is sometimes called the *under side of the crown*, the highest point of the extrados being the *crown*. Arches are designated in various ways, as from their shape (circular, elliptic, &c.), or from the resemblance of the whole contour of the curve to some familiar object (lancet arch, horse-shoe arch), or from the method used in describing the curve, as equilateral, three-centred, four-centred, ogee, and the like; or from the style of architecture to which they belong, as Roman,



pointed, and Saracenic arches.—*Triumphal arch*, originally a simple decorated arch under which a victorious Roman general and army passed in triumph. At a later period the triumphal arch was a richly-sculptured, massive, and permanent structure, having an archway passing through it, with generally a smaller arch on either side. The name is sometimes given to an arch, generally of wood decorated with flowers or evergreens, erected on occasion of some public rejoicing, &c.

Archæan (är-kē'an) **Rocks** (Gr. *archaios*, ancient), the oldest rocks of the earth's crust, mostly crystalline in character, and embracing granites, gneisses, mica-schists, &c., all devoid of fossil remains. These rocks underlie a group of stratified and igneous masses that are usually distinguished from them as Huronian; the first beds with a well-marked fauna (lowest Cambrian) lie above the Huronian, and the Huronian and the Archæan groups are often conveniently classed together as pre-Cambrian, and are separated from the stratified and fossiliferous formations, which indeed have chiefly taken origin from them. The core of the Malvern range, and the rocks of N.W. Sutherland, are examples of Archæan masses in Great Britain.

Archæol'ogy (Gr. *archaios*, ancient, and *logos*, a discourse), the study of antiquity, or the science which takes cognizance of the history of nations and peoples as evinced by the remains, architectural, implemental, or otherwise, which belong to the earlier epoch of their existence. In a more extended sense the term embraces every branch of knowledge which bears on the origin, religion, laws, languages, science, arts, and literature of ancient peoples. It is to a great extent synonymous with *prehistoric annals*, as a large if not the principal part of its field of study extends over those periods in the history of the human race in regard to which we possess almost no information derivable from written records. Archæology divides the primeval period of the human race, more especially as exhibited by remains found in Europe, into the *stone*, the *bronze*, and the *iron* ages, these names being given in accordance with the materials employed for weapons, implements, &c., during the particular period. The *stone* age has been subdivided into the *palæolithic* and *neolithic*, the former being that older period, in which the stone implements were not polished as they are in the latter and more recent period. The *bronze* age, which admits of a similar subdivision, is that in which implements were of copper or bronze. In this

age the dead were burned and their ashes deposited in urns or stone chests, covered with conical mounds of earth or cairns of stones. Gold and amber ornaments appear in this age. The *iron* age is that in which implements, &c., of iron begin to appear, although stone and bronze implements are found along with them. The word *age* in this sense (as explained under *Age*) simply denotes the stage at which a people has arrived. The phrase stone age, therefore, merely marks the period before the use of bronze, the bronze age that before the employment of iron, among any specific people. See *Excavations; Crete; Egypt; &c*—BIBLIOGRAPHY: Sir J. Evans, *Stone Implements of Great Britain*; Boyd-Dawkins, *Early Man in Britain*; J. Geikie, *Prehistoric Europe*; R. Munro, *Lake Dwellings of Europe*; Sir W. Ridgeway, *Early Age of Greece*; H. R. Hall, *Ægean Archæology*; W. M. Flinders Petrie, *Methods and Aims in Archæology*; A. P. F. Michaelis, *A Century of Archæological Discoveries*.

Archæopteryx (är-kē-op'te-riks), a fossil bird from the oolitic limestone of Solenhofen, of the size of a rook, and differing from all known birds in having two free claws representing the thumb and forefinger projecting from the wing, and about twenty tail vertebræ free and prolonged as in mammals.



Archæopteryx macrura, a fossil lizard-tailed bird

Archangel (ärk'ān-jel; Gr. prefix, *arch-*, denoting chief), an angel of superior or of the highest rank. The only archangel mentioned by name in Scripture is Michael in the *Epistle of Jude*.

Archangel (ärk-ān'jel), a seaport, capital of the Russian government of same name, on the right bank of the Northern Dvina, about 20 miles above its mouth in the White Sea. Below the town the river divides into several branches and forms a number of islands, on one of which, called Sollenbole, is the harbour. The houses are mostly of wood; the place has some manufactures and an important trade, exporting linseed, flax, tow, tallow, train-oil, mats, timber, pitch and tar, &c. The port is closed for six months by ice. Archangel, founded in 1584, was long the only port which Russia possessed. Pop. 43,388.—The province, which before the Russian revolution extended from the Ural Mountains to Finland, had an area of

326,063 sq. miles. Pop. 483,500.—For the Archangel Expedition of 1918, see *Murmansk, Russia*.

Archangel'ica. See *Angelica*.

ARCHAEOLOGY: ANTIQUITIES OF THE STONE, BRONZE, AND IRON AGES



OLDER STONE AGE: 1, Flint Pick. 2, Carved Mammoth Tusk. 3, Double Scraper. 4, Barbed Harpoon Heads. LATER STONE AGE: 5, Pick of Deer Antler. 6, Flint and Pyrites. 7, Stone Celt in Haft. 8, Arrowhead. 9, Bowl. BRONZE AGE:

10, Celt. 11, Drinking-cup. 12, Ornamental Pin. 13, Spear-head. 14, Bronze Tweezers. 15, 16, Gold Bracelets. 17, Engraved Pin. 18, Short Sword. 19, Spectacle Brooch. 20, Razor. EARLY IRON AGE: 21, Bronze Brooch. 22, Bone Hand-comb for weaving. 23, Bronze Mirror. 24, Bronze Jug. 25, Bronze Spoon. 26, Iron Currency Bars. 27, Bronze Brooch.

Archbishop (ärch-), a chief bishop, or bishop over other bishops; a metropolitan prelate. The establishment of this dignity is to be traced up to an early period of Christianity, when the bishops and inferior clergy met in the capitals to deliberate on spiritual affairs, and the bishop of the city where the meeting was held presided. In England there are two archbishops—those of Canterbury and York; the former styled *Primate of all England*, the latter *Primate of England*. The Archbishop of Canterbury is the first peer of the realm, having precedence before all great officers of the Crown and all dukes not of royal birth. He crowns the sovereign, and when he is invested with his archbishopric he is said to be enthroned. He can grant special licences to marry at any time or place, and can confer degrees otherwise to be obtained only from the universities. He is addressed by the titles of *your grace* and *most reverend father in God*, and writes himself *by divine providence*, while the Archbishop of York and the bishops only write *by divine permission*. The first Archbishop of Canterbury was Augustine, appointed A.D. 598 by Ethelbert. Next in dignity is the Archbishop of York, between whom and the Archbishop of Canterbury the Lord High-Chancellor of England has his place in precedence. The first Archbishop of York was Paulinus, appointed in 622. The incomes of the sees are £15,000 and £10,000 respectively. An Archbishop of Wales was first appointed in 1920. Scotland had two archbishops—St. Andrews and Glasgow. Ireland had four, but the Episcopal Church has but two—Armagh and Dublin, the former being *Primate of all Ireland*, the latter *Primate of Ireland*. There are four Roman Catholic archbishops in England and Wales—Westminster, Cardiff, Birmingham, and Liverpool; two in Scotland—St. Andrews and Edinburgh, and Glasgow; four in Ireland—Armagh, Dublin, Cashel, and Tuam.

Archdeacon (ärch-), in England, an ecclesiastical dignitary next in rank below a bishop, having a certain jurisdiction over a part of the diocese. From two to four archdeacons are appointed by the bishops, under whom they perform their duties, and they hold courts which decide cases subject to an appeal to the bishop.

Archduke, a title peculiar to the royal family of Austria—the Habsburgs, who ruled until 1918.

Archelaus (a`r-kē-lā`us), the name of several personages in ancient history, one of whom was the son of Herod the Great. He received from Augustus the sovereignty of Judea, Samaria, and Idumea. The people, tired of his tyrannical and bloody reign, accused him before Augustus, who banished him to Gaul.

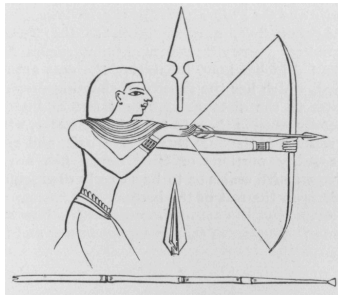
Archer, William, journalist and miscellaneous writer, born at Perth, Scotland, in 1856. Educated at Edinburgh University, he went to London after some experience of journalism at Edinburgh, and after a visit to Australia was called to the bar, and was dramatic critic for *The World* from 1884 to 1905. Subsequently he has been dramatic critic for *The Tribune* and *The Nation*. He has done much to introduce Ibsen to the English public, by translating his dramas and otherwise, and has written *English Dramatists of To-day*; *A Life of Macready*; *About the Theatre: Essays and Studies*; *Masks or Faces?: a Study on the Psychology of Acting*; *The Theatrical World* (a collection of his dramatic criticisms) (5 vols.); *Study and Stage*; *America To-Day* (the result of a visit in 1900); *Poets of the Younger Generation*; *Real Conversations* (the result of a series of interviews with persons of note); *Through Afro-America* (1910); *The Life, Trial, and Death of Francisco Ferrer* (1911); *Play-Making* (1912); *The Thirteen Days* (1915); *India and the Future* (1917); *War is War*.

Archer-fish, a name given to the *Toxotes jaculātor*, a scaly-finned, acanthopterygian fish, about 6 inches long, inhabiting the seas around Java, which has the faculty of shooting drops of water to the distance of 3 or 4 feet at insects, thereby causing them to fall into the water, when it seizes and devours them. The soft, and even the spiny portions of their dorsal fins are so covered with scales as to be scarcely distinguishable from the rest of the body.

Arch'ery, the art of shooting with a bow and arrow. The use of these weapons in war and the chase dates from the earliest antiquity. Ishmael, we learn from *Gen.* xxi, "became an archer". The Egyptians, Assyrians, Persians, Parthians, excelled in the use of the bow; and while the Greeks and Romans themselves made little use of it, they employed foreign archers as mercenaries. Coming to much more recent times, we find the Swiss famous as archers, but they generally used the arbalist or cross-bow, and were no match for their English rivals, who preferred the long-bow. (See *Bow*.) The English victories of Cressy, Poitiers, and Agincourt, gained against apparently overwhelming odds, may be ascribed to the bowmen. Archery disappeared gradually as firearms came into use, and as an instrument of war or the chase the bow is now confined to the most savage tribes of both hemispheres. But though the bow has been long abandoned among civilized nations as a military weapon, it is still cherished as an instrument of healthful recreation, encouraged by archery clubs or societies, which have been established in many parts of Britain. The oldest, and by far the most historically important of these societies, is the Royal Company of Archers, called also the King's Body-guard for Scotland, formed originally, it is said, by James I, but constituted in its present form by an Act of the Privy Council of Scotland, in 1676, and having its head-quarters in Edinburgh, counting among its members many of the nobility and gentry of the northern kingdom, and holding annual meetings, where prizes are competed for. In recent years a number of clubs have been formed in the United States. Archery has the merit of forming a sport open to women as well as men.—BIBLIOGRAPHY: William Garrard, *The Arte of Warre*; E. S. Morse, *Archery, Ancient and Modern*; H. A. Ford, *The Theory and Practice of Archery*.



Assyrian Archer



Egyptian Archer with
arrow-heads and stone-
tipped reed arrow

Arches, Court of, the chief and most ancient consistory court, belonging to the archbishopric of Canterbury, for the debating of spiritual causes. It is named from the church in London, St. Mary le Bow, or Bow Church (so called from a fine *arched* crypt), where it was formerly held. The jurisdiction of this court extends over the province of Canterbury. The office of president or dean is now merged in that of the judge appointed by the Public Worship Regulation Act (1874). The court now sits in the library of Lambeth Palace.

Archil, or **Orchil** (a´r´kil, or´kil), a red, violet, or purple colouring matter obtained from various kinds of lichens, the most important of which are the *Rocella tinctoria* and the *R. fuciformis*, natives of the rocks of the Canary and Cape Verde Islands, Mozambique and Zanzibar, South America, &c., and popularly called dyer's-moss. The dye is used for improving the tints of other dyes, as from its want of permanence it cannot be employed alone; but the aniline colours have largely superseded it. Cudbear and litmus are of similar origin.

Archilochus (a´r-kil´o-kus) **of Paros**, one of the earliest Ionian lyric poets, the first Greek poet who composed iambic verses according to fixed rules. He flourished about 700 B.C. His iambic poems were renowned for force of style, liveliness of metaphor, and a powerful but bitter spirit of satire. In other lyric poems of a graver character he was also considered as a model. All his works are lost but a few fragments.

Archiman´drite, in the Greek Church, an abbot or abbot-general, who has the superintendence of many abbots and convents. The title dates from the

fourth century.

Archimē'dean Screw, a machine for raising water, said to have been invented by Archimedes. It is formed by winding a tube spirally round a cylinder so as to have the form of a screw, or by hollowing out the cylinder itself into a double or triple-threaded screw and enclosing it in a water-tight case. When the screw is placed in an inclined position and the lower end immersed in water, by causing the screw to revolve, the water may be raised to a limited extent.

Archimedes (a ´ r-ki-mē´dēz), a celebrated ancient Greek physicist and geometrician, born at Syracuse, in Sicily, about 287 B.C. He devoted himself entirely to science, and enriched mathematics with discoveries of the highest importance, upon which the moderns have founded their admeasurements of curvilinear surfaces and solids. Archimedes is the only one among the ancients who has left us anything satisfactory on the theory of mechanics and on hydrostatics. He first taught the hydrostatic principle to which his name is attached, "that a body immersed in a fluid loses as much in weight as the weight of an equal volume of the fluid", and determined by means of it that an artist had fraudulently added too much alloy to a crown which King Hiero had ordered to be made of pure gold. He discovered the solution of this problem while bathing; and it is said to have caused him so much joy that he hastened home from the bath undressed, and crying out, *Eurēka! Eurēka!* 'I have found it, I have found it!' Practical mechanics also received a great deal of attention from Archimedes, who boasted that if he had a fulcrum or stand-point he could move the world. He is the inventor of the compound pulley, probably of the endless screw, the Archimedean screw, &c. During the siege of Syracuse by the Romans he is said to have constructed many wonderful machines with which he repelled their attacks, and he is stated to have set on fire their fleet by burning-glasses. At the moment when the Romans gained possession of the city by assault (212 B.C.), tradition relates that Archimedes was slain while sitting in the market-place contemplating some mathematical figures which he had drawn in the sand.

Archipel´ago, a term originally applied to the Ægean, the sea lying between Greece and Asia Minor, then to the numerous islands situated therein, and subsequently to any cluster of islands. In the Grecian

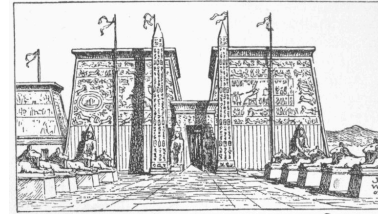
Archipelago the islands nearest the European coast lie together almost in a circle, and for this reason are called the *Cyclades* (Gr. *kyklos*, a circle); those nearest the Asiatic, being farther from one another, the *Sporades* ('scattered'). (See these articles, and *Negropont*, *Scio*, *Samos*, *Rhodes*, *Cyprus*, &c.) The Malay, Indian, or Eastern Archipelago, on the east of Asia, includes Borneo, Sumatra, and other large islands.

Architec'ture, in a general sense, is the art of designing and constructing houses, bridges, and other buildings for the purposes of civil life; or, in a more limited but very common sense, that branch of the fine arts which has for its object the production of edifices not only convenient for their special purpose, but characterized by unity, beauty, and often grandeur.—The first habitations of man were such as nature afforded, or cost little labour to the occupant—caves, huts, and tents. But as soon as men rose in civilization and formed settled societies they began to build more commodious and comfortable habitations. They bestowed more care on the materials, preparing bricks of clay or earth, which they at first dried in the air, but afterwards baked by fire; and subsequently they smoothed stones and joined them at first without, and at a later period with, mortar or cement. After they had learned to build houses, they erected temples for their gods on a larger and more splendid scale than their own dwellings. The Egyptians are the most ancient nation known to us among whom architecture had attained the character of a fine art. Other ancient peoples among whom it had made great progress were the Babylonians, whose most celebrated buildings were temples, palaces, and hanging gardens; the Assyrians, whose capital, Nineveh, was rich in splendid buildings; the Phœnicians, whose cities, Sidon, Tyre, &c., were adorned with equal magnificence; and the Israelites, whose temple was a wonder of architecture. But comparatively few architectural monuments of these nations have remained till our day.

This is not the case with the architecture of Egypt, however, of which we possess ample remains in the shape of pyramids, temples, sepulchres, obelisks, &c. Egyptian chronology is far from certain, but the greatest of the architectural monuments of the country, the pyramids of Ghizeh, are at least as old as 2800 or 2700 B.C. The Egyptian temples had walls of great thickness and sloping on the outside from bottom to top; the roofs were flat, and composed of blocks of stone reaching from one wall or column to another. The columns were numerous, close, and very thick, generally

without bases, and exhibiting great variety in the designs of their capitals. The principle of the arch, though known, was not employed for architectural purposes. Statues of enormous size, sphinxes carved in stone, and on the walls sculptures in outline of deities and animals, with innumerable hieroglyphics, are the decorative objects which belong to this style.

The earliest architectural remains of Greece are of unknown antiquity, and consist of massive walls built of huge blocks of stone. In historic times the Greeks developed an architecture of noble simplicity and dignity. The discoveries in Crete and Argolis have shown that Greek architecture owes much less than was supposed to Egyptian and Chaldæan architecture. It is considered to have attained its greatest perfection in the age of Pericles, or about 460-430 B.C. The great masters of this period were Phidias, Ictinus, Callicrates, &c. All the extant buildings are more or less in ruins. The style is characterized by beauty, harmony, and simplicity in the highest degree. Distinctive of it are what are called the *orders* of architecture, by which term are understood certain modes of proportioning and decorating the column and its superimposed entablature. The Greeks had three orders, called respectively the *Doric*, *Ionic*, and *Corinthian*. (See articles under these names.) Greek buildings were abundantly adorned with sculptures, and painting was extensively used, the details of the structures being enriched by different colours or tints. Lowness of roofs and the absence of arches were distinctive features of Greek architecture, in which, as in that of Egypt, horizontality of line is another characteristic mark. The most remarkable public edifices of the Greeks were temples, of which the most famous is the Parthenon at Athens. Others exist in various parts of Greece as well as in Sicily, Southern Italy, Asia Minor, &c., where important Greek communities were early settled. Their theatres were semicircular on one side and square on the other, the semicircular part being usually excavated in the side of some convenient hill. This part, the auditorium, was filled with concentric seats, and might be capable of containing 20,000 spectators. A number exist in Greece, Sicily, and Asia Minor, and elsewhere. By the end of the Peloponnesian War (c. 400 B.C.) the best period of Greek



Egyptian—Restoration of
Temple of Luxor

architecture was over; a noble simplicity had given place to excess of ornament. After the death of Alexander the Great (323 B.C.) the decline was still more marked.

Among the Romans there was no original development of architecture as among the Greeks, though they early took the foremost place in the construction of such works as aqueducts and sewers, the arch being in early and extensive use among this people. As a fine art, however, Roman architecture had its origin in copies of the Greek models, all the Grecian orders being introduced into Rome, and variously modified. Their number, moreover, was augmented by the addition of two new orders—the *Tuscan* and the *Composite*. The

Romans became acquainted with the architecture of the Greeks soon after 200 B.C., but it was not till about two centuries later that the architecture of Rome attained (under Augustus) its greatest perfection. Among the great works now erected were temples, aqueducts, amphitheatres, magnificent villas, triumphal arches, monumental pillars, &c. The *amphitheatre* differed from the theatre in being a completely circular or rather elliptical building, filled on all sides with ascending seats for spectators and leaving only the central space, called the *arena*, for the combatants and public shows. The Coliseum is a stupendous structure of this kind. The *thermæ*, or baths, were vast structures in which multitudes of people could bathe at once. Magnificent tombs were often built by the wealthy. Remains of private residences are numerous, and the excavations at Pompeii in particular have thrown great light on the internal arrangements of the Roman dwelling-house. Almost all the successors of Augustus embellished Rome more or less, erected splendid palaces and temples, and adorned, like Hadrian, even the conquered countries with them. But after the period of Hadrian (A.D. 117-138) Roman architecture is considered to have been on the decline. The refined and noble style of the Greeks was neglected, and there was an attempt to embellish the beautiful more and more. This decline was all the more rapid at a later time owing to the disturbed state of the Empire and the incursions of the barbarians.

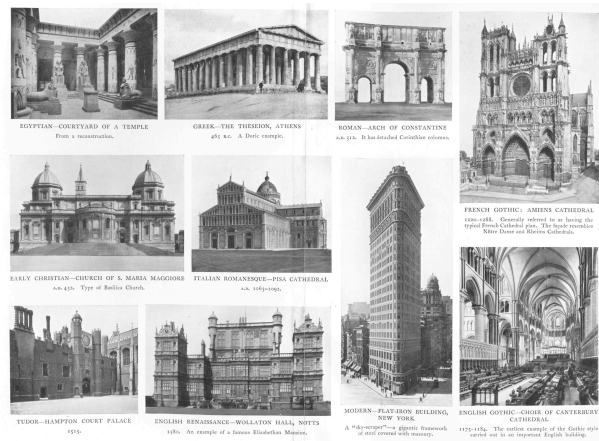


Byzantine—Church of St. Sophia, Constantinople

In Constantinople, after its virtual separation from the Western Empire, arose a style of art and architecture which was practised by the Greek Church during the whole of the Middle Ages. This is called the Byzantine style. The church of St. Sophia at Constantinople, built by Justinian (reigned 527-565), offers the most typical specimen of the style, of which the fundamental principle was an application of the Roman arch, the dome being the most striking feature of the building. In the most typical examples the dome or cupola rests on four pendentives.

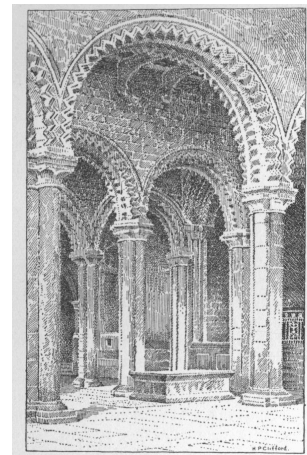
After the dismemberment of the Roman Empire the beautiful works of ancient architecture were almost entirely destroyed by the Goths, Vandals, and other barbarians in Italy, Greece, Asia, Spain, and Africa; or what was spared by them was ruined by the fanaticism of the Christians. A new style of architecture now arose, two forms of which, the Lombard and the Norman Romanesque, form important phases of art. The Lombard prevailed in North Italy and South Germany from the eighth or ninth to the thirteenth century (though the Lombard rule came to an end in 774); the Norman Romanesque flourished, especially in Normandy and England, from the eleventh to the middle of the thirteenth century. The semicircular arch is the most characteristic feature of this style. With the Lombard Romanesque were combined Byzantine features, and buildings in the pure Byzantine style were also erected in Italy, as the church of St. Mark at Venice.

ARCHITECTURE



The conquests of the Moors introduced a fresh style of architecture into Europe after the eighth century—the Moorish or Saracenic. This style accompanied the spread of Mahommedanism after its rise in Arabia in the seventh century. The edifices erected by the Moors and Saracens in Spain, Egypt, and Turkey are distinguished, among other things, by a peculiar form of the arch, which forms a curve constituting more than half a circle or ellipse. A peculiar flowery decoration, called *arabesque*, is a common ornament of this style, of which the building called the Alhambra (q.v.) is perhaps the chief glory.

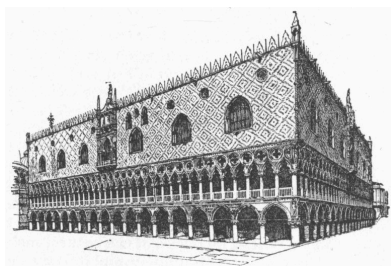
The Germans were unacquainted with architecture until the time of Charlemagne. He introduced into Germany the Byzantine and Romanesque styles. Afterwards the Moorish or Arabian style had some influence upon that of the Western nations, and thus originated the mixed style which maintained itself till the middle of the thirteenth century. Then began the modern Gothic style, which grew up in France, England, and Germany. Its striking characteristics are its pointed arches, its pinnacles and spires, its large buttresses, clustered pillars, vaulted roofs, profusion of ornament, and, on the whole, its lofty, bold character. Its most distinctive feature, as compared with the Greek or the Egyptian style, is the predominance in it of perpendicular or rising lines, producing forms that convey the idea of soaring or mounting upwards. Its greatest capabilities have been best displayed in ecclesiastical edifices. The Gothic style is divided into four principal epochs: the Early Pointed, or general style of the thirteenth century; the Decorated, or style of the fourteenth century; the Perpendicular, practised during the fifteenth and early part of the sixteenth centuries; and the Tudor, or general style of the sixteenth century. This style lasted in England up to the seventeenth century, being gradually displaced by that branch of the Renaissance or modified revival of ancient Roman architecture which is known as the *Elizabethan style*, and which is perhaps more purely an English style than any other that can be named.



Norman Romanesque—
Galilee Chapel, Durham
Cathedral

The rise of the Renaissance style in Italy is the greatest event in the history of architecture after the introduction of the Gothic style. The Gothic style

had been introduced into the country and extensively employed, but had never been thoroughly naturalized. The Renaissance is a revival of the classic style based on the study of the ancient models; and having practically commenced in Florence about the beginning of the fifteenth century, it soon spread with great rapidity over Italy and the greater part of Europe. The most illustrious architects of this early period of the style were Brunelleschi, who built at Florence the dome of the cathedral, the Pitti Palace, &c., besides many edifices at Milan, Pisa, Pesaro, and Mantua; Alberti, who wrote an important work on architecture, and erected many beautiful churches; Bramante, who began the building of St. Peter's, Rome, and Michael Angelo, who erected its magnificent dome. On St. Peter's were also employed Raphael, Peruzzi, and San Gallo. The noblest building in this style of architecture in Britain is St. Paul's, London, the work of Sir Christopher Wren.

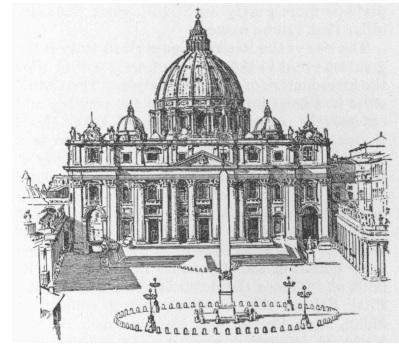


Italian Gothic—Doges' Palace,
Venice

Since the Renaissance period there has been no architectural development requiring special note. In buildings erected at the present day some one of the various styles of architecture is employed according to taste. Modern dwelling-houses have necessarily a style of their own as far as stories and apartments and windows and chimneys can give them one. In general the Grecian style, as handed down by Rome and modified by the Italian architects of the Renaissance, from its right angles and straight entablatures, is more convenient, and fits better with the distribution of our common buildings, than the pointed and irregular Gothic. But the occasional introduction of the Gothic outline and the partial employment of its ornaments has undoubtedly an agreeable effect both in public and private edifices; and we are indebted to it, among other things, for the spire, a structure exclusively Gothic, which, though often misplaced, has become an object of general approbation and a pleasing landmark to cities and villages. The works most characteristic of the present day are the large bridges, viaducts, &c., in many of which iron is the sole or most characteristic portion of the material.

A few words may be added on the architecture of India and China. Although many widely-differing styles are to be found in India, the oldest

and only true native style of Indian ecclesiastical architecture is the Buddhist, the earliest specimens dating from 250 B.C. Among the chief objects of Buddhist art are *stupas* or *topes*, built in the form of large towers, and employed as *dágobas* to contain relics of Buddha or of some noted saint. Other works of Buddhist art are temples or monasteries excavated from the solid rock, and supported by pillars of the natural rock left in their places.



Renaissance—St. Peter's, Rome

Buddhist architecture is found in Ceylon, Tibet, Java, &c., as well as in India. The most remarkable Hindu or Brahmanical temples are in Southern India. They are pyramidal in form, rising in a series of stories. The Saracenic or Mohammedan architecture afterwards introduced into India is, of course, of foreign origin. The Chinese have made the *tent* the elementary feature of their architecture; and of their style any one may form an idea by inspecting the figures which are depicted upon common chinaware. Chinese roofs are concave on the upper side, as if made of canvas instead of wood. (For further information see *Greek, Roman, Gothic, English, French, Russian Architecture*; and *Building, Fine Arts, Arch, Column, Aqueduct, Corinthian, Doric, Ionic, Theatre, &c.*)—BIBLIOGRAPHY: J. Ruskin, *Seven Lamps of Architecture*; E. A. Freeman, *History of Architecture*; Viollet-le-Duc, *How to build a House*; J. T. Micklethwaite, *Modern Parish Churches*; H. H. Statham, *Architecture for General Readers and Critical History of Architecture*; J. Fergusson, *History of Architecture in all Countries*; F. M. Simpson, *A History of Architectural Development*; *Cyclopedia of Architecture*.

Architrave (är'ki-träv), in architecture, the part of an entablature which rests immediately on the heads of the columns, being the lowest of its three principal divisions, the others being the *frieze* and the *cornice*.

Archives (är'kīvz). See *Records*.

Archivolt (är'ki-volt), in architecture, the ornamental band of mouldings on the face of an arch and following its contour.

Archons (är'konz), the chief magistrates of ancient Athens, chosen to superintend civil and religious concerns. They were nine in number; the first was properly the *archōn*, or *archōn epōnŷmos*, by whose name the year was distinguished in the public records; the second was called *archōn basileus*, or king archon, who exercised the functions of high priest; the third, *polemarchos*, or general of the forces. The other six were called *thesmothētai*, or legislators.

Archytas (a' r-kī'tas), an ancient Greek mathematician, statesman, and general, who flourished about 400 B.C., and belonged to Tarentum in Southern Italy. The invention of the analytic method in mathematics is ascribed to him, as well as the solution of many geometrical and mechanical problems. He constructed various machines and automata, among the most celebrated of which was his flying pigeon. He was a Pythagorean in philosophy, and Plato and Aristotle are said to have been both deeply indebted to him. Only inconsiderable fragments of his works are extant.

Arcis-sur-Aube (a' r-sē-su' r-ōb), a small town of France, department Aube, at which, in 1814, was fought a battle between Napoleon and the Allies, after which the latter marched to Paris. Pop. 3000.

Arc-light, a certain kind of electric light in which the illuminating source is the current of electricity passing between two sticks of carbon kept a short distance apart, one of them being in connection with the positive, the other with the negative terminal of a battery or dynamo.

Arco, a town of Tyrol, near Lake Garda, a favourite winter resort of invalids. Pop. 3800.

Arcole (a' r'ko-la'), a village in North Italy, 15 miles S.E. of Verona, celebrated for the battles of 15th, 16th, and 17th Nov., 1796, fought between the French under Bonaparte and the Austrians, in which the latter were defeated with great slaughter.

Arcos' de la Fronte'ra, a city of Spain, 30 miles E. by N. from Cadiz, on the Guadalete, here crossed by a stone bridge, on a sandstone rock 570 feet above the level of the river. On the highest part of the rock stands the castle of the dukes of Arcos, partly in ruins. The principal manufactures are leather, hats, and cordage. Pop. 13,980.

Ar'cot, two districts and a town of India, within the Presidency of Madras. —*North Arcot* is an inland district with an area of 7616 sq. miles. The country is partly flat and partly mountainous, where intersected by the Eastern Gháts. Pop. 2,200,000.—*South Arcot* lies on the Bay of Bengal, and has two seaports, Cuddalor and Porto Novo. Area 5217 sq. miles. Pop. 2,170,000.—The town *Arcot* is in North Arcot, on the Palar, about 70 miles w. by s. of Madras. There is a military cantonment at 3 miles' distance. The town contains handsome mosques, a nabob's palace in ruins, and the remains of an extensive fort. Arcot played an important part in the wars which resulted in the ascendancy of the British in India. It was taken by Clive, 31st Aug., 1751, and heroically defended by him against an apparently overwhelming force under Raja Sahib. Pop. 11,475.

Arctic (a'rk'tik), an epithet given to the north pole from the proximity of the constellation of the Bear, in Greek called *arktos*. The *Arctic Circle* is an imaginary circle on the globe, parallel to the equator, and $23^{\circ} 28'$ distant from the north pole. This and its opposite, the *Antarctic*, are called the two polar circles.

Arctic Expeditions. See *North Polar Expeditions*.

Arctic Ocean, that part of the water surface of the earth which surrounds the north pole, and washes the northern shores of Europe, Asia, and America; its southern boundary roughly coinciding with the Arctic Circle (lat. $66^{\circ} 30'$ N.). It encloses many large islands, and contains large bays and gulfs which deeply indent the northern shores of the three continents. Its great characteristic is ice, which is perpetual nearly everywhere.

Arctic Regions, the regions round the north pole, and extending from the pole on all sides to the Arctic Circle in lat. $66^{\circ} 30'$ N. The Arctic or North Polar Circle just touches the northern headlands of Iceland, cuts off the southern and narrowest portion of Greenland, crosses Fox's Strait north of Hudson's Bay, whence it goes over the American continent to Behring's Strait. Thence it runs to Obdorsk at the mouth of the Obi, then crossing Northern Russia, the White Sea, and the Scandinavian Peninsula, returns to Iceland. Though much skill and heroism have been displayed in the exploration of this portion of the earth, there is still an area round the pole estimated at 2,500,000 sq. miles, which is a blank to geographers. Many

have adopted the belief in the existence of an open polar sea about the north pole. But this belief is not supported by any positive evidence. Valuable minerals, fossils, &c., have been discovered within the Arctic regions. In the archipelago north of the American continent excellent coal frequently occurs. The mineral cryolite is mined in Greenland. Fossil ivory is obtained in islands at the mouth of the Lena. In Scandinavia, parts of Siberia, and north-west America, the forest region extends within the Arctic Circle. The most characteristic of the natives of the Arctic regions are the Esquimaux. The most notable animals are the white-bear, the musk-ox, the reindeer, and the whalebone whale. Fur-bearing animals are numerous. The most intense cold ever registered in those regions was 74° below zero F. The aurora borealis is a brilliant phenomenon of Arctic nights. See *North Polar Expeditions*.

Arc'tium. See *Burdock*.

Arc'tomys. See *Marmot*.

Arctu'rus, or **Alpha Boötis**, a fixed star of the first magnitude in the constellation of Boötes (the Ploughman), is one of the brightest stars in the northern heavens, yellow in colour. Its light is believed to be intrinsically at least 140 times as brilliant as the sun's, and to take over 40 years to reach us. It is notable as having a comparatively large proper motion.

Ardahan', a small fortified town about 6400 feet above the sea, between Kars and Batúm in Armenia. It was captured by the Russians in 1877, and ceded to them by the Berlin Treaty, 1878. It was handed over to Turkey by the Treaty of Brest-Litovsk in 1918, but the Turkish troops evacuated it after the armistice in 1918.

Ar'dea, the genus to which the heron belongs, type of the family Ardēidæ, which includes also cranes, storks, bitterns, &c.

Ar'debil, or **Ardabil**, a Persian town, province of Azerbaijan, near the Karasu, a tributary of the Aras, about 40 miles from the Caspian, in an elevated and healthy situation; it has mineral springs and a considerable trade. Pop. 16,000.

Ardèche (a'r-dāsh), a department in the south of France (Languedoc), on the west side of the Rhone, taking its name from the River Ardèche, which

rises within it, and falls into the Rhone after a course of 46 miles; area, 2144 sq. miles. It is generally of a mountainous character, and contains the culminating point of the Cevennes. Silk and wine are produced. Annonay is the principal town, but Privas is the capital. Pop. (1921), 294,308.

Arden, Forest of, a wood in Warwickshire. Shakespeare is supposed to have used it as a setting for *As You Like It*.

Ardennes (a`r-den´), an extensive tract of hilly land stretching over a large portion of the north-east of France and south-west of Belgium. Anciently the whole tract formed one immense forest (*Arduenna Silva* of Cæsar); but large portions are now occupied by cultivated fields and populous towns.

Ardennes (a`r-den´), a frontier department in the north-east of France; area, 2027 sq. miles, partly consisting of the Forest of Ardennes. There are extensive slate-quarries, numerous ironworks, and important manufactures of cloth, ironware, leather, glass, earthenware, &c. It was the scene of many battles during the European War (1914-8). Chief towns, Mézières (the capital), Rocroi, and Sedan. Pop. 277,791.

Ardnamurchan (-mur´han) **Point**, the most westerly point of the Island of Great Britain, in Argyllshire, having a lighthouse, 180 feet above sea-level, visible 18 to 20 miles off.

Ar´doch, a parish in south Perthshire, celebrated for its Roman remains, one, a camp, being the most perfect existing in Scotland.

Ardross´an, a seaport of Scotland, in Ayrshire, on the Firth of Clyde. It has a large harbour and shipbuilding yards, and is a centre of steamship services with Arran, Ireland, and Douglas, I.O.M. Pop. (1921), 7214.

Ards´ley, East and West, an urban district or town of England, W. Riding of Yorkshire, several miles north-west of Wakefield, with collieries, iron-mines, ironworks, &c. Pop. (1921), 7058.

Are (är), the unit of the French land measure, equal to 100 sq. metres, or 1076.44 English sq. feet. A *hectare* is 100 ares, equal to 2.47 English acres. The tenth part of an are is called a *déciare*, and a hundredth part a *centiare*.

A´rea, the superficial content of any figure or space, the quantity of surface it contains in terms of any unit. See *Mensuration*.

Are´ca, a genus of lofty palms with pinnated leaves, and a drupe-like fruit enclosed in a fibrous rind. *A. Catēchu* of the Coromandel and Malabar coasts is the common areca palm which yields areca or betel-nuts, and also the astringent juice catechu. *A. oleracēa* is the cabbage tree or cabbage palm of the West Indies. With lime and the leaves of the betel-pepper, the areca-nuts when green form the celebrated masticatory of the East. They are an important article in Eastern trade.

Arecibo (a´-re-thē´bō), a seaport town on the north coast of the Island of Porto Rico. Pop. 9612.

Areiopagus. See *Areopagus*.

Are´na, the enclosed space in the central part of the Roman amphitheatres, in which took place the combats of gladiators or wild beasts. It was usually covered with sand or saw-dust to prevent the gladiators from slipping, and to absorb the blood. See *Amphitheatre*.

Arenaceous Rocks include all sediments in which quartz sand is the most important constituent. Owing to its resistance to comminution and to chemical attack, quartz accumulates in sea-beaches while other mineral matter becomes removed. Hence sands gather near a shore and ultimately become consolidated by various natural cements into sandstones, those cemented by silica being styled quartzites. Sand-dunes in deserts or on coasts are unconsolidated arenaceous rock-masses.

Ar´endal a seaport of Southern Norway, exporting timber, wood pulp, and iron, and owning numerous ships. It is a well-built place, having been rebuilt since the great fire of 1868. Wood pulp, paper, and cotton are manufactured. Pop. 11,000.

Arenenberg Castle (mediæval, **Narrenberg**), a castle and estate in the Swiss Canton Thurgau, the possession of Queen Hortense, who died there in 1837. In 1855 it became the property of the Empress Eugénie.

Arenga, a term sometimes used as the generic name of the areng or gomuti palm, which is then botanically designated *Arenga saccharifera*. See

Gomuti.

Arenic'ola. See *Lobworm*.

Are'olar Tissue, an assemblage of fibres in bundles, pervading almost every part of the animal structure, and connected with each other so as to form innumerable small cavities, the whole serving as a means by which the various organs and parts of organs are connected together. It is called also *Cellular Tissue* and *Connective Tissue*. The fibres are of two kinds—white fibrous tissue and yellow elastic fibrous tissue, and interspersed among the bundles or occupying the cellular cavities are cells and corpuscles of several kinds. It is a tissue found in large quantities under the skin, covering the muscles, the blood-vessels, and nerves, and in various parts forming a kind of protective covering for delicate and important organs. It is because of its general distribution, and because of its binding various structures together, that it is called *connective*.—In botany the term is sometimes applied to the *non*-vascular substance, composed entirely of untransformed cells, which forms the soft substance of plants.

Areom'eter (from Gr. *araios*, thin, *metron*, a measure), an instrument for measuring the specific gravity of liquids; a *hydrometer* (q.v.).

Areop'agus, the oldest of the Athenian courts of justice, an assembly having a position more august than an ordinary court, and in its best days exercising a general supervision over public morals. It obtained its name from its place of meeting, on the Hill of Ares (Mars), near the Acropolis or citadel of Athens. Another explanation connects the word with *Aræ* (Curses), commonly known as *Semnæ* (Awful Goddesses), who were the guardians of the hill. It existed from very remote times, and the crimes tried before it were wilful murder, poisoning, robbery, and arson, while it had under its control also dissoluteness of morals, and innovations in the State and in religion. Its meetings were held in the open air, and its members were selected from those who had held the office of archon. The tribunal eventually lost many of its powers, but it continued to exist in name at least as late as the time of Cicero or later, having had an existence of seven or eight hundred years.

Arequipa (a'-rā-kē'pa'), a city of Peru, capital of a province of same name, situated in a fertile valley, 7850 feet above sea-level, at a distance of about

55 miles from the coast and on the railway which runs from its port Mollendo inland to Puno on Lake Titicaca. Behind the city rises the volcano of Arequipa, or Peak of Misti (20,328 feet). The climate is healthy but the locality is liable to earthquakes, one of which almost completely destroyed the town in 1868, after which it was rapidly rebuilt. A considerable trade is carried on through Mollendo, there being a large transit trade with the interior, and the town carries on various industries, manufacturing cotton and woollen goods, &c. It was founded in 1540. Pop. 35,000 to 40,000.—The province has an area of 21,947 sq. miles, and a pop. of 229,007.

Ares (ā'rēz). See *Mars*.

Arethu'sa, in Greek mythology, a daughter of Nereus and Doris, a nymph changed by Artēmis into a fountain in order to free her from the pursuit of the river-god Alpheus. This fountain was said to exist in the small Island of Ortygia, near Syracuse, and was fabled to have a subterranean connection with the River Alpheus in Greece.

Aretino (ä-rā-tē'nō), Guido. See *Guido*.

Aretino, Pietro, Italian poet, born at Arezzo 1492, died at Venice 1556, the natural son of a nobleman called Luigi Bacci. He early displayed a talent for satirical poetry, and when still a young man was banished from Arezzo on account of a sonnet against indulgences. He went to Perugia, and thence to Rome (1517), where he secured the papal patronage, but subsequently lost it through writing licentious sonnets. Through the influence of the Medici family he found an opportunity to insinuate himself into the favour of Francis I. In 1527 Aretino went to Venice, where he acquired powerful friends, among them the Bishop of Vicenza. By his devotional writings he regained the favour of the Roman Court. He called himself 'the divine', and 'the scourge of princes', but he was also their abject flatterer, and that solely to obtain money. The obscenity of some of his writings was such that his name has become proverbial for licentiousness. Among them are five comedies and a tragedy.

Arezzo (a'-ret'sō; ancient **Arretium**), a city of Central Italy, capital of a province of the same name in Tuscany, near the confluence of the Chiana with the Arno. It has a noble cathedral, containing some fine pictures and monuments; remains of an ancient amphitheatre, &c. It was one of the

twelve chief Etruscan towns, and in later times fought long against the Florentines, to whom it had finally to succumb. It is the birthplace of Mæcenas, Petrarch, Pietro Aretino, Redi, and Vasari. Pop. 50,093.—The province of Arezzo contains 1274 sq. miles, and 292,763 inhabitants (1915).

Ar'gal, **Argol**, or **Tartar**, a hard crust formed on the sides of vessels in which wine has been kept, red or white according to the colour of the wine. It is an impure bitartrate of potassium.

Ar'gali, a species of wild sheep (*Caprŏvis Argăli* or *Ovis ammon*) found on the mountains of Siberia, Central Asia, and Kamtchatka. It is 4 feet high at the shoulders, and proportionately stout in its build, with horns nearly 4 feet in length measured along the curve, and at their base about 19 inches in circumference. It lives in small herds. This true argali must not be confounded with the North-African wild sheep, called the bearded argali and known as the arni, the Algerian moufflon, and the Barbary sheep.



Argali (*Ovis ammon*)

Ar'gall, Sir Samuel (1572-1639), one of the early English adventurers to Virginia. He planned and executed the abduction of Pocahontas, the daughter of the Indian chief Powhattan, in order to secure the ransom of English prisoners. He was deputy-governor of Virginia (1617-9), and was accused of many acts of rapacity and tyranny. In 1620 he served in an expedition against Algiers, and was knighted by James I.

Ar'gand Lamp, a lamp named after its inventor, Aimé Argand, a Swiss chemist and physician (born 1755, died 1803), the distinctive feature of which is a burner forming a ring or hollow cylinder covered by a chimney, so that the flame receives a current of air both on the inside and on the outside.

Argaum (a' r-ga' u'm), a village of India, in Berar, celebrated for the victory of General Wellesley (Duke of Wellington) over the Mahrattas under Scindia and the Rajah of Berar, 29th Nov., 1803.

Ar'gelander, Friedrich Wilhelm August, German astronomer, born in 1799. He added to the knowledge of the progressive motion of the solar system in space, and published a catalogue of 560 stars having 'proper motion'. His works include: *Atlas des nördlichen gestirnten Himmels* (1857), *Neue Uranometrie* (1843), &c. He died in 1875.

Argemone (a`r-jem´o-nē), a small genus of ornamental American plants of the poppy order. From the seeds of *A. mexicāna* is obtained an oil very useful to painters. The handsomest species is *A. grandiflōra*, which has large flowers of a pure white colour.

Argensola (a`r-hen-sō´la`), Lupercio and Bartolomé Leonardo de, brothers, born at Barbastro, in Aragon, the former in 1565, died in 1613; the latter born in 1566, died in 1631. Lupercio produced tragedies and lyric poems; Bartolomé a number of poems and a historical work, *The Conquest of the Moluccas*. Their writings are singularly alike in character, and are reckoned among the Spanish classics. The tragedies are of the heavy Senecan type, but the satirical writings of both brothers are full of pungent wit of a pleasing kind.

Argenson (a`r-zha`n´-sōn´), Marc Pierre de Voyer, Comte d', celebrated French statesman, born in 1696, died 1764. After holding a number of subordinate offices he became minister for foreign affairs, and succeeded in bringing about the Congress of Breda, which was the prelude to that of Aix-la-Chapelle. He was present at the battle of Fontenoy, and was exiled to his estate for some years through the machinations of Madame Pompadour. His *Considérations sur le Gouvernement de la France* was a very advanced study on the possibility of combining with a monarchic form of government democratic principles and local self-government. *Les Essais, ou Loisirs d'un Ministre d'État*, published in 1785, is a collection of characters and anecdotes in the style of Montaigne.

Ar'gent, in coats of arms, the heraldic term expressing silver: represented in engraving by a plain white surface.

Argentan (a`r-zha`n´-tän´), a French town, department of Orne (Normandy), with an old castle and some manufactures. Pop. 6300.

Argenteuil (a'r-zha'n'-teu-yè), a town in France, department Seine-et-Oise, 7 miles below Paris; it has an active trade in wine, fruit, and vegetables. Pop. 24,282.

Argentie'ra, or **Kimōli** (ancient, **Cimōlus**), an island in the Grecian Archipelago, one of the Cyclades, about 18 miles in circumference, rocky and sterile. It produces a detergent chalk called *Cimolian earth* (q.v.), used in washing and bleaching. Pop. 1337.

Ar'gentine, a silvery-white slaty variety of calc-spar, containing a little silica with laminæ usually undulated. It is found in primitive rocks and frequently in metallic veins.—Argentine is also the name of a small British fish (*Scopēlus boreālis*) less than 2 inches long and of a silvery colour.

Ar'gentine Republic, formerly called the United Provinces of **La Plata**, a vast country of South America, the extreme length of which is 2300 miles, and the average breadth a little over 500 miles, the total area 1,153,119 sq. miles. It consists of fourteen provinces, ten territories, and one federal district. It is bounded on the N. by Bolivia; on the E. by Paraguay, Brazil, Uruguay, and the Atlantic; on the S. by the Antarctic Ocean; and on the W. by the Andes. It comprises four great natural divisions: (1) the Andine region, containing the provinces of Mendoza, San Juan, Rioja, Catamarca, Tucuman, Salta, and Jujuy; (2) the Pampas, containing the provinces of Santiago, Santa Fé, Cordova, San Luis, and Buenos Ayres, with the territories Formosa, Pampa, and Chaco; (3) the Argentine 'mesopotamia', between the Rivers Paraná and Uruguay, containing the provinces of Entre Rios and Corrientes, and the territory Misiones; (4) Patagonia, including the eastern half of Tierra del Fuego. With the exception of the N.W., where lateral branches of the Andes run into the plain for 150 or 200 miles, and the province of Entre Rios, which is hilly, the characteristic feature of the country is the great monotonous and level plains called 'pampas'. In the north these plains are partly forest-covered, but all the central and southern parts present vast treeless tracts, which afford pasture to immense herds of horses, oxen, and sheep, and are varied in some places by brackish swamps, in others by salt steppes. The great water-course of the country is the Paraná, having a length of fully 2000 miles from its source in the mountains of Goyaz, Brazil, to its junction with the Uruguay, where begins the estuary of La Plata. The Paraná is formed by the union of the Upper Paraná and

Paraguay Rivers, near the N.E. corner of the State. Important tributaries are the Pilcomayo, the Vermejo, and the Salado. The Paraná, Paraguay, and Uruguay are valuable for internal navigation. Many of the streams which tend eastward terminate in marshes and salt lakes, some of which are rather extensive. Not connected with the La Plata system are the Colorado and the Rio Negro, the latter formerly the southern boundary of the State, separating it from Patagonia. The source of the Negro is Lake Nahuel Huapi, in Patagonia (area, 1200 sq. miles), in the midst of magnificent scenery. The level portions of the country are mostly of tertiary formation, and the river and coast regions consist mainly of alluvial soil of great fertility. In the pampas clay have been found the fossil remains of extinct mammalia, some of them of colossal size.

European grains and fruits, including the vine, have been successfully introduced, and large areas are now under wheat, maize, flax, and other crops, another source of wealth consisting in the countless herds of cattle and horses and flocks of sheep, which are pastured on the pampas, and which multiply there very rapidly. Gold, silver, nickel, copper, tin, lead, and iron, besides marble, jasper, precious stones, and bitumen, are found in the mountainous districts of the N.W., while petroleum wells have been discovered on the Rio Vermejo; but the development of this mineral wealth has hitherto been greatly retarded by the want of proper means of transport. As a whole there are not extensive forests in the State except in the region of the Gran Chaco (which extends also into Bolivia), where there is known to be 60,000 sq. miles of timber. Thousands of square miles are covered with thistles, which grow to a great height in their season. Cacti also forms great thickets. Peach and apple trees are abundant in some districts. The native fauna includes the puma, the jaguar, the tapir, the llama, the alpaca, the vicuña, armadillos, the rhea or nandu, a species of ostrich, &c. The climate is agreeable and healthy, 97° being about the highest temperature experienced. The rainfall is very scanty in some districts, and is nowhere very large.

As a whole this vast country is very thinly inhabited, some parts of it as yet being very little known. The native Indians were never very numerous, and have given little trouble to the European settlers. Tribes of them yet in the savage state still inhabit the less-known districts, and live by hunting and fishing. Some of the Gran Chaco tribes are said to be very fierce, and

European travellers have been killed by them. The European element is strong in the republic, more than half the population being Europeans or of pure European descent. Large numbers of immigrants arrive from Southern Europe, the Italians having the preponderance among those of foreign birth. The typical inhabitants of the pampas are the *Gauchos*, a race of half-breed cattle-rearers and horse-breakers; they are almost continually on horseback, galloping over the plains, collecting their herds and droves, taming wild horses, or catching and slaughtering cattle. In such occupations they require a marvellous dexterity in the use of the lasso and bolas.

The River La Plata was discovered in 1512 by the Spanish navigator Juan Diaz de Solis, and the La Plata territory had been brought into the possession of Spain by the end of the sixteenth century. In 1810 the territory cast off the Spanish rule, and in 1816 the independence of the United States of the Rio de la Plata was formerly declared, but it was long before a settled government was established. The present constitution dates from 1853, being modified in 1866 and 1898. The executive power is vested in a president—elected by the representatives of the fourteen provinces for a term of six years. A national congress of two chambers—a Senate and a House of Deputies—wields the legislative authority, and the republic is making rapid advances in social and political life. The national revenue for 1918 amounted to about £32,860,306, while the expenditure amounted to £34,407,074; the public debt was, at the end of 1916, about £120,000,000. There are about 22,000 miles of railway opened. The external commerce is important, the chief exports being beef and mutton, wheat, maize, and linseed, wool, skins and hides, tallow. The imports are chiefly manufactured goods. The trade is largely with Britain and France, and is increasing rapidly, the exports having advanced from £9,000,000 in 1876 and £73,200,000 in 1908 to £201,360,000 in 1920. The imports in 1920 were £170,820,000. The chief denomination of money is the dollar or *peso*, value (in gold) 4s. Buenos Ayres (or Aires) is the capital. Other towns are Rosario, Cordova, La Plata, Tucuman, Mendoza, and Santa-Fé. The population of the republic, which is rapidly increasing by immigration, was, in 1905, 5,678,197, and 8,284,266 in 1918; of the capital, 1,637,155 (1918).—BIBLIOGRAPHY: C. E. Akers, *History of South America, 1854 to 1904*; W. H. Hudson, *The Naturalist in La Plata*; Keane and Markham, *Central and South America* (in Stanford's *Compendium of Geography and Travel*);

Martinez and Lewandowski, *Argentina in the Twentieth Century*; Sir John Foster Fraser, *The Amazing Argentine*; H. Stephens, *Illustrated Descriptive Argentina*; *The Argentine Year Book*.

Ar'gentite, sulphide of silver, a blackish or lead-grey mineral, a valuable ore of silver found in the crystalline rocks of many countries.

Argentometer. See *Hydrometer*.

Argillaceous Rocks are rocks in which clay prevails (including shales and slates).

Argives (a'r'jīvz), or **Argivi**, the inhabitants of Argos; used by Homer and other ancient authors as a generic appellation for all the Greeks.

Ar'go. See *Argonauts*.

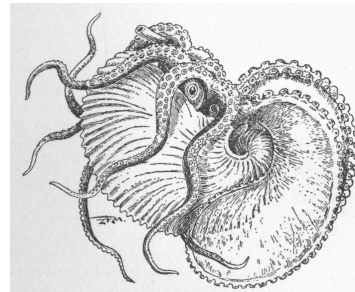
Argol. See *Argal*.

Argolis. See *Argos*.

Argon, a gas which is fairly widely distributed in the free state and is a constant constituent of the atmosphere, of which it forms about 1 per cent by volume. It was discovered by Lord Rayleigh and Professor Ramsay in 1894. During their determinations of the density of nitrogen they noticed that the density of nitrogen derived from the air differed from the density of nitrogen derived from ammonia and other compounds of nitrogen, and after a series of very careful experiments they succeeded in isolating a new gas, which they named Argon. The gas occurs in sea and river water, in plants, in the blood of animals, and the gases issuing from volcanoes and mineral springs. It is always in the free state and never in combination, and is associated with nitrogen. It is colourless, odourless, and tasteless, and may be liquefied and solidified. It is heavier than air, and is chemically a very inert substance. It is usually referred to as one of the *rare* gases of the atmosphere. Argon is manufactured in fairly large quantity from air, making use of the inertness of the substance compared to oxygen and nitrogen, the chief constituents of the air. Several methods are in use, e.g. nitrogen may be removed by passing it repeatedly over red-hot magnesium; thus the nitrogen is absorbed and the argon left. When oxygen and nitrogen of air are absorbed by a mixture of 90 per cent calcium carbide and 10 per cent

calcium chloride previously heated to redness *in vacuo*, a gas becoming richer and richer in argon is obtained. Another method of preparing argon is by fractionation of liquid air. It is used for filling electric bulbs.

Ar'gonaut, a molluscous animal of the genus *Argonauta*, belonging to the dibranchiate or two-gilled cuttle-fishes, distinguished by the females possessing a single-chambered external shell, not organically connected with the body of the animal. The males have no shell and are of much smaller size than the females. The shell is fragile, translucent, and boat-like in shape; it serves as the receptacle of the ova or eggs of the female, which sits in it with the respiratory tube or 'funnel' turned towards the carina or 'keel'. This famed mollusc swims only by ejecting water from its funnel, and it can crawl in a reversed position, carrying its shell over its back like a snail. The account of its floating on the surface of the sea, with its sail-shaped arms extended to catch the breeze, and with the six other arms as oars, is a mere fable. The argonaut, or *paper-nautilus*, must be carefully distinguished from the *pearly-nautilus* or nautilus proper (*Nautilus Pompilius*).



Argonaut—Female

Argonauts, in the legendary history of Greece, those heroes who performed a hazardous voyage to Colchis, a far-distant country at the eastern extremity of the Euxine (Black Sea), with Jason in the ship *Argo*, for the purpose of securing a golden fleece, which was preserved suspended upon a tree, and under the guardianship of a sleepless dragon. By the aid of Medea, daughter of the King of Colchis, Jason was enabled to seize the fleece, and, after many strange adventures, to reach his home at Iolcos in Thessaly. Among the Argonauts were Hercules, Castor and Pollux, Orpheus and Theseus.—**BIBLIOGRAPHY:** A. R. Hope Moncrieff, *Classic Myth and Legend*; Kingsley, *The Heroes*; N. Hawthorne, *The Wonder-book*.

Argo-Navis, the southern constellation of the Ship, is almost entirely invisible in Britain. It contains Canopus, next to Sirius the brightest fixed star. In the great nebula in Argo is situated the remarkable star Eta Argûs. It is variable, generally faint, but in 1837 it became temporarily one of the brightest stars in the sky.

Argonne, a district of France, between the Rivers Meuse, Marne, and Aisne, celebrated for the campaign of Dumouriez against the Prussians in 1792, and for the military movements and actions which took place therein previous to the battle of Sedan, in 1870.

Argonne, Battle of. When, in the autumn of 1914, the Allies retreated towards the Marne, the German Crown Prince's army endeavoured to invest Verdun. His right wing advanced through the thick and extensive forest of Argonne, but took precipitate flight after the battle of the Marne before the army of General Sarrail. In the summer of 1915 the Crown Prince endeavoured to hack his way through the French Argonne lines, using much heavy artillery, poison-gas, liquid-fire, and tear-shells. A final effort to break through was made in September, between Le Four de Paris and Vienne-le-Château, but, after gaining a footing in the first line of French trenches, the Germans were hurled back by a dashing counter-attack. The great salient from the Argonne to St. Mihiel was the salvation of Verdun.

Ar'gos, a town of Greece, in the north-east of the Peloponnesus, between the Gulfs of Ægina and Nauplia or Argos. This town and the surrounding territory of Argolis were famous from the legendary period of Greek history onwards. Here, besides Argos, was Mycenæ, where Agamemnon ruled. Modern Argos is a straggling place of 10,000 inhabitants, with some ancient remains. The territory Argolis forms a nomarchy of Greece. Pop., Argolis and Corinthia, 153,172. The capital is Nauplia.

Argos'toli, a city of the Ionian Islands, capital of Cephalonia, and the residence of a Greek bishop. Pop. 14,000.

Ar'gosity, a poetical name for a large merchant-vessel; derived from *Ragusa*, a port which was formerly more celebrated than now, and whose vessels did a considerable trade with England. It is popularly but erroneously connected with the ship *Argo* in which Jason sailed. See *Argonauts*.

Argot (Fr.; a`r-gō), the jargon, slang, or peculiar phraseology of a class or profession; originally the conventional slang of thieves and vagabonds, invented for the purpose of disguise and concealment. Some of François Villon's poems are written in argot.—Cf. W. von Knoblauch, *Dictionary of Argot*.

Arguim, or **Arguin** (a'r-gwim', a'r-gwin'), a small island on the west coast of Africa, not far from Cape Blanco, formerly a centre of trade. Its possession was violently disputed between the Portuguese, Dutch, English, and French.

Ar'gument, a term sometimes used as synonymous with the *subject* of a discourse, but more frequently appropriated to any kind of method employed for the purpose of confuting or at least silencing an opponent. Logicians have reduced arguments to a number of distinct heads, such as the *argumentum ad judicium*, which founds on solid proofs and addresses to the judgment; the *argumentum ad verecundiam*, which appeals to the modesty or bashfulness of an opponent by reminding him of the great names or authorities by whom the view disputed by him is supported; the *argumentum ad ignorantiam*, the employment of some logical fallacy towards persons likely to be deceived by it; and the *argumentum ad hominem*, an argument which presses a man with consequences drawn from his own principles and concessions, or his own conduct. See *Fallacy*, *Logic*.

Argument of the People, the document set forth by the Council of the Army on 15th Jan., 1649, fifteen days before the execution of King Charles I. See *Levellers*.

Ar'gus, in Greek mythology, a fabulous being, said to have had a hundred eyes, placed by Juno to guard Io. Hence 'argus-eyed', applied to one who is exceedingly watchful.

Argus-pheasant (*Argus gigantēus*), a large, beautiful, and very singular species of pheasant, found native in the south-east of Asia, more especially in Sumatra and some of the other islands. The males measure from 5 to 6 feet from the tip of the beak to the extremity of the tail, which has two greatly-elongated central feathers. The plumage is exceedingly beautiful, the secondary quills of the wings, which are longer than the primary feathers, being each adorned with a series of ocellated or eye-like spots (whence the name—see *Argus*) of



Argus-pheasant (*Argus*)

brilliant metallic hues. The general body *gigantēus*
plumage is brown.

Argyll, or **Argyle** (a`r-gīl´), an extensive county in the south-west of the Highlands of Scotland, consisting partly of mainland and partly of islands belonging to the Hebrides group, the chief of which are Islay, Mull, Jura, Tiree, Coll, Luing, Lismore, and Colonsay, with Iona and Staffa. On the land side the mainland is bounded north by Inverness; east by Perth and Dumbarton; elsewhere surrounded by the Firth of Clyde and its connections and the sea; area, 3255 sq. miles (or over 2,000,000 acres), of which the islands comprise about 1000 sq. miles. It is greatly indented by arms of the sea, which penetrate far inland, the most important of these being Loch Sunart, Loch Linnhe (the extremities of which are Loch Eil and Loch Leven), Loch Etive, Loch Fyne, Loch Tarbert, Loch Riddon, Loch Striven, and Loch Long. The mainland is divided into six districts of Northern Argyle, Lorn, Argyle, Cowal, Knapdale, and Kintyre; the two first being subdivided into the sub-districts of Lochiel, Ardgour, Sunart, Ardnamurchan, Morven, and Appin. The county is exceedingly mountainous, the chief summits being Bidean-nam-Bian (3766 feet), Ben Laoigh (3708 feet), Ben Cruachan (3611 feet), Benmore, in Mull (3185 feet), the Paps of Jura (2565 feet), and Ben Arthur or the Cobbler (2891 feet). There are several lakes, the principal of which is Loch Awe. Cattle and sheep are reared in numbers, and fishing is largely carried on, as is also the making of whisky. There is but little arable land. The chief minerals are slate, marble, limestone, and granite. County town, Inveraray; others, Campbeltown, Oban, and Dunoon. Pop. (1921), 76,856.

Argyll, Campbells of, a historic Scottish family, raised to the peerage in the person of Sir Duncan Campbell of Lochow, in 1445. The more eminent members are: Archibald, 2nd Earl, killed at the battle of Flodden, 1513.—Archibald, 5th Earl, attached himself to the party of Mary of Guise, and was the means of averting a collision between the Reformers and the French troops in 1559; was commissioner of regency after Mary's abdication, but afterwards commanded her troops at the battle of Langside; died 1575.—Archibald, 8th Earl and 1st Marquess, born 1598: a zealous partisan of the Covenanters; created a marquess by Charles I. It was by his persuasion that Charles II visited Scotland, and was crowned at Scone in 1651. At the Restoration he was committed to the Tower, and afterwards sent to

Scotland, where he was tried for high treason, and beheaded in 1661.—Archibald, 9th Earl, son of the preceding, served the king with great bravery at the battle of Dunbar, and was excluded from the general pardon by Cromwell in 1654. On the passing of the Test Act in 1681 he refused to take the required oath except with a reservation. For this he was tried and sentenced to death. He managed to escape to Holland, from whence he returned with a view of aiding the Duke of Monmouth. His plan, however, failed, and he was taken and conveyed to Edinburgh, where he was beheaded in 1685.—Archibald, 10th Earl and 1st Duke, son of the preceding, died 1703; took an active part in the Revolution of 1688-9, which placed William and Mary on the throne, and was rewarded by several important appointments and the title of duke.—John, 2nd Duke and Duke of Greenwich, son of the above, born 1678, died 1743; served under Marlborough at the battles of Ramillies, Oudenarde, and Malplaquet, and assisted at the sieges of Lisle and Ghent. He incurred considerable odium in his own country for his efforts in promoting the union. In 1712 he had the military command in Scotland, and in 1715 he fought an indecisive battle with the Earl of Mar's army at Sheriffmuir, near Dunblane, and forced the Pretender to quit the kingdom. He was long a supporter of Walpole, but his political career was full of intrigue. He is the Duke of Argyll in Scott's *Heart of Midlothian*.—George Douglas Campbell, K.G., K.T., &c., 8th Duke (of United Kingdom, 1892), was born in 1823. He early took a part in politics, especially in discussions regarding the Presbyterian Church of Scotland. In 1852 he became Lord Privy Seal under Lord Aberdeen, and again under Lord Palmerston in 1859; Postmaster-General in 1860; Secretary for India from 1868 to 1874; again Lord Privy Seal in 1880, but retired, being unable to agree with his colleagues on their Irish policy. He died in 1900. He wrote *The Reign of Law, Scotland as it Was and as it Is, &c.*—John Douglas Sutherland Campbell, 9th Duke of, son of the 8th Duke and a daughter of the 2nd Duke of Sutherland, was born in 1845, and succeeded his father in 1900. He completed his education at Trinity College, Cambridge, sat in Parliament as member for Argyllshire, 1868-78, was Governor-General of Canada from 1878 to 1883, and again sat in Parliament as member for South Manchester from 1895 to 1900, as a Liberal-Unionist. He married the Princess Louise of Great Britain, fourth daughter of Queen Victoria, in 1871. He died in 1914. His works include: *The United States after the War, Imperial Federation, Canadian Pictures,*

Memories of Canada and Scotland, Life of Lord Palmerston, Tales and Poems, The Psalms in English Verse, Life and Times of Queen Victoria, Yesterday and To-day in Canada, &c.

Argyll and Sutherland Highlanders (Princess Louise's), raised by the 5th Duke of Argyll (1794), received their present title in 1872. The regiment served under Sir David Baird in Cape Colony, and at Balaklava made itself immortal as Sir Colin Campbell's 'thin red line'. It was further distinguished in the European War at Gheluvelt, Le Cateau, and the second battle of Ypres.

Argyro-Castro (a `r'gi-rō-), a town in Albania, 40 miles north-west of Janina; built on three ridges intersected by deep ravines, across which are several bridges. It was occupied by the Greeks in 1916. Pop. about 10,000.

Argyropu'los, Johannes, one of the principal revivers of Greek learning in the fifteenth century. Born in Constantinople 1415, died at Rome 1486.

Aria, in music. See *Air*.

Ariadne (a-ri-ad'ne), in Greek mythology, a daughter of Minos, King of Crete. She gave Theseus a clue of thread to conduct him out of the labyrinth after his defeat of the Minotaur. Theseus abandoned her on the Isle of Naxos, where she was found by Dionysus, who married her.

Aria'na, the ancient name of a large district in Asia, forming a portion of the Persian Empire; bounded on the north by the provinces of Bactriana, Margiana, and Hyrcania; east by the Indus; south by the Indian Ocean and the Persian Gulf; west by Media.

Ariano (ä-rē-ä'nō), a town in South Italy, province of Avellino, 44 miles north-east of Naples, the seat of a bishop, with a handsome cathedral. Pop. 17,650.

Ar'ians, the adherents of the Alexandrian presbyter Arius, who, about A.D. 318, promulgated the doctrine that Christ was a created being inferior to God the Father in nature and dignity, though the first and noblest of all created beings; and also that the Holy Spirit is not God, but created by the power of the Son. Arianism has been defined as an attempt to determine the relations of the Persons of the Trinity on a basis of distinction and

subordination. It does not seem to have sprung from any strong ethical impulse; its philosophy was pagan, and the object of the leaders political rather than religious. The doctrines were condemned by the Council of Nicæa in 325. Arius died in 336, and after his death his party gained considerable accessions, including several emperors, and for a time held a strong position. Since the middle of the seventh century, however, the Arians have nowhere constituted a distinct sect, although similar opinions have been advanced by various theologians in modern times. The Arian controversy was revived in England during the eighteenth century by William Whiston and Dr. Samuel Clarke.—BIBLIOGRAPHY: H. M. Gwatkin, *Studies of Arianism*; J. H. Newman, *Arians of the Fourth Century*; J. H. Colligan, *Arian Movement in England*.

Arica (a'-rē'ka'), a seaport of Chile, 30 miles S. of Tacna; previous to 1880 it belonged to Peru. It has suffered frequently from earthquakes, being in 1868 almost entirely destroyed, part of it being also submerged by an earthquake wave. Pop. about 4000. It has a wireless station.

Arica. See *Tacna-Arica Dispute*.

Arichat (-shat'), a seaport town and fishing station of Nova Scotia, on a small bay, south coast of Madame Island. Pop. about 2500.

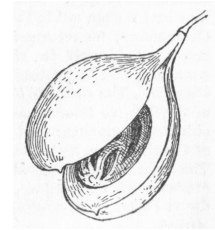
Ariège (a'-rē-āzh), a mountainous department of France, on the northern slopes of the Pyrenees, comprising the ancient countship of Foix and parts of Languedoc and Gascony. The principal rivers are the Ariège, Arize, and Salat, tributaries of the Garonne. Sheep and cattle are reared; the arable land is small in quantity. Chief town, Foix. Area, 1892 sq. miles. Pop. (1921), 172,851.

A'riel, a symbolic name for Jerusalem in the Old Testament; in the demonology of the later Jews a spirit of the waters. In Shakespeare's *Tempest*, Ariel was the "tricksy spirit" whom Prospero had in his service.

Aries (ā'ri-ēz; Lat.), the Ram, a northern constellation. It is the first of the twelve signs of the zodiac, which the sun enters at the vernal equinox, about the 21st of March. The "First Point in Aries" is where the equator cuts the ecliptic in the ascending node, from which point the right ascensions of heavenly bodies are reckoned on the equator, and their longitudes upon the

ecliptic. Owing to the precession of the equinoxes the sign Aries no longer corresponds with the constellation Aries, which it did 2000 years ago.

Ar'il, or **Aril'lus**, in some plants, as in the nutmeg, an extra covering of the seed, outside of the true seed-coats, proceeding from the placenta, partially investing the seed, and falling off spontaneously. It is either succulent or cartilaginous, coloured, elastic, rough, or knotted. In the nutmeg it is known as *mace*.



Aril, Fruit of
Nutmeg

Arimas'pians, in ancient Greek traditions, a people who lived in the extreme north-east of the ancient world. They were said to be one-eyed and to carry on a perpetual war with the gold-guarding griffins, whose gold they endeavoured to steal. Cf. Milton, *Paradise Lost*, II, 943.

Arimathæ'a, a town of Palestine, identified with the modern *Ramleh*, 22 miles W.N.W. of Jerusalem.

Ari'on, an ancient Greek poet and musician, born at Methymna, in Lesbos, flourished about 625 B.C. He lived at the Court of Periander of Corinth, and afterwards visited Sicily and Italy. Returning from Tarentum to Corinth with rich treasures, the avaricious sailors resolved to murder him. Apollo, however, having informed him in a dream of the impending danger, Arion in vain endeavoured to soften the hearts of the crew by the power of music. He then threw himself into the sea, when one of a shoal of dolphins, which had been attracted by his music, received him on his back and bore him to land. The sailors, having returned to Corinth, were confronted by Arion and convicted of their crime. The lyre of Arion, and the dolphin which rescued him, became constellations in the heavens. A fragment of a hymn to Poseidon, ascribed to Arion, is extant.

Arios'to, Ludovi'co, one of the most celebrated poets of Italy, was born at Reggio, in Lombardy, 8th Sept., 1474, of a noble family; died 6th June, 1533. His lyric poems in the Italian and Latin languages, distinguished for ease and elegance of style, introduced him to the notice of the Cardinal Ippolito d'Este, son of Duke Ercole I of Ferrara. In 1503 Ippolito employed him in his service and used his counsel in the most important affairs. In this service he began and finished, in ten or eleven years, his immortal poem,

the *Orlando Furioso*, which was published in 1515, and immediately became highly popular. He afterwards entered the service of Alfonso I, Duke of Ferrara, the cardinal's brother, a lover of the arts, who put much confidence in him. After quelling disturbances that had broken out in the wild and mountainous Garfagnana, he returned to Ferrara, where he employed himself in the composition of his comedies, and in putting the last touches to his *Orlando*. The *Orlando Furioso* is a continuation of the *Orlando Innamorato* of Bojardo, details the chivalrous adventures of the paladins of the age of Charlemagne, and extends to forty-six cantos. The best English translation is that of Rose (1823). Cf. E. Gardner, *Ariosto: the Prince of Court Poets*; J. S. Nicholson, *Life and Genius of Ariosto*.

Arish. See *El Arish*.

Aristæus, in Greek mythology, son of Apollo and Cyrene, the introducer of bee-keeping. Cf. Virgil, *Georgics*, IV, 315-558.

Aristarchus (a-ris-tär'kus), an ancient Greek grammarian, born at Samothrace 220 B.C., died at Cyprus 143 B.C. He edited Homer's poems with the greatest acuteness and ability, endeavouring to restore the text to its genuine state, and to clear it of all interpolations and corruptions; hence the phrase, Aristarchian criticism. His edition of Homer furnished the basis of all subsequent ones.

Aristarchus, an ancient Greek astronomer belonging to Samos, flourished about 155 B.C., and first asserted the revolution of the earth about the sun; also regarded as the inventor of the sun-dial.

Aris'teas, a personage of ancient Greek legend, represented to have lived over many centuries, disappearing and reappearing by turns.

Aristides (a-ris-tī'dēz), a statesman of ancient Greece, for his strict integrity surnamed the *Just*. He was one of the ten generals of the Athenians when they fought with the Persians at Marathon, 490 B.C. Next year he was eponymous archon, and in this office enjoyed such popularity that he excited the jealousy of Themistocles, who succeeded in procuring his banishment by ostracism (about 483). Three years after, when Xerxes invaded Greece with a large army, the Athenians hastened to recall him, and Themistocles now admitted him to his confidence and councils. In the battle

of Plataea (479) he commanded the Athenians, and had a great share in gaining the victory. To defray the expenses of the Persian war he persuaded the Greeks to impose a tax, which should be paid into the hands of an officer appointed by the States collectively, and deposited at Delos. The confidence which was felt in his integrity appeared in their entrusting him with the office of apportioning the contribution. He died at an advanced age about 468 B.C., so poor that he was buried at the public expense.

Aristip'pus (c. 425-366 B.C.), a disciple of Socrates, and founder of a philosophical school among the Greeks, which was called the *Cyrenaic*, from his native city Cyrēnē, in Africa; flourished 380 B.C. His moral philosophy differed widely from that of Socrates, and was a science of refined voluptuousness. His fundamental principles were—that all human sensations may be reduced to two, pleasure and pain. Pleasure is a gentle, and pain a violent emotion. All living beings seek the former and avoid the latter. Happiness is nothing but a continued pleasure, composed of separate gratifications; and as it is the object of all human exertions, we should abstain from no kind of pleasure. Still we should always be governed by taste and reason in our enjoyments. His doctrines were taught only by his daughter Arētē, and by his grandson Aristippus the younger, by whom they were systematized. Other Cyrenaics compounded them into a particular doctrine of pleasure, and are hence called *Hedonici*. His writings are lost.

Aristoc'rary (Gr. *aristos*, best, *kratos*, rule), a form of government by which the wealthy and noble, or any small privileged class, rules over the rest of the citizens. The term has now become almost entirely social in meaning, and is mostly applied to the nobility or chief persons in a State.

Aristogeiton (-gī'ton), a citizen of Athens, whose name is rendered famous by a conspiracy (514 B.C.) formed in conjunction with his friend Harmodius against the tyrants Hippias and Hipparchus, the sons of Pisistratus. Both Aristogeiton and Harmodius lost their lives through their attempts to free the country, and were reckoned martyrs of liberty.

Aristolochia (-lō'ki-a), a genus of plants, the type of the ord. Aristolochiaceae, which consists of dicotyledonous monochlamydeous plants, with an inferior 3-6-celled fruit, found for the most part in the hotter parts of the world, and in many cases used medicinally on account of their

tonic and stimulating properties. The genus has emmenagogic qualities, especially the European species *A. rotunda*, *A. longa*, and *A. Clematītis*. *A. bracteāta* is used in India as an anthelmintic; *A. odoratissima*, a West Indian species, is a valuable bitter and alexipharmic. *A. serpentaria* is the Virginian snake-root, popularly regarded as a remedy for snake bites.

Aristophanes (-tof´a-nēz), the greatest comic poet of ancient Greece, born at Athens probably about the year 455 B.C., died 375 B.C. Little is known of his life. He appeared as a poet in 427 B.C., and having indulged in some sarcasms on the powerful demagogue Cleon, was ineffectually accused by the latter of having unlawfully assumed the title of an Athenian citizen. He afterwards revenged himself on Cleon in his comedy of the *Knights*, in which he himself acted the part of Cleon, because no actor had the courage to do it. Of fifty-four (or forty-four) comedies attributed to him, eleven only remain; believed to be the flower of the ancient comedy, and distinguished by wit, humour, and poetry, as also by grossness. In them there is constant reference to the manners, actions, and public characters of the day, the freedom of the old Greek comedy allowing an unbounded degree of personal and political satire. The names of his extant plays are *Acharnians*, *Knights*, *Clouds*, *Wasps*, *Peace*, *Birds*, *Lysistrata*, *Thesmophoriazusæ*, *Frogs*, *Ecclesiasuzæ*, and *Plutus*.—BIBLIOGRAPHY: B. B. Rogers, *Complete Works of Aristophanes, with verse translation* (by far the best translation); Hookham-Frere, *Translation* (five plays only); Couat, *Aristophane et l'ancienne comédie attique*.

Ar´istotle (Gr. *Aristot´eles*), a distinguished philosopher and naturalist of ancient Greece, the founder of the Peripatetic school of philosophy, was born in 384 B.C. at Stagira, in Macedonia; died at Chalcis, 322 B.C. His father, Nicomachus, was physician to Amyntas II, King of Macedonia, and claimed to be descended from Æsculapius. Aristotle had lost his parents before he came, at about the age of seventeen, to Athens to study in the school of Plato. With that philosopher he remained for twenty years, became pre-eminent among his pupils, and was known as the 'Intellect of the School'. Upon the death of Plato, 348 B.C., he took up his residence at Atarneus, in Mysia, on the invitation of his former pupil Hermeias, the ruler of that city, on whose assassination by the Persians, 343 B.C., he fled to Mitylene with his wife Pythias, a near relative of Hermeias. During his residence at Mitylene he received an invitation from Philip of Macedon to

superintend the education of his son Alexander, then in his fourteenth year. This relationship between the great philosopher and the future conqueror continued for five or six years, during which the prince was instructed in grammar, rhetoric, poetry, logic, ethics, and politics, and in those branches of physics which had even then made some considerable progress. On Alexander succeeding to the throne Aristotle continued to live with him as his friend and councillor till he set out on his Asiatic campaign (334 B.C.). He returned to Athens and established his school in the Lyceum, a gymnasium attached to the temple of Apollo Lyceus, which was assigned to him by the State. He delivered his lectures in the wooded walks of the Lyceum while walking up and down with his pupils. From the action itself, or more probably from the name of the walks (*peripatoi*), his school was called Peripatetic. Pupils gathered to him from all parts of Greece, and his school became by far the most popular in Athens. The statement that he had two circles of pupils, the *exoteric* and the *esoteric* has given rise to much controversy. By some it has been held that Aristotle published during his lifetime popular discourses with a view to make way for his doctrines in Athenian society, then impregnated with Platonic theories, and that these are called exoteric in contradistinction to those in which are embodied his matured opinions. It was during the time of his teaching at Athens that Aristotle is believed to have composed the great bulk of his works. But it is not possible to speak with any certainty about the chronology of his writings, as the references may be additions of editors. On the death of Alexander a revolution occurred in Athens hostile to the Macedonian interests with which Aristotle was identified. He therefore retired to Chalcis, where he soon after died. Sir Charles Walston, in 1891, opened a tomb near Eretria which he supposed to be that of Aristotle. According to Strabo he bequeathed all his works to Theophrastus, who, with other disciples of Aristotle, amended and continued them. They afterwards passed through various hands, till, about 50 B.C., Andronicus of Rhodes put the various fragments together and classified them according to a systematic arrangement. Many of the books bearing his name are spurious, others are of doubtful genuineness. The whole are generally divided into logical, theoretical, and practical. The logical works are comprehended under the title *Organon* (Instrument). The theoretical are divided into physics, mathematics, and metaphysics. The physical works (including those on natural history) are on the *General Principles of Physical Science*, The

Heavens, Generation and Destruction, Meteorology, Natural History of Animals, On the Parts of Animals, On the Generation of Animals, On the Locomotion of Animals, On the Soul, On Memory, Sleep and Waking, Dreams, Divination. In mathematics there are two treatises, *On Indivisible Lines* and *Mechanical Problems*. The *Metaphysics* consist of fourteen books; the title (*Ta meta ta Physika*, 'the things following the Physics',) is the invention of an editor. The practical works embrace ethics, politics, economics, and treatises on art, and comprise the *Nicomachæan Ethics* (so called because dedicated to his son, Nicomachus), *The Politics*, *Æconomics*, *Poetry*, and *Rhetoric*. Among the lost works are the dialogues and others termed exoteric. A treatise *On the Constitution of Athens* was discovered in 1891. His style is devoid of grace and elegance. His works were first printed in a Latin translation, with the commentaries of Averroes, at Venice in 1489; the first Greek edition was that of Aldus Manutius (5 vols., 1495-8). See *Peripatetic Philosophy*.—BIBLIOGRAPHY: Blakesley, *Life of Aristotle*; S. H. Butcher, *Poetics* (with translation and excursus); R. Shute, *History of the Aristotelian Writings*; J. C. Wilson, *Aristotelian Studies*; E. Zeller, *Aristotle and the Earlier Peripatetics*; E. Barker, *Political Thought of Plato and Aristotle*.

Aristoxenus, an ancient Greek musician and philosopher of Tarentum, born about 324 B.C. He studied music under his father Mnesias, and philosophy under Aristotle, whose successor he aspired to be. He endeavoured to apply his musical knowledge to philosophy, and especially to the science of mind, but it only appears to have furnished him with far-fetched analogies and led him into a kind of materialism. We have a work on the *Elements of Harmony* by him.

Arithmetic (Gr. *arithmos*, number) is primarily the science of numbers. As opposed to algebra it is the practical part of the science. Although the processes of arithmetical operations are often highly complicated, they all resolve themselves into the repetition of four primary operations—addition, subtraction, multiplication, and division. Of these the two latter are only complex forms of the two former, and subtraction again is merely a reversal of the process of addition. Little or nothing is known as to the origin and invention of arithmetic. Some elementary conception of it is in all probability coeval with the first dawn of human intelligence. In consequence of their rude methods of numeration, the science made but

small advance among the ancient Egyptians, Greeks, and Romans, and it was not until the introduction of the decimal scale of notation and the Arabic, or rather Indian, numerals into Europe that any great progress can be traced. In this scale of notation every number is expressed by means of the ten digits, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, by giving each digit a local as well as its proper or natural value. The value of every digit increases in a tenfold proportion from the right towards the left; the distance of any figure from the right indicating the power of 10, and the digit itself the number of those powers intended to be expressed: thus $3464 = 3000 + 400 + 60 + 4 = 3 \times 10^3 + 4 \times 10^2 + 6 \times 10 + 4$. The earliest arithmetical signs appear to have been hieroglyphical, but the Egyptian hieroglyphics were too diffuse to be of any arithmetical value. The units were successive strokes to the number required, the ten an open circle, the hundred a curled palm-leaf, the thousand a lotus flower, ten thousand a bent finger. The letters of the alphabet afforded a convenient mode of representing figures, and were used accordingly by the Chaldeans, Hebrews, and Greeks. The first nine letters of the Hebrew alphabet represented the units, the second nine tens, the remaining four together with five repeated with additional marks, hundreds; the same succession of letters with added points was repeated for thousands, tens of thousands, and hundreds of thousands. The Greeks followed the same system up to tens of thousands. They wrote the different classes of numbers in succession as we do, and they transferred operations performed on units to numbers in higher places; but the use of different signs for the different ranks clearly shows a want of full perception of the value of place as such. They adopted the letter M as a sign for 10,000 and by combining this mark with their other numerals they could note numbers as high as 100,000,000. The Roman numerals, which are still used in marking dates or numbering chapters, were almost useless for purposes of computation. From one to four were represented by vertical strokes **I**, **II**, **III**, **IIII**, five by **V**, ten by **X**, fifty by **L**, one hundred by **C**, afterwards **D**, five hundred by **D**, a thousand by **M**. These signs were derived from each other according to particular rules, thus **V** was the half of **X**, **Λ** being also used; **L** was likewise the half of **C**. **M** was artistically written **M** and **cIo** and **Io**, afterwards **D**, became five hundred. **ccI** represented 5000, **ccIo** 10,000, **Io** 50,000, **cccIo** 100,000. They were also compounded by addition and subtraction, thus **IV** stood for four, **VI** for six, **XXX** for thirty, **XL** for forty, **LX** for sixty. Arithmetic is divided into *abstract* and *practical*: the

former comprehends notation, numeration, addition, subtraction, multiplication, division, measures and multiples, fractions, powers and roots; the latter treats of the combinations and practical applications of these and the so-called rules, such as reduction, compound addition, subtraction, multiplication, and division, proportion, interest, profit and loss, &c. Another division is *integral* and *fractional* arithmetic, the former treating of integers, or whole numbers, and the latter of fractions. Decimal fractions were invented in the sixteenth century, and logarithms, embodying the last great advance in the science, in the seventeenth century.

Arithmet'ical, pertaining to arithmetic or its operations.—*Arithmetical mean*, the middle term of three quantities in arithmetical progression, or half the sum of any two proposed numbers; thus 11 is the arithmetical mean to 8 and 14.—*Arithmetical progression*, a series of numbers increasing or decreasing by a common difference, as 1, 3, 5, 7, &c.—*Arithmetical signs*, certain symbols used in arithmetic, and indicating processes or facts. The common signs used in arithmetic are the following: $+$ signifying that the numbers between which it is placed are to be added; $-$ that the second is to be subtracted from the first; $\times$ that the one is to be multiplied by the other; $\div$ that the former is to be divided by the latter; $=$ signifies that the one number is equal to the other; $:$ $::$ $:$ are the signs placed between the members of a proportional series, as $4 : 6 :: 8 : 12$. A small figure placed on the right hand of another at the top signifies the corresponding power of the number beside which it is placed, as 5^2 , 4^3 , meaning the square of 5 and the cube of 4. $\sqrt{}$ placed before or over a number signifies the square root of that number; with a figure it signifies the root of a higher power, as $\sqrt[3]{}$, which means cube root. A period placed to the left of a series of figures indicates that they are decimal fractions.

A'rius, the originator of the Arian heresy. See *Arians*.

Arizo'na, a former territory of the United States, admitted into the union as a sovereign State on 14th Feb., 1912, is bounded south by Mexico, west by California and Nevada (the River Colorado forming the greater part of the boundary), north by Utah, and east by New Mexico; area, 113,956 sq. miles. The surface is generally mountainous, but many fertile and well-watered valleys lie between the ridges. Part of the surface consists of deserts often entirely destitute of vegetation. The territory belongs to the

basin of the Colorado, which passes through a portion of it, besides forming the boundary; while the Gila and Little Colorado, tributaries of the Colorado, traverse it from east to west. The canyons of the Colorado form a wonderful feature, the river flowing for hundreds of miles in a deep rocky channel with walls rising perpendicularly to the height of 1500 to 6000 feet. In some parts timber is plentiful. The rainfall is small, and irrigation has been employed for agricultural purposes. Large tracts of elevated land have been found excellently adapted as pastures for sheep and cattle. The territory is rich in copper, gold, silver, lead, and other minerals, and mining is largely carried on, with much copper smelting and refining. The capital is Phoenix. Pop. 29,053. The Southern and the Santa Fé Pacific Railways traverse it. Pop. (1920), 333,903 (of which 171,468 are white), exclusive of Apaches and other Indians on reservations (area, 29,017 sq. miles; pop. (1920), 42,400).

Arjish Dagh, the loftiest peak of the peninsula of Asia Minor, at the western extremity of the Anti-Taurus Range, 13,150 feet; an extinct volcano; on the N. and N.E. slopes are extensive glaciers.

Ark, the name applied in our translation of the Bible to the boat or floating house in which Noah resided during the flood or deluge; to the floating vessel of bulrushes in which the infant Moses was laid; and to the chest in which the tables of the law were preserved—the *ark of the covenant*. This last was made of shittim-wood, overlaid within and without with gold, about 3¾ feet long by 2¼ feet high and broad, and over it were placed the golden covering or mercy-seat and the two cherubim. It was placed in the sanctuary of the temple of Solomon; before his time it was kept in the tabernacle, and was moved about as circumstances dictated. At the captivity it appears to have been either lost or destroyed.

Arkansas (ar'kan-sa'' or ar-kan'sas), one of the United States of America, bounded north by Missouri; east by the Mississippi, which separates it from the States of Mississippi and Tennessee; south by Louisiana and Texas; and west by the Indian Territory and Texas; area, 53,335 sq. miles. The surface in the east is low, flat, and swampy, densely wooded, and subject to frequent inundations from the numerous streams which water it. Towards the centre it becomes more diversified, presenting many undulating slopes and hills of moderate elevation. In the west it rises still higher, being

traversed by a range of hills called the Ozark, which attains a height of 2000 feet, some peaks rising to 3000. In various parts the prairies are of great extent; the forests also are extensive, principally of oak, hickory, ash, cotton, linden, maple, locust, and pine. Coal and other minerals are worked. The principal rivers, all tributaries of the Mississippi, are the Arkansas, the Red River, the St. Francis, and the Washita. Near the centre of the State are warm springs, much resorted to for chronic rheumatic and paralytic affections. The climate is subject to great extremes of heat and cold, and in the lower districts is unhealthy to new settlers. The staple products are cotton and maize; fruit is tolerably abundant. Many districts are admirably adapted for grazing, and great numbers of excellent cattle are reared. Arkansas was colonized as early as 1685 by the French. As part of Louisiana it was purchased by the United States in 1803. It was made into a separate territory in 1819, and admitted into the Union in 1836. It was one of the seceding States. The capital is Little Rock. The enumerated population in 1920 was 1,750,995.

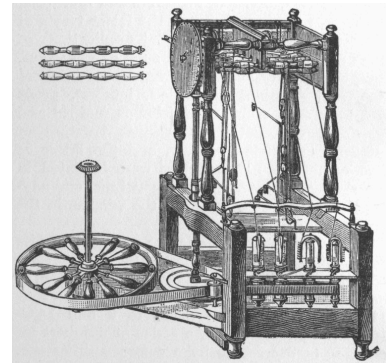
Arkansas, a river of the United States, which gives its name to the above State, the largest affluent of the Mississippi after the Missouri. It rises in the Rocky Mountains, about lat. 39° N., long. 107° W., flows in a general southeasterly direction through Colorado, Kansas, the Indian Territory, and lastly through the State of Arkansas, and after a course of 2170 miles enters the Mississippi. During greater part of the year it is navigable for steamboats for 800 miles.

Arkite. See *Explosives*.

Ark'low, a town in Ireland, County Wicklow, on the right bank of the Avoca, which falls into the sea about 500 yards below the town; the scene of a severe fight during the rebellion of 1798. Fishing is the chief industry. Pop. 5042.

Ark'wright, Sir Richard, famous for his inventions in cotton-spinning, was born at Preston, in Lancashire, in 1732, died 1792. The youngest of thirteen children, he was bred to the trade of a barber. When about thirty-five years of age he gave himself up exclusively to the subject of inventions for spinning cotton. The thread spun by Hargreaves' jenny could not be used except as weft, being destitute of the firmness or hardness required in the

longitudinal threads or warp. But Arkwright supplied this deficiency by the invention of the *spinning-frame*, which spins a vast number of threads of any degree of fineness and hardness, leaving the operator merely to feed the machine with cotton and to join the threads when they happen to break. His invention introduced the system of spinning by rollers, the carding, or *roving* as it is technically termed (that is, the soft, loose strip of cotton), passing through one pair of rollers, and being received by a second pair, which is made to revolve with (as the case may be) three, four, or five times the velocity of the first pair. By this contrivance the roving is drawn out into a thread of the desired degree of tenuity and hardness. His inventions being brought into a pretty advanced state, Arkwright removed to Nottingham in 1768 in order to avoid the attacks of the same lawless rabble that had driven Hargreaves out of Lancashire. Here his operations were at first greatly fettered by a want of capital; but two gentlemen of means having entered into partnership with him, the necessary funds were obtained, and Arkwright erected his first mill, which was driven by horses, at Nottingham, and took out a patent for spinning by rollers in 1769. As the mode of working the machinery by horse-power was found too expensive, he built a second factory on a much larger scale at Cromford, in Derbyshire, in 1771, the machinery of which was turned by a water-wheel. Having made several additional discoveries and improvements in the processes of carding, roving, and spinning, he took out a fresh patent for the whole in 1775, and thus completed a series of the most ingenious and complicated machinery. Notwithstanding a series of law-suits in defence of his patent rights, and the destruction of his property by mobs, he amassed a large fortune. He was knighted by George III in 1786.



Arkwright's Water Frame

Arlberg (a`rl`berh), a branch of the Rhætian Alps, in the west of Tyrol, between it and Vorarlberg, pierced by the fourth longest railway tunnel in the world. It is 6½ miles long, and was finished in Nov., 1883, and connects the valley of the Inn with that of the Rhine, and the Austrian railway system with the Swiss railways.

Ar'lecdon, an urban district of England, in Cumberland, 4 miles east of Whitehaven, with coal and iron mines. Pop. (1921), 5152.

Arles (a `rl; ancient, **Arelāte**), a town of Southern France, department Bouches du Rhône, 17 miles south-east of Nismes. It was an important town at the time of Cæsar's invasion, and under the later emperors it became one of the most flourishing towns on the farther side of the Alps. It still possesses numerous ancient remains, of which the most conspicuous are those of a Roman amphitheatre, which accommodated 24,000 spectators. It has a considerable trade, manufactures of silk, &c., and furnishes a market for the surrounding country. Pop. 16,746.

Ar'lington, Henry Bennet, Earl of, member of the Cabal ministry, and one of the scheming creatures of Charles II, born 1618, died 1685. He is supposed to have lived and died a Roman Catholic.

Ar'lon, a Belgian town, capital province of Luxemburg, a thriving town, with manufactures of ironware, leather, tobacco, &c. Pop. 12,012.

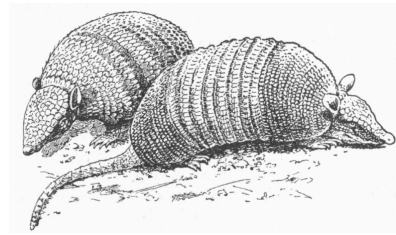
Arm, the upper limb in man, connected with the thorax or chest by means of the scapula or shoulder-blade, and the clavicle or collar-bone. It consists of three bones, the arm-bone (*humērus*), and the two bones of the fore-arm (*radius* and *ulna*), and it is connected with the bones of the hand by the *carpus* or wrist. The head or upper end of the arm-bone fits into the hollow called the *glenoid cavity* of the scapula, so as to form a joint of the ball-and-socket kind, allowing great freedom of movement to the limb. The lower end of the humerus is broadened out by a projection on both the outer and inner sides (the outer and inner *condyles*), and has a pulley-like surface for articulating with the fore-arm to form the elbow-joint. This joint somewhat resembles a hinge, allowing of movement only in one direction. The ulna is the inner of the two bones of the fore-arm. It is largest at the upper end, where it has two processes, the *coronoid* and the *olecranon*, with a deep groove between to receive the humerus. The radius—the outer of the two bones—is small at the upper and expanded at the lower end, where it forms part of the wrist-joint. The muscles of the upper arm are either *flexors* or *extensors*, the former serving to bend the arm, the latter to straighten it by means of the elbow-joint. The main flexor is the *biceps*, the large muscle which may be seen standing out in front of the arm when a weight is raised.

The chief opposing muscle of the biceps is the *triceps*. The muscles of the fore-arm are, besides flexors and extensors, *pronators* and *supinators*, the former turning the hand palm downwards, the latter turning it upwards. The same fundamental plan of structure exists in the limbs of all vertebrate animals.

Arma'da, the Spanish name for any large naval force; usually applied to the Spanish fleet vaingloriously designated the *Invincible Armada*, intended to act against England A.D. 1588. It was under the command of the Duke of Medina-Sidonia, and consisted of 130 great war vessels, larger and stronger than any belonging to the English fleet, with 30 smaller ships of war, and carried 19,295 marines, 8460 sailors, 2088 slaves, and 2630 cannon. It had scarcely quitted Lisbon on 29th May, 1588, when it was scattered by a storm, and had to be refitted in Corunna. It was to co-operate with a land force collected in Flanders under the Prince of Parma, and to unite with this it proceeded through the English Channel towards Calais. In its progress it was attacked by the English fleet under Lord Howard, who, with his lieutenants, Drake, Hawkins, and Frobisher, endeavoured by dexterous seamanship and the discharge of well-directed volleys of shot to destroy or capture the vessels of the enemy. The great lumbering Spanish vessels suffered severely from their smaller opponents, which most of their shot missed. Arrived at length off Dunkirk, the armada was becalmed, thrown into confusion by fire-ships, and many of the Spanish vessels destroyed or taken. The Duke of Medina-Sidonia, owing to the severe losses, at last resolved to abandon the enterprise, and conceived the idea of reconveying his fleet to Spain by a voyage round the north of Great Britain; but storm after storm assailed his ships, scattering them in all directions, and sinking many. Some went down on the cliffs of Norway, others in the open sea, others on the Scottish coast, others on the coast of Ireland. In all, seventy-two large vessels and over 10,000 men were lost.—BIBLIOGRAPHY: J. A. Froude, *Spanish Story of the Armada*; Sir J. K. Laughton, *State Papers relating to the Defeat of the Spanish Armada*; J. R. Hale, *Story of the Great Armada*.

Armadales, a town of Scotland, Linlithgowshire, in coal and iron district. Pop. 4739.

Armadi'lo (genus *Dasypus*), an edentate mammal peculiar to South America, consisting of various species, belonging to a family intermediate between the sloths and ant-eaters. They are covered with a hard bony shell, divided into belts, composed of small separate plates like a coat of mail, flexible everywhere except on the forehead, shoulders, and haunches, where it is not movable. The belts are connected by a membrane which enables the animal to roll itself up like a hedgehog. These animals burrow in the earth, where they lie during the daytime, seldom going abroad except at night. They are of different sizes: the largest, *Dasypus gigas*, being 3 feet in length without the tail, and the smallest only 10 inches. They subsist chiefly on fruits and roots, sometimes on insects and flesh. They are inoffensive, and their flesh is esteemed good food.—There is a genus of isopodous Crustacea called Armadillo, consisting of animals allied to the wood-lice, capable of rolling themselves into a ball.



Armadillos—Left, Hairy Armadillo. Right, Kappler's Armadillo

Armageddon (-ged'don), the great battlefield of the Old Testament, where the chief conflicts took place between the Israelites and their enemies—the table-land of Esdraelon in Galilee and Samaria, in the centre of which stood the town of Megiddo, on the site of the modern Lejjun: used figuratively in the *Apocalypse* to signify the place of 'the battle of the great day of God'. It may, however, be *har migdo*, his fruitful mountain, 'the mountain land of Israel'. The phrase 'an Armageddon' expresses any great slaughter or final conflict, and has been frequently applied to the Great War of 1914-8. During this war severe fighting took place in Sept., 1918, on the field of Armageddon, the entrance to the passes of Megiddo. The battle ended in an overwhelming victory for General Allenby's armies. See *Megiddo*.

Armagh (a 'r-mä'), a county of Ireland, in the province of Ulster; surrounded by Monaghan, Tyrone, Lough Neagh, Down, and Louth; area, 328,086 acres, of which about a half is under tillage. The north-west of the county is undulating and fertile. The northern part, bordering on Lough Neagh, consists principally of extensive bogs. On the southern border is a range of barren hills. The chief rivers are the Blackwater, which separates it from Tyrone; the Upper Bann, which discharges itself into Lough Neagh;

and the Callan, which falls into the Blackwater. There are several small lakes. The manufacture of linen is carried on very extensively. Armagh, Lurgan, and Portadown are the chief towns. The county sends three members to Parliament. Pop. 120,291.—The county town, *Armagh*, is situated partly on a hill, about half a mile from the Callan. It has a Protestant cathedral crowning the hill, a Gothic building dating from the eighth century, repaired and beautified recently; a new Roman Catholic cathedral in the pointed Gothic style, and various public buildings. It is the see of an archbishop of the Protestant Episcopal Church, who is primate of all Ireland, and is a place of great antiquity. Pop. 7356.

Armagnac (a`r-ma`-nya`k), an ancient territory of France, in the province of Gascony, some of the counts of which hold prominent places in the history of France. Bernard VII, son of John II, surnamed the Hunchback, succeeded his brother, John III, in 1391, and was called to Court by Isabella of Bavaria, with the view of heading the Orleans in opposition to the Burgundian faction, where he no sooner gained the ascendancy than he compelled the queen to appoint him Constable of France. He showed himself a merciless tyrant, and became so generally execrated that the Duke of Burgundy, to whom Isabella had turned for help, found little difficulty in gaining admission into Paris, and even seizing the person of Armagnac, who was cast into prison in 1418, when the exasperated populace burst in and killed him and his followers. John V, grandson of the above, who succeeded in 1450, made himself notorious for his crimes. He was assassinated in his castle of Lectoure in 1473 by an agent of Louis XI, against whom he was holding out.

Ar'mature, a term applied to the piece of soft iron which is placed across the poles of permanent or electro-magnets for the purpose of receiving and concentrating the attractive force. In the case of permanent magnets it is also important for preserving their magnetism when not in use, and hence it is sometimes termed the *keeper*. It produces this effect in virtue of the well-known law of induction, by which the armature, when placed near or across the poles of the magnet, is itself converted into a temporary magnet with reversed poles, and these, reacting upon the permanent magnet, keep its particles in a state of constant magnetic tension, or, in other words, in that constrained position which is supposed to constitute magnetism. A horse-shoe magnet should therefore never be laid aside without its armature; and

in the case of straight bar-magnets two should be placed parallel to each other, with their poles reversed, and a keeper or armature across them at both ends. The term is also applied to the core and coil of the electro-magnet, which revolves before the poles of the permanent magnet in the magneto-electric machine.

Arme Blanche, a term applied to the rapier and duelling-foil, and frequently also to all weapons other than fire-arms. The phrase is particularly applied to the sabres and lances carried by cavalry, but also to the bayonet.

Arme'nia, a mountainous country of Western Asia, of great historical interest as the original seat of one of the oldest civilized peoples in the world. The name Armenia occurs in the *Vulgate*, but the Hebrew name is Ararat. It has an area of about 120,000 sq. miles, and is intersected by the Euphrates, which divides it into the ancient divisions, Armenia Major and Armenia Minor. The country is an elevated plateau, enclosed on several sides by the ranges of Taurus and Anti-Taurus, and partly occupied by other mountains, the loftiest of which is Ararat. Several important rivers take their rise in Armenia, namely, the Kur or Cyrus, and its tributary the Aras or Araxes, flowing east to the Caspian Sea; the Halys or Kizil-Irmak, flowing north to the Black Sea; and the Tigris and Euphrates, which flow into the Persian Gulf. The chief lakes are Van and Urumiyah. The climate is rather severe. The soil is on the whole productive, though in many places it would be quite barren were it not for the great care taken to irrigate it. Wheat, barley, tobacco, hemp, grapes, and cotton are raised; and in some of the valleys apricots, peaches, mulberries, and walnuts are grown. The inhabitants are chiefly of the genuine Armenian stock, a branch of the Aryan or Indo-European race; but besides them, in consequence of the repeated subjugation of the country, various other races have obtained a footing. The total number of Armenians is estimated at 2,900,000, of whom one-half are in Armenia. The remainder, like the Jews, are scattered over various countries, and are generally engaged in commercial pursuits. They everywhere retain, however, their distinct nationality.

Little is known of the early history of Armenia, but it was a separate State as early as the eighth century B.C., when it became subject to Assyria, as it also did subsequently to the Medes and the Persians. It was conquered by

Alexander the Great in 325 B.C., but regained its independence about 190 B.C. Its king, Tigranes, son-in-law of the celebrated Mithridātes, was defeated by the Romans under Lucullus and Pompey between 69-66 B.C., but was left on the throne. Since then its fortunes have been various under the Romans, Parthians, Byzantine emperors, Persians, Saracens, and Turks. Until quite recently Armenia had no political existence, having been partitioned between Turkey, Persia, and Russia, the last acquiring considerable portions in 1829 and 1878. The hope of the Armenians to see their country formed into an autonomous province administered by Christians was frustrated by the Treaty of Berlin in 1878. The advanced party amongst the Armenians, therefore, determined to obtain their object by the production of disturbances and the spread of a revolutionary movement. The Porte retaliated by wholesale massacres of the Armenians in 1896, 1904, and 1908. The Armenian revolutionary and national parties in the meantime continued their activity and propaganda. Armenia proclaimed its independence in Aug., 1918. In Jan., 1920, the Supreme Council of the Allied Governments recognized the Armenian Republic of Erivan. A mandate for Armenia was also offered to the United States of America, but it was refused by the American Senate in May, 1920. On 18th March, 1922, Soviet Russia concluded a treaty with Turkey, giving to the latter most of Armenia. Batum was attached to Georgia. See *Erivan, Russia, Turkey*.

The Armenians received Christianity at an early date, most probably at the beginning of the third century, although native historians maintain that several of the apostles preached in Armenia. The real apostle of Armenia was Gregory the Illuminator, in the third century. During the Monophysitic disputes they held with those who rejected the twofold nature of Christ, and being dissatisfied with the decisions of the Council of Chalcedon (451) they separated from the Greek Church in 536. The Popes had at different times attempted to gain them over to the Roman Catholic faith, but have not been able to unite them permanently and generally with the Roman Church. There are, however, small numbers here and there of United Armenians, who acknowledge the spiritual supremacy of the Pope, agree in their doctrines with the Catholics, but retain their peculiar ceremonies and discipline. But the far greater part are yet Monophysites, and have remained faithful to their old religion and worship. Their doctrine differs from the

orthodox chiefly in their admitting only one nature in Christ, and believing the Holy Spirit to proceed from the Father alone. Their sacraments are seven in number. They adore saints and their images, but do not believe in purgatory. Their hierarchy differs little from that of the Greeks. The *Catholicus*, or head of the Church, has his seat at Etchmiadzin, a monastery near Erivan, the capital of former Russian Armenia, on Mount Ararat.

The Armenian language belongs to the Indo-European family of languages, and is most closely connected with the Iranic group. The Old Armenian or Haikan language, which is still the literary and ecclesiastical language, is distinguished from the New Armenian, the ordinary spoken language, which contains a large intermixture of Persian and Turkish elements. The most flourishing period of Armenian literature extended from the fourth to the fourteenth century. It then declined, but a revival began in the seventeenth century, and at the present day wherever any extensive community of Armenians has settled they have set up a printing-press. The Armenian Bible, translated from a Syriac version, and revised by means of the Septuagint, by Isaac the Great and St. Mesrop, early in the fifth century, is a model of the classic style.—BIBLIOGRAPHY: E. N. and H. Buxton, *Travel and Politics in Armenia*; N. T. Gregor, *History of Armenia*; W. L. Williams, *Armenia, Past and Present*.

Armentières (a`r-ma`n`-tyār), a town in France, department Nord, 10 miles W.N.W. of Lille, on the Lys. The town had extensive manufactures of linen and cotton goods and an extensive trade. The Germans captured Armentières by massed assault early in April, 1918, after methodically shelling the town for about two years and destroying almost every building in it. The enemy's offensive was intended to achieve greater results. Indeed, its object was to break through to the Channel ports. It began on 9th April, after artillery preparation, from La Bassée to Armentières. When, however, the battle of Armentières died down, the enemy plan to break through to the coast had been definitely and finally frustrated. German losses were extremely heavy, their attacks having been made with massed troops. (See *Ypres*.) Pop. 28,086.

Arm´felt, Gustav Moritz, Count of, Swedish soldier, born 1757, died 1814. Though he had been highly favoured and loaded with honours by Gustavus III, he incurred the enmity of the Duke of Sudermania, guardian of the

young king, Gustavus IV, and was deprived of all his titles and possessions. He was restored to his fortune and honours in 1799, when Gustavus IV attained his majority, and held several high military posts. Ultimately, however, he entered the Russian service, was made count, chancellor of the University of Abo, president of the department for the affairs of Finland, member of the Russian Senate, and served in the campaign against Napoleon in 1812.

Armida (a ´ r-mē´da ´), a beautiful enchantress in Tasso's *Jerusalem Delivered*, who succeeds in bringing the hero Rinaldo, with whom she had fallen violently in love, to her enchanted gardens. Here he completely forgets the high task to which he had devoted himself, until messengers from the Christian host having arrived at the island, Rinaldo escapes with them by means of a powerful talisman. In the sequel Armida becomes a Christian.

Ar´millary Sphere (Lat. *armilla*, a hoop), an astronomical instrument consisting of an arrangement of rings, all circles of one sphere, intended to represent the principal circles of the celestial globe, the rings standing for the meridian of the station, the ecliptic, the tropics, the arctic and antarctic circles, &c., in their relative positions. Its main use is to give a representation of the apparent motions of the celestial bodies.

Armin´ians, a sect or party of Christians, so called from Jacobus *Arminius* or Harmensen. (See *Arminius*.) They were called also *Remonstrants*, from their having presented a *remonstrance* to the States-general in 1610. The Arminian doctrines are: (1) Conditional election and reprobation, in opposition to absolute predestination. (2) Universal redemption, or that the atonement was made by Christ for all mankind, though none but believers can be partakers of the benefit. (3) That man, in order to exercise true faith, must be regenerated and renewed by the operation of the Holy Spirit, which is the gift of God; but that this grace is not irresistible and may be lost, so that men may relapse from a state of grace and die in their sins. Arminianism being a revolt against certain aspects of Calvinism, especially the absolutism of the eternal decrees, its doctrines were vehemently attacked by the Calvinists of Holland, and were condemned by the Synod of Dort in 1619. The Arminians, in consequence, were treated with great severity; many of them fled to, and spread in, other countries, and though

there is no longer any particular sect to which the name is exclusively applied, many bodies are classed as Arminians, as being opposed to the Calvinists on the question of predestination.—BIBLIOGRAPHY: Regenboog, *Historie der Remonstranten*; Caspar Brandt, *Life of Arminius* (English translation by J. Guthrie); W. B. Pope, *Compendium of Christian Theology* (3 vols.).

Armin'ius, an ancient German hero celebrated by his fellow-countrymen as their deliverer from the Roman yoke, born about 18-16 B.C., assassinated A.D. 19. Having been sent as a hostage to Rome, he served in the Roman army, and was raised to the rank of *eques*. Returning home, he found the Roman governor, Quintilius Varus, making efforts to Romanize the German tribes near the Rhine. Placing himself at the head of the discontented tribes he completely annihilated the army of Varus, consisting of three legions, in a three days' battle fought in the Teutoburg Forest. For some time he baffled the Roman general Germanicus, and after many years' resistance to the vast power of the Empire he drew upon himself the hatred of his countrymen by aiming at the regal authority, and was assassinated. A national monument to his memory was inaugurated on the Grotenburg, near Detmold, in 1875.—BIBLIOGRAPHY: see Tacitus, *Annals* (translated by Murphy); O. Kemmer, *Arminius*; F. W. Fischer, *Armin und die Römer*.

Arminius, Jacobus (properly Jakob Harmensen), founder of the sect of Arminians or Remonstrants, born in South Holland in 1560, died 1609. He studied at Utrecht, in the University of Leyden, and at Geneva, where his chief preceptor in theology was Theodore Beza (1582). On his return to Holland he was appointed minister of one of the churches in Amsterdam, and chosen to undertake the refutation of a work which strongly controverted Beza's doctrine of predestination; but he happened to be convinced by the work which he had undertaken to refute. Elected in 1603 professor of divinity at Leyden, he openly declared his opinions, and was involved in harassing controversies, especially with his fellow professor Gomarus. These contests, with the continual attacks on his reputation, at length impaired his health and brought on a complicated disease, of which he died. See *Arminians*.

Ar'mitage, Edward, English historical painter, born 1817, died 1896. He studied under Delaroche at the École des Beaux Arts, Paris, was one of the

ablest pupils of that painter, and in 1842 exhibited at the Salon (in the Louvre) a picture of *Prometheus Bound*. At the exhibition of cartoons for historical pictures in Westminster Hall (1843) he obtained a premium of £300 for his design of *Cæsar's First Invasion of Britain*. Other similar premiums were gained by his *Spirit of Religion* (1845), and *Battle of Meeanee* (1847—£500). He now went to study at Rome, and exhibited at the Academy in 1848 his *Henry VIII and Katherine Parr*, and his *Trafalgar (Death of Nelson)*. He had pictures in most of the subsequent Academy exhibitions up nearly to the time of his death. In 1867 he was elected an associate, and in 1872 a full academician. He did much for the restoration of fresco painting in England. A large number of his pictures were biblical in subject, such as *Ahab and Jezebel*, *Esther's Banquet*, *The Remorse of Judas*, *Joseph and Mary*, *Herod's Birthday Feast*, &c. As professor of painting to the Royal Academy he delivered lectures on painting, which were published in 1883. In 1898 appeared a volume of his *Pictures and Drawings*.

Armor'ica (from two Celtic words signifying 'upon the sea'), a name anciently applied to all north-western Gaul, afterwards limited to what is now Brittany. Hence *Armoric* is one name for Breton or the language of the inhabitants of Brittany, a Celtic dialect closely allied to Welsh.

Armour. See *Arms*.

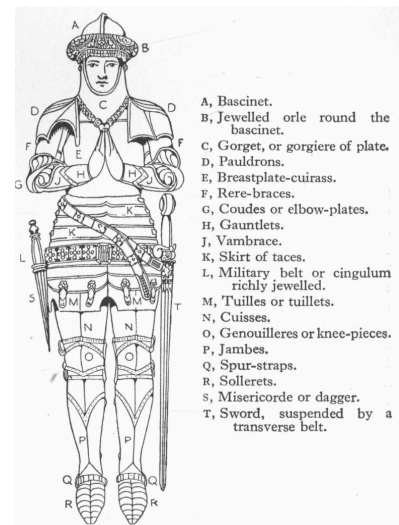
Armoured Car, a self-propelled car completely protected by bullet-proof armour-plating. Such a car is a stage in the development of mechanical warfare, i.e. warfare by means of a self-propelled, armed, and manned machine. The idea is a very ancient one, some form of protected vehicle having been in use among the Chinese in almost prehistoric times. The modern armoured car is constructed on a strongly-engined chassis, and is provided with a bullet-proof armour-plating both for engine and crew. This armour is continued low down over the wheels. The armament of such a car consists of two heavy machine-guns, firing through slits in the armour-plating of the body of the car.

Armour-plates, iron or steel plates with which the sides of vessels of war are covered with the view of rendering them shot-proof. See *Iron-clad Vessels*.

Arms, Coat of, or Armorial Bearings, a collective name for the devices borne on shields, on banners, &c., as marks of dignity and distinction, and, in the case of family and feudal arms, descending from father to son. They were first employed by the Crusaders, and became hereditary in families at the close of the twelfth century. They took their rise from the knights painting their banners or shields each with a figure or figures proper to himself, to enable him to be distinguished in battle when clad in armour. See *Heraldry*.

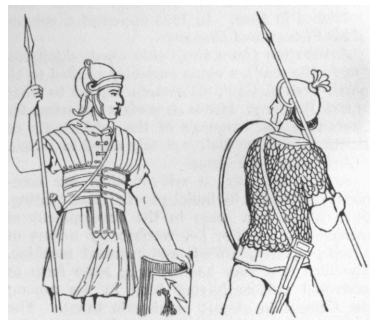
Arms, College of. See *Herald*.

Arms and Armour. The former term is applied to weapons of offence, the latter to the various articles of defensive covering used in war and military exercises, especially before the introduction of gunpowder. Weapons of offence are divisible into two distinct sections—firearms, and arms used without gunpowder or other explosive substance. The first arms of offence would probably be wooden clubs, then would follow wooden weapons made more deadly by means of stone or bone, stone axes, slings, bows and arrows with heads of flint or bone, and afterwards various weapons of bronze. Subsequently a variety of arms of iron and steel was introduced, which comprised the sword, javelin, pike, spear or lance, dagger, axe, mace, chariot scythe, &c.; with a rude artillery consisting of catapults, ballistæ, and battering-rams. From the descriptions of Homer we know that almost all the Grecian armour, defensive and offensive, in his time was of bronze; though iron was sometimes used. The lance, spear, and javelin were the principal weapons of this age among the Greeks. The bow is not often mentioned. Among ancient nations the Egyptians seem to have been most accustomed to the use of the bow, which was the principal weapon of the Egyptian infantry. Peculiar to the Egyptians was a defensive weapon intended to catch and break the sword of the enemy. With the Assyrians the bow was a favourite weapon; but with them lances, spears, and javelins were in more common use than with the Egyptians. Most of the large engines of war—chariots with scythes projecting at each side from the axle, catapults, and ballistæ—seem to have been of Assyrian origin. During the historical age of Greece the characteristic weapon was a heavy spear from 21 to 24 feet in length. The sword used by the Greeks was short, and was worn on the right side. The Roman sword was from 22 to 24 inches in length, straight, two-edged, and obtusely pointed, and as by the Greeks was worn on the right side. It was used principally as a stabbing weapon. It was originally of bronze. The most characteristic weapon of the Roman legionary soldier, however, was the *pilum*, which was a kind of pike or



Armour, from the effigy of Sir Richard Peyton, in Tong Church, Shropshire

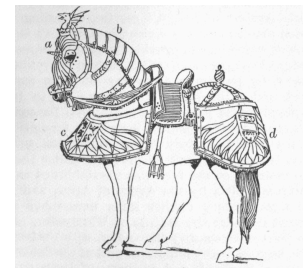
javelin, some 6 feet or more in length. The pilum was sometimes used at close quarters, but more commonly it was thrown. The favourite weapons of the ancient Germanic races were the battle-axe, the lance or dart, and the sword. The weapons of the Anglo-Saxons were spears, axes, swords, knives, and maces or clubs. The Normans had similar weapons, and were well furnished with archers and cavalry. The cross-bow was a comparatively late invention introduced by the Normans. Gunpowder was not used in Europe to discharge projectiles till the beginning of the fourteenth century. Cannon are first mentioned in England in 1338, and there seems to be no doubt that they were used by the English at the siege of Cambrai in 1339. The projectiles first used for cannon were of stone. Hand fire-arms date from the fifteenth century. At first they required two men to serve them, and it was necessary to rest the muzzle on a stand in aiming and firing. The first improvement was the invention of the match-lock, about 1476; this was followed by the wheel-lock, and about the middle of the seventeenth century by the flint-lock, which was in universal use until it was superseded by the percussion-lock, the invention of a Scottish clergyman early in the nineteenth century. The needle-gun dates from 1838. The only important weapon not a fire-arm that has been invented since the introduction of gunpowder is the bayonet, which is believed to have been invented about 1650. See *Cannon, Musket, Rifle, &c.*



Roman Armour—Soldiers wearing Cuirass



Chain Armour



Horse-armour of Maximilian I of Germany *a*, Chamfron. *b*, Manefaire. *c*, Poitrinal, poitrel, or breastplate. *d*, Croupiere or buttock-piece.



Greek Armour

Some kind of defensive covering was probably of almost as early invention as weapons of offence. The principal pieces of defensive armour used by the ancients were shields, helmets, cuirasses, and greaves. In the earliest ages of Greece the shield is described as of immense size, but in the time of the Peloponnesian War (about 420 B.C.) it was much smaller. The Romans had two sorts of shields: the *scutum*, a large oblong rectangular highly-convex shield, carried by the legionaries; and the *parma*, a small round or oval flat shield, carried by the light-armed troops and the cavalry. In the declining days of Rome the shields became larger and more varied in form. The helmet was a characteristic piece of armour among the Assyrians, Greeks, Etruscans, and Romans. Like all other body armour it was usually made of bronze. The helmet of the historical age of Greece was distinguished by its lofty crest. The Roman helmet in the time of the early emperors fitted close to the head, and had a neck-guard and hinged cheek-pieces fastened under the chin, and a small bar across the face for a visor. Both Greeks and Romans wore cuirasses, at one time of bronze, but afterwards of flexible materials. Greaves for the legs were worn by both, but among the Romans usually on one leg. The ancient Germans had large shields of plaited osier covered with leather; afterwards their shields were small, bound with iron, and studded with bosses. The Anglo-Saxons had round or oval shields of wood, covered with leather, and having a boss in the centre; and they had also corselets, or coats of mail, strengthened with iron rings. The Normans were well protected by mail; their shields were somewhat triangular in shape, their helmets conical. In Europe generally metal armour was used from the tenth to the eighteenth century, and at first consisted of a tunic made of iron



Allecret (Light Plate) Armour, A.D. 1540

rings firmly sewn flat upon strong cloth or leather. The rings were afterwards interlinked one with another so as to form a garment of themselves, called *chain-mail*. Another variety of this flexible armour was known as *banded-mail*. This consisted of rings sewn upon a fabric foundation, the whole being covered with leather. In addition to this, 'scale armour', which had been in use from the very earliest periods of history, was still in common fashion in the thirteenth century. By degrees the suit of mail was reinforced by the addition of pieces of plate on the breast, knees, elbows, and arms, and by the end of the fourteenth century the full suit of plate had been evolved, the mail being only worn as a skirt round the waist or as a coif attached to the helmet. The golden age of plate armour is the middle of the fifteenth century, when the design was light and graceful, and at the same time fully protective. In the sixteenth century, when 'shock tactics' of cavalry were the order of the day, the 'war harness' became heavier. This was particularly noticeable in the armour for the joust or tournament, in which sport the aim of the contestants was to score points and not to inflict injury. Many of these jousting armours weigh over 80 lb. The weapons in use through the whole of the plate-armour period were the lance, the sword, the axe or war-hammer, the long-bow, and the cross-bow. The introduction of fire-arms in the fourteenth century was one of the causes which led to the increase of weight in armour, for the armourer was continually improving and strengthening his products to make them proof against musket and pistol, and he generally succeeded, but by doing so increased the weight till it became insupportable. In the seventeenth century leg armour was abandoned, and by the end of the civil war the popular defence was the steel cap and breastplate. In the eighteenth century armour entirely disappeared, except for ceremonial, and was thought to be entirely obsolete till it was revived in the recent war in the form of the steel shrapnel-helmet, which was favoured by all the Allies and also by the enemy. The German troops occasionally used heavy body armour. Daggers and clubs, weapons likewise thought to be obsolete, were frequently used by all combatants, especially on raids.—BIBLIOGRAPHY: A. Hutton, *The Sword of the Centuries*; H. S. Cowper, *The Art of Attack*; C. ffoulkes, *Armour and Weapons*; C. H. Ashdown, *British and Foreign Arms and Armour*; C. Hall, *Modern Weapons of War by Land*.

Armstrong, John, Scottish poet and physician, born about 1709, died 1779. After studying medicine in Edinburgh he settled in London. In 1744 he published his chief work, the *Art of Preserving Health*, a didactic poem. This work raised his reputation to a height which his subsequent efforts scarcely sustained. In 1746 he became physician to a hospital for soldiers, and in 1760 he was appointed physician to the forces which went to Germany. After his return to London he published a collection of his *Miscellanies*, which contained, however, nothing valuable. He afterwards visited France and Italy, and published an account of his tour under the name of Lancelot Temple. His last production was a volume of *Medical Essays*.

Armstrong, William George, Lord, engineer and mechanical inventor, born at Newcastle-on-Tyne, 10th Nov., 1810. He was trained as a solicitor, and practised as such for some time. Among his early inventions were the hydro-electric machine, a powerful apparatus for producing frictional electricity, and the hydraulic crane. In 1847 the Elswick works, near Newcastle, were established for the manufacture of his cranes and other heavy iron machinery, and these works are now among the most extensive of their kind. Here the first rifled ordnance gun which bears his name was made in 1854. His improvements in the manufacture of guns and shells led to his being appointed engineer of rifled ordnance under Government, and he was knighted in 1858. This appointment came to an end in 1863, since which time his ordnance has taken a prominent place in the armaments of different countries. He was made a peer, as Baron Armstrong, in 1887. He died 27th Dec., 1900.

Armstrong Gun, a kind of cannon, so called from its inventor. It has an inner tube or core of steel, rifled with numerous shallow grooves, the tube being surrounded by a jacket of spirally-coiled bars of wrought iron, so disposed as to bring the metal into the most favourable position for the strain to which it is to be exposed. His first guns were small, but larger ones were soon made, and afterwards those of the very highest calibre. The breech-loading principle was also adopted in them, and special provision to effect this satisfactorily was invented by him. The improved shells introduced by him were of the elongated and pointed type now so well known, the charge being inserted in a special chamber behind the bore.

Army, a collection of bodies of men armed, disciplined, and organized for war. The essence of a modern army is that it shall be composed of organized units each under its own commander, grouped in formations of ever-increasing size, and owing allegiance through these commanders to one supreme head. Discipline and organization are essential, or such a force becomes merely a collection of armed men.

In the early days of our history every able-bodied man was, to a greater or lesser extent, a possible fighting man, and all had arms of one kind or another. Consequently, when an army was required, landowners and county authorities were ordered to provide the troops necessary. Every free landowner between the ages of sixteen and sixty was liable to service, which was limited to two months in a year. This was the Saxon 'fyrd' system. Later it was improved on by the institution of 'Thane's Service', which made it incumbent on the more considerable landowners to appear fully armed and mounted, and to serve for the whole campaign. The horse, however, was only used as a means of locomotion: for fighting purposes their riders dismounted, as did the dragoons of the seventeenth century and the mounted infantry of still more modern times. The fyrd was an unorganized and undisciplined force and entirely ephemeral in its nature, so that we find the Danish kings of England casting about for some more permanent force, which came into existence under the title of the 'House Carles', or Royal Guard. With the Norman Conquest the fyrd was largely supplanted by the feudal system of knight's service, according to which the country was divided into knight's fees, each of which had to provide its quatum of men. The gradual appearance of the custom of avoiding service by payments of money—in time regulated under the name of scutage—led to the employment of paid mercenaries, who for some two centuries were almost invariably foreigners. In the twelfth century it was found that sufficient troops could not be provided under these two systems, so the fyrd was re-established as a National Militia by the Assize of Arms, and in the next century further steps were taken to protect it under the Statute of Winchester. In the fourteenth century the archer, with his longbow, became a very important part of the fighting forces of England, and an army of those days consisted of the heavily-armed and armoured knights and men-at-arms for shock action, and the unarmoured archers for 'volley action', to use a later term. With the gradual disappearance of the foreign mercenaries,

it became the custom for the king to issue indents to certain influential subjects for the raising of paid troops. From this custom arose the free companies, which, in time, became nothing more or less than commercial undertakings. The indents were accepted, and the men enlisted primarily for what could be got out of the business of fighting, either in the shape of ransom or the sack of towns. Some attempt was also made at tactical organization, and an army of the period was divided into vanguard, battle, and rearguard. Artillery also was beginning to be developed in Germany for siege purposes. The sixteenth century saw the first formation of companies into regiments, though as yet of no fixed strength. Arms were also modernized, and by the end of the century muskets, 18-foot pikes, and swords, were the arms of infantry instead of the varied assortment of halberds, pikes, muskets, harquebuses, and longbows common at the beginning. Elizabeth introduced the press-gang as an aid to recruiting, and abolished the white coat of the soldier in favour of a long red or blue cassock. In the next century Cromwell's new model army became the first standing army of England, and, though it was disbanded by Act of Parliament at the Restoration, one of its regiments—Monk's—remained, and is now the Coldstream Guards. After this regiments were raised from time to time on one pretext or another, and the nucleus of a standing army became a *fait accompli*, though it was for a long time considered more as an appanage of the king than as a national institution. With the standing army came the first beginnings of civilian control, a Secretary-at-War being appointed in 1660. He had, however, no responsibility, and was subordinate to the commander-in-chief, and it was not till 1710 that he assumed his present responsibility to Parliament. During the eighteenth century the strength of the army rose or fell according to the state of the military barometer and the success or otherwise of the various recruiting expedients, among which was the first attempt at a short-service system in 1703. In 1871-2 the old numbering in regiments was abolished and a territorial designation substituted. According to this scheme, the first twenty-five regiments, all of which had already two battalions, were grouped together, the rest being joined arbitrarily to form new regiments under county designations. With these regiments were affiliated the militia and volunteer battalions, which have now been amalgamated into the Special Reserve and the Territorial Force.

For the requirements of the war of 1914-8 the Empire, as a whole, including India, raised and maintained a total of 8,654,467 men, of which the contribution of the United Kingdom was over 6,000,000. Casualties for the whole Empire were 3,060,616, of which the United Kingdom has for her share nearly 2,500,000, including 666,083 killed, 1,644,786 wounded, and 140,312 missing.

During 1918 the combatant strength of all arms of the British army in France fluctuated between 1,293,000 in March and 1,164,790 in November, while the rifle or infantry strength was from 616,000 to 416,748 during the same periods. From the date of the armistice to 31st Dec., 1919, the following number of demobilizations and discharges were effected:—

Demobilized.—Officers, 144,144; other ranks, 3,332,882.

Discharged as medically unfit.—Officers, 23,476; other ranks, 207,500.

Discharged from reserves.—Other ranks, 143,603.

The modern British army is governed by the Army Council (instituted 1904), presided over by the Secretary of State for War. This Council, which consists of five military and five civilian members, including the president, works through the War Office, of which the principal departments are in charge of one or other of the members of the Council. On the military side these departments are those of the Chief of the Imperial General Staff, the Adjutant-General to the Forces, the Quartermaster-General to the Forces, and the Master-General of the Ordnance. For administrative and training purposes the United Kingdom is divided into seven Commands and the London District. When necessary, Commands are further subdivided into Districts. The army, generally speaking, consists of (1) the Regular Army, (2) the Territorial Force, and (3) the Reserves. The service battalions, which formed such a large and important part of the army in the war, do not, properly speaking, form part of the permanent military forces, though the organization of the army as a whole is such that it is capable of expansion to any extent by the process of raising new battalions and affiliating them to existing regular or territorial units. The regular army comprises the Household Cavalry, Cavalry of the Line, the Royal Regiment of Artillery, the Corps of Royal Engineers, the Brigade of Guards, and Infantry of the Line. In addition there are administrative troops and services such as the

Royal Army Service Corps, the Royal Army Medical Corps, with its allied service Queen Alexandra's Imperial Military Nursing Service, the Royal Army Ordnance Corps, and others. Other corps brought into existence during the war, such as the Tank and Machine-gun Corps, have at present no permanent status. In the future, however, machine-gun companies will form an integral portion of each battalion of the regular army.

Under the conditions of the war the old national method of voluntary recruitment was found to be insufficient, and recourse was had to the principle of universal military service. Under the Military Service Acts the age limit was gradually raised till it finally included all men up to the age of fifty. Exceptions were made in the case of munition-workers, or those employed on work of national importance. Since the signature of the Treaty of Peace the army so raised was gradually demobilized till, by 31st March, 1920, it had decreased to 400,000, including 100,000 Indians paid by the Imperial Government. Concurrently with demobilization, voluntary enlistment was reintroduced, and the post-war army is once more a voluntary one, in which men serve under very much improved conditions as to pay and prospects. The period of service under this system is twelve years, of which seven normally are with the colours and five in the reserve: in certain cases modifications of these periods are allowed, and, in addition, a soldier may be allowed to extend his colour service to the full twelve years, or, in exceptional cases, to complete twenty-one years for pension. Discharge or transfer to the reserve is ordinarily granted on completion of the agreed period of service. Pay of all ranks was very materially improved in 1919. Whereas formerly a private soldier on enlistment received 1s. a day, he now receives 2s. 9d., and after two years' service 3s. 6d. To this last amount is added, under very reasonable conditions, a further daily sum of 6d. proficiency pay. A sergeant now gets 7s. a day instead of from 2s. 4d. to 3s. 4d., and a regimental sergeant-major 14s. instead of 5s. or 6s. Add to these rates of pay free rations, free housing, free medical attendance, and, in the future, doubtless free education, and it must be admitted that the present-day soldier is not badly paid. The rate of pay is a flat rate for all arms, special allowances being given where necessary.

The Household Cavalry comprises the 1st and 2nd Life Guards and the Horse Guards (Blues). In peace-time they serve only in London and Windsor. They alone retain the old cavalry rank of corporal of horse instead

of sergeant. Cavalry of the line consists of dragoon guards, dragoons, hussars, and lancers. The dragoon guards are numbered separately from 1 to 7, while dragoons, hussars, and lancers run consecutively from 1 to 21. A regiment of cavalry is commanded by a lieutenant-colonel and consists of 25 officers and 497 other ranks. Each regiment is organized in three squadrons commanded by majors, while a squadron is divided into four troops, each under a subaltern officer, troops being further subdivided into sections under non-commissioned officers. Cavalry regiments, except hussars, carry guidons or standards for ceremonial purposes. These differ from the colours of infantry in that they are not consecrated and are carried by non-commissioned officers instead of by officers. Hussars carry no standards. There are six cavalry depots for recruiting and preliminary-training purposes, i.e. for lancers at Woolwich, hussars at Scarborough, Bristol, and Dublin, and dragoons at Newport (Mon.) and Dunbar. The Cavalry Special Reserve consists of the Irish Horse and King Edward's Horse, and during the war reserve cavalry regiments were maintained.

The Royal Regiment of Artillery comprises batteries of horse artillery designated by letters of the alphabet, and batteries of field, siege, heavy, and mountain by numbers. A battery, with some exceptions in the case of the heaviest type, consists of six guns or howitzers, horse artillery having 13-pounder guns, and field artillery 18-pounder guns or 4.5-inch howitzers. Horse artillery is intended to act with cavalry, and is therefore provided with a rather lighter gun. It can go anywhere that cavalry can go, and all the gun detachments are mounted. Field artillery works with infantry, and the gun detachments either walk or are carried on the limbers, only the officers, certain non-commissioned officers, and specialists such as scouts, range-finders, and trumpeters being mounted.

A battery is commanded by a major, with a captain as second-in-command, and is organized in three sections of two guns each under a subaltern. These sections are again subdivided into subsections of one gun each under a sergeant. Each gun is drawn by six horses, the driver of the leading pair being responsible for direction and pace. A corporal in the Royal Artillery is known as a bombardier, and the rank and file as gunners or drivers, according to their special duties, though drivers are also trained to some extent as gunners.

Heavy and siege artillery have come into their own in the late war, and consist roughly of all armament heavier than that of field artillery. Sixty-pounders and 4.7-inch howitzers form heavy batteries, while guns of 6 inch and upwards drawn by mechanical transport or mounted on railway trucks are known as siege batteries. Mountain artillery, of which most of the batteries are in India, is armed with 2.95-inch screw guns capable of being dismantled and carried piecemeal on mules. Another form is found on the west coast of Africa, where carriers take the place of mules. These guns are brought into action very quickly, but their shell-power is small.

The corps of Royal Engineers is responsible for the construction and maintenance of barracks, fortifications, and other military works, and for the personnel required for search-lights and electrical communications of the coast and anti-aircraft defences. With few exceptions the personnel of the corps is recruited entirely from skilled tradesmen and artisans. For service in the field, Royal Engineer units known as field squadrons and field companies accompany the fighting troops, and carry a certain amount of bridging material and tools. More highly specialized units carry out such services as mining, heavy bridging, railway, survey, and sound-ranging work. An important feature of Royal Engineer work in war is the supply of materials and stores, for which purpose an elaborate organization is provided in addition to the units already noted.

The Brigade of Guards—the infantry of the household troops—comprises the five regiments of foot-guards. These are the Grenadier, the Coldstream, the Scots, the Irish, and the Welsh Guards of from one to three battalions each. Being household troops these regiments are subject to certain special regulations and have certain privileges. As a general rule they serve only in London, Windsor, or Aldershot, and only leave England for active service, though individual battalions have served in the past both in Cairo and Gibraltar.

The infantry, of which there are sixty-eight regiments of from two to four battalions each, provides the bulk of the army. Infantry is formed into regiments for recruiting and territorial distribution purposes, but the battalion is the actual unit both for fighting and administration. In many cases the Army List gives the name of an officer holding the appointment of colonel of the regiment: this is in all cases a purely honorary appointment

and entails no duties or responsibilities. An infantry battalion is commanded by a lieutenant-colonel and consists of 32 officers and 1000 other ranks. Both in peace and war a battalion is divided into head-quarters and four companies, each of the latter having six officers, including the company commander, who is a major or senior captain. Subalterns command platoons. For recruiting purposes for infantry of the line the country is divided into Regimental Districts, in which are located the depots of the regiment concerned: these depots are commanded by a senior officer of one of the battalions of the regiment with the necessary staff for training purposes. Recruits are usually first trained at the depot and later transferred to the battalion requiring them. The Regimental Districts are again combined into larger districts in which are situated the Record Offices dealing with the regiments of the district. The denomination of the district dealing with any particular regiment is shown in the Army List in brackets. The principles of interior organization are the same throughout the army, and as they can be best illustrated with the example of an infantry battalion a short description of this organization follows. Owing to the continual growth of military science, the improvement in arms and means of destruction generally, and the confusion and noise inseparable from a modern battle, the size of the personally-controlled unit has gradually decreased till, in the present day, in the British army, it is accepted as an axiom that no larger number of men than six can be conveniently controlled in battle by one man. In former days companies, battalions, and even larger formations were both controlled and received their executive orders direct from their commanders—and to such an extent was this carried that Fortescue, in his *History of the British Army*, notes that Marlborough was in the habit of putting his whole army through the platoon exercise by means of flags and bugle-calls. This, of course, was not actually in face of the enemy, but the principle is the same. The stress of modern war now makes individual control of large bodies impossible, and the British army is therefore organized both for peace and war in a series of units of ever-increasing size, each under its own commander, who is responsible to his immediate superior for the well-being, training, and leading of his command. Taking the infantry organization as an example, we find that in the lowest stage, that of the 'section', the command is both personal and direct, in that the corporal controls and commands the six men composing his fighting unit personally and directly by word of mouth. In peace-time,

and for administrative and training purposes, the section may reach to ten men, who live, work, and play together. In the next stage—that of the 'platoon', consisting of four sections—we find the control is rather less personal and direct, in that the platoon commander, a subaltern, controls his command largely through his subordinates, the section commanders. A further stage is that of the 'company', which consists of four platoons and company head-quarters. A company is commanded by a major or senior captain, has a captain as second-in-command, and a company sergeant-major and quartermaster-sergeant to assist in running it. Here again the control is less direct though still personal. The next stage is the amalgamation of companies into a battalion, consisting of a head-quarters and four companies. Battalion head-quarters consist of a lieutenant-colonel commanding, a major second-in-command, an adjutant, and a quartermaster. Certain other officers, when required, and the regimental sergeant-major and quartermaster-sergeant, and various other ranks make up the total of some 130. The commanding officer of a battalion is directly responsible for the well-being of his command, for its training, discipline, equipment, and general efficiency. In carrying on his duties he works through his company commanders, and with the assistance of the regimental staff mentioned above, so that we have a direct chain of command and responsibility from the corporal commanding a section of six to ten men through platoons and companies to the lieutenant-colonel commanding a battalion of some thousand men. A detail of armament made possible by the enormous increase of machine-guns necessitated during the war is interesting. Thirty-two Lewis-guns are now provided for each infantry battalion, and are distributed to alternate sections in a platoon. Thus in each platoon two sections are known as rifle sections and two as Lewis-gun sections, and these arms are normally used by the respective sections; but men of all sections are trained in the use of both rifle and Lewis-gun.

When we come to formations larger than a battalion, we find the system of control and command becoming less and less personal and direct, as in all such formations the commander works to a less or greater extent through his staff. Roughly speaking, the staff is of two divisions, the one consisting of the general staff branch and the other of the branch of the adjutant and quartermaster-general. Again speaking very generally, the general staff is charged with duties bearing directly on military operations, while officers

of the adjutant and quartermaster-general's branch deal more with administrative questions. Officers of the general staff are known as general staff officers, while those of the other branch are called, for example, assistant or deputy-assistant adjutant or quartermaster-general, according to their several duties.

The formation in which distinct and separate units are first collected under one superior commander is known as a brigade. This, according to present establishment, consists of three battalions and a trench-mortar battery, the whole under a general officer called a brigadier-general, assisted by a staff of two officers—a brigade-major and a staff-captain. Since March, 1920, however, the title of brigadier-general has been altered to 'colonel-commandant'. The strength of a brigade is something over 3000 of all ranks. In a division, which is the next highest formation, and which is commanded by a major-general with a staff of three general staff officers and three officers belonging to the A.G. and Q.M.G. branch, we find the first appearance of a mixed force. It is not a force of 'all arms', as cavalry is not included, but, in addition to infantry (three brigades), it has a considerable strength in artillery, besides engineers and the necessary administrative troops. Two or more divisions, together with a cavalry regiment and certain other troops, form an 'army corps', and two or more corps go to make up an 'army'. These are not at present peace-time formations of the British army.

Of the administrative troops and services already mentioned, the Royal Army Service Corps provides for the material wants of the army both in the way of food and transport. It is organized in companies designated by numerals.

The Royal Army Medical Corps provides the personnel and organization for the medical and sanitary services of the army. In peace-time this service is organized on a garrison basis, hospitals being established where required for the use of all troops in that particular garrison. For war purposes medical officers are still attached to regiments, and in addition the corps provides the personnel and organization necessary for field ambulances, casualty clearing-stations, hospital trains and ships, and various classes of fixed hospitals. The corps is organized in numbered companies, and the rank and file are trained in first aid and ambulance duties generally. It is

administered by a director-general of Army Medical Services with the rank of lieutenant-general, who is an officer of the adjutant-general's department.

The other departments and administrative services of the army consist of the Royal Army Ordnance Corps, dealing generally with munitions of war; the Army Pay Department; the Royal Army Chaplains' Department; and the Royal Army Veterinary Corps, of which the functions are sufficiently designated by their title. In addition, there are manufacturing establishments at Woolwich Arsenal and elsewhere.

The Army Reserve consists of men who have completed their term of colour service, or service with a unit, and have thus passed into civil life, though still remaining liable for a period of years to be recalled to the colours if mobilization is ordered.

The Special Reserve was formed under the Territorial and Reserve Forces Act of 1907 out of the old militia. It consists generally of one battalion to each regiment of infantry, and is numbered consecutively with regular battalions of the regiment. It will, in the future, probably be again known as the Militia.

The Royal Marines—artillery and infantry, or the 'blue' and the 'red' marines, Kipling's "soldier and sailor too"—are not part of the army proper, as they are administered entirely by the Admiralty. They are, however, amenable to the Army Act when serving ashore. The term of service is for twelve years, which may be extended to make up twenty-one. Men may be transferred to or from the army at their own request.

The Territorial Force, or, as it is to be called in future, the Territorial Army, is raised entirely on a county or territorial basis. It was originally created by the Territorial and Reserve Forces Act of 1907 out of a nucleus of the old yeomanry and volunteers. It is raised and administered by County Associations in each county and principal city. These associations consist of a president, chairman, military representative, and co-opted members. The administration of the Territorial Army is carried out by the County Associations in accordance with schemes provided by the Army Council, while all questions of training are reserved to the War Office. The Territorial Army consists of all arms, including machine-gun corps and the necessary administrative services: its full war establishment is fixed at approximately

345,000 of all ranks, though, for the present, only some 60 per cent of them are to be enlisted. The rejuvenated Territorial Army is to be in all respects a true second line of imperial defence, self-contained and self-supporting, while the regular army and its special reserve of militia battalions form the first line. Defence entails a certain amount of offence to bring it to a successful issue: it has therefore been decided that the new Territorial Army will not be relegated merely to the duty of guarding the country from invasion, but will, in a national emergency, be entitled to take its place under its own organization in the fighting line in any part of the world where its services may be required. This will entail enlistment for general service, but the interest of the force and of individuals composing it are safeguarded by the proviso that before the Territorial Army can be ordered out of the country an Act authorizing the movement be passed by Parliament. It is further provided that the Territorial Army will on no account be called on to supply drafts for regular regiments, and that in case fresh regiments have to be raised on the lines of the New Army, the machinery of the Territorial Army will be used to organize them. Enlistment will be for three or four years, according to whether a man has served during the European War (1914-8) or not; age limits are normally between 18 and 38. The army is to be organized in one cavalry (yeomanry) division of 12 regiments, and 14 infantry divisions each under a selected general officer, either regular or territorial. Pay and allowances during training periods will be as in the regular army, and in addition certain bounties will be obtainable. Training periods will be fifteen days in camp annually, besides a minimum number of drills and a musketry course. On completion of colour service a man will pass to the Territorial Reserve.

The New Army, consisting of the 'service battalions' of existing regiments, is a product of the war. When, on the outbreak of war, many new regiments were rapidly raised, they were affiliated to regular regiments with consecutive numbers after the territorial battalions, and this organization was continued and extended to cope with the personnel obtained under the Military Service Acts.

Educational establishments connected with the army include the Staff College, the Royal Military College at Sandhurst, and the Royal Military Academy at Woolwich. At the Staff College officers obtain a course of instruction and study to fit them for service on the staff of the army. At

Sandhurst, where the course of instruction is two years, some 700 gentlemen cadets are trained for commissions in the guards, cavalry, and infantry of the line, and the Indian army. At Woolwich gentlemen cadets desirous of entering the Royal Artillery or the Royal Engineers receive their training. In addition to these there are schools of gunnery and engineering, the Small Arms School at Hythe, the School of Physical Training at Aldershot, and many others: while, for sons and orphans of soldiers, there are the Duke of York's Royal Military School and the Royal Hibernian School. The Royal Hospital, Chelsea, and the Royal Hospital, Kilmainham, Dublin, are institutions for the care of old and distressed soldiers. In every garrison there are garrison schools for soldiers under fully qualified army schoolmasters, while in the future there is likely to be a very great increase in educational facilities of all kinds for the rank and file of the army. The training of the British army for war now embraces a variety of subjects, and at the Royal Military College and Academy the gentlemen cadets are not only taught the principles and practice of their future profession, but are also instructed in the methods of imparting their knowledge to others. For example, the course of training at the Royal Military College embraces drill and weapon training—under which head is included musketry (both theory and practice) and bayonet work—physical training, and riding. As part of the physical-training course they receive instruction as to the best methods of organizing regimental assaults-at-arms and sports. Among the more academic subjects are military history and tactics, field sketching and topography, field engineering, military law and administration, and elementary hygiene. A great deal of practical work is done, and the course of two years is designed to fit a young officer, on joining his regiment, to undertake the entire charge and training of his troop or platoon. At the Royal Military Academy extra subjects, such as artillery work and more advanced engineering, are also taught.

As to the soldier's training generally, drill is insisted on as an aid to discipline, which it undoubtedly is, and exact performance of the various movements ordered is expected. In other branches of training more individuality is allowed, and the days when the bayonet-exercise was performed by a battalion to the music of the regimental band having passed, considerable latitude as to positions and execution is permitted in this particular branch, attention being principally concentrated on inculcating

the 'offensive spirit'. The modern soldier also learns how to use a Lewis-gun, to throw or fire a grenade, what to do in case of a gas attack, the rudiments of field engineering, and how to keep himself healthy.

In addition to the more generally-known units of the army there are certain corps which, though raised in the colonies, still form part of the army, and which are administered by the imperial authorities. Under this head are the Royal Malta Artillery (local service); the West India Regiment (two battalions) and the West African Regiment, both for general service and both administered by the War Office. Among other colonial corps maintained by the imperial Government, though not forming part of the regular army, are the West African Frontier Force (Nigeria) and the King's African Rifles (East Africa). Both these are administered by the Colonial Office.

Dominions.—The military forces of the self-governing dominions are raised and organized under the laws of such dominions.

Those of the Commonwealth of Australia are organized on a system of compulsory military training for all males between the ages of twelve and twenty-six. In the earlier stages boys are trained in cadet corps, from which they pass to the Citizen Army, and from there, having attained the age of twenty-six, to recognized rifle clubs. The annual period of training in the Citizen Army is sixteen days. When the scheme is in full working order this force will consist of twenty-three 4-battalion brigades of infantry, twenty-three regiments of light horse, fifty-six 4-gun batteries, and the necessary complement of engineers and administrative troops. During the war this organization was in abeyance, and regiments were raised as required for overseas service, and, though proposals for conscription were negatived, the commonwealth still managed to send some 330,000 men to the various theatres of war out of 417,000 raised. The casualties, killed, wounded, and missing, were 210,724.

The Commonwealth also maintains a small permanent force of trained professional soldiers.

The New Zealand forces are also organized on the principle of universal training for all males. The details differ somewhat from those in favour in Australia, but the principle is the same, i.e. that every male should be

trained for home defence. Boys of from twelve to eighteen years of age are trained in cadet corps, from which they pass to regiments of the Territorial Force, and from twenty-five to thirty belong to the reserve. Cadets do annually a specified number of drills, while the territorial training extends to seven clear days, a musketry course, and certain drills every year. For the purposes of the war, conscription was introduced in 1916, and 220,000 men were raised between 1914 and 1918, out of which the casualties were nearly 57,000.

Canada, unlike Australia and New Zealand, has no system of graduated military training. The military forces of the dominion are organized as a militia under a Minister of Militia and Defence working with a Council. This militia is recruited by voluntary enlistment, and, on the outbreak of war, consisted of a permanent force of 3000 and some 60,000 men who had received militia training. This made possible the rapid dispatch to France of a division which, by 1916, had increased to a corps of four divisions and a cavalry brigade. Like the Mother Country and New Zealand, Canada introduced conscription in 1917, and during the war raised nearly 641,000 men and suffered 206,149 casualties, of which 56,110 were killed, 149,733 wounded, and 306 missing.

The Union of South Africa divides its military forces into the permanent force and the citizen force. There is also a coast-defence force. The permanent force consists of the five regiments of the South African Mounted Rifles. South Africa's greatest military effort during the war was directed towards German South-West and East Africa, but some 27,000 men were enlisted for and sent to Europe out of a total number of 136,000 raised. This total does not include coloured troops. The casualties were 18,000.

In other self-governing portions of the Empire troops were raised as required, and in the West and East African colonies the existing formations of native troops were considerably increased for service in suitable portions of the various theatres of war. The official statement of troops raised shows under the heading of 'other colonies' 134,837, including coloured troops from South Africa and the West Indies. The casualties among them amounted to 7519.

The Army in India.—The military forces in India consist of those units of British cavalry, artillery, and infantry temporarily serving in the country, and the Indian army proper, consisting of regiments recruited from among the native inhabitants and normally serving there. Enlistment is voluntary and for general service, one of the promises made by a man on enrolment being "to go wherever ordered by land and sea and not to allow caste usages to interfere with his duties as a soldier". The Indian army, as a disciplined and organized force, dates from the years between 1748 and 1758. In 1748 Major Stringer Lawrence arrived in Madras with a commission from the Company as commander-in-chief. His first act was to form the existing European independent companies into regiments; his second to raise certain native independent companies. In 1758 he formed these companies in their turn into battalions, which he designated 'coast sepoy', and which still exist under their present names of the 61st Pioneers and following numbers. His system was extended to the other presidencies, and at the period of the mutiny, in 1857, the native army in India consisted of some 230,000 regular troops, besides irregulars. When the post-mutiny reconstruction took place, the army was reorganized on an irregular basis instead of as regular regiments on the British model. According to this new system, the number of British officers in a regiment was considerably reduced; native officers were given command of troops and companies, while the British officer's command became the squadron, or wing. Native artillery, with the exception of certain mountain batteries, was abolished, and cavalry was reconstituted on the Silladar system, whereby, in consideration of a larger monthly pay than was given to the infantry sepoy, the trooper, or sowar, provided his own horse and sword. The system thus introduced virtually remains to the present day, though it has been modified and improved to suit later conditions. The infantry officer's command has decreased from the wing of four companies to the double company of two, and it is now known as a company and is organized in four platoons on the British service model, platoons being commanded by Indian officers. Of late years the number of British officers with an Indian regiment has been increased to twelve, and at the present time a committee is sitting in India to deliberate on the future construction of the army. It is therefore impossible to give details of its future strength. This, just before the war, was some 160,000, organized in 38 regiments of cavalry, the corps of guides, 3 regiments of sappers and miners, 118 regiments of infantry of 1 battalion each, and 10

regiments of Gurkhas of 2 battalions each. There were also 13 mountain batteries. The 'Imperial Service Troops', of which many contingents took part in the war, are raised, paid, and maintained by princes and chiefs as a contribution to the defence of the country, while their training is supervised by British inspecting officers. The 'Indian Defence Force', which has lately replaced the volunteers, and in which service is compulsory for Europeans, is available for home defence only. During the war India, by voluntary enlistment, provided 1,401,350 men. Of these many new regiments were formed, and second, third, and fourth battalions added to existing regiments. Casualties were very nearly 114,000, including some 48,000 killed. Native Indian officers of cavalry are known as *ressaldars*, *ressaiders*, and *jemadars*, while those of the infantry are called *subadars* and *jemadars*. In each regiment the senior Indian officer is called *ressaldar* or *subadar-major*.

The army in India, by which is meant all military forces in India, is administered by a commander-in-chief, who is a member of council. The head-quarter staff includes a military secretary, the chief of the general staff, an adjutant and a quartermaster-general, director-general of ordnance and military works, and a director of medical services.—BIBLIOGRAPHY: Hon. J. W. Fortescue, *History of the British Army*; C. W. C. Oman, *A History of the Art of War: Middle Ages*; C. H. Firth, *Cromwell's Army*; C. Walton, *History of the British Standing Army, 1660-1700*; War Office, *Army Book for the British Empire*; F. N. Maude, *Evolution of Modern Strategy*; G. F. R. Henderson, *The Science of War*; C. Romagny, *Histoire générale de l'armée nationale*; Heimann, *L'Armée allemande*.

Army Act. See *Military Law*.

Army Worm, the very destructive larva of the moth *Heliothrips* or *Leucania unipuncta*, so called from its habit of marching in compact bodies of enormous number, devouring almost every green thing it meets. It is about 1½ inches long, greenish in colour, with black stripes, and is found in various parts of the world, but is particularly destructive in North America. The larva of *Sciara militaris*, a European two-winged fly, is also called army worm.



Army Worm

Arnat´to, or **Annotta**. See *Annatto*.

Arnauld (a´r-nō), the name of a French family, several members of which greatly distinguished themselves.—Antoine, an eminent French advocate, was born 1560, died 1619. Distinguished as a zealous defender of the cause of Henry IV, and for his powerful and successful defence of the University of Paris against the Jesuits in 1594. His family formed the nucleus of the sect of the Jansenists (see *Jansenius*) in France.—His son Antoine, called the *Great Arnauld*, was born 6th Feb., 1612, at Paris, died 9th Aug., 1694, at Brussels. He devoted himself to theology, and was received in 1641 among the doctors of the Sorbonne. He engaged in all the quarrels of the French Jansenists with the Jesuits, the clergy, and the Government, was the chief Jansenist writer, and was considered their head. Excluded from the Sorbonne, he retired to Port Royal, where he wrote, in conjunction with his friend Nicole, a celebrated system of logic (hence called the *Port Royal Logic*). On account of persecution he fled, in 1679, to the Netherlands. His works, which are mainly controversies with the Jesuits or the Calvinists, are very voluminous.—His brother Robert, born 1588, died 1674, retired to Port Royal, where he wrote a translation of Josephus, and other works.—Robert's daughter Angélique, born 1624, died 1684, was eminent in the religious world, and was subjected to persecution on account of her unflinching adherence to Jansenism.

Ar´nauts. See *Albania*.

Arndt (a´rnt), Ernst Moritz, German patriot and poet, was born 1769, died 1860. He was appointed professor of history at Greifswald in 1806, and stirred up the national feeling against Napoleon in his work *Geist der Zeit* (*Spirit of the Time*). In 1812-3 he zealously promoted the war of independence by a number of pamphlets, poems, and spirited songs, among which it is sufficient to refer to his *Was ist des Deutschen Vaterland?*, *Der Gott, der Eisen wachsen liess*, and *Was blasen die Trompeten? Husaren, heraus!*, which were caught up and sung from one end of Germany to the other. In 1817 he married a sister of the theologian Schleiermacher, and settled at Bonn in order to undertake the duties of professor of history. He was, however, suspended till 1840 on account of his liberal opinions, when he was restored to his chair on the accession of Frederick William IV.

Arndt, Johann, celebrated German mystic theologian, born 1555, died 1621. His principal work, *Wahres Christenthum* (True Christianity), is still popular in Germany, and has been translated into almost all European languages. Another of his publications is *Paradiesgaertlein*, translated into English (The Garden of Paradise).

Arne (a`rn), Thomas Augustine, English composer, born at London 1710, died 1778. His first opera, *Fair Rosamond*, was performed in 1733 at Lincoln's-Inn Fields, and was received with great applause. Then followed a version of Fielding's *Tom Thumb*, altered into *The Opera of Operas*, a musical burlesque. His style in the *Comus* (1738) is still more original and cultivated. To him we owe the national air *Rule, Britannia*, originally given in a popular piece called the *Masque of Alfred*. After having composed two oratorios and several operas he received the degree of Doctor of Music at Oxford. He composed, also, music for several of the songs in Shakespeare's dramas, and various pieces of instrumental music.

Arnee´, one of the numerous Indian varieties of the buffalo (*Bubălus arni*), remarkable as being the largest animal of the ox kind known. It measures about 7 feet high at the shoulders, and from 9 to 10½ feet long from the muzzle to the root of the tail. It is found chiefly in the forests at the base of the Himalayas.

Arn´hem, or **Arnheim**, a town in Holland, province of Gelderland, 18 miles south-west of Zutphen, on the right bank of the Rhine. Pleasantly situated, it is a favourite residential resort, and it contains many interesting public buildings; manufactures cabinet wares, mirrors, carriages, mathematical instruments, &c.; has paper-mills, and its trade is important. In 1795 it was stormed by the French, who were driven from it by the Prussians in 1813. Pop. 70,664 (1917).

Arnhem Land, a portion of the northern territory of S. Australia, lying west of the Gulf of Carpentaria, and forming a sort of peninsula.

Ar´ni, a town of Madras, on the Cheyair River, 16 miles south of Arcot; formerly a large military station; stormed by Clive in 1751, and scene of defeat of Hyder Ali by Sir Eyre Coote in 1782. Pop. 5050.

Ar'nica, a genus of plants, nat. ord. Compositæ, containing eighteen species, one of which is found in Central Europe, *A. montāna* (leopard's bane or mountain tobacco), but is not a native of Britain. It has a perennial root, a stem about 2 feet high, bearing on the summit flowers of a dark golden yellow. In every part of the plant there is an acrid resin and a volatile oil, and in the flowers an acrid bitter principle called *arnicin*. The root contains also a considerable quantity of tannin. A tincture of it is employed as an external application to wounds and bruises. It was introduced into English gardens about the middle of the eighteenth century.

Ar'nim, Elisabeth von, a German writer, also known as Bettina, wife of Louis Achim von Arnim, and sister of the poet Clemens Brentano; born at Frankfort in 1785, died at Berlin 1859. Even in her childhood she manifested an inclination towards eccentricities and poetical peculiarities of many kinds. She entered into correspondence with Goethe, for whom she entertained a violent passion, although he was then in his sixtieth year. In 1835 she published Goethe's *Briefwechsel mit einem Kinde* (Goethe's Correspondence with a Child), containing, among others, the letters that she alleged to have passed between her and Goethe. Her later writings dealt with subjects like the emancipation of the Jews, and the abolition of capital punishment. Her husband, Ludwig Achim von Arnim, born at Berlin in 1781, died 1831, distinguished himself as a writer of novels. In concert with her brother, Clemens Brentano, he published a collection of popular German songs and ballads entitled *Des Knaben Wunderhorn*.—Her daughter, Gisela von Arnim, is known in literature by her *Dramatische Werke* (3 vols., 1857-63).

Ar'no (ancient **Arnus**), a river of Italy which rises in the Etruscan Apennines, makes a sweep to the south and then flows westwards, divides Florence into two parts, washes Pisa, and falls, 4 miles below it, into the Tuscan Sea, after a course of 130 miles.

Arno'bīus, an early Christian writer, was a teacher of rhetoric at Sicca Veneria, in Numidia, and in 303 became a Christian; he died about 326. He wrote seven books of *Disputationes adversus Gentes* (or *Adversus Nationes*), in which he refuted the objections of the heathens against Christianity. This work betrays a defective knowledge of Christianity, but is rich in materials for the understanding of Greek and Roman mythology.

Arnold, an urban district or town of England, Nottinghamshire, 3 miles north-east of Nottingham, with lace and hosiery manufactures, &c. It has a church built in the twelfth century, and a tower dating from the fifteenth century and restored in 1868 and 1877. Pop. 11,800.

Ar'nold, Benedict, a general in the American army during the War of Independence, born in 1741. He rendered his name infamous by his attempt to betray the strong fortress of West Point, with all the arms and immense stores which were there deposited, into the hands of the British. The project failed through the capture of Major André, when Arnold made his escape to the British lines. He received a commission as brigadier-general in the British army, and took part in several marauding expeditions. He subsequently settled in the West Indies, and ultimately came to London, where he died in 1801.

Ar'nold, Sir Edwin, K.C.I.E., poet, Sanskrit scholar, and journalist, born 1832. Educated at Oxford, where he took the Newdigate prize for a poem entitled the *Feast of Belshazzar* in 1852, he was successively second master in King Edward VI's College at Birmingham, and principal of the Sanskrit College at Poonah in Bombay. In 1861 he joined the editorial staff of the *Daily Telegraph*, with which he was henceforth connected. He died in 1904. He was author of *Poems, Narrative and Lyrical*; translations from the Greek and Sanskrit; *The Light of Asia*, a poem on the life and teaching of Buddha; *The Light of the World*; *Pearls of the Faith*; *Lotus and Jewel*, &c.

Ar'nold, Matthew, English critic, essayist, and poet, was born at Laleham, near Staines, 1822, being a son of Dr. Arnold of Rugby. He was educated at Winchester, Rugby, and Oxford, and became a Fellow of Oriel College. He was private secretary to Lord Lansdowne, 1847-51; appointed inspector of schools, 1851; professor of poetry at Oxford, 1858; published *A Strayed Reveller and other poems*, 1848; *Empedocles on Etna*, 1853; *Merope*, 1858; *Essays in Criticism*, 1865; *On the Study of Celtic Literature*, 1867; *Schools and Universities on the Continent*, 1868; *St. Paul and Protestantism*, 1870; *Literature and Dogma*, 1873; *Last Essays on Church and Religion*, 1877; *God and the Bible*, 1878; *Discourses on America*, 1885, &c. He received the degree of LL.D. from Edinburgh, and that of D.C.L. from Oxford, and lectured in Britain and in America. He died in 1888. A complete edition of his works in 15 vols. appeared in 1905.—BIBLIOGRAPHY: H. W. Paul,

Matthew Arnold (English Men of Letters Series); G. Saintsbury, *Matthew Arnold* (Modern English Writers Series); G. W. E. Russell, *Matthew Arnold* (Literary Lives Series); F. Bickley, *Matthew Arnold and his Poetry*.

Ar'nold, Thomas, headmaster of Rugby School, and professor of modern history in the University of Oxford, born at Cowes, in the Isle of Wight, in 1795, died 1842. He entered Oxford in his sixteenth year, and in 1815 he was elected Fellow of Oriel College, and both in that year and 1817 he obtained the chancellor's prize for Latin and English essays. After taking deacon's orders he settled at Laleham, near Staines, where he employed himself in preparing young men for the universities. In 1828 he was appointed headmaster of Rugby School, and devoted himself to his new duties with the greatest ardour. While giving due prominence to the classics, he deprived them of their exclusiveness by introducing various other branches into his course, and he was particularly careful that the education which he furnished should be in the highest sense moral and Christian. His success was remarkable. Not only did Rugby School become crowded beyond any former precedent, but the superiority of Dr. Arnold's system became so generally recognized that it may be justly said to have done much for the general improvement of the public schools of England. In 1841 he was appointed professor of modern history at Oxford, and delivered his introductory course of lectures with great success. His chief works are his edition of Thucydides, his *History of Rome* (unhappily left unfinished), and his *Sermons*. There is an admirable memoir of him by A. P. Stanley, Dean of Westminster (London, 2 vols., 1845).—Cf Lytton Strachey, *Eminent Victorians*.

Ar'nold of Brescia, an Italian religious and political reformer and martyr of the twelfth century. He was one of the disciples of Abélard, and attracted a considerable following by preaching against the corruption of the clergy. Excommunicated by Innocent II, he withdrew to Zürich, but soon reappeared in Rome, where he was taken and burned (1155).

Arnold-Forster, Hugh Oakeley, grandson of Dr. Arnold of Rugby, and adopted son of the late W. E. Forster, M.P., whose wife was his aunt, was born in 1855, died in 1909. He was educated at Rugby and University College, Oxford. He sat as member of Parliament for West Belfast from 1892, for Croydon from 1906, as a Liberal Unionist, was Parliamentary

Secretary to the Admiralty from 1900 to 1903, then Secretary of State for War, a position which he held till Dec., 1905, having put forward sweeping proposals for the improvement of our army. He wrote on various subjects, especially books for popular instruction, including *How to Solve the Irish Land Question*, *The Citizen Reader*, *This World of Ours*, *Things New and Old*, *In a Conning Tower*, *A History of England*, *English Socialism of To-day*, *Military Needs and Military Policy*, &c.

Ar'non, a river in Palestine, the boundary between the country of the Moabites and that of the Amorites, afterwards of the Israelites, a tributary of the Dead Sea. It is now called Wady-el-Mojib.

Ar'not, or **Ar'nut**, a name of the agreeably flavoured farinaceous tubers of the earth-nut or pig-nut (*Bunium flexuōsum* and *B. Bulbocastānum*). See *Earth-nut*.

Ar'nott, Dr. Neil, an eminent physician and physicist, was born at Arbroath, 1788, died 1874. Having graduated as M.A. at Aberdeen, he then studied medicine, and was appointed a surgeon in the East India Company's naval service. In 1811 he commenced practice in London. In 1837 he was appointed extraordinary physician to the queen. In 1827 he published *Elements of Physics*, and in 1838 a treatise on *Warming and Ventilation*, &c. He is widely known as the inventor of a stove which is regarded as one of the most economical arrangements for burning fuel, a ventilating chimney-valve, and his water-bed for the protection of the sick against bed-sores. In 1869 he gave £1000 to each of the four Scottish universities and £2000 to London University for the promotion of the study of physics. He was a strong advocate of a scientific as opposed to a purely classical education.

Arnprior, a town of Canada, province Ontario, 35 miles west of Ottawa, on the right bank of the River Ottawa, where it is joined by the Madawaska, and with important railway connections. Pop. 4405.

Arnsberg (a`rnz`berh), a town in Prussia, province Westphalia, capital of the district of same name, on the Ruhr. Pop. 10,256.—The district of Arnsberg has an area of 2972 sq. miles, and a population of 2,400,000.

Arnstadt (a`rn`sta`t), a town of Germany, in Schwarzburg-Sondershausen, 11 miles south by west of Erfurt, upon the Gera, which divides it into two

parts. It has manufactures in leather, &c., and a good trade in grain and timber. Pop. 17,907.

Arnsvalde (a`rnz`va`l-de), a town of Prussia, province Brandenburg, 39 miles south-east of Stettin. Pop. 8730.

Ar`nulf, great-grandson of Charlemagne, elected King of Germany in A.D. 887; invaded Italy, captured Rome, and was crowned emperor by the Pope (896); died A.D. 898.

Aroi`deæ. See *Araceæ*.

Ar`olsen, a German town, capital of Waldeck. Pop. 2793.

Aromat`ics, drugs, or other substances which yield a fragrant smell, and often a warm pungent taste, as calamus (*Acōrus Calāmus*), ginger, cinnamon, cassia, lavender, rosemary, laurel, nutmegs, cardamoms, pepper, pimento, cloves, vanilla, saffron. Some of them are used medicinally as tonics, stimulants, &c.

Aromatic Vinegar, a very volatile and powerful perfume made by adding the essential oils of lavender, cloves, &c., and often camphor, to crystallizable acetic acid. It is a powerful excitant in fainting, languor, and headache.

Aro`na, an ancient Italian town near the south extremity of Lago Maggiore. Pop. 4474. In the vicinity is the colossal statue of San Carlo Borromeo, 70 feet in height, exclusive of pedestal 42 feet high.

Aroos`took, a river of the north-eastern United States and New Brunswick, a tributary of the St. John, length 120 miles.

Arou`ra**, or **Aru`ra, an ancient Egyptian measure of surface, according to Herodotus the square of 100 cubits, containing, 21,904 sq. feet.

Arpad, founder of the Magyar monarchy, born about 870, died 907. See *Hungary*.

Arpeggio (a`r-pej`ō), the distinct sound of the notes of an instrumental chord; the striking the notes of a chord in rapid succession, as in the manner of touching the harp instead of playing them simultaneously.

Arpent (a`r-pän`), formerly a French measure of land, equal to five-sixths of an English acre; but it varied in different parts of France; the Parisian arpent contained 32,400 sq. feet, the common arpent 40,000 sq. feet.

Arpino (a`r-pē`nō; ancient **Arpinum**), a town of Southern Italy, province of Caserta, celebrated as the birthplace of Gaius Marius and Cicero. It manufactures woollens, linen, paper, &c. Pop. 10,309.

Arqua (a`r`kwa`), a village of Northern Italy, about 13 miles south-west of Padua, where the poet Petrarch died, 18th July, 1374. A monument has been erected over his grave. Pop. 1700.

Ar`quebus, a hand-gun; a species of fire-arm of the sixteenth century, resembling a musket. It was fired from a forked rest, and sometimes cocked by a wheel, and carried a ball that weighed nearly 2 ounces. A larger kind used in fortresses carried a heavier shot.

Arraca`cha. See *Aracacha*.

Arracan`. See *Aracan*.

Ar`rack. See *Arack*.

Ar`ragon. See *Aragon*.

Ar`rah, a town of British India, in Shahabad district, Bengal, rendered famous during the mutiny of 1857 by the heroic resistance of a body of twenty civilians and fifty Sikhs, cooped up within a detached house, to a force of 3000 sepoy, who were ultimately routed and overthrown by the arrival of a small European reinforcement. Pop. 46,170.

Arraignment (ar-rān`-), the act of calling or setting a prisoner at the bar of a court to plead guilty or not guilty to the matter charged in an indictment or information. In Scots law the term is *calling the diet*.—The *Clerk of Arraignment* is an officer attached to assize courts and to the Old Bailey, who assists in the arraignment of prisoners, and puts formal questions to the jury.

Ar`ran, an island of Scotland, in the Firth of Clyde, part of Bute county; length, north to south, 20 miles; breadth, about 10 miles; area, 165 sq. miles, or 105,814 acres, of which about 15,000 are under cultivation. It is of a wild and romantic appearance, particularly the northern half, where the

island attains its loftiest summit in Goatfell, 2866 feet high. The coast presents several indentations, of which that of Lamlash, forming a capacious bay, completely sheltered by Holy Island, is one of the best natural harbours in the west of Scotland. On the small island of Pladda, about a mile from the south shore, a lighthouse has been erected. The geology of Arran has attracted much attention, as furnishing within a comparatively narrow space distinct sections of the great geological formations; while the botany possesses almost equal interest, both in the variety and the rarity of many of its plants. Among objects of interest are relics of Danish forts, standing stones, cairns, &c. Lamlash and Brodick are villages. The island is a favourite resort of summer visitors, and is reached by steamer from Ardrossan. Pop. 8294.

Arran, Earls of. See *Hamilton, Family of*.

Arrangement, in music, the adaptation of a composition to voices or instruments for which it was not originally written; also, a piece so adapted.

Ar´ran Islands. See *Aran*.

Arraro´ba. See *Araroba*.

Arras (a´-rä), a town of France, capital of the department Pas-de-Calais, well built, with several handsome squares and a citadel, cathedral, public library, botanic garden, museum, and numerous flourishing industries. In the Middle Ages it was famous for the manufacture of tapestry, to which the English applied the name of the town itself (arrazo). The battle of Arras was fought and Vimy Ridge taken by the Allies on 9th April, 1917. Pop. 24,200.

Arrest´ is the apprehending or restraining of one's person, which, in civil cases, can take place legally only by process in execution of the command of some court or officers of justice; but in criminal cases any man may arrest without warrant or precept, and every person is liable to arrest without distinction, but no man is to be arrested unless charged with such a crime as will at least justify holding him to bail when taken. *Magna Charta* and the *Habeas Corpus Act* are the two great statutes for securing the liberty of the subject against unlawful arrests and suits.

Arrest´ment, in Scots law, a process by which a creditor may attach money or movable property which a third party holds for behoof of his debtor. In

1870 an Act was passed for Scotland which provides that only that part of the weekly wages of labourers, and of workpeople generally, which is in excess of 20s. is liable to arrestment for debt.

Arrest of Judgment, in law, the staying or stopping of a judgment after verdict, for causes assigned. Courts have power to arrest judgment for intrinsic causes appearing upon the face of the record; as when the declaration varies from the original writ; when the verdict differs materially from the pleadings; or when the case laid in the declaration is not sufficient in point of law to found an action upon.

Arre'tium. See *Arezzo*.

Arrhenath'erum, a genus of oat-like grasses, of which *A. elatius*, sometimes called French rye-grass, is a valuable fodder plant.

Arrhenius, Svante August, famous Swedish physicist and chemist, born 19th Feb., 1859, at Wyk, near Upsala. He was educated at the Universities of Upsala (1876-81) and Stockholm (1881-4), spent two years in travelling, and after doing much original research was appointed professor of physics at the University of Stockholm. To him is due the establishment of the theory of electrolytic dissociation, supplying a reasonable explanation of many chemical phenomena otherwise insoluble. He subsequently extended the application of the electrolytic theory to the phenomena of atmospheric electricity. His dissertation *Sur la conductibilité galvanique des électrolytes* appeared in 1884. Among his other works is *Worlds in the Making* (English translation, 1908).

Ar'ria, the heroic wife of a Roman named Cæcīna Pætus. Pætus was condemned to death in A.D. 42 for his share in a conspiracy against the emperor Claudius, and was encouraged to suicide by his wife, who stabbed herself and then handed the dagger to her husband with the words, 'It does not hurt, Pætus!'

Ar'rian, or **Flavius Arrianus**, a Greek historian, native of Nicomedia, flourished in the second century, under the emperor Hadrian and the Antonines. He was first a priest of Ceres; but at Rome he became a disciple of Epictetus, was honoured with the citizenship of Rome, and was advanced to the senatorial and even consular dignities. His extant works are: *The*

Expedition of Alexander, in seven books; a book *On the Affairs of India*; an *Epistle to Hadrian*; a *Treatise on Tactics*; a *Periplus of the Euxine Sea*; a *Periplus of the Red Sea*; and his *Enchiridion*, a moral treatise, containing the discourses of Epictetus.

Ar'ris, in architecture, the line in which the two straight or curved surfaces of a body, forming an exterior angle, meet each other.

Arro'ba (Spanish), a weight formerly used in Spain, and still used in the greater part of Central and South America. In the States of Spanish origin its weight is generally equal to 25.35 lb. avoirdupois; in Brazil it equals 32.38 lb.—Also a measure for wine, spirits, and oil, ranging from 2¾ gallons to about 10 gallons.

Arrøe, Danish island. See *Aerøe*.

Arrondissement. See *France*.

Arrow. See *Archery*, *Bow*.

Arrowhead (*Sagittaria*), a genus of aquatic plants found in all parts of the world within the torrid and temperate zones, nat. ord. Alismaceæ, distinguished by possessing barren and fertile flowers, with a three-leaved calyx and three coloured petals. The common arrowhead (*S. sagittifolia*), the only native species in Britain, is known by its arrow-shaped leaves with lanceolate straight lobes.

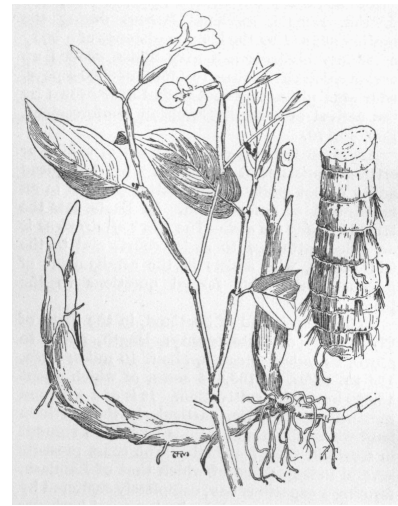
Arrowheaded Characters. See *Cuneiform Writing*.

Arrow Lake, an expansion of the Columbia River, in British Columbia, Canada; about 95 miles long from N. to S.; often regarded as forming two lakes—Upper and Lower Arrow Lake.

Arrowrock Dam. See *Dams* and *Reservoirs*.

Arrow-root, a starch largely used for food and for other purposes. Arrow-root proper is obtained from the rhizomes or rootstocks of several species of plants of the genus *Maranta* (nat. ord. Marantaceæ), and perhaps owes its name to the scales which cover the rhizome, which have some resemblance to the point of an arrow. Some, however, suppose that the name is due to the fact of the fresh roots being used as an application against wounds inflicted

by poisoned arrows, and others say that *arrow* is a corruption of *ara*, the Indian name of the plant. The species from which arrow-root is most commonly obtained is *M. arundinācĕa*, hence called the *arrow-root plant*. Brazilian arrow-root, or tapioca meal, is got from the large fleshy root of *Manihot utilissima*, after the poisonous juice has been got rid of; East Indian arrow-root, from the large rootstocks of *Curcūma angustifolia*; Chinese arrow-root, from the creeping rhizomes of *Nelumbium speciōsum*; English arrow-root, from the potato; Portland arrow-root, from the corms of *Arum maculātum*; and Oswego arrow-root, from Indian corn. Analyses made in 1902 and 1906 show that the idea generally held of the nourishing qualities of arrow-root is a delusion, and that the proteids, which are true muscle-builders, are present in an extremely small extent. Arrow-root, however, mixed with eggs, milk, and flavourings, is largely used in the dietary of invalids.



Arrow-root (*Maranta arundinācĕa*)

Arrowsmith, Aaron, a distinguished English cartographer, born 1750, died 1823; he raised the execution of maps to a perfection it had never before attained.—His nephew, John, born 1790, died 1873, was no less distinguished in the same field; his *London Atlas of Universal Geography* may be specially mentioned.

Arroyo (a`r-rō´yo), the name of two towns of Spain, in Estremadura, the one, called Arroyo del Puerco (pop. 5727), about 10 miles west of Caceres; the other, called Arroyo Molinos de Montanches, about 27 miles south-east of Caceres, memorable from the victory gained by Lord Hill over a French force under General Gerard, 28th Oct., 1811.

Ar´ru (or **Aroo**) **Islands**, a group belonging to the Dutch, south of western New Guinea, and extending from north to south about 127 miles. They are composed of coralline limestone, nowhere exceeding 200 feet above the sea, and are well wooded and tolerably fertile. The natives belong to the Papuan race, and some of them are Christians. The chief exports are

trepane, tortoise-shell, pearls, mother-of-pearl, and edible birds'-nests. Pop. of group about 20,000.

Arsa'ces, the founder of a dynasty of Parthian kings (256 B.C.), who, taking their name from him, are called Arsacidæ. There were thirty-one in all. See *Parthia*.

Ar'samas, a manufacturing town in the Russian government of Nijni-Novgorod, on the Tessa, 250 miles east of Moscow, with a cathedral and large convent. Pop. 12,000.

Ar'senal, a royal or public magazine or place appointed for the making, repairing, keeping, and issuing of military stores. An arsenal of the first class should include factories for guns and gun-carriages, small-arms, small-arms ammunition, harness, saddlery, tents, and powder; a laboratory and large store-houses. In arsenals of the second class workshops take the place of the factories. The Royal Arsenal, Woolwich, which manufactures warlike implements and stores for the army and navy, was formed about 1720, and comprises factories, laboratories, &c., for the manufacture and final fitting up of almost every kind of arms and ammunition. Great quantities of military and naval stores are kept at the dockyards of Chatham, Portsmouth, Plymouth, and Pembroke. In France there are various arsenals or depots of war-material, which is manufactured at Mézières, Toulouse, Besançon, &c.; the great naval arsenals are Brest and Toulon. Until 1919 the chief German arsenals were at Spandau, Strassburg, and Dantzig. The chief Austrian arsenal was the immense establishment at Vienna, which included gun-factory, laboratory, small-arms and carriage factories, &c. Russia had her principal arsenal at Petrograd, with supplementary factories of arms and ammunition at Briansk, Kiev, and elsewhere. In Italy Turin is the centre of the military factories. There are a number of arsenals in the United States, but individually they are of little importance.

Ar'senic (symbol **As**, atomic weight 75), a common element usually found combined with metals as arsenides, the commonest of which is arsenical pyrites, FeAsS. It has a steel colour and high metallic lustre, and tarnishes on exposure to the air, first changing to yellow, and finally to black. In hardness it equals copper; it is extremely brittle, and very volatile, beginning to sublime before it melts. It burns with a blue flame, and emits a

smell of garlic. Its specific gravity is 5.76. It forms compounds with most of the metals. Combined with sulphur it forms orpiment and realgar, which are the yellow and red sulphides of arsenic. Orpiment is the true *arsenicum* of the ancients. With oxygen arsenic forms two compounds, the more important of which is arsenious oxides or arsenic trioxide (As_2O_3), which is the *white arsenic*, or simply *arsenic* of the shops. It is usually seen in white, glassy, translucent masses, and is obtained by sublimation from several ores containing arsenic in combination with metals, particularly from arsenical pyrites. Of all substances arsenic is that which has most frequently occasioned death by poisoning, both by accident and design. The best remedies against the effects of arsenic on the stomach are ferric hydroxide or magnesian hydroxide, or a mixture of both, with copious draughts of bland liquids of a mucilaginous consistence, which serve to procure its complete ejection from the stomach. Oils and fats generally, milk, albumen, wheat-flour, oatmeal, sugar or syrup, have all proved useful in counteracting its effect. Like many other virulent poisons it is a safe and useful medicine, especially in skin diseases, when judiciously employed. It is used as a flux for glass, and also for forming pigments. The arsenite of copper (Scheele's green) and a double arsenite and acetate of copper (emerald green) were formerly largely used to colour paper-hangings for rooms; but as poisonous gases are liable to be given off, the practice has been to a great extent abandoned. Arsenic compounds have been used for colouring confectionery, and other articles, bright green, but their chief industrial use is in the preparation of insecticides. Arsenic is found in crude oil of vitriol, and occasionally in products such as grape-sugar, beer, &c., in the manufacture of which oil of vitriol is employed. Plants die when placed in a solution of arsenic, but corn is often steeped in such a solution, previous to planting, for preventing smut, and the growth of the future plant is not injured thereby.

Arshin (a'r-shēn'), a Russian measure of length equal to 28 inches.

Arsin'oe, a city of ancient Egypt on Lake Mœris, said to have been founded about 2300 B. C., but renamed after Arsinoë, wife and sister of Ptolemy II of Egypt, and called also *Crocodilopolis*, from the sacred crocodiles kept at it.

Ar'sis, a term applied in prosody to that syllable in a measure where the emphasis is put; in elocution, the elevation of the voice, in distinction from *thesis*, or its depression. *Arsis* and *thesis*, in music, are the strong position and weak position of the bar, indicated by the down-beat and up-beat in marking time.

Ar'son, in English law, the malicious burning of a dwelling-house or outhouse of another man, which by the common law is felony, and which, if any person is therein, is capital. Also, the wilful setting fire to any church, chapel, warehouse, mill, barn, agricultural produce, ship, coal-mine, and the like. In Scotland it is called *wilful fire-raising*, and in both England and Scotland it is a considerable aggravation of the crime if the burning is to defraud insurers.

Art, in its most extended sense, as distinguished from nature on the one hand and from science on the other, has been defined as every regulated operation or dexterity by which organized beings pursue ends which they know beforehand, together with the rules and the result of every such operation or dexterity. Science consists in *knowing*, art in *doing*. In this wide sense it embraces what are usually called the useful arts. In a narrower and purely æsthetic sense it designates what are more specifically termed the fine arts, as architecture, sculpture, painting, music, and poetry. The useful arts have their origin in positive practical needs, and restrict themselves to satisfying them. The fine arts minister to the sentiment of taste through the medium of the beautiful in form, colour, rhythm, or harmony. See *Fine Arts, Painting, Sculpture, &c.*—In the Middle Ages it was common to give certain branches of study the name of arts.—Cf. A. C. R. Carter, *History of Art, The Year's Art*.

Art Collections. See *Collections, Artistic*.

Art, Teaching of. With the advent of the present industrial age the teaching of art has undergone a profound change. The fine and the industrial arts have been equally affected. In mediæval times, and in the earlier classic ages, the system of apprenticeship prevailed, and all teaching of the arts and of the artistic crafts was given by masters of the various arts or trades to the apprentices who worked under their guidance as assistants. Standards of excellence were maintained by trade guilds, who enforced rules as to

workmanship as well as rules for the economic conditions of each trade or craft. The painter of pictures, or of mural decorations, was trained in the same way as any other craftsman, working as an apprentice under a master.

When, in the last century, machinery driven by steam-power took the place of hand labour in industry, the small independent workshops gradually disappeared, as the industrial centres increased in those localities where coal or raw material was most easily obtained; and, as the processes of each trade or craft became more and more subdivided and specialized, the old system of apprenticeship, which had become unnecessary, broke down. The teaching and tradition of the small independent craft workshops had no counterpart in the new centralized industrial systems. Even the painters of pictures needed no longer to prepare their own materials, for special industries arose, and mechanical processes were developed, for the work which formerly had been done in the artists' workshops by apprentices. The fine arts in this way suffered the loss of their old systems of teaching and instruction.

To meet the need for a revival of art teaching in the crafts and other industries, there arose a movement towards the centralization of teaching in schools of art during the latter half of the nineteenth century. Following the impulse given to that movement by the great exhibition in 1850, the British Government founded the schools of science and art in London and in most of the important provincial towns. Earlier in the century bodies of artists had founded national academies for the teaching of art; and the teaching of drawing was gradually adopted as a part of ordinary school education. Step by step training in schools of art or technical schools took the place of the teaching formerly given during apprenticeship in every craft workshop. The ancient guilds were replaced by the new trades unions, but these took no part in the maintenance of artistic standards nor of quality in workmanship.

At the present time the teaching of art begins with the early school lessons in drawing, and is carried on in special technical classes or schools of art, where teachers of the 'fine arts' and of the artistic crafts give instruction to students preparing for professional work. In a few of the artistic trades the system of apprenticeship still survives, but the teaching given by that means is usually supplemented by attendance at a school of art or technical school.

Under the Education Act of 1918 attendance at technical classes in the daytime became compulsory for apprentices in all industrial trades.

The subject of art teaching was formerly disregarded by the universities, but has become definitely within their province since the founding of the Slade professorships at Oxford, Cambridge, and London Universities, and of the professorship of fine art at the University of Edinburgh.

The chief schools of art in Great Britain are the schools of the Royal Academy in London, the Slade School at University College, London, and the Royal College of Art at South Kensington, also the Central School of Arts and Crafts, and the other large metropolitan schools of the London County Council.

In most of the English provincial towns are municipal or other schools of art under the control of the Board of Education.

In Scotland the chief schools are the four central institutions—the Edinburgh College of Art, and the schools of art of Glasgow, Aberdeen, and Dundee. In each of these a diploma is given on the satisfactory completion of a prescribed course of study.

In Ireland the chief schools are those of the Royal Hibernian Academy in Dublin, and the provincial schools of art under the Department of Agriculture and Technology.

On the Continent the chief centre of artistic training has for many years been in Paris. Advanced students from most European and American art schools spend some of the later period of their study in the schools of Paris, in painting, in sculpture, or in architecture. There is, naturally, a valuable incentive and stimulation due to this gathering together of advanced students from all countries, as well as to the high academic tradition and sense of style of the French.

The 'atelier' system, which is followed in the French schools, is simple and personal. The expenses of an 'atelier', or studio, are borne by a group of students by the consent and under the guidance of an artist of reputation, who visits the 'atelier' at stated intervals, but is not concerned in its administration. In this way the relations between the professor and his

pupils are extremely direct and personal. Most of the distinguished artists of Paris are attached to some 'atelier' to which chosen pupils are admitted.

There are also in Paris excellent schools for training in the applied arts, schools for furniture-making, printing, jewellery, and other artistic trades. These are on private foundations, but also receive State aid.

The teaching of art that is given at the present day as a part of ordinary general education attempts little more than a training in the elements of drawing, with some practice in the use of colour. The purpose of the drawing lesson is the attainment of skill in the representation of objects rather than the training of the æsthetic sense, or of artistic judgment or taste.

In the schools of art opportunities are provided for training in drawing and painting, sculpture, and architecture, and in the general principles of design in these arts, and in many of the artistic crafts and industrial processes. In some localities, where particular industries or artistic trades are concentrated, special schools for artistic and technical training are provided. The present tendency is towards the development of special schools for particular artistic trades or professions.

A complete system of training in any art must of necessity include: (1) actual technical practice; (2) teaching of the canons of workmanship of the art; (3) acquaintance with its historical development, especially with the notable examples and the highest achievements of past masters in the art.

Arta (ancient **Ambracia**), a gulf, town, and river of north-western Greece. The town was transferred by Turkey to Greece in 1881 (pop. 8000). It stands on the River Arta, which for a considerable distance above its mouth formed a part of the boundary between Greece and Turkey.—The province of Arta has an area of 395 sq. miles, and a pop. of 52,400.

Artaxerxes (Old Pers. *Artakhsathra*, 'the mighty'), the name of several Persian kings:—1. **Artaxerxes**, surnamed Longimanus, succeeded his father Xerxes I, 465 B.C. He subdued the rebellious Egyptians, terminated the war with Athens, governed his subjects in peace, and died 425 B.C.—2. **Artaxerxes**, surnamed Mnemon, succeeded his father Darius II in the year 405 B.C. After having vanquished his brother Cyrus he made war on the Spartans, who had assisted his enemy, and forced them to abandon the

Greek cities and islands of Asia to the Persians. On his death, 359 B.C., his son Ochus ascended the throne under the name of—3. **Artaxerxes Ochus** (359 to 339 B.C.). After having overcome the Phœnicians and Egyptians, and displayed great cruelty in both countries, he was poisoned by his general Bagoas.

Arte'di, Peter, a Swedish naturalist, born 1705, drowned at Amsterdam 1735. He studied at Upsala, turned his attention to medicine and natural history, and was a friend of Linnæus. His *Bibliotheca Ichthyologica* and *Philosophia Ichthyologica*, together with a life of the author, were published at Leyden in 1738.

Artel, a name for co-operative associations in Russia. These associations were known in ancient Russia as *drushina* or *wataga*. The artels originally consisted of bodies of men associating for the purpose of jointly undertaking a piece of work and dividing the profits. Artels were formed in the thirteenth and fourteenth centuries for the co-operation of their members in hunting and fishing. Inspired and stimulated by the Schulze-Delitzsch associations established in Germany, the Russian artels have extended their activity to various branches of industrial life. There are now consum artels, credit artels, and insurance artels, but the most important are the artisan and industrial artels. Some of the artels, however, are little more than trade guilds with mutual responsibility.

Ar'tēmis, an ancient Greek divinity, identified with the Roman Diana. She was the daughter of Zeus (Jupiter) and Leto or Latona, and was the twin sister of Apollo, born in the Island of Delos. She is variously represented as a huntress, with bow and arrows; as a goddess of the nymphs, in a chariot drawn by four stags; and as the moon-goddess, with the crescent of the moon above her forehead. She was a maiden divinity, never conquered by love, except when Endymion made her feel its power. She demanded the strictest chastity from her worshippers, and she is represented as having changed Actæon into a stag, and caused him to be torn in pieces by his own dogs, because he had secretly watched her as she was bathing. The Artemisia was a festival celebrated in her honour at Delphi. The famous temple of Artemis at Ephesus was considered one of the wonders of the world, but the goddess worshipped there was very different from the

huntress goddess of Greece, being of Eastern origin, and regarded as the symbol of fruitful nature.

Artemi'sia, Queen of Caria, in Asia Minor, about 352-350 B.C., sister and wife of Mausōlus, to whom she erected in her capital, Halicarnassus, a monument, called the Mausolēum, which was reckoned among the seven wonders of the world.

Artemi'sia, a genus of plants of numerous species, nat. ord. Compositæ, comprising mugwort, southernwood, and wormwood. Certain alpine species are the flavouring ingredient in absinthe. See *Wormwood*.

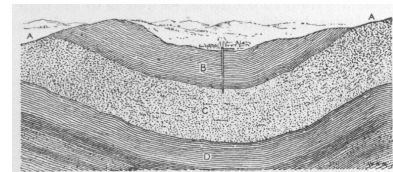
Artemi'sium, a promontory in Eubœa, an island of the Ægean, near which several naval battles between the Greeks and Persians were fought, 480 B.C.

Ar'temus Ward. See *Browne, Charles Farrar*.

Ar'teries, the system of cylindrical vessels or tubes, membranous, elastic, and pulsatile, which convey the blood from the heart to all parts of the body, by ramifications which, as they proceed, diminish in size and increase in number, and terminate in minute capillaries uniting the ends of the arteries with the beginnings of the veins. There are two principal arteries or arterial trunks: the *aorta*, which rises from the left ventricle of the heart and ramifies through the whole body, sending off great branches to the head, neck, and upper limbs, and downwards to the lower limbs, &c.; and the *pulmonary artery*, which conveys venous blood from the right ventricle to the lungs, to be purified in the process of respiration.

Arteriot'omy, the opening or cutting of an artery for the purpose of blood-letting, as, for instance, to relieve pressure of the brain in apoplexy.

Arte'sian Wells, so called from the French province of Artois, where they appear to have been first used on an extensive scale, are perpendicular borings into the ground through which water rises to the surface of the soil, producing a constant flow or stream, the ultimate sources of supply being higher than the mouth of the boring, and the water thus rising by the well-known law. They are generally sunk



Artesian Well. A. A. Outcrops of pervious stratum (C) acting as collecting areas. B and D. Impervious stratum.

in valley plains and districts where the lower pervious strata are bent into basin-shaped curves. The rain falling on the outcrops of these saturates the whole porous bed, so that when the bore reaches it the water by hydraulic pressure rushes up towards the level of the highest portion of the strata. The supply is sometimes so abundant as to be used extensively as a moving power, and in arid regions for fertilizing the ground, to which purpose artesian springs have been applied from a very remote period. Thus many artesian wells have been sunk in the Algerian Sahara, which have proved an immense boon to the district. The water of most of these is potable, but a few are a little saline, though not to such an extent as to influence vegetation. The hollows in which London and Paris lie are both perforated in many places by borings of this nature. At London they were first sunk only to the sand, but afterwards into the chalk. One of the most celebrated artesian wells is that of Grenelle, near Paris, 1798 feet deep, completed in 1841, after eight years' work. Artesian wells are now common in many countries, and have been sunk to the depth of a mile or more. As the temperature of water from great depths is invariably higher than that at the surface, artesian wells have been made to supply warm water for heating manufactories, greenhouses, hospitals, fish-ponds, &c. They have also been made in the United States and Australia for the purpose of irrigation. Petroleum wells are generally of the same technical description. Artesian wells are now made with larger diameters than formerly, and altogether their construction has been rendered much more easy in modern times. See *Boring*.

Arteveld, or **Artevelde** (a' r'te-velt, a' r'te-vel-de), the name of two men distinguished in the history of the Low Countries.—1. Jacob van, a brewer of Ghent, born about 1300, was selected by his fellow-townsmen to lead them in their struggles against Count Louis of Flanders. In 1338 he was appointed captain of the forces of Ghent, and for several years exercised a sort of sovereign power. A proposal to make the Black Prince, son of Edward III of England, Governor of Flanders led to an insurrection in which Arteveld lost his life (1345).—2. Philip, son of the former, at the head of the forces of Ghent gained a great victory over the Count of Flanders, Louis II, and for a time assumed the state of a sovereign prince. His reign proved short-lived. The Count of Flanders returned with a large French force, fully disciplined and skilfully commanded. Arteveld was rash

enough to meet them in the open field at Roosebeke, between Courtrai and Ghent, in 1382, and fell with 25,000 Flemings.

Arthri'tis (Gr. *arthron*, a joint), any inflammatory distemper that affects the joints, particularly chronic rheumatism or gout.

Arthro'dia, a species of articulation, in which the head of one bone is received into a shallow socket in another; a ball-and-socket joint.

Arthrop'oda, one of the two primary divisions (Anarthropoda being the other) into which modern naturalists have divided the sub-kingdom Annulosa, having the body composed of a series of segments, some always being provided with articulated appendages. The division comprises Crustaceans, Spiders, Scorpions, Centipedes, and Insects.

Arthrozo'a, a name sometimes given to all articulated animals, including the arthropoda and worms.

Arthur, Chester Alan, twenty-first President of the United States, born 1830, died 1886, was the son of Scottish parents, his father being pastor of Baptist churches in Vermont and New York. He chose law as a profession, and practised in New York. As a politician he became a leader in the Republican party. During the civil war he was energetic as quarter-master-general of New York in getting troops raised and equipped. He was afterwards collector of customs for the port of New York. In 1880 he was elected Vice-President, succeeding as President on the death of Garfield in 1881.

Ar'thur, King, an ancient British hero of the sixth century, son of Uther Pendragon and the Princess Igera, wife of Gorlois, Duke of Cornwall. He married Guinevere, or Ginevra; established the famous order of the Round Table; and reigned, surrounded by a splendid Court, twelve years in peace. After this, as the poets relate, he conquered Denmark, Norway, and France, slew the giants of Spain, and went to Rome. From thence he is said to have hastened home on account of the faithlessness of his wife, and Modred, his nephew, who had stirred up his subjects to rebellion. He subdued the rebels, but died in consequence of his wounds, on the Island of Avalon. The story of Arthur is supposed to have some foundation in fact, and has ever been a favourite subject with our romanticists and our poets. It is generally

believed that Arthur was one of the last great Celtic chiefs who led his countrymen from the west of England to resist the settlement of the Saxons in the country. But many authorities regard him as a leader of the Cymry of Cumbria and Strath-Clyde against the Saxon invaders of the east coast and the Picts and Scots north of the Forth and the Clyde. See *Grail*, *Merlin*, *Round Table*.—BIBLIOGRAPHY: J. Rhys, *Studies in the Arthurian Legend*; W. Lewis Jones, *King Arthur in History and Legend*.

Arthur's Seat, a picturesque hill within the King's Park in the immediate vicinity of Edinburgh; has an altitude of 822 feet; descends rollingly to the N. and E. over a base each way of about five furlongs; presents an abrupt shoulder to the S., and breaks down precipitously to the W. It is composed of a diversity of eruptive rocks, with some interposed and up-tilted sedimentary ones; and derives its name somehow from the legendary King Arthur.

Ar'tiad (Gr. *artios*, even-numbered), in chemistry, a name given to an element of even equivalency, as a dyad, tetrad, &c.: opposed to a perissad, an element of uneven equivalency, such as a monad, triad, &c.

Artichoke (*Cynāra Scolymus*), sometimes called 'the Globe Artichoke', a well-known plant of the nat. ord. Compositæ, somewhat resembling a thistle, with large divided prickly leaves. The erect flower-stem terminates in a large round head of numerous imbricated oval spiny scales which surround the flowers. The fleshy bases of the scales with the large receptacle are the parts that are eaten. Artichokes were introduced into England early in the sixteenth century. The Jerusalem artichoke (a corruption of the It. *girasole*, a sunflower), or *Helianthus tuberōsus*, is a species of sunflower, whose roots are used like potatoes; it was introduced into England in the early part of the seventeenth century.

Article, in grammar, a part of speech used before nouns to limit or define their application. In English *a* or *an* is usually called the indefinite article (the latter form being used before a vowel sound), and *the*, the definite article, but they are also described as adjectives. *An* was originally the same as *one*, and *the* as *that*. In Latin there were no articles, and Greek has only the definite article.

Articles, Lords of the, in Scottish history, a committee chosen equally from each estate or division of Parliament to prepare the various measures, which, when completed, were laid before the Parliament for adoption or rejection. They were first appointed in 1369, and gradually became a recognized part of the Scottish legislative machinery. Abolished 1690.

Articles, The Six, in English ecclesiastical history, articles imposed by a statute (often called the Bloody Statute) passed in 1541, the thirty-third year of the reign of Henry VIII. They decreed the acknowledgment of transubstantiation, the sufficiency of communion in one kind, the obligation of vows of chastity, the propriety of private masses, celibacy of the clergy, and auricular confession. Acceptance of these doctrines was made obligatory on all persons under the severest penalties; the Act, however, was relaxed in 1544, and repealed in 1549.

Articles, The Thirty-nine, of the Church of England, a statement of the particular points of doctrine, thirty-nine in number, maintained by the English Church; first promulgated by a convocation held in London in 1562-3, and confirmed by royal authority; founded on and superseding an older code issued in the reign of Edward VI. The first five articles contain a profession of faith in the Trinity; the incarnation of Jesus Christ, His descent to hell, and His resurrection; the divinity of the Holy Ghost. The three following relate to the canon of the Scripture. The eighth article declares a belief in the Apostles', Nicene, and Athanasian creeds. The ninth and following articles contain the doctrine of original sin, of justification by faith alone, of predestination, &c. The nineteenth, twentieth, and twenty-first declare the Church to be the assembly of the faithful; that it can decide nothing except by the Scriptures. The twenty-second rejects the doctrine of purgatory, indulgences, the adoration of images, and the invocation of saints. The twenty-third decides that only those lawfully called shall preach or administer the sacraments. The twenty-fourth requires the liturgy to be in English. The twenty-fifth and twenty-sixth declare the sacraments effectual signs of grace (though administered by evil men), by which God excites and confirms our faith. They are two: baptism and the Lord's supper. Baptism, according to the twenty-seventh article, is a sign of regeneration, the seal of our adoption, by which faith is confirmed and grace increased. In the Lord's supper, according to article twenty-eight, the bread is the communion of the body of Christ, the wine the communion of his blood, but only through faith

(article twenty-ninth); and the communion must be administered in both kinds (article thirty). The twenty-eighth article condemns the doctrine of transubstantiation, and the elevation and adoration of the host; the thirty-first rejects the sacrifice of the mass as blasphemous; the thirty-second permits the marriage of the clergy; the thirty-third maintains the efficacy of excommunication. The remaining articles relate to the supremacy of the king, the condemnation of Anabaptists, &c. They were ratified anew in 1604 and 1628. All candidates for ordination must subscribe these articles, but they are not binding upon laymen, except judges and certain university officials. This formulary is now accepted by the Episcopalian Churches of Scotland, Ireland, and America.

Articles of Association are the rules framed by a company for the administration of its affairs. Public companies usually have separate articles of their own, but this is not essential. When a registered company has no articles, its business procedure is regulated by the statutory form, found in Schedule 1 of the Companies (Consolidation) Act, 1908, and known as Table A. When separate articles are used they must be printed, signed, stamped, and filed along with the memorandum of association. Unlike the memorandum, the articles may be altered at any time, by special resolution, provided the alteration is within the powers given by the memorandum. Thus, where preferred shares are created by the memorandum, their privileges are more secure than if merely issued under the articles. The articles are intended merely for internal administration, and, while binding in questions between the company and its members as such, they do not affect third parties, unless the company has acted *ultra vires* and this was discoverable from the articles, which the public can inspect at a nominal fee.—BIBLIOGRAPHY: Sir F. B. Palmer, *Company Law*; A. Coles, *Guide for the Company Secretary*; A. F. Topham, *Principles of Company Law*.

Articles of War. See *Military Law*.

Articula'ta, the third great section of the animal kingdom according to the arrangement of Cuvier, applied to invertebrates such as insects and worms, in which the body displays a jointed structure. The name is now obsolete. See *Arthropoda*.

Articulation, in anatomy a joint; the joining or juncture of the bones. This is of three kinds: (1) *Diarthrōsis*, or a movable connection, such as the ball-and-socket joint; (2) *Synarthrōsis*, immovable connection, as by suture, or junction by serrated margins; (3) *Symphysis*, or union by means of another substance, by a cartilage, tendon, or ligament.

Artificial Limbs. Artificial limbs of a primitive kind have been in use from very early times; but, as the material of which they were made was perishable, few specimens have been preserved. In the museum of the Royal College of Surgeons in London there is a good specimen of a Roman artificial leg which is believed to date back to the third century B.C. This leg is made of wood, covered with thin bronze; it has an iron sole-plate, and was fastened on by means of a waist-belt.

In Irish legend we are told of Nuada, who led the tribe of the Dananns back to Ireland, having an artificial hand made of silver; from this he received the name of Argetlam or Silver-handed. In mediæval times Goetz von Berlichingen, who lost his right hand at the siege of Landshut (1505), had a hand made of iron to supply its place.

Great improvements have been made in the manufacture of artificial limbs during the last fifty years. After the Franco-Prussian war the French Government adopted an artificial arm and hand known as the 'Beaufort' and issued it to the soldiers who required it. This hand had a movable thumb controlled by a cord passing up the arm and fastened to a loop band over the opposite shoulder. An Englishman named Heather Bigg invented a hand with a movable thumb which was worked by a piston and controlled by a rubber ball fastened under the arm-pit. These designs have been improved upon from time to time, and an artificial hand can now be made which is capable of opening and closing at will, and of lifting and holding light articles.

Artificial legs vary in design, from the simple pin leg to ingenious contrivances such as the 'Anglesey' leg, which is made of seasoned willow wood with steel ankles and knee-joints. This 'Anglesey' leg is the standard best-quality limb in Great Britain. The German artificial-limb makers use leather instead of wood. These leather legs fit comfortably, but are too heavy, as they have to be supported by steel bands.

Artillery, all ordnance such as guns and howitzers as opposed to small arms and machine-guns. The term is also used for the troops who serve these arms.

Generally speaking, artillery is divided into field, heavy, and siege artillery. For details of organization see *Army*. The earliest form of artillery was a metal tube which was placed in a convenient position on the ground. The charge of gunpowder was ignited by placing a match to a hole bored at the closed end, and the resulting explosion forced the projectile—a stone—more or less in the required direction. Later on this primitive weapon was provided with wheels. In another form one man fired it while a second man supported it on his shoulder. In the later Middle Ages guns of various calibres were known by the names of birds of prey or reptiles; among such were falcons and falconets, culverins and demi-culverins. The fourteenth century saw the development of artillery for siege purposes—chiefly by the Germans—and in the next century it began to be employed in open warfare; while in 1537 the present Honourable Artillery Company was formed in London to encourage the use of all 'weapons of volley'. These weapons of volley were not even confined to fire-arms, but included bows and cross-bows. The earliest English troops raised as artillery personnel were called the Regiment of Firelocks. Of late years artillery science has made almost inconceivable progress, thanks to which, *inter alia*, it is no longer necessary for the target to be visible from the gun-position. This fact, combined with the use of smokeless powder, makes the locating of hostile batteries exceedingly difficult. All field artillery, by which is meant guns and howitzers, which accompany mobile troops are designed on the quick-firing principle, by which the inevitable recoil at the moment of firing is absorbed by an arrangement known as the recoil-carriage, thus preventing any movement of the gun-carriage proper, and avoiding unnecessary labour for the gun-detachments, while at the same time allowing the men composing it to remain under cover of the shield with which the guns are provided. With the quick-firing gun, propellant and projectile are combined in one cartridge similar to that in use with small arms; with the howitzer they are separated. This difference is due to the fact that whereas guns are designed for a flat trajectory with a deep zone of fire-effect, howitzers are intended for high-angle fire with an almost vertical fall of shrapnel-bullets. This effect is produced by varying the charge for different ranges. A field battery consists

of 6 guns or howitzers and 12 ammunition-wagons. Both guns and wagons are of the limbered type, i.e. in two detachable parts, and the weight behind the team of an English gun is approximately 2 tons.

As to tactical principles, it is recognized that the function of artillery is to assist the other arms, that it cannot by itself win a battle, and that its true *métier* is to prepare the way for and assist the infantry. During 1914-18 some 700,000 officers and men served with the Royal Regiment.

Artillery Company, The Honourable, the oldest existing body of volunteers in Great Britain, instituted in 1537, revived in 1610. It comprises six companies of infantry, besides artillery, and furnishes a guard of honour to the sovereign when visiting the city of London. Previous to 1842 the Company elected their own officers, but since that date they have been appointed by the Crown.

Artillery Schools, institutions established for the purpose of giving a special training to the officers, and in some cases the men, belonging to the artillery service. In Great Britain the artillery schools are at Woolwich and Shoeburyness. The Department of Artillery studies at Woolwich give artillery officers the means of continuing their studies after they have completed the usual course at the Royal Military College, and of qualifying for appointments requiring exceptional scientific attainments. The School of Gunnery at Shoeburyness gives instruction in gunnery to officers and men, and conducts all experiments connected with artillery and stores. The sands at the mouth of the Thames afford ample opportunity for artillery practice and firing at long ranges. The Royal Artillery Institution at Woolwich contains a museum, lecture-room, and printing-press, from which professional and scientific papers are periodically issued.

Artiodac'tyla (Gr. *artios*, even numbered, *daktȳlos*, a finger or toe), a section of the Ungulata or hoofed mammals, comprising all those in which the number of the toes is even (two or four), including the ruminants, such as the ox, sheep, deer, &c., and also a number of non-ruminating animals, as the hippopotamus and the pig.

Artisans' and Labourers' Dwellings Act, an English Act of Parliament passed in 1868 to empower town councils and other local authorities to demolish or improve dwellings unfit for human habitation, and to build and

maintain better dwellings in lieu thereof. Other Acts for the same object were passed in 1875, 1879, and 1882. See *Housing*.

Artocarpa'ceæ, a nat. ord. of plants, the bread-fruit order, by some botanists ranked as a sub-order of the Urticaceæ or nettles. They are trees or shrubs, with a milky juice, which in some species hardens into caoutchouc, and in the cow-tree (*Brosimum Galactodendron*) is a milk as good as that obtained from the cow. Many of the plants produce an edible fruit, of which the best known is the bread-fruit (*Artocarpus*).

Artois (a ´ r-twä), a former province of France, anciently one of the seventeen provinces of the Netherlands, now almost completely included in the department of Pas-de-Calais.

Artois, The Battle of. See *European War*.

Arts, the name given to certain branches of study in the Middle Ages, originally called the 'liberal arts' to distinguish them from the 'servile arts' or mechanical occupations. These arts were usually given as grammar, dialectics, rhetoric, music, arithmetic, geometry, and astronomy. Hence originated the terms 'art classes', 'degrees in arts', 'Master of Arts', &c., still in common use in universities, the faculty of arts being distinguished from those of divinity, law, medicine, or science. See *University*.

Artsybashev, Mikhail, Russian author, born in 1878. After a number of short stories he wrote, at the age of twenty-five, a novel entitled *Sanin*, published in 1907, which placed him at once among the most famous contemporary Russian authors. Whatever opinion literary critics may hold of this novel, his wonderful realism, simplicity of style, and psychological analysis cannot be disputed.

Art Unions, associations for encouraging art, an object which they mainly pursue by disposing of pictures, sculptures, &c., by lottery among subscribers. They seem to have originated in France during the time of Napoleon I. They soon afterwards took root in Germany, where they have been very successful. The first art union established in Britain was that at Edinburgh in 1834. Art unions were legalized by the Art Unions Act, 1846 See *Lottery*.

Artvin, a town in the Republic of Georgia, in the Caucasus, about 35 miles inland from Batum. Pop. 6720.

Aruba (a'-rö'ba'), an island off the north coast of Venezuela, belonging to Holland (a dependency of Curaçoa), about 30 miles long and 7 broad; surface generally rock, quartz being abundant, and containing considerable quantities of gold; a phosphate which is exported for manure is also abundant. The climate is healthy. Pop. 9481 (1916).

Aru Islands. See *Arru Islands*.

A´rum, a genus of plants, nat. ord. Araceæ. *A. maculātum* (the common wake-robin, lords-and-ladies, or cuckoo pint) is abundant in woods and hedges in England and Ireland, but is rare in Scotland. It has acrid properties, but its corm yields a starch which is known by the name of Portland sago or arrowroot. At one time this was prepared to a considerable extent in Portland Island. All the species of this genus develop much heat during flowering.



Cuckoo Pint or Wake Robin
(*Arum maculātum*). 1, Spadix.
2, Stamen. 3, Female flower. 4,
Fruit.

Ar´undel, a town in Sussex, England, on the River Arun, 4 miles from its mouth, the river being navigable to the town for vessels of 250 tons. The castle of Arundel, the chief residence of the dukes of Norfolk, stands on a knoll on the north-east side of the town. Pop. (1921), 2741.

Ar´undel, Thomas, third son of Richard FitzAlan, Earl of Arundel, born 1352, died 1413. He was Chancellor of England and Archbishop of Canterbury. He concerted with Bolingbroke to deliver the nation from the oppression of Richard II, and was a bitter persecutor of the Lollards and followers of Wycliffe.

Arundelian Marbles, a series of ancient sculptured marbles discovered by William Petty, who explored the ruins of Greece at the expense of and for Thomas Howard, Earl of Arundel, who lived in the time of James I and Charles I, and was a liberal patron of scholarship and art. After the

Restoration they were presented by the grandson of the collector to the University of Oxford. Among them is the *Parian Chronicle*, a chronological account of the principal events in Grecian, and particularly in Athenian, history, during a period of 1186 years, from the reign of Cecrops (1450 B.C.) to the archonship of Diognetus (264 B.C.).

Arun´do. See *Phragmites*.

Aruspices (a-rus´pi-sēz), or **Haruspices**, a class of priests in ancient Rome, of Etrurian origin, whose business was to inspect the entrails of victims killed in sacrifice, and by them to foretell future events.

Aruwimi, a large river of equatorial Africa, a tributary of the Congo, on the north bank.

Arval Brothers (*Fratres Arvāles*), a college or company of twelve members elected for life from the highest ranks in ancient Rome, so called from offering annually public sacrifices for the fertility of the fields (Lat. *arvum*, a field).

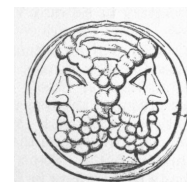
Arve (a´rv), a river rising in the Savoyan Alps, passes through the valley of Chamonix, and falls into the Rhone near Geneva, after a course of about 50 miles.

Arvic´ola, a genus of rodent animals, sub-ord. Muridæ or Mice. There are three British species. *A. amphibia* is the water-vole (or water-rat), and *A. agrestis* is the field-vole or short-tailed field-mouse. They are prolific animals, having three or four litters in the year, each consisting of from four to ten young.

A´ryan, or **Indo-European Family of Languages**. See *Indo-European Family*.

As, a Roman weight of 12 ounces, answering to the libra or pound, and equal to 237.5 grains avoirdupois, or 327.1873 grammes French measure. In the most ancient times of Rome the copper or bronze coin which was called *as* actually weighed an *as*, or a pound, but in 264 B.C. it was reduced to 2 ounces, in 217 to 1 ounce, and in 191 to ½ ounce.

A'sa, great grandson of Solomon and third King of Judah; he ascended the throne at an early age, and distinguished himself by his zeal in rooting out idolatry with its attendant immoralities. He died after a prosperous reign of forty-one years.



As (half real size)—
Specimen in
British
Museum

Asafe'tida, or **Asafœtida**, a fetid inspissated sap from Central Asia, the solidified juice of the *Narthex Asafetida*, a large umbelliferous plant. It is used in medicine as an anti-spasmodic, and in cases of flatulency, in hysteric paroxysms, and other nervous affections. Notwithstanding its very disagreeable odour it is used as a seasoning in the East, and sometimes in Europe. An inferior sort is the product of certain species of *Ferula*.

Asagræ'a. See *Sabadilla*.

Asa'ma, an active volcano of Japan, about 50 miles north-west of Tokio, 8260 feet high.

A'saph, a Levite and psalmist appointed by David as leading chorister in the divine services. His office became hereditary in his family, or he founded a school of poets and musicians, which were called, after him, "the sons of Asaph". See *Psalms*.

Asaph, St., a small cathedral city and bishop's see in Wales, 15 miles north-west of Flint; founded about 550 by St. Kentigern or St. Mungo, bishop of Glasgow, and named after his disciple St. Asaph, from whom both the diocese and town took their name. The cathedral was built about the close of the fifteenth century; it consists of a choir, a nave, two aisles, and a transept. Pop. 1833.

Asarabac'ca, a small hardy European plant, nat. ord. Aristolochiaceæ (*Asărum europæum*). Its leaves are acrid, bitter, and nauseous, and its root is extremely acrid. Both the leaves and root were formerly used as an emetic. It entered into the composition of medicated snuffs recommended in cases of headache.

As'arum. See *Asarabacca*.

Asben, Air, or Ahir, a kingdom of Africa, in the Sahara. It consists of a succession of mountain groups and valleys, with a generally western slope, and attains in its highest summits a height of over 5000 feet. The valleys are very fertile, and often of picturesque appearance. The inhabitants are Tuaregs or Berbers, with an admixture of negro blood. They live partly in villages, partly as nomads. The country is nominally ruled over by a sultan, who resides in the capital, Agades. Pop. about 60,000.

Asbes'tos, or Asbestus, a remarkable and highly-useful mineral, a fibrous variety of several members of the hornblende family, composed of separable filaments, with a silky lustre. The fibres are sometimes delicate, flexible, and elastic; at other times stiff and brittle. It is incombustible, and anciently was wrought into a soft, flexible cloth, which was used as a shroud for dead bodies. In modern times it has been manufactured into incombustible cloth, gloves, felt, paper, &c.; is employed in gas-stoves; is much used as a covering to steam boilers and pipes; is mixed with metallic pigments, and used as a paint on wooden structures, roofs, partitions, &c., to render them fire-proof, and is employed in various other ways, the manufacture having recently greatly developed. Some varieties are compact, and take a fine polish, others are loose, like flax or silky wool. *Ligniform asbestos*, or *mountain-wood*, is a variety presenting an irregular filamentous structure, like wood. *Rock-cork*, *mountain-leather*, *fossil-paper*, and *fossil-flax* are varieties. Asbestos is found in many parts of the world, but is chiefly supplied by Italy, Canada, and Australia. Mineralogically it is distinct from chrysotile, which is used for similar purposes.

Asbjörnsen (a ' s'byeurn-sen), Peter Kristen, born 1812, died 1885, a distinguished Norwegian naturalist and collector of the popular tales and legends, fairy stories, &c., of his native country.

Asbury Park, a small town on the coast of New Jersey, United States, a great summer resort. Its population increases during the summer months from 10,000 to 100,000.

As'calon, or Ash'kelon, a ruined town of Palestine, on the sea-coast, 40 miles w.s.w. of Jerusalem. It was occupied by the Crusaders under Richard I after a great battle with Saladin (1192) and by General Allenby's troops in Nov., 1917.

Asca'nus, the son of Æneas and Creusa, and the companion of his father's wanderings from Troy to Italy.

As'caris. See *Nematoda*.

Ascen'sion (discovered on Ascension Day), an island of volcanic origin belonging to Britain, near the middle of the South Atlantic Ocean, about lat. 7° 55' s.; long. 14° 25' w.; 700 miles north-west of St. Helena; area, about 34 sq. miles. Pop. 196 (1914). It is retained by Britain as a naval sanatorium, coaling and victualling station, and store depot. It has a steam factory, naval and victualling yards, hospitals, and a coal depot. It is celebrated for its turtle, which are the finest in the world. Wild goats are plentiful, and oxen, sheep, pheasants, guinea-fowl, and rabbits have been introduced, and thrive well. The village of Georgetown, the seat of government, stands on the west side of the island, which is governed under the admiralty by a naval officer.

Ascension, Right, of a star or other heavenly body, in astronomy, the arc of the equator intercepted between the first point of Aries and that point of the equator which comes to the meridian at the same instant with the star.

Ascension Day, the day on which the ascension of the Saviour is commemorated, often called *Holy Thursday*: a movable feast, always falling on the Thursday but one before Whitsuntide.

Ascet'icism and **Ascet'ics** (from the Gr. *askēsis*, meaning practice, bodily exercise, or athletic training). The early Christians, who devoted themselves to severe exercises of piety and strove to distinguish themselves from the world by abstinence from sensual enjoyments and by voluntary penances, adopted the name to signify the practice of spiritual things. Ascetics and asceticism have played an important part in the Christian Church, but the principle of striving after a higher and more spiritual life by subduing the animal appetites and passions has no necessary connection with Christianity. Thus there were ascetics among the Jews previous to Christ, and asceticism was inculcated by the Stoics, while in its most extreme form it may still be seen among the Brahmans and Buddhists. Monasticism was but one phase of asceticism. It must also be borne in mind that in the history of asceticism, pagan, Christian, Jewish, and Mohammedan, we are often dealing not only with religious but distinctly abnormal temperaments. See *Monasticism*.

Asch (a´sh), a town of Czecho-Slovakia, in the extreme north-western corner of former Bohemia, with manufactures of cotton, woollen, and silk goods, bleachfields, dyeworks, &c. Pop. 21,583.

Aschaffenburg (a´-sha´f´en-börh), a town of Bavaria, on the Main and Aschaff, 26 miles E.S.E. of Frankfort. The chief edifice is the castle of Johannisberg, built between 1605 and 1614, and for centuries the summer residence of the Elector. There are manufactures of coloured paper, tobacco, liqueurs, &c. Pop. 29,891.

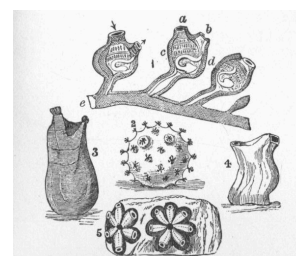
Ascham (as´kam), Roger, a learned Englishman, born in 1515 of a respectable family in Yorkshire, died 1568. He was entered at St. John's College, Cambridge, 1530, and was elected fellow in 1534 and tutor in 1537. He was Latin secretary to Edward VI and also to Mary. He was tutor to Princess Elizabeth during her girlhood, and he became her secretary after she ascended the throne. In 1544 he wrote his *Toxophilus, or Schole of Shooting*, in praise of his favourite amusement and exercise—archery. Between 1563 and 1568 he wrote his *Scholemaster*, a treatise on the best method of teaching children Latin. Some of his writings, including many letters, were in Latin. He wrote the best English style of his time. His life was written by Dr. Johnson to accompany an edition of his works published in 1769.

Aschersleben (a´sh´érz-lā-ben), a town of Prussian Saxony, in the district of Magdeburg, near the junction of the Eine with the Wipper. Industries: woollens, machinery and metal goods, sugar, paper, &c. Pop. 28,968.

Ascid´ia (Gr. *askos*, a wine-skin), the name given to the 'Sea-squirts' or main section of the Tunicata, a class of animals of low grade, resembling a double-necked bottle, of a leathery or gristly nature, found at low-water mark on the sea-beach, and dredged from deep water attached to stones, shells, and fixed objects. One of the prominent openings admits the food and the water required in respiration, the other is the excretory aperture. A single *ganglion* represents the nervous system, placed between the two apertures. Male and female reproductive organs exist in each ascidian. They pass through peculiar phases of development, the young ascidian appearing like a tadpole-body. They may be *single* or *simple*, *social* or *compound*. In *social ascidians* the peduncles of a number of individuals are united into a

common tubular stem, with a partial common circulation of blood. In these animals evolutionists see a link between the Mollusca and the Vertebrata.

Asclepiada'ceæ, an order of gamopetalous Dicotyledons, the distinguishing characteristic of which is that the anthers adhere to the five stigmatic processes, the whole sexual apparatus forming a single mass. The pollination arrangements are peculiar, recalling those of orchids. The members of this order are shrubs, or sometimes herbaceous plants, occasionally climbing, almost always with a milky juice. Many of them are employed as purgatives, diaphoretics, tonics, and febrifuges, and others as articles of food. *Asclepias* is the typical genus. See *Asclepias*, *Calotropis*, *Stapelia*, *Stephanotis*.



Ascidians

1, *Perophora*: *a*, mouth; *b*, vent; *c*, intestinal canal; *d*, stomach; *e*, common tubular stem. 2, *Ascidia echinata*. 3, *Ascidia virginea*. 4, *Cynthia quadrangularis*. 5, *Botryllus violaceus*.

Ascle'piades (-dēz), the name of a number of ancient Greek writers—poets, grammarians, &c—of whom little is known, and also of several ancient physicians, the most celebrated of whom was *Asclepiades*, of Bithynia, who acquired considerable repute at Rome about the beginning of the first century B.C.

Ascle'pias, or **Swallow-wort**, a genus of plants, the type and the largest genus of the nat. ord. *Asclepiadaceæ*. Most of the species are North American herbs, having opposite, alternate, or verticillate leaves. Many of them possess powerful medicinal qualities. *A. decumbens* is diaphoretic and sudorific, and has the singular property of exciting general perspiration without increasing in any sensible degree the heat of the body; *A. curassavica* is emetic, and its roots are frequently sent to England as ipecacuanha; the roots of *A. tuberōsa* are famed for diaphoretic properties. Many other species are also used as medicines, and several are cultivated for the beauty of their flowers.

Asclepios. See *Æsculapius*.

As'coli, or **Ascoli Piceno** (ancient, **Ascŭlum**), a province in Central Italy.—The capital of the province, also called *Ascoli Piceno*, episcopal see of the Marches (the ancient Ascŭlum), is situated 90 miles north-east of Rome and contains, among several handsome new buildings, the remains of temples, an ancient theatre, &c. It has also many fine pre-Renaissance buildings, such as the Gothic Church S. Francesco and the Palazzo del Commune. At

Castel Trosino, near Ascoli, a necropolis of the seventh century was discovered in 1895. Population of the town, 28,882; of the province, 261,835 (1915).

As'coli Satriano (ancient, **Ascŭlum Apŭlum**), a town of S. Italy, province Foggia. Pop. 9700.

Ascomyce'tes (-tēz), one of the main subdivisions of the Eumycetes or Higher Fungi, distinguished by their principal spores being produced in organs called *asci*. Typically, an *ascus* is a cylindrical or club-shaped structure containing at maturity eight *ascospores*, which are usually liberated explosively and thereafter dispersed by the wind. As a rule numerous asci are massed together in a layer or *ascus-hymenium*, which is variously disposed on a more or less massive fruit-body, according to the form and structure of which the group is further subdivided into a number of sections and families, the chief being Erysiphales, Plectascineæ, Pyrenomycetes, Discomycetes (q.v.).

Asco'nus (Quintus A. Pedianus), a Roman writer of the first century A.D., who wrote a life of Sallust, a reply to the detractors of Virgil, and commentaries on Cicero's orations, some of which are extant.

As'cot, an English race-course adjacent to the s.w. extremity of the great park of Windsor. The races, which take place in the second week in June, constitute, for value of stakes and quality of horses, the best meeting of the year, as it is the most fashionable.

As'gard (literally, gods' yard, or the abode of the gods), in Scandinavian mythology the home of the gods or *Æsir*, rising, like the Greek Olympus, from *midgard*, or the middle world, that is, the earth. It was here that Odin and the rest of the gods, the twelve *Æsir*, dwelt—the gods in the mansion called Gladsheim, the goddesses dwelling in Vingulf. Walhalla, in which heroes slain in battle dwelt, was also here. Below the boughs of the ash tree Yggdrasill the gods assembled every day in council.

Asgill (as'gil), John, an eccentric English writer, a lawyer by profession, born 1659, died 1738. In 1699 he published a pamphlet to prove that Christians were not necessarily liable to death, death being the penalty imposed for Adam's sin and Christ having satisfied the law. Having crossed

over to Ireland, he was beginning to get into a good practice, and was elected to the Irish House of Commons, when his pamphlet was ordered to be burned by the public hangman, and he himself was expelled the House. His whole subsequent life was passed in pecuniary and other troubles, mostly in the Fleet or within the rules of the King's Bench.

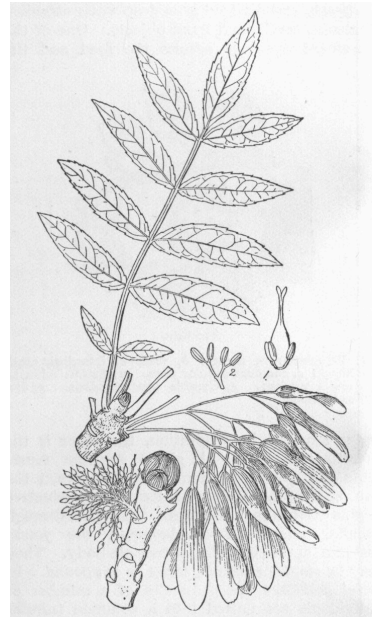
Ash (*Fraxinus*), a genus of deciduous trees belonging to the nat. ord. Oleaceæ, having imperfect flowers and a seed-vessel prolonged into a thin wing at the apex (called a *samara*). There are a good many species, chiefly indigenous to North America. The common ash (*F. excelsior*), the only species indigenous to Central and Northern Europe, has a smooth bark, and grows tall and rather slender. The branches are flattened; the leaves have five pairs of pinnæ, terminated by an odd one, dark-green in colour; lanceolate, with serrated edges. The flowers are produced in loose spikes from the sides of the branches, and are succeeded by flat seeds which ripen in autumn. It is one of the most useful of British trees on account of the excellence of its hard tough wood and the rapidity of its growth, but often suffers greatly from a canker caused by the fungus *Nectria ditissima*. There are many varieties of it, as the weeping-ash, the curled-leaved ash, the entire-leaved ash, &c. The flowering or manna ash (*F. Ornus*), by some placed in a distinct genus (*Ornus*), is a native of the south of Europe and Palestine. It yields the substance called manna, which is obtained by making incisions in the bark, when the juice exudes and hardens. Among American species are the white ash (*F. americana*), with lighter bark and leaves; the red or black ash (*F. pubescens*), with a brown bark; the black ash (*F. sambucifolia*), the blue ash, the green ash, &c. They are all valuable trees. The mountain-ash or rowan belongs to a different order.

Ash, or **Ashes**, the incombustible residue of organic bodies (animal or vegetable) remaining after combustion; in common usage, any incombustible residue of bodies used as fuel; as a commercial term, the word generally means the ashes of vegetable substances, from which are extracted the alkaline matters called potash, pearl-ash, kelp, barilla, &c.

Ashan'go, a region in the interior of Southern Africa between lat. 1° and 2° s., and between the Ogowe and the Lower Congo, a mountainous country in the French territory. The inhabitants belong to the Bantu stock, and among

them are a dwarfish people, the Obongo, said to be about 4½ feet high at most.

Ashanti', a British territory in West Africa belonging to the Gold Coast Colony, of which it forms a large inland portion, under a chief commissioner; area about 20,000 sq. miles. It is in great part hilly, well-watered, and covered with dense tropical vegetation, and rich forests with excellent timber trees. The country round the towns, however, is carefully cultivated. The crops are chiefly rice, maize, millet, sugar-cane, cocoa, and yams, the last forming the staple vegetable food of the natives. Rubber is also a product. The domestic animals are cows, horses of small size, goats, and a species of hairy sheep. The wild animals include the elephant, rhinoceros, buffalo, lion, hippopotamus, &c. Birds are numerous, and crocodiles and other reptiles abound. Gold is abundant, and mining concessions are now successfully worked. The Ashantis make excellent cotton and silk cloths, articles in gold and other metals, earthenware, leather, &c. Ashanti used to form a kingdom ruled despotically, human sacrifices being very common. It is now under British administration, and attention is given to education, sanitation, agriculture, means of communication, &c. There are Government schools at Coomassie and Sunyani and a number of missionary schools. The chief town is Coomassie (or Kumassi), now reached by a railway from the coast (length 168 miles); its population is 24,000. The British first came in contact with the Ashantis in 1807, and hostilities continued off and on till 1826, when they were driven from the sea-coast. Immediately after the transfer of the Dutch settlements on the Gold Coast to Britain in 1872—when the entire coast remained in British hands—the Ashantis reclaimed the sovereignty of the tribes round the settlement of Elmina. This brought on a war, leading to a British expedition in 1874, in which Coomassie was captured. In 1896 the country became a British protectorate. In 1901 a rebellion had to be put down, and next year Ashanti



Common Ash (*Fraxinus excelsior*)

1, Hermaphrodite flower. 2, Anthers of male flower.

was fully annexed. Pop. 287,814 (1911).—Cf. R. A. Freeman, *Travels and Life in Ashantee and Jaman*.

Ash'bourne, a town of England, in Derbyshire, 12 miles N.W. of Derby, with manufactures of cottons and lace. Pop. 4039.

Ash'burton, a town in Devonshire, England, 16 miles S.W. of Exeter, a parliamentary borough till 1868, and till 1918 giving name to a parliamentary division. Pop. (1921), 2362.

Ash'burton, Alexander Baring, Lord, a British statesman and financier, born 1774, died 1848. A younger son of Sir Francis Baring, he was bred to commercial pursuits, which for some years kept him in the United States and Canada, and in 1810 he became head of the great firm of Baring Brothers & Co. He sat in Parliament from 1806 to 1835, when he was raised to the peerage, after being a member of Peel's Government (1834-5).

Ash'burton Treaty, a treaty concluded at Washington, 1842, by Alexander Baring, Lord Ashburton, and the President of the United States; it defined the boundaries between the States and Canada, &c.

Ashby-de-la-Zouch (ash'bi-del-a-zöch'), a town in Leicestershire, England, on the borders of Derbyshire, with manufactures of hosiery, leather, &c. Pop. (1921), 4983.

Ash'dod, a place on the coast of Palestine, formerly one of the chief cities of the Philistines, now an insignificant village.

A'sher, one of Jacob's sons, and founder of the tribe called after him, who occupied a fertile territory in Palestine along the coast between Carmel and Lebanon.

Ashe'ra, an ancient Semitic goddess whose symbol was the phallus. In the Revised Version of the Old Testament this word is used to translate what in the Authorized Version is translated "grove", as connected with the idolatrous practices into which the Jews were prone to fall.

Ash'ford, a thriving town of England, in Kent, situated near the confluence of the upper branches of the River Stour, with large locomotive and

railway-carriage works. It gives name to a parliamentary division of the county. Pop. (1921), 14,355.

Ashington, an urban district or town of England, Northumberland, north-east of Morpeth, and about 2 miles from the sea, in a district of collieries. Pop. (1921), 29,406.

Ashi'ra, a native race or people of Western Equatorial Africa, to the south of the Ogowe River, in the French Congo Territory.

Ash'land, a city of the United States, in Wisconsin. Pop. (1920), 11,334.—Also a city of Kentucky. Pop. (1920), 14,729.

Ash'lar, masonry consisting of stones squared and smoothed in front and built in regular courses.

Ashley, Lord. See *Shaftesbury, First Earl of*.

Ashmead-Bartlett, Sir Ellis, English politician, born in 1849. He entered Parliament in 1880, and was Civil Lord of the Admiralty from 1885 to 1892. He served in the Græco-Turkish and South African wars, and was knighted in 1892. He died in 1902.

Ash'mole, Elias, English antiquary, born 1617, died 1692. He became a chancery solicitor in London, but afterwards studied at Oxford, taking up mathematics, physics, chemistry, and particularly astrology. He published *Theatrum Chymicum* in 1652. On the Restoration he received the post of Windsor Herald, and other appointments both honourable and lucrative. In 1672 appeared his *History of the Order of the Garter*. He presented to the University of Oxford his collection of rarities, to which he afterwards added his books and MSS., thereby commencing the Ashmolean Museum.

Ash'taroth. See *Astarte*.

Ashton-in-Makerfield, a town of Lancashire, England, 4 miles from Wigan, with collieries, cotton-mills, &c. Pop. (1921), 22,489.

Ashton-under-Lyne, a municipal and parliamentary borough of Lancashire, England, 6 miles east of Manchester, on the north bank of the River Tame, a well-built place, with handsome streets and public buildings. The chief employment is the cotton manufacture, but there are also

collieries and ironworks, which employ a great many persons. Pop. 51,080; (municipal borough) (1921), 43,333.

Ashton-upon-Mersey, a town or urban district of England, Cheshire, on the south side of the Mersey, several miles south-west of Manchester. Pop. (1921), 7780.

Ashura'da, a small island in the s.e. corner of the Caspian, formed by Russia into a trading station.

Ash-Wednesday, the first day of Lent, so called from a custom in the Western Church of sprinkling ashes that day on the heads of penitents, then admitted to penance. The period at which the fast of Ash-Wednesday was instituted is uncertain; but it probably dates from the eighth century at least. In the Roman Catholic Church the ashes are now strewn on the heads of all the clergy and people present. In the Anglican Church Ash-Wednesday is regarded as an important fast day.

Asia, the largest of the great divisions of the earth; length, from the extreme south-western point of Arabia, at the Strait of Bab-el-Mandeb, to the extreme north-eastern point of Siberia—East Cape, or Cape Vostochni, in Behring's Strait—6900 miles; breadth, from Cape Chelyuskin, in Northern Siberia, to Cape Romania, the southern extremity of the Malay Peninsula, 5300 miles; area estimated at 17,250,000 sq. miles, about a third of all the land of the earth's surface. On three sides, N., E., and S., the ocean forms its natural boundary, while in the W. the frontier is marked mainly by the Ural Mountains, the Ural River, Caspian Sea, the Caucasus, the Black Sea, the Mediterranean, the Suez Canal, and the Red Sea. There is no proper separation between Asia and Europe, the latter being really a great peninsula of the former. Asia, though not so irregular in shape as Europe, is broken in the S. by three great peninsulas, Arabia, Hindustan, and Farther India, while the east coast presents peninsular projections and islands, forming a series of sheltered seas and bays, the principal peninsulas being Kamtshatka and Corea. The principal islands are those forming the Malay or Asiatic Archipelago, which stretch round in a wide curve on the S.E. of the continent. Besides the larger islands—Sumatra, Java, Borneo, Celebes, Mindanao, and Luzon (in the Philippine group)—there are countless smaller islands grouped round these. Other islands are Ceylon, in the S. of

India; the Japanese Islands and Sakhalin on the E. of the continent; Formosa, s.e. of China; Cyprus, s. of Asia Minor; and New Siberia and Wrangell, in the Arctic Ocean.

The mountain systems of Asia are of great extent, and their culminating points are the highest in the world. The greatest of all is the Himálaya system, which lies mainly between long. 70° and 100° E. and lat. 28° and 37° N. It extends, roughly speaking, from north-west to south-east, its total length being about 1500 miles, forming the northern barrier of Hindustan. The loftiest summits are Mount Everest, 29,002 feet high, Godwin-Austen, 28,265, and Kanchinjinga, 28,156. The principal passes, which rise to the height of 18,000 to 20,000 feet, are the highest in the world. A second great mountain system of Central Asia, connected with the north-western extremity of the Himálaya system by the elevated region of Pamir (about long. 70° - 75° E., lat. 37° - 40° N.), is the Thian-Shan system, which runs north-eastward for a distance of 1200 miles. In this direction the Altai, Sayan, and other ranges continue the line of elevations to the north-eastern coast. A north-western continuation of the Himálaya is the Hindu Kush, and farther westward a connection may be traced between the Himálaya mass and the Elburz range (18,460) feet, south of the Caspian, and thence to the mountains of Kurdistan, Armenia, and Asia Minor.

There are vast plateaux and elevated valley regions connected with the great central mountain systems, but large portions of the continent are low and flat. Tibet forms the most elevated table-land in Asia, its mean height being estimated at 15,000 feet. On its south is the Himálayan range, while the Kuen-Lun range forms its northern barrier. Another great but much lower plateau is that which comprises Afghanistan, Baluchistan, and Persia, and which to the north-west joins into the plateau of Asia Minor. The principal plain of Asia is that of Siberia, which extends along the north of the continent and forms an immense alluvial tract sloping to the Arctic Ocean. Vast swamps or peat-mosses called *tundras* cover large portions of this region. South-west of Siberia, and stretching eastward from the Caspian, is a low-lying tract consisting to a great extent of steppes and deserts, and including in its area the Sea of Aral. In the east of China there is an alluvial plain of some 200,000 sq. miles in extent; in Hindustan are plains extending for 2000 miles along the south slope of the Himálaya; and between Arabia and Persia, watered by the Tigris and Euphrates, is the plain of

Mesopotamia or Assyria, one of the richest in the world. Of the deserts of Asia the largest is that of Gobi (long. 90°-120° E., lat. 40°-48° N.), large portions of which are covered with nothing but sand or display a surface of bare rock. An almost continuous desert region may also be traced from the desert of North Africa through Arabia (which is largely occupied by bare deserts), Persia, and Baluchistan to the Indus.

POLITICAL MAP OF ASIA



Some of the largest rivers of Asia flow northward to the Arctic Ocean—the Obi, the Yenisei, and the Lena. The Hoang-Ho and Yang-tse, and the Amoor, are the chief of those which flow into the Pacific. The Ganges, Brahmaputra, Irawadi, and Indus flow into the Indian Ocean. The Persian Gulf receives the united waters of the Euphrates and the Tigris. There are several systems of inland drainage, large rivers falling into lakes which have no outlet.

The largest lake of Asia (partly also European) is the Caspian Sea, which receives the Kur from the Caucasus (with its tributary the Aras from Armenia), and the Sefid Rud and other streams from Persia (besides the Volga from European Russia, and the Ural, which is partly European, partly Asiatic). The Caspian lies in the centre of a great depression, being 83 feet below the level of the Sea of Azov. East from the Caspian is the Sea of Aral,

which, like the Caspian, has no outlet, and is fed by the Rivers Amu-Darya (Oxus) and Syr-Darya. Still farther east, to the north of the Thian-Shan Mountains, and fed by the Ili and other streams, is Lake Balkhash, also without an outlet and very salt. Other lakes having no communication with the ocean are Lob Nor, in the desert of Gobi, receiving the River Tarim, and the Dead Sea, far below the level of the Mediterranean, and fed by the Jordan. The chief fresh-water lake is Lake Baikal, in the south of Siberia, between long. 104° and 110° E., a mountain lake from which the Yenisei draws a portion of its waters.

Geologically speaking, large areas of Asia are of comparatively recent date, the lowlands of Siberia, for instance, being submerged during the tertiary period, if not more recently. Many geologists believe that subsequently to the glacial period there was a great sea in Western Asia, of which the Caspian and Aral Seas are the remains. The desiccation of Central Asia is still going on, as is also probably the upheaval of a great part of the continent. The great mountain chains and elevated plateaux are of ancient origin, however, and in them granite and other crystalline rocks are largely represented. Active volcanoes are only met with in the extreme east (Kamtchatka) and in the Eastern Archipelago. From the remotest times Asia has been celebrated for its mineral wealth. In the Altai and Ural Mountains gold, iron, lead, and platinum are found; in India and other parts rubies, diamonds, and other gems are, or have been, procured; salt in Central Asia; coal in China, India, Central Asia, &c.; petroleum in the districts about the Caspian and in Burmah; bitumen in Syria; while silver, copper, sulphur, &c., are found in various parts.

Every variety of climate may be experienced in Asia, but as a whole it is marked by extremes of heat and cold and by great dryness, this in particular being the case with vast regions in the centre of the continent and distant from the sea. The great lowland region of Siberia has a short but very hot summer, and a long but intensely cold winter, the rivers and their estuaries being fast bound with ice, and at a certain depth the soil is hard frozen all the year round. The northern part of China to the east of Central Asia has a temperate climate with a warm summer, and in the extreme north a severe winter. The districts lying to the south of the central region, comprising the Indian and Indo-Chinese Peninsulas, Southern China, and the adjacent islands, present the characteristic climate and vegetation of the southern

temperate and tropical regions modified by the effects of altitude. Some localities in Southern Asia have the heaviest rainfall anywhere known. As the equator is approached the extremes of temperature diminish till at the southern extremity of the continent they are such as may be experienced in any tropical country. Among climatic features are the monsoons of the Indian Ocean and the eastern seas, and the cyclones or typhoons, which are often very destructive.

The plants and animals of Northern and Western Asia generally resemble those of similar latitudes in Europe (which is really a prolongation of the Asiatic continent), differing more in species than in genera. The principal mountain trees are the pine, larch, and birch; the willow, alder, and poplar are found in lower grounds. In the central region European species reach as far as the Western and Central Himálaya, but are rare in the Eastern. They are here met by Chinese and Japanese forms. The lower slopes of the Himálaya are clothed almost exclusively with tropical forms. Higher up, between 4000 and 10,000 feet, are found all the types of trees and plants that belong to the temperate zone, there being extensive forests of conifers. Here is the native home of the deodar cedar. The south-eastern region, including India, the Eastern Peninsula, and China, with the islands, contains a vast variety of plants useful to man and having here their original habitat, such as the sugar-cane, rice, cotton, and indigo, pepper, cinnamon, cassia, clove, nutmeg, and cardamoms, banana, coco-nut, areca, and sago palms; the mango and many other fruits, with plants producing a vast number of drugs, caoutchouc, and gutta-percha. The forests of India and the Malay Peninsula contain oak, teak, sâl, and other timber woods, besides bamboos, palms, sandal-wood, &c. The palmyra palm is characteristic of Southern India; while the talipot palm flourishes on the western coast of Hindustan, Ceylon, and the Malay Peninsula. The cultivated plants of India and China include wheat, barley, rice, maize, millet, sugar-cane, tea, coffee, indigo, cotton, jute, opium, tobacco, &c. In North China and the Japanese Islands large numbers of deciduous trees occur, such as oaks, maples, limes, walnuts, poplars, and willows, the genera being European but the individual species Asiatic. Among cultivated plants are wheat, and in favourable situations rice, cotton, the vine, &c. Coffee, rice, sugar, &c., are extensively grown in some of the islands of the Asiatic Archipelago. In Arabia and the warmer valleys of Persia, Afghanistan, and Baluchistan aromatic shrubs are

abundant. Over large parts of these regions the date palm flourishes and affords a valuable article of food. Gum-producing acacias are, with the date palm, the commonest trees in Arabia. African forms are found extending from the Sahara along the desert region of Asia.

Nearly all the mammals of Europe are found in Northern Asia, with numerous additions to the species. Central Asia is the native land of the horse, the ass, the ox, the sheep, and the goat. Both varieties of the camel, the single and the double humped, are Asiatic. To the inhabitants of Tibet and the higher plateaux of the Himálaya the yak is what the reindeer is to the tribes of the Siberian plain, almost their sole wealth and support. The elephant, of a different species from that of Africa, is a native of tropical Asia. The Asiatic lion, which inhabits Arabia and Persia, and still exists in the north-west of India, is smaller than the African species. Bears are found in all parts, the white bear in the far north, and other species in the more temperate and tropical parts. The tiger is the most characteristic of the larger Asiatic Carnivora. It is found in Armenia and throughout the entire continent, being absent, however, from the greater portion of Siberia and from the high table-land of Tibet; it is found also in Sumatra, Java, and Bali. In South-Eastern Asia and the islands we find the rhinoceros, buffalo, ox, deer, squirrels, porcupines, &c. In birds nearly every order is represented. Among the most interesting forms are the hornbills, the peacock, the Impey pheasant, the tragopan or horned pheasant, and other gallinaceous birds, the pheasant family being very characteristic of South-Eastern Asia. It was from Asia that the common domestic fowl was introduced into Europe. The tropical parts of Asia abound in monkeys, of which the species are numerous. Some are tailed, others, such as the orang, are tailless, but none have prehensile tails like the American monkeys. In the Malay Archipelago marsupial animals, so characteristic of Australia, first occur in the Moluccas and Celebes, while various mammals common in the western part of the Archipelago are absent. A similar transition towards the Australian type takes place in the species of birds. (See *Wallace's Line*.) Of marine mammals the dugong is peculiar to the Indian Ocean; in the Ganges is found a peculiar species of dolphin. At the head of the reptiles stands the Gangetic crocodile, frequenting the Ganges and other large rivers. Among the serpents are the cobra de capello, one of the most deadly snakes in existence; there are also large boas and pythons, besides sea and fresh-water

snakes. The seas and rivers produce a great variety of fish. The Salmonidæ are found in the rivers flowing into the Arctic Ocean. Two rather remarkable fishes are the climbing perch and the archer-fish. The well-known goldfish is a native of China.

Asia is mainly peopled by races belonging to two great ethnographic types, the Caucasian or fair type, and the Mongolic or yellow. To the former belong the Aryan, or Indo-European, and the Semitic races, both of which mainly inhabit the south-west of the continent; to the latter belong the Malays and Indo-Chinese in the S.E., as well as the Mongolians proper (Chinese, &c.), occupying nearly all the rest of the continent. To these may be added certain races of doubtful affinities, as the Dravidians of Southern India, the Cingalese of Ceylon, the Ainos of Yesso, and some negro-like tribes called Negritos, which inhabit Malacca and the interior of several of the islands of the Eastern Archipelago. The total population is estimated at 823,000,000, or more than half that of the whole world, of which 330,000,000 inhabit Chinese territory, 302,000,000 British, and 25,000,000 Russian. Portions of Asia are under the control of European Powers (Russia, Great Britain, Holland, France), of the United States of America, China, and Japan. The chief States are China, Japan, Corea, Siam, Afghanistan, Persia, and Arabia. The chief religions are the Brahmanism of India, the Buddhism of Burmah, China, &c., the creeds of Confucius and Lao-tse in China, and the various forms of Mahommedanism in Arabia, Persia, India, &c. More than a half of the whole population profess some form of Buddhism. Several native Christian sects are found in India, Armenia, Kurdistan, and Syria.

Asia is generally regarded as the cradle of the human race. It possesses the oldest historical documents, and, next to the immediately contiguous kingdom of Egypt, the oldest historical monuments in the world. The Old Testament contains the oldest historical records which we have of any nation in the form of distinct narrative. The period at which Moses wrote was probably 1500 or 1600 years before the Christian era. His and the later Jewish writings confine themselves almost exclusively to the history of the Hebrews; but in Babylonia, as in Egypt, civilization had made great advances long before this time. The earliest seat of the Aryan race some assign to the banks of the Oxus. Hence, perhaps from the pressure of the Mongolian tribes to the north, they spread themselves to the south-east and south-west, finally occupying Northern India, Persia, and other parts of

Western Asia, and spreading into Europe, perhaps about 2000-1500 B.C. In China authentic history extends back probably to about 1000 B.C., with a long preceding period of which the names of dynasties are preserved without chronological arrangement. The kingdoms of Assyria, Babylonia, Media, and Persia predominated by turns in South-Western Asia. In regard to the history of these monarchies, much light has been obtained from the decipherment of the cuneiform inscriptions. The arms of the Pharaohs extended into Asia, but their conquests there were short-lived. From Cyrus (559 B.C.), who extended the empire of Persia from the Indus to the Mediterranean, while his son, Cambyses, added Egypt and Libya to it, to the conquest of Alexander (330 B.C.) Persia was the dominant Power in Western Asia. Alexander's great empire became broken up into separate kingdoms, which were finally absorbed in the Roman Empire, and this ultimately extended to the Tigris. Soon after the most civilized portions of the three continents had been reduced under one empire, the great event took place which forms the dividing-line of history, the birth of Christ and the spread of Christianity. In A.D. 226 a protracted struggle began between the newer Persian Empire and the Romans, which lasted till the advent of Mahomet, and the conquests of the Arabians. Persia was the first great conquest of Mahomet's followers. Syria and Egypt soon fell before their arms, and within forty years of the celebrated flight of Mahomet from Mecca (the *Hejra*), the sixth of the caliphs, or successors of the Prophet, was the most powerful sovereign of Asia. The Mongols next became the dominant race. In 999 Mahmud, whose father, born a Turkish slave, became Governor of Ghazni, conquered India, and established his rule. The dynasty of the Seljuk Tartars was established in Aleppo, Damascus, Iconium, and Kharism, and was distinguished for its struggles with the Crusaders. Othman, an emir of the Seljuk sultan of Iconium, established the Ottoman Empire in 1300. About 1220 Genghis Khan, an independent Mongol chief, made himself master of Central Asia, conquered Northern China, overran Turkestan, Afghanistan, and Persia; his successors took Bagdad and abolished the caliphate. In Asia Minor they overthrew the Seljuk dynasty. One of them, Timur or Tamerlane, carried fire and sword over Northern India and Western Asia, defeated and took prisoner Bajazet, the descendant of Othman (1402), and received tribute from the Greek emperor. The Ottoman Empire soon recovered from the blow inflicted by Timur, and Constantinople was taken and the Eastern Empire finally overthrown by the

Sultan Mohammed II in 1453. China recovered its independence about 1368 and was again subjected by the Manchu Tartars (1618-45), soon after which it began to extend its empire over Central Asia. Siberia was conquered by the Cossacks on behalf of Russia (1580-4). The same country effected a settlement in the Caucasus about 1786, and has since continued to make steady advances into Central Asia. The discovery by the Portuguese of the passage to India by the Cape of Good Hope led to their establishment on the coast of the peninsula (1498). They were speedily followed by the Spanish, Dutch, French, and British. The struggle between the two last Powers for the supremacy of India was completed by the destruction of the French settlements (1760-5). At present the forms of government in Asia range from the primitive rule of the nomad sheik to the constitutional monarchy of Japan.—BIBLIOGRAPHY: Sven Hedin, *Through Asia*; H. F. Blanford, *Elementary Geography of India, Burma, and Ceylon*; Max. Müller, *The Sacred Books of the East*; A. Little, *The Far East*; R. Cobbold, *Innermost Asia*; Colonel A. Durand, *The Making of a Frontier*; J. G. C. Chamberlain, *Continents and their Peoples*; E. Huntington, *The Pulse of Asia*; E. C. Hannah, *Eastern Asia*.

Asia, Central, a designation loosely given to the regions in the centre of Asia east of the Caspian, also called Turkestan, and formerly Tartary. The eastern portion belongs to China, the western to Russia. Russian Central Asia comprises the Kirghiz Steppe (Uralsk, Turgai, Akmolinsk, Semipalatinsk, &c.), and what was the government-general of Turkestan till 1918, besides the territory of the Turkomans, or Transcaspia and Merv. See *Turkestan, Republic of*.

Asia Minor, the most westerly portion of Asia, being the peninsula lying west of the Upper Euphrates, and forming part of Asiatic Turkey. It forms an extensive plateau, with lofty mountains rising above it, the most extensive ranges being the Taurus and Anti-Taurus, which border it on the south and south-east, and rise to over 10,000 feet. There are numerous salt- and freshwater lakes. The chief rivers are the Kizil-Irmak (Halys), Sakaria (Sangarius), entering the Black Sea; and the Sarabat (Hermus) and Menderes (Mæander), entering the Ægean. The coast regions are generally fertile, and have a genial climate; the interior is largely arid and dreary. Valuable forests and fruit-trees abound. Smyrna is the chief town. *Anatolia* is an equivalent name. See *European War; Turkey*.

Asiago, a town in Italy in the province of Vicenza, capital of the Seven Communes (Sette Comuni). In the great European War several battles were fought on the Asiago Plateau. The town was evacuated by the Italians on 28th May, 1916, but retaken on 25th June, 1916. See *European War*.

Asiatic Societies, learned bodies instituted for the purpose of collecting information respecting the different countries of Asia, such as the Asiatic Society of Bengal, founded in 1784 by Sir William Jones; and the Royal Asiatic Society of Great Britain and Ireland, established by Colebrooke, and opened in 1823. There are similar societies on the European Continent and in America, such as the Société Asiatique at Paris, founded in 1822; the Oriental Society of Germany (Deutsche Morgenländische Gesellschaft), founded in 1845; and the Oriental Society at Boston, founded in 1842.

Asiphona'ta, or **Asiphon'ida**, an order of lamellibranchiate, bivalve molluscs, destitute of the siphon or tube through which, in the Siphonata, the water that enters the gills is passed outwards. It includes the oysters, the scallop-shells, the pearl-oyster, the mussels, and in general the most useful and valuable molluscs.

Asir. See *Hejaz*.

Askabad', the administrative centre of the Russian province of Transcaspia, situated in the Akhal Tekke oasis, and occupied by Skobelev in Jan., 1881, after the sack of Geok Tepé. Its distance from Merv is 232 miles, from Herat 388 miles. Pop. 54,000.

As'kew, Anne, a victim of religious persecution, born 1521, martyred 1546. She was a daughter of Sir William Askew of Lincolnshire, and was married to a wealthy neighbour named Kyme, who, irritated by her Protestantism, drove her from his house. In London, whither she went probably to procure a divorce, she spoke against the dogmas of the old faith, and, being tried, was condemned to death as a heretic. Being put to the rack to extort a confession concerning those with whom she corresponded, she continued firm, and was then taken to Smithfield, chained to a stake, and burned.

Askja (a'sk'ya'), a volcano near the centre of Iceland, first brought into notice by an eruption in 1875. Its crater is 17 miles in circumference,

surrounded by a mountain-ring from 500 to 1000 feet high, the height of the mountain itself being between 4000 and 5000 feet.

As'mannshausen (-hou-zn), a Prussian village on the Rhine, in the district of Wiesbaden, celebrated for its wine. Many judges prefer the red wine of Asmannshausen to the best Burgundy, but it retains its merits for three or four years only.

Asmo'dai, or **Asmo'deus**, an evil spirit, who, as related in the book of Tobit, slew seven husbands of Sara, daughter of Raguel, but was driven away into the uttermost parts of Egypt by the young Tobias under the direction of the angel Raphael. Asmodai signifies a desolater, a destroying angel, identical with the demon Aēshma of the Zend-Avesta. He is represented in the Talmud as the prince of demons who drove King Solomon from his kingdom.

Asmonæ'ans, a family of high priests and princes who ruled over the Jews for about 130 years, from 153 B.C., when Jonathan, son of Mattathias, the great-grandson of Chasmon or Asmonæus, was nominated to the high-priesthood.

Asnières (än-yār), a town on the Seine, a N.W. suburb of Paris, a favourite boating resort of the Parisians. Pop. 42,583.

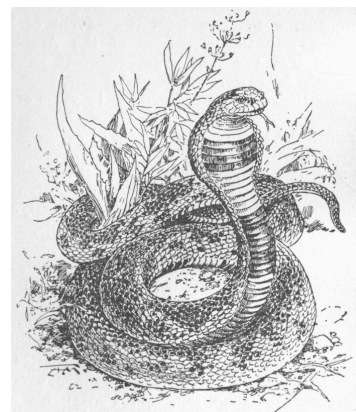
Aso'ka, an Indian sovereign who reigned from 264 to 228 B.C. over the whole of Northern India, grandson of Chandragupta or Sandracottus. He embraced Buddhism, and forced his subjects also to become converts. Many temples and *stupas*, or brick cupolas, still remaining are attributed to him.

Aso'ka (*Jonesia asōca*), an Indian tree, nat. ord. Leguminosæ, having a lovely flower, showing orange, scarlet, and bright-yellow tints; sacred to the god Siva, and often mentioned in Indian literature.

Aso'pus, the name of several rivers in Greece, of which the most celebrated is in Bœotia.

Asp, or **Aspic** (*Naja*, or *Vipěra haje*), a species of viper found in Egypt, resembling the cobra de capello or spectacle-serpent of the East Indies, and having a very venomous bite. When approached or disturbed it elevates its

head and body, swells out its neck, and appears to stand erect to attack the aggressor. Hence the ancient Egyptians believed that the asps were guardians of the spots they inhabited, and the figure of this reptile was adopted as an emblem of the protecting genius of the world. The balancing motions made by it in the endeavour to maintain the erect attitude have led to the employment of the asp as a dancing serpent by the African jugglers. The "deaf adder that stoppeth her ear" of *Psalm lvi*, 4, 5 is translated asp in the margin, and seems to have been this species. Cleopatra is said to have committed suicide by means of an asp's bite, but the incident is generally associated with the Cerastes or horned viper, not with the haje. The name asp is also given to a viper (*Vipera aspis*) common on the continent of Europe.



Asp (*Naja haje*)

Asparagine, or **Aminosuccinamic Acid**, CH_2CONH_2 , $\text{CH}(\text{NH}_2)\text{COOH}$, is a white crystalline substance of unpleasant acid taste found in the shoots of asparagus, in potato and dahlia tribes, and in many other plants, from which it may be extracted by means of water.

Aspar'agus (*Asparāgus officinālis*), a plant of the order Liliaceæ, the young shoots of which, cut as they are emerging from the ground, are a favourite culinary vegetable. In Greece, and especially in the southern steppes of Russia and Poland, it is found in profusion; and its edible qualities were esteemed by the ancients. Pliny states that asparagus was in his time cultivated in gardens, particularly at Ravenna. The best asparagus is grown in gardens near the sea, and hundreds of acres are devoted to its cultivation in Holland and Belgium. It grows wild in Essex and Lincolnshire, but does not attain nearly to the size of the cultivated plant. It is usually raised from seed; and the plants should remain three years in the ground before they are cut; after which, for several years, they will continue to afford a regular annual supply. The beds are protected by straw or litter in winter. Its diuretic properties are ascribed to the presence of a crystalline substance found also in the potato, lettuce, &c.

Aspa'sia, a celebrated woman of ancient Greece, was born at Miletus, in Ionia, but passed a great part of her life at Athens, where her house was the general resort of the most distinguished men in Greece. She won the affection of Pericles, who united himself to Aspasia as closely as was permitted by the Athenian law, which declared marriage with a foreign woman illegal. Her power in the State has often been exaggerated, but it is beyond question that her genius left its mark upon the administration of Pericles. In 432-431 B.C. she was accused of impiety, and was only saved from condemnation by the eloquence and tears of Pericles. After his death (429 B.C.) Aspasia is said to have attached herself to a wealthy but obscure cattle-dealer of the name of Lysicles, whom she raised to a position of influence in Athens. Nothing more is known of her life. She had a son by Pericles, who was legitimated (430 B.C.) by a special decree of the people. There is a bust bearing her name in the Pio Clementino Museum in the Vatican.

Aspatria, a town (urban district) of England, Cumberland, 8 miles north-east of Maryport, with an agricultural college. Pop. 3340.

As'pe, a town of Southern Spain, province of Alicante. Pop. (1921), 3525.

As'pect, in astrology, denotes the situation of the planets with respect to each other. There are five different major aspects: the sextile, when the planets are 60° distant; quartile, when they are 90° distant; trine, when 120° distant; opposition, when 180° distant; and conjunction, when both are in the same longitude. The aspects were classed by astrologers as *benign*, *malignant*, or *indifferent*, according to their fancied influences upon human affairs.

Aspect of Land. See *Exposure*.

As'pen, or trembling poplar (*Pōpŭlus tremŭla*), a species of poplar indigenous to Britain and to most mountainous regions throughout Europe and Asia. It is a beautiful tree of rapid growth and extremely hardy, with nearly circular toothed leaves, smooth on both sides, and attached to footstalks so long and slender as to be shaken by the slightest wind; wood light, porous, soft, and of a white colour, useful for various purposes.

Asper, or **Aspre**, a small Turkish coin, of which there are 120 in the piastre, value $\frac{1}{54}d$.

Aspergill'us, the brush used in Roman Catholic churches for sprinkling holy water on the people. It is said to have been originally made of hyssop.

As'pern and **Esslingen** (or **Essling**) (es'ling-en), two villages east of Vienna, and on the opposite bank of the Danube; celebrated as the chief contested positions in the bloody but indecisive battle fought between the Archduke Charles and Napoleon I, 21st and 22nd May, 1809, when it was estimated that the Austrians lost a third of their army, and the French no less than half.

Asper'ula, the woodruff genus of plants.

Asphalt, or **Asphal'tum**, the most common variety of bitumen; also called mineral pitch. Asphalt is a compact, glossy, brittle, black or brown mineral, which breaks with a polished fracture, melts easily with a strong pitchy odour when heated, and when pure burns without leaving any ashes. It is found in the earth in many parts of Asia, Europe, and America, and in a soft or liquid state on the surface of the Dead Sea, which, from this circumstance, was called *Asphaltites*. It is of organic origin, the asphalt of the great Pitch Lake of Trinidad being derived from bituminous shales, containing vegetable remains in the process of transformation. Asphalt is produced artificially in making coal-gas. During the process much tarry matter is evolved and collected in retorts. If this be distilled, naphtha and other volatile matters escape, and asphalt is left behind. It is sometimes called *Jew's Pitch*.

Asphalte (or **Asphalt**) **Rock**, a limestone impregnated with bitumen, found in large quantities in various localities in Europe, as in the Val de Travers, Neufchâtel, Switzerland; in the department of Ain in France; in Alsace, Hanover, Holstein, Sicily, &c. These rocks contain a variable quantity of bitumen (from 7 or 8 to 20 or 30 per cent) naturally diffused through them. The Val de Travers asphalt was discovered in 1710. In 1837 an English patent was taken out for its application to roads, pavements, terraces, areas, roofs, &c. Since then other asphalte-rocks, as well as artificial preparations made by mixing bitumen, gas-tar, pitch, or other materials with sand, chalk, &c., have been brought into competition with it.

Asphodel (*Asphodelus*), a genus of plants, ord. Liliaceæ, consisting of perennials, with fasciculated fleshy roots, flowers arranged in racemes, six stamens inserted at the base of the perianth, a sessile almost spherical ovary with two cells, each containing two ovules; fruit a capsule with three cells, in each of which there are, as a rule, two seeds. Two species are cultivated in Britain as garden flowers, the yellow asphodel (*Asphodelus luteus*) and the white asphodel (*Asphodelus albus*). The English word 'daffodil' is a perversion of asphodel. The *Asphodelus ramosus*, which attains a height of 5 feet, is cultivated in Algeria and elsewhere, its tubercles yielding a very pure alcohol, and the residue, together with the stalks and leaves, being used in making pasteboard and paper. The asphodel was a favourite plant among the ancients, who were in the habit of planting it round their tombs. In Greek religion it is associated with Persephone, the dead, and the underworld.

Asphyxia, literally, the state of a living animal in which no pulsation can be perceived, but the term is more particularly applied to a suspension of the vital functions from causes hindering respiration. The normal accompaniments of death from asphyxia are dark fluid blood, a congested brain and exceedingly congested lungs, the general engorgement of the viscera, and an absence of blood from the left cavities of the heart while the right cavities and pulmonary artery are gorged. The restoration of asphyxiated persons has been successfully accomplished at long periods after apparent death. The attempt should be made to maintain the heat of the body and to secure the inflation of the lungs as in the case of the apparently drowned. See *Respiratory System*.

Asphyxiating Gas. See *Poison Gas*.

Aspic, a dish consisting of a clear savoury meat jelly, containing fowl, game, fish, &c.

Aspidistra, a genus of plants of the lily family, comprising three or four species, natives of China and Japan, being plants with large smooth oblong lanceolate leaves, rising from an underground rhizome, and with campanulate flowers of a dull purplish or brownish colour. They are now very common in Britain, being especially cultivated as indoor plants.

Aspid'ium, a genus of ferns, nat. ord. Polypodiaceæ, comprising the shield-fern and male-fern.

As'pinwall. See *Colon*.

As'pirate, a name given to any sound like our *h*, to the letter *h* itself, or to any mark of aspiration, as the Greek rough breathing ('). Such characters or sounds as the Sanskrit *kh*, *gh*, *bh*, and the Greek *ch*, *th*, *ph*, are called *aspirates*.

As'pirator, an instrument used to promote the flow of a gas from one vessel into another by means of a liquid. The simplest form of aspirator is a cylindrical vessel containing water, with a pipe at the upper end which communicates with the vessel containing the gas, and a pipe at the lower end also, with a stopcock and with its extremity bent up. By allowing a portion of the water to run off by the pipe at the lower part of the aspirator a measured quantity of air or other gas is sucked into the upper part.

Asple'nium, a genus of ferns, of the nat. ord. Polypodiaceæ. Nine species are found in Britain, among them the well-known Wall-rue.

Aspromon'te, a mountain of Italy in the south-west of Calabria, where Garibaldi was wounded and taken prisoner with the greater part of his army, in Aug., 1862.

Aspropot'amo. See *Achelous*.

Aspull, a town (urban district) of England, Lancashire, 2 or 3 miles north-east of Wigan, with large collieries and other works. Pop. 7851.

Asquith, Herbert Henry, prominent politician of the Liberal party, born in 1852, educated at City of London School and Balliol College, Oxford, where he graduated with high distinction. Called to the Bar in 1876, he became Q.C. in 1890; in 1886 was elected member of Parliament for East Fife, and held his seat for this constituency uninterruptedly until Dec., 1918, when he was defeated. From 1892 to 1895 he was Home Secretary, being also made a Privy Councillor in the former year. Both in regard to the South African War and various other questions, when out of office, he spoke more in harmony with the views of Lord Rosebery than with those of Sir H. Campbell-Bannerman, but under the latter he accepted the post of

Chancellor of the Exchequer in the ministry formed in Dec., 1905. On the retirement of Sir Henry in 1908 he succeeded him as Prime Minister. He at once hastened to draw up the Liberal reform programme, the list of measures including the Lloyd George Budget (1909), the Parliament Act (1911), the Insurance Act, and the Irish Home Rule Bill. In May, 1915, a cabinet crisis having resulted from disagreements, Asquith formed a Coalition Government, eight Unionists being admitted. Towards the end of 1916 there was a feeling in the country not only that the Coalition cabinet of twenty-three ministers was unwieldy, but that Mr. Asquith's Government was not sufficiently energetic and showed too much hesitation in dealing with the vital problems of the war. Mr. Asquith therefore resigned on 5th Dec., 1916, and Mr. Lloyd George formed a new ministry. Unseated in the General Election of 1918, Mr. Asquith accepted the invitation to stand for Paisley in 1920. He was returned by a majority of 2834.

As´rael, the Mahommedan angel of death, who takes the soul from the body.

Ass (*Equus asinus*), a species of the horse genus, supposed by Darwin to have sprung from the wild variety (*Asinus tæniopus*) found in Abyssinia; by some writers to be a descendant of the *onāger* or wild ass, inhabiting the mountainous deserts of Tartary, &c.; and by others to have descended from the kiang or djiggetai (*A. hemionus*) of South-Western Asia. The ass was used in Egypt long before the horse, and it played an important part in Homeric Greece. According to Aristotle, however, it was unknown in his time in Pontus, Scythia, and in the land of the Celts. The ass seems to have been introduced into England in the days of Ethelred, but did not become common before the end of the seventeenth century. Both in colour and size the ass is exceedingly variable, ranging from dark grey and reddish brown to white, and from the size of a Newfoundland dog in North India to that of a good-sized horse. In the south-western countries of Asia and in Egypt, in some districts of Southern Europe, as in Spain, and in Kentucky and Peru, great attention has been paid to selection and interbreeding, with a result no less remarkable than in the case of the horse. Thus in Syria there appear to be four distinct breeds: a light and graceful animal used by ladies, an Arab breed reserved for the saddle, an ass of heavier build in use for ploughing and draft purposes, and the large Damascus breed. The efforts made to raise the deteriorated British breed have only been partially successful. The male

ass is mature at two years of age, the female still earlier. The she-ass carries her young eleven months. The teeth of the young ass follow the same order of appearance and renewal as those of the horse. The life of the ass does not usually exceed thirty years. It is in general much healthier than the horse, and is maintained in this condition by a smaller quantity and coarser quality of food; it is superior to the horse in its ability to carry heavy burdens over the most precipitous roads, and is in no respect its inferior in intelligence, despite the reputation for stupidity which it has borne from very ancient times. The skin is used as parchment to cover drums, &c., and in the East is made into shagreen. The hybrid offspring of the horse and the female ass is the hinny, that of the ass and the mare is the mule; but the latter is by far the larger and more useful animal. Asses' milk, long celebrated for its sanative qualities, more closely resembles that of a woman than any other. It is very similar in taste, and throws up an equally fluid cream, which is not convertible into butter.

Assa. See *Piave, Battles of the; European War.*

Assab', a bay of Africa on the south-west coast of the Red Sea, belonging to the Italian territory of Eritrea, which has been acquired since Italy established here a colony and free port in 1881.

Assafœtida. See *Asafetida.*

Assai-palm (as-ī; *Euterpe oleracœa*), a native of tropical S. America, only about 4 inches in diameter and 60 or 80 feet high, with a crown of leaves, beneath which a small fruit grows on branched horizontal spadices. The pulp of the fruit mixed with water is used as a beverage.

Assal', a salt lake in North-Eastern Africa, in Adal.

Assam', one of the fifteen provinces of British India, separated from Eastern Bengal and reconstituted in 1912; area, 53,015 sq. miles. It forms a series of fertile valleys watered by the Brahmaputra and its tributaries, the valley of the Brahmaputra, which is the main one, consisting of rich alluvial plains, either but little elevated above the river, or so low that large extents of them are flooded for three or four days once or twice in the year, while the course of the river often changes. The climate is marked by great humidity, and malarious diseases are common in the low grounds;

otherwise it is not unhealthy. The whole province, except the cultivated area, may be designated as forest, the trees including teak, sâl, sissoo, the date and sago palm, the areca palm (the betel-nut tree), the Indian fig tree, &c. The article of most commercial importance is tea, which was first exported in 1838, and the yield of which is now over 100,000,000 pounds annually. Other crops raised are rice, Indian corn, pulse, oil-seeds, sugar-cane, hemp, jute, potatoes, &c. In the jungles and forests roam herds of elephants, the rhinoceros, tiger, buffalo, leopard, bear, wild hog, jackal, fox, goat, and various kinds of deer. Among serpents are the python and the cobra. Pheasants, partridges, snipe, wild peacock, and many kinds of water-fowl abound. Coal, petroleum, and limestone are found in abundance; iron is smelted to a small extent; gold-dust is met with; lime is exported to Bengal. There is no single Assamese nationality, and the Assamese language is merely a modern dialect of Bengali. Pop. 6,713,635, 3,637,828 of whom are Hindus, 1,886,528 Mahommedans, 66,430 Christians, 10,506 Buddhists, the rest being chiefly hill tribes of aboriginal faiths. The labourers in the tea-gardens are mostly drawn from Bengal. In 1826 Assam became a possession of Britain, being taken from the Burmese, who had made themselves masters of it about the end of the eighteenth century. The largest town is Sylhet (pop. 14,000).—Cf. E. A. Gait, *History of Assam*.

As'sapan (*Sciuropterus volucella*), the flying-squirrel of N. America, a little animal with folds of skin along its sides which enable it to take leaps of 40 or 50 yards.

Assass'ins (from *hashshāshīn*, drinkers of *hashish*), an Asiatic order or society having the practice of assassination as its most distinctive feature, founded by Hassan Ben Sabbah, the Himyarite, a *dai* or missionary of the heterodox Mahommedan sect the Ismailites. The society grew rapidly in numbers, and in 1090 the Persian fortress of Alamut fell into their hands. Other territories were added, and the order became a recognized military power. Its organization comprised seven ranks, at its head being the Sheikh-al-Jebala or 'Old man of the mountains'. Upon a select band fell the work of assassination, to which they were stimulated by the intoxicating influence of *hashish*. For nearly two centuries they maintained their power under nine sheikhs. Hassan, after a long and prosperous reign, died in 1124. Most of his successors died violent deaths at the hands of relatives or dependents. After proving themselves strong enough to withstand the powerful sultans

Noureddin and Saladin, and making themselves feared by the Crusaders, the *Assassins* were overcome by the Tartar leader Hulaku. The last chief, Rokneddin, was killed for an act of treachery subsequent to his capture, and his death was followed by a general massacre of the assassins, in which 12,000 perished. Dispersed bands led a roving life in the Syrian mountains, and it is alleged that in the Druses and other small existing tribes their descendants are still to be found. See *Crusades*; *Khoja*.

Assault, in law, an attempt or offer, with force and violence, to do a corporal hurt to another, as by striking at him with or without a weapon. If a person lift up or stretch forth his arm and offer to strike another, or menace anyone with any staff or weapon, it is an assault in law. Assault, therefore, does not necessarily imply a hitting or blow, because in trespass for assault and battery a man may be found guilty of the assault and acquitted of the battery. But every battery includes an assault.

Assaye, or **Assye** (as-sī'), a village in Southern India, in Hyderabad, where Wellington (then Major-General Wellesley) gained a famous victory in 1803. With only 4500 troops at his disposal he completely routed the Mahratta force of 50,000 men and 100 guns. The victory, however, cost him more than a third of his men.

Assaying, the estimation of the amount of pure metal present in an ore or an alloy. The term was originally applied to testing of gold and silver only. It is now usually applied to the determination of the quantity of valuable metal in an ore or alloy, and is also sometimes applied to the estimation of any element which may affect the value of the ore.

Assaying, therefore, means the estimation of one or more metals in an ore or alloy. Before an assay can be made, an average sample of the material must be obtained. If an ore, pieces of material are taken from different parts of the vein or, if already mined, from different parts of the dump. The pieces are crushed up finely and divided into four equal parts. Two of these parts are then mixed and divided into four again, and so on until an average sample has been obtained. In the case of metals in ingots or bars, samples are obtained by drilling and chipping corners or edges. Coins, which are never homogeneous, may be rolled out into a thin sheet and cut into small pieces. A preliminary examination is made to determine the constituents.

Finally an assay of the substance is made. The methods used are determined by the metals and the proportions of these present in the ore or alloy. Originally the term assaying was applied to dry methods, i.e. the substance was heated in a special crucible with a suitable flux, and a bead of metal was obtained which was weighed. An assay now may be carried out in various ways, for example, by fusing with a reducing agent and obtaining a bead of metal, or by dissolving the substance to be assayed in suitable solvent and precipitating the metal as an insoluble salt, or volumetric methods may be used. Dry assaying is still used for gold. The assay depends on first heating the gold ore or alloy with lead in a porous crucible, that is, *cupelling* it. Lead oxidizes on heating in a furnace; part volatilizes, and part of the oxide is absorbed by the cupel and carries with it oxides of other metals with the exception of gold and silver. The proportion of lead must be regulated, depending on the metals alloyed with gold. A bead is obtained containing gold and silver. This is beaten out into a thin plate, and then rolled out until it is thin enough to be rolled up by hand. The gold alloy is rolled up in the fingers into a cornette and treated with nitric acid. This dissolves silver, leaving a brittle cornette of gold, which is thoroughly washed, dried, and weighed. All gold alloys and silver alloys must be assayed, and their fineness stamped on them. The Goldsmith's Company of London is the statutory assay-master of England.—BIBLIOGRAPHY: C. and J. J. Beringer, *A Text-book of Assaying*; E. A. Wraight, *Assaying in Theory and Practice*; J. Park, *A Text-book of Practical Assaying*.

As'segai (from Ar. *as-zahayah*), a spear used as a weapon among the Kaffirs of S. Africa, made of hard wood tipped with iron, and used for throwing or thrusting.

Assembly, General, the supreme ecclesiastical court of the Established Church of Scotland, consisting of delegates from every presbytery, university, and royal burgh in Scotland. It has the countenance of a representative of the king, styled the Lord High Commissioner, who is always a nobleman. It holds its meetings annually and (according to the present practice) in the month of May, usually sitting for ten or twelve days. In its judicial capacity as a court of review, and as the court of last resort, the General Assembly has a right to determine finally every question brought from the inferior courts, by reference, complaint, or appeal. It possesses, besides, a general superintendence of the discipline of the

Church, of the management of the inferior courts, of the conduct of the clergy, and of the morals of the people. In its legislative capacity it has the power of enacting statutes with regard to every subject of ecclesiastical cognizance, which are binding on the Assembly itself, on the inferior courts, and on the individual members of the Church. But by an Act of Assembly in 1697, from its substance and design named the Barrier Act, every proposition for a new law must first be considered in the form of an overture; and though it should be approved of by the Assembly it cannot be enacted as a statute till it has been first transmitted to the several presbyteries of the Church for their consideration, and has received the sanction of at least a majority of the presbyteries. The United Free Church of Scotland has also a General Assembly similar in its constitution and functions to that of the Established Church, and the same is the case with the Presbyterian Churches of Ireland and America.

Assembly, National (France), a body set up in France on the eve of the Revolution. Upon the convocation of the States-General by Louis XVI the privileged nobles and clergy refused to deliberate in the same chamber with the commons or *tiers-état* (third estate). The latter, therefore, on the proposition of the Abbé Siéyès, constituted themselves an *assemblée nationale*, with legislative powers (17th June, 1789). They bound themselves by oath not to separate until they had furnished France with a constitution, and the Court was compelled to give its assent. In the 3250 decrees passed by the Assembly were laid the foundations of a new epoch, and, having accomplished this task, it dissolved itself, 30th Sept., 1791.

Assembly of Divines. See *Westminster Assembly*.

Assembly, The Right of, is an essential principle of popular government, as understood by the British and American constitutions. The right of British citizens to assemble peaceably for any purpose which is not strictly prohibited by law is implied in the right of petition, as affirmed in the Bill of Rights (q.v.). Unlawful assembly, which is a criminal offence, is distinguished from the offence of riot. Whilst the latter is an actual attempt to carry an unlawful purpose into effect, the former is defined as "an assembly with intent to carry out a common purpose which may lead to a breach of the peace".

As'sen, chief town of the province of Drenthe, Holland. Pop. 13,000.

Assent', The Royal, is the approbation given by the sovereign in Parliament to a Bill which has passed both Houses, after which it becomes a law. It may either be done in person, when the sovereign comes to the House of Peers and the assent (in Norman French) is declared by the Clerk of Parliament; or it may be done by letters-patent under the great seal, signed by the sovereign.

As'ser, John, a learned British ecclesiastic, originally a monk of St. David's, distinguished as the instructor, companion, and biographer of Alfred the Great, who appointed him abbot of two or three different monasteries, and finally Bishop of Sherborne, where he died in 908 or 910. His life of Alfred, written in Latin in 893 (*Annales Rerum Gestarum Ælfredi Magni*), is of very great value, though its authenticity has been questioned. There are several English translations of it.

Assessed Taxes, taxes charged upon persons by means of a schedule or paper sent to each, and strictly including such taxes as the income-tax, the house-tax, local rates, &c. In Britain the so-called assessed taxes include those upon servants, carriages, dogs, armorial bearings, &c., though these are really excise licence duties.

Asses'sor, a person appointed to ascertain and fix the amount of taxes, rates, &c.; or a person who sits along with the judges in certain courts, and assists them with his professional knowledge.

As'sets (Fr. *assez*, enough), property or goods available for the payment of a bankrupt or deceased person's obligations. Assets are personal or real, the former comprising all goods, chattels, &c., devolving upon the executor as saleable to discharge debts and legacies. In commerce and bankruptcy the term is often used as the antithesis of 'liabilities', to designate the stock in trade and entire property of an individual or an association.—*Intangible* (or fictitious) assets are those not represented by any existing value, e.g. goodwill; *liquid* assets are cash, investments, or other immediately available funds.

Asside'ans, Haside'ans, or Hasidim ('the pious'), one of the two great sects into which, after the Babylonish captivity, the Jews were divided with

regard to the observance of the law—the Hasidim accepting it in its later developments, the Zadikim professing adherence only to the law as given by Moses. See *Pharisees, Talmudists, Rabbinites*.

Assien'to, the permission of the Spanish Government to a foreign nation to import negro slaves from Africa into the Spanish colonies in America, for a limited time, on payment of certain duties. It was accorded to the Netherlands about 1552, to the Genoese in 1580, and to the French Guinea Company (afterwards the Assiento Company) in 1702. In 1713 the celebrated *Assiento Treaty* with Britain for thirty years was concluded at Utrecht. By this contract the British obtained the right to send yearly a ship of 500 tons, with all sorts of merchandise, to the Spanish colonies. This led to frequent abuses and contraband trade; acts of violence followed, and in 1739 a war broke out between the two Powers. At the peace of Aix-la-Chapelle, in 1748, four years more were granted to the British; but by the Treaty of Madrid, two years later, £100,000 sterling were promised for the relinquishment of the two remaining years, and the contract was annulled.

Assignats (a's-ē-nya'), the name of the national paper currency in the time of the French Revolution. Assignats to the value of 400,000,000 francs were first struck off by the Constituent Assembly, with the approbation of the king, 19th April, 1790, to be redeemed with the proceeds of the sale of the confiscated goods of the Church. On the 27th Aug. of the same year Mirabeau urged the issuing of 2,000,000,000 francs of new assignats, which caused a dispute in the Assembly. Vergasse and Dupont, who saw that the plan was an invention of Clavière for his own enrichment, particularly distinguished themselves as the opponents of the scheme. Mirabeau's exertions, however, were seconded by Péthion, and 800,000,000 francs more were issued. They were increased by degrees to 45,578,000,000, and their value rapidly declined. In the winter of 1792-3 they lost 30 per cent, and, in spite of the law to compel their acceptance at their nominal value, they continued to fall, till in the spring of 1796 they had sunk to one three hundred and forty-fourth their nominal value. This depreciation was due partly to the want of confidence in the stability of the Government, partly to the fact that the coarsely-executed and easily-counterfeited assignats were forged in great numbers. They were withdrawn by the Directory from the currency, and at length redeemed by mandats at one-thirtieth of their nominal value.

Assignee', a person appointed by another to transact some business, or exercise some particular privilege or power. Formerly the persons appointed under a commission of bankruptcy, to manage the estate of the bankrupt on behalf of the creditors, were so called, but now they are called *trustees*.

Assign'ment is a transfer by deed of any property, or right, title, or interest in property, real or personal. Assignments are usually given for leases, mortgages, and funded property.

Assiniboi'a, the smallest of the four districts into which that portion of the north-western territories of Canada now forming the provinces of Saskatchewan and Alberta was divided in 1882. It lay on the west of Manitoba, with Saskatchewan on the north and Alberta on the west, the United States on the south. The name is now given to an electoral district of the province of Saskatchewan. The region contains much good wheat land. Regina was the capital, as it now is of the new province.

Assiniboine, a river of Canada, which flows through Manitoba and joins the Red River at Winnipeg, about 40 miles above the entrance of the latter into Lake Winnipeg, after a somewhat circuitous course of about 500 miles from the west and north-west. Steamers ply on it for over 300 miles.

Assiout. See *Siout*.

Assisi (a's-sē'sē), a small town in Italy, in the province of Umbria, 20 miles north of Spoleto, the see of a bishop, and famous as the birthplace of St. Francis d'Assisi. The splendid church built over the chapel where the saint received his first impulse to devotion is one of the finest remains of mediæval Gothic architecture.

Assi'zes, a term chiefly used in England to signify the sessions of the courts held at Westminster prior to Magna Charta, but thereafter appointed by successive enactments to be held annually in every county. Twelve judges, who are members of the highest courts in England, twice in every year perform a *circuit* into all the counties into which the kingdom is divided (the counties being grouped into seven circuits), to hold these assizes, at which both civil and criminal cases are decided. Occasionally this circuit is performed a third time for the purpose of jail-delivery. In London and Middlesex, instead of circuits, courts of *nisi prius* are held. At the assizes all

the justices of the peace of the county are bound to attend. Special commissions of assize are granted for inquest into certain causes. In Scotland the term *assize* is still applicable to the jury in criminal cases.

Among the more important historic uses of the term *assize* are its application to any sitting or deliberative council, and its transference thence to their ordinances, decrees, or assessments. In the latter sense we have the Assizes of Jerusalem, a code of feudal laws formulated in 1099 under Godfrey of Bouillon; the Assizes of Clarendon (1166), of Northampton (1176), and of Woodstock (1184); also the *assisæ venalium* (1203), for regulating the prices of articles of common consumption; the Assize of Arms (1181), an ordinance for organizing the national militia, &c.

Assmanshausen. See *Asmannshausen*.

Associated Counties, a term applied to Essex, Cambridge, Norfolk, Suffolk, and Hertford, with, subsequently, Huntingdon and Lincoln. The association was formed in 1642 to raise an army for the Parliament and keep the war out of their own districts. The successive leaders were Lord Grey of Wark, the Earl of Manchester, and Cromwell.

Association of Ideas, a doctrine of both psychological and philosophical import. In psychology the term is used to comprise the conditions under which one idea is able to recall another to consciousness. It is, therefore, the doctrine which deals with the reproduction of past experience by a present object of consciousness. The phrase 'association of ideas' was first introduced by Locke, and dealt with by Berkeley and Hartley, who became the founder of the so-called *Associationist School*.

Ass´onance, in poetry, a term used when the terminating words of lines have the same vowel sound but make no proper rhyme. Such verses, having what we should consider false rhymes, are regularly employed in Spanish poetry; but cases are not wanting in leading British poets. Mrs. Browning not only used them frequently, but justified the use of them.

Assouan (a´ s-sō-a´ n´), or **Aswan** (Syēnē), a town of Upper Egypt, on the east bank of the Nile, below the first cataract, opposite the Island of Elephantine. The granite quarries of the Pharaohs are here. Pop. 15,000.

Assouan Dam, a great dam constructed across the Nile in Upper Egypt, near Assouan, at the foot of the first cataract. It is about a mile and a quarter long, and is provided with a large number of sluices in two tiers. It was originally built to a height of about 96 feet between 1898 and 1902, and raised to a height of 112 feet above bed-rock between 1907 and 1911. It is intended to regulate the supply of water for irrigation purposes to the country lower down, the water being stored up at the time when the river is high, and allowed to escape when it is required for the crops. When the reservoir is full it forms a lake about 130 miles long. The dam was planned by Sir William Willcocks, and the work carried out under Sir William Garstin and Sir Benjamin Baker, at a cost of £5,000,000 (Egyptian).

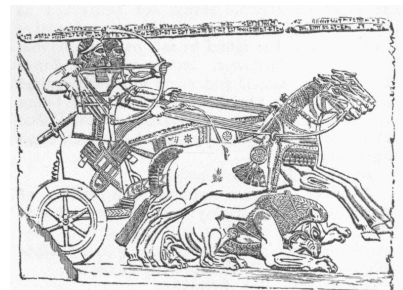
Assump'sit, in English law, an action to recover compensation for the non-performance of a *parole* promise; that is, a promise not contained in a deed under seal. Assumpsits are of two kinds, *express* and *implied*. The former are where the contracts are actually made in word or writing; the latter are such as the law implies from the justice of the case; e.g. employment to do work implies a promise to pay.

Assumption. See *Asuncion*.

Assumption, Feast of, the ecclesiastical festival celebrating the miraculous ascent into Heaven of the Virgin Mary's body as well as her soul, kept on the 15th of August. The legend first appeared in the third or fourth century, and the festival was instituted some three centuries later.

Assurance. See *Insurance*.

Assyr'ia (the **Asshur** of the Hebrews, **Athurâ** of the ancient Persians), an ancient monarchy in Asia, intersected by the upper course of the Tigris, and having the Armenian Mountains on the north and Babylonia on the south; area, about 50,000 sq. miles; surface partly mountainous, hilly, or undulating, partly a portion of the fertile Mesopotamian plain. The numerous remains of ancient habitations show how thickly this vast flat must have once been peopled; now, for the most part, it is a mere



Assyrian bas-relief from the Palace of Nimrûd showing Lion-hunting about 884 B.C.

wilderness. Geographically and historically, however, Assyria and Babylonia are interdependent, and the Assyrians and Babylonians are ethnographically and linguistically the same race. Whereas, however, the classical authors speak of Assyria to the exclusion of Babylonia, the decipherment of the inscriptions has proved that Babylonia was the mother-country, and that Assyria, except during a period of eight centuries, was a dependency of the former. This discovery coincides with the contents of the tenth chapter of *Genesis*. See *Babylonia*.

Ast, Georg Anton Friedrich, German philosopher, 1776-1841. He wrote on æsthetics and the history of philosophy, but is best known as an editor of Plato, whose works he published with a Latin translation and commentary.

As'tacus. See *Crayfish*.

Astar'te, a Syrian goddess, probably corresponding to the *Ashtaroth* of the Hebrews, and representing the productive power of nature. She was a moon-goddess. Some regard her as corresponding to *Hera* (*Juno*), and others identify her with *Aphroditē*.

Astatic needle, a magnetic needle having another needle of the same intensity fixed parallel to it, the poles being reversed, so that the needles neutralize one another, and are unaffected by the earth's magnetism; used in the *astatic galvanometer*.

As'ter, a genus of plants, nat. ord. Compositæ, comprehending several hundred species, scattered over Europe and Asia, but mostly natives of North America. Many are cultivated as ornamental plants. One, *A. Tripolium*, is native in Britain, and is found in salt marshes, having a pretty purple flower. Asters generally flower late in the season, and some are hence called Michaelmas or Christmas Daisies. The China Aster, not an aster proper (*Aster* or *Callistēphus chinensis*), is a very showy annual, of which there are many varieties.

Asterabad'. See *Astrabad*.

Aste'ria, a name applied to a variety of corundum, which displays an opalescent star of six rays of light when cut with certain precautions; and also to the *cat's-eye*, which consists of quartz, and is found especially in Ceylon.

Aster'idaë. See *Asteroidea*.

As'terisk, the figure of a star, thus *, used in printing and writing, as a reference to a passage or note in the margin, or to fill the space when a name, or the like, is omitted.

Asteroi'dea, the ord. of the Echinodermata to which the star-fishes belong. See *Star-fishes*.

As'teroids, Planetoids, or Minor Planets, a numerous group of very small planets revolving round the sun, in the great majority of cases at mean distances, intermediate between those of Mars and Jupiter, in orbits of large eccentricity at considerable inclination to the ecliptic. The diameter of the largest is not supposed to exceed 450 miles, while most of the others are very much smaller. Over one thousand are known, and new members are being constantly discovered. The first to be discovered was Ceres, on 1st Jan., 1801, and within seven years more Pallas, Juno, and Vesta were seen. The diminutive size of these four bodies, and resemblances in their orbits, gave rise to the opinion that they were but the fragments of a planet that had formerly existed and had been brought to an end by some catastrophe. For nearly forty years investigations were carried on, but no more planets were discovered till 8th Dec., 1845, when a fifth planet in the same region of the solar system was discovered. The rapid succession of discoveries that followed was for a time taken as a corroboration of the disruption theory, but the breadth of the zone occupied makes the hypothesis of a shattered planet more than doubtful. In recent years a few have been discovered which are at times considerably within the orbit of Mars, the nearest perihelia being less than 15 million miles beyond the earth's orbit. Another group, known as the 'Trojan Planets', has been found, whose mean distances are practically identical with that of Jupiter. The total mass of the asteroids cannot exceed one-fourth that of the earth, and is probably much less. See *Planets*.

Asterol'epis, a genus of primitive ganoid fishes, found only in a fossil state in the Old Red Sandstone. They were about 1 foot long, and the head and body were enclosed in armour of strong bony plates.

Asthma (ast'ma), difficulty of respiration, returning at intervals, with a sense of stricture across the chest and in the lungs, a wheezing, hard cough

at first, but more free towards the close of each paroxysm, with a discharge of mucus, followed by a remission. Asthma is essentially a spasm of the muscular tissue which is contained in the smaller bronchial tubes. It generally attacks persons advanced in years, and seems, in some instances, to be hereditary. The exciting causes are various—accumulation of blood or viscid mucus in the lungs, noxious vapours, a cold and foggy atmosphere, or a close, hot air, flatulence, accumulated fæces, violent passions, organic diseases in the thoracic viscera, &c. In recent years a treatment first used by Dr. Alexander Francis has come into prominence. By far the most important part of the treatment consists in obviating or removing the several exciting causes. It seldom proves fatal except as inducing dropsy, consumption, &c.

Asti (a`s'tē), a town of Northern Italy, province of Alessandria, 28 miles E.S.E. of Turin, the see of a bishop, with an old cathedral. In the Middle Ages it was one of the most powerful republics of Northern Italy. It was the birthplace of Alfieri, the poet, whose statue adorns the principal square. There is also an equestrian statue of King Humbert. The industries comprise silk, matches, gold, mosaic wares, &c. A favourite wine is produced in the neighbourhood. Asti, anciently *Asta*, was a place of some importance under the Roman emperors, and in the Middle Ages was an independent republic. Pop. 41,252.

Astig'matism (Gr. *a*, not, *stigma*, spot, mark), a malformation or imperfection, congenital or accidental, of the globe of the eye, in consequence of which the individual does not see objects clear and distinct, but with a blurred outline. It is due to the cornea or transparent outer coat of the eye not being regularly spherical, but having different degrees of curvature in different directions. Usually the degree of convexity is not the same horizontally as it is vertically, so that the rays from an object, instead of converging into one focus, meet in more than one. If a person with this defect is looking at vertical lines crossed by horizontal ones he will see the one set more distinctly than the other, though a slight movement will enable him to see the other distinctly also, but not at the same time. Almost all eyes are more or less astigmatic, but persons only become aware of it when it is excessive. Special lenses are required to correct it—usually lenses plane in one direction and concave or convex in the other. Short sight or long sight is often associated with astigmatism, so that suitable spectacles cannot be very easily provided.

Astle, Thomas, English antiquary, born 1735, died 1803. He was a trustee of the British Museum and keeper of the public records in the Tower. His chief work, *The Origin and Progress of Writing*, appeared in 1784, and the portion dealing with mediæval handwriting is still of value. He formed a famous collection of MSS., the most valuable portion of which is now in the British Museum.

Astom'ata, one of the two groups into which the Protozoa are divided with regard to the presence or absence of a mouth, of which organ the Astomata are destitute. The group comprises two classes, Gregarinida and Rhizopoda. See *Stomatoda*.

Aston Manor, formerly a municipal and parliamentary borough of Warwickshire, England, situated about 1½ miles E.N.E. of Birmingham, and engaged in similar branches of industry. It was incorporated with Birmingham in 1911 and gives its name to one of its parliamentary divisions. Pop. 75,029.

Astor, John Jacob, born near Heidelberg, Germany, 1703, died at New York, 1848. In 1783 he emigrated to the United States, settled at New York, and became extensively engaged in the fur trade. In 1811 the settlement of Astoria, founded by him, near the mouth of the Columbia River, was formed to serve as a central depot for the fur trade between the lakes and the Pacific. He subsequently engaged in various speculations, and died worth £4,000,000, leaving £80,000 to found the Astor Library in New York. This institution is contained in a splendid building, enlarged in 1859 at the cost of his son, and comprises about 260,000 volumes. Since 1895 it has formed part of the New York public library.—His great-grandson, William Waldorf Astor, born in 1848, died in 1919, naturalized in England in 1899, was made a baron in 1916 and a viscount in 1917.

Astor, Lady. Nancy Witcher, Viscountess Astor, married the second Viscount Astor in 1906. She is a daughter of the late Colonel Chiswell Dabney Langhorne, of Virginia, United States. In Nov., 1919, she was elected member of Parliament for the Sutton division of Plymouth, and was the first woman to take a seat in the House of Commons.

Astor'ga, a city of Spain, province of Leon; the *Asturica Augusta* of the Romans. It figured prominently during the Peninsular War; it was taken by

the French after an obstinate defence, 1810, and retaken by the Spaniards, 1812. Pop. 5682.

Asto´ria, a town of Oregon, United States, on the Columbia River, with numerous salmon-canning establishments. Pop. 10,595. See *Astor*.

Astrabad´, a town of Persia, province of same name, about 24 miles E. of the Caspian. It was formerly the residence of the Kajar princes, the ancestors of the present Persian dynasty. It is very unhealthy, but is still the centre of a considerable trade. Pop. estimated at from 10,000 to 30,000. The province of Astrabad has an area of 5800 sq. miles, and a pop. of 150,000.

Astræ´a, in Greek mythology, the daughter of Zeus and Themis, the goddess of justice. During the golden age she dwelt on earth, but on that age passing away she withdrew from the society of men and was placed among the stars, where she forms the constellation Virgo. The name was given to one of the asteroids, discovered in 1845. It revolves round the sun in 1511.10 solar days, and is about 2½ times the distance of the earth from the sun.

As´tragal, in architecture, a small semicircular moulding, with a fillet beneath it, which surrounds a column in the form of a ring, separating the shaft from the capital.

Astrag´alus, a genus of papilionaceous plants, herbaceous or shrubby, and often spiny. *A. gummifer* yields gum tragacanth.

Astrag´alus, the upper bone of the foot supporting the tibia; the buckle, ankle, or sling bone. It is a strong irregularly-shaped bone, and is connected with the others by powerful ligaments.

Astrakhan (a´ s-tra´ -ha´ n'), a Russian city, capital of government of same name, on an elevated island in the Volga, about 30 miles above its mouth in the Caspian, communicating with the opposite banks of the river by numerous bridges. It is the seat of a Greek archbishop and has a large cathedral, as well as places of worship for Mahommedans, Armenians, &c. The manufactures are large and increasing, and the fisheries (sturgeon, &c.) very important. It is the chief port of the Caspian, and has regular steam communication with the principal towns on its shores. In 1919 the town was made a naval base by the Bolshevist Government of Moscow. Pop. 163,800,

composed of various races.—The government (or province) has an area of 91,042 sq. miles. It consists almost entirely of two vast steppes, separated from each other by the Volga, and forming for the most part arid sterile deserts. In 1918 the district of Astrakhan proclaimed its autonomy and independence of Moscow. Pop. 1,427,500.

Astrakhan, a name given to sheepskins with a curled woolly surface obtained from a variety of sheep found in Búkhara, Persia, and Syria; also a rough fabric with a pile in imitation of this.

Astralite. See *Explosives*.

Astral Spirits, spirits formerly believed to people the heavenly bodies or the aerial regions. In the Middle Ages they were variously conceived as fallen angels, souls of departed men, or spirits originating in fire, and belonging neither to heaven, earth, nor hell. Paracelsus regarded them as demoniacal in character.

Astrin´gent, a medicine which contracts the organic textures and canals of the body, thereby checking or diminishing excessive discharges. The chief astringents are the mineral acids, alum, lime-water, chalk, salts of copper, zinc, iron, lead, silver; and among vegetables catechu, kino, oak-bark, and galls.

Astroca´ryum, a genus of tropical American palms, species of which yield oil and valuable fibre. Tucum oil and tucum thread are obtained from *A. vulgāre*.

As´trolabe, an instrument formerly used for taking the altitude of the sun or stars, now superseded by the quadrant and sextant. The name was also formerly given to an armillary sphere.—Cf. Chaucer, *Treatise on the Astrolabe*.

Astrolabe Bay, an inlet on the N.E. coast of Australian New Guinea.

Astrol´ogy, literally, the science or doctrine of the stars. The name was formerly used as equivalent to astronomy, but is now restricted in meaning to the pseudo-science which pretends to enable men to judge of the effects and influences of the heavenly bodies on human and other mundane affairs, and to foretell future events by their situations and conjunctions. As usually

practised, the whole heavens, visible and invisible, were divided by great circles into twelve equal parts, called *houses*. As the circles were supposed to remain immovable, every heavenly body passed through each of the twelve houses every twenty-four hours. The portion of the zodiac contained in each house was the part to which chief attention was paid, and the position of any planet was settled by its distance from the boundary circle of the house, measured on the ecliptic. The houses had different names and different powers, the first being called the house of life, the second the house of riches, the third of brethren, the sixth of marriage, the eighth of death, and so on. The part of the heavens about to rise was called the *ascendant*, the planet within the house of the ascendant being *lord of the ascendant*. The different *aspects* of the planets were of great importance. To *cast a person's nativity* (or *draw his horoscope*) was to find the position of the heavens at the instant of his birth, which being done, the astrologer, who knew the various powers and influences possessed by the sun, the moon, and the planets, could predict what the course and termination of that person's life would be. The temperament of the individual was ascribed to the planet under which he was born, as *saturnine* from *Saturn*, *jovial* from *Jupiter*, *mercurial* from *Mercury*, &c., words which are now used with little thought of their original meaning. The virtues of herbs, gems, and medicines were supposed to be due to their ruling planets. The history of astrology, which was the foster-sister of astronomy, goes back to the early days of the human race. Egyptians and Hindus, as well as the nations on the Euphrates and Tigris, were zealous astrologers. The Christian Church strongly opposed the teachings of astrology, but its study spread among Jews and Arabs during the Middle Ages. Francis Bacon abused the astrologers of his day, and Swift wrote against them his famous *Prediction for the Year 1708*, by Isaac Bickerstaff, Esq.—BIBLIOGRAPHY: E. H. Bennet, *Astrology*; G. Wilde, *Chaldean Astrology Up-to-date*; A. Maury, *La Magie et l'astrologie à l'antiquité et au moyen âge*; A. J. Pearce, *Textbook of Astrology*.

Astron´omy (from Gr. *astron*, a heavenly body, and *nemein*, to classify or arrange) is that science which investigates the motions, distances, magnitudes, and various phenomena of the heavenly bodies. The science may be divided into several branches. *Descriptive astronomy* denotes merely a presentation of astronomical facts in a systematic but popular

form; *practical astronomy* treats of the instruments used in observing the celestial bodies, the methods of their employment, and the manner of deducing results from the observations; investigation of the causes of the motions of these bodies was formerly termed *physical astronomy*, but now generally *dynamical* or *gravitational astronomy*; *physical astronomy* or *astro-physics* is the comparatively modern branch which deals with their physical conditions, radiation, temperature, and chemical constitution. Recent years have added two new fields of investigation which are full of promise for the advancement of astronomical science. The first of these—*celestial photography*—has furnished us with invaluable light-pictures of the sun, moon, and other bodies, and has recorded the existence of myriads of stars invisible even to the best telescopes; while the second, *spectrum analysis*, now employed by many scientists, reveals to us a knowledge of the physical constituents of the universe, telling us for instance that in the sun (or his atmosphere) there exist many of the elements familiar to us on the earth. It is also applied to the determination of the velocities with which stars are approaching, or receding from, our system; and to the measurement of movements taking place within the solar atmospheric envelopes. From analysis of some of the unresolved nebulae the inference is drawn that they are not star-swarms but simply incandescent gas; whence a second inference results favourable to the hypothesis of the gradual condensation of nebulae, and the successive evolutions of suns and systems.

The most remote period to which we can go back in tracing the history of astronomy refers us to a time about 2500 B.C., when the Chinese are said to have recorded the simultaneous conjunction of Saturn, Jupiter, Mars, Mercury, and the moon. This remarkable phenomenon is found, by calculating backward, to have taken place 2460 B.C. Astronomy has also an undoubtedly high antiquity in India. The mean annual motion of Jupiter and Saturn was observed as early as 3062 years B.C.; tables of the sun, moon, and planets were formed, and eclipses calculated. In the time of Alexander the Great, the Chaldeans or Babylonians had carried on astronomical observations for 1900 years. They regarded comets as bodies travelling in extended orbits, and predicted their return; and there is reason to believe that they had correct ideas regarding the solar system. The priests of Egypt gave astronomy a religious character; but their knowledge of the science is testified to only by their ancient zodiacs and the position of their pyramids

with relation to the cardinal points. It was among the Greeks that astronomy took a more scientific form. Thales of Miletus (born 639 B.C.) predicted a solar eclipse, and his successors held opinions which are in many respects wonderfully in accordance with modern ideas. Pythagoras (500 B.C.) and his followers formed theories of the planetary system. They taught the sphericity and revolution of the earth, but placed an imaginary 'Central Fire', not the sun itself, at the centre of the system. Great progress was made in astronomy under the Ptolemies, and we find Timochares and Aristyllus employed about 300 B.C. in making useful planetary observations. But Aristarchus of Samos (born 267 B.C.) is said, on the authority of Archimedes, to have far surpassed them, by teaching the double motion of the earth around its axis and around the sun. A hundred years later Hipparchus determined more exactly the length of the solar year, and the eccentricity of the ecliptic, discovered the precession of the equinoxes, and even undertook a catalogue of the stars. It was in the second century after Christ that Claudius Ptolemy, a famous mathematician of Pelusium in Egypt, propounded the system that bears his name, viz., that the earth was the centre of the universe, and that the sun, moon, and planets revolved around it in the following order: nearest to the earth was the sphere of the moon; then followed the spheres of Mercury, Venus, the Sun, Mars, Jupiter, and Saturn; then came the sphere of the fixed stars; these were succeeded by two *crystalline* spheres and an outer sphere named the *primum mobile* or first movable, which last was again circumscribed by the *cælum empyreum*, of a cubic shape, wherein happy souls found their abode. The Arabs began to make scientific astronomical observations about the middle of the eighth century, and for 400 years they prosecuted the science with assiduity. Ibn-Yunis (A.D. 1000) made important observations of the perturbations and eccentricities of Jupiter and Saturn. In the sixteenth century Nicolaus Copernicus, born in 1473, introduced the system that bears his name, and which recognized the sun's central place in the solar system, and that all the other bodies, the earth included, revolve around it. This arrangement of the universe (see *Copernicus*) came at length to be generally received on account of the simplicity it substituted for the complexities and difficulties of the theory of Ptolemy. The observations and calculations of Tycho Brahe, a Danish astronomer, born in 1546, continued over many years, were of the highest value, and secured for him the title of regenerator of practical astronomy. His assistant and pupil, Kepler, born in 1571, was enabled,

principally from the data provided by his master's labours, to arrive at those laws which have made his name famous: 1. That the planets move, not in circular, but in elliptical orbits, of which the sun occupies a focus. 2. That the radius vector, or imaginary straight line joining the sun and any planet, moves over equal spaces in equal times. 3. That the squares of the times of the revolutions of the planets are as the cubes of their mean distances from the sun. Galileo, who died in 1642, advanced the science by his observations and by the new revelations he made through his telescopes, which established the truth of the Copernican theory. Newton, born in 1642, carried physical astronomy suddenly to comparative perfection. Accepting Kepler's laws as a statement of the facts of planetary motion, he deduced from them his theory of gravitation. The science was enriched towards the close of the eighteenth century by the discovery by Herschel of the planet Uranus and its satellites, the resolution of the Milky Way into myriads of stars, and the investigation of nebulae and of double and triple stars. The splendid analytical researches of Lalande, Lagrange, Delambre, and Laplace mark the same period. The nineteenth century opened with the discovery of the first four minor planets; and the existence of another planet (Neptune), more distant from the sun than Uranus, was, in 1845, independently predicted by Leverrier and Adams. Of late years the sun has attracted a number of observers, the spectroscope and photography having been especially fruitful in this field of investigation. By various methods the sun's mean distance has been ascertained within very small limits of error, and found to be nearly 93,000,000 miles. Many additions have been made to the known secondary planets or satellites, including some with retrograde motions. A vast number of asteroids has been discovered, and the width of the zone occupied by them found to be much more extensive. Much success has been achieved in ascertaining the parallax of fixed stars.

The objects with which astronomy has chiefly to deal are the earth, the sun, the moon, the planets, the fixed stars, comets, nebulae, and meteors. The stellar universe is composed of an unknown host of stars, many millions in number. Those visible to the naked eye were in ancient times grouped into the constellations still recognized. The nebulae are cloud-like patches of light scattered all over the heavens. Some of them have been resolved into star-clusters, but many of them are masses of incandescent gas. Of the so-called fixed stars, many form binary or multiple systems, the members

revolving in orbits under each other's attractions, while other more scattered groups are moving clusters, travelling in parallel paths through space like flocks of birds. Variable stars and extinct or dark stars are also known. The fixed stars preserve, at least to unaided vision, an unalterable relation to each other, because of their vast distance from the earth. Their apparent movement from east to west is the result of the earth's revolution on its axis in twenty-four hours from west to east. The planets have not only an apparent, but also a real and proper motion, since, like our earth, they revolve around the sun in their several orbits and periods. The nearest of these bodies to the sun is *Mercury*. *Venus*, the second planet from the sun, is to us the brightest and most beautiful of all the planets. The *Earth* is the first planet accompanied by a satellite or moon. *Mars*, the next planet, has two satellites, discovered in 1877. Its surface has a variegated character, and the existence of land, water, snow, and ice has been inferred. The *Asteroids*, of which over 1000 are known, form a broad zone of small bodies, at distances from a little beyond the earth's to that of Jupiter. *Jupiter*, the largest planet, has at least nine satellites, of which the two outermost have retrograde motion. Its surface is diversified by spots, markings, and bands parallel to its equator. *Saturn*, with its nine or more satellites and broad thin rings in its equatorial plane, is, perhaps, the most striking telescopic object in the heavens. *Urānus*—discovered by Herschel in 1781—is accompanied by four satellites. *Neptune*, the farthest removed from the sun, has one satellite, the motion of which is retrograde. Besides the planets, quite a number of comets are known to be members of the solar system. The physical constitution of these bodies is still one of the enigmas of astronomy. The observation of meteors has recently attracted much attention. They are seen in largest numbers in the autumn months. Meteor streams are supposed to represent the results of the disintegration of comets. Among the more modern astronomers we may mention: Gustav Kirchhoff, G. B. Donati, Christian Doppler, H. C. Vogel, Sir William Huggins, Simon Newcomb, and Sir David Gill. See *Earth, Sun, Moon, Planet, Comet, Stars, Asteroids, Celestial Photography, Spectrography, &c.*—BIBLIOGRAPHY: Sir J. N. Lockyer, *Dawn of Astronomy*; Sir G. C. Lewis, *Historical Survey of the Astronomy of the Ancients*; Sir F. W. Dyson, *Astronomy*; Sir R. Ball, *Atlas, and Popular Guide to the Heavens*; G. P. Serviss, *Astronomy with an Opera-glass*; *The Pleasures of the Telescope*; A. M. Clerke, *History of Astronomy during the 19th Century*, H. Macpherson, *Romance of Modern*

Astronomy; C. A. Young, *General Astronomy*; G. F. Chambers, *Handbook of Astronomy* (3 vols.); E. W. Maunder, *Astronomy of the Bible*; A. C. D. Crommelin, *The Star World*; Agnes Giberne, *Sun, Moon, and Stars* (popular).

Astropalia, an island in the Ægean Sea. It was occupied during the Balkan war of 1912 by the Italians under Admiral Presbitero and General d'Ameglio.

Astrophysics. See *Spectroscopy*.

Astur. See *Goshawk*.

Astu'ria, or **The Asturias**, a Spanish principality, now forming the province of Oviedo, on the north coast of Spain; an alpine region, with steep and jagged mountain ridges, valuable minerals, luxuriant grazing lands, and fertile well-watered valleys. The heir apparent of Spain has borne since 1388 the title of Prince of the Asturias. See *Spain*.

Asty'ages (-jēz), last king of the Medes, 593-558 B.C., deposed by Cyrus, an event which transferred the supremacy from the Medes to the Persians.

Asuncion (a' -su'n-thē-on'), or **Nuestra Señora de la Asuncion** (Eng. *Assumption*), the chief city of Paraguay, on the River Paraguay, picturesquely situated and with good public buildings. It was founded in 1537 on the feast of the Assumption. Its trade is mostly in the yerba tea, hides, tobacco, oranges, &c. It was taken and plundered by the Brazilians in 1869. A railway runs for a short distance into the interior. Pop. (1920), 99,836.

As'wail, the native name for the sloth-bear (*Ursus labiātus*) of the mountains of India, an uncouth, unwieldy animal, with very long black hair, inoffensive when not attacked. Its usual diet consists of roots, bees'-nests, grubs, snails, ants, &c. Its flesh is in much favour as an article of food. When captured young it is easily tamed.



Aswail (*Ursus labiātus*)

Asy'lum, a sanctuary or place of refuge, where criminals and debtors sheltered themselves from justice, and from which they could not be taken

without sacrilege. Temples were anciently asylums, as were Christian churches in later times. (See *Sanctuary*.) The term is now usually applied to an institution for receiving, maintaining, and, so far as possible, ameliorating the condition of persons labouring under certain bodily defects or mental maladies; sometimes also a refuge for the unfortunate.

Asylum, Right of. See *Extradition*.

Asymptote (as'im-tōt), in geometry, a line which is continually approaching a curve, but never meets it, however far either of them may be prolonged. This may be conceived as a tangent to a curve at an infinite distance. See *Conic Sections*.

Asyn'deton, a figure of speech by which connecting words are omitted; as 'I came, I saw, I conquered', or Cicero's 'Abiit, excessit, evasit, erupit'.

Atacama (a'-ta'-kă'ma'), a desert region on the west coast of S. America belonging to Chile, partly in the province of Atacama, partly in the territory of Antofagasta. It mainly consists of a plateau extending from Copiapó northward to the River Loa, and lies between the Andes and the sea. It forms the chief nitrate district of Chile, there being also rich silver-mines, while gold is also found, as well as argentiferous lead, copper, nickel, cobalt, and iron; with guano on the coast. The northern portion belonged to Bolivia until 1904. The Chilean province of Atacama has an area of 30,711 sq. miles, and a pop. of 63,893.

Ataca'mite, a combination of the hydroxide and chloride of copper, occurring abundantly in some parts of South America, as at Atacama, whence it has its name. It is worked as an ore in South America, and is exported to England.

Atahual'pa, the last of the Incas, succeeded his father in 1529 on the throne of Quito, whilst his brother Huascar obtained the kingdom of Peru. They soon made war against each other, when the latter was defeated, and his kingdom fell into the hands of Atahualpa. The Spaniards, taking advantage of these internal disturbances, with Pizarro at their head invaded Peru, and advanced to Atahualpa's camp. Here, while Pizarro's priest was telling the Inca how the Pope had given Peru to the Spaniards, fire was opened on the

unsuspecting Peruvians, Atahualpa was captured, and, despite the payment of a vast ransom in gold, was executed (1533).

Atalan'ta, in Greek mythology, a famous huntress of Arcadia. She was to be obtained in marriage only by him who could outstrip her in a race, the consequence of failure being death. One of her suitors obtained from Aphrodītē (Venus) three golden apples, which he threw behind him, one after another, as he ran. Atalanta stopped to pick them up, and was not unwillingly defeated. There was another Atalanta belonging to Bœotia, who cannot very well be distinguished, the same stories being told about both.

Ataman. See *Hetman*.

At'avism (Lat. *atāvus*, an ancestor), in biology, the tendency to reproduce the ancestral type in animals or plants which have become considerably modified by breeding or cultivation; the reversion of a descendant to some peculiarity of a more or less remote ancestor. See *Mendelism*, *Natural Selection*, *Evolution*, *Heredity*. The term *atavism* is also frequently used in sociological literature, in the sense of reversion to more primitive types, as explanation of criminal instincts and pathological phenomena.

Ataxy, or **Ataxia**, in medicine, irregularity in the animal functions, or in the symptoms of a disease. See *Locomotor Ataxy*.

Atba'ra, the most northerly tributary of the Nile. It rises in the Abyssinian highlands, receives several large tributaries, and enters the Nile about 18° N. The town of Atbara is situated about 380 miles S.E. of Wadi Halfa. The battle of Atbara, between the British under Earl Kitchener (then Sir Herbert), and the followers of the Mahdi, was fought on 8th April, 1898.

Atchafalay'a ('Lost Water'), a river of the United States, an outlet of the Red River which strikes off before the junction of that river with the Mississippi, flows southward, and enters the Gulf of Mexico by Atchafalaya Bay. Its length is about 225 miles.

Atcheen'. See *Acheen*.

Atch'ison, a city of Kansas, United States, on the Missouri, about 30 miles from Leavenworth, an important railway centre with an increasing trade. Pop. (1920), 12,630.

A'tē, among the Greeks, the goddess of hate, injustice, crime, and retribution, daughter of Zeus according to Homer, but of Ēris (Strife) according to Hesiod.

At'eles, a genus of American monkeys. See *Spider-monkey*.

Ateliers Nationaux (a' -tl-yā na' -syo-nō), or national workshops, were established by the French Provisional Government in 1848. They interfered much with private trade, as about 100,000 workmen threw themselves on the Government for work. The breaking up of the system led to disorders, but it was abolished in July, 1848.

Atella'næ Fab'ulæ (called also **Oscan plays**), a kind of light interlude, in ancient Rome, performed not by the regular actors, but by freeborn young Romans; it originated from the ancient *Atella*, a city of the Oscans. They were the origin of the Italian *commedie dell'arte*. Cf. Munk, *De Fabulis Atellanis*.

Atesh'ga (the place of fire), a sacred place of the Guebres or Persian fire-worshippers, on the Peninsula of Apsheron, on the W. coast of the Caspian, visited by large numbers of pilgrims, who bow before the sacred flames which issue from the bituminous soil.

Ath (ät), a fortified town of Belgium, in the province of Hainaut, on the Dender; it carries on weaving, dyeing, and printing cottons. It was the scene of fighting in Nov., 1918. Pop. 11,108.

Athabas'ca, a river and lake of Canada. The river rises on the eastern slopes of the Rocky Mountains not far from Mount Hooker, in the province of Alberta, flows N.E. and N., and falls into Lake Athabasca after a course of about 600 miles.—*Lake Athabasca*, or Lake of the Hills, is about 190 miles S.S.E. of the Great Slave Lake, to which its waters are carried by means of the Slave River. It is about 200 miles in length from east to west, and 35 miles wide where widest, but narrows to a point at either extremity.—The former district of *Athabasca*, in 1905 divided between the two new provinces of Alberta and Saskatchewan, had British Columbia on the west, Keewatin on the east, Alberta and Saskatchewan districts on the south, the parallel of 60° on the north, being crossed by the Athabasca and the Peace Rivers. Lake Athabasca is partly in Alberta, partly in Saskatchewan.

Athali'ah, daughter of Ahab, King of Israel, and wife of Joram, King of Judah. After the death of her son Ahaziah, she opened her way to the throne by the murder of forty-two princes of the royal blood. She reigned six years; in the seventh the high-priest Jehoiada placed Joash, the young son of Ahaziah, who had been secretly preserved, on the throne of his father, and Athaliah was slain. Cf. *2 Kings*, xi. The story of Athaliah supplied Racine with the plot of one of his most famous tragedies.

Athana'sian Creed, a creed or exposition of Christian faith, supposed formerly to have been drawn up by St. Athanasius, though this opinion is now generally rejected, and the composition often ascribed to Hilary, Bishop of Arles (about 430). It is an explicit avowal of the doctrines of the Trinity (as opposed to Arianism, of which Athanasius was a great opponent) and of the incarnation, and contains what are known as the 'damnatory clauses', in which it declares that damnation must be the lot of those who do not believe the true and catholic faith. It is contained in the *Book of Common Prayer*, to be read on certain occasions.—BIBLIOGRAPHY: F. J. A. Hort, *Two Dissertations*; G. D. W. Ommanney, *Critical Dissertation on the Athanasian Creed*; J. A. Robinson, *The Athanasian Creed*; E. C. S. Gibson, *The Three Creeds*; R. O. P. Taylor, *Athanasian Creed in the Twentieth Century*.

Athana'sius, St., Archbishop of Alexandria, a renowned father of the Church, born in that city about A.D. 296, died 373. While yet a young man he attended the Council at Nice (325), where he gained the highest esteem of the fathers by the talents which he displayed in the Arian controversy. He had a great share in the decrees passed here, and thereby drew on himself the hatred of the Arians. Shortly afterwards he was appointed Archbishop of Alexandria. The complaints and accusations of his enemies at length induced the Emperor Constantine to summon him in 334 before the Councils of Tyre and Jerusalem, when he was suspended, and afterwards banished to Trèves. The death of Constantine put an end to this banishment, and Constantius recalled the holy patriarch. His return to Alexandria resembled a triumph. Deposed again in 340, he was reinstated in 342. Again in 355 he was sentenced to be banished, when he retired into those parts of the desert which were entirely uninhabited. He was followed by a faithful servant, who, at the risk of his life, supplied him with the means of subsistence. Here Athanasius composed many writings, full of eloquence, to

strengthen the faith of the believers, or expose the falsehood of his enemies. When Julian the Apostate ascended the throne, toleration was proclaimed to all religions, and Athanasius returned to his former position at Alexandria. His next controversy was with the heathen subjects of Julian, who excited the emperor against him, and he was obliged to flee in order to save his life. The death of the emperor and the accession of Jovian (363) again brought him back; but Valens becoming emperor, and the Arians recovering the superiority, he was once more compelled to flee. He concealed himself in the tomb of his father, where he remained four months, until Valens allowed him to return. From this period he remained undisturbed in his office till he died. Of the forty-six years of his official life he spent twenty in banishment, and the greater part of the remainder in defending the Nicene Creed. Athanasius was not so much a speculative theologian as a great Christian pastor (cf. L. Duchesne, *Histoire ancienne. de l'Église*, 1907). His writings, which are in Greek, are on polemical, historical, and moral subjects. The polemical treat chiefly of the doctrines of the Trinity, the incarnation of Christ, and the divinity of the Holy Spirit. The historical ones are of the greatest importance for the history of the Church. See *Athanasian Creed*.

A'theism (Gr. *a*, priv., and *Theos*, God), the disbelief of the existence of a God or supreme intelligent being; the doctrine opposed to *theism* or *deism*. The term has been often loosely used as equivalent with *infidelity* generally, with deism, with pantheism, and with the denial of immortality. The most famous exponents of atheism were La Mettrie, Holbach, Feuerbach, and Carl Vogt; whilst Comte and Haeckel have put forward systems of thought essentially atheistic.—BIBLIOGRAPHY: R. Flint, *Anti-theistic Theories*; J. S. Blackie, *Natural History of Atheism*; F. A. Lange, *History of Materialism*.

Ath'eling, a title of honour among the Anglo-Saxons, meaning one who is of noble blood. The title was gradually confined to the princes of the blood royal, and in the ninth and tenth centuries was used exclusively for the sons or brothers of the reigning king.

Atheling, Edgar. See *Edgar Atheling*.

Ath'elney, formerly an island in the midst of fens and marshes, now drained and cultivated in Somersetshire, England, about 7 miles southeast

of Bridgwater. Alfred the Great took refuge in it during a Danish invasion, and afterwards founded an abbey there.

Ath´elstan, King of England, born 895, died 941, succeeded his father, Edward the Elder, in 925. He was victorious in his wars with the Danes of Northumberland, and the Scots, by whom they were assisted. After a signal overthrow of his enemies at Brunanburgh he governed in peace and with great ability.

Athē´na, or **Athēnē**, a Greek goddess, identified by the Romans with Minerva, the representative of the intellectual powers; the daughter of Zeus (Jupiter) and Mētis (that is, wisdom or cleverness). According to the legend, before her birth Zeus swallowed her mother, and Athena afterwards sprang from the head of Zeus with a mighty war shout and in complete armour. In her character of a wise and prudent warrior she was contrasted with the fierce Ares (Mars). In the wars of the giants she slew Pallas and Enceladus. In the wars of the mortals she aided and protected heroes. She is also represented as the patroness of the arts of peace. The sculptor, the architect, and the painter, as well as the philosopher, the orator, and the poet, considered her their tutelar deity. She is also represented among the healing gods. In all these representations she is the symbol of the thinking faculty, the goddess of wisdom, science, and art; the latter, however, only in so far as invention and thought are comprehended. In the images of the goddess a manly gravity and an air of reflection are united with female beauty in her features. As a warrior she is represented completely armed, her head covered with a gold helmet. As the goddess of peaceful art she appears in the dress of a Grecian matron. To her insignia belong the Ægis, the Gorgon's head, the round Argive buckler; and the owl, the cock, the serpent, an olive branch, and a lance were sacred to her. All Attica, but particularly Athens, was sacred to her, and she had numerous temples there. Her most brilliant festival at Athens was the Panathenæa.

Athenæ´um, the temple of Athena or Minerva, at Athens, frequented by poets, learned men, and orators. The same name was given at Rome to the school which Hadrian established on the Capitoline Mount for the promotion of literary and scientific studies. In modern times the same name is given to literary clubs and establishments connected with the sciences. It is also the title of several literary periodicals.

Athenæ'us, a Greek rhetorician and grammarian, who lived at the end of the second and beginning of the third century after Christ, author of an encyclopædic work, in the form of conversation, called *The Professors at the Dinner-table* (*Deipnosophistæ*), which is a rich but ill-arranged treasure of historical, antiquarian, philosophical, grammatical, &c., knowledge.

Athenag'oras, a Platonic philosopher of Athens, a convert to Christianity, who wrote a Greek *Apology for the Christians*, addressed to the Emperor Marcus Aurelius, in 177, one of the earliest that appeared.

Ath'ens (Gr. *Athēnai*, Lat. *Athēnæ*), anciently the capital of Attica and centre of Greek culture, now the capital of the kingdom of Greece. It is situated in the central plain of Attica, about 4 miles from the Saronic Gulf or Gulf of Ægina, an arm of the Ægean Sea running in between the mainland and the Peloponnesus. It is said to have been founded about 1550 B.C. by Cecrops, the mythical Pelasgian hero, and to have borne the name Cecropia until under Erechtheus it received the name of Athens in honour of Athēnē. The Acropolis, an irregular oval crag 150 feet high, with a level summit 1000 feet long by 500 in breadth, was the original nucleus of the city, which, according to tradition, was extended by Theseus when Athens became the head of the confederate Attic States. The three chief eminences near the Acropolis—the Areopagus to the north-west, the Pnyx to the south-west, and the Museum to the south of the Pnyx—were thus included within the city boundary as the sites of its chief public buildings, the city itself, however, afterwards taking a northerly direction. On the east ran the Ilissus and on the west the Cephissus, while to the south-west lay three harbours—Phalerum, the oldest and nearest; the Piræus, the most important; and Munychia, the Piræan Acropolis. At the height of its prosperity the city was connected with its harbours by three massive walls (the 'long walls'). The architectural development of Athens may be dated from the rule of the Pisistratids (560-510 B.C.), who are credited with the foundation of the huge temple of the Olympian Zeus, completed by Hadrian seven centuries later, the erection of the Pythium or temple of the Pythian Apollo, and of the Lyceum or temple of Apollo Lyceus—all near the Ilissus; and to whom were due the enclosure of the Academy, a gymnasium and gardens to the north of the city, and the building of the Agora with its Portico or Stoa, Bouleuterium or Senate-house, Tholus, and Prytaneum. With the foundation of Athenian democracy under Clisthenes, the Pnyx or place of public

assembly, with its semicircular area and cyclopean wall, first became of importance, and a commencement was made of the Dionysiac theatre (theatre of Dionysus or Bacchus) on the south side of the Acropolis. After the destruction wrought by the Persians in 480 B.C., Themistocles reconstructed the city upon practical lines and with a larger area, enclosing the city in new walls 7½ miles in circumference, erecting the north wall of the Acropolis, and developing the maritime resources of the Piræus; while Cimon added to the southern fortifications of the Acropolis, placed on it the temple of Wingless Victory, planted the Agora with trees, laid out the Academy, and built the Theseum on an eminence north of the Areopagus; his brother-in-law, Peisianax, erected the famous Stoa Poecilē, a hall with walls covered with paintings (whence the *Stoics* got their name). Under Pericles the highest point of artistic development was reached. An Odeum was erected on the east of the Dionysiac theatre for the recitations of rhapsodists and musicians; and with the aid of the architects Ictīnus, Callicrates, and Mnesicles, and of the sculptor Phidias, the Acropolis was perfected. Covering the whole of the western end rose the Propylæa, the splendid structure through which the Acropolis was entered, constructed of Pentelic marble and consisting of a central gateway portico with two wings in the form of Doric temples. Just outside the Propylæa was the small temple of Wingless Victory. A short distance within the entrance stood the bronze statue of Athena Promachus, a colossal work of Phidias, 66 feet high, showing the goddess in complete armour and leaning on a lance. Beyond it to the left was the Erechtheum, the shrine of Athena Polias, guardian of the city, containing a very ancient and sacred statue of Athena in olive-wood; while to the right, on the highest part of the Acropolis, was the marble Parthenon or temple of Athena, the crowning glory of the whole. This renowned structure, still glorious in its ruins, was built under the auspices of Pericles, Phidias being the sculptor and artistic adviser, and Ictīnus and Callicrates the architects. It is in the Doric style, and among its numerous sculptures were fifty life-size statues, while in the interior was a chryselephantine (gold and ivory) figure of the goddess, 39 feet high. (See *Parthenon*.) Minor statues and shrines occupied the rest of the area of the Acropolis, which was for the time wholly appropriated to the worship of the guardian deities of the city. The Acropolis museum, a building of recent date, contains an interesting and valuable collection of works of art found here. In the interval between the close of the Peloponnesian War and the

battle of Chæronea few additions were made to the city. But the long walls and Piræus, destroyed by Lysander, were restored by Conon, and under the orator Lycurgus the Dionysiac temple was completed, the Panathenaic stadium commenced, and the choragic monuments of Lysicrates and Thrasyllus erected. Later on Ptolemy Philadelphus gave Athens the Ptolemæum near the Theseum, Attalus I the stoa north-east of the Agora, Eumenes II that near the great theatre, and Antiochus Epiphanes carried on the Olympieum. Under the Romans it continued a flourishing city, Hadrian in the second century adorning it with many new buildings, and constructing an aqueduct, finished by his son Antoninus Pius. At this time also a wealthy citizen, Herodes Atticus, did much to beautify the city, and in particular constructed an Odeum, the ruins of which are still conspicuous. Indeed Athens was at no time more splendid than under the Antonines, when Pausanias visited and described it. But after a time Christian zeal, the attacks of barbarians, and robberies of collectors made sad inroads among the monuments. About A.D. 420 paganism was totally annihilated at Athens, and when Justinian closed even the schools of the philosophers, the reverence for buildings associated with the names of the ancient deities and heroes was lost. The Parthenon was turned into a church of the Virgin Mary, and St. George stepped into the place of Theseus. Finally, in 1456, the place fell into the hands of the Turks. The Parthenon became a mosque, and in 1687 was greatly damaged by an explosion at the siege of Athens by the Venetians. Enough, however, remains of it and of the neighbouring structures to attest the splendour of the Acropolis; while of the other buildings of the city, the Theseum, or temple of Theseus, and the Horologium, or temple of the Winds, are admirably preserved, as are also structures belonging to the Pnyx, Panathenaic stadium (restored and again used for games), &c. The Theseum, indeed, is said to be the best preserved building of all ancient Greece, and is hardly less imposing than the Parthenon. Of more than a hundred columns that belonged to the Olympieum or temple of the Olympian Zeus, completed by Hadrian, only fifteen are still standing. Soon after the commencement of the war of liberation in 1821 the Turks surrendered Athens, but captured it again in 1826-7. The Great Powers now intervened to bring about the independence of Greece. The Turks evacuated Athens in 1833, and the troops of King Otho then entered the city. In 1835 it became the royal residence, and it soon began to make rapid progress, though its natural position is by no

means advantageous. The modern city mostly lies north, north-east, and north-west of the Acropolis, and consists mainly of straight and well-built streets. Among the principal buildings are the royal palace, a stately building with a façade of Pentelic marble (completed 1843), the university, the academy of science, national museum of archaeology, public library, exhibition building, polytechnic institute, theatre, and observatory. There are two universities, the National University, opened in 1836, and the Capodistrian University with 3250 students. There are valuable museums, in particular the National Museum and that in the Polytechnic School, which contains the Schliemann collection, &c. These are constantly being added to by excavations. There are four foreign archæological schools or institutes, the French, German, American, and British. The Zappeion or exhibition building is a handsome structure, erected at the expense of the brothers Zappas to exhibit Greek industries. Tramways have been made in the principal streets, and the city is connected by tramway and railway (6 miles) with its port, the Piræus. Athens has also railway connection with the north and west of the kingdom as well as with the Peloponnesus. The Piræus is the chief Greek centre of trade and industry. Water is brought from Mount Pentelicus on the north-east, the aqueduct begun by Hadrian being utilized in supplying the city. Pop. 167,479, and including the Piræus 241,058.—BIBLIOGRAPHY: E. A. Gardner, *Ancient Athens*; J. E. Harrison, *Mythology and Monuments of Ancient Athens*; W. Warde Fowler, *The City-State*, chapter vi; W. M. Leake, *Topography of Athens and the Demi*; C. H. Weller, *Athens and its Monuments*.

Athens, the name of many places in the United States, the chief being in Georgia, and containing the Georgia University and the State college of agriculture. It carries on the cotton manufacture, has manufactures of agricultural implements, &c., and is a centre of trade. It was founded in 1801. Pop. 14,913.

Athérine (*Atherīna*), the name of a genus of small fishes abundant in the Mediterranean and caught in British waters, especially on the coasts of the south of England, some of them being highly esteemed as food. They are also known as *Sand-smelts*. There are two British species.

Athero'ma, in pathology, a term applied to a change that may take place in the inner coat of an artery, consisting in a kind of fatty degeneration,

leading to an aneurism or bursting. Also an encysted tumour containing matter of a curdy appearance.

Ath'erstone, a town in Warwickshire, England, 8 miles S.E. of Tamworth, and equidistant (100 miles) from London, Liverpool, and Lincoln. It has manufactures of hats, and is the reputed birthplace of the poet Drayton. Pop. (1921), 20,849 (rural district).

Ath'erton, town of England, Lancashire, 13 miles north-west of Manchester; cotton-factories, collieries, and ironworks give chief employment to the inhabitants. Pop. (1921), 19,863.

Athletes (ath'lēts; Gr. *athlētai*, from *athlos*, a contest, *athlon*, a prize), originally, in ancient Greece, combatants who took part and contended for a prize (*athlon*) in the public games. The profession was an honourable one; tests of birth, position, and character were imposed, and crowns, statues, special privileges, and pensions were among the rewards of success. (See *Games*.) The word is used in a similar sense at the present day, but is more especially applied to persons who can exhibit feats of strength. Games and athletic competitions, if they do not hold such an honourable position to-day as they did in antiquity, are still practised with great enthusiasm and excite the keenest interest in their patrons.

Athletic Sports, a general name for certain physical exercises demanding a special natural ability, and embodying a variety of events which conventionally include not only running and jumping but such feats of strength as putting the weight and throwing the hammer. The selection of these events at any athletic meeting is a somewhat arbitrary one, and the inclusion of those which require strength and skill rather than speed and agility rests more on a traditional than a logical basis. A particular feature which distinguishes these exercises as athletic sports is the presence of the idea of competition; thus running and walking, as isolated exercises, can be called 'sports' only when men compete against one another, although the factor of competition may be only indirectly present, as when an athlete endeavours to beat a record.

In this country athletic sports have long been a national characteristic, and records, more or less authentic, have been handed down for the last hundred years or more. Until comparatively recently, such sports have been the

prerogative of the British Isles; but during the last thirty years the United States have adopted them with enormous enthusiasm and success, and more recently still the vogue has extended throughout the Continent, and good results have been obtained by representatives from France, Germany, Italy, Austria, Sweden, and Finland, as well as from all the Colonies. In this country no school, no matter how small or how humble its pretensions, fails to hold its athletic meeting annually. The same applies to all colleges of the leading universities, Oxford and Cambridge, the best representatives of which compete against one another, whilst the smaller universities hold similar competitions. In addition, a large number of clubs are in existence throughout the country for the promotion and encouragement of sports, the whole system of athletics being under the Amateur Athletic Association (founded in 1880), whose rules and regulations for the correct maintenance of athletics in the best interests of amateurism are regarded as a standard throughout the world. Under its auspices an annual meeting—the Amateur Championships—is held. This meeting is open to the whole world, and many of the championships have been held at one time or another by distinguished visitors from America, the Colonies, and the Continent. At the time of writing, the association is considering the project of holding two distinct annual meetings, one of which shall, as hitherto, be unrestricted, the other confined to residents in the British Isles. This, which is the most important meeting of the year, has taken place uninterruptedly since its origin in 1866 with the exception of the military interval, 1915-8, and has been successfully resumed in 1919. For the past fourteen or fifteen years the meeting has been held in London on the first Saturday in July, and this practice will probably be a permanent one, although hitherto the venue was, in rotation, London, the Midlands, and the North. The university and inter-university meetings are held before Easter, the former at the respective university towns, the latter at Queen's Club, London. Among other important representative contests may be mentioned the Public Schools' Championships (usually in April), the United Hospitals' Championships, the Irish, the Scottish, the Welsh, the Midland and the Northern Counties' Championships. During the war, athletics were practically restricted to the services, and the Army Athletic Championships, held in Aug., 1919, was a successful reunion of soldier athletes from the various theatres of war, and included, for the first time in history, coloured troops.

The standard inter-university meeting comprises ten events, namely, flat races—100 yards, $\frac{1}{4}$ mile, $\frac{1}{2}$ mile, 1 mile, and 3 miles; 120-yards hurdle race; the high jump and long jump; putting the weight and throwing the hammer. These events appear in the programme of the Amateur Championship meeting, with the substitution of a 4-miles race for the 3 miles, and the addition of a 220-yards race, a 2-miles walking race, a 2-miles steeplechase (representing a miniature cross-country event), and the pole jump. A relay race, in which four representatives from each club run half a mile, a quarter of a mile, 220 yards, and 220 yards respectively, is also included as a standard event; whilst at the 1919 meeting a race of 440 yards over hurdles appeared for the first time, and will probably occupy a permanent place in the programme. Two additional Amateur Championship events, 7-miles walk and 10-miles flat race, are usually held at a separate meeting in the spring.

In addition to the preceding British meetings, a great International contest, the Olympic Games, is held every four years in a country selected by the Olympic committee. This meeting is truly international, the last before 1914 having taken place at Stockholm in 1912, when representatives from the most distant parts of the world competed with representatives from every country in Europe in a remarkably elaborate programme, which included, in addition to the preceding, such feats as throwing the javelin and the discus, and the classic 'Marathon race' over the traditional distance of $26\frac{1}{4}$ miles. The 1920 Olympic meeting was arranged to take place at Antwerp.

Although many excellently arranged athletic sports are held successfully upon a grass course, at any important meeting the races are contested upon a properly-constructed cinder-path, a quarter or a third of a mile in length, and in shape an oval flattened on two sides so as to include as much straight as possible. The width of the running path is variable, but 18 to 24 feet may be regarded as an average. The centre is of grass, and spaces are prepared for the hurdle race, for the jumps, and the other events which are described as the *field*, as opposed to *track*, events.

Flat races are classified as 'sprint races', 'middle distance' and 'long distance' races, although the distinction between these is somewhat arbitrary. Whether or no a man is actually capable of running the whole distance in question at full speed, the term 'sprint' is applied to those distances in which

an attempt is made to put forth a continuous maximum effort. The limit is, by general consent, fixed at 300 yards. At any good meeting the 100-yards race will be run in 10 seconds; at the very best meeting this time will be beaten; and many runners have been credited with $9\frac{4}{5}$ seconds, a few, under exceptional conditions, with $9\frac{3}{5}$ seconds. The record for 220 yards is $21\frac{1}{5}$ seconds. Middle-distance running includes races from a quarter mile to a mile, and races are held at 440 yards, 600 yards, 880 yards, 1000 yards, 1 mile, and very occasionally at $\frac{3}{4}$ mile. With modern specialization, however, it is rare to find any one runner capable of supremacy at more than one of these distances. The $\frac{1}{4}$ mile is the common ground for the sprinter and the middle-distance runner, and 48 seconds has been beaten on several occasions, although it may be said that anything inside 50 seconds is a first-class performance. The record for 600 yards is 1 minute 11 seconds. The $\frac{1}{2}$ -mile race has demonstrated latterly, perhaps, the greatest advance of all; and whilst anything under 2 minutes may still be regarded as a good performance, a championship event will most always be won in 3 or 4 seconds faster time; whilst at an Olympic meeting the wonderful record of 1 minute $52\frac{1}{2}$ seconds has been made. The mile, which was originally regarded as a long-distance event, is now legitimately considered as within the capacity of a middle-distance runner. At any first-class meeting 4 minutes 20 seconds will be accomplished, and any diminution of this time may be regarded as of superlative merit. The record, which has stood since 1886, is 4 minutes $12\frac{3}{4}$ seconds, although a recent performance in America, which is a tiny fraction of a second faster, has yet to be passed. Over 1 mile, long-distance running begins, and, as considerable staying-power is required, it is not unusual to find one man prove champion at 4 miles and 10 miles, and even the 1-mile race in the same year. No runner has yet achieved the capacity of running 12 miles within the hour, although two or three have been within a few hundred yards of this distance. About ten years ago the fashion became a craze of contesting 'Marathon races' in which all sorts of distances, quite independent of the classic $26\frac{1}{4}$ miles, were employed. At rare intervals very long-distance running, such as 50 miles, is indulged in. For any distance over 20 miles a special form of endurance is called for, rather than orthodox running in good style. As a competition 'walking' is an unsatisfactory exercise, because of the extreme difficulty in deciding when the athlete is still fulfilling the orthodox

regulation as to what constitutes fair 'heel and toe', inasmuch as the style of a man who is ostensibly walking, yet actually progressing at a rate faster than 9 miles an hour (faster than the average untrained person can run), is exceedingly difficult to analyse. About fifteen years ago long-distance walking became exceedingly popular, and hundreds of competitors attempted the classic walk to Brighton and back.

The usual hurdle race is over 120 yards, with ten flights of hurdles 10 yards apart, so that a distance of 15 yards separates the start from the first flight, and the same distance the last flight from the finish. The hurdles are 3½ feet high, with perfectly-level top rails. In correct 'hurdling' the 'three-stride method' is essential, that is to say, three strides are taken on the flat between the hurdles, and the athlete rises 6 feet from the obstacle, taking it in his stride, so that retardation of speed is reduced to a minimum. The skill and accuracy of an accomplished hurdler is remarkable, and the race is frequently run inside 16 seconds; a record of 15 seconds, and even a trifle less, has been accomplished.

In this country, running long jump and high jump (and to a less extent the pole jump) alone are practised to any extent, although as occasional events the standing high and long jumps are contested, and, still less frequently, the old-fashioned hop, step, and jump. In the long jump the athlete employs all the impetus he can acquire by a sprint of about 30 yards. The ideal aimed at is to run at the fastest speed which is consistent with reaching the taking-off board with accuracy, and then to leap as high as possible. A fraction of an inch under 25 feet has been cleared on two occasions, although it may be said that anything over 24 feet is exceptional, and that any jumper capable of 23 feet consistently has a good chance to win an Amateur Championship. The high jump requires skill of a very peculiar character. Whilst the novice regards this feat as dependent on momentum, and takes a correspondingly long run to acquire speed, the crack performer employs his capacity of manipulating his body and limbs so as to cross the bar in a horizontal position. In this way the prodigious height of 6 feet 7 inches has been cleared. In general it may be said that first-class jumping begins at 6 feet.

Pole jumping, a particularly pretty event to watch, has never been practised to any great extent in this country, and, in fact, does not even appear to be so popular here as twenty years ago. The pole employed is of light but

strong bamboo about 14 feet long, with a sharp ferrule at one end, which is stuck firmly into the ground. By the help of the pole, which is firmly grasped near the other end, the jumper elevates himself to the bar, over which he throws his legs and his body, finally relinquishing his hold of the pole, to fall on the opposite side. Recent years have witnessed the development of great skill in this event, particularly by the Americans, Canadians, and Swedes, and 13 feet has been cleared.

The weight or shot is an iron ball weighing 16 lb., which must be put with one hand only from the shoulder within a circle of 7 feet diameter. Although great strength is essential, skill in utilizing the whole of the body plays a very important part. Over 50 feet has been put on several occasions.

The 'hammer' is a ball of lead or iron attached by a wire to a handle. The total length must not exceed four feet; the weight of the whole must be at least 16 lb. The performer grasps the handle with one or both hands, and, standing within a 7-foot circle, swings the ball round and round to acquire impetus, which is then increased by rapid rotatory movements of his body. Once again skill and co-ordination must be wedded to strength. A crack performer has thrown over 175 feet.

Throwing the javelin and discus are classical rather than popular events, and their cultivation is fashionable only when an Olympic contest is imminent. Among other 'strong-men' contests, which have long been favourite sports in Scotland, are tossing the *caber* and putting the stone. The latter is usually a very heavy implement weighing about 56 lb.; the 'caber' is the trunk of a fir or other tree, freed from branches, which is held upright close to the chest by the smaller end, and thrown so as to alight on the heavier end.—BIBLIOGRAPHY: *Encyclopædia of Sports and Games*; *Annual Sporting and Athletic Register*; F. A. M. Webster, *The Evolution of the Olympic Games, 1829 B.C.-A.D. 1914*; G. Le Roy, *Athlétisme*; E. W. Hjertberg, *Athletics in Theory and Practice*; P. Withington, *The Book of Athletics*.

Athlone, a town of Ireland, divided by the Shannon into two parts, one in Westmeath, the other in Roscommon; about 76 miles west of Dublin. Its position has made it one of the chief military depots, and a centre of trade

by river, canal, and railway. It manufactures woollen goods, linens, &c. Up to 1885 it sent one member to Parliament. Pop. 7500.

Ath'oll, or **Athole**, a mountainous and romantic district in the north of Perthshire, Scotland, giving the title to a duke of the Murray family who owns a large area there.

Athor, **Hathor**, or **Het-her**, an Egyptian goddess, identified with Aphrodītē or Venus. Her symbol was the cow bearing on its head the solar disc and hawk-feather plumes. Her chief temple was at Denderah. From her the third month of the Egyptian year derived its name.

A'thos (now **Hagion Oros** or **Monte Santo**, that is, Holy Mountain), a mountain 6700 feet high, terminating the most eastern of the three peninsulas of Macedonia that jut in parallel directions into the Archipelago. The name, however, is frequently applied to the whole peninsula, which is about 30 miles long by 5 broad. It is covered with forests, and plantations of olive, vine, and other fruit-trees. Both the surface and coast-line are irregular. The Persian fleet under Mardonius was wrecked here in 493 B.C., and to avoid a similar calamity Xerxes caused a canal, of which traces may yet be seen, to be cut through the isthmus that joins the peninsula to the mainland. On the peninsula there are situated about twenty monasteries and a multitude of hermitages, which contain from 6000 to 8000 monks and hermits of the order of St. Basil. The libraries of the monasteries are rich in literary treasures and manuscripts. Every nation belonging to the Greek Church has here one or more monasteries of its own, which are annually visited by pilgrims. After having passed in the fifteenth century from the sovereignty of the Greek Emperors of Byzantium to that of the Sultans, it fell again into the hands of the Greeks, who occupied it in Nov., 1912. Each of the twenty monasteries is a little republic in itself, and until 1912 they used to pay an annual tribute of nearly £4000 to the Turks, and were governed by a synod of twenty monastic deputies and four presidents meeting weekly. They are now ruled either by abbots chosen for life, or by a board of overseers elected for a certain number of years. The revenue of the community is derived from pilgrims, and from a considerable trade in amulets, rosaries, crucifixes, images, and wooden furniture.

Athy´, a town in Ireland, county of Kildare, 37 miles south-west of Dublin, on the Barrow, which is here joined by the Grand Canal. Its chief trade is in corn. Pop. 3535.

Atit´lan, a lake and mountain of Central America in Guatemala. The lake is about 24 miles long and 10 broad; the mountain is an active volcano 12,160 feet high.

Atlan´ta, a city in the United States, capital of Georgia, on an elevated ridge, 7 miles south-east of the Chattahoochee River. It is an important railway centre; carries on a large trade in grain, paper, cotton, flour, and especially tobacco, and possesses flour-mills, paper-mills, ironworks, &c. Here are Atlanta University for negro men and women, a theological college, a medical college, &c. Atlanta suffered severely during the Civil War, and a battle was fought there on 22nd July, 1864. A fire which broke out on 21st May, 1917, caused damage estimated at more than £1,000,000. Pop. (1920), 200,600.

Atlan´tes, or **Telamōnes**, in architecture, male figures used in place of columns or pilasters for the support of an entablature or cornice. Female figures so employed are termed *caryatides*.

Atlantic City, a fashionable watering-place of the United States, on the coast of New Jersey. It is an important air port, and has an aerodrome covering about 160 acres. Pop. 50,682.

Atlantic Ocean, the vast expanse of sea lying between the west coasts of Europe and Africa and the east coasts of North and South America, and extending from the Arctic to the Antarctic Ocean; greatest breadth, between the west coast of Northern Africa and the east coast of Florida, 4150 miles; least breadth, between Norway and Greenland, 930 miles. The total area of the North Atlantic (including the inland seas) is 13,262,000 sq. miles; the area of the South Atlantic is 12,627,000 sq. miles. The principal inlets and bays are Baffin's and Hudson's Bays, the Gulf of Mexico, the Caribbean Sea, the North Sea, the Bay of Biscay, and the Gulf of Guinea. The principal islands north of the equator are Iceland, the Faroe and British Islands, the Azores, Canaries, and Cape Verde Islands, Newfoundland, Cape Breton, and the West India Islands; and south of the equator, Ascension, St. Helena, and Tristan da Cunha.

The great currents of the Atlantic are the Equatorial Current (divisible into the Main, Northern, and Southern Equatorial Currents), the Gulf Stream, the North African and Guinea Current, the Southern Connecting Current, the Southern Atlantic Current, the Cape Horn Current, Rennel's Current, and the Arctic Current. The current system is primarily set in motion by the trade-winds which drive the water of the intertropical region from Africa towards the American coasts. The Main Equatorial Current, passing across the Atlantic, is turned by the S. American coast, along which it runs at a rate of 30 to 50 miles a day, till, having received part of the North Equatorial Current, it enters the Gulf of Mexico. Issuing thence between Florida and Cuba under the name of the Gulf Stream, it flows with a gradually-expanding channel nearly parallel to the coast of the United States. It then turns north-eastward into the mid-Atlantic, the larger proportion of it passing southward to the east of the Azores to swell the North African and Guinea Current created by the northerly winds off the Portuguese coast. The Guinea Current, which takes a southerly course, is divided into two on arriving at the region of the north-east trades, part of it flowing east to the Bight of Biafra and joining the South African feeder of the Main Equatorial, but the larger portion being carried westward into the North Equatorial drift. Rennel's Current, which is possibly a continuation of the Gulf Stream, enters the Bay of Biscay from the west, curves round its coast, and then turns north-west towards Cape Clear. The Arctic Current runs along the east coast of Greenland (being here called the Greenland Current), doubles Cape Farewell, and flows up towards Davis' Strait; it then turns to the south along the coasts of Labrador and the United States, from which it separates the Gulf Stream by a cold band of water. Immense masses of ice are borne south by this current from the Polar seas. In the interior of the North Atlantic there is a large area comparatively free from currents, called the Sargasso Sea, from the large quantity of sea-weed (of the genus *Sargassum*) which drifts into it. A similar area exists in the South Atlantic. In the South Atlantic the portion of the Equatorial Current which strikes the American coast below Cape St. Roque flows southward at the rate of from 12 to 20 miles a day along the Brazil coast under the name of the Brazil Current. It then turns eastward and forms the South Connecting Current, which, on reaching the South African coast, turns northward into the Main and Southern Equatorial Currents. Besides the surface currents, an under current

of cold water flows from the poles to the equator, and an upper current of warm water from the equator towards the poles.

The greatest depth as yet discovered is north of Porto Rico, in the West Indies, namely 27,360 feet. Cross-sections of the North Atlantic between Europe and America show that its bed consists of two great valleys lying in a north-and-south direction, and separated by a ridge, on which there is an average depth of 1800 fathoms. The mean depth of the North Atlantic is 2047 fathoms, that of the South Atlantic 2067 fathoms. A ridge, called the *Wyville-Thomson Ridge*, with a depth of little more than 200 fathoms above it, runs from near the Butt of Lewis to Iceland, cutting off the colder water of the Arctic Ocean from the warmer water of the Atlantic. The South Atlantic, of which the greatest depth yet found is over 3000 fathoms, resembles the North Atlantic in having an elevated plateau or ridge in the centre with a deep trough on either side. The saltness and specific gravity of the Atlantic gradually diminish from the tropics to the poles, and also from within a short distance of the tropics to the equator. In the neighbourhood of the British Isles the salt has been stated at one thirty-eighth of the weight of the water. The North Atlantic is the greatest highway of ocean traffic in the world. It is also a great area of submarine communication, by means of the telegraphic cables that are laid across its bed. See *Oceanography*.

Atlantic Telegraph. See *Telegraph*.

Atlan'tides (-dēz), a name given to the Pleiades, which were fabled to be the seven daughters of Atlas or of his brother Hesperus.

Atlan'tis, an island which, according to Plato, existed in the Atlantic over against the Pillars of Hercules (Straits of Gibraltar), was the home of a great nation, and was finally swallowed up by the sea. The legend has been accepted by some as fundamentally true; but others have regarded it as the outgrowth of some early discovery of the New World.

Atlan'tosaurus, a gigantic fossil reptile, ord. Dinosauria, obtained in the upper Jurassic strata of the Rocky Mountains, attaining a length of 110 feet or more.

Atlas, an extensive mountain system in North Africa, starting near Cape Nun on the Atlantic Ocean, traversing Morocco, Algiers, and Tunis, and

terminating on the coast of the Mediterranean; divided generally into two parallel ranges, running w. to e., the Greater Atlas lying towards the Sahara and the Lesser Atlas towards the Mediterranean. The principal chain is about 1500 miles long, and the principal peaks rise above or approach the line of perpetual congelation, Miltsin in Morocco being 11,400 feet high, and Tizi Likumpt being 13,150. The highest elevation is perhaps Tizi Tamyurt, estimated at fully 15,000 feet. Silver, antimony, lead, copper, iron, &c., are among the minerals. The vegetation is chiefly European in character, except on the low grounds and next the desert.

Atlas, in Greek mythology, the name of a Titan whom Zeus condemned to bear the vault of heaven.—The same name is given to a collection of maps and charts, and was first used by Gerard Mercator in the sixteenth century, the figure of Atlas bearing the globe being given on the title-pages of such works.

Atlas, in anatomy, is the name of the first vertebra of the neck, which supports the head. It is connected with the occipital bone in such a way as to permit of the nodding movement of the head, and rests on the second vertebra or *axis*, their union allowing the head to turn from side to side.

At'las, a kind of silk or silk-satin fabric of Eastern manufacture.

Atmidom'eter, an instrument for measuring the evaporation from water, ice, or snow. It somewhat resembles Nicholson's hydrometer, being constructed so as to float in water and having an upright graduated stem, on the top of which is a metal pan. Water, ice, or snow is put into the pan, so as to sink the zero of the stem to a level with the cover of the vessel, and as evaporation goes on the stem rises, showing the amount of evaporation in grains.

Atmom'eter, an instrument for measuring the amount of evaporation from a moist surface in a given time. It is often a thin hollow ball of porous earthenware in which is inserted a graduated glass tube. The cavity of the ball and tube being filled with water and the top of the tube closed, the instrument is exposed to the free action of the air; the relative rapidity with which the water transuding through the porous substance is evaporated is marked by the scale on the tube as the water sinks.

Atmosphere, primarily the gaseous envelope which surrounds the earth; but the term is applied to that of any orb. Twilight effects show that the atmosphere is sufficiently dense up to a height of 40 miles to scatter or reflect to an appreciable degree the sun's rays, while the phenomena of meteors, which are rendered luminous through friction, show that it extends, though in extremely attenuated form, to 100 or even 200 or more miles. It exerts on every part of the earth's surface a pressure of about 15 (14.73) lb. per sq. inch. The existence of this atmospheric pressure was first proved by Torricelli, who thus accounted for the rush of a liquid to fill a vacuum, and who, working out the idea, produced the first barometer. The average height of the mercurial column counterbalancing the atmospheric weight at the sea-level is a little less than 30 inches; but the pressure varies from hour to hour, and, roughly speaking, diminishes in geometrical progression with arithmetical increase in altitude. Of periodic variations there are two maxima of daily pressure, occurring when the temperature is about the mean of the day, and two minima, when it is at its highest and lowest respectively; but the problems of diurnal and seasonal oscillations have yet to be fully solved. The pressure upon the human body of average size is no less than 14 tons, but as it is exerted equally in all directions no inconvenience is caused by it. It is sometimes convenient to take the atmospheric pressure as a standard for measuring other fluid pressures; thus the steam pressure of 30 lb. per sq. inch on a boiler is spoken of as a pressure of two atmospheres.

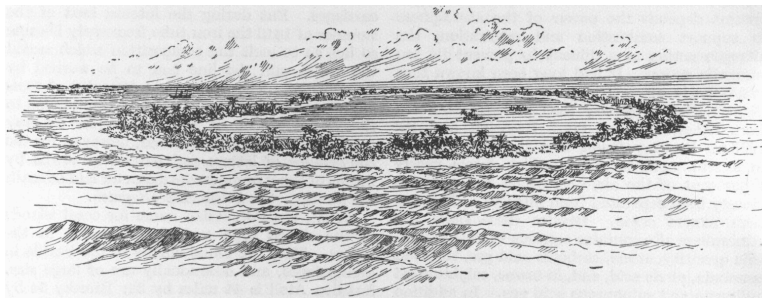
The atmosphere, first subjected to analysis by Priestley and Scheele in the latter part of the eighteenth century, consists practically of oxygen and nitrogen in the almost constant proportion of 20.81 volumes of oxygen to 79.19 volumes of nitrogen, or, by weight, 23.01 parts of oxygen to 76.99 of nitrogen. The gases are associated together, not as a chemical compound, but as a mechanical mixture. Upon the oxygen present depends the power of the atmosphere to support combustion and respiration, the nitrogen acting as a diluent to prevent its too energetic action. It had long been known that atmospheric nitrogen appeared to have a very slightly greater density than nitrogen obtained from other sources. Lord Rayleigh and Sir William Ramsay found that the fact was due to a still more inert gas which forms nearly 1 per cent of the air, and which had not previously been separated from nitrogen. This has been named *argon*. Besides these gases,

the atmosphere also contains aqueous vapour in variable quantity, ozone, carbonic acid gas, traces of ammonia, nitric acid, and, in towns, sulphuretted hydrogen and sulphurous acid gas. In addition to its gaseous constituents the atmosphere is charged with dust, bacteria, &c. For other gases which are present in traces, see *Neon*. See *Climate*; *Meteorology*.—BIBLIOGRAPHY: C. Flammarion, *L'Atmosphère*; Sir Napier Shaw, *The Weather Map*.

Atmospheric Engine, name given by early inventors to engines in which the piston is restored to the bottom of its stroke by atmospheric pressure.

Atmospheric Railway, so called in consequence of the motive power being derived from the pressure of the atmosphere, or from compressed air. The idea of thus obtaining motion was first suggested by the French engineer Papin, about 200 years ago. In 1810, and again in 1827, Medhurst published a scheme for 'propelling carriages through a close-fitting air-tight tunnel by forcing in air behind them'; and in 1825 a similar project was patented by Vallance of Brighton. About 1835 H. Pinkus, an American residing in England, patented a pneumatic railway. The carriages were to travel on an open line of rails, along which a cast-iron tube of between 3 and 4 feet diameter was to be laid, having a longitudinal slit from 1 to 2 inches wide and closed by a flexible valve along its upper side, through which a connection could be formed between the leading carriage and a piston working within the tube. This method was improved by Messrs. Clegg & Samuda, who in 1840 tried some experiments on a portion of the West London Railway with sufficient success to induce the Government to advance a loan to the Dublin and Kingstown Railway Company, for the construction of a pneumatic line from Kingstown to Dalkey. It was opened for passenger traffic at the end of 1843, and was worked for many months. The London and Croydon Company subsequently obtained powers for laying down an atmospheric railway by the side of their other line from London to Croydon, and in experimental trips in 1845 a speed of 30 miles an hour was obtained with sixteen carriages, and of 70 miles with six carriages. But during the intense heat of the summer of 1846 the iron tube frequently became so hot as to melt the composition which sealed the valve, and the line had to be worked by locomotives. The mechanical difficulty of commanding a sufficient amount of rarefaction led to the abandonment of the system for railway purposes. It has been revived, however, for the

conveyance of letters and parcels in towns by means of tubes of moderate diameter laid beneath the streets. See *Pneumatic Dispatch*.



Atoll

Atoll', the Polynesian name for coral islands of the ringed type enclosing a lagoon in the centre. They are found chiefly in the Pacific in archipelagos, and occasionally are of large size. Suadiva Atoll is 44 miles by 34; Rimsky 54 by 20. See *Coral*.

Atomic Theory, a theory as to the existence and properties of atoms (see *Atoms*); especially, in chemistry, the theory accounting for the fact that in compound bodies the elements combine in certain constant proportions, by assuming that all bodies are composed of ultimate atoms, the weight of which is different in different kinds of matter. It is associated with the name of Dalton, who systematized and extended the imperfect results of his predecessors. On its practical side the atomic theory asserts three *Laws of Combining Proportions*: (1) The Law of Constant or Definite Proportions, teaching that in every chemical compound the nature and relative weights of the constituent elements are definite and invariable; thus water invariably consists of 8 parts by weight of oxygen to 1 part by weight of hydrogen; (2) The Law of Multiple Proportions, according to which the several proportions in which one element unites with a given weight of another invariably bear towards each other a simple relation; thus 1 part by weight of hydrogen unites with 8 parts by weight of oxygen to form water, and with 16 (i.e. 8×2) parts of oxygen to form peroxide of hydrogen; (3) The Law of Combination in Reciprocal Proportions, that the proportions in which two elements combine with a third also represent the proportions in which, or in some simple multiple of which, they will themselves combine; thus in olefiant gas hydrogen is present with carbon in the proportion of 1 to 6, and in carbonic oxide, oxygen is present with carbon in the proportion of 8 to 6, 1 to 8 being also the proportions in which hydrogen and oxygen combine with each other. The theory that these *proportional numbers* are, in

fact, nothing else but the relative weights of atoms so far accounts for the phenomena that the existence of these laws might have been predicted by the aid of the atomic hypothesis long before they were actually discovered by analysis. In themselves, however, the laws do not prove the theory of the existence of ultimate particles of matter of a certain relative weight; and although many chemists, even without expressly adopting the atomic theory itself, have followed Dalton in the use of the terms *atom* and *atomic weight*, in preference to *proportion*, *combining weight*, *equivalent*, and the like, yet in using the word *atom* it should be held in mind that it merely denotes the combining weights of the elements. These will remain the same whether the atomic hypothesis which suggested the employment of the term be true or false. Dalton supposed that the atoms are spherical, and invented certain symbols to represent the mode in which he conceived they might combine. The latest atomic hypothesis is one which assigns an electrical structure to the atom. See *Chemistry*; *Electricity*; *Matter*. Cf. H. E. Roscoe and A. Harden, *New View of Dalton's Atomic Theory*; Sir J. J. Thomson, *Atomic Theory*.

Atomic Weights. See *Chemistry*; *Molecular Weights*.

Atomists. See *Atoms*.

Atoms, for many years regarded as the ultimate indivisible particles of the chemical elements. The idea originated with some of the ancient philosophers (the atomists), more especially Democritus (450 B.C.), Epicurus, and Lucretius (99-55 B.C.), and was developed into a definite theory by Dalton (1804). According to Dalton the atoms of any one element are alike in all their properties, but differ from the atoms of other elements, and when chemical combination occurs it takes place between the atoms of the combining elements (see *Chemistry*). Various views have been held with regard to the nature of atoms. Newton regarded them as hard, ponderable particles, perfectly unalterable, and concluded that the difference between substances was due to different kinds of atoms. Lord Kelvin propounded the view that the properties of atoms might be explained by those of vortices or vortex rings in a homogeneous frictionless fluid. As a result of the researches of British and French physicists on radium, the latest view is that matter and electricity are closely connected, that atoms are not indivisible, but complex aggregates containing positive and negative electrons, the

differences between the atoms depending mainly on the numbers of these electrons and their velocity. See *Radium*.

Atonement, in Christian theology, the special work of Christ effected by His life, sufferings, and death. The first explicit exposition of the evangelical doctrine of the atonement is ascribed to Anselm, Archbishop of Canterbury, in 1093.

Atrato (a´-trä´tō), a river of S. America, in the north-west of Colombia, emptying itself by nine mouths into the Gulf of Darien; it is navigable by steamers of some size for 250 miles, and has long been the subject of schemes for establishing water-communication between the Atlantic and Pacific.

Atrauli, a town of India, United Provinces, Aligarh district, clean, well built, and with a good trade. Pop. 16,560.

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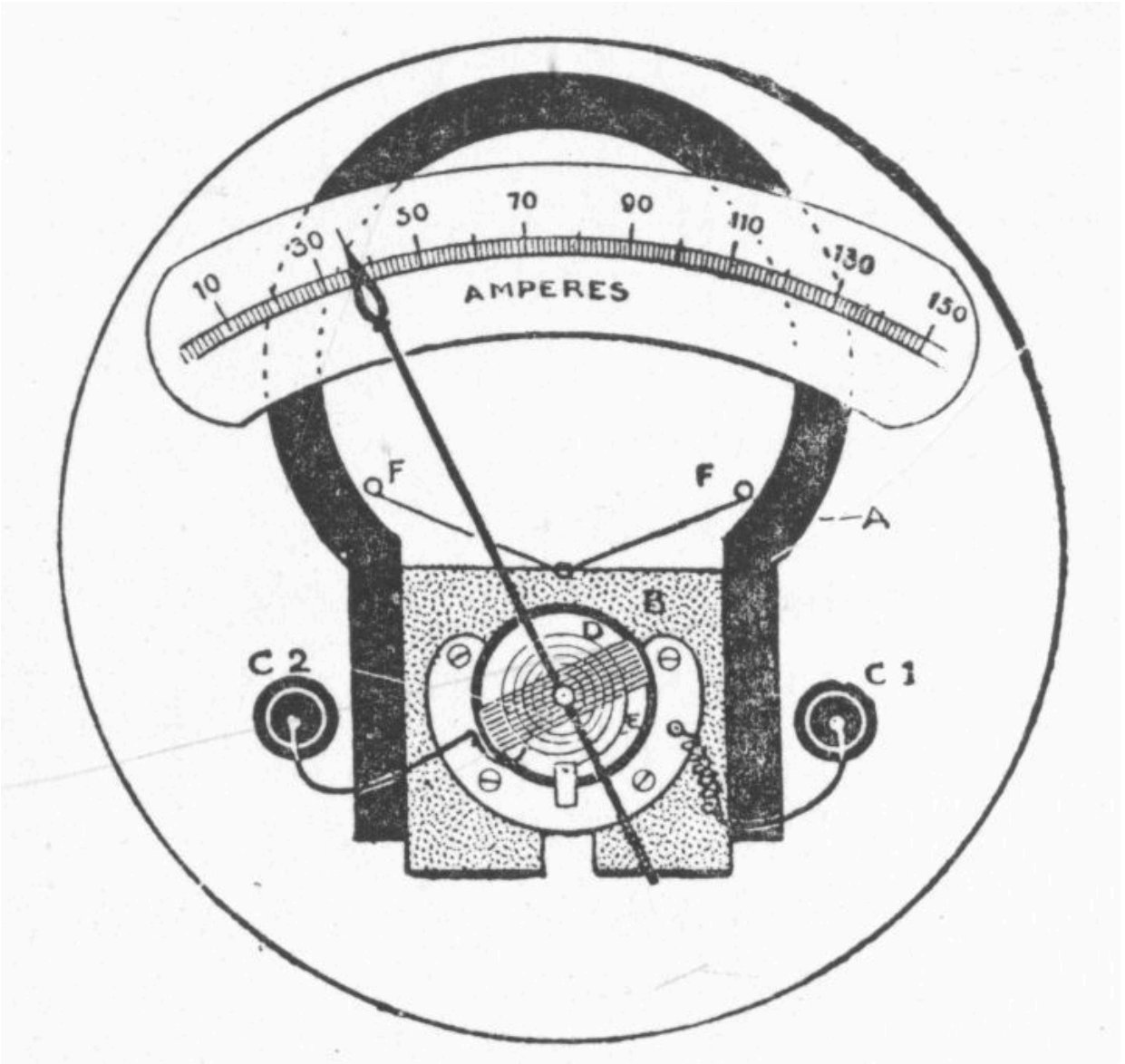
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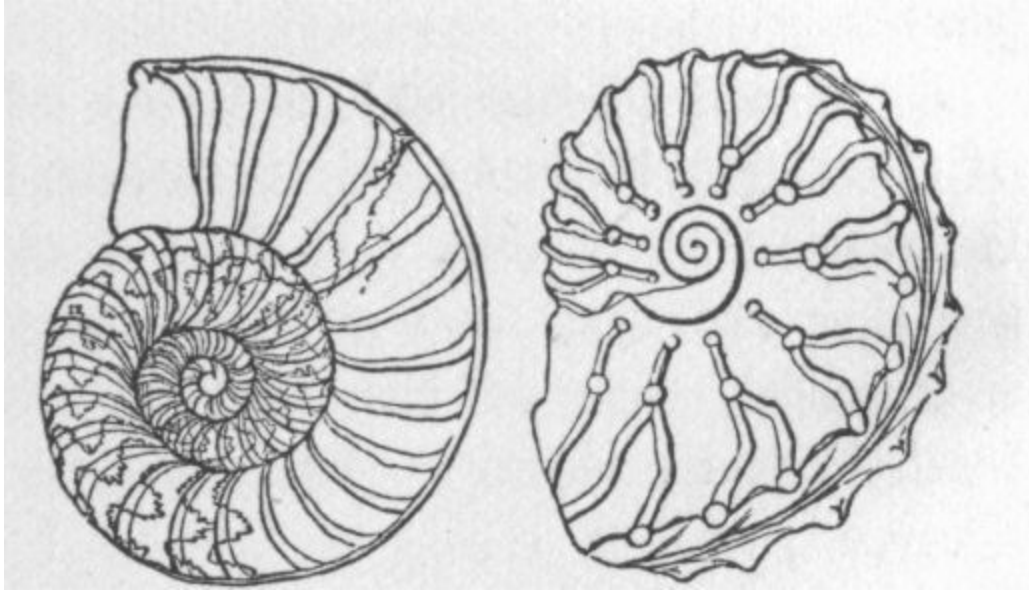
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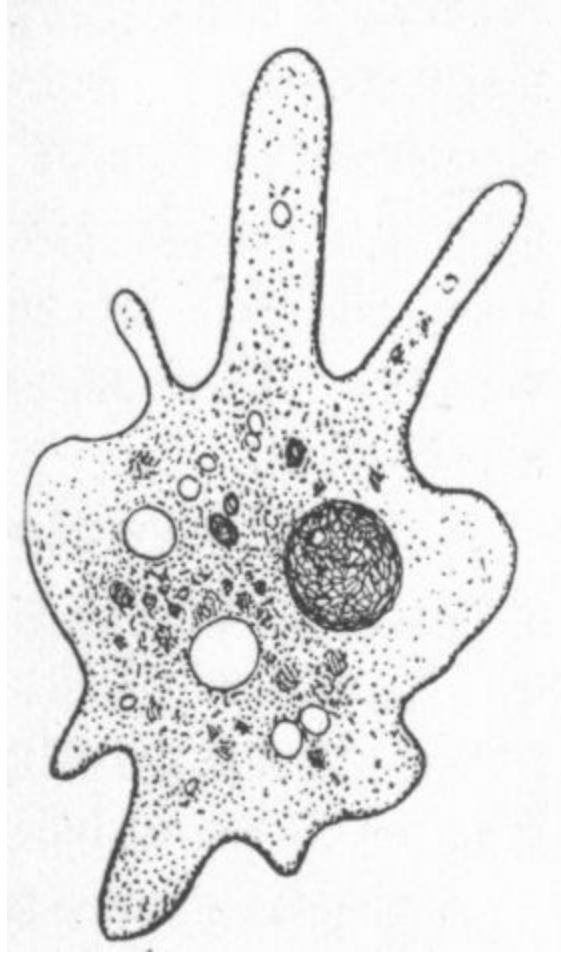
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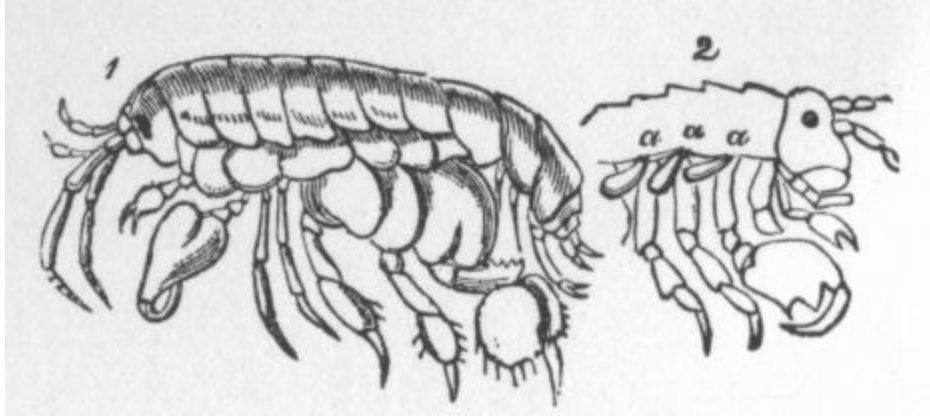
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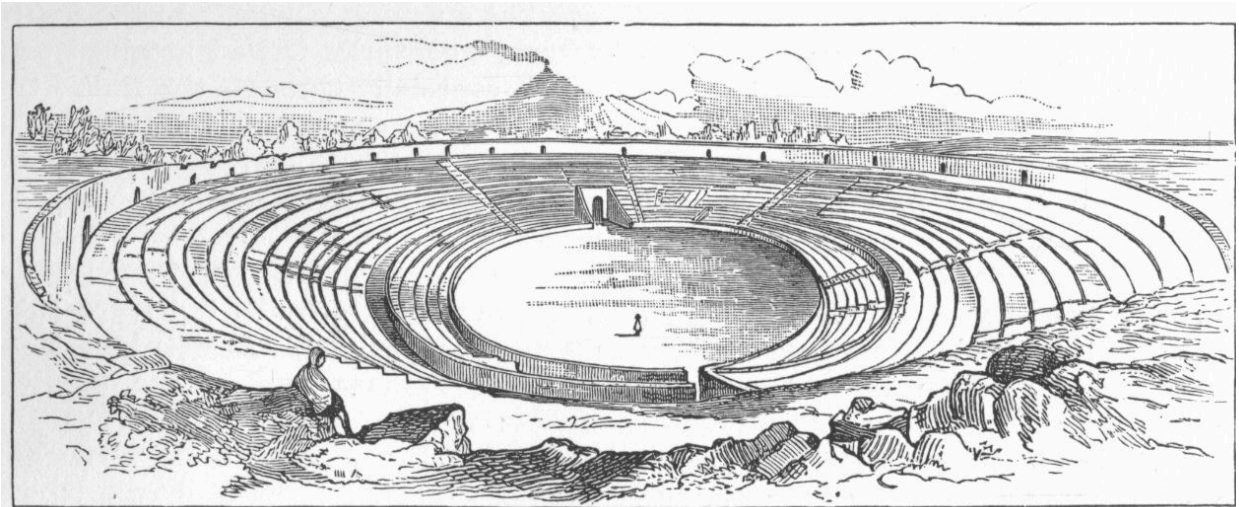
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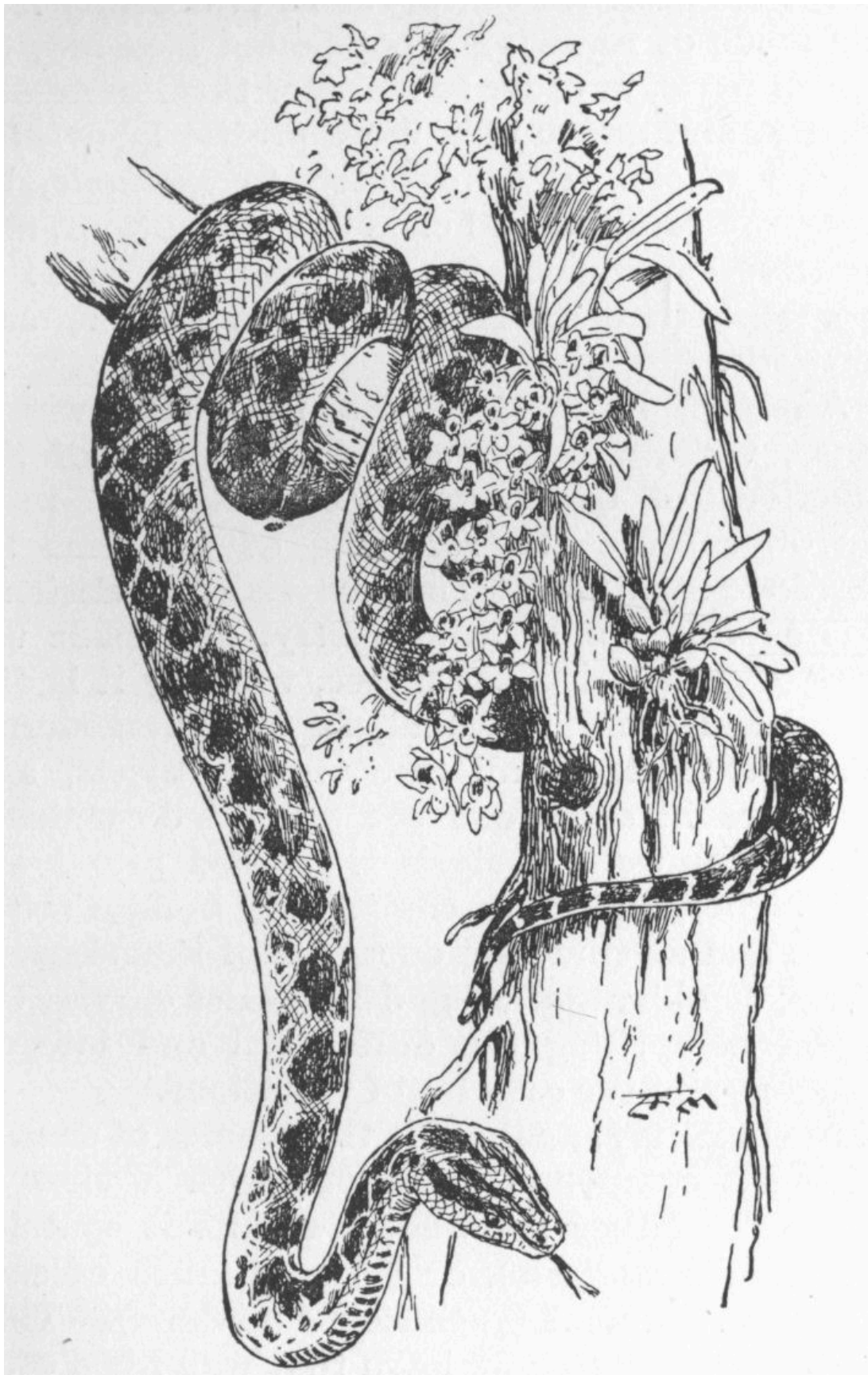
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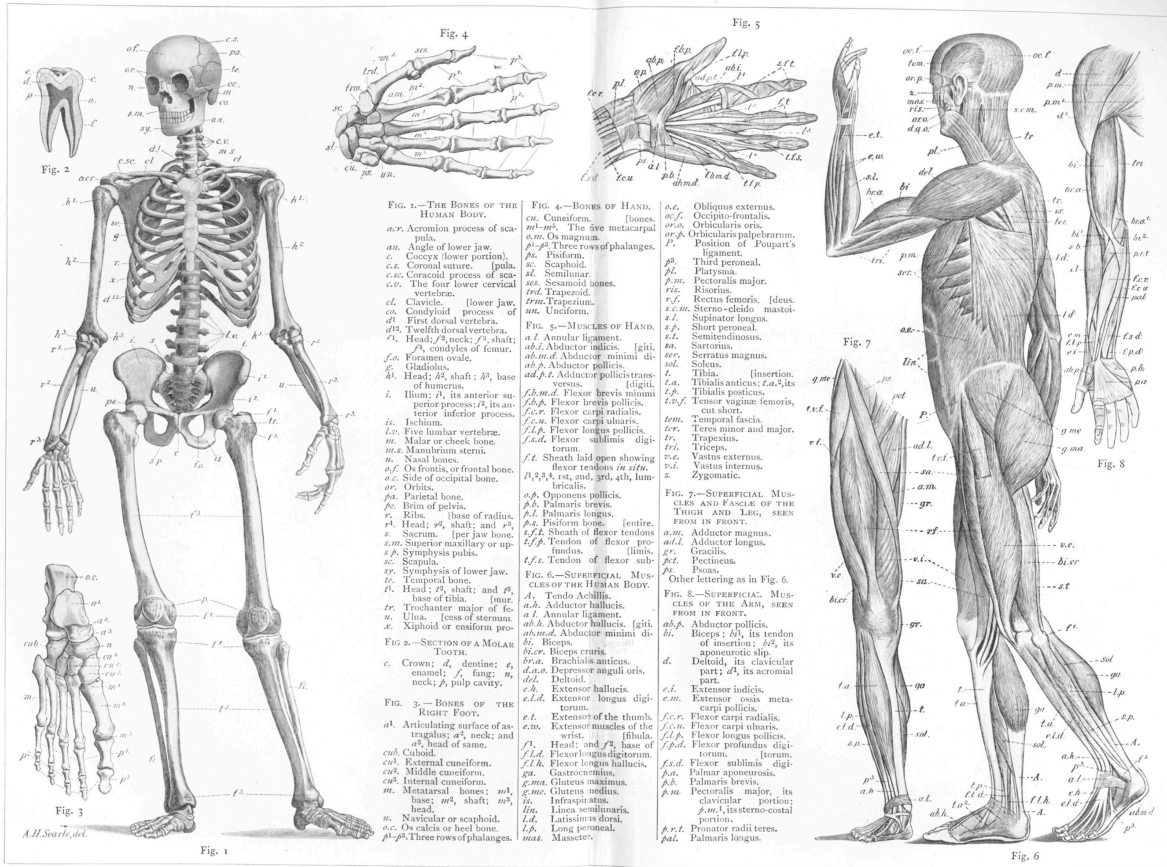
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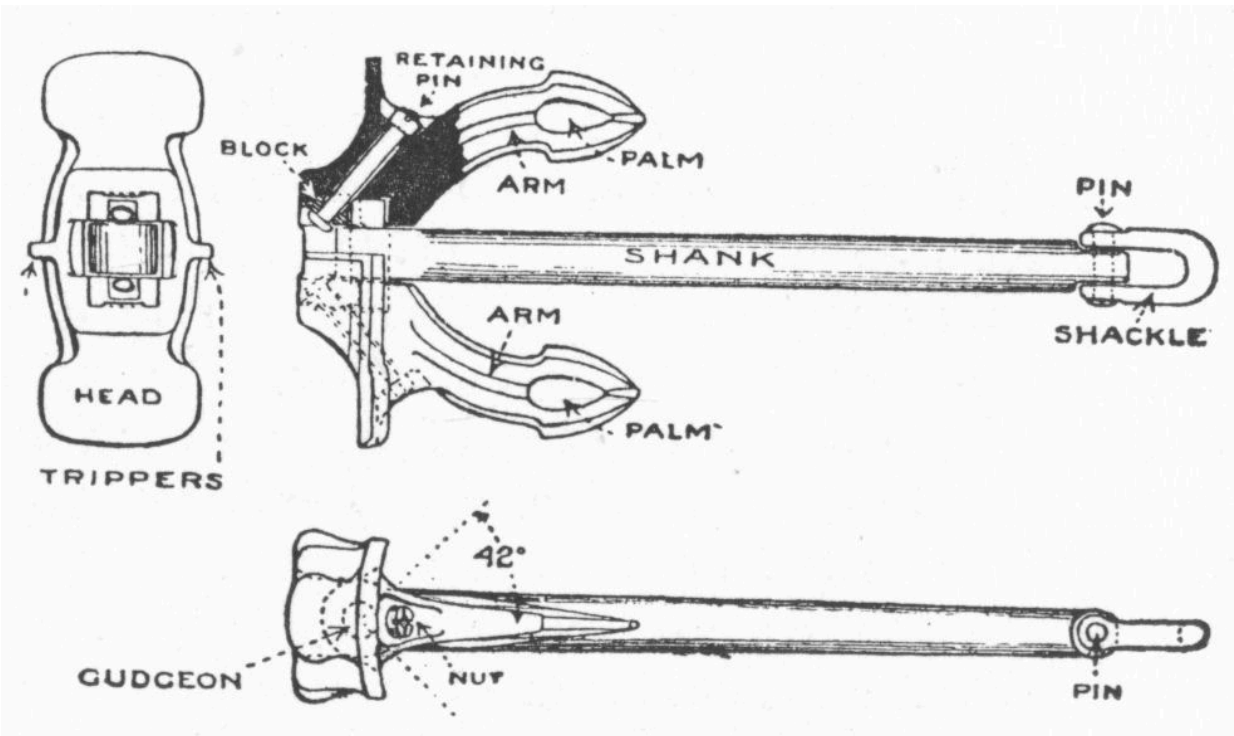
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— — — — —
The As-syr-ian came down like the wolf on the fold.
—(Byron's *The Destruction of Sennacherib*.)

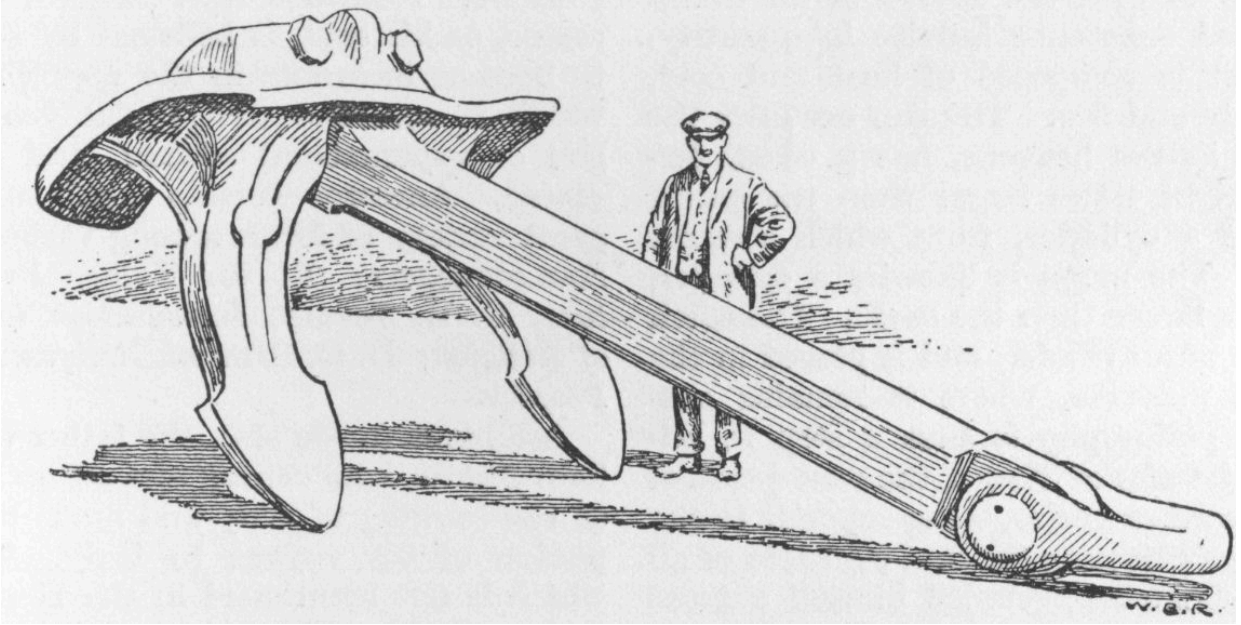
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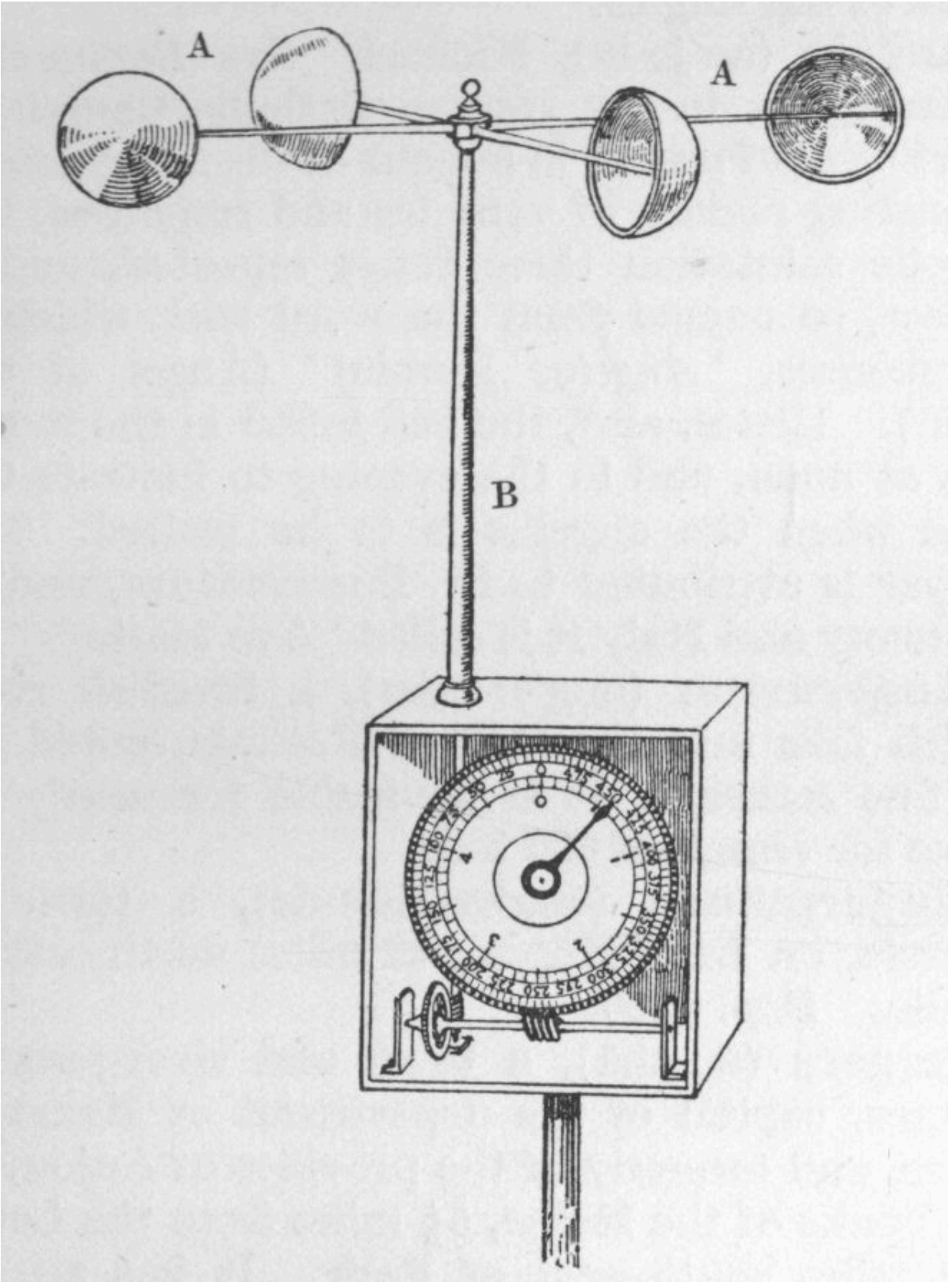
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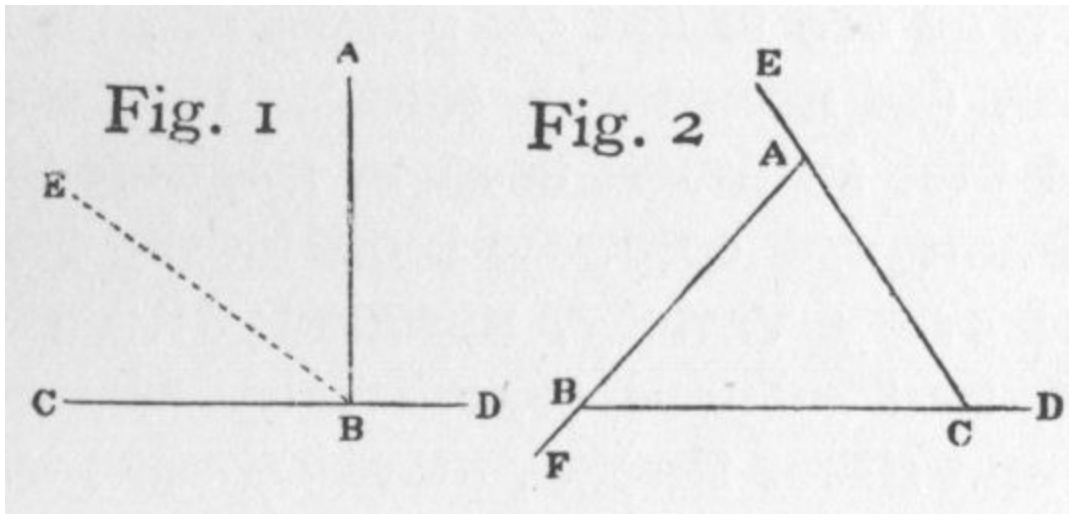
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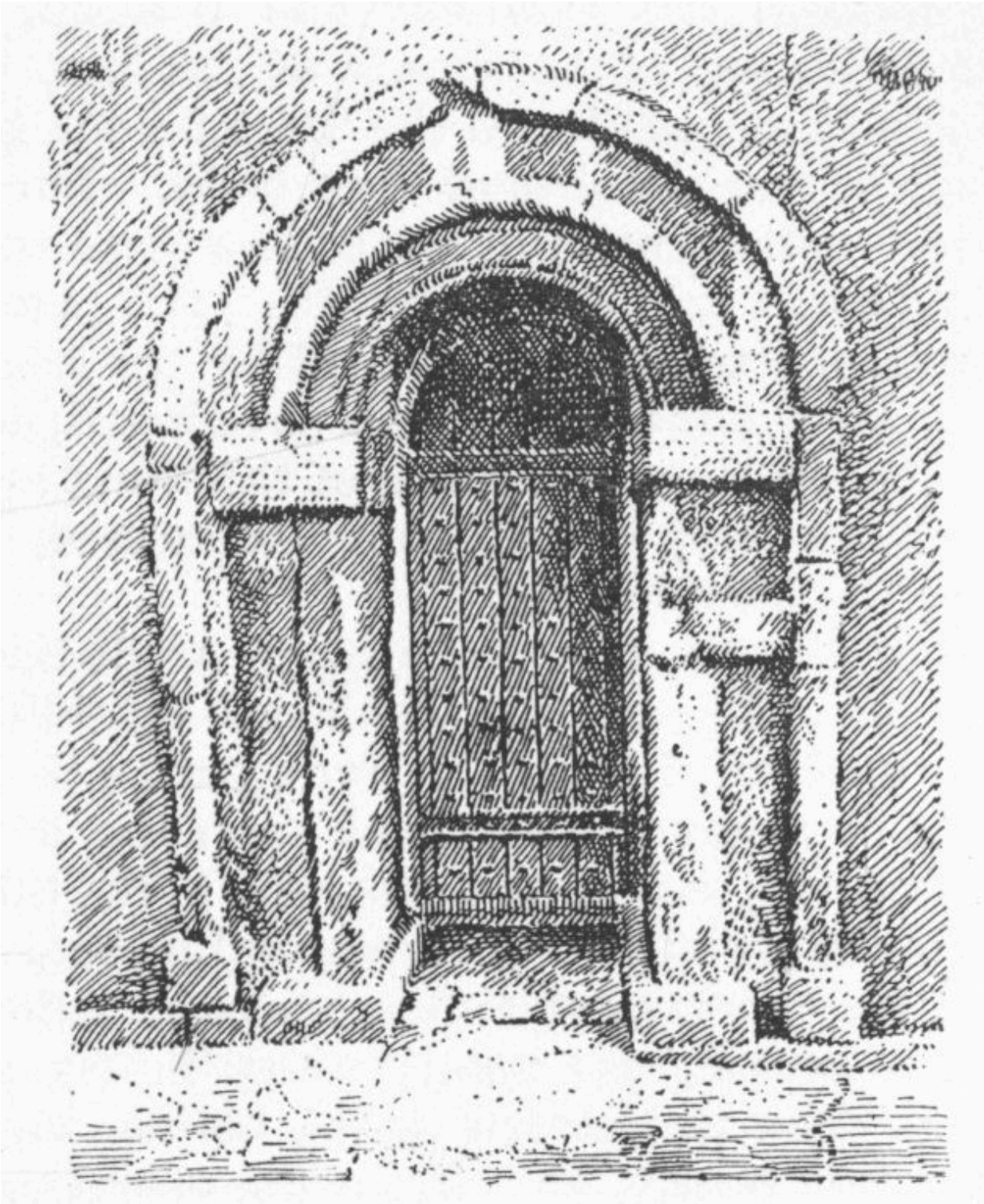
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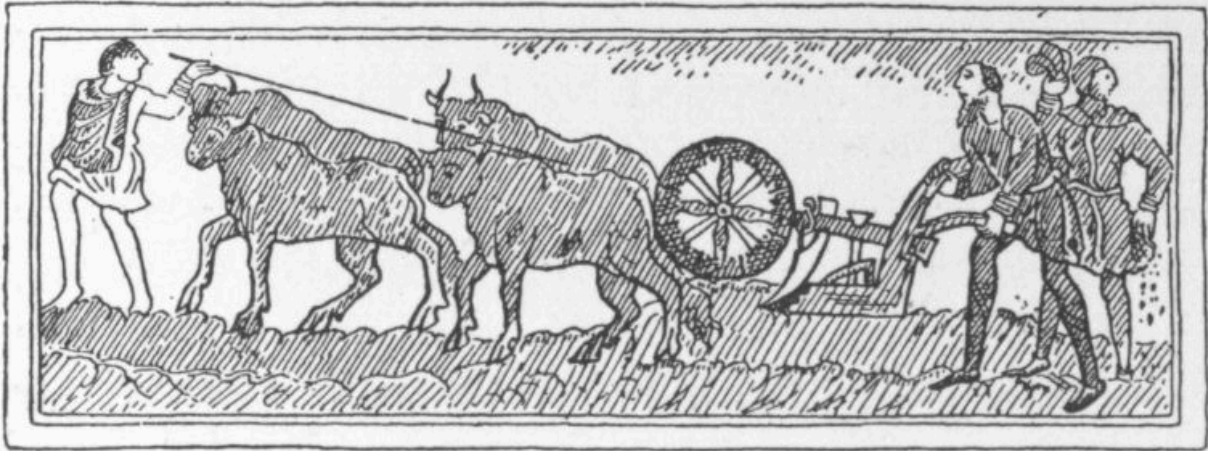
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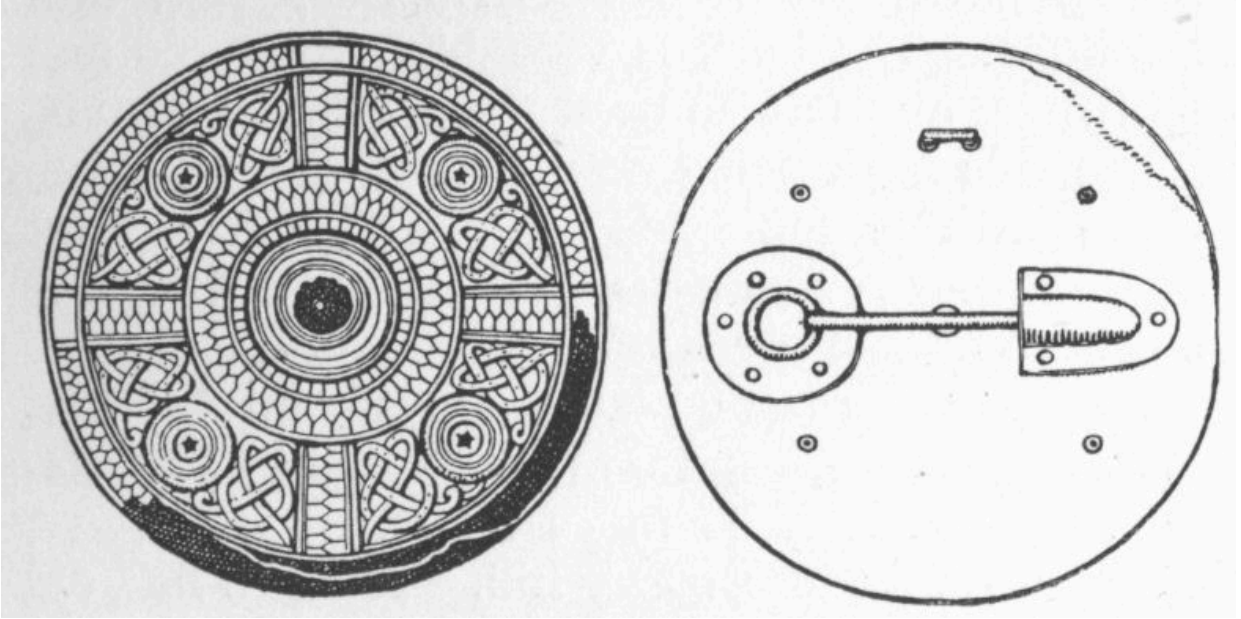
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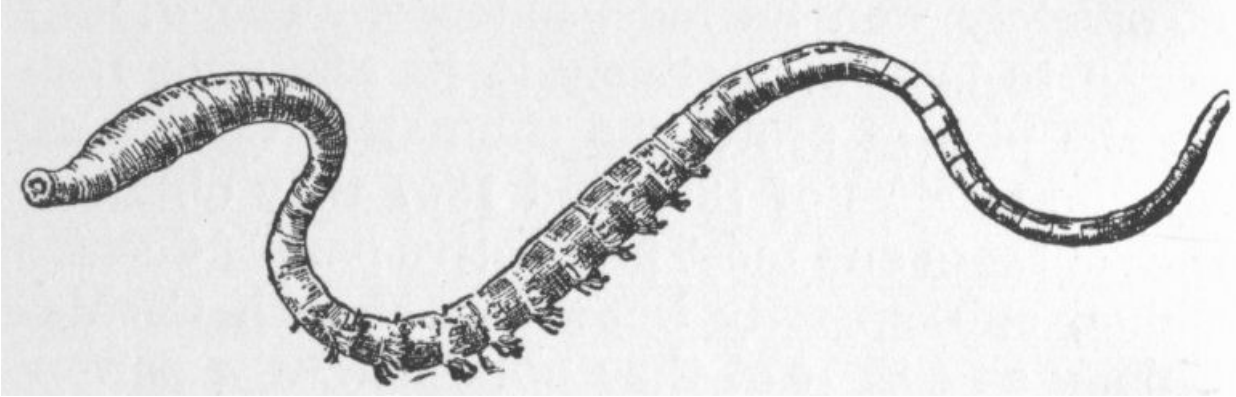
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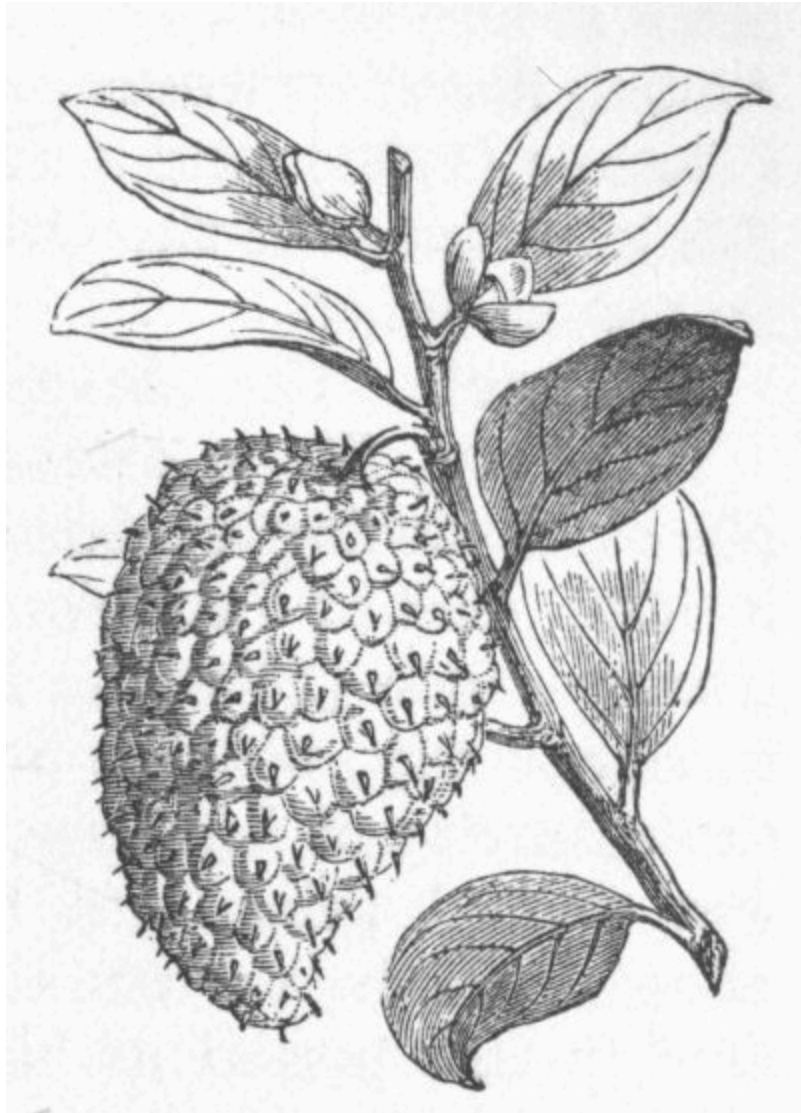
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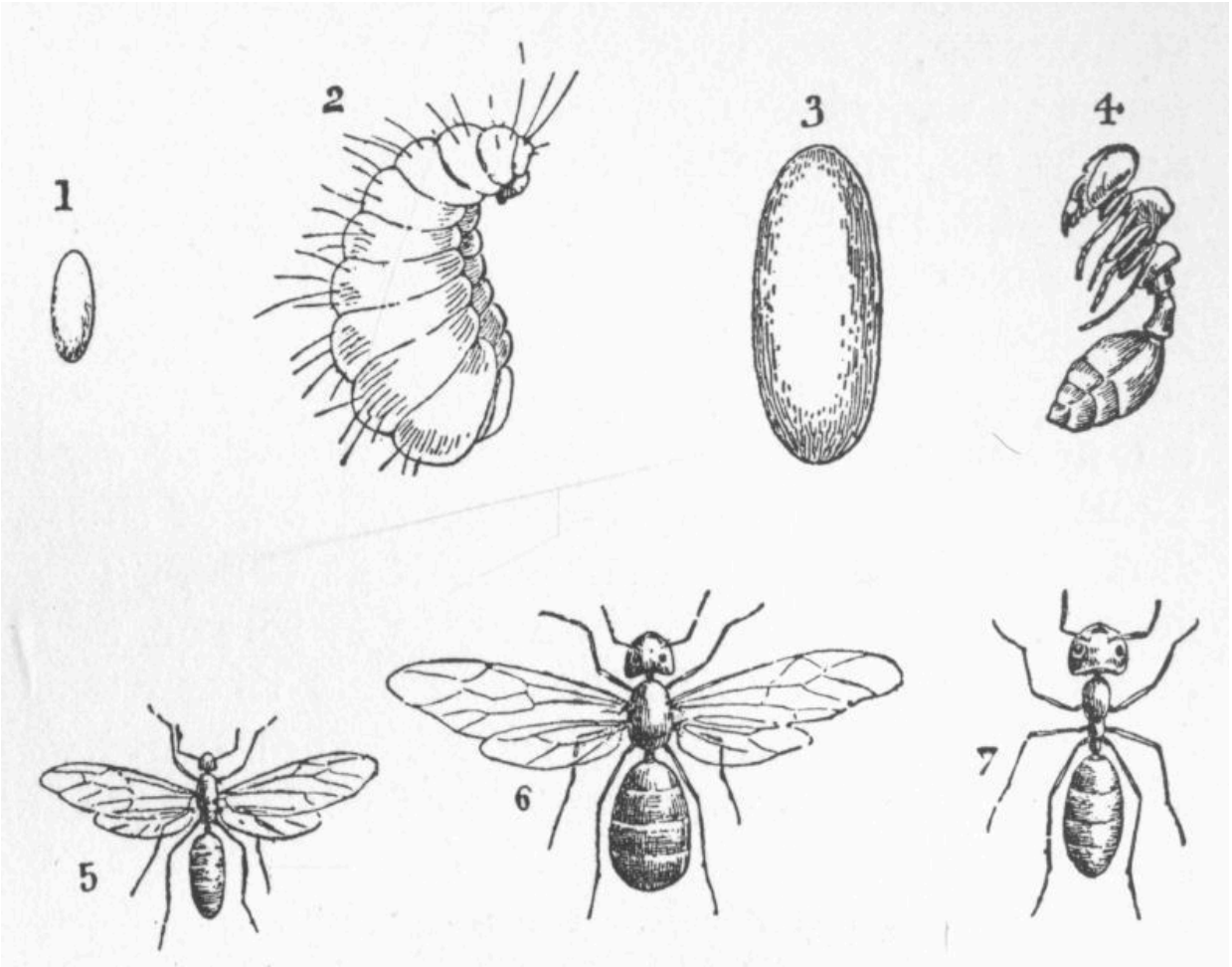
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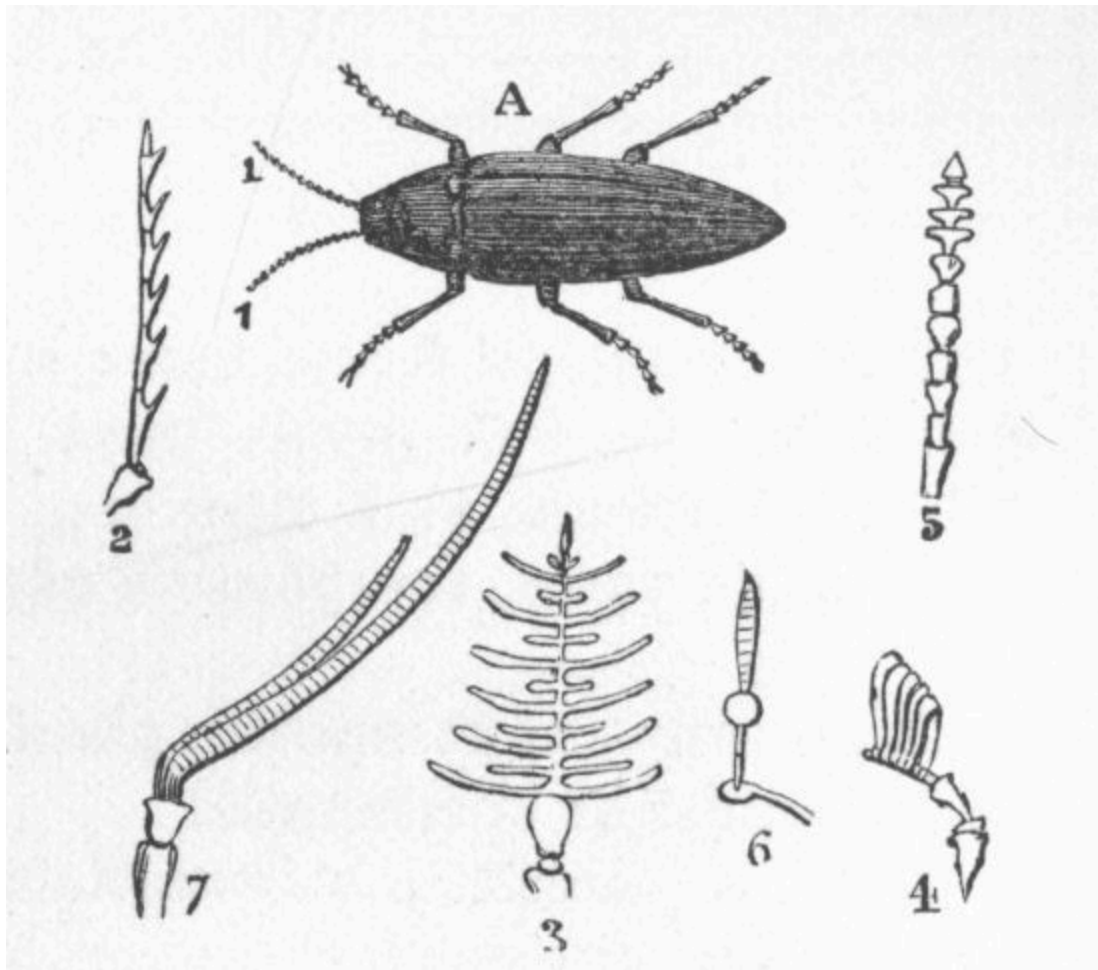
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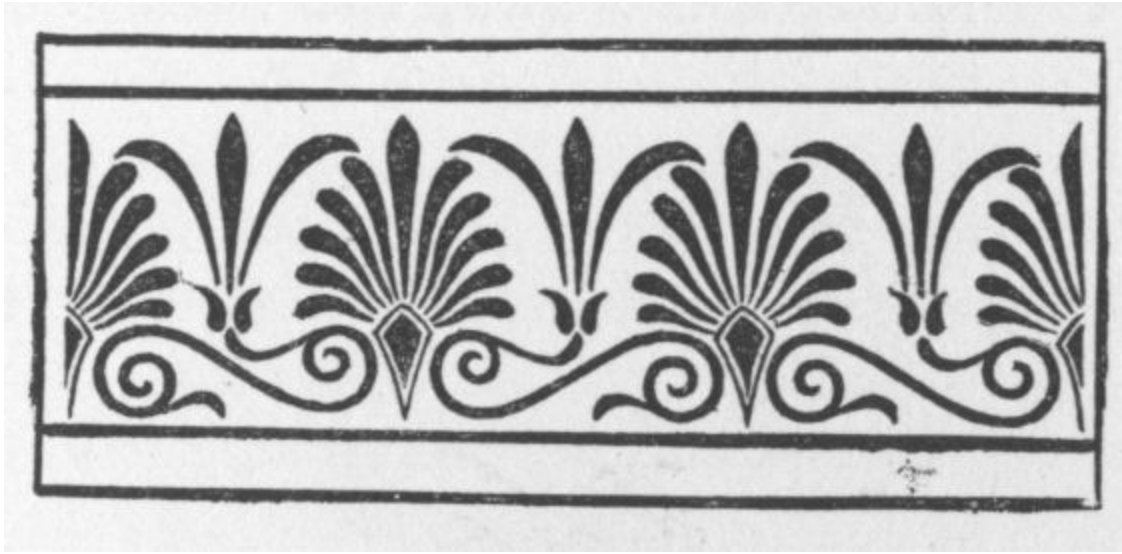
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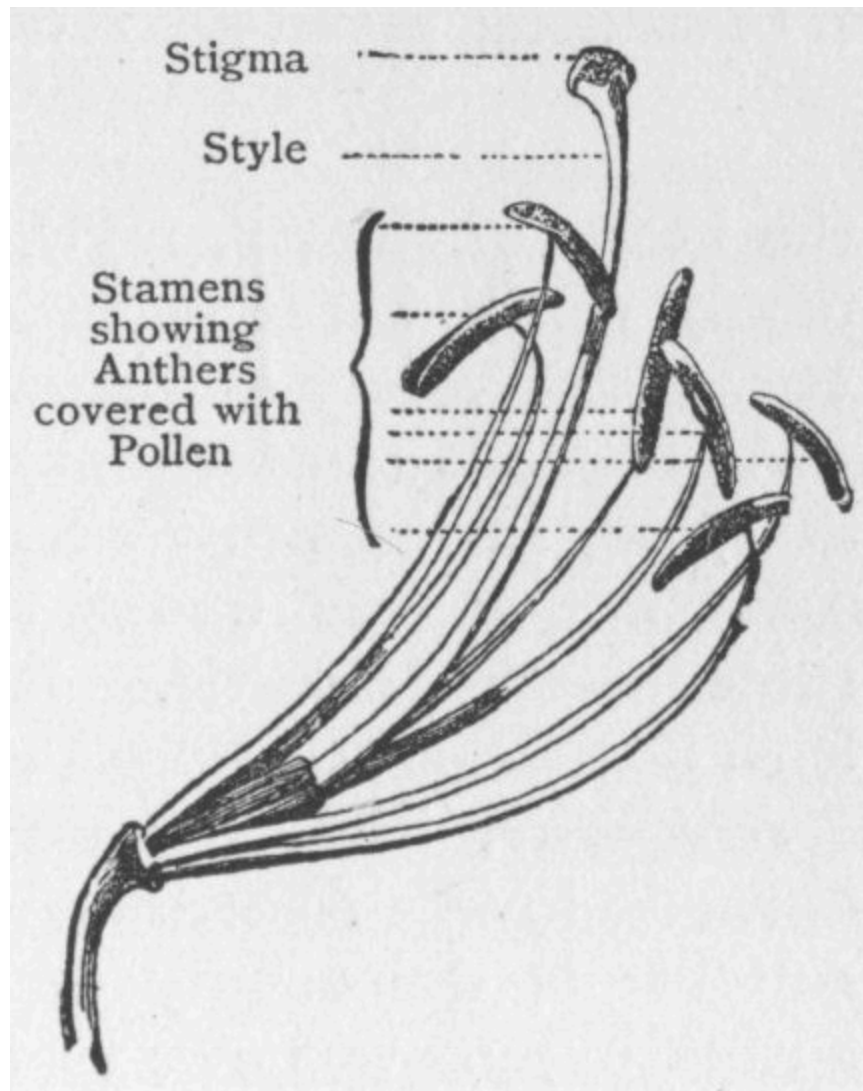
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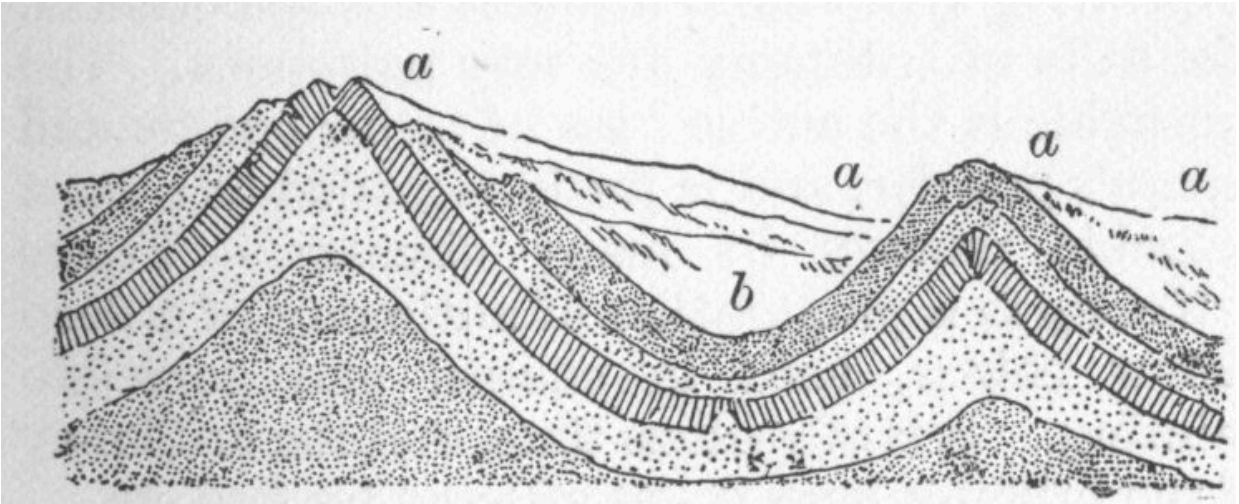
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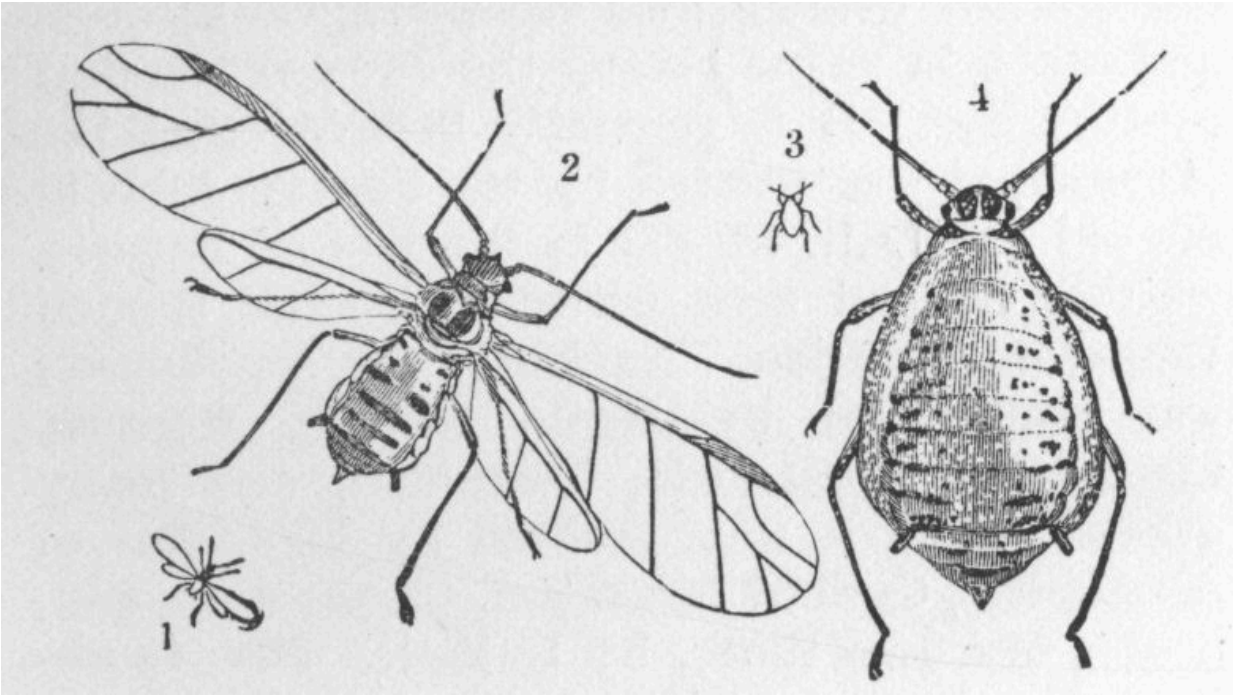
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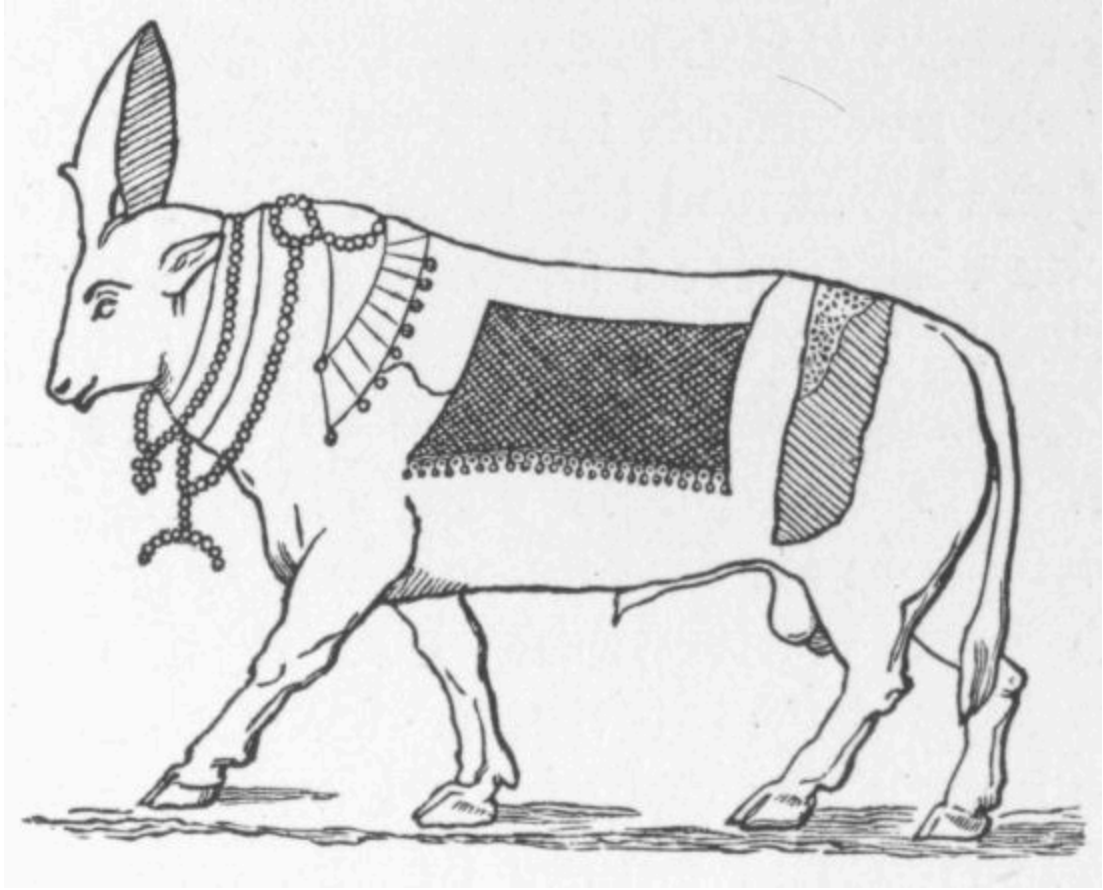
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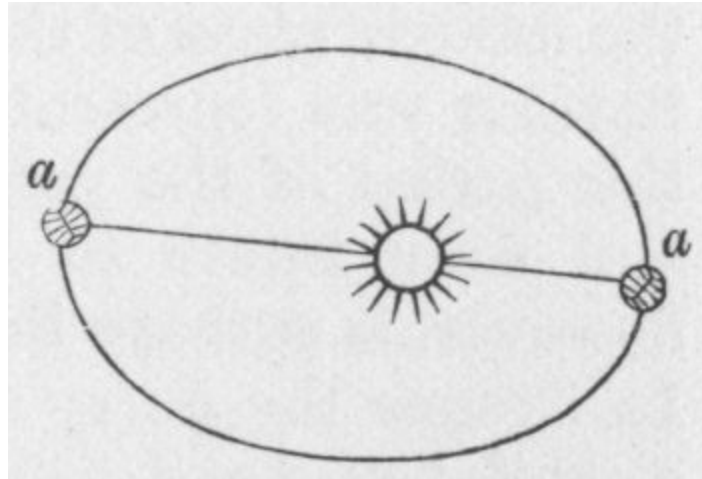
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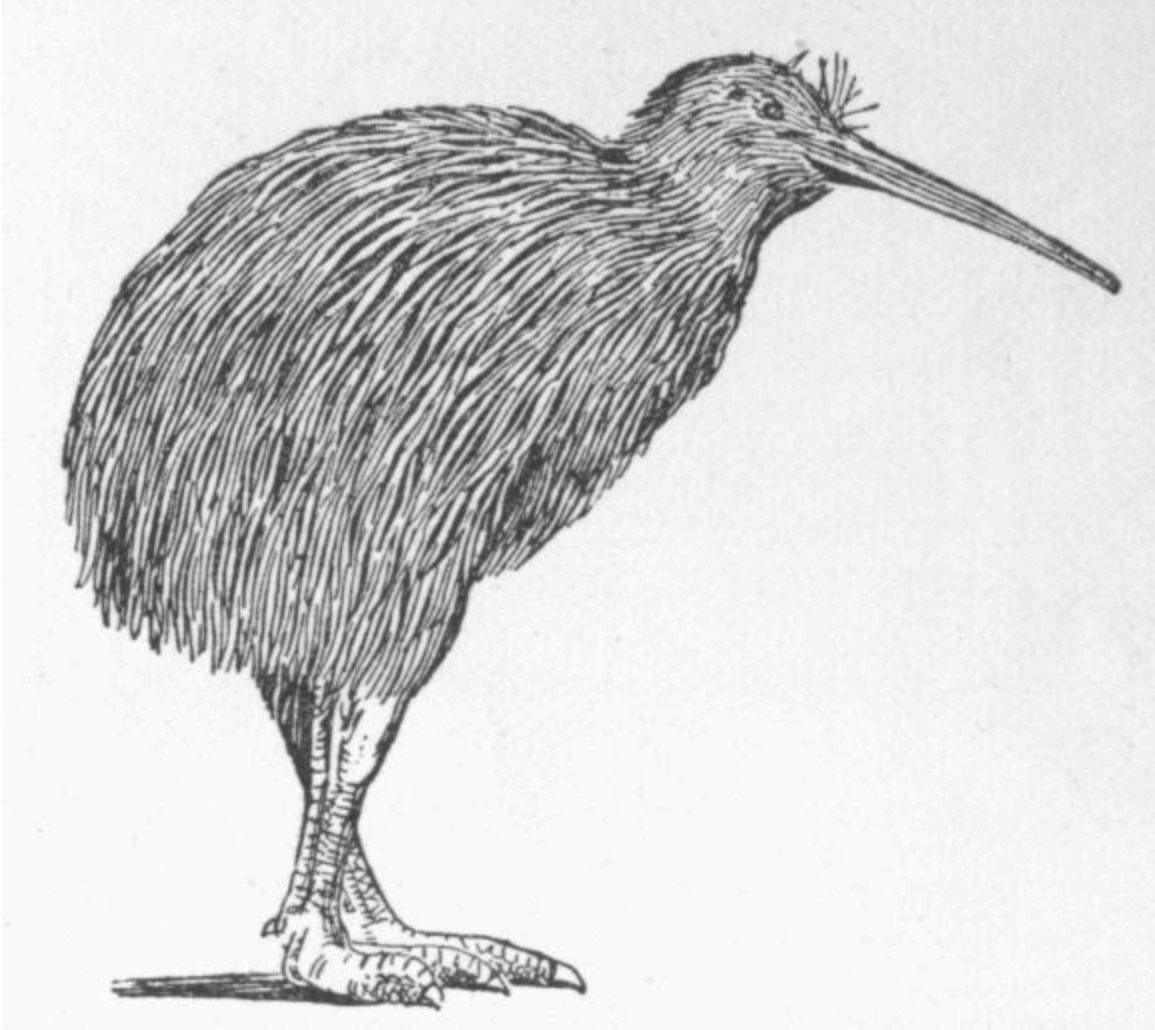
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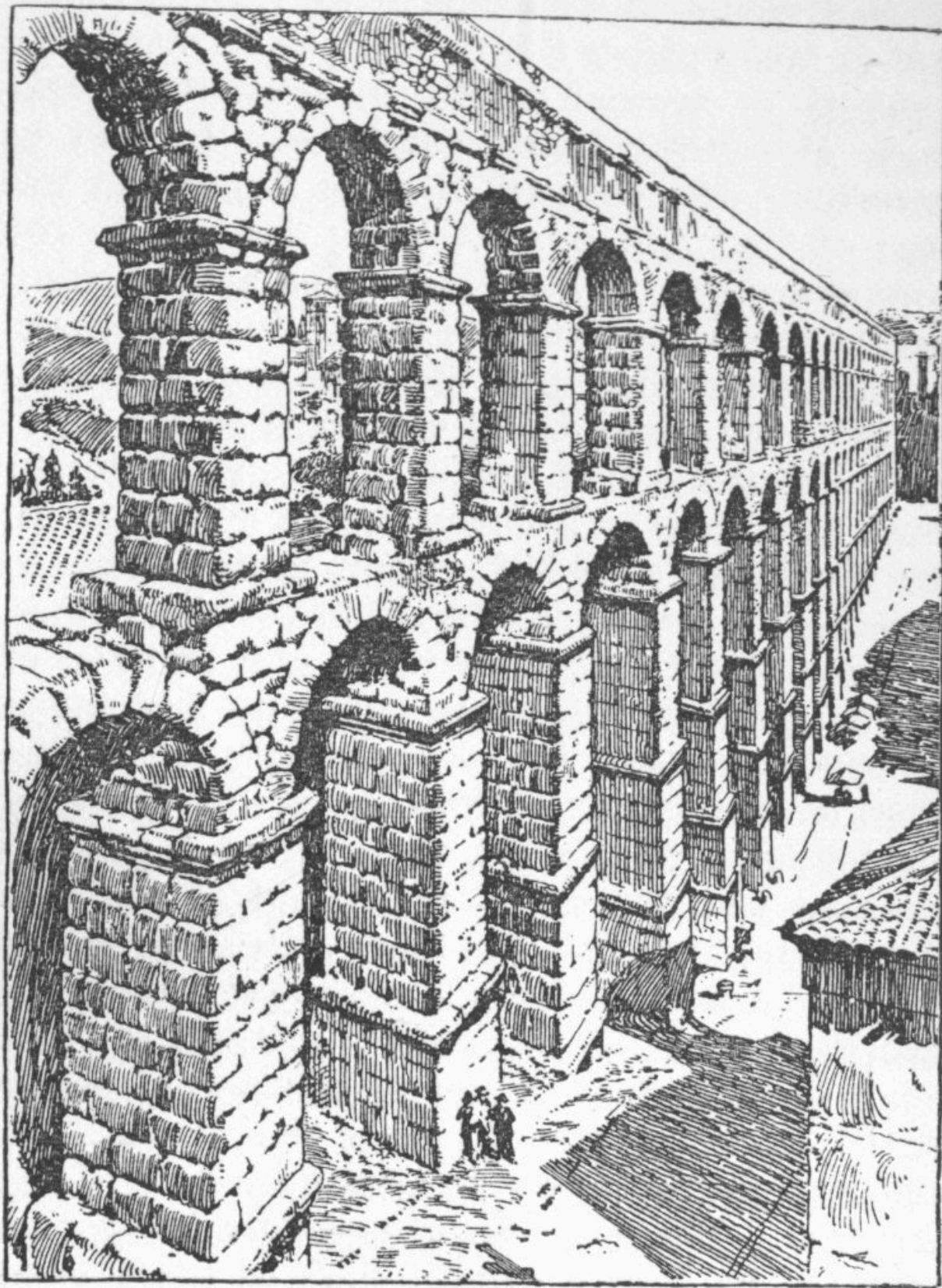
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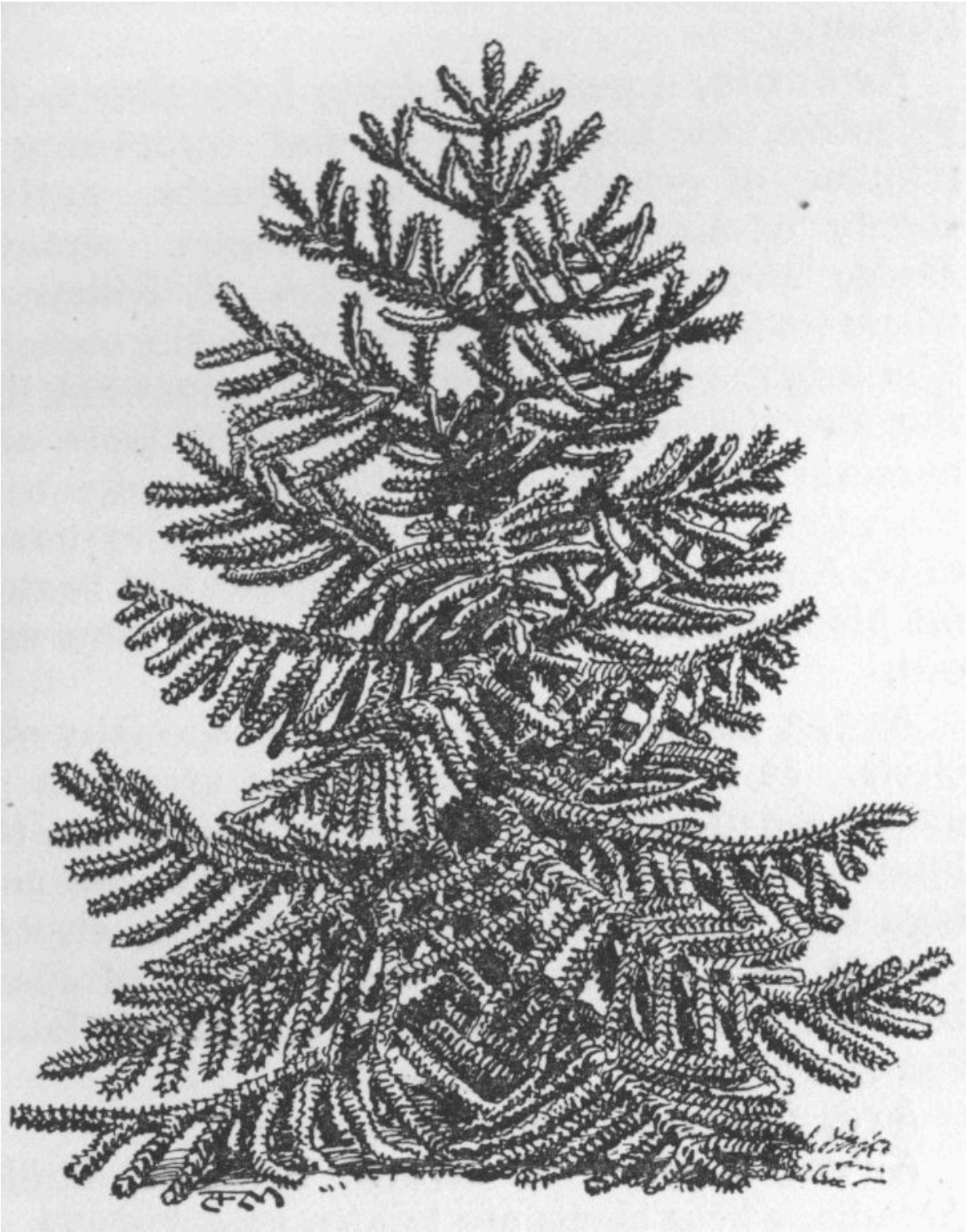
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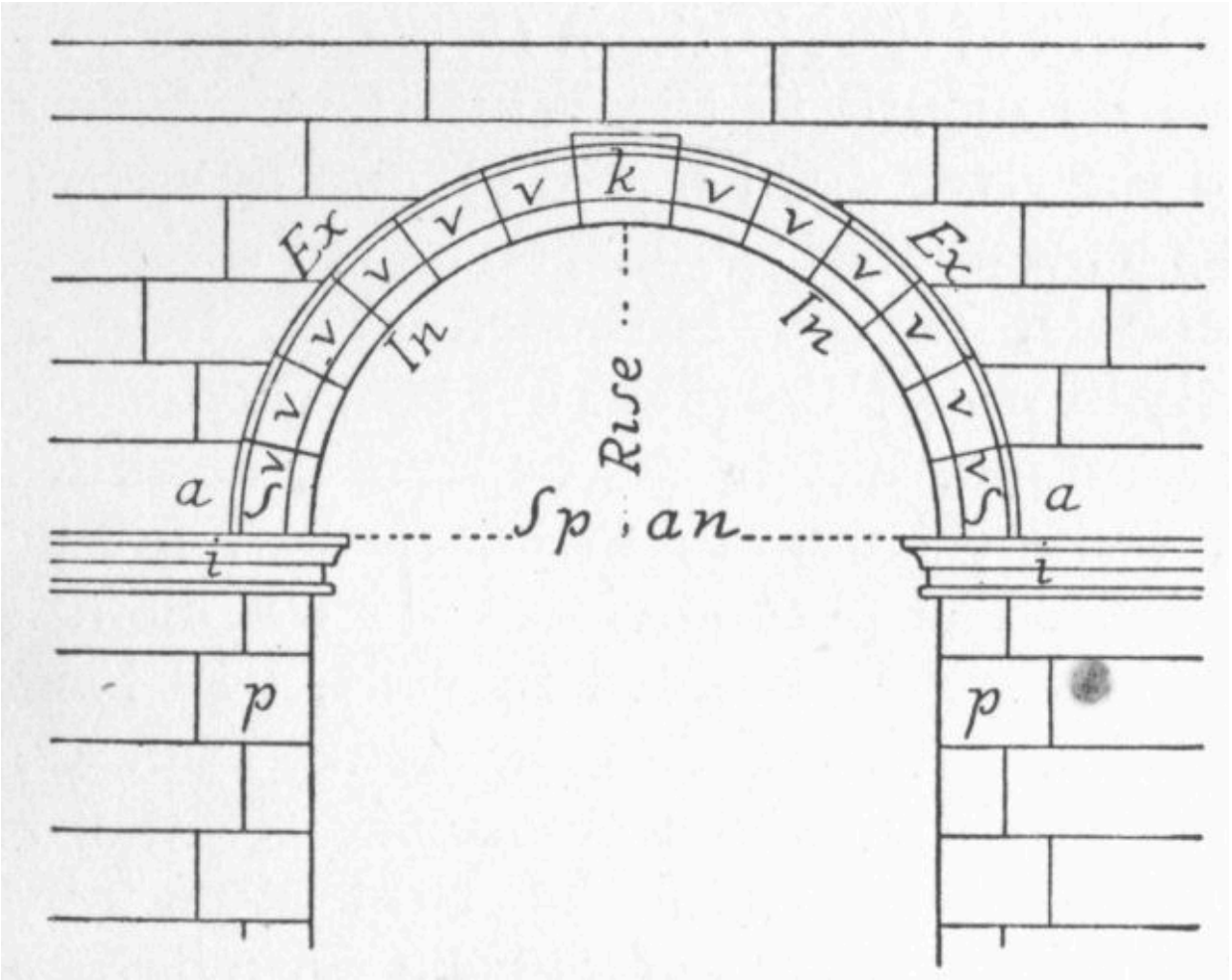
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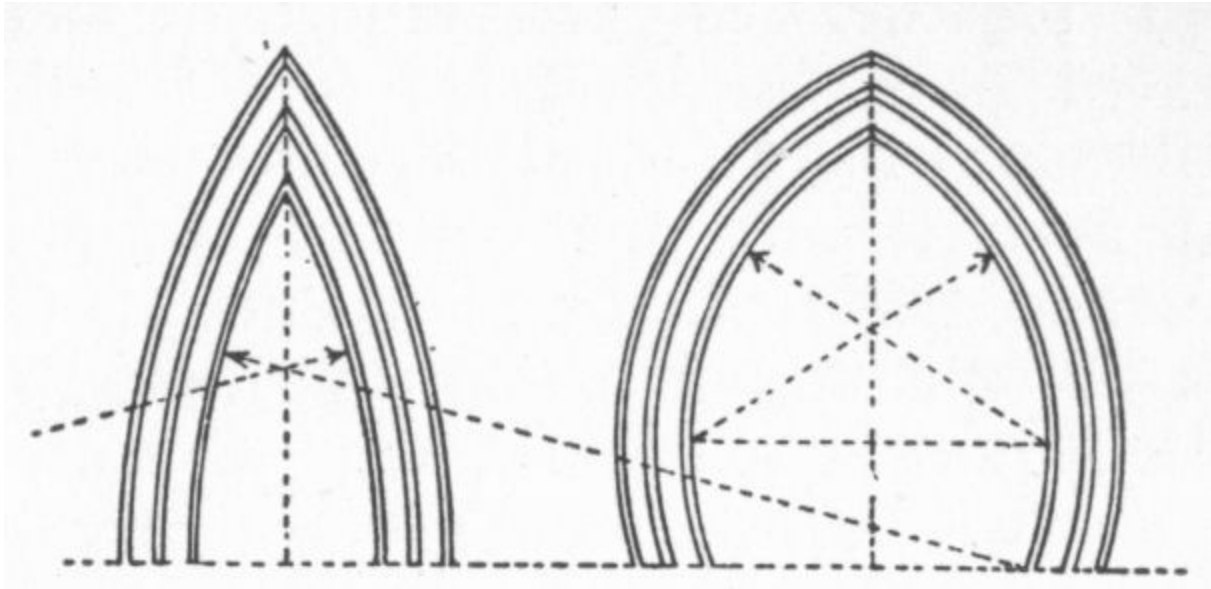
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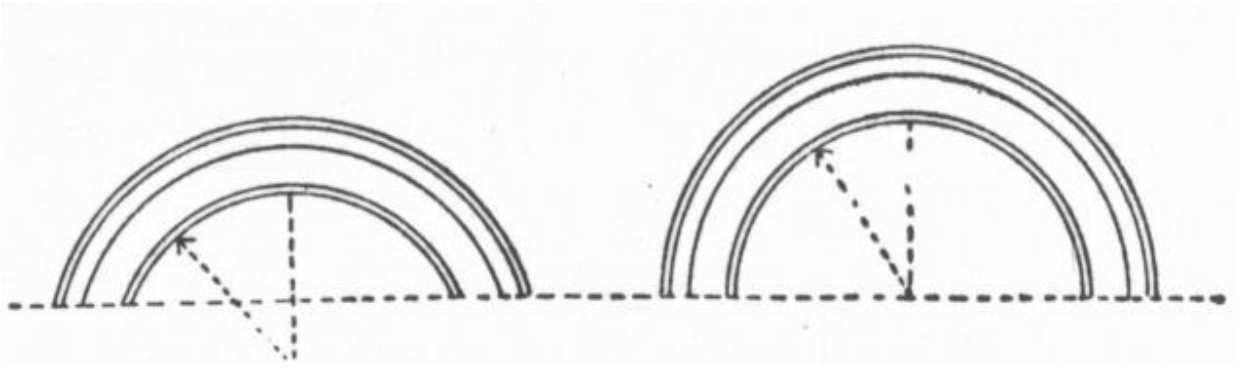
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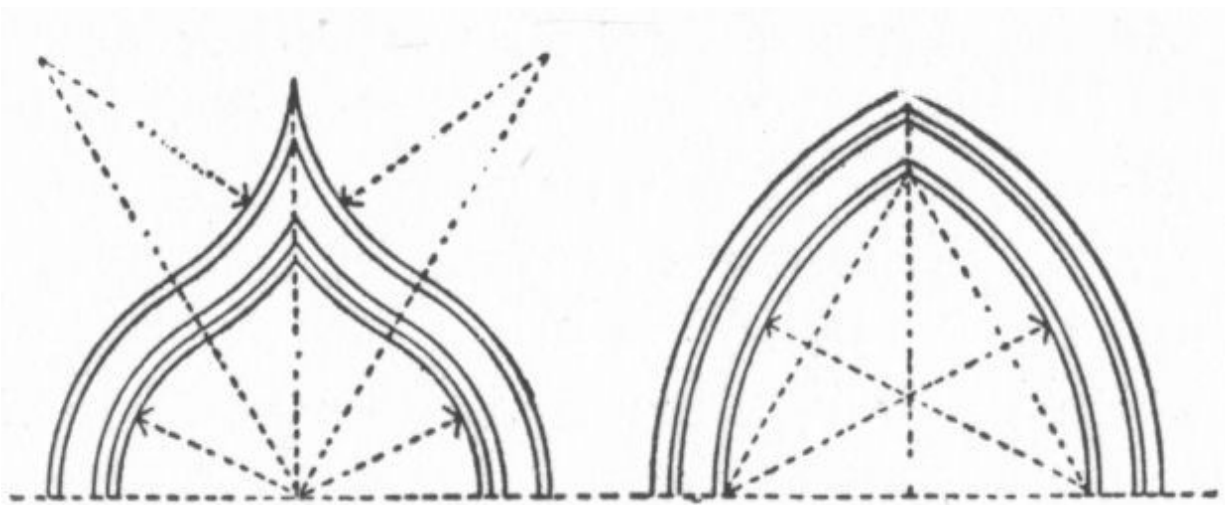
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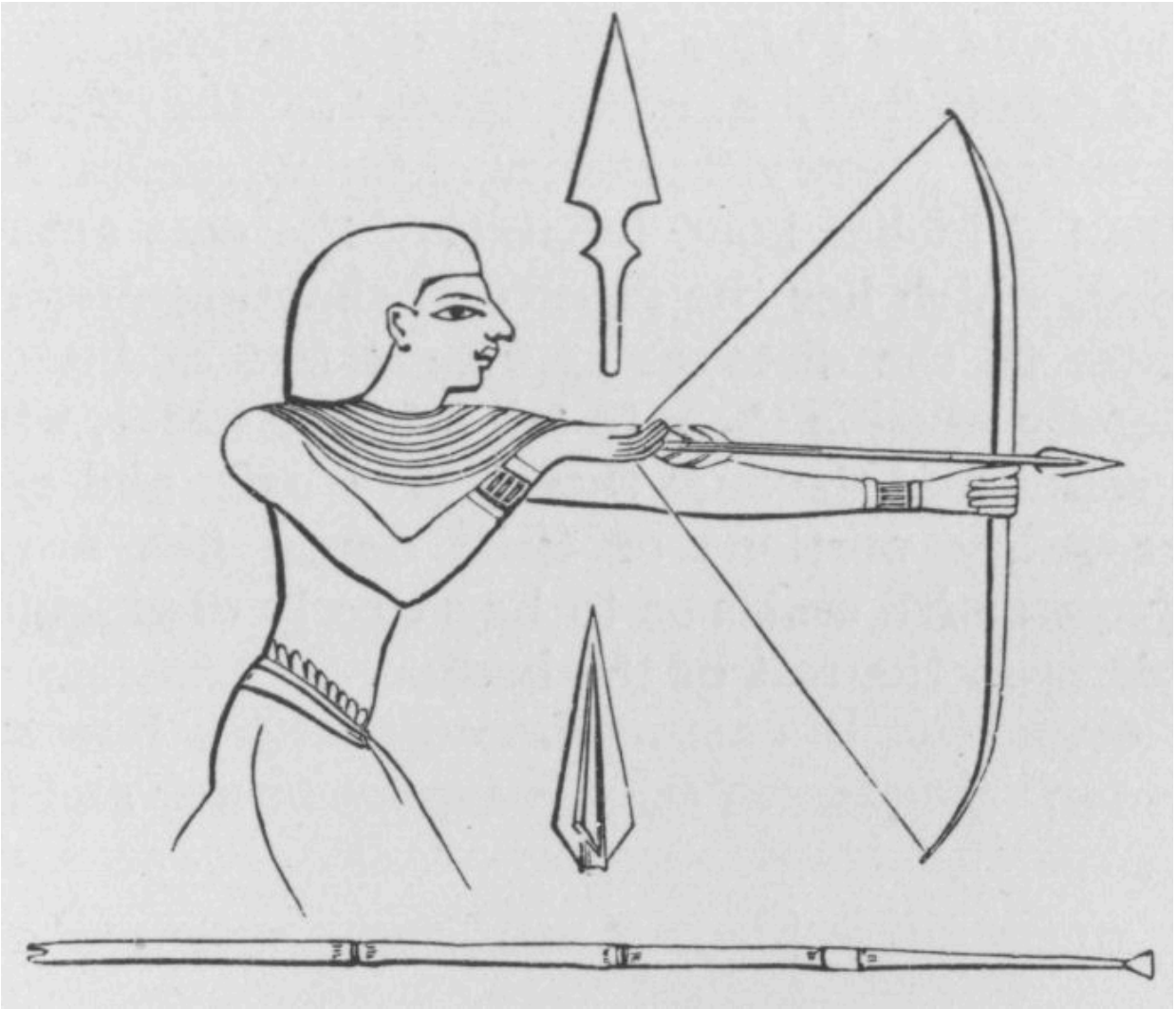
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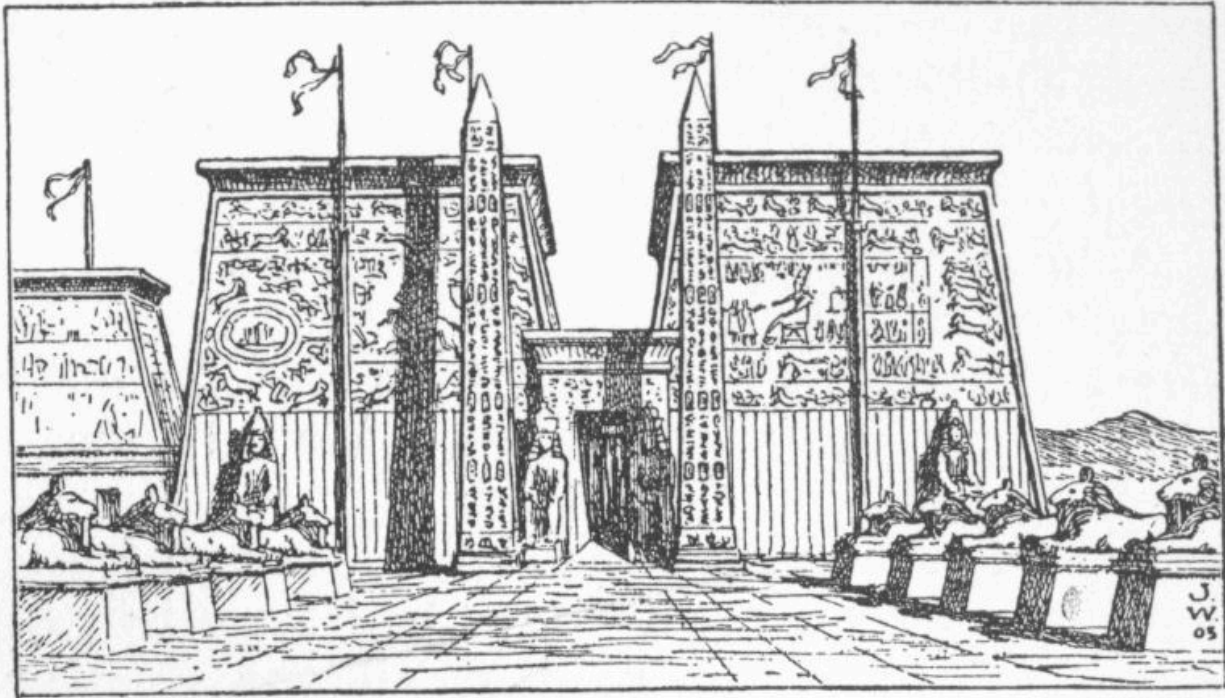
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EGYPTIAN—COURTYARD OF A TEMPLE
From a reconstruction.



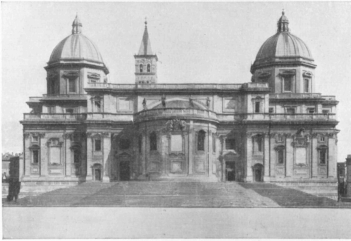
GREEK—THE THESEION, ATHENS
465 B.C. A Doric example.



ROMAN—ARCH OF CONSTANTINE
A.D. 312. It has detached Corinthian columns.



FRENCH GOTHIC: AMIENS CATHEDRAL
1220–1288. Generally referred to as having the typical French Cathedral plan. The façade resembles Notre Dame and Rheims Cathedrals.



EARLY CHRISTIAN—CHURCH OF S. MARIA MAGGIORE
A.D. 432. Type of Basilica Church.



ITALIAN ROMANESQUE—PISA CATHEDRAL
A.D. 1063–1092.



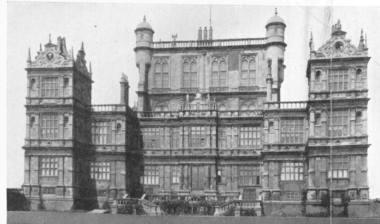
MODERN—FLAT-IRON BUILDING,
NEW YORK
A “sky-scraper”—a gigantic framework of steel covered with masonry.



ENGLISH GOTHIC—CHOIR OF CANTERBURY CATHEDRAL
1175–1184. The earliest example of the Gothic style carried out in an important English building.

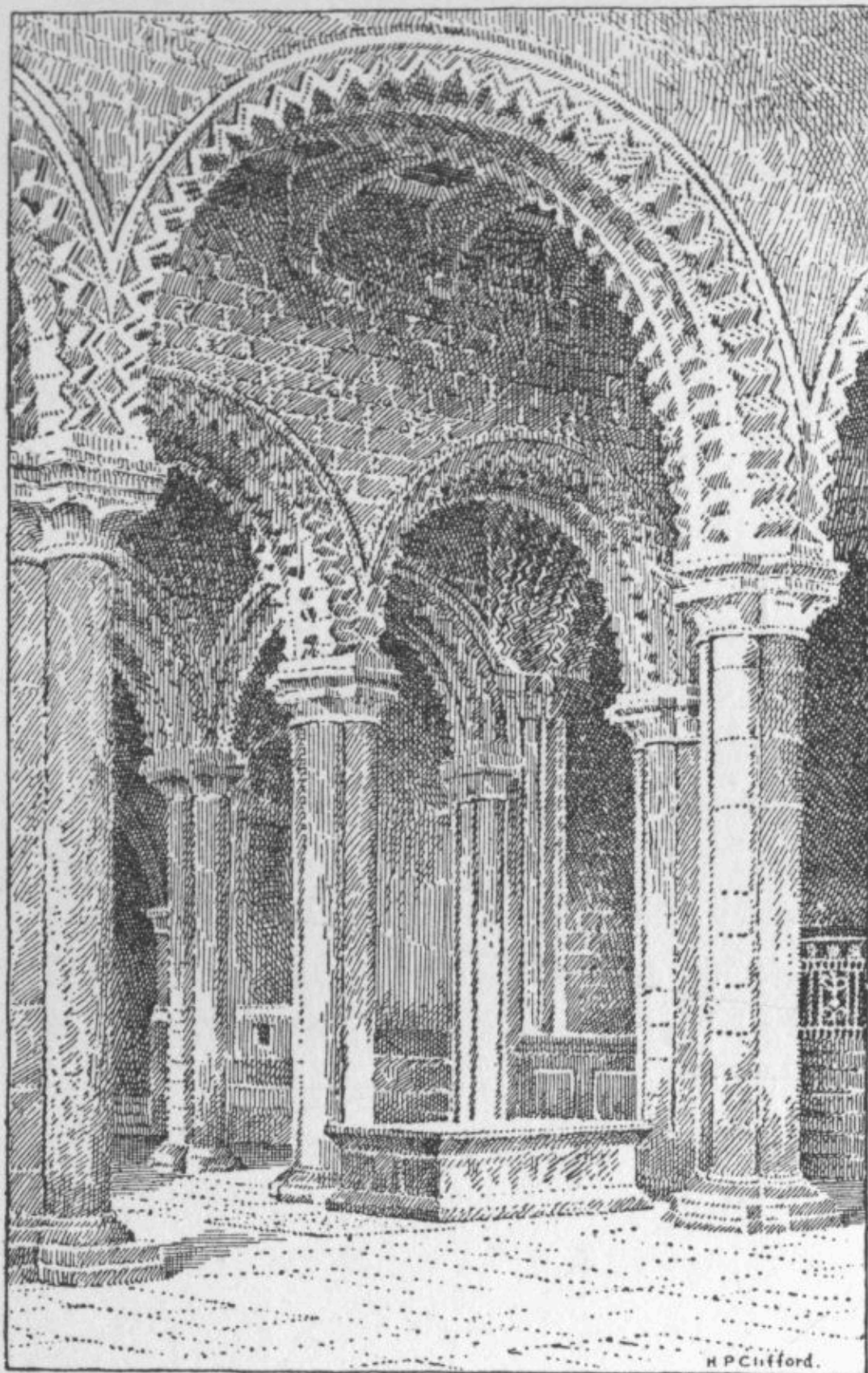


TUDOR—HAMPTON COURT PALACE
1515.

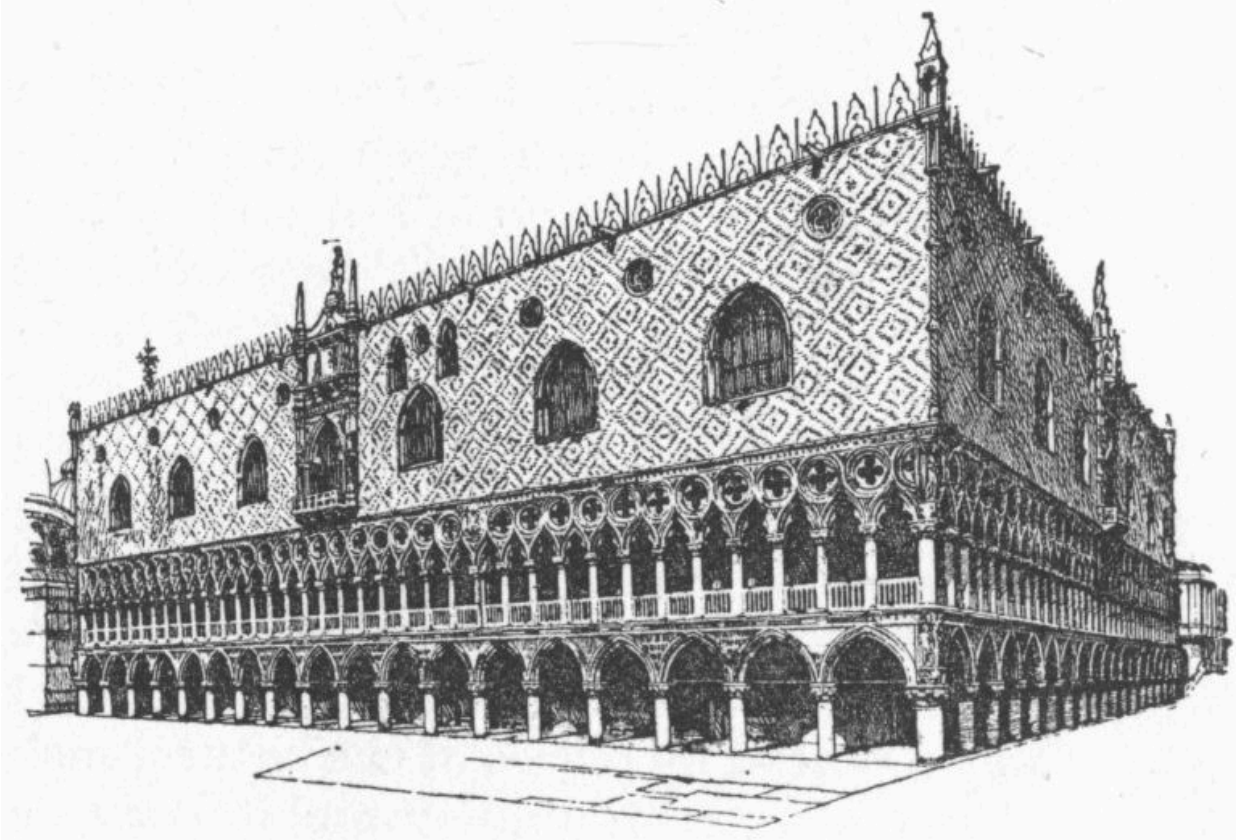


ENGLISH RENAISSANCE—WOLLATON HALL, NOTTS
1580. An example of a famous Elizabethan Mansion.

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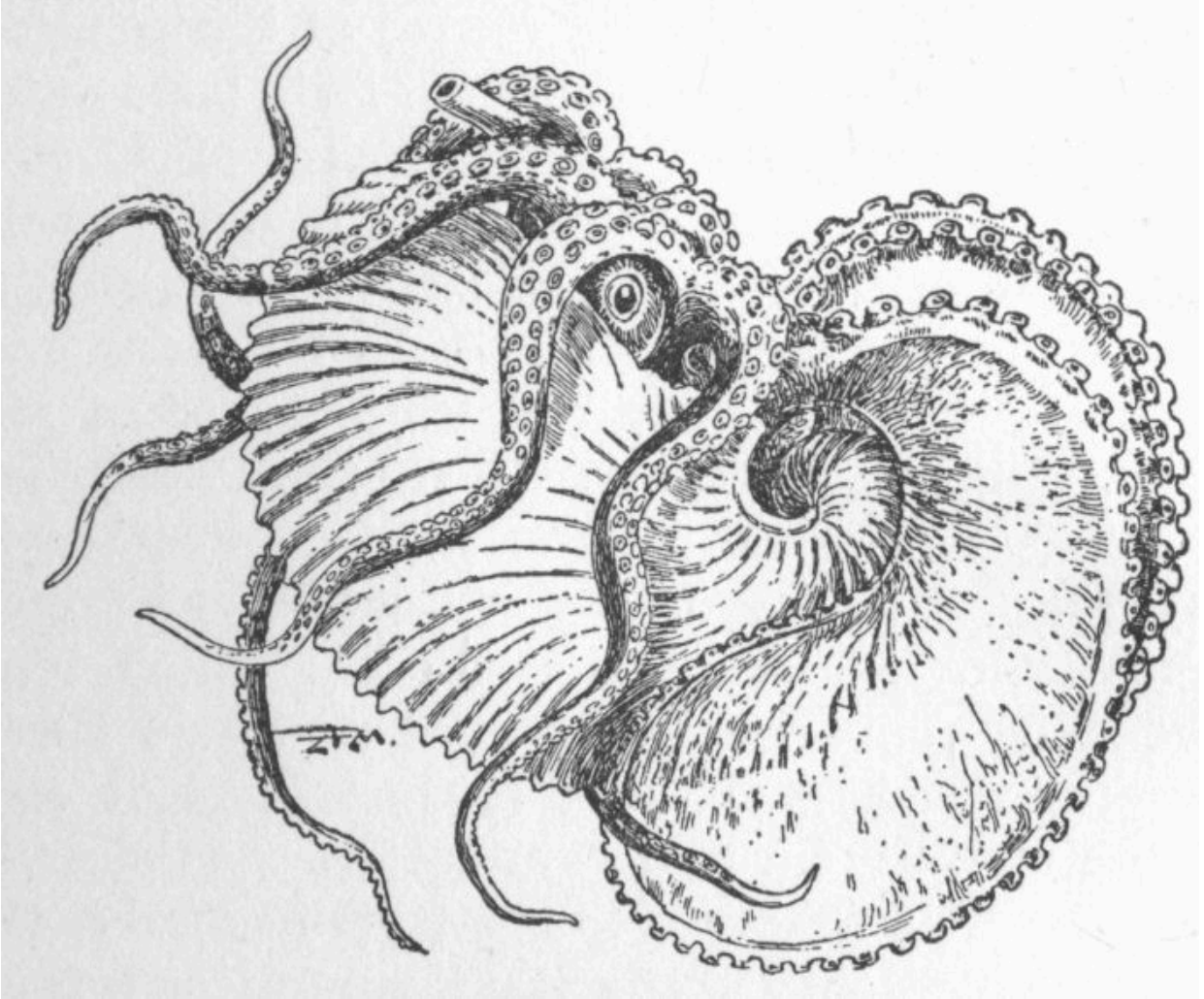
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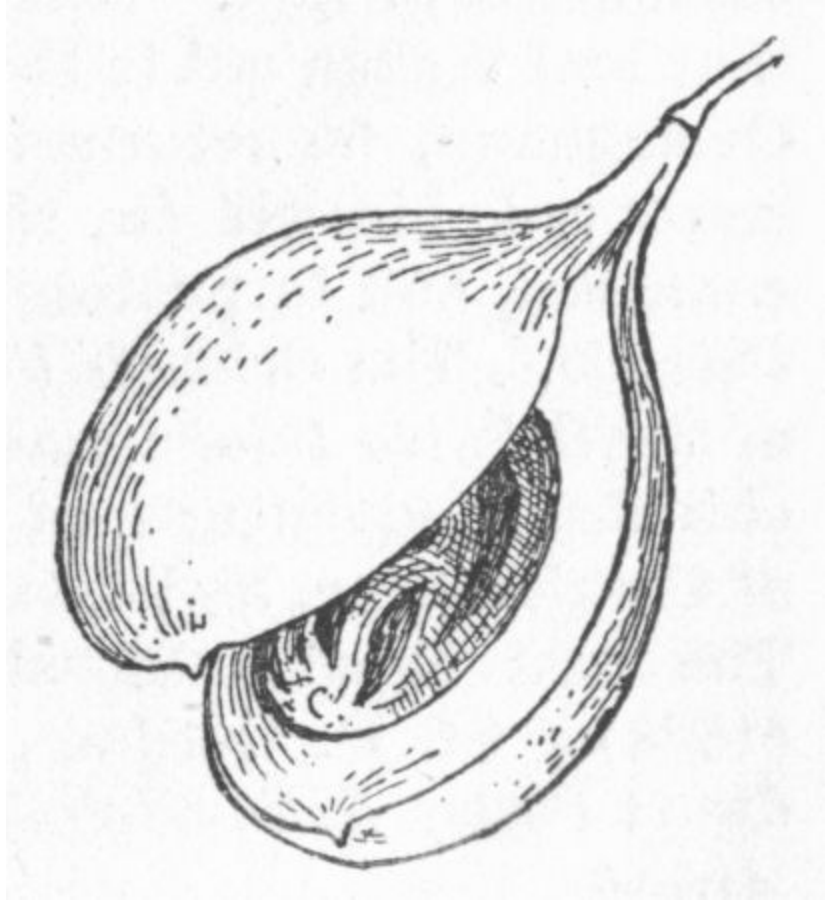
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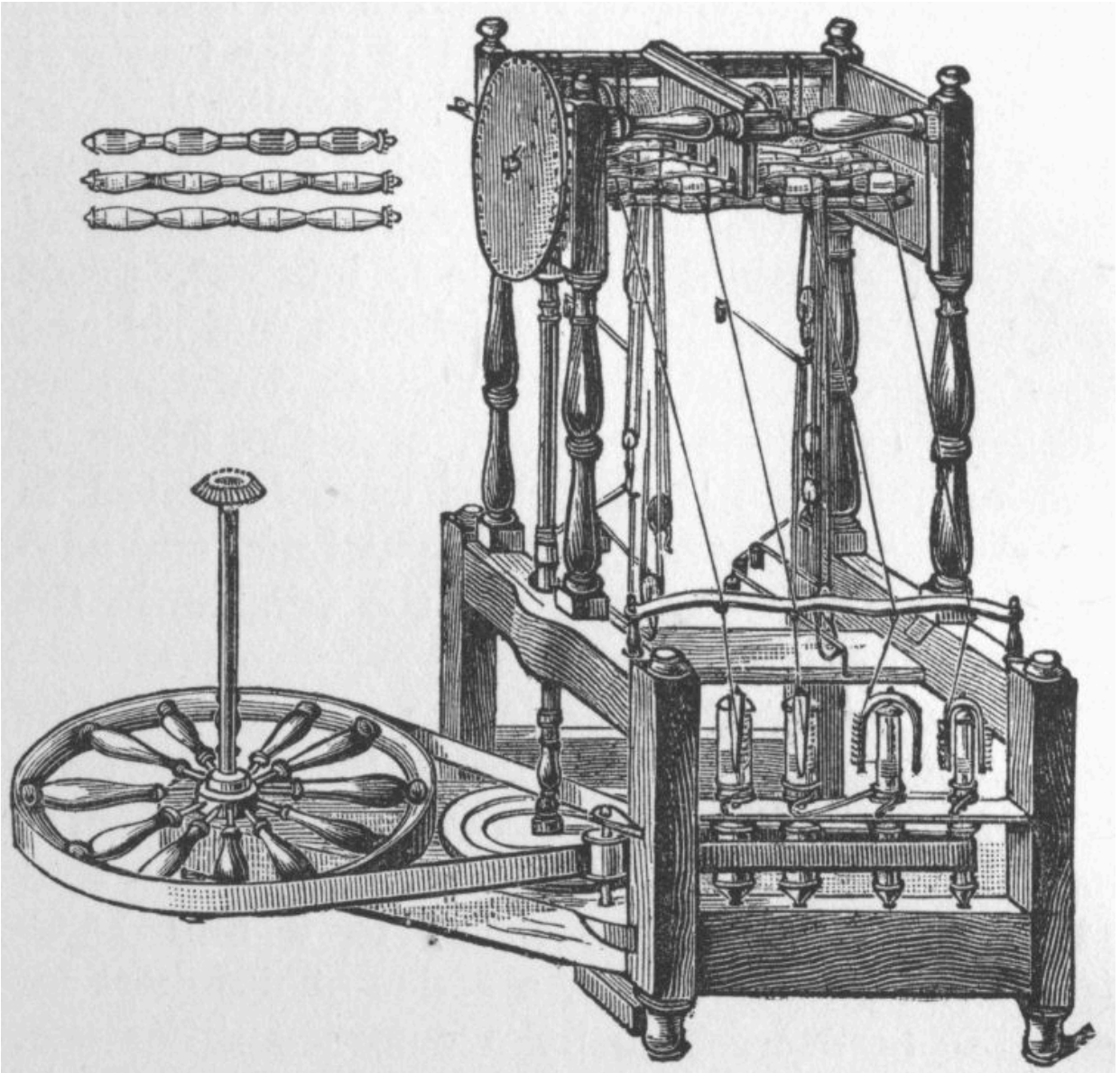
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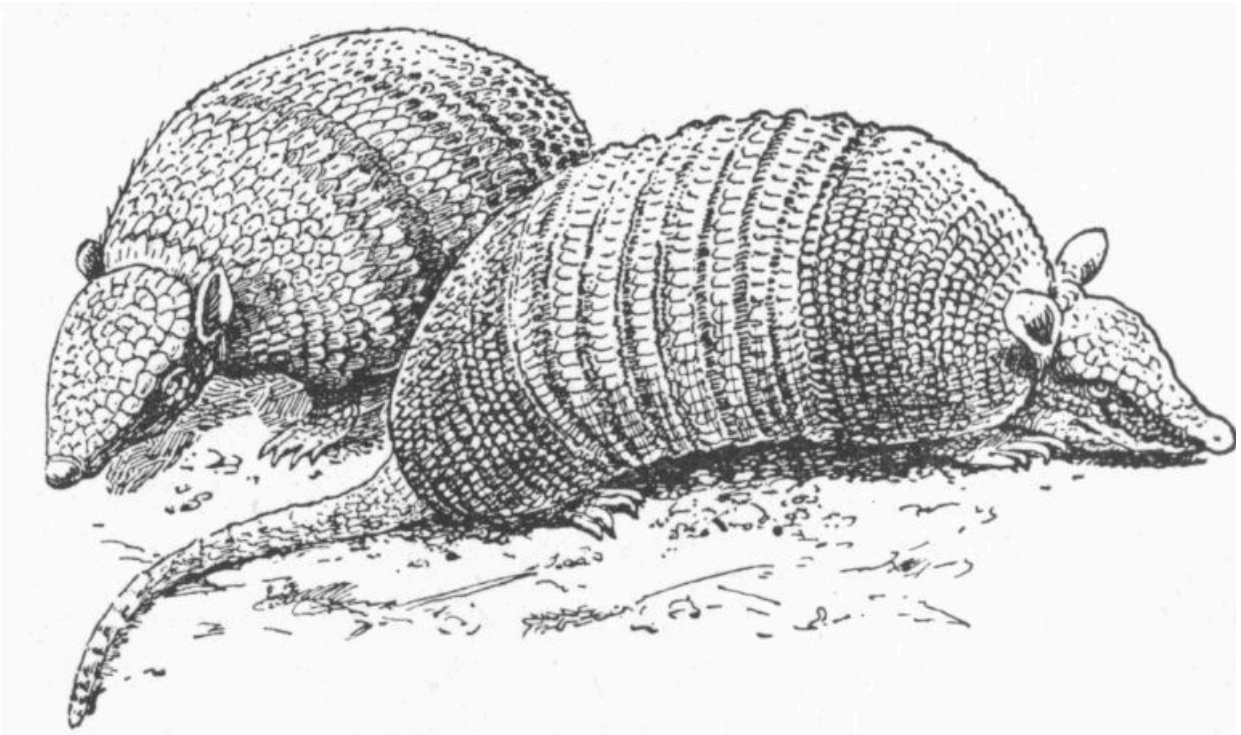
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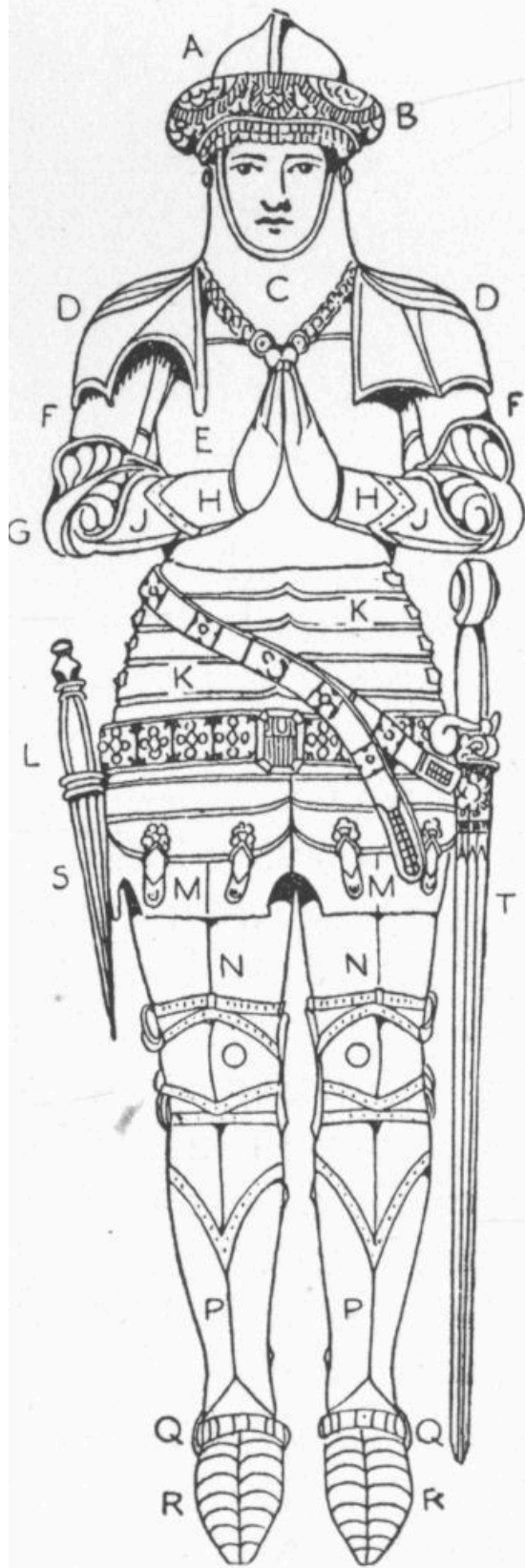
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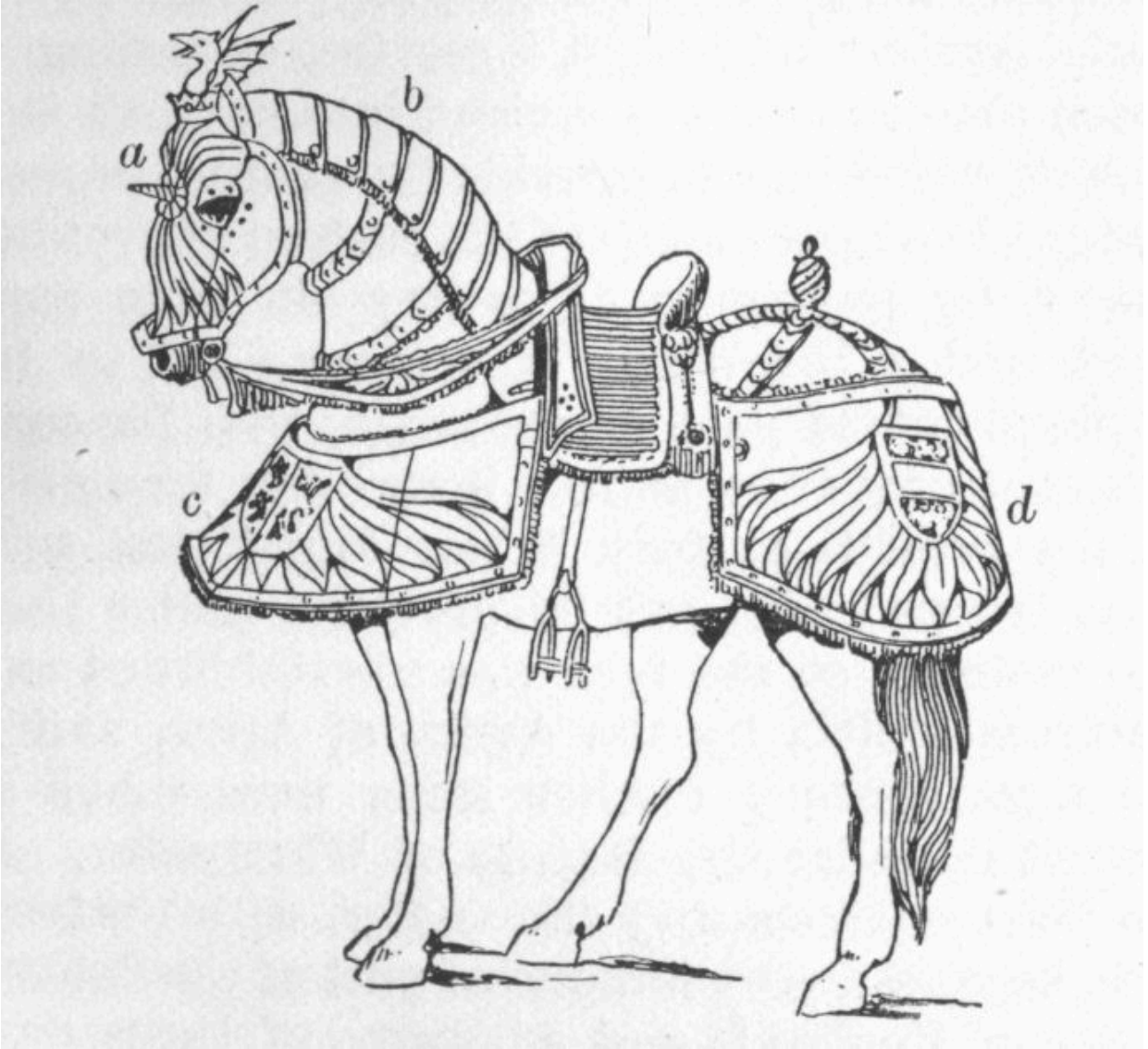
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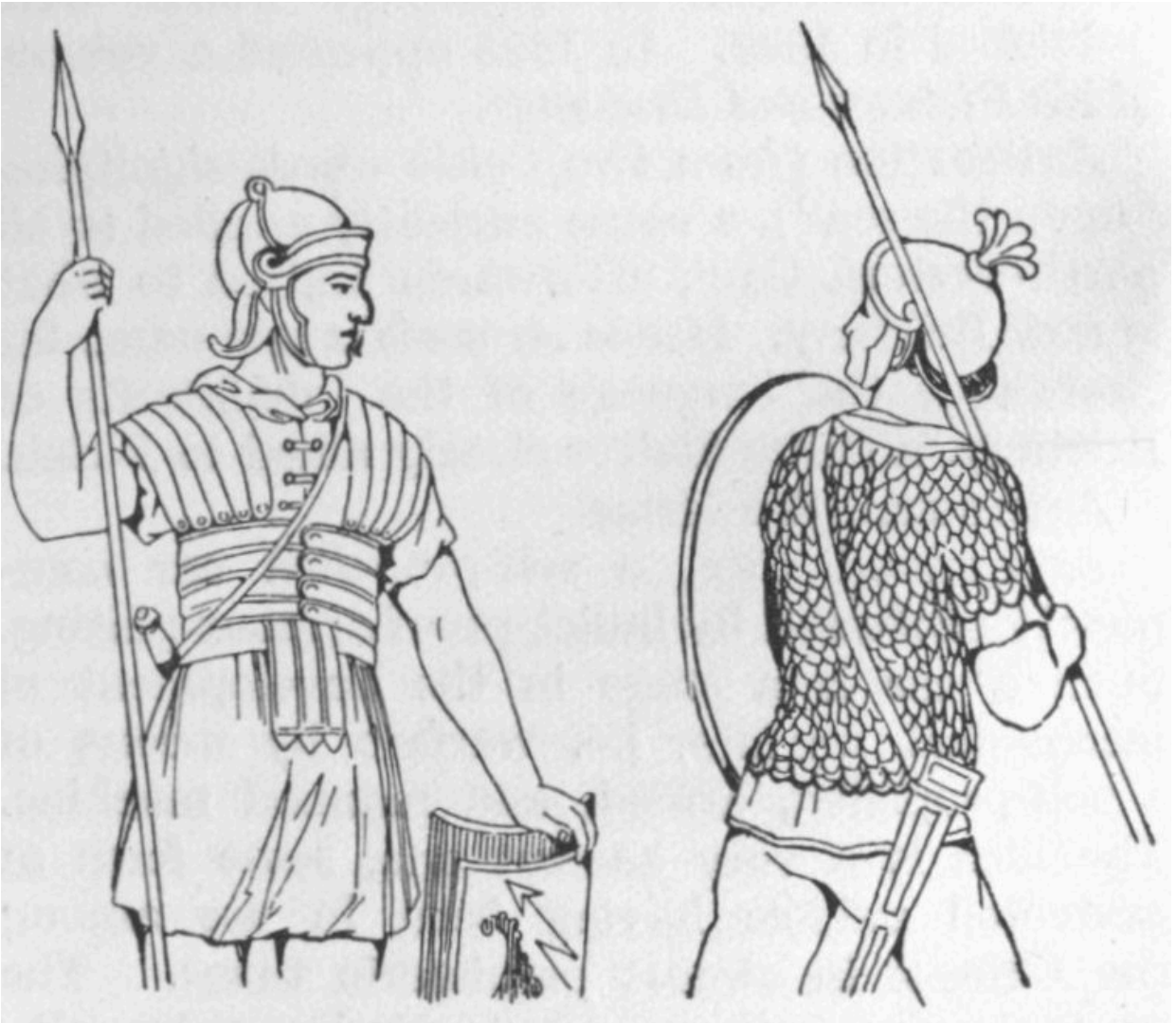
- A, Bascinet.
- B, Jewelled orle round the bascinet.
- C, Gorget, or gorgiere of plate.
- D, Pauldrons.
- E, Breastplate-cuirass.
- F, Rere-braces.
- G, Coudes or elbow-plates.
- H, Gauntlets.
- J, Vambrace.
- K, Skirt of taces.
- L, Military belt or cingulum richly jewelled.
- M, Tuilles or tuillets.
- N, Cuisses.
- O, Genouilleres or knee-pieces.
- P, Jambes.
- Q, Spur-straps.
- R, Sollerets.
- S, Misericorde or dagger.
- T, Sword, suspended by a transverse belt.



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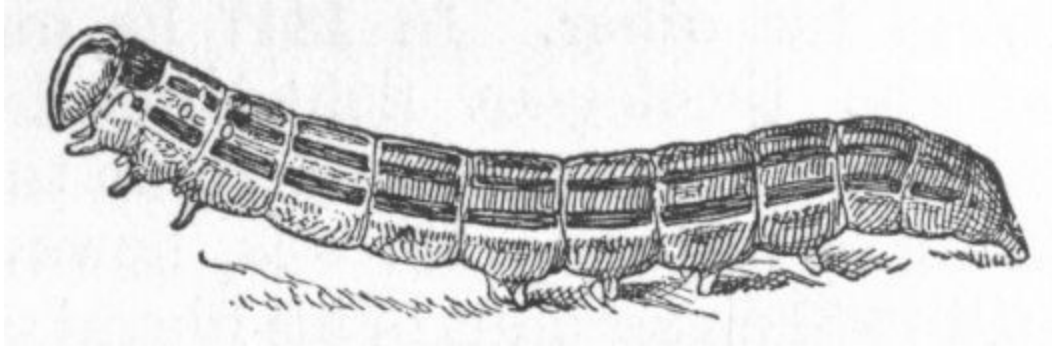
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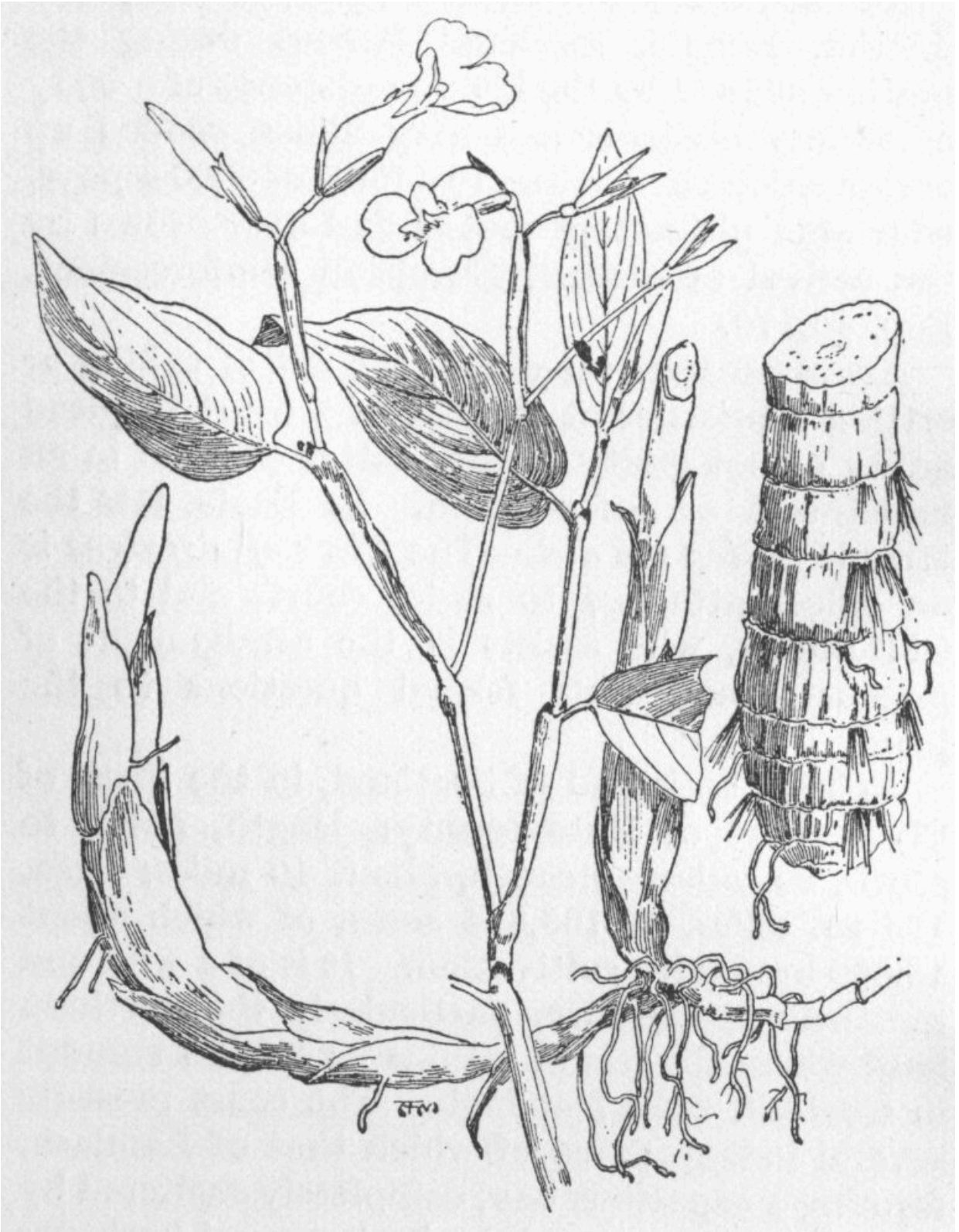
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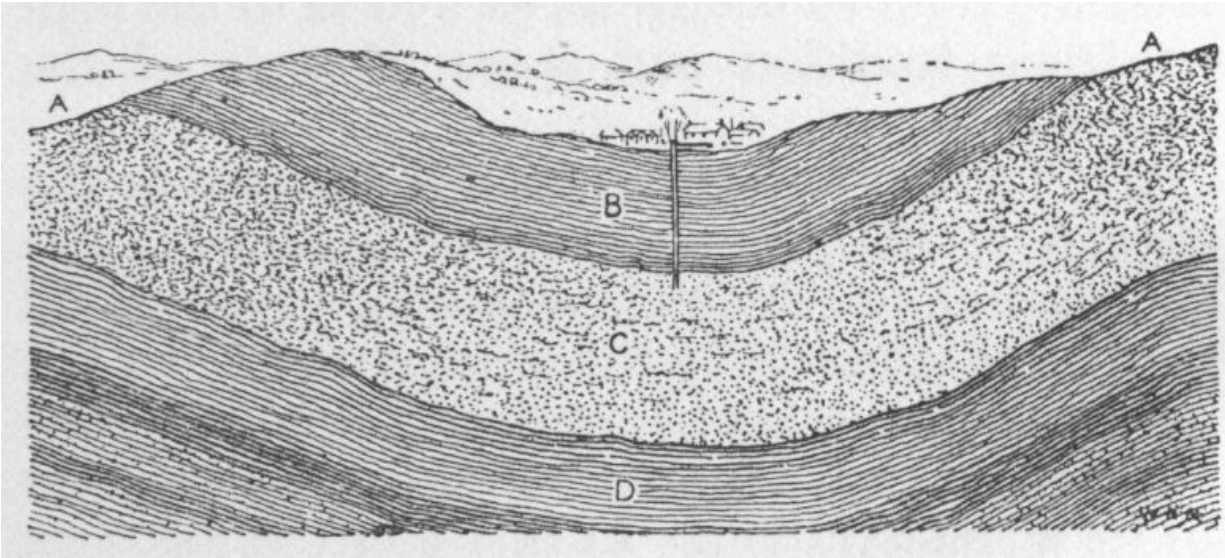
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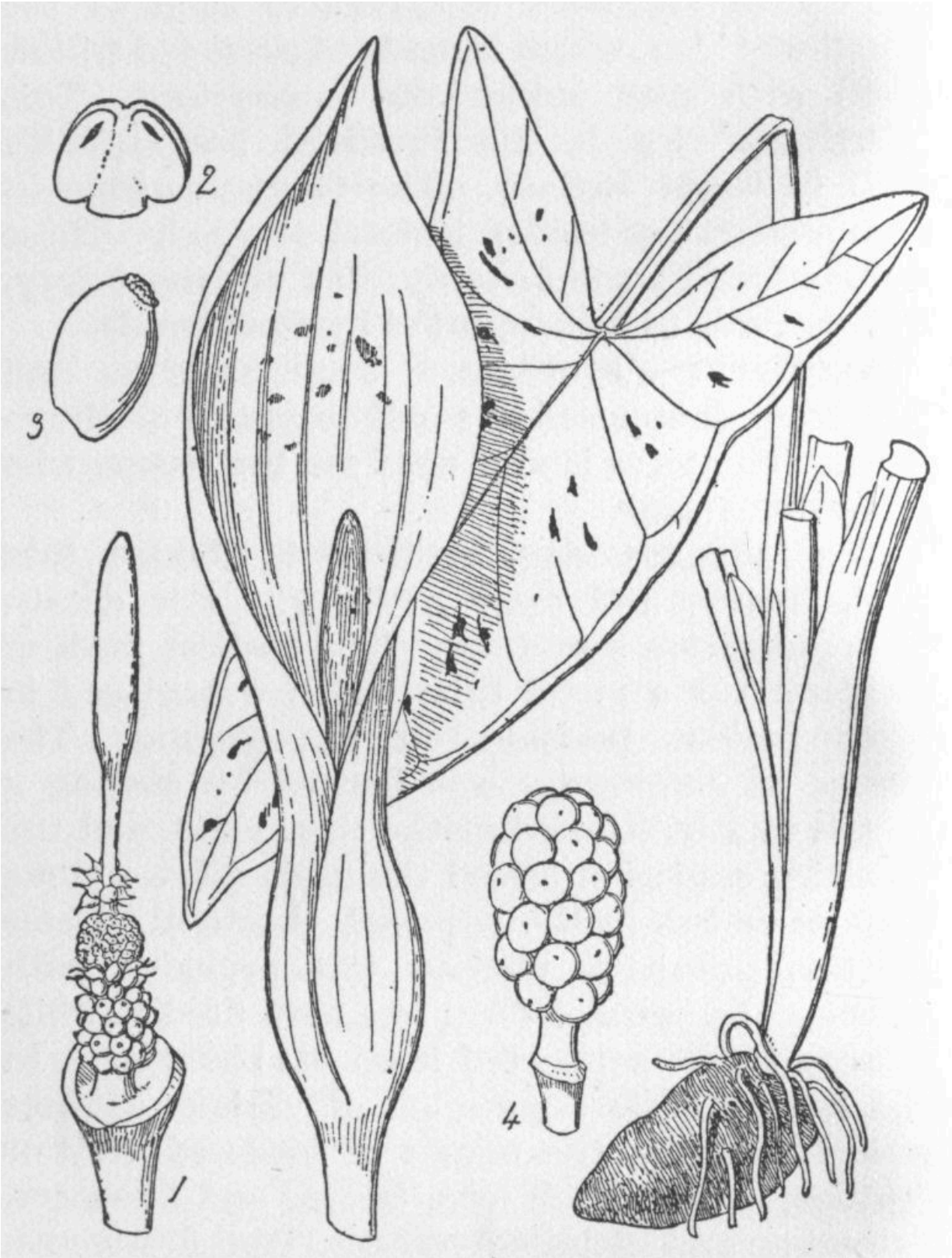
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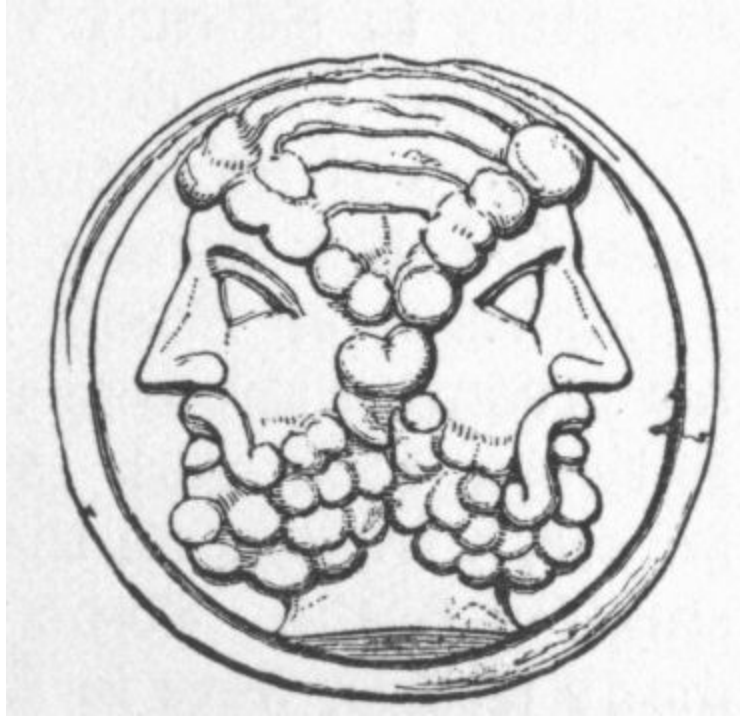
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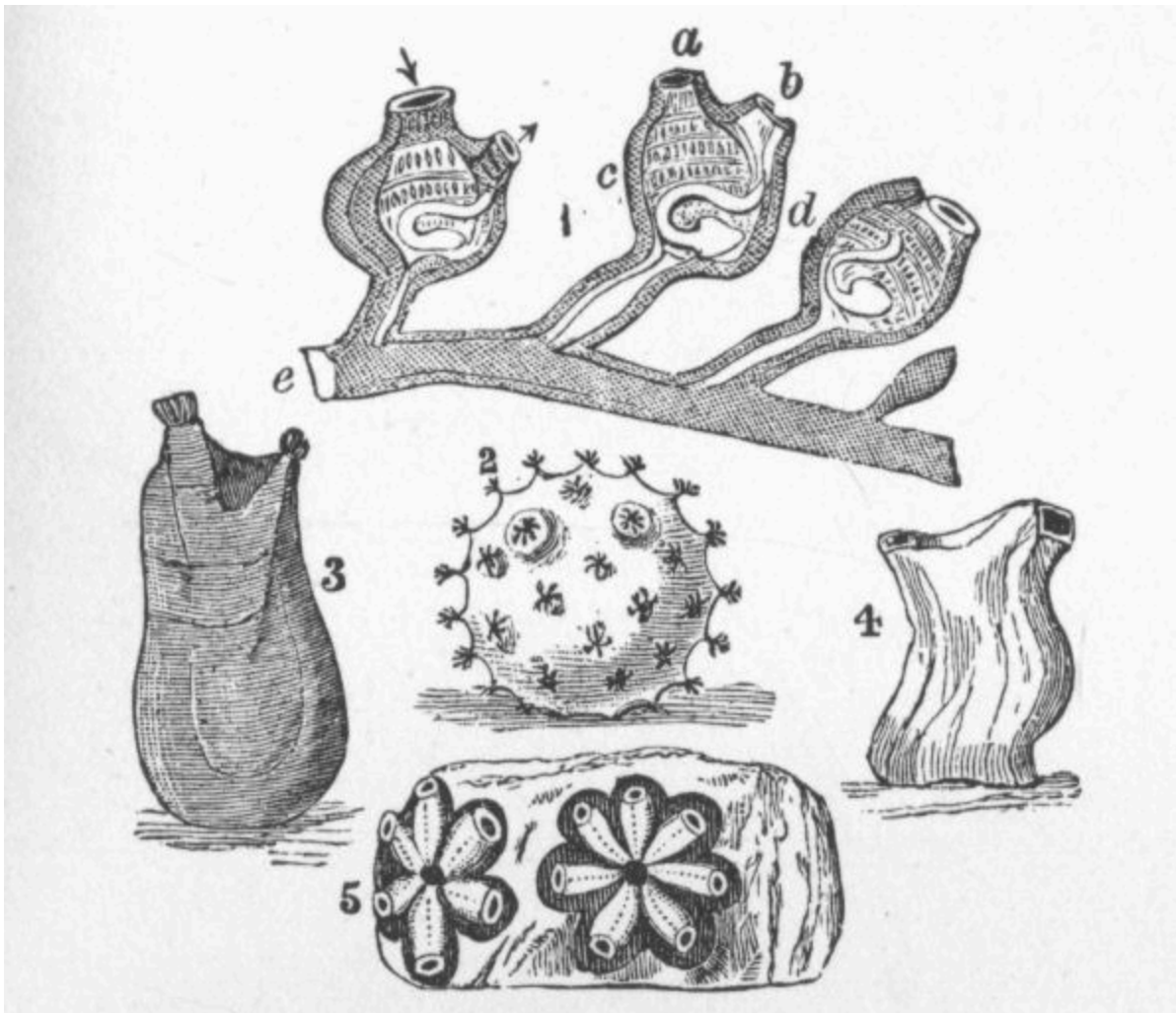
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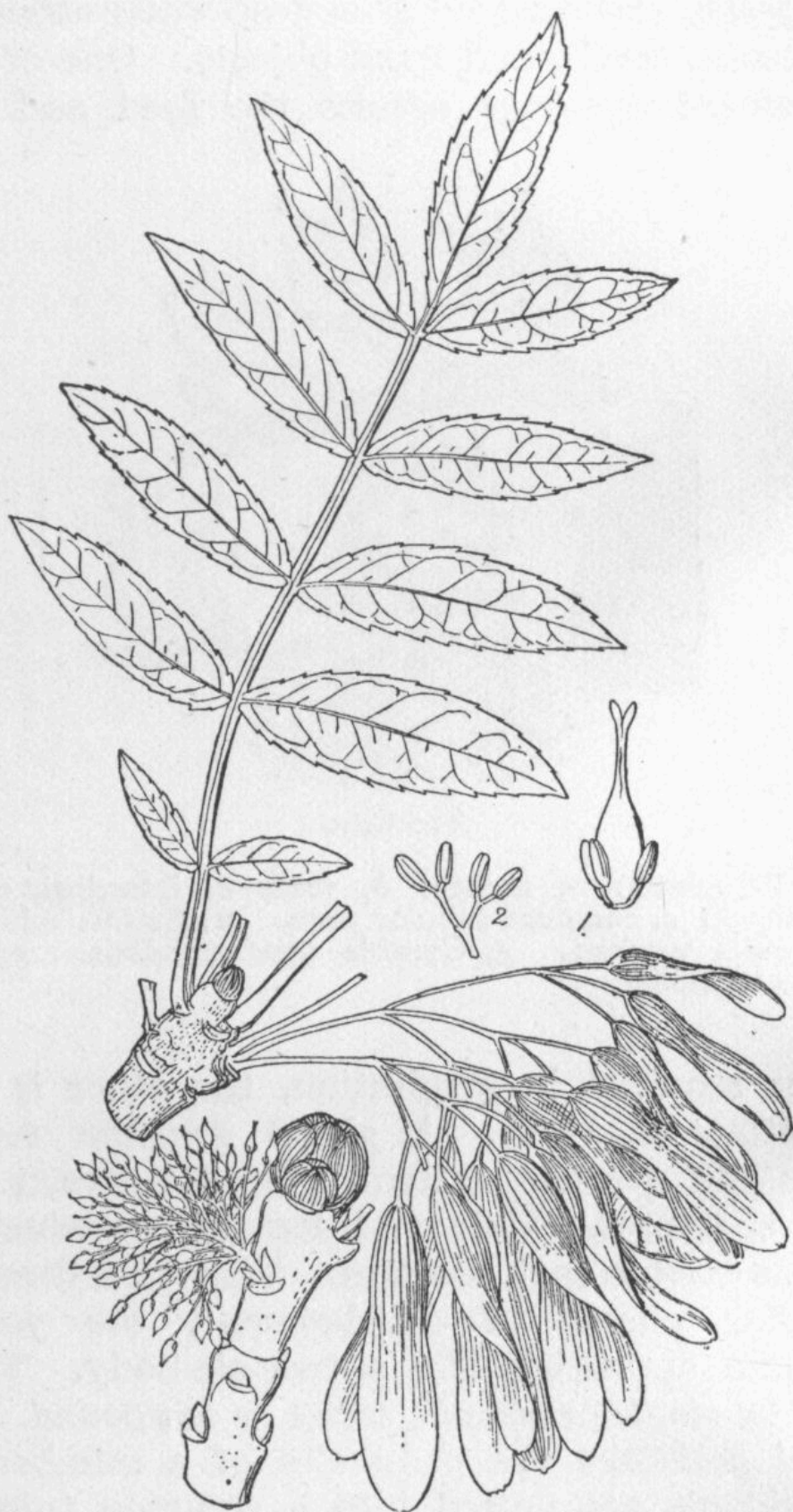
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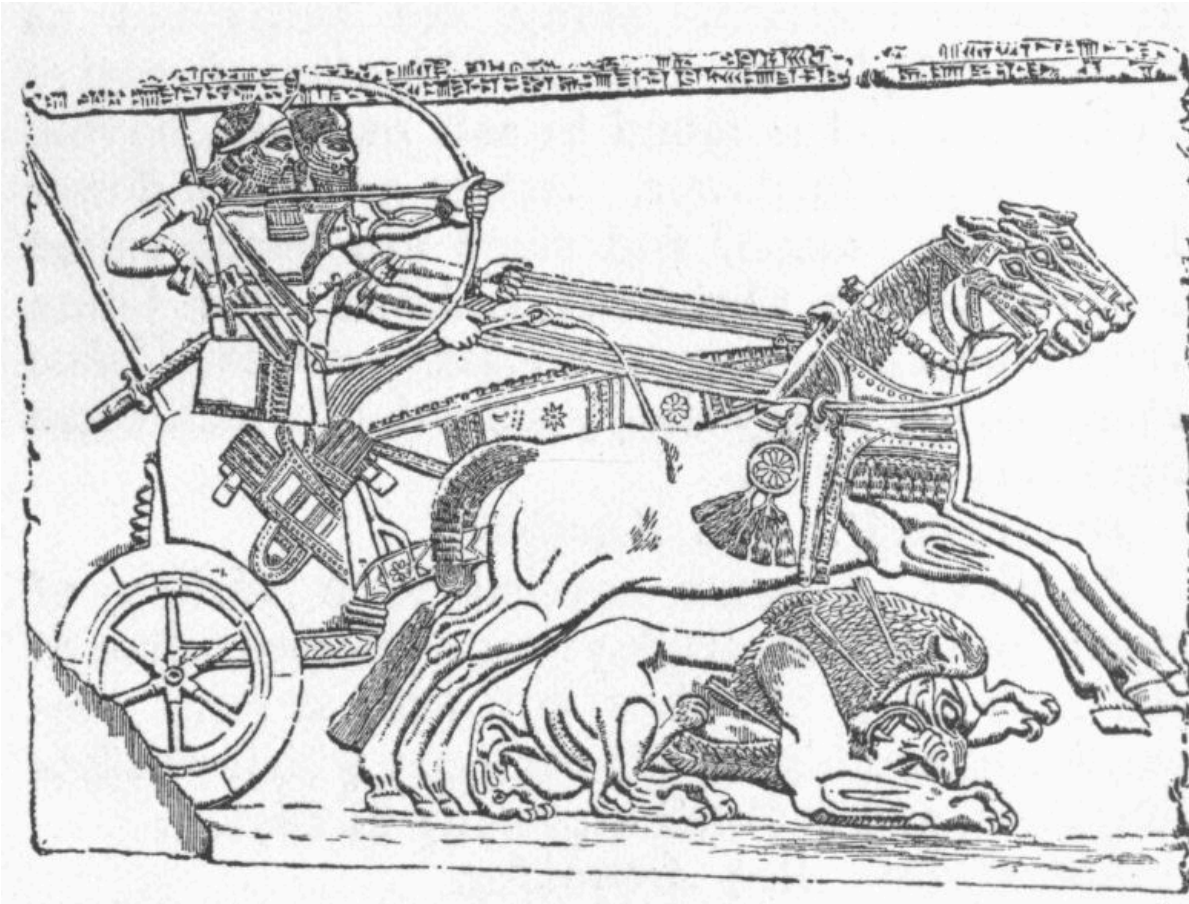
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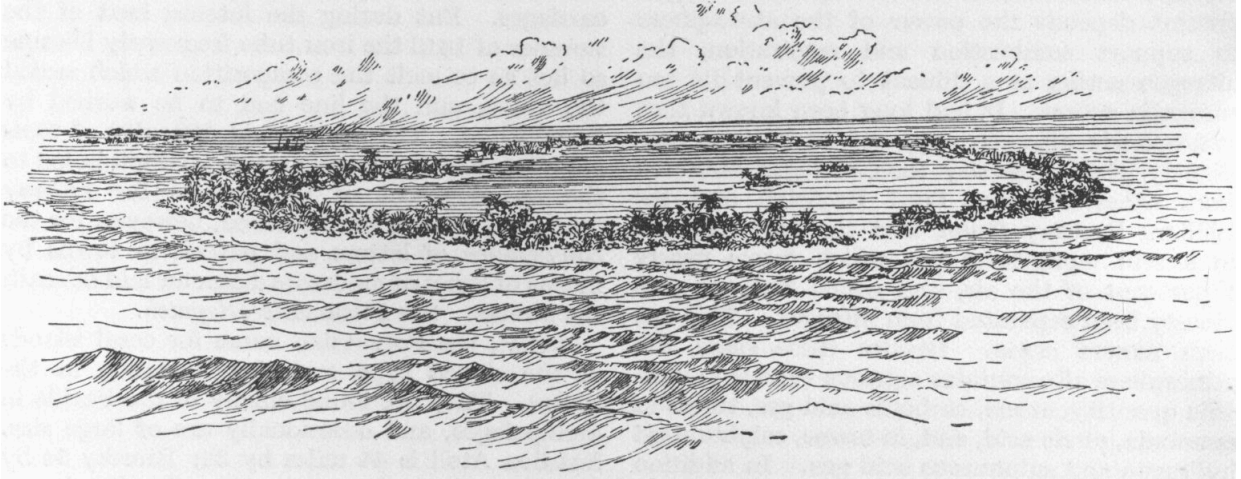
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